



TIA STANDARD

Project 25

Inter-RF Subsystem Interface Messages and Procedures for Voice Services, Mobility Management, and RFSS Capability Polling Services

TIA-102.BACA-B

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(This foreword is not part of this standard.)

This document has been submitted to APCO/NASTD/FED by the Telecommunications Industry Association (TIA), as provided for in a Memorandum of Understanding (MOU) dated December, 1993. That MOU provides that APCO/NASTD/FED will devise a Common System Standard for digital public safety communications (the Standard), and that TIA shall provide technical assistance in the development of documentation for the Standard. This document has been developed by TR8.19 (Wireline Interfaces) with inputs from the APCO Project 25 Interface Committee (APIC), the APIC ISSI Task Group, and TIA Industry members. The symbol "P25" or the wording "Project 25" appearing on the cover sheet of this document when published or in the title of any TIA-102 document or Telecommunications Service Bulletin (TSB) referenced herein, that indicates the APCO/NASTD/FED Project 25 Steering Committee has adopted the document as part of the Standard. The appearance of the abbreviation "P25" or the wording "Project 25" on the cover sheet of this document or in the title of any document referenced herein does not limit the applicability of the information contained in this document to "P25" or "Project 25" implementations exclusively. This document is being published to provide technical information on the emerging digital techniques for Land Mobile Radio Service.

This document defines an Inter-RF Subsystem Interface (ISSI) supporting TIA-102 trunked voice, trunked mobility management, and RFSS capability polling services. The ISSI is the interface between Radio Frequency Subsystems (RFSSs) enabling relevant Project 25 user requirements to be satisfied (e.g., as described in the Inter-RF Subsystem Interface Overview (TSB-102.BACC-B). Standardization efforts are planned to continue developing TIA standards defining compliance tests (e.g., measurement methods, performance recommendations, and interoperability tests) applicable to equipment implementing the TIA-102 ISSI protocols defined in this and other related TIA standards.

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DOCUMENT REVISION HISTORY

Version	Date	Description
Original	August 2006	Specify ISSI messages and procedures for trunked voice and mobility management services, replacing TSB102.BACA (December 1996 Version).
Revision A	January 2009	Incorporate TIA-102.BACA-1 (Addendum 1 Erratum 1 to fix errors and omissions), TIA-102.BACA-2 (Addendum 2 – trunked console ISSI), and TIA-102.BACA-3 (Addendum 3 - supplementary data ISSI mobility management), plus other corrections/clarifications.
Revision B	October 2012	Incorporate TIA-102.BACA-A-1 (Addendum 1 - packet data ISSI mobility management), TIA-102.BACA-A-2 (Addendum 2 – RFSS service capability polling), and TIA-102.BACA-A-3 (Addendum 3 Erratum 1 to fix errors and omissions). Modify ISSI messages and procedures to also support 2-slot TDMA. Remove/Replace “P25” throughout the document, plus other corrections/clarifications.

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1. Introduction

The Inter-RF Subsystem Interface (ISSI) supports selected services and functions that span more than a single Radio Frequency Subsystem (RFSS) to provide an extended service coverage area. To satisfy these needs, the ISSI defines a means to track and locate desired subscriber units (SUs) belonging to a given LMR system [T102], and a means to route service support to these SUs. This interface is available for LMR systems at the G reference point between RFSSs (see Figure 1 [102BACC]).

The G reference point comprises a Home part, hereafter referred to as Home RFSS, and a Serving part, hereafter referred to as Serving RFSS.

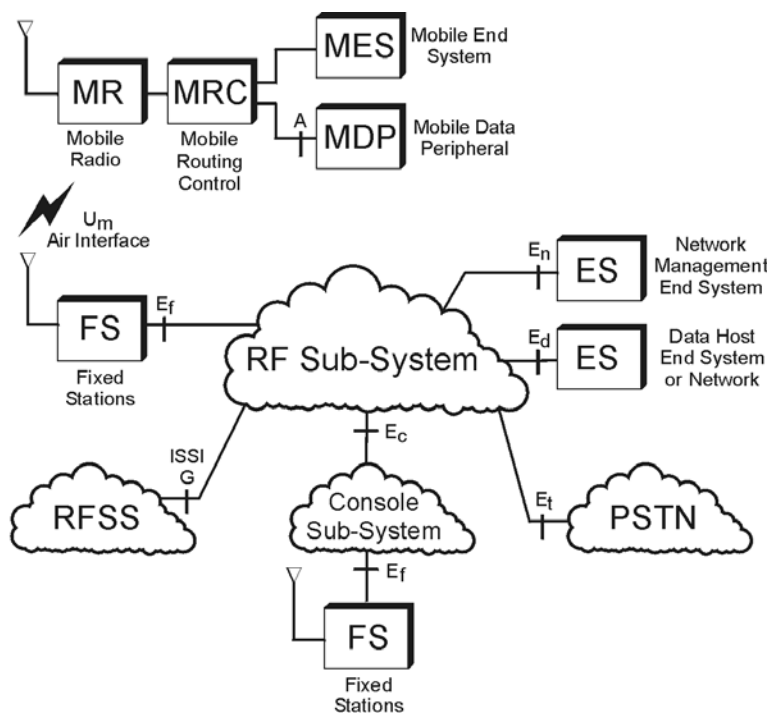


Figure 1 - Typical System Model Depicting ISSI (G Interface)

[102BACC] provides an overview of the ISSI, and documents the basic ISSI users' requirements.

1.1 Scope

This document provides a detailed specification of the messages and procedures required between RFSSs within a given LMR system, and across LMR systems, in order to provide mandatory voice services and SU mobility management.

This document focuses on protocol and message support requirements for the upper layers of the ISSI Internet Protocol Suite (the Network Layer and above), and does not address the lower layers (corresponding to the Physical and Data Link Layers of the OSI Protocol Reference Model).

This document describes the architecture, protocol suite, messages, parameters, state models, and procedures to support the following services among RFSSs over the ISSI:

Voice services:

- Group voice service for both confirmed and unconfirmed group calls,
- SU-to-SU voice service for both direct calls and calls requiring prior RF availability check.

Mobility management functions:

- Mobile SU registration,
- Mobile SU tracking,
- Group affiliation by mobile SUs and consoles,
- Group tracking,
- Authentication credential distribution²,
- SNDCP Context registration.

Note that mobility management provides functionality to support voice, supplementary data, and packet data services.

Optional ISSI support functions:

- RFSS service capability polling

In certain scenarios, the Serving RFSS needs to determine an SU's Home RFSS ID and unique IP address. This is done based on the WACN ID and System ID of the SUID as sent by the SU in the Unit Registration Request (U_REG_REQ in [102AABC]) during registration. The process by which the Serving RFSS obtains

² Until such time as ISSI security services are standardized, the protection of the Authentication credentials while traversing the ISSI is outside the scope of the ISSI. It is left up to the System owners to coordinate a method for the protected transfer of authentication credentials (as well as any other information) between their systems on the ISSI.

this information is implementation-dependent, and therefore not documented in this or other ISSI standards.

Conceptual models that define primitives and states MAY be utilized to describe TIA-102 interface protocols. The primitive exchanges and state transitions described by these models are to illustrate a conceptual design for a stack that implements the protocol and are informative. The boundaries defined by conceptual models are internal protocol stack layer boundaries defined to clarify the protocol description and are not testable boundaries.

Trunked and conventional operational details for these services and functions might differ [T102].

This document specifies ISSI which enables mobile SU and wireline console participation in trunked voice and mobility management services.

Network configuration, network engineering, and manufacturer-dependent implementation details are beyond the scope of this document.

The use of “p25” (or “P25”) in a domain name, a radical name, a parameter, an ABNF rule name (e.g., “p25-condition-code”), or a literal text string (e.g., “X-TIA-P25-HalfRate”) in TIA-102 ISSI messages and procedures throughout this document does not indicate such uses are limited to Project 25 implementations exclusively.

1.2 Terminology

In this document, the key words "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "MAY", "NEED NOT", "CAN", and "CANNOT", whether all-capitalized or not, are to be interpreted as described in Annex C of [TIA_STYLE].

The terms “protected”, “secure”, and “encrypted” are used interchangeably, depending on the immediate contexts, throughout this document.

1.3 Document Structure

Acronyms, abbreviations, and list of references appear in the following subsections. Section 2 provides an overview of the ISSI architecture and Internet Protocol Suite for voice services over the ISSI. Sections 3 through 8 contain detailed specification of the messages, state models, and procedures for mobility management, call control, Push-to-Talk (PTT) management, and RFSS service capability polling.

There are nine annexes in this specification. Annex A and Annex B are normative. Annex C through Annex I are informative and not considered a part of this specification.

Annex A lists generic SIP responses over ISSI. Annex B lists the timers and constants used in this specification. Annex C identifies TIA-102 vocoder error mitigation data. Annex D and Annex E respectively contain SU-to-SU and group call flows. Annex F and Annex G respectively contain SU-to-SU and group call flows involving multiple vocoder modes. Annex H highlights topics for future considerations. Annex I explains console features.

1.4 Definitions

For the purposes of the present document, the following terms and definitions apply.

Called RFSS: The Serving RFSS of the called SU

Called SU: The called party of an SU-to-SU Call

Calling RFSS: The Serving RFSS of the calling SU

Calling SU: The calling party of an SU-to-SU call

Console: This term is used to denote a fixed console device. A console has one SUID and can be a member of multiple SGIDs for its trunking system.

Console Patch: The console patch feature connects two or more resources that normally do not communicate with each other. Patch is not a conference call; one member of the group of resources that are patched together is allowed to speak at a time. Patch operates in a “one-to-many mode”, the same as a group call. There are two types of console patch: audio patch and re-grouped patch. Audio patch is when the console subsystem performs a switching function to connect resources that normally cannot communicate with each other. For instance, the console subsystem can cause calls on a conventional station to be heard on a trunking talk group. A re-grouped patch is when trunking resources are patched together by temporarily re-grouping all of the members of the different groups into the same talk group. This type of patch uses air interface resources in an efficient manner.

Console Resource: Anything with audio that the console position has access to: Trunking talk group, Unit-to-Unit call, conventional channel, conventional talk group, or a phone line.

Console SU: The subscriber unit of a console as a fixed device.

Conventional Complexity 1: The conventional system architecture where a console has control of the fixed station. The console can change station settings such as current channel or repeat mode. There is no mobility (fixed or static) and no air interface changes are required to enable it. If two stations are connected

together it is through console intervention only. This architecture does not use the protocol defined in this document. This architecture uses the protocol defined in [102BACE].

Conventional Complexity 2: This conventional system architecture is sometimes called “fixed mobility” because the location of conventional groups and units are fixed and known a priori. For each conventional fixed station conventional IDs (UIDs and GIDs) are mapped to fully qualified SUIDs and SGIDs. The Home RFSSs and Serving RFSSs are preconfigured with these mappings as needed. This complexity utilizes the ISSI protocol defined in this document for inter-RFSS connections. Conventional groups and units behave, from an ISSI perspective, like trunking groups and units. Control of the fixed station (selecting the channel turning repeat mode on or off, for instance) is performed by the station’s RFSS. The console does not have the ability to control the conventional fixed station under complexity 2. From a console user perspective the user interface is the same as the trunking user interface. Additional information regarding how group calls and SU-to-SU calls are made using Complexity 2 can be found in [102BACE].

Conventional Complexity 3: This conventional system architecture is similar to Complexity 2 and adds support for full dynamic mobility of conventional units. This architecture is anticipated to use the protocol defined in this document for inter-RFSS connections and requires changes to the CAI and Fixed Station Interface to support dynamic conventional mobility. This architecture is not fully defined.

Direct Call: A direct call is an SU-to-SU voice call in which the Called Serving RFSS grants an incoming call request over the ISSI without prior target availability check with the Called SU.

Ec: The TIA-102 console subsystem interface. Ec is the entire suite of standardized protocols between a console subsystem and an RFSS. The trunking version of Ec is defined herein.

FNE Home RFSS: The RFSS acting as the home of a General FNE address.

FNE Serving RFSS: An RFSS that has expressed interest in receiving Supplementary Data addressed to a General FNE address.

Full Rate: The voice coding method defined in [102BABA].

Group Call Segment: A connection between a Serving RFSS and the SG’s Home RFSS after group call set up. This connection includes group call control information exchanged between the two RFSSs during call set up time. A Full Rate, Half Rate or Native vocoder mode RTP session is not necessarily set up after a call segment has been set up.

Group Home RFSS: The Home RFSS of a subscriber group (see also the definition for “**Home RFSS**”).

Group Serving RFSS: An RFSS that has registered for a group.

Half Rate: The voice coding method defined in [102BABA-1].

Home RFSS: An RFSS which has access to directory information of an SU or a SG. For an SU, the Home RFSS tracks the current location of the SU. For an SG, the Home RFSS tracks the list of RFSSs currently serving SUs that are affiliated to the SG.

Listening (Serving) RFSS: An RFSS that has expressed interest in listening to (monitoring) supplementary data transactions of a particular unit or group.

LMR System: A collection of one or more RFSSs supporting a set of communication services across a defined coverage area. An LMR system is uniquely identified by its WACN and System ID. Because Registration Areas are always equated to System IDs, when an SU roams from one LMR system to another, it is required to perform a full registration with the new system.

Local Policy: A set of rules for behaviors that are left to the manufacturer or system operator to define.

Location Registration Area (LRA): One or more Sites of a Registration Area. There might be one or more Location Registration Areas within a Registration Area. Typically, the LRA equates to the complete coverage area of a single RFSS, but it might comprise the coverage area of multiple RFSSs. When an SU roams from one LRA to another within the same System, it is required to perform an Extended Location Registration, and might be assigned a new WUID and WGID. When an SU roams from one RFSS to another within the LRA, it is required only to perform a short-form Location Registration and maintains its WUID and WGID.

Media Tree: A set of RTP sessions being used to convey voice for an SU-to-SU call or a group call with the ISSI vocoder mode in any given RTP session being Full Rate, Half Rate, or Native.

Mobile SU: This term is used to denote a mobile radio device. A Mobile is portable and can roam. A Mobile has one SUID and can be a member of an SGID for its trunking system.

Native Vocoder Mode: The use of Native Vocoder Mode for talk spurts on a call segment means the audio is left in its original format. For example, both RFSSs have the means to send and receive Half Rate or Full Rate audio on a talk spurt by talk spurt basis.

NULL Address: A subscriber unit identity of zero (i.e., not identifying any subscriber unit within an LMR system).

P25 Condition Code: A two-hexadecimal-digit code carried in a SIP response that indicates a specific TIA-102 ISSI operational condition encountered by the responding RFSS.

Radio SU: Subscriber Unit of a radio terminal

Registration Area (RA): A region in which a subscriber unit registers for service, and over which region a registered subscriber unit might roam without subsequent registration. A Registration Area is equated to the System Identity [102AABD].

RF Sub-System (RFSS): A basic element in the network infrastructure to compose an LMR system. An RFSS is uniquely identified by its WACN, System ID and RFSS ID. An RFSS is typically associated with one or more RF sites in a system, but need not be. For example, a Home RFSS might have no associated RF sites and might operate strictly as a clearing house and voice distribution device for system elements connected to it.

Serving RFSS: An RFSS that has executed a Successful registration for a Subscriber unit. Same definition applies to a Subscriber group.

Subscriber Group (SG): A group of subscriber units, identified by an SGID, within an LMR system.

Subscriber Group Identity (SGID): A 48-bit number assigned to a group of subscriber units by the system operator. It consists of the Wide Area Communication Network Identity, System Identity, and Group Identity.

Subscriber Unit (SU): An SU can be a fixed or portable unit. A Subscriber Unit is identified by a Subscriber Unit Identity (SUID). A console is considered a fixed SU. A Mobile is considered a portable SU."

Subscriber Unit Identity (SUID): A 56-bit number assigned to a subscriber by the system operator. It consists of the Wide Area Communication Network Identity, System Identity, and Unit Identity.

SU-to-SU Call Segment: A connection between two adjacent RFSSs following an SU-to-SU call setup. This connection includes SU-to-SU call control information exchanged between the two RFSSs during call setup, and an RTP session with Full Rate, Half Rate, or Native vocoder mode.

System Identity: The 12-bit designation for an LMR system within a Wide Area Communication Network (WACN).

Talk Spurt: Voice information sent by an SU to an RFSS after being granted the right to talk.

Vocoder Mode: The audio format used by a particular talk spurt on an ISSI call segment. Audio on the ISSI can be vocoded using either the Full Rate or Half Rate vocoder mode. This document also defines the Native Vocoder Mode, and is defined above.

Wide Area Communication Network (WACN): A collection of one or more LMR systems uniquely identified by a 20-bit WACN ID.

Working Group Identity (WGID): A local (i.e., within a Location Registration Area) significant number used to identify or address a group over the CAI. The WGID is an unsigned 16-bit value corresponding to the “Group Address” defined in other TIA-102 specifications. WGIDs are assigned consistent with the reserved values of Section 2.5 (Talk Group ID) of [102BAAC]. The WGID of a group homed to the system of the RFSS assigning the WGID SHALL be the 16-bit Group Identity of its SGID.

Working Unit Identity (WUID): A local (within a Location Registration Area) significant number used to identify or address a subscriber unit over the CAI. The WUID is an unsigned 24-bit value corresponding to the “Source Address” or “Destination Address” defined in other TIA-102 specifications. WUIDs are assigned consistent with the reserved values of Section 2.4 (Source and Destination IDs) of [102BAAC]. An SU operating in its Home System SHALL be assigned a WUID which is identical to the Unit Identity portion of its SUID.

1.5 Abbreviations

ABNF	Augmented Backus-Naur Form
AGID	Announcement Group Identity
AlgID	Algorithm ID
ANSI	American National Standards Institute
APCO	Association of Public safety Communications Officials
APIC	APCO Project 25 Interface Committee
APN	Access Point Name
AVP	Audio/Video Profile
CAI	Common Air Interface
CC	Call Control
CF	Control Function
CGC	Confirmed Group Call
CRLF	Carriage-Return Line-Feed
CSRC	Contributing SouRCe
DAC	Data Access Control
DS	Differentiated Services

DSCP	Differentiated Services CodePoint
DSUT	Data Subscriber Unit Type
ES	End System
ESS	Encryption Synchronization Signaling
ESW	Encryption Sync Word
FACCH	Fast Associated Control CHannel
FED	FEDeral Government
FR	Full Rate
FS	Fixed Station
GCCF	Group Call Control Function
GMMF	Group Mobility Management Function
HR	Half Rate
ID	Identity
IEEE	Institute of Electrical & Electronics Engineers
IETF	Internet Engineering Task Force
IHL	Internet Header Length
IMBE	Improved MultiBand Excitation
IP	Internet Protocol
IPv4	Internet Protocol version 4
IPv6	Internet Protocol version 6
IS	Information Services
ISSI	Inter-RF SubSystem Interface
KeyID	Key ID
LCF	Link Control Format
LCO	Link Control Opcode
LCW	Link Control Word
LDU	Logical link Data Unit
LMR	Land Mobile Radio
LRA	Location Registration Area
LSD	Low Speed Data
M	Marker
MAC	Media Access Control layer
MAC	Message Authentication Code
MBE	Multi-Band Excitation
MDP	Mobile Data Peripheral
MDPCO	MDP Configuration Options
MES	Mobile End System
MFID	ManuFacturers ID
MIPV4	Mobile Internet Protocol Version 4
MM	Mobility Management
MMF	Master Media Function
MMP	Mobility Management Protocol
MoU	Memorandum of Understanding
MR	Mobile Radio
MRC	Mobile Routing Control
MSB	Most Significant Bit
MSC	Message Sequence Chart
MSO	Manufacturer Specific Options

MTU	Maximum Transmission Unit
N/A	Not Applicable (or Not typical used)
NASTD	National Association of State Telecommunications Directors
NAT	Network Address Translation
NIST	National Institute of Standards & Technology
NSAPI	Network Service Access Point Identifier
OSI	Open System Interconnection
OTAR	Over The Air Rekeying
P	Protected flag
P25	Project 25
PCO	Protocol Configuration Options
PD	Packet Data
PDIG	Packet Data ISSI Gateway
PDMMF	Packet Data Mobility Management Function
PDSF	Packet Data Service Function
PDU	Protocol Data Unit
PDUPM	PDU Priority Maximum
PHB	Per-Hop Behavior
PSTN	Public Switched Telephone Network
PT	Payload Type
PTT	Push-To-Talk
QoS	Quality of Service
RA	Registration Area
RCPF	Radio service Capability Polling Function
RF	Radio Frequency
RFC	Request For Comment
RFSS	RF SubSystem
RP	Radio service capability Polling
RTCP	RTP Control Protocol
RTP	Real-time Transport Protocol
SACCH	Slow Associated Control CHannel
SAP	Service Access Point
SCTP	Stream Control Transmission Protocol
SD	Supplementary Data
SDP	Session Description Protocol
SF	Superframe
SG	Subscriber Group
SGID	Subscriber Group IDentity
SIP	Session Initiation Protocol
SMF	Subordinate Media Function
SN	Sub-Network dependent convergence protocol
SN Context	"SNDP Context" mobility object
SNC	"SNDP Context" mobility object
SNDP	Sub-Network Dependent Convergence Protocol
SoR	Statement of Requirements
SSRC	Synchronization SouRCe
SU	Subscriber Unit
SUC	SU-to-SU Call

SUID	Subscriber Unit IDentity
TBD	To Be Defined
TC	Transmission Control
TCP	Transmission Control Protocol
TGID	Talk Group ID
TIA	Telecommunications Industry Association
TOS	Type Of Service
TS	TimeSlot
TSB	Telecommunications System Bulletin
TTL	Time To Live
UA	User Agent
UAC	User Agent Client
UAS	User Agent Server
UCCF	Unit Call Control Function
UDP	User Datagram Protocol
UGC	Unconfirmed Group Call
UMMF	Unit Mobility Management Function
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
UTF-8	Universal Transformation Format 8 (character set)
V	Version
VBB	Voice Block Bundling
VM	Vocoder Mode
VoIP	Voice over IP
VPN	Virtual Private Network
WACN	Wide Area Communication Network
WGID	Working Group IDentity
WUID	Working Unit Identity
X	eXtension

1.6 References

1.6.1 Normative References

The following documents contain provisions that, through reference in this text, constitute provisions of this document. At the time of publication, the editions indicated were valid. All standards are subject to revision, and parties to agreements based on this document are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below. American National Standards Institute (ANSI), IEEE, IETF, and TIA maintain registers of currently valid standards published by them.

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2. Overview of Architecture and Protocol Suite

This section describes the architecture of the ISSI (reference point G of the general system model), as illustrated in Figure 2 and is primarily informative in nature. Normative statements (i.e., statements containing SHALL, SHOULD, and MAY, in capitals or not) contained herein, however, are intended to be a part of this specification.

2.1 Architecture Overview

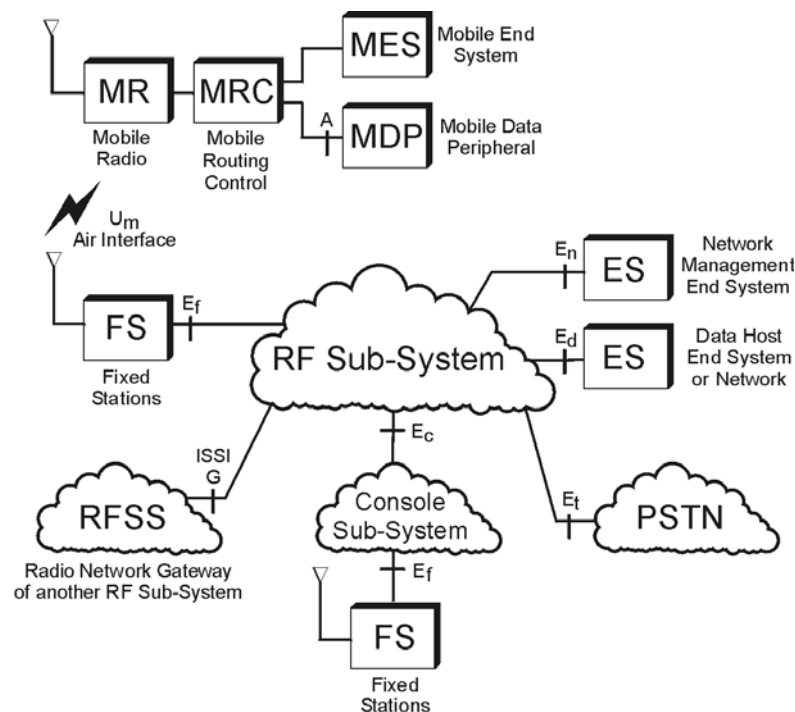


Figure 2 - General System Model

The ISSI enables inter-operability between RFSSs of various manufacturers in a wide area network. It is assumed, but is not actually required, that an RFSS includes one or more RF sites. At any given time an RF site can support calls that use the FDMA air interface, the 2-slot TDMA air interface, or both air interfaces, depending on its implementation and configuration. Subscribers located in the coverage area of an RF site can support an FDMA air interface or both FDMA and 2-slot TDMA air interfaces (Note that a subscriber at an RF site transmits using one mode for any given talk spurt). Therefore an RFSS's desired

vocoder (audio) mode for a particular talk spurt on an ISSI call segment (group or SU-to-SU) when it has SU(s) locally active in the call could be based on the capabilities of the subscribers and the capabilities of the RF sites that they are located at.

This specification models a console subsystem as an RFSS which includes consoles. An RFSS with consoles can optionally also include RF sites. An ISSI that supports consoles can then become the trunking version of Ec because it can be used to connect Trunked console subsystems to other Trunked RFSSs. Thus this document, in describing the ISSI messages and procedures to support consoles, is also describing the trunking Ec. Depending on implementation, an RFSS could support, over the ISSI, Full Rate, Half Rate, or Native vocoding modes separately or in combinations³, for its consoles. Therefore an RFSS's desired vocoder (audio) mode for a particular talk spurt on an ISSI call segment (group or SU-to-SU) could also be based on the supported ISSI vocoder mode (Full rate, Half rate, or Native vocoding modes, separately or in combinations) for the RFSS's consoles participating in the call.

An RFSS might desire that the talk spurts on a call segment be sent and received without conversion between modes. In this case the RFSS can accept Full Rate or Half Rate on a talk spurt by talk spurt basis for the ISSI call segment (group or SU-to-SU). In other words the RFSS prefers that the audio be sent and received in the Native format that the source encoded.

Thus, the ISSI might be involved between two RFSSs within the same LMR system or two RFSSs in two separate LMR systems where the systems themselves might even be in separate WACN's.

The ISSI serves three different types of RFSSs:

- Those with RF sites only
- One or more console subsystems
- One or more console subsystems with RF sites.

The internal make-up of a console subsystem is not defined by this specification. It is a collection of consoles and related equipment from a single manufacturer. The entire console subsystem interfaces to an RFSS through the ISSI. The ISSI enables different manufacturers' console subsystems to be used by the same System/RFSS.

Throughout this interface, RFSSs take on one of two roles: "home" or "serving" that are defined based on the relationship of the RFSS to the group or units involved in the particular interface function.

³ If Full Rate, Half Rate, and Native mode are offered in a combination (more than one are offered) the Answering RFSS responds with a single mode.

Generally speaking, a “Serving RFSS” is responsible for tracking a unit or group within its particular serving area, and for relaying information from or about that unit or group to its Home RFSS. A console’s subsystem is the console’s Serving RFSS. The “Home RFSS” has responsibility for keeping track of which Serving RFSS a particular unit is using, or group is active on, and for serving as the nexus of talk spurt transmissions for that group or unit. A console’s subsystem is also the console’s Home RFSS. For group calls, the Home RFSS agrees on the vocoder mode on each call segment with the Serving RFSSs. The Home RFSS therefore has the capability to limit the vocoder mode to a single mode to ensure audio is not decrypted and then re-encrypted.

Any particular RFSS might take on either role, or both roles, with respect to a particular “mobility object” (group or unit). For example, an RFSS might be strictly a Home RFSS for some particular collection of objects, or it might be strictly a Serving RFSS (i.e. home for no objects at all). An RFSS might also be a home for some objects, and yet take on a serving role for other objects. All such combinations are possible.

The ISSI employs two distinct protocol suites. The SIP protocol [RFC3261] is used for “mobility management” (tracking the virtual location of units and groups), and for “call control” (setting up interconnections between RFSSs for the later conveyance of voice). The RTP protocol [RFC3550] is used for arbitration of individual talk spurts within a call, and for the conveyance of voice between RFSSs. Both protocols operate on top of UDP [RFC0768] and IP [RFC0791].

One RTP session is set up for a group call to a console subsystem. It is the responsibility of the console subsystem to fan out the RTP packets to each affiliated console in the console subsystem.

In this document the term SU is used to denote a device which can be either a console or a mobile radio. The term console is used to denote a wireline console or similar device which is used to dispatch one or more resources in a radio system. Mobile is used to denote a subscriber radio which can roam. The term RFSS is used to denote any of the three types of RFSSs: those with RF sites, console subsystems, or console subsystems with RF sites.

2.1.1 Call Models

2.1.1.1 Call Models for LMR System

LMR systems and the TIA-102 ISSI support a variety of call types and subtypes. First among these types are “group calls” and “unit-to-unit” (or “SU-to-SU”) calls. Group calls connect collections of units that are affiliated with a particular group, identified in the ISSI by fully qualified group identifiers or SGIDs. Unit-to-unit calls connect a particular pair of units, identified by fully qualified unit identifiers, or SUIDs.

Group calls might be either “transmission trunked” or “message trunked”, as illustrated in Figure 3. In a transmission trunked call, RF resources are allocated at the beginning of a transmission (talk spurt), and de-allocated immediately after the transmission is complete. When two transmissions overlap, for example, when a console interrupts a talking SU, the call persists until the final talker completes. In this case (override), “losing audio” might be created which needs to be routed, at a minimum, to the console that is interrupting the ongoing talker.

Similar to transmission trunked calls, message trunked calls begin with the beginning of a transmission. Message trunked calls, however, persist for a “hang time” beyond the end of a transmission. Message trunked transmissions can also be overridden, creating similar losing audio scenarios.

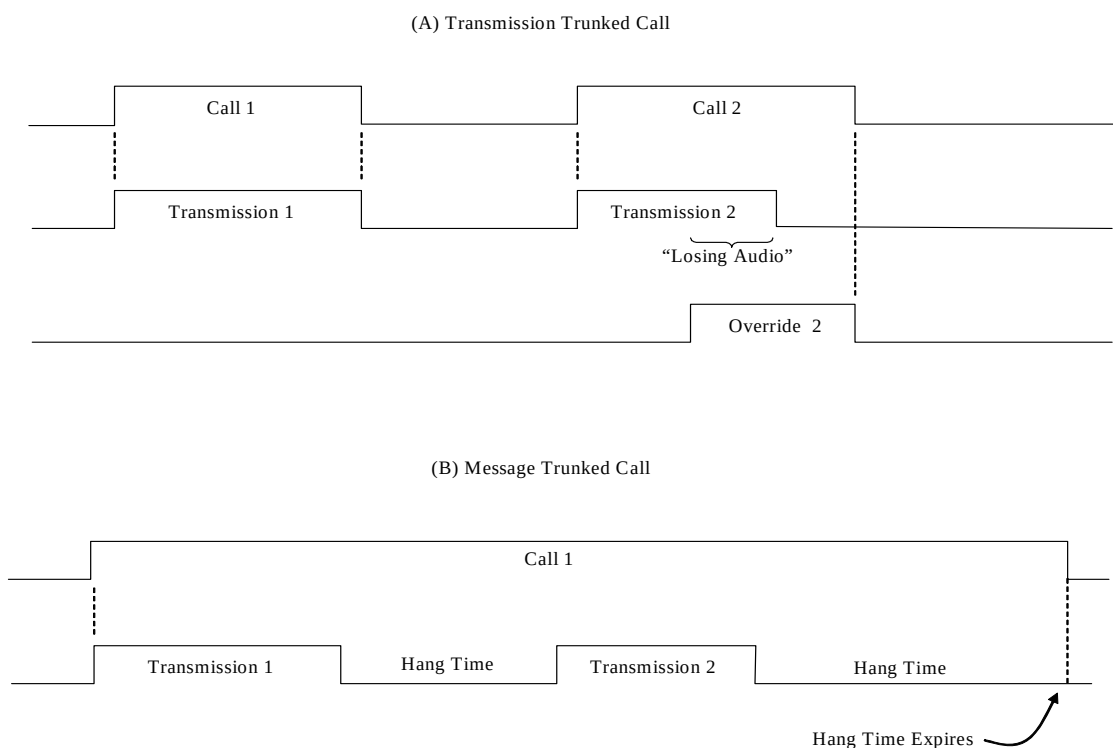


Figure 3 - Typical Call Models for LMR System

In addition to being message or transmission trunked, group calls might also be “confirmed” or “unconfirmed”. In a confirmed call, the goal is to make a “best effort” to include particular system elements (RFSSs, sites or users) in the call, i.e. to make sure that particular elements will receive the voice. When a confirmed call is setup, the initial talker is queued until all of the required resources are available, or a timeout occurs. Confirmed calls might be

transmission or message trunked. An unconfirmed call makes no particular effort to guarantee the participation of particular RFSSs, sites or users. The call begins immediately upon the beginning of the first transmission.

Likewise, in addition to being message or transmission trunked, unit-to-unit calls might (or might not) employ the optional availability check. When the availability check is employed, the system attempts to locate the destination SU and waits for a response from the user before the call proceeds. The system then makes a determination of which SU will be the initial talker in the call, depending on implementation. In some systems the called party talks first, like in a phone call, in other systems the calling party talks first like in a group call. A “direct” unit-to-unit call does not employ the availability check, and, depending on the system implementation, might behave like a confirmed or unconfirmed group call.

2.1.1.2 ISSI Call Models

The ISSI is intended to support all types of TIA-102 voice calls.

Figure 4 illustrates transmission and message trunked calls carried over the ISSI. Note that in each case, the ISSI Hang-time is independent of the hang-time for the underlying TIA-102 voice call. Thus, though the TIA-102 voice call in (a) is transmission trunked, the ISSI resources persist through a non-zero hang-timer. Similarly, in the message trunked case (b) the ISSI hang-timer is greater than that of the TIA-102 voice call. This independence from the underlying call type allows RFSSs to manage ISSI resources (SIP sessions and RTP ports) efficiently.

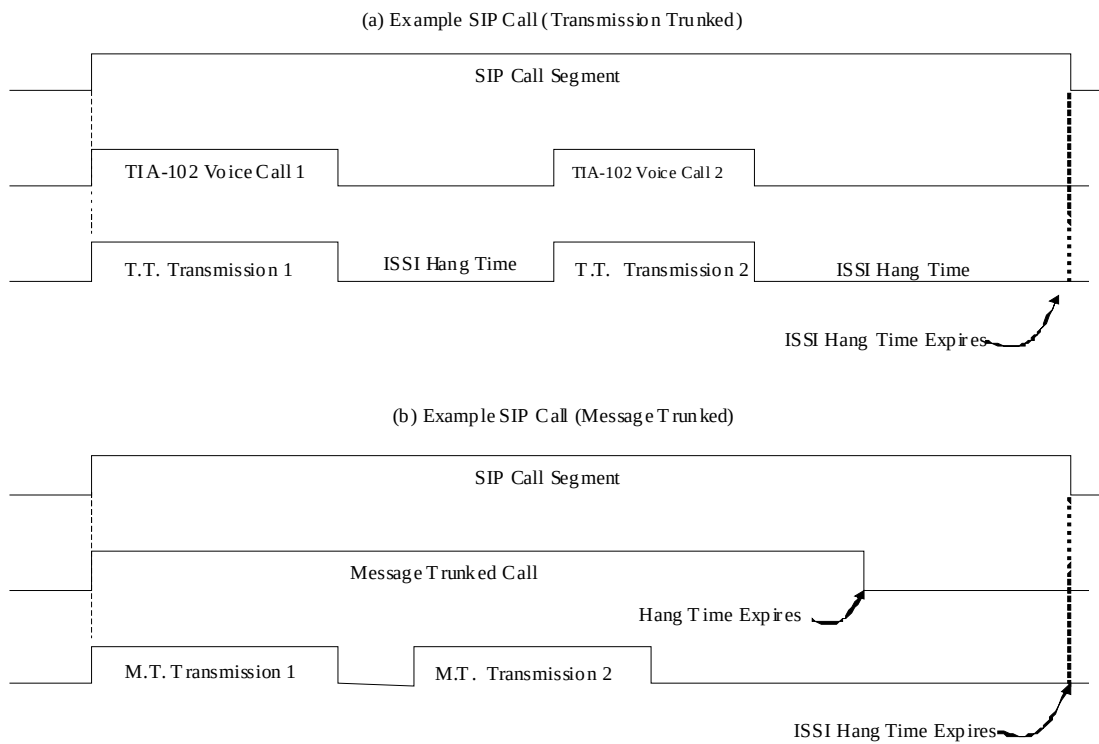


Figure 4 - ISSI Call Model

2.1.2 Subsystem Topologies

Figure 5 illustrates a typical interconnection of RFSSs for a group call. During the call, call segments are established between the Home RFSS and all of the Serving RFSSs that are involved in the call. Each segment is independent, and is conjoined to the others at the Home RFSS. Figure 5 also illustrates the voice flow for a talk spurt from Serving RFSS 1. The voice is copied and distributed (“fanned out”) by the Home RFSS to other interested Serving RFSSs. With the addition of the U_{m2} air interface, the voice flow from Serving RFSS to Home RFSS and from Home RFSS to interested Serving RFSSs can be Full Rate or Half Rate audio. It is possible, but not required, for an RFSS to be able to convert one audio format to the other. For group call the Home RFSS chooses the allowed vocoder mode(s)⁴ for the talk spurts on each particular call segment based on the home’s desired vocoder mode(s) and the desired vocoder mode(s) of the interested Serving RFSS. If the Home RFSS and an interested Serving RFSS agree upon the Native vocoder mode then the vocoder mode of the talk spurts on the associated ISSI call segment might change from one talk spurt to another.

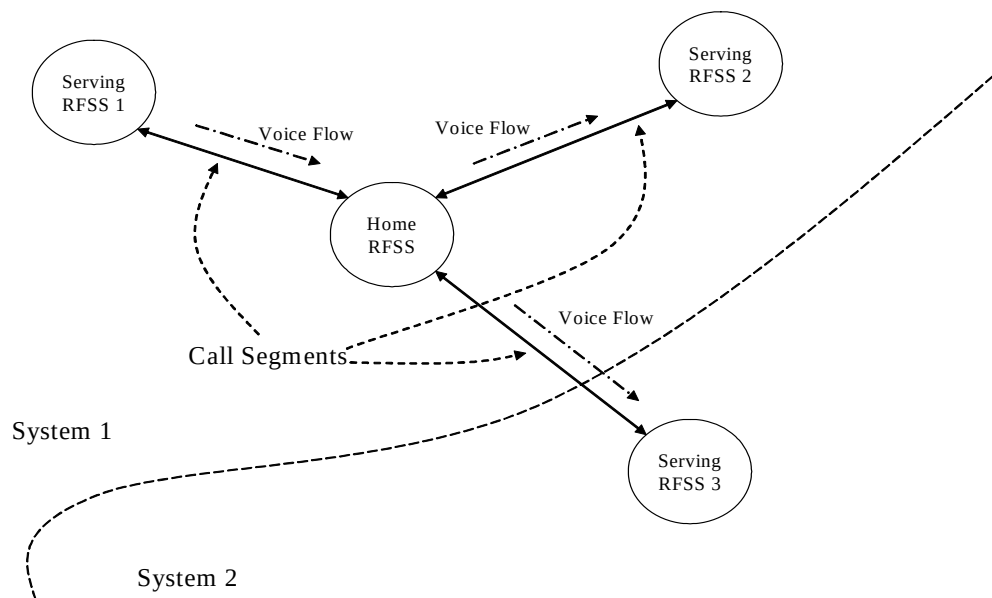


Figure 5 - Typical Group Call Interconnection Model

Figure 6 illustrates the interconnection in a Unit-to-Unit call. Here, three independent call segments are involved. One between the Serving RFSS of the party that initiates the call (the “Calling Serving”) and the Home RFSS of the

⁴ The term vocoder mode refers to the format of the audio. Audio on the ISSI can be vocoded using either the Full Rate vocoder or the Half Rate vocoder.

initiating party (the “Calling Home RFSS”), another between the Calling Home RFSS and the Home RFSS of the destination of the call (the “Called Home RFSS”) and one between the Called Home RFSS and the Serving RFSS of the destination of the call (the “Called Serving RFSS”). For unit-to-unit calls, the Calling Home RFSS plays dual roles, that of a home with respect to the Calling Serving RFSS, and that of a serving with respect to the Called Home RFSS. All of the other RFSSs operate in their normal roles. Again the voice flow is illustrated proceeding from the source Serving RFSS, through both Home RFSSs to the destination Serving RFSS. With the addition of the U_{m2} air interface, the voice flow on each of the call segments from the source Serving RFSS to the destination Serving RFSS can be Full Rate or Half Rate audio. It is possible, but not required, for an RFSS to be able to convert one audio format to the other. For Unit-to-Unit call the Calling Home RFSS and Called Home RFSS negotiate the allowed vocoder mode for talk spurts on each call segment taking into account the Calling Serving RFSS’s offered mode and the Called Serving RFSS’s answer. If a call segment’s associated RFSSs agree upon the Native mode then the vocoder mode of the talk spurts on the associated ISSI call segment might change from one talk spurt to the next.

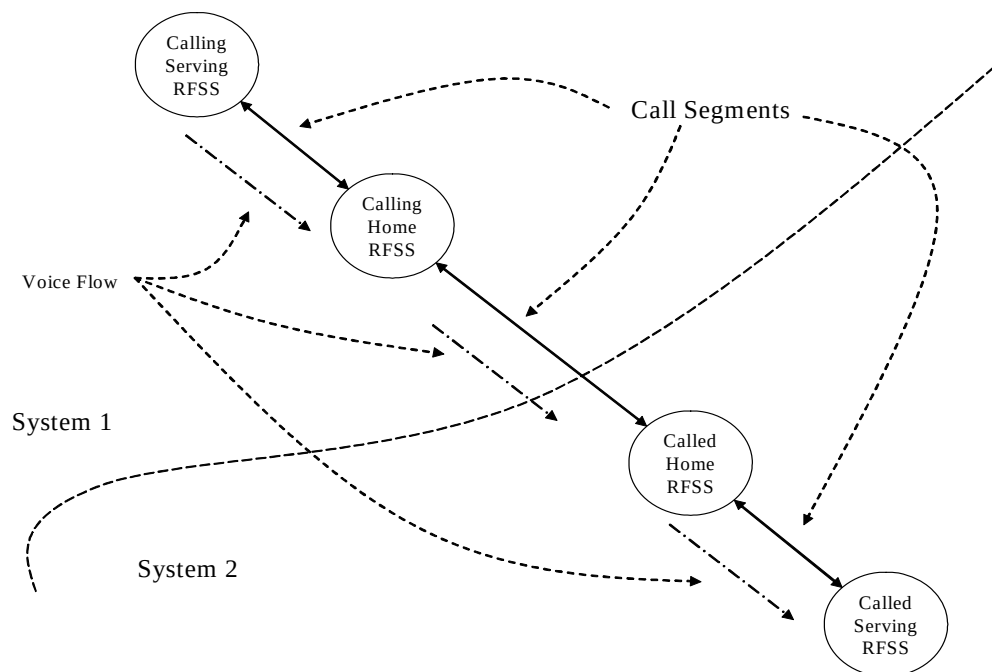


Figure 6 - Typical Unit-to-Unit Call Interconnection Model

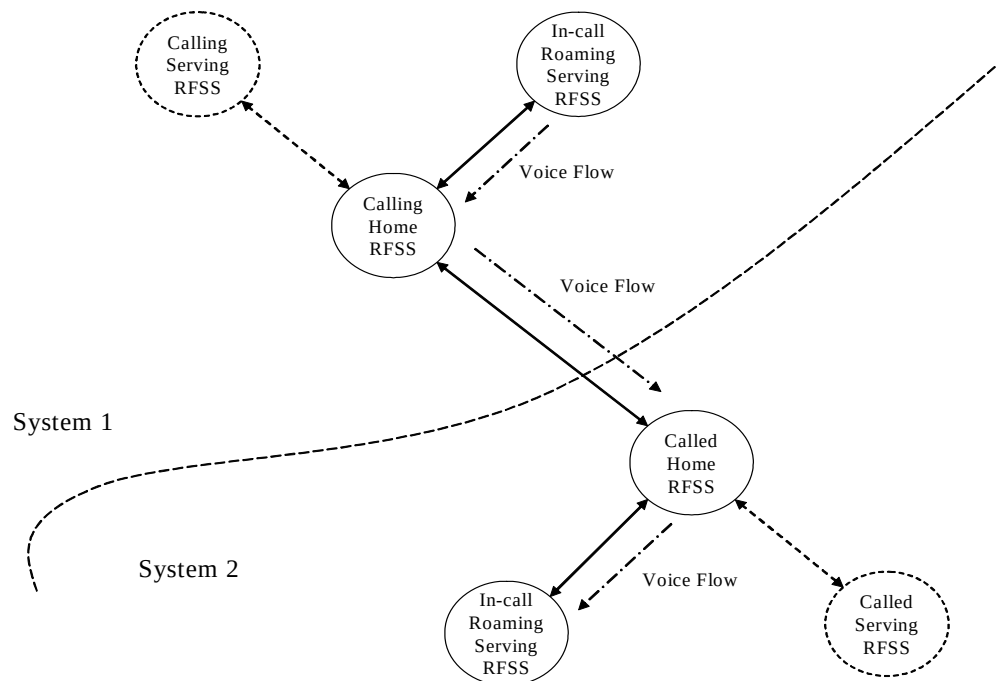


Figure 7 - Unit-to-Unit Call after In-call Roaming

Figure 7 illustrates the connections following “in-call roaming”. In-call roaming occurs when an SU that is participating in a Unit-to-unit call enters and successfully registers in a new serving domain. In the illustrated case, both SU’s have roamed to new domains. The calling and called Serving RFSSs no longer have any part in the call. An In-call roaming Serving RFSS behaves nearly identically to a Called RFSS in its relations with its Home RFSS. At various points throughout this document, however, important distinctions will be made between the two.

As suggested by the previous figures, an ISSI Call consists of the collection of call segments involved in the call, all joined at an RFSS. This organization is shown in Figure 8. Call Segments are created and destroyed through the “Call Control” functional service. During the life of the call segment, 0 or more “Talk Spurts” occur, identified by “Transmission Sequence Number” and managed through the “media” or “transmission control” functional service via RTP.

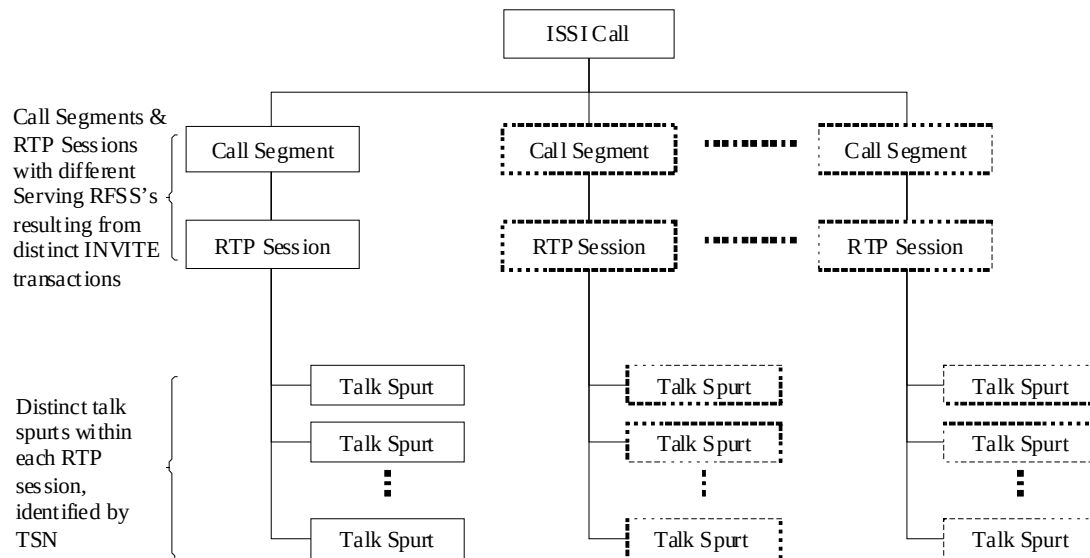


Figure 8 - General ISSI Call Model

Note also that the possible existence of “losing audio” implies that a particular call segment might have zero, one or more than one active talk spurts in each direction at any particular time. The ISSI provides mechanisms for managing these talk spurts.

2.1.3 Functional Model

In this release, the ISSI supports only inter-working between RFSSs in trunked mode using the following functional services, illustrated in Figure 9:

- 1) Mobility Management
- 2) Call Control
- 3) Media Processing (Transmission Control)
- 4) RFSS Service Capability Polling

“Mobility Management” is a set of functional services corresponding to the registration, affiliation and authentication functions of the TIA-102 air interface. Mobility Management for a console corresponds to a console’s group affiliation. Mobility management comprises two basic functional entities, the Unit Mobility Management Function (UMMF) and the Group Mobility Management Function (GMMF). An additional functional entity, the Packet Data Mobility Management Function (PDMMF), is required for TIA-102 Packet Data mobility management,

as is detailed in Section 5 of this document. Mobility Management is provided via the SIP protocol.

“Call Control” is a set of functional services related to the control signaling for ISSI calls. It includes mechanisms for call setup, call teardown, management of RTP resources, determination of the allowed vocoder mode for talk spurts on an ISSI call segment, and the coordination of confirmed calls. Call control comprises two functional entities, the Unit Call Control Function (UCCF) and the Group Call Control Function (GCCF). Call Control is also provided via the SIP protocol.

“Media Processing” (or “Transmission Control”) is concerned with the management of talk spurts within an ISSI call. It includes mechanisms for arbitrating simultaneous or nearly simultaneous call requests, for the management of losing audio, and for the conveyance of voice information. When consoles participate in group calls, media processing also includes console priority which enables higher priority console transmission requests to take precedence over a talk spurt of a lower priority console transmission request. Media Processing comprises two functional entities, the Master Media Function (MMF) and the Subordinate Media Function (SMF). Which media function is active for a particular call is determined by the role of the RFSS in the call. Media processing is provided via RTP.

“RFSS Service Capability Polling” is an optional ISSI support function that allows an RFSS to:

- Determine if it can communicate with another RFSS connected using the ISSI.
- Obtain the other RFSS's service capability information if the other RFSS supports sending this information.
- Update the other RFSS with any change in the originating RFSS' service capability information.

It includes a mechanism for an RFSS to request service capability information from another ISSI-connected RFSS. RFSS Service Capability Polling comprises one functional entity, the RFSS Service Capability Polling Function (RCPPF). RFSS Service Capability Polling is provided via the SIP protocol.

These functional services are described in detail in the following sections.

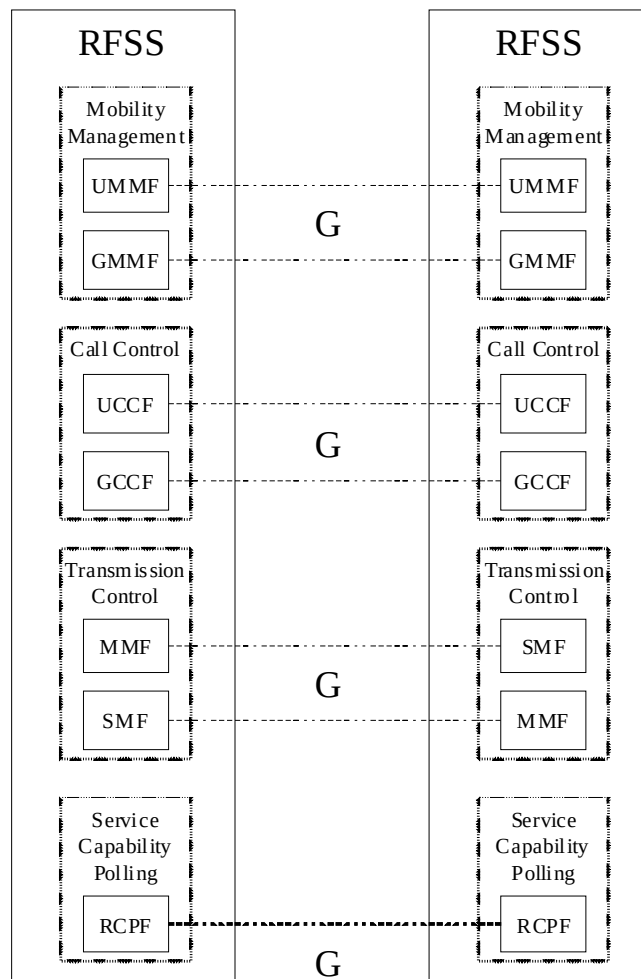


Figure 9 - ISSI Functional Entities

2.1.3.1 Mobility Management

2.1.3.1.1 Unit Mobility Management

For the purposes of this specification, each SU is uniquely identified by one and only one fully qualified SUID at any given time. Each SU of a network has a Home RFSS that has access to the directory information associated with the particular SU.

Unit mobility management services are listed below:

- SU registration
- SU tracking
- SU roaming
- SU authentication
- SU Supplementary Data Interest management
- SU Supplementary Data Monitoring Interest management
- SU Packet Data SMDCP Context management

Section 2.1.3.1.3 describes Mobility Management functionality for Supplementary Data in more detail.

Within each RFSS, the **Unit Mobility Management Function (UMMF)** functional entity is responsible for unit mobility management. The UMMF function supports different functions depending on the role of the RFSS with respect to the unit in question.

When the RFSS is a Serving RFSS relative to the unit, the UMMF is responsible for the following functions:

- Initiation of registration for Voice, Supplementary Data, and Packet Data with the Home RFSS when the SU registers within the serving domain of the RFSS.
- Authentication of the SU when the SU triggers a link layer authentication or when the service profile of the SU requires authentication
- Tracking of the location of the SU within the serving domain of the RFSS
- Removal of obsolete location information upon request of the Home RFSS such as when the Home RFSS detects that the SU has roamed into a different serving domain.
- Requesting the SU's service profile information on the ISSI only when required to support registration and authentication of the SU within the serving domain of the RFSS
- Responding to the Home RFSS when the Home RFSS requests to confirm that the RFSS still has an interest in being registered for an SU
- Forcing an SU to re-register, and re-affiliate on the air interface when the Home RFSS requests the Serving RFSS to do so
- De-registering the SU when the SU initiates de-registration with the RFSS
- Registering interest in General-FNE addressed Supplementary Data from an SU (becoming a General FNE Serving RFSS)
- Registering interest in General-FNE addressed Supplementary Data for a System (becoming a General FNE Serving RFSS) by using the General-FNE SUID for that system
- Registering interest in Supplementary Data Monitoring of an SU (becoming a Listening Serving RFSS)
- Registering interest in Supplementary Data Monitoring of a System (becoming a Listening Serving RFSS) by using the General-FNE SUID for that system
- For TIA-102 Packet Data Services:
 - o Receiving all SMDCP context parameters values from the Home RFSS upon successful registration of an SU in the Serving RFSS.

- o Providing local indication to the PDMMF that an SU is registered in this Serving RFSS with activated Sndcp contexts.
- o Providing local indication to the PDMMF when the SU has been deregistered in this Serving RFSS.
- o Subsequently updating the Home RFSS on selective Sndcp context parameter values.

When the RFSS is a Home RFSS relative to the unit, the UMMF is responsible for the following functions:

- Provisioning of authentication parameters and other service profile information to suitably authorized Serving RFSSs upon the request of the Serving RFSS
- Prohibiting registration of an SU in serving domains for which it is not authorized
- Tracking of the Serving RFSS within which the SU is currently registered (SU tracking)
- Requesting a Serving RFSS to remove obsolete location information for the SU upon registration of the SU in another RFSS (SU migration)
- Checking whether the Serving RFSS still has an interest in being registered for an SU, in accordance with local policy
- Forcing a Serving RFSS to re-register, re-affiliate, and re-authenticate an SU on the air interface in accordance with local policy
- Maintaining a distribution list of Listening Serving RFSSs with Supplementary Data Monitoring Interest for each SUID homed to the RFSS.
- Maintaining a distribution list of FNE Serving RFSSs for any General FNE address homed to the RFSS.
- For TIA-102 Packet Data Services:
 - o Retrieving Sndcp context parameter values from the PDMMF entity.
 - o Delivering all Sndcp context parameter values to the Serving RFSS as part of the registration response for the roaming SU.
 - o Providing local indication to the PDMMF when an SU is no longer registered.

2.1.3.1.2 Group Mobility Management

For the purposes of this specification, each group is identified by one and only one SGID address. Each group (SG) of a network has a Home RFSS that has access to the directory information associated with this SG.

Group mobility management services are listed below:

- Group affiliation
- Group tracking
- Group Supplementary Data Interest management
- Group Supplementary Data Monitoring Interest management

Section 2.1.3.1.3 describes Mobility Management functionality for Supplementary Data in more detail.

Within each RFSS, the **Group Mobility Management Function (GMMF)** is responsible for group mobility management. The GMMF supports different functions depending on the role of the RFSS with respect to the group in question.

When the RFSS is the Serving RFSS relative to the group, the GMMF is responsible for the following functions:

- Registration of the group with the Home RFSS when the RFSS detects it has SUs currently affiliated with the group in its serving area,
- As part of registration reporting to the Home RFSS its supported ISSI vocoder mode(s) for the group listed in preferred order (possible modes include Full Rate, Half Rate, and Native mode).
- De-registration of the group or announcement subscriber group with the Home RFSS when the RFSS determines that it no longer has affiliated members for the group,
- Responding to the Home RFSS when the Home RFSS requests confirmation that the RFSS still has an interest in being registered for the group,
- Requesting service profile information for the group on the ISSI only when the RFSS is currently registered to the SG, or is in the process of registering with the group.
- Registration of the announcement subscriber group associated with a group that the Serving RFSS has registered if it has not previously registered that announcement subscriber group.
- Registering interest in Supplementary Data for Group (becoming a Group Serving RFSS).
- Registering interest in Supplementary Data addressed to General-FNE from any SU affiliated with a Group (becoming a General FNE Serving RFSS).
- Registering interest in Supplementary Data Monitoring of Group (becoming a Listening Serving RFSS).

When the RFSS is the Home RFSS relative to the group, the GMMF is responsible for the following function:

- Providing the group service profile information to suitably authorized Serving RFSSs that request the information.
- Providing its supported vocoder modes listed in preferred order (possible modes include Full Rate, Half Rate, and Native mode) for the SG to suitably authorized Serving RFSSs that request the information.
- Prohibiting registration of the group in serving domains for which it is not authorized.

- Maintaining a list of Serving RFSSs of the SG and their supported vocoder modes (possible modes include Full Rate, Half Rate, and Native mode), if known.
- Checking whether a Serving RFSS still has an interest in being registered for the SG, in accordance with local policy.
- Maintaining a distribution list of Listening Serving RFSSs with Supplementary Data Monitoring Interest for each SGID homed to the RFSS.
- Maintaining a distribution list of Group Serving RFSSs with interest in Supplementary Data for each group homed to the RFSS.

NOTE – The list of affiliated subscriber units to a group in a Serving RFSS is not known at the SG's Home RFSS. As a consequence, the SG's Home RFSS might only control the right of a Serving RFSS to be in a group call coverage or not.

2.1.3.1.3 Mobility Management for Supplementary Data

Supplementary Data messages MAY be addressed in three ways when sent by an SU: to an individual SU, a Group, or to General FNE.

Messages addressed to an individual SU are routed to that SU's Serving RFSS, so the Serving RFSS's registration with the SU's Home RFSS is sufficient to allow Supplementary Data routing to occur.

Messages addressed to a Group are delivered to all Group Serving RFSSs that register interest in Supplementary Data for that Group with the Group's Home RFSS, so a means is necessary to register interest in a Group for Supplementary Data.

An SU sends Supplementary Data addressed to a General FNE address when the SU doesn't know the SUID of its dispatcher. The General FNE address for a given System is an SUID assigned to the fixed network equipment in that system; the SUs and the RFSS are configured with the General FNE address. The General FNE address is only used as a destination address, and never as an originating address by a mobile SU. The General FNE address is a fully qualified SUID.

Messages addressed to a General FNE address are delivered to all FNE Serving RFSSs that register interest in Supplementary Data for General FNE with the General FNE address' Home RFSS. There are three flavors of General FNE interest that can be registered:

- interest in all General-FNE-addressed Supplementary Data sent by a particular SU (also known as FNE Unit Supplementary Data)
- interest in all General FNE addressed Supplementary Data sent by SUs affiliated with a particular Group (also known as FNE Group Supplementary Data)
- interest in all General FNE addressed Supplementary Data for a particular System (also known as FNE System Supplementary Data)

A means is necessary to register interest in these three flavors of General-FNE-addressed Supplementary Data.

Finally, it is necessary to be able to monitor the Supplementary Data transactions that involve an SU, a Group, General FNE from SUs affiliated with a Group, and General FNE for a System. To register interest in Supplementary Data Monitoring, a Listening Serving RFSS registers interest with the Home RFSS of the SU, the Group, or the General FNE address, so a means is necessary to register interest for Supplementary Data Monitoring.

Therefore, the ISSI Mobility Management function supports signaling for:

- a. Supplementary Data Interest in SD addressed to
 - i. A Group
 - ii. An SU
- b. SD Monitoring Interest in
 - i. A Group, for either
 - 1. Group addressed SD, or
 - 2. General-FNE addressed SD from SUs affiliated with a Group
 - ii. SU addressed SD
 - iii. General-FNE addressed SD for a System
- c. Supplementary Data Interest in SD addressed to General FNE
 - i. From SUs affiliated with a Group
 - ii. From an SU
 - iii. For a System

The messages and procedures for Supplementary Data and Supplementary Data Monitoring over the ISSI are found in [102BACD].

2.1.3.1.4 Packet Data Mobility Management Function

The Packet Data Mobility Management Function extends mobility management by adding an “SN Context” mobility object to support the activation, deactivation, and transfer of SMDCP contexts of a roaming SU over the ISSI.

From the ISSI perspective, each SN Context object is uniquely identified by the SUID and the NSAPI in the SMDCP context activation request received from the SU.

Packet Data Mobility Management extends the use of the optional ISSI specific “SU Presence” parameter, initially defined for SU registration in this document, to the SIP registration of the “SN Context” mobility object. “SU Presence” indicates whether or not the Serving RFSS has actually confirmed the presence of the SU at the Serving RFSS immediately prior to sending the SIP registration request for the SN Context Object.

The Serving RFSS includes the SU Presence parameter with a value of “Confirmed” in the event that:

- The Serving RFSS is notifying the Home RFSS of the Sndcp context parameters provided by the SU for a new Sndcp context or,
- The Serving RFSS is notifying the Home RFSS of a change to the Sndcp context parameters for an existing Sndcp context.

The Serving RFSS includes the SU Presence parameter with a value of “Unconfirmed” when the purpose of the notification is to refresh a SIP Register timer or in accordance with local policy.

When the RFSS is a Serving RFSS for an SU, the PDMMF performs the following functions:

- Upon local indication from the UMMF that an SU object has SIP registered with at least one active Sndcp Context, create an SN Context object for each of the active contexts.
- Upon receipt of an SU request to activate a new Sndcp context, create an SN Context object with appropriate Sndcp context parameters provided by the SU. Then SIP Register the SN Context Object with the Home RFSS with the SU Presence parameter set to Confirmed
- Providing local indication to the Packet Data Service Function (PDSF) when a given Sndcp context no longer exists.
- Querying the Home RFSS for an SN Context in accordance with local policy.
- Unsolicited delivery of updated Sndcp context parameters to the Home RFSS any time after SIP Registration of an SN Context object. The SIP Register request includes the SU Presence parameter with a value of Confirmed
- Responding to a query received from the Home RFSS for an SN context object.
- Upon receipt of a request from the Home RFSS to activate a specific Sndcp context, send a context activation command to the SU.
- Upon receipt of a request from the Home RFSS to deactivate a specific Sndcp context, send a context deactivation request to the SU.
- Upon receipt of an SU request to deactivate an Sndcp context, SIP Deregister the SN Context object with the Home RFSS.
- Removing obsolete SN Context object(s) upon local indication from the UMMF that the SU is no longer registered in the Serving RFSS.
- Send a SIP Register request for an SN Context object to the Home RFSS with the SU Presence parameter set to “NOT Confirmed” in accordance with local policy to refresh the SIP Register Timer.

When the RFSS is the Home RFSS of an SU, the PDMMF performs the following functions:

- Prohibiting SIP registration of SN Context object(s) for an SU in a Serving RFSS for which it is not authorized.
- Upon receipt of a SIP Registration request for an SN Context object that does not already exist from a Serving RFSS, create an SN Context object with the SNDCP context parameters provided and complete the SIP Registration.
- Assigning mobile host address to the SU or MDP associated with the SN Context, if requested by the Serving RFSS (dynamic address assignment).
- Checking whether the IP address supplied by the SU or MDP is consistent with that provisioned by the Home RFSS, if requested by the Serving RFSS (static IP address assignment)
- Upon receipt of a SIP Register request for an SN Context object that already exists with the SU Presence parameter set to Confirmed from the Serving RFSS, update the SNDCP context parameters.
- Responding to a query received from a Serving RFSS for an SN context object.
- Querying the Serving RFSS for an SN Context in accordance with local policy.
- Requesting a Serving RFSS to activate a specific SNDCP Context in accordance with local policy.
- Requesting a Serving RFSS to deactivate a specific SNDCP Context in accordance with local policy.
- Removing obsolete SN Context object(s) upon local indication from the UMMF entity that the corresponding SU is no longer registered.
- Upon receipt of a SIP Deregistration request from the Serving RFSS, remove the SN Context object.
- Upon receipt of a SIP Register request for an SN Context object from the Serving RFSS with the SU Presence parameter set to NOT Confirmed, refresh the SIP Register Timer for the SN Context object.

2.1.3.2 Call Control

The Call Control functional service is responsible for call control signaling between RFSSs at set up and at call release. In this model, separate functional entities are defined for Unit-to-Unit call control and Group Call control, which are described in detail in the following sections.

The SIP protocol (See **[RFC3261]**) is used for call control over the ISSI.

RF Resources, for the purposes of a console, is to be construed throughout the document as referring to analogous resources (e.g. talk-paths or link bandwidth) of a console. No change has been made to the sections of this document which discuss the indication of the availability of RF Resources. When consoles are

considered, RF resources refer to the console's similar, limited availability resources.

2.1.3.2.1 Unit Call Control

Unit-to-Unit call control comprises the following supported services:

- Negotiation of SU-to-SU call type (with or without the optional availability check)
- Negotiation of the vocoder mode that can be used with the talk spurts on each SU-to-SU ISSI call segment
- Setup of SU-to-SU calls that employ the optional availability check
- Setup of SU-to-SU calls that do not employ the optional availability check
- SU-to-SU call termination
- Call transfer during SU roaming
- Identification of the Initial Talking Radio

The **Unit Call Control Function (UCCF)** is responsible for SU-to-SU call control, including but not limited to the management of RTP resources within the RFSS. The UCCF supports different functions depending on the role of the RFSS in the unit-to-unit call. Figure 10 illustrates the four possible roles. In the figure, the "Calling RFSS" is the Serving RFSS that provides services to the SU that is the initiator of the call. The "Calling Home" RFSS is the Home RFSS of the SU that is the initiator of the call. The "Called Home" RFSS is the Home RFSS of the destination of the call, and the "Called RFSS" is the Serving RFSS providing services to the destination SU. Regardless of the role of the RFSS, it is the UCCFs of the participating RFSSs that communicate for SU-to-SU Call Control.

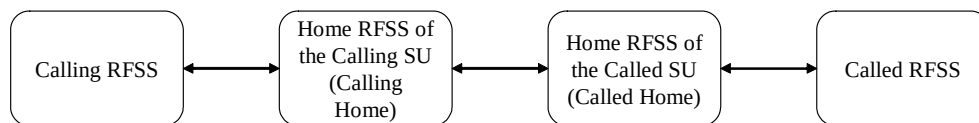


Figure 10 - Unit-to-unit Call Functional Architecture

Figure 10 illustrates the case where there are four distinct RFSSs involved in the call. In general, adjacent UCCFs might be co-located in the same RFSS. In such a case the interface between these two co-located functions are internal to the RFSS and as such is beyond the scope of the ISSI definition. Similarly, non-adjacent UCCFs might be co-located in the same RFSS. In that case, both functions are executed without any direct interaction between them.

The UCCF of the calling RFSS is responsible for the following functions:

- Initiation of an SU-to-SU call set up with the calling Home RFSS when an SU initiates an SU-to-SU call.
- Participation in the determination of the type of unit-to-unit call (with or without availability check)
- Participation in the determination of the allowed ISSI vocoder mode for the talk spurts on the call segment to the Calling Home RFSS.
- Propagation of call preferences.
- Propagation of SU-to-SU call set up from the Calling Home RFSS to the Calling SU after an In-Call Roaming of the Calling SU
- Release of the call upon request by the Calling Home RFSS
- Requesting release of the call to the Calling Home RFSS if the Calling SU requests to release the call.
- Release of the call upon loss of RTP connectivity, on RTP resource preemption, RF resource preemption, or other local policy.

The UCCF of the Calling Home RFSS is responsible for the following functions:

- Propagation of an SU-to-SU call set up from the Calling Serving RFSS to the Called Home RFSS,
- Participation in the determination of the type of unit-to-unit call (with or without availability check)
- Participation in the determination of the allowed ISSI vocoder mode for the talk spurts on the call segments to the Called Home RFSS and to the Calling Serving RFSS.
- Propagation of call preferences.
- Propagation of session progress information from the Called Home RFSS to the Calling Serving RFSS
- Bridging of the RTP session set up with the Calling Serving RFSS and the RTP session set up with the Called Home RFSS
- Propagation of the call release between the Called Home RFSS and the Calling Serving RFSS,
- Releasing of the call on loss of RTP connectivity, on RTP resource preemption or other local policy.
- Establishment of new call segments when the calling SU, after having successfully set-up an SU-to-SU Call, roams to a new Serving RFSS, and release of the segment related to the previous Calling Serving RFSS

The UCCF of the Called Home RFSS is responsible for the following functions:

- Locating the Serving RFSS currently supporting the Called SU
- Determination of the type of unit-to-unit call (with or without availability check)
- Determining the transmitting SU and holding off the start of the call until RF resources are available on both Serving RFSSs
- Participation in the determination of the allowed ISSI vocoder mode for the talk spurts on the call segments to the Called Serving RFSS and to the Calling Home RFSS.
- Propagation of call preferences.

- Propagation of SU-to-SU call set up from the Calling Home RFSS to the Called Serving RFSS.
- Propagation of session progress information (such as ringing, queued and grant information) between the Called Serving RFSS and the Calling Home RFSS,
- Bridging of the RTP sessions set up with the Calling Home RFSS and the Called Serving RFSS,
- Propagation of the call release between the Calling Home RFSS and the Called Serving RFSS.
- Release of the call when requested by the Calling Home RFSS or the Called Serving RFSS.
- Release of the call on loss of RTP connectivity, on RTP resource preemption or other local policy.
- Establishment of new call segments when the called SU, after having successfully set-up an SU-to-SU Call, roams to a new Serving RFSS, and release of the segments related to the previous Called Serving RFSS

The UCCF of the called RFSS is responsible for the following functions:

- Propagation of SU-to-SU call set up from the Called Home RFSS to the Called SU.
- Acceptance of the type of unit-to-unit call (with or without availability check)
- Participation in the determination of the allowed ISSI vocoder mode for the talk spurts on the call segment to the Called Home.
- Transfer of information about the session progress such as ringing, queued and grant information to the Called Home RFSS.
- Release of the call when the Home RFSS of the Called SU requests to release the call
- Requesting release of the call to the Called Home RFSS if the Called SU requests to release the call.
- Releasing of the call on loss of RTP connectivity, RTP resource preemption, RF resource preemption, or other local policy.

2.1.3.2.1.1 Media Tree Topology for SU-to-SU Calls

The Home RFSS of the Calling SU and the Home RFSS of the Called SU allocates media resource (RTP ports, processing capability and so on) in order to support supplementary services such as recording or discreet listening from a console.

After an SU-to-SU Call set up, if the four functional entities are involved on the ISSI, 3 RTP sessions are set up and used to convey voice information between the Calling SU and Called SU.

Consequently, if the four functional entities are involved on the ISSI, the media tree topology for an SU-to-SU call is as illustrated in Figure 11.

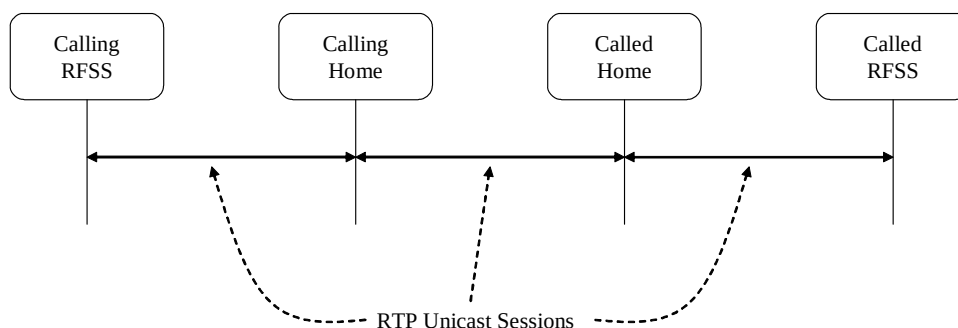


Figure 11 - Unit-to-unit Call Media Tree Topology

2.1.3.2.1.2 RTP resource management for SU-to-SU Calls

As noted above, the UCCF is responsible for RTP resource management of the underlying call segments. For SU-to-SU calls, if no RTP resource is available to set up a call segment, or if at any time the previously established RTP resources are pre-empted and reallocated for some other call, the SU-to-SU call is not setup, or is released.

2.1.3.2.1.3 RF resource management for SU-to-SU Calls

The UCCF isn't responsible for RF resource management directly, but it is responsible for delaying the start of the call until both Serving RFSSs have RF resources to service the call, and for tearing the call down if allocated RF resources are preempted.

2.1.3.2.1.4 Vocoder Mode Selection for SU-to-SU Calls

As noted above, the UCCF is responsible for determining the allowed vocoder mode associated with the talk spurts on each individual call segment for the SU-to-SU call. The UCCFs in the adjacent RFSSs negotiate the allowed vocoder mode for the talk spurts on their call segment based on the offered vocoder mode(s) and their own capabilities and requirements. If a common mode between any two adjacent RFSSs involved is not available then the call is not set up.

2.1.3.2.2 Group Call Control

Group Call Control comprises the following services:

- Group call set up
- Group call segment set up, including determining the allowed vocoder mode for a call segment's talk spurts

- Group call segment termination
- Group call termination
- Confirmed Call Management

The **Group Call Control Function (GCCF)** is the functional entity activated in the RFSS which is in charge of the Call Control for the group call. The GCCF supports different functions depending on the role of the RFSS with respect to the group in question.

The GCCF of the Serving RFSS is responsible for the following functions:

- Requesting a group call set up when an SU triggers a group call request in its area for a group for which no call currently exists
- Informing the Home RFSS of its offered vocoder mode(s) when it requests that a group call be set up
- Participating in the establishment of new call segments in response to requests from the Home RFSS
- Releasing a group call segment with the Home RFSS on loss of RTP connectivity or other local policy condition
- Once a group call segment has been set up, tearing down or setting up RTP sessions upon preemption or Home RFSS request.
- Communicating to the Home RFSS the RTP session parameters
- Notifying the Home RFSS of the availability or lack of availability of RF resources required for a confirmed call
- Requesting a change of group call priority based on local policy
- Requesting a change of its call segment's allowed vocoder mode based on local policy

The GCCF of the Home RFSS is responsible for the following functions:

- Establishing group call segments in response to group call setup requests from Serving RFSSs or SU PTT triggers within its own area, including:
 - to get the list of registered Serving RFSSs from the GMMF
 - Collaborating with the registered Serving RFSSs to determine the allowed vocoder mode for the talk spurts on each call segment
 - to initiate group call set ups with the registered Serving RFSSs
 - to bridge RTP sessions which are set up with the Serving RFSSs
 - to handle INVITE and re-INVITE request collisions
- On-going management of group call segments during a call, including
 - To initiate group call segment set ups with any registered Serving RFSSs not already joined to the call (late call entry)
 - To inform a registered Serving RFSS of a requested change of the allowed vocoder modes for the talk spurts on the call segment, based on the Serving RFSS's request and/or based on local policy
 - To handle a group call segment set up when it receives a group call set up request from any Serving RFSS
 - Once group call segment has been set up, tearing down or setting up RTP

sessions upon preemption or Serving RFSS request.

- Releasing the group call on ISSI hangtime⁵ expiry or other local policy condition
- Releasing a group call segment with a Serving RFSS on loss of RTP connectivity or other local policy condition
- Communicating to each Serving RFSS the RTP session parameters associated with its call segment
- Managing the establishment and maintenance of confirmed group calls
- Changing the group call priority upon Serving RFSS request or upon its own local policy.

2.1.3.2.2.1 Group Call Topology

Figure 12 illustrates the media tree topology for group calls. Call segments, and consequentially RTP unicast sessions within the call segments, are created between the Home RFSS of the group and the Serving RFSSs registered to the group. After a group call is set up and RTP resources are available, the Home RFSS bridges the RTP sessions it has set up with each Serving RFSS.

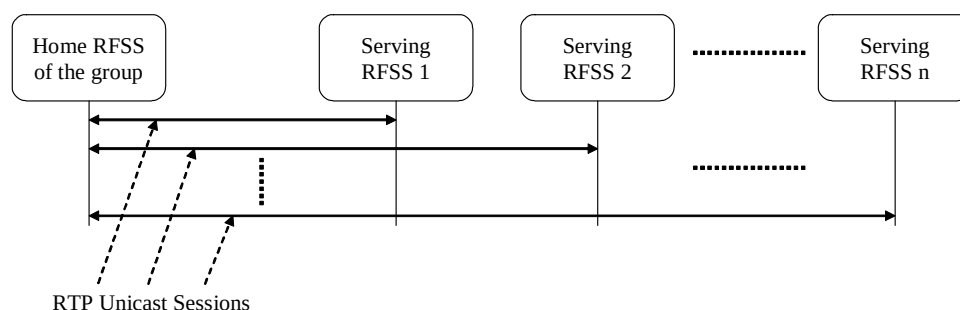


Figure 12 - Group Call Media Tree Topology

2.1.3.2.2.2 Group Call RTP Resource Management

As noted above, the GCCF is responsible for the management of RTP resources within the context of the call segments that comprise the calls. The following rules govern the management of these resources.

If no RTP resource is available for a given call segment, and no other resource can be preempted for the benefit of this segment, the group call segment is set up but without an RTP session. The RFSS (the Home RFSS or the Serving RFSS) which does not have resources for the call segment is thus able to track the progress of the call and, upon new resources becoming available, to

⁵ Hangtime on the ISSI is not correlated to the RF resource hangtime. Generally, hangtime on the ISSI is longer than RF resource hangtime.

communicate with its peer RFSS to set up the RTP session associated with this call. The peer RFSS, in turn, waits for the RFSS which is not able to serve this call to have available RTP resources.

Likewise, if an RTP resource of a group call segment is preempted at an RFSS, the resources might be re-allocated while the group call segment persists. In this case, the same mechanism as described above enables an RFSS to limit the total number of call requests it needs to service.

2.1.3.2.2.3 Group Call RF Resource Management

For confirmed group call, the Home RFSS needs to be informed of RF resource availability in the Serving RFSS and start the call when the requisite resources are available. For unconfirmed group call, no consideration of RF resource availability is required.

2.1.3.2.2.4 Group Call Vocoder Mode Selection

As noted above, the GCCF is responsible for the selection of the allowed vocoder mode within the context of the talk spurts on each call segment that comprise the group call. Each call segment's allowed vocoder mode can be negotiated independently of the other call segments. The following rules govern the selection of the allowed vocoder mode.

The GCCF in the initiating RFSS (Home or Serving) indicates in its INVITE which mode or modes it can support for the call segment's talk spurts, taking into account the supported vocoder mode information that it possibly received from the receiving RFSS during group registration over the ISSI. The GCCF in the receiving RFSS (Serving or Home) responds back with its choice of the offered modes. The mode agreed upon becomes the allowed mode for the talk spurts on the call segment. The Home and Serving RFSSs, when initially sending their INVITE, can use their possible knowledge of the supported vocoder mode(s) of the other RFSS associated with the call segment. The knowledge of the supported mode can be determined before the initial call request through the Serving RFSS's SG registration. The GCCF in one RFSS informs the other when it wishes to change the allowed vocoder modes for the call segment. The receiving RFSS takes action on this information based on its capability and local policy.

2.1.3.3 Media processing

Media processing is a functional service which is responsible for:

- Push-To-Talk (PTT) management of competing talk spurts, and
- Conveyance of Full Rate, Half Rate, or Native mode voice information and other signaling between RFSSs participating in a call.

Media processing employs RTP **[RFC3550]**. Each media tree established on the ISSI is composed of a set of RTP unicast sessions. The functional entities

involved in the Media processing on the ISSI are located at the RTP unicast session endpoints and are responsible for proper execution of the ISSI Voice protocol.

Two functional entities are involved in media processing, the **Master Media Function (MMF)** and the **Subordinate Media Function (SMF)**.

The MMF is responsible for the following functions:

- Arbitration of competing talk spurts coming from different SMFs, including arbitrating between a console and mobiles and between multiple consoles.
- Fanout (bridging) of RTP sessions between SMFs, including copying RTP packets from a source SMF to all other SMFs participating in the call.
- Fanout of preempted talk spurts to all SMFs interested in receiving losing audio (for Group Calls).
- Monitoring of RTP session connectivity and reporting to the GCCF or UCCF in case of loss of connectivity.
- Monitoring of voice payloads and reporting to the GCCF or UCCF for ISSI hang-time management. In SU-to-SU calls, this function only applies for the Called Home RFSS.

The SMF is responsible for the following functions:

- Arbitration of competing talk spurts within its serving area, including arbitrating between a console and mobiles and between multiple consoles.
- Propagation of PTT Transmit Requests, voice activity and progress to the MMF
- Management of MMF responses to arbitration requests
- Fanout of voice activity within its serving area
- Monitoring RTP session connectivity and reporting to the GCCF or UCCF in case of loss of connectivity.

2.1.3.3.1 Media Processing for SU-to-SU Calls

Figure 13 illustrates media processing for SU-to-SU calls. The Calling and Called RFSSs each employ SMFs and the Called Home RFSS employs an MMF. In a unit-to-unit call, the Calling Home RFSS bridges a media session between itself and the Calling RFSS (in which it uses the transmission control protocols of the MMF) with another media session between itself and the Called Home RFSS (wherein it takes the role of an SMF). In essence the Calling Home RFSS serves as a proxy for the Calling RFSS as far as the transmission control protocols are concerned. For Unit-to-unit calls, talk spurts are transmitted transparently from the Calling RFSS to the Called RFSS in full duplex mode.

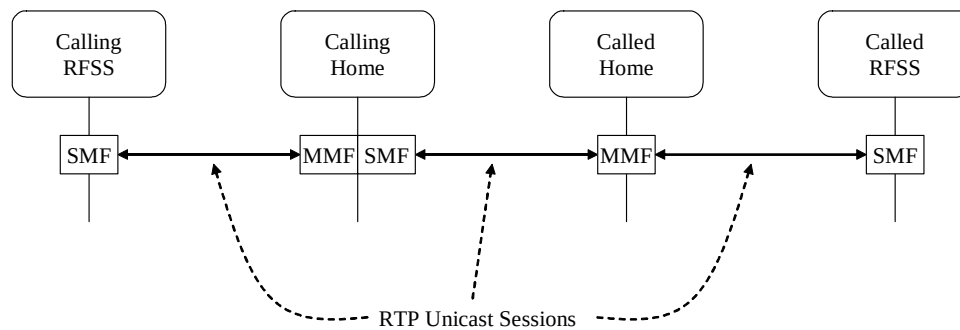


Figure 13 - Unit-to-Unit Call Media Processing

2.1.3.3.2 Media Processing for Group Calls

Figure 14 illustrates media processing for group calls. For group calls, the Home RFSS of the group provides the MMF and the Serving RFSSs employ SMFs.

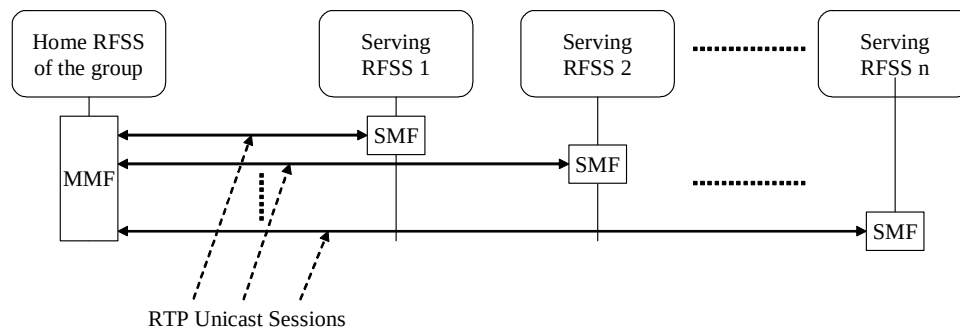


Figure 14 - Group Call Media Processing

2.1.3.4 RFSS Service Capability Polling

RFSS Service Capability Polling is a function that allows an RFSS to determine if it can communicate with another RFSS connected using the ISSI, and if it can, to understand the service capability of another RFSS connected by the ISSI. An RFSS (the polling RFSS) uses the SIP OPTIONS method (See [RFC3261]) to request the service capabilities of another RFSS (the polled RFSS). The other RFSS acknowledges the request with either a successful response (200 OK) or a SIP Final Error response (according to [RFC3261], response to the SIP OPTIONS method is mandatory). A response indicates that the polling RFSS has basic ISSI connectivity to the polled RFSS, and a successful response

optionally carries more detailed RFSS service capability information for use at the polling RFSS according to local policy. If no response is received the polling RFSS is able to assume that that it has no ISSI connectivity to the polled RFSS whose response failed.

The **RFSS Service Capability Polling Function (RCPF)** is the functional entity responsible for:

- Requesting the service capabilities of another RFSS
- Replying to the service capability requests of other RFSSs
- Communicating changes in the service capabilities of an RFSS to other RFSSs

2.1.4 Service Access Point Model

For the purposes of this specification, the relationship between the ISSI protocols is as illustrated in

Figure 15. An “ISSI Control Function” (CF) uses the Mobility Management Service Access Point (MM-SAP), the Call Control Service Access Point (CC-SAP), the Transmission Control Service Access Point (TC-SAP) and the RFSS Service Capability Polling Service Access Point (RP-SAP) to access the capabilities of each protocol, and to receive information from each protocol. The Control Function is responsible for coordinating activities between the protocols, for managing resources, and for other similar functions.

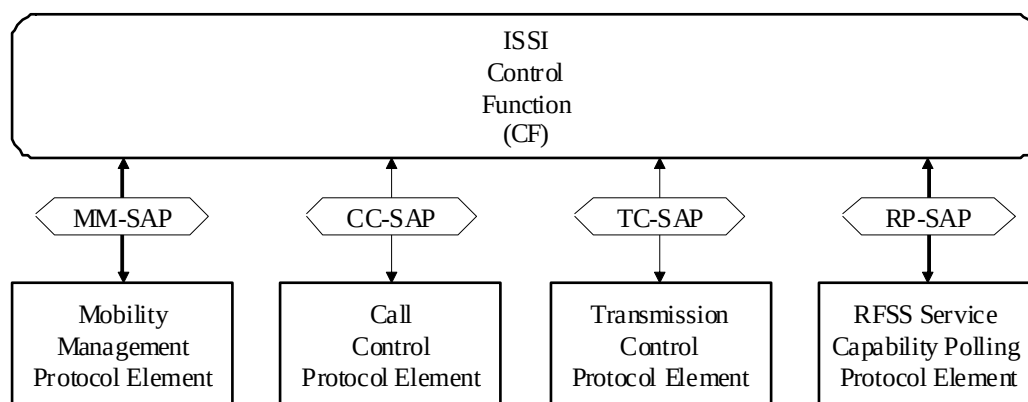


Figure 15 - Service Access Point Model

Each access point consists of functions initiated by the Control Function (Requests) and functions initiated by the Protocol (Indications). In turn, both

requests and indications have three transaction styles, which are illustrated in Figure 16.

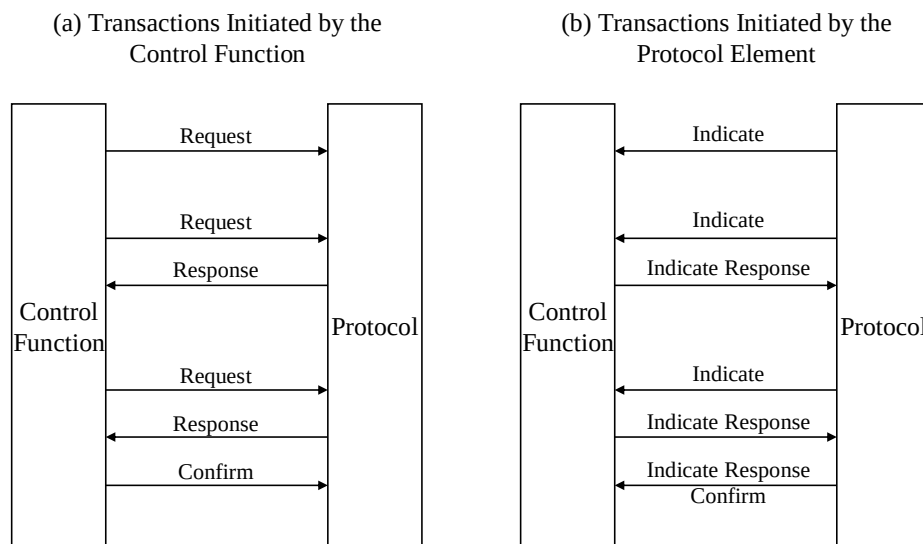


Figure 16 - SAP Transactions

As can be seen from the figure, some requests from the control function require no feedback from the protocol layer. Others require a simple “Response”, and still others require a response that is in turn “Confirm(ed)” by the control function. Similarly, transactions “Indicate(d)” by the Protocol function might stand alone, or might require an “Indicate Response” or an “Indicate Response” followed by an “Indicate Response Confirm(ation)”.

The primitives for each SAP represent operations performed on objects of the ISSI, such as creating, modifying, or deleting such objects. For the MM-SAP, the objects are subscriber unit (SU), talk group (SG), and Sndcp Context (SN Context). They are identified by the SU identifier, group identifier, and (SU identifier, NSAPI) tuple, respectively. For the CC-SAP, the objects are RTP connections and SIP dialogs. A RTP connection is identified by the IP addresses and UDP port numbers for both end-points. The SIP dialog, once established, is identified by the SIP call-ID and tags. For the TC-SAP, objects are talk spurts and they are identified by the pair: (RTP connection, talk spurt transmission number). For the RP-SAP, objects are RFSSs whose service capability is being polled. Each primitive contains a number of parameters, which will be described in their respective sections on the protocols.

Throughout this document, SAP primitives are identified in a notation consisting of a two letter prefix indicating the SAP (MM, CC, TC), followed by an underscore, followed by a primitive name (e.g. “Registration”), followed by a

period, followed by a primitive type (Request, Response, etc.). For example, "MM_Registration.Request" is a request from the control function to the mobility management protocol for "Registration", and "MM_Registration.Response" is the resulting response from the mobility management protocol to the control function. Where the context is obvious, the leading prefix (MM, CC, TC) is sometimes dropped for conciseness. Likewise, abbreviated forms of the primitive names might be used by reducing the suffix as follows: ".Rqst" for "Request", ".Resp" for "Response", ".Conf" for ".Confirm", ".Ind" for ".Indicate", ".IndResp" for ".IndicateResponse", and ".IndRConf" for ".IndicateResponseConfirm". Thus, "CC_Modify.IndicateResponseConfirm" could be abbreviated as "_Modify.IndRConf" within the call control section.

In the case of the transmission control SAP, two additional primitives (TC_Create and TC_Destroy) are defined which are unique to that SAP (see Section 2.1.4.3).

2.1.4.1 Mobility Management SAP

The Mobility Management SAP operates on nine mobility objects:

1. Group for Voice
2. Unit for Voice, Supplementary Data, and Packet Data
3. FNE-addressed Unit Supplementary Data
4. Group Supplementary Data
5. FNE-addressed Group Supplementary Data
6. Monitoring Interest in Unit Supplementary Data
7. Monitoring Interest in Group Supplementary Data
8. Monitoring Interest in FNE-addressed Group Supplementary Data
9. SN Context for Packet Data

The first two mobility objects address mobility management for voice services for Groups and Units.

The nine mobility objects address the nine cases of the functional model for Supplementary Data found in Section 2.1.3.1.3 as follows:

1. The Group mobility object is not applicable for Supplementary Data
2. The Unit mobility object expresses interest in unit-to-unit SD addressed to a particular SU (case a(ii)).
3. The FNE-addressed Unit Supplementary Data mobility object expresses interest in SD addressed to the General FNE address either from a particular unit (case c(ii)) or for an entire system (case c(iii)).
4. The Group Supplementary Data mobility object expresses interest in SD addressed to a group (case a(i)).
5. The FNE-addressed Group Supplementary Data mobility object expresses interest in SD addressed to the General FNE address from units affiliated with a group (case c(i)).
6. The Monitoring Interest in Unit Supplementary Data mobility object expresses monitoring interest for a unit or system (cases b(ii) and b(iii)).

7. The Monitoring Interest in Group Supplementary Data mobility object expresses monitoring interest for a group (case b(i.1)).
8. The Monitoring Interest in FNE-addressed Group Supplementary Data mobility object expresses monitoring interest in FNE-addressed SD from units affiliated with a group (case b(i.2)).
9. The SN Context mobility object is not applicable for Supplementary Data.

The Registration, Serving De-registration and Home De-registration MM transactions apply to all nine mobility objects. The Query transaction only applies to the Group, Unit, and SN Context mobility objects.

With respect to unit mobility management for TIA-102 Packet Data Services, the MM-SAP transactions in Section 2.1.4.1 are extended to also carry:

- Appropriate SND CP context parameter values for an SU, from the Home RFSS to the Serving RFSS (_Registration.IndicateResponse, Registration.Response).
- The SN Context parameter values for an SU supplied by the Serving RFSS to the Home RFSS (_Registration.Request, _Registration.Indicate). When an updated SN Context parameter value needs to be sent to the Home RFSS any time after the SN Context Object has been SIP Registered, the Serving RFSS initiates a SIP Register request with the SU Presence parameter set to Confirmed, which also contains the updated SND CP context parameters.
- The Home PDIG Address to be used for subsequent MIPv4 registration (_Registration.Response). Note that the Home PDIG Address **[102BACF]** can be different from the IP address of the Home RFSS used for SU registration as specified in this document.

With respect to packet data mobility management, the MM-SAP transactions in Section 2.1.4.1 apply to the “SN Context” object:

- The MM-SAP functions dedicated to the MM_Registration procedure (_Registration.Request, _Registration.Indicate, Registration.IndicateResponse, _Registration.Response) apply to both the “SU Presence Confirmed” and “SU Presence Not Confirmed” SND CP Context Activation.
- The MM-SAP functions dedicated to the MM_Deregistration procedure (_DeRegistration.Request, _DeRegistration.Indicate, DeRegistration.IndicateResponse, _DeRegistration.Response) apply to SU initiated SND CP Context Deactivation.

- The MM-SAP functions dedicated to the MM_Query procedures (_Query.Request, _Query.Indicate, _Query.IndicateResponse, Query.Response) apply to the Sndcp Context Activation Command requested by the Home RFSS or initiated by the SU, and synchronization of the Serving RFSS or the Home RFSS in accordance with respective local policies.
- The MM SAP functions dedicated to the MM_Roamed procedures apply to Home RFSS initiated Sndcp Context Deactivation.

The Mobility Management SAP consists of four transactions, as summarized in Table 1 and illustrated in Figure 17. Detailed definitions of each of the transactions and primitives supporting each function are provided below. Note that when a unit is a console the MM_Registration message does not correspond to an air link message. Somehow, internal to the console subsystem, a console has informed its console subsystem that it wants to perform group registration. The methods of console registration within an RF subsystem are beyond the scope of this specification and not part of the ISSI.

Table 1 - Mobility Management SAP Functions

Transaction	Type	Purpose
MM_Registration	Rqst/Resp, Ind/IResp	Initiated by a serving CF to Register a mobility object with a Home RFSS, corresponding to unit registration and group affiliation on the airlink, or to registration of SD or SN Context interest.
MM_ServDereg	Rqst/Resp, Ind/IResp	Initiated by a serving CF to de-register a mobility object from a Home RFSS.
MM_HomeDereg	Rqst/Resp, Ind/IResp	Initiated by a home CF to inform a Serving RFSS that a unit has roamed into a different serving domain, or to remove interest in any mobility object according to local policy.
MM_Query	Rqst/Resp, Ind/IResp	Initiated by a control function to query a peer RFSS for information on the group, unit, or SN Context. Also used, for SN Context, to trigger an Sndcp Context activation command.

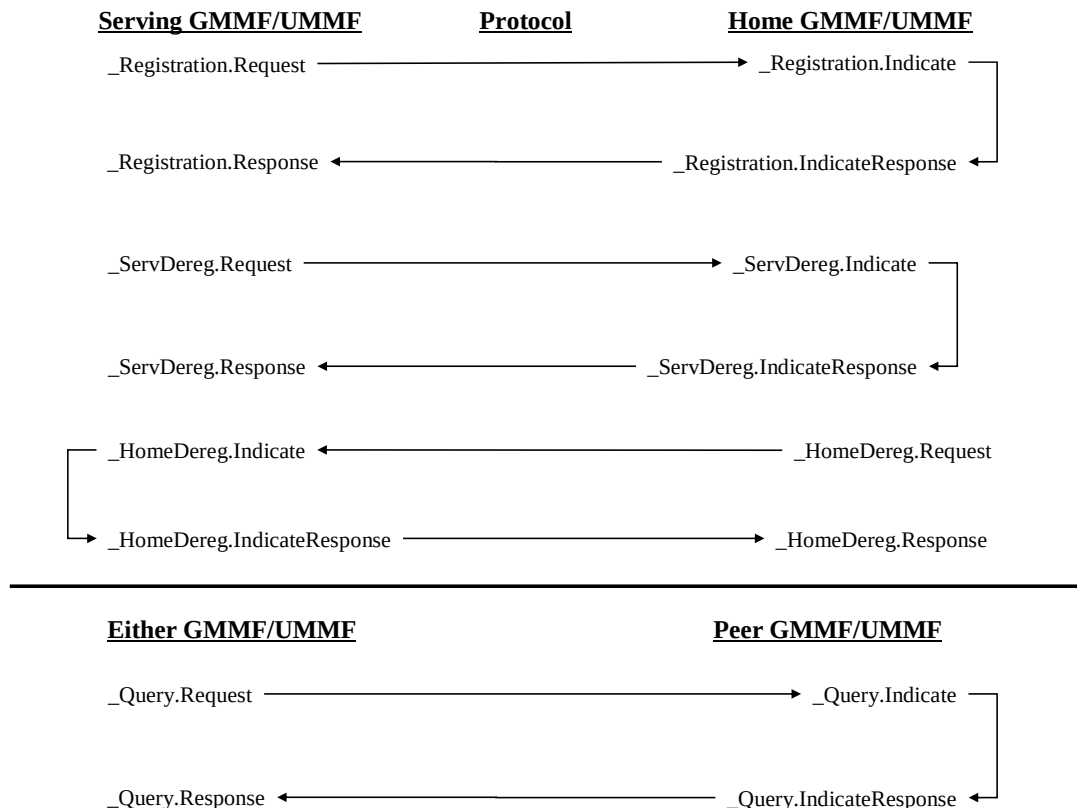


Figure 17 - MM SAP/Message Model

2.1.4.1.1 MM Registration Transaction

The Registration Transaction provides the means for a serving control function to register a mobility object with a Home RFSS. The following primitives comprise the transaction.

MM_Registration.Request is used in a serving CF to initiate the registration of a mobility object. It provides the Mobility Management Protocol (MMP) with the identity of the object to be registered and requires the *MM_Registration.Response* to complete the transaction. For group registration requests the serving CF can include its supported vocoder modes.

MM_Registration.Response is used by the MMP to indicate to the CF the final status of a registration attempt. It provides the CF with an indication of whether the registration was Successful, Unsuccessful or Failed, where “Successful” indicates a successful registration, “Unsuccessful” indicates the status of the response received from the home (and might be further qualified with a reason for the rejection of the registration), and “Failed” indicates that the registration

attempt could not be completed. Also, for group registrations the home CF can use this response to indicate its supported vocoder mode(s) for the group.

MM_Registration.Indicate is used by the MMP to indicate to a home CF that a serving CF is attempting to register a mobility object. It provides the CF with the identity of the object being registered, and the identity of the RFSS attempting to register the object. Upon receipt of the indication, the home CF is expected to validate the request and reply with *MM_Registration.IndicateResponse*. For unit registrations, the control function is expected to note the new location of the unit, and, if the unit was previously registered to another RFSS, to initiate roaming procedures (*MM_HomeDereg.Request*) with the previous RFSS. Subsequent unit based transactions (e.g. unit-to-unit calls) will be routed to the new Serving RFSS. For group registrations, the control function is expected to include the requesting RFSS in any group calls for the subject group, and to accept group calls from the RFSS for the subject group. Also, it notes the Serving RFSS's supported vocoder mode(s), if provided. For Supplementary Data object registrations, the control function maintains a list of interested Serving RFSSs for the object, to provide to the SD and SDM SAP as needed. The CF initiates a corresponding PD SAP transaction to manipulate the MIP tunnel when the SN Context object registration occurs.

MM_Registration.IndicateResponse provides the MMP with the final status (Successful, Unsuccessful) of a registration attempt. The MMP is expected to pass the status to the requesting RFSS. For group registrations, the home CF could include its possible vocoder modes for the group in the response. If this information is not included the serving CF acts according to local policy.

2.1.4.1.2 MM Serving Deregistration Transaction

The Serving Deregistration Transaction allows a Serving RFSS to notify the home that it no longer has interest in a mobility object. It is implemented by four primitives described in the following paragraphs.

MM_ServDereg.Request is used by a serving CF to deregister a mobility object and includes the identity of the object to deregister. It anticipates the receipt of *MM_ServDereg.Response* to complete the transaction.

MM_ServDereg.Response provides the CF with the final result of a deregistration attempt as Successful, Unsuccessful or Failed. The "Successful" response indicates that the transaction was successfully completed. "Unsuccessful", which might be further qualified with a reason, indicates that the Home RFSS rejected the request.

MM_ServDereg.Indicate deregisters a mobility object, effectively undoing *MM_Registration.Indicate*. It includes the identity of the object being deregistered and the identity of the RFSS requesting deregistration. The home CF might not reject the indication unless the object is not homed to the receiving

RFSS. *MM_ServDereg.Indicate* anticipates *MM_ServDereg.IndicateResponse* to complete the transaction.

MM_ServDereg.IndicateResponse provides the means for the CF to accept a deregistration request, or indicate that the referenced object is not homed to the RFSS. It anticipates that the response (Successful, Not Found) will be forwarded to the requesting RFSS.

2.1.4.1.3 MM Home Deregistration Transaction

The Home Deregistration Transaction provides a means for a Home RFSS to notify a Serving RFSS that a particular unit has registered in another domain and is therefore no longer registered in the serving domain, or to inform a Serving RFSS that registration of a mobility object is removed. *MM_HomeDereg.Request* initiates the transaction and provides the protocol with the mobility object's identifier and destination (Serving) RFSS. It results in *MM_HomeDereg.Indicate* at the receiving RFSS, notifying the receiving control layer of the roamed SU or the removed mobility object. The response (OK, Not Found) from the Serving RFSS is provided through the *MM_HomeDereg.IndicateResponse* and *MM_HomeDereg.Response* primitives.

2.1.4.1.4 MM Query Transaction

The Query Transaction is used by a Serving RFSS to request service profile information for a particular unit from a Home RFSS, and by a Home RFSS to verify that a Serving RFSS still has interest in a particular unit, group, or SN Context. It is implemented by the four primitives described below.

MM_Query.Request is used to ask another RFSS for the status of a unit or group. It includes the identity of the object in question, the identity of the RFSS being queried, and the type of query being requested which might be "Serving", "Normal Home", "Forced", or "Confirmed". It anticipates *MM_Query.Response* to complete the transaction.

MM_Query.Response provides the control function an answer to *MM_Query.Request*. Regardless of the type of query, the answer includes a status indication (Successful, Not Found, Forbidden, or Failed). The answer to a successful serving query includes the service profile information for the unit or group. The answer to a successful normal home, forced or confirmed query indicates whether or not the subject RFSS still has need for the object to be registered.

MM_Query.Indicate includes the identity of the unit or group being queried, the identity of the requesting RFSS, and the type of query being requested. It requires *MM_Query.IndicateResponse* in reply. A Serving RFSS that receives a "Forced" or "Confirmed" query from the Home RFSS of the subject object is expected to take particular steps (described in Section 5) to validate the registration of the object prior to responding to the query.

MM_Query.IndicateResponse provides the reply to *MM_Query.Indicate*. It includes a status (Successful, Not Found, Forbidden or Failed) that is expected to be relayed to the requesting RFSS, and, for serving queries only, might include service profile information to be communicated with the status.

2.1.4.2 Call Control SAP

The Call Control SAP consists of three transactions, as summarized in Table 2 and illustrated in Figure 18. Detailed definitions of each of the supporting transactions and primitives are provided below.

Table 2 - Call Control SAP Functions

Transaction	Type	Purpose
CC_Setup	Rqst/Resp/Conf, Ind/IResp/IRConf	Provides the control function the ability to setup a call segment.
CC_Teardown	Rqst, Ind	Provides the control function with the ability to teardown a call segment.
CC_Modify	Rqst/Resp/Conf, Ind/IResp/IRConf	Provides the ability to modify a call segment.

The CC_Setup transaction involves all six primitive types (Request, Response, Confirm, Indicate, Indicate Response, and IndicateResponseConfirm), as does the CC_Modify transaction. The CC_Teardown transaction only includes two primitive types, Request and Indicate. Figure 18 illustrates the transactions based on these primitives.

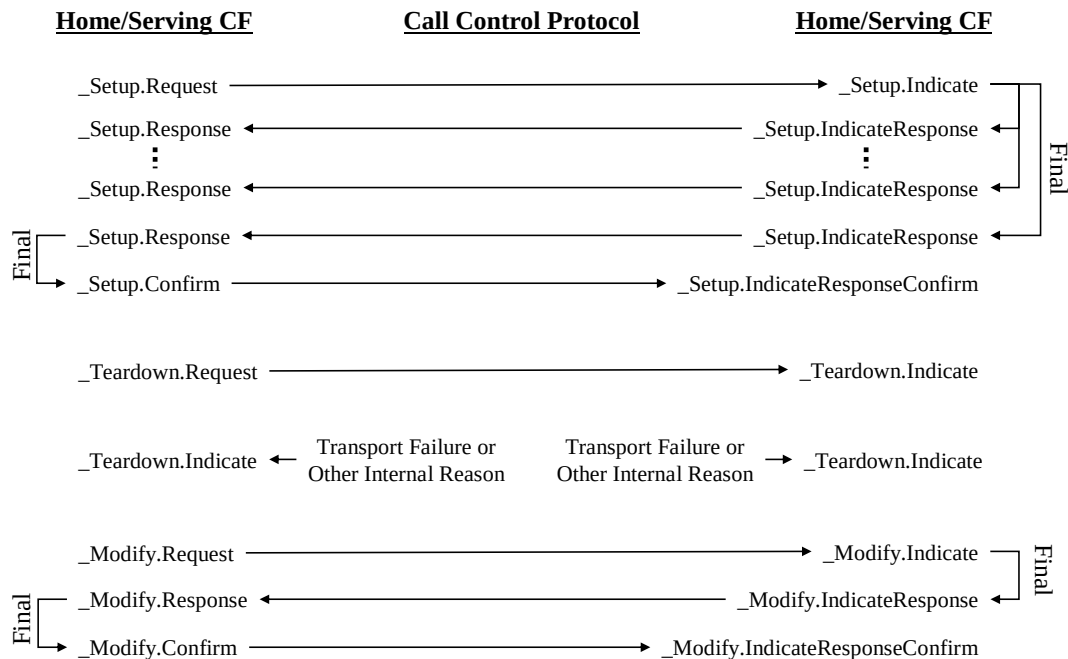


Figure 18 - CC SAP/Message Model

2.1.4.2.1 CC Call Segment Setup Transaction

The Call Segment Setup transaction is used by the control function to set up a call segment between itself and a peer RFSS, regardless of the role of either RFSS in the call. As illustrated in Figure 18, the transaction is initiated with the *CC_Setup.Request* primitive. Information from the request is propagated by the call control protocol to the peer, resulting in the invocation of a *CC_Setup.Indicate* primitive. The resulting response (*CC_Setup.IndicateResponse*) propagates back to the initiator of the segment setup as a provisional or final response to the initial request (*CC_Setup.Response*). Finally, upon receipt of a final response, the initiator confirms receipt of the response (*CC_Setup.Confirm*) and the confirmation is likewise propagated through the protocol to the peer (*CC_Setup.IndicateResponseConfirm*). Each of these primitives carries information and assumes behavior that is described in the following paragraphs.

CC_Setup.Request allows the control function to setup a call segment, providing such information as the type of call (unit or group, confirmed or unconfirmed, behavior preferences etc.), the role of the RFSS pursuant to the call (Home RFSS, Serving RFSS, Called Home RFSS, etc.), the call parameters (group or unit identifier, and the offered vocoder mode(s)), and so on to the protocol for transport to the peer RFSS. The setup request implicitly creates a new context for the call segment.

CC_Setup.Indicate passes the setup information to the control function of the peer RFSS. The peer CF examines its role relative to the object in question, validates the setup request, allocates RTP resources for the call segment, and takes such other action as might be appropriate. In the case of a unit-to-unit call, for example, a Calling Home RFSS would initiate a call segment setup to the Called Home RFSS. This might result in zero or more “provisional” responses, followed ultimately by a “final” response, both sent through the *CC_Setup.IndicateResponse* primitive. The peer CF also might determine how to respond to the offered vocoder mode(s). As with the setup request, the setup indication implicitly establishes a new context for the call segment.

CC_Setup.IndicateResponse passes the peer CF response to the request to the initiator of the request. The response might be provisional or final. The final response always includes information about RTP Resources Availability and, for confirmed group calls, also includes RF Resource Availability. The final response can also include information about the agreed upon vocoder mode. A negative final response implicitly destroys the call-segment context, and obviates the need for a *CC_Setup.IndicateResponseConfirm* to complete the transaction.

Upon receipt of the response information, the protocol at the initiator provides the *CC_Setup.Response* to communicate the response type, status, the agreed upon vocoder mode, and resource availabilities to the initiating control function. As with the indication response, a negative final status obviates the need for subsequent use of the *CC_Setup.Confirm* primitive by the control function.

In an unconfirmed group call, the response is immediately acknowledged with *CC_Setup.Confirm*, which causes *CC_Setup.IndicateResponseConfirm* in the peer RFSS. For a unit-to-unit call, the *CC_Setup.Confirm* is held off until RF Resources are available. For confirmed group calls, the *CC_Setup.Confirm* includes information about RF resource availability.

2.1.4.2.2 CC Call Segment Teardown Transaction

At any time, a control function might use the *CC_Teardown.Request* primitive to tear down a call segment and, if possible, notify the peer of the teardown. Likewise, the *CC_Teardown.Indicate* primitive tears down the call-segment as a result of some condition detected in the peer or local to the call control protocol. *CC_Teardown.Indicate* includes the reason for the call segment teardown (peer or self). Both primitives implicitly destroy the call-segment context.

2.1.4.2.3 CC Call Segment Modification Transaction

The Call Segment Modification Transaction operates identically to the call segment setup, and is used to change the characteristics of a call and notify the peer RFSS. In particular, the modification request allows the control function to request a change in the priority of group calls, and to notify the peer of the pre-emption or subsequent availability of RTP resources, and to report the RF resource state of the call (for confirmed group calls). The Call Segment

Modification Transaction is also used by the control function to request a change in the agreed upon vocoder mode for the talk spurts associated with the segment. The six primitives corresponding in function to those of the call segment setup are named: *CC_Modify.Request*, *CC_Modify.Response*, *CC_Modify.Confirm*, *CC_Modify.Indicate*, *CC_Modify.IndicateResponse*, and *CC_Modify.IndicateResponseConfirm*.

2.1.4.3 Transmission Control SAP

The Transmission Control SAP consists of ten transactions, as summarized in Table 3 and illustrated (with the exception of the TC and TC_Error Functions) in Figure 19, and Figure 20. Detailed definitions of each of the transactions and primitives are provided below.

Table 3 - Transmission Control SAP Functions

Transaction	Type	Purpose
TC	Create/Destroy	Create or Destroy a Transmission Control Context
TC_SpurtRqst/ TC_Decision	Rqst/Ind/IResp, Rqst/Ind	Request from an SMF CF to start a talk spurt.
TC_DecisionTimeout	Ind/IResp	Indication to the control function of a request timeout and request for the policy to apply.
TC_IsUnstoppable	Rqst	Request from a SMF CF to proceed with the spurt regardless of whether or not it is granted by the MMF.
TC_SetLosing	Rqst/Ind	Indicates to the protocol that the losing audio status of a spurt has changed.
TC_SpurtStart	Rqst/Ind	Request from an MMF CF to start a spurt.
TC_SpurtEnd	Rqst/Ind	Request from a CF to end a spurt.
TC_Audio	Rqst/Ind	Request to transmit audio (or other content such as header information).
TC_Mute	Rqst/Ind	Request to mute (or unmute) receipt of audio from a peer
TC_Error	Ind	Passes error status from the protocol entity to the control function.

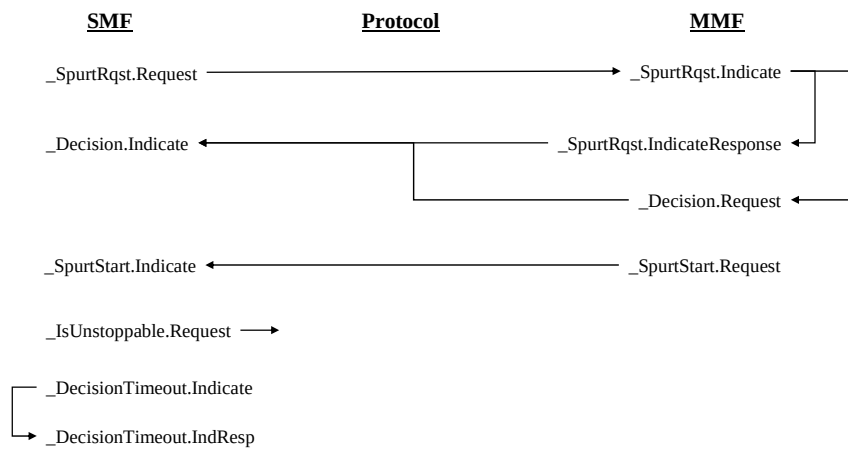


Figure 19 - TC SAP/Message Model (Setup)

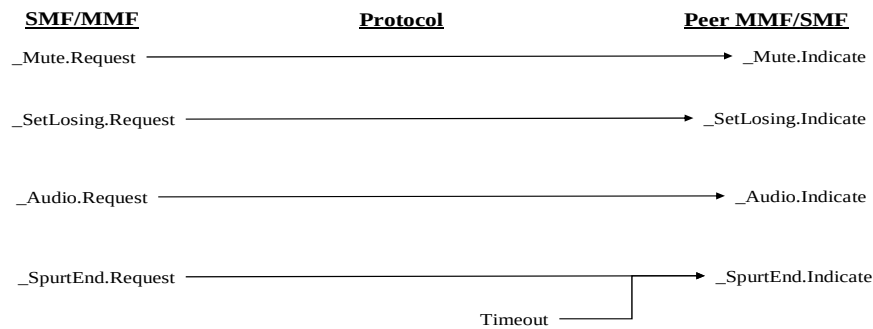


Figure 20 - TC SAP/Message Model (In-call)

2.1.4.3.1 TC Context Management Transactions

TC.Create and *TC.Destroy* are primitives provided to the control function to create and destroy transmission control contexts.

2.1.4.3.2 TC Spurt Request/Decision Transaction

TC_SpurtRqst.Request is used by a serving CF (SMF) to request arbitration of a talk spurt. It includes the type of group (if applicable), unit (if available), and priority of the spurt, and implicitly creates a new transmission. Requests might be either Normal or Unstoppable. A normal request requires arbitration by the MMF of the unit or group, and promises to be responsive to the result of the arbitration. An unstoppable request is not responsive to arbitration. For example, when a half-duplex radio keys on a conventional channel, its call might be able to be preempted outbound, but the user cannot be notified that it was. This is an unstoppable request. *TC_SpurtRqst.Request* has no response primitive, but anticipates eventual receipt of one or more *TC_Decision.Indicate* primitives to notify the CF of the result of the arbitration.

TC_SpurtRqst.Indicate notifies a home CF (MMF) of a request received from a Serving RFSS. It contains the same information as the *TC_SpurtRqst.Request* primitive. As a result of receiving *TC_SpurtRqst.Indicate* the home CF arbitrates the request, responding with the *TC_SpurtRqst.IndicateResponse*. Arbitration in the home might have additional consequences to other spurts within the same call, such as, for example, modifying the losing audio state (with *TC_SetLosing.Request*), affecting the response to other arbitration requests, and similar behaviors.

TC_SpurtRqst.IndicateResponse is the reply of the home CF to *TC_SpurtRqst.Indicate* and carries a status which might be Grant, Deny, or Decision Deferred. In the case of a deferred decision, it implicitly implies that a subsequent *TC_Decision.Request* will be invoked by the control function to complete the arbitration of the request.

TC_Decision.Request is used by a home CF to notify the protocol of the final result of a deferred arbitration decision and carries either a Grant or Deny indication for the spurt.

TC_Decision.Indicate is used by the protocol to notify a serving CF of the decision of the home, or of an internal decision made by the protocol. It carries a status that might be Grant, Deny, Deferred, or SelfGrant. Following a deferred indication, the serving CF anticipates that a subsequent Grant, Deny or SelfGrant decision will be provided.

2.1.4.3.3 TC Spurt Start Transaction

TC_SpurtStart.Request is used by the CF of an MMF to initiate a talk spurt. It includes the same information as *TC_SpurtRqst.Request*. Like a spurt request, it implicitly creates a new talk spurt.

TC_SpurtStart.Indicate notifies a serving CF of a new spurt, again with similar content to *TC_SpurtRqst.Indicate*. In contrast to a spurt request, however, the SMF does not arbitrate spurt starts.

2.1.4.3.4 TC Unstoppable Request Transaction

TC_IsUnstoppable.Request is used by a serving CF to mark a particular talk spurt as “unstoppable” in the sense described above.

2.1.4.3.5 TC Decision Timeout Transaction

TC_DecisionTimeout.Indicate is employed by the protocol in a serving CF to determine the local policy of the CF in response to an arbitration timeout. It anticipates a *TC_DecisionTimeout.IndicateResponse* that will subsequently result in a corresponding *TC_Decision.Indicate*.

TC_DecisionTimeout.IndicateResponse indicates the local policy of the CF for an arbitration timeout as either Transmit or Fail. In the event of a Transmit policy, the protocol is expected to employ a *TC_Decision.Indicate* primitive with status of SelfGrant. Otherwise, it is expected to use the same primitive with a status of Deny.

2.1.4.3.6 TC Mute Transaction

The *TC_Mute.Request* and *TC_Mute.Indicate* allow the control function to mute or unmute audio, and provide notification that a spurt has been muted or unmuted. In a typical application, Mute would be employed at an RFSS that has no demand for losing audio when it receives a spurt that has been marked as losing audio.

2.1.4.3.7 TC Set Losing Audio Transaction

TC_SetLosing.Request is used by a CF to change the status of a talk spurt to “losing” or “preferred”. The losing audio status reflects the preference of the source RFSS of one talk spurt over all others.

TC_SetLosing.Indicate indicates to a CF that the status of a talk spurt has changed from “losing” to “preferred” or visa versa. When multiple talk spurts are simultaneously active for a given transmission, an RFSS is expected to take account of the losing audio status of the spurt (as provided by the source of the spurt) in its decisions about routing the spurt to other RFSSs (in the case of a Home RFSS) or to end-points within its own domain.

2.1.4.3.8 TC Audio Transaction

TC_Audio.Request and *TC_Audio.Indicate* are the primitives responsible for the conveying of voice payloads (including ISSI header information, low speed data and so on) between the control functions and the protocols.

2.1.4.3.9 TC Spurt End Transaction

TC_SpurtEnd.Request and *TC_SpurtEnd.Indicate* serve to terminate a talk spurt. The request is initiated by the CF of either the home or serving CF, and the indication by the protocol element. The indication might be generated as a result of receiving an explicit indication from the peer of the end of a talk spurt, or as a result of a timeout. The indication might have other implications in the receiving RFSS, for instance, it might result in the changing of the losing audio state or other talk spurt, or of the tear-down of one or more call segments using primitives from the call control SAP.

2.1.4.3.10 TC_Error Transaction

TC_Error.Indicate serves to notify the control function of problems detected by the transmission control entity, including the failure of RTP heartbeats, and other implementation specific errors.

2.1.4.4 RFSS Service Capability Polling SAP

The RFSS Service Capability Polling SAP consists of one transaction, as summarized in Table 4 and illustrated in Figure 21. Detailed definitions of the supporting transaction and primitives are provided below.

Table 4 - RFSS Service Capability Polling SAP Functions

Transaction	Type	Purpose
RP_Query	Rqst/Resp, Ind/IResp	Initiated by a CF to poll the service capabilities of another RFSS.

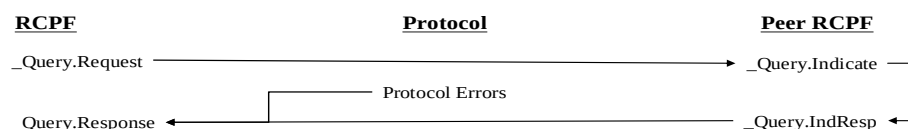


Figure 21 - RP SAP/Message Model

2.1.4.4.1 RP Query Transaction

The RP Query Transaction consists of 4 primitives.

RP_Query.Request allows the CF to request an RFSS service capability polling request. The primitive carries any parameters describing detailed service capabilities of the RFSS provided by the CF.

RP_Query.Indicate notifies the CF of the peer RFSS of the RFSS service capability polling request. The primitive carries any parameters describing detailed service capabilities of the RFSS provided by the CF.

RP_Query.IndicateResponse allows the CF of the peer RFSS to reply to the RFSS service capability polling request. The primitive carries any parameters describing detailed service capabilities of the RFSS provided by the CF.

RP_Query.Response provides the CF of the polling RFSS with the reply from the peer RFSS. The primitive carries any parameters describing detailed service capabilities of the RFSS provided by the peer RFSS.

2.2 ISSI Internet Protocol Suite

Figure 22 depicts the general protocol suite relevant to the scope of this document.

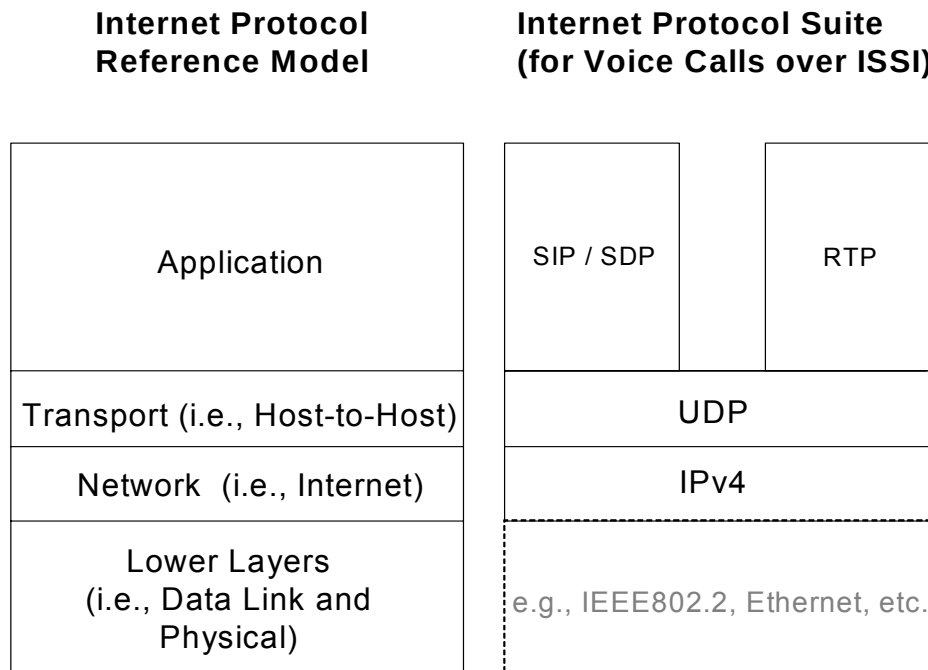


Figure 22 - ISSI Internet Protocol Suite (for Voice Services)

As shown in Figure 22, the ISSI is a set of application protocols that operates on top of Transport Layer, Network Layer, and various lower layers as defined in the following subsections. As indicated in Section 1.1, the scope of this document is limited to protocol specifications of the Network Layer and above, therefore not addressing protocol issues for the Physical and Data Link Layers of the Internet Protocol Reference Model.

2.2.1 Lower Layers

As stated in [102BACC], these layers correspond to the physical and data link layers of the OSI Protocol Reference Model. Network access protocols commonly used in these layers include Ethernet [IEEE802.3] and IEEE 802.2 [IEEE802.2].

The actual protocols to be used might vary depending on whether two RFSSs are connected over the ISSI via dedicated links, or via a customer's private telecommunications network (e.g., virtual private network) which conforms to the

ISSI. Detailed specification for the lower protocol layers of the ISSI is beyond the scope of this document.

2.2.2 Network Layer

There are currently two versions of IP: IPv4 [RFC0791] and IPv6 [RFC2460].

The ISSI SHALL support IPv4 and future migration to IPv6. The goal is for ISSI to be IPv4 and IPv6 neutral, supporting, where possible, IPv4 features which are compatible with IPv6, therefore facilitating forward migration to and interoperability with IPv6 in the future.

2.2.2.1 Internet Protocol Version 4 (IPv4)

2.2.2.1.1 IPv4 Header Format

For IPv4:

- The IP header length is typically 20 octets.
- The minimum MTU for each IP packet (IP header & body) SHALL be 1500 octets.

Figure 23 shows the header format of a typical IPv4 packet.

1																2																3															
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1																
Version				IHL				Type of Service								Total Length																															
Identification																Flags				Fragment Offset																											
Time to Live								Protocol								Header Checksum																															
Source Address																																															
Destination Address																																															
Options																								Padding																							

Figure 23 - Typical IPv4 Datagram Header Format

2.2.2.1.2 Header Field Definitions

Detailed generic definitions of the IPv4 header fields are contained in **[RFC0791]**.

For voice services over the ISSI, the IPv4 header fields SHALL be set as follows (all numbers are in decimal):

Version (4 bits):	4
IHL (4 bits):	(see [RFC0791])
Type of Service (8 bits):	(see [RFC0791]) ⁶

⁶ Editor's Note: RFC 2474 defines a replacement header field, called the DS (Differentiated Services) field, superseding the existing definitions of the IPv4 TOS (Type of Service) octet. This new field includes a 6-bit DSCP (Differentiated Services Codepoint) which could be flexibly mapped to particular forwarding treatment, or per-hop behavior (PHB), at each Differentiated Services compliant network node (e.g., routers, etc.) along its path, for efficient routing of SIP messages and RTP packets to their destinations, according to the urgency of the call request (e.g., preemptive priority call). Appropriate assignment and mapping of DSCPs for ISSI applications are for further study.

Total Length (16 bits):	(see [RFC0791])
Identification (16 bits):	0
Flags (3 bits):	0 or 1 (i.e., allow fragmentation)
Fragment Offset (13 bits):	0
Time to Live (8 bits):	(see [RFC0791])
Protocol (8 bits):	17 (i.e., UDP, see [RFC3232])
Header Checksum (16 bits):	(see [RFC0791])
Source Address (32 bits):	IPv4 address of the source RFSS
Destination Address (32 bits):	IPv4 address of the destination RFSS
Options (variable in length):	(see [RFC0791])
Padding (variable in length):	(used to ensure that the data begins on 32 bit word boundary. The padding is zero.)

NOTE - The assignment and coordination of IP addresses for the RFSSs are at the discretion of the user community, and are therefore beyond the scope of this document.

2.2.2.2 Internet Protocol Version 6 (IPv6)

The ISSI will support IPv6 [RFC2460] in the future.

2.2.3 Transport Layer

2.2.3.1 User Datagram Header Format

The UDP header is 8 octets long. Figure 24 shows the header format of a user datagram.

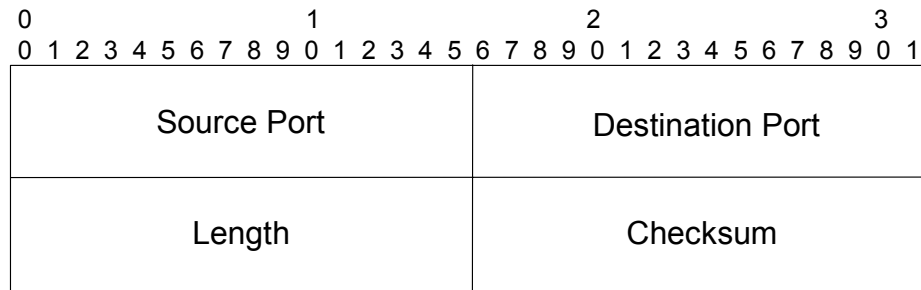


Figure 24 - User Datagram Header Format

2.2.3.2 Header Field Definitions

Detailed generic definitions of the user datagram header fields are contained in **[RFC0768]**.

As specified in Section 6 of **[RFC4566]** and Section 11 of **[RFC3550]**, even UDP ports are used for RTP.

For voice services over the ISSI, the user datagram header fields SHALL be set as follows (all numbers are in binary):

Source Port (16 bits):	Port of sending process at originating RFSS
Destination Port (16 bits):	Port of receiving process at destination RFSS
Length (16 bits):	(see [RFC0768])
Checksum (16 bits):	16-bit one's complement of the one's complement sum of a pseudo IP header (comprising the source address, the destination address, the protocol, and the UDP length), the UDP header, and the SIP message or RTP packet. See [RFC0768] .

NOTE - The assignment and coordination of UDP ports for the RFSSs are at the discretion of the user community, and are therefore beyond the scope of this document.

2.2.4 Application Layer

Two application layer protocols are used to support the ISSI:

1. The Session Initiation Protocol (SIP) is used to track the mobility of SUs, and to set up, maintain, and release SU-to-SU and group voice calls.
2. The Real-Time Transport Protocol (RTP) is used for the conveyance of voice, and to control voice transmissions within an RTP session being set up by SIP.

Each SIP message is encapsulated within a UDP datagram, which, in turn, is encapsulated within an IP packet. The SIP message comprises a SIP header, an optional Session Description Protocol (SDP) body that specifies new or modified SIP session characteristics, and/or an optional ISSI-specific body that carries service profiles and other ISSI-specific attributes for SU-to-SU and group calls, and/or an optional manufacturer-specific body.

Similarly, each RTP packet is encapsulated within a UDP datagram, which, in turn, is encapsulated within an IP packet.

2.3 ISSI Data Transmission Order

The order of transmission of the header and data described in this document is resolved to the octet level, as detailed in **[RFC0791]**. Whenever a diagram shows a group of octets, the order of transmission of those octets is the normal order in which they are read in English. For example, in Figure 25, the octets are transmitted in the order they are numbered.

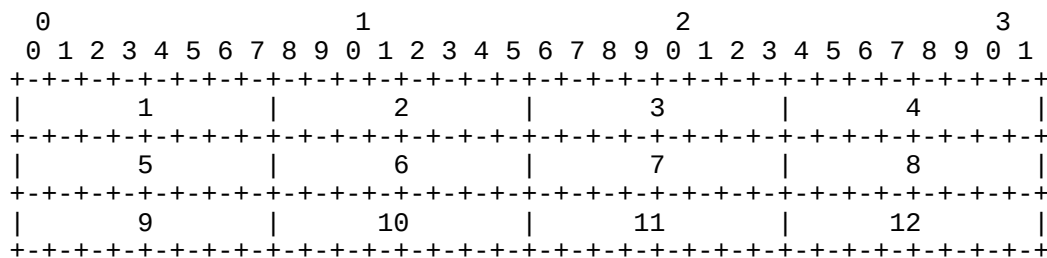


Figure 25 - Transmission Order of Octets

Whenever an octet represents a numeric quantity the left most bit in the diagram is the high order or most significant bit. That is, the bit labeled 0 is the most significant bit and is transmitted first. For example, Figure 26 represents the value 170 (decimal).

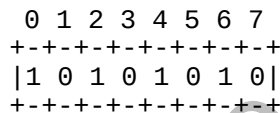


Figure 26 - Significance of Bits

Similarly, whenever a multi-octet field represents a numeric quantity the left most bit of the whole field is the most significant bit. When a multi-octet quantity is transmitted, the most significant octet is transmitted first.

3. SIP Messages and Parameters Definition for ISSI

3.1 Preamble: Grammar

The grammar used to describe SIP parameters and headers is expressed in Augmented Backus- Naur Form (ABNF) as defined [RFC2234] and used for defining SIP in [RFC3261] (Section 25) and for defining SDP in [RFC4566].

The following ABNF definitions are used:

```
integer          =  POS-DIGIT * (DIGIT)
                  ; integer without a leading 0

DIGIT           =  "0" / POS-DIGIT

POS-DIGIT       =  "1" / "2" / "3" / "4" / "5" / "6" / "7" / "8" / "9"

octet-string    =  1*unreserved
unreserved      =  alphanum / mark
alphanum        =  ALPHA / DIGIT
ALPHA           =  %x41-5A / %x61-7A ; A-Z / a-z
mark            =  "-" / "_" / "." / "!" / "~" / "*" / "'" / "(" / ")"
```

In addition, the following definitions are used:

```
SP              =  %x20 ; the space character

CRLF            =  %x0D %x0A ; internet standard newline

DQUOTE          =  %x22 ; the double quote character

boolean         =  "0" / "1"

int             =  "0" / integer

HEX-DIGIT       =  DIGIT / "A" / "B" / "C" / "D" / "E" / "F"
```

A number formed by concatenating "n" HEX-DIGIT is expressed as nHEX-DIGIT.

3.2 SIP Transport

As discussed in Section 2.2 of this document, UDP SHOULD be used as the transport layer protocol for SIP message exchanges over the ISSI. The corresponding port number SHALL be 5060.

3.3 Request Line Construction

A SIP Message is constructed as described in Section 7 of [RFC3261].

The request line is defined as follows:

```
request-line = method SP sip-uri-for-ISSI SP sip-version CRLF
method       = ("INVITE" / "ACK" / "BYE" / "CANCEL" / "REGISTER"
               / "MESSAGE" / "OPTIONS")
```

sip-uri-for-ISSI is defined in the next subsection.

```
sip-version  = "SIP/2.0"
```

3.4 SIP URI and related addressing logic

3.4.1 General Characteristics of SIP URIs

The SIP URIs of the ISSI use a limited form of SIP URIs as described in [RFC3261]. Specifically, they have the following form:

```
sip-uri-for-ISSI = "sip:" [radical-name "@" ] domain-name [parameters]
```

where domain-name is described in Section 3.4.1.1, radical-name is described in Section 3.4.1.2, and parameters, if present, are as described in Section 3.4.1.3. The notation [...] indicates a field that is optional.

"sip:" and "@" are always present in the indicated positions in ISSI SIP URIs except for the Request URI of the SIP register message, where radical-name and @ are never present.

All fields in ISSI SIP URIs SHALL be case insensitive.

3.4.1.1 ISSI Domain Names

An ISSI domain name is composed of the concatenation of up to four string fields, as follows:

```
domain-name = [rfss-name / system-name] "p25dr"
rfss-name   = rfss-id "." system-id "." wacn-id "."
system-name = system-id "." wacn-id "."
rfss-id     = 2HEX-DIGIT / 4HEX-DIGIT
```

system-id	=	3HEX-DIGIT
wacn-id	=	5HEX-DIGIT

The literal string "p25dr" is always present in an ISSI domain name, as are periods (".") included in the syntax. Furthermore:

- rfss-id is expressed as two or four hexadecimal characters for the 8-bit / 16 bit RFSS ID of the referenced object,
- the value of the rfss-id SHALL be unique within a system and values \$00 and \$FF SHALL be reserved (unused) as per [102AABD],
- the 4HEX-DIGIT rfss-id SHALL only be used when the RFSS ID is greater than 255,
- any RFSS (particularly console subsystems) that does not have RF sites MAY be assigned a 4HEX-DIGIT rfss-id greater than 255,
- any RFSS (including console subsystems) that has RF sites SHALL be assigned a 2HEX-DIGIT rfss-id,
- system-id is expressed as three hexadecimal characters for the 12-bit system ID of the referenced object,
- wacn-id is expressed as five hexadecimal characters for the 20-bit WACN ID of the referenced object.

For an RFSS the domain name SHALL be in the form of rfss-name; and for a System, the domain name SHALL be in the form of system-name.

Examples:

If the RFSS ID is equal to 1, rfss-id is equal to 01
 If the RFSS ID is equal to 12, rfss-id is equal to 0C
 If the RFSS ID is equal to 120, rfss-id is equal to 78

3.4.1.2 ISSI Radical Names

The radical-name is defined as follows:

radical-name =	("TIA-P25-Groupcall" / "TIA-P25-U2Uorig" / "TIA-P25-U2Udest" / "TIA-P25-SDMsap" / su-id / sg-id / snc-id / "TIA-P25-RFSSCapability")
----------------	--

su-id	=	wacn-id system-id u-id
sg-id	=	wacn-id system-id g-id
snc-id	=	wacn-id system-id u-id nsapi-id
wacn-id	=	5HEX-DIGIT
system-id	=	3HEX-DIGIT
u-id	=	6HEX-DIGIT
g-id	=	4HEX-DIGIT
nsapi-id	=	1HEX-DIGIT

The nsapi-id value 0 SHALL be used in the PD SNC Home Deregistration procedure to deactivate all Sndcp contexts associated with an SU. This nsapi-id value SHALL NOT be used for any other purposes. Valid nsapi-id values to be used are as defined in **[102BAEB]** for the NSAPI field of the Sndcp PDU.

ISSI radical names SHALL be employed as described in Table 5, Table 6, Table 7, and Table 8. The “TIA-P25-SDMsap” radical is employed as described in **[102BACD]**.

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Table 5 - ISSI Radical Names for Group Call Control Procedures

ISSI radical Name	Syntax & Format description	Use	Role	Direction ⁷	URI Example
Group Call	“TIA-P25-Group call” Literal string.	Route header field of the SIP INVITE for a group call set-up.	Enables the receiving RFSS to recognize that it is requested to set up a group call.	S $\Rightarrow$ H	sip:TIA-P25-Groupcall@01.002.00003.p25dr;lr
Group Identifier	sg-id String composed of the 12 hexadecimal characters of the fully qualified SG number	From and To header field of the SIP message for a group call. Request URI of a SIP INVITE, SIP ACK, SIP BYE and SIP CANCEL for a group call.	Enables the receiving RFSS to retrieve the fully qualified group identifier of the SG.	Any	sip:001002F24623@p25dr;user=TIA-P25-SG

⁷ H $\Rightarrow$ S means the header field is used in a SIP Message sent from the Home RFSS to the Serving RFSS
S $\Rightarrow$ H means the header field is used in a SIP Message sent from the Serving RFSS to the Home RFSS

Table 6 - ISSI Radical Names for Unit-to-Unit Call Control Procedures

Radical Name	Syntax & Format description	Use	Role	Direction ⁸	URI Example
SU-to-SU Origination	"TIA-P25-U2Uorig" Literal String	Route header field in the SIP messages for an SU-to-SU call.	Enables the receiving RFSS to recognize that it is requested to process a SIP message of a unit-to-unit call as Calling Home RFSS.	Any	sip:TIA-P25-U2Uorig@01.002.00003.p25dr;lr
		Record-Route header field in the SIP messages for an SU-to-SU call.	Used to record that the Calling Home RFSS is on the route. This is used to route the SIP ACK requests.		
Unit Identifier	su-id String made of the 14 hexadecimal character of the fully qualified SU number	Request URI of a SIP INVITE, SIP ACK, SIP BYE and SIP CANCEL associated with an SU-to-SU call.	Enables the receiving RFSS to retrieve the fully qualified identifier of the Called SU for SU- to-SU Call Set-up initiated by the Calling Serving RFSS, SU-to-SU Call termination in the direction towards the Called SU, and Call segment set-up after an In-Call Roaming of the Called SU. Enables the receiving RFSS to retrieve the fully qualified identifier of the Calling SU for SU-to-SU Call termination in the direction towards the Calling SU, and Call segment set-up after an In-Call Roaming of the Calling SU..	Any	sip:00001002F24623@p25dr;user=TIA-P25-SU
		From and To header field of the SIP messages associated with an SU-to-SU call.	Enables the receiving RFSS to retrieve the fully qualified identifier of the Calling SU and the Called SU for an SU-to-SU call.	Any	sip:00001002F24623@p25dr;user=TIA-P25-SU
SU-to-SU Destination	"TIA-P25-U2Udest" Literal string	Route header field of the SIP messages for an SU-to-SU call.	Enables a receiving RFSS to recognize that it is requested to process a SIP message of an SU-to-SU call as the Called Home RFSS.	Any	sip:TIA-P25-U2Udest@01.002.00003.p25dr;lr
		Record-Route header field in the SIP messages for an SU-to-SU call.	Used to record that the Called Home RFSS is on the route. This is used to route the SIP ACK requests.		

⁸ H ⇒ S means the header field is used in a SIP Message sent from the Home RFSS to the Serving RFSS
 S ⇒ H means the header field is used in a SIP Message sent from the Serving RFSS to the Home RFSS

Table 7 - ISSI Radical Names for mobility management procedures

ISSI radical Name	Syntax & Format description	Use	Role	Direction ⁹	URI Example
Unit Identifier	su-id String composed of the 14 hexadecimal characters of the fully qualified SU identifier	From and To header fields of the SIP message for unit mobility object and supplementary data mobility objects relative to a unit.	Enables the receiving RFSS to retrieve the fully qualified unit identifier.	Any	sip: 00001002F24623@p25dr;user=TIA-P25-SU sip: 00001002F24623@p25dr;user=TIA-P25-SD-SU sip: 00001002F24623@p25dr;user=TIA-P25-SDMON-SU
Group Identifier	sg-id String composed of the 12 hexadecimal characters of the fully qualified SG identifier	From and To header fields of the SIP message for group mobility object and supplementary data mobility objects relative to a group or to the general FNE.	Enables the receiving RFSS to retrieve the fully qualified group identifier.	Any	sip: 00001002F246@p25dr;user=TIA-P25-SG sip: 00001002F246@p25dr;user=TIA-P25-SD-SG sip: 00001002F246@p25dr;user=TIA-P25-SDMON-SG sip: 00001002F246@p25dr;user=TIA-P25-SD-SGFNE sip: 00001002F246@p25dr;user=TIA-P25-SDMON-SGFNE
SNC Identifier	snc-id String composed of the 15 hexadecimal characters of the fully qualified SNC identifier	From and To header fields of the SIP message for a SN registration.	Enables the receiving RFSS to retrieve the fully qualified SNC identifier.	Any	sip: 00001002F246231@p25dr;user=TIA-P25-PD-SNC

⁹ H ⇒ S means the header field is used in a SIP Message sent from the Home RFSS to the Serving RFSS
S ⇒ H means the header field is used in a SIP Message sent from the Serving RFSS to the Home RFSS

Table 8 - ISSI Radical Names for RFSS Service Capability Polling Procedures

Radical Name	Syntax & Format description	Use	Role	Direction	URI Example
RFSS to RFSS Service Capability Polling Request	Always set to "TIA-P25-RFSSCapability"	Request URI of the SIP messages associated with RFSS Service Capability Polling.	Enables the polled RFSS to recognize that the polling RFSS is requesting its current service capability information	Any	sip:TIA-P25-RFSSCapability@01.002.00003.p25dr; user=TIA-P25-RFSS
		From and To header fields of the SIP messages associated with RFSS Capability Polling.			

3.4.1.3 ISSI Parameters

The parameters are defined in the following way:

parameters	=	numplan-param / lr-param / branch-param
numplan-param	=	“;user=TIA-P25-SU” / “;user=TIA-P25-SG” / “;user=TIA-P25-SD-SU” / “;user=TIA-P25-SD-SG” / “;user=TIA-P25-SD-SGFNE” / “;user=TIA-P25-SDMON-SU” / “;user=TIA-P25-SDMON-SG” / “;user=TIA-P25-SDMON-SGFNE” / “;user=TIA-P25-PD-SNC” / “;user=TIA-P25-RFSS”
lr_param	=	“;lr”
branch-param	=	”;branch=z9hG4bK” branch-id
branch-id	=	token

where token is defined in Section 25.1 of **[RFC3261]**.

As stated in **[RFC3261]** Section 8.1.1, the branch-param parameter begins with the magic cookie “z9hG4bK”, followed by a globally unique branch-id parameter. The construction of the branch-id is manufacturer-dependent. However an example construction method is provided below for information only.

ISSI parameters have the following form, as shown in Table 9 below:

Table 9 - ISSI Parameters in the SIP URIs

ISSI Parameter	Syntax & Format description	Use
SU Numbering Plan	"user=TIA-P25-SU"	In any SIP URI where the su-id radical name is used
SG Numbering Plan	"user=TIA-P25-SG"	In any SIP URI where the sg-id radical name is used
SD Numbering Plans	"user=TIA-P25-SD-SU" "user=TIA-P25-SD-SG" "user=TIA-P25-SD-SGFNE"	For mobility management procedures only, to identify supplementary data interest in the given mobility object.
SD Monitoring Numbering Plans	"user=TIA-P25-SDMON-SU" "user=TIA-P25-SDMON-SG" "user=TIA-P25-SDMON-SGFNE"	For mobility management procedures only, to identify supplementary data monitoring interest in the given mobility object.
PD SNC Numbering Plan	"user=TIA-P25-PD-SNC"	For mobility management procedures only, to identify the Sndcp context of a given SU.
RFSS Service Capability Polling Numbering Plans	"user=TIA-P25-RFSS"	For RFSS service capability polling procedures only
lr parameter	"lr" Refer to Section 19.1.1 of the [RFC3261]	In any SIP URI of the Record-Route and Route header fields
Branch parameter	"branch=z9hG4bK<branch-id>" Where <branch-id> is the hexadecimal representation of the variable part of the branch ID.	In SIP URI of the Via header field Refer to Section 8.1.1.7 of [RFC3261]
Empty	"	In any other SIP URI

Following is an example method for calculating the branch-param (provided here for information only):

The branch-id is described as follows:

```

branch-id    =    loop-detector "." branch-number
loop-detector =    1*HEX-DIGIT
branch-number =    1*HEX-DIGIT

```

As recommended in the [RFC3261], loop-detector is the hexadecimal representation of the 128-bit value, calculated according to MD5 Message-Digest algorithm defined in the RFC 1321 (The MD5 Message-Digest Algorithm). The following "input" value is used for MD5:

```

input      =    To-field From-field Call-ID-field Request-URI CSeq-number

```

where:

- To-field, From-field, Call-ID-field are respectively the To, From, Call-ID header fields of the received request. For each header field xyz (located at line N in the request, with N>0), the xyz-field string used is located between the CRLF of line (N-1) and the CRLF of line N, CRLF excluded.
- Request-URI: The Request-URI of the received request
- CSeq-number is the sequence number from the CSeq header field.

The components of "input" are strung together - in the order given above, with no space or any other character between them - to form a sequence of UTF-8 characters and thus a sequence of bits.

branch-number is a counter starting at 1 for the first branch and incremented by one for the next branch.

Refer to Via header field in Section 3.5.17 for an example of a branch parameter ID.

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3.4.2 Request URI formatting Summary

The Request URI field format is as summarized in Table 10, Table 11, and Table 12 below:

Table 10 - Summary of Request URI formats for various SIP Requests for an SU-to-SU Call

Request type	ISSI Object	Direction	Request URI form
SIP Register	Unit	Home RFSS $\Rightarrow$ Serving RFSS	sip:domain-name where domain-name is the domain name of the Serving RFSS.
SIP Register	Unit	Serving RFSS $\Rightarrow$ Home RFSS of the Unit	sip:domain-name where domain-name is the domain name of the Home RFSS.
SIP Invite/SIP Cancel	Unit	Calling Serving RFSS $\Rightarrow$ Calling Home RFSS	sip:su-id@p25dr;user=TIA-P25-SU where su-id is the identifier of the Called SU.
SIP Invite/SIP Cancel (In-Call roaming)	Unit	Calling Home RFSS $\Rightarrow$ Calling Serving RFSS	sip:su-id@domain-name;user=TIA-P25-SU where su-id is the identifier of the Calling SU and domain-name is the domain name of the Calling Serving RFSS.
SIP Invite/SIP Cancel	Unit	Calling Home RFSS $\Rightarrow$ Called Home RFSS	sip:su-id@domain-name;user=TIA-P25-SU where su-id is the identifier of the Called SU and domain-name is the domain name of the Called Home RFSS.
SIP Invite/SIP Cancel	Unit	Called Home RFSS $\Rightarrow$ Called Serving RFSS	sip:su-id@domain-name;user=TIA-P25-SU where su-id is the identifier of the Called SU and domain-name is the name of the Called Serving RFSS.
SIP ACK	Unit	Calling Serving RFSS $\Rightarrow$ Calling Home RFSS Calling Home RFSS $\Rightarrow$ Called Home RFSS Called Home RFSS $\Rightarrow$ Called Serving RFSS	sip: su-id@p25dr;user=TIA-P25-SU where su-id is the identifier of the Called SU
SIP ACK (In-Call roaming)	Unit	Calling Home RFSS $\Rightarrow$ Calling Serving RFSS	sip: su-id@p25dr;user=TIA-P25-SU where su-id is the identifier of the Calling SU
SIP BYE	Unit	Calling Serving RFSS $\Rightarrow$ Calling Home RFSS Calling Home RFSS $\Rightarrow$ Called Home RFSS Called Home RFSS $\Rightarrow$ Called Serving RFSS	sip: su-id@p25dr;user=TIA-P25-SU where su-id is the identifier of the Called SU
SIP BYE	Unit	Called Serving RFSS $\Rightarrow$ Called Home RFSS Called Home RFSS $\Rightarrow$ Calling Home RFSS Calling Home RFSS $\Rightarrow$ Calling Serving RFSS	sip: su-id@p25dr;user=TIA-P25-SU where su-id is the identifier of the Calling SU

Table 11 - Summary of Request URI formats for various SIP Requests for a group call

Request type	ISSI Object	Direction	Request URI form
SIP Register	Group	Home RFSS $\Rightarrow$ Serving RFSS	sip:domain-name where domain-name is the name of the Serving RFSS.
SIP Register	Group	Serving RFSS $\Rightarrow$ Home RFSS of the Group	sip:domain-name where domain-name is the name of the Home RFSS.
SIP Invite/SIP Cancel	Group	Serving RFSS $\Rightarrow$ Home RFSS	sip:sg-id@p25dr;user=TIA-P25-SG where sg-id is the identifier of the SG
SIP Invite/SIP Cancel	Group	Home RFSS $\Rightarrow$ Serving RFSS	sip:sg-id@domain-name;user=TIA-P25-SG where sg-id is the identifier of the SG and domain-name is the name of the Serving RFSS
SIP ACK	Group	Any	sip:sg-id@p25dr;user=TIA-P25-SG where sg-id is the identifier of the SG
SIP BYE	Group	Any	sip:sg-id@p25dr;user=TIA-P25-SG where sg-id is the identifier of the SG

Table 12 - Summary of Request URI formats for RFSS Service Capability Polling

Request type	ISSI Object	Direction	Request URI form
SIP OPTIONS	RFSS	Polling RFSS $\Rightarrow$ Polled RFSS	sip:TIA-P25-RFSSCapability@domain-name;user=TIA-P25-RFS S where domain-name is the name of the polled RFSS.

Additional Request URIs for Supplementary Data are described in [102BACD].

3.5 SIP Header Encoding

General syntax for header fields is covered in Section 20 of [RFC3261] and in [RFC1035].

This section provides additional requirements.

Table 13 lists the header fields used in the SIP messages sent by the RFSSs over the ISSI. This table gives restriction rules on a similar table given in Section 20 of [RFC3261].

The “where” Column describes the request and response types in which the header field can be used. Possible values in this column are:

Request: header field MAY only appear in requests
 Response: header field MAY only appear in responses
 2xx, 4xx, etc: A numerical value or range indicating response codes in which the header field can be used
 Copied: header field is copied from the request to the response

An empty entry in the “where” column indicates that the header field MAY be present in all requests and responses.

The “SIP Method” columns describe the requirement specific to the ISSI definition. Possible values are:

C-ISSI: The header field is conditional for ISSI and depends on the context of the procedure.
 C-RFC: Same as the “c” condition in Table 2 (Summary of header fields) in Section 20 of [RFC3261].
 M-ISSI: The header field is mandatory for ISSI.
 M-RFC: Mandatory as defined in Table 2 (Summary of header fields) in the Section 20 of [RFC3261].
 NO-ISSI: The header field is never sent out on ISSI.
 O-ISSI: The header field is optional for ISSI.
 -: The void “-” identifier is used to indicate the header field is not applicable for the SIP Method as defined in Section 20 of [RFC3261].

If an RFSS receives a SIP Request or SIP Response that contains a SIP header field specified as being “NO-ISSI” in this document, the RFSS SHOULD ignore this SIP header field. Upon receipt of the SIP Request, the RFSS SHOULD return a GEN_IGNORE_NON_ISSI_HEADER 2xx response (refer to Annex A of this document).

The RFSS SHOULD simply discard any additional, irrelevant SIP header fields received, and it SHOULD continue with the SIP transaction.

Table 13 SIP-message header fields used on the ISSI

SIP Header Field	Where	SIP Methods					
		ACK	BYE	CANCEL	INVITE	REGISTER	OPTIONS
Accept	Request	-	NO-ISSI	-	M-ISSI	M-ISSI	M-ISSI
Accept	2xx	-	-	-	NO-ISSI	NO-ISSI	M-ISSI
Accept	415	-	-	-	M-ISSI	C-ISSI	M-ISSI
Allow	Request	-	M-ISSI	-	M-ISSI	M-ISSI	M-ISSI
Allow	2xx	-	M-ISSI	-	M-ISSI	M-ISSI	M-ISSI
Allow	Response	-	M-ISSI	-	M-ISSI	M-ISSI	-
Call-ID	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Contact	Request	NO-ISSI	-	-	M-RFC	C-ISSI(1)	NO-ISSI
Contact	1xx	-	-	-	NO-ISSI	-	-
Contact	2xx	-	-	-	M-RFC	C-ISSI(1)	NO-ISSI
Content-Disposition		C-ISSI(7)	C-ISSI(7)	-	C-ISSI(7)	C-ISSI(7)	C-ISSI(7)
Content-Length		M-ISSI	M-ISSI	M-ISSI	M-ISSI	M-ISSI	M-ISSI
Content-Type		C-ISSI(2)	C-ISSI(2)	-	C-ISSI(2)	C-ISSI(2)	C-ISSI(2)
Cseq	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Expires	Request, 2xx response	-	-	-	NO-ISSI	C-ISSI(1)	-
Expires	Non-2xx response	-	-	-	NO-ISSI	NO-ISSI	-
From	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Max-Forwards	Request	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
MIME-Version		C-ISSI(3)	C-ISSI(3)	-	C-ISSI(3)	C-ISSI(3)	C-ISSI(3)
Priority	Request	-	-	-	M-ISSI	-	-
Record-Route	Request	NO-ISSI	NO-ISSI	NO-ISSI	C-ISSI(4)	-	NO-ISSI
Record-Route	2xx, 18x	-	NO-ISSI	NO-ISSI	C-ISSI(4)	-	NO-ISSI
Route	Request	C-RFC(5)	NO-ISSI(5)	NO-ISSI(5)	C-RFC(5)	NO-ISSI(5)	NO-ISSI
To	Copied(1)	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Via	Copied	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC	M-RFC
Warning	Response	-	NO-ISSI	NO-ISSI	C-ISSI(6)	C-ISSI(6)	C-ISSI(6)

- (1): Refer to Section 5
 (2): Refer to Section 3.5.7
 (3): Refer to Section 3.5.12
 (4): Refer to Section 3.5.14
 (5): Refer to Section 3.5.15
 (6): Refer to Section 3.5.18
 (7): Refer to Section 3.5.5

If a compact form exists for a header field it SHOULD be used (Refer to Section 20 of [RFC3261]). However, examples are given in the extended form for more clarity in this document.

An RFSS SHALL be able to decode both the compact and extended forms for any SIP header field.

SIP Header Encoding for the MESSAGE method for Supplementary Data is described in **[102BACD]**.

3.5.1 Accept

This header field indicates the acceptable bodies in the response:

- SDP, P25-ISSI, and P25-SNDP bodies¹⁰, as described in Sections 3.7, 3.8, and 3.9, respectively.
- Bodies for Supplementary Data, are defined in **[102BACD]**.
- Manufacturer-specific body, if acceptable, are as described in Section 3.5.7.

The Accept header field carries a list of body type descriptors, following the syntax and semantics defined in **[RFC2616]**, and as referenced in sections 7 and 20 of **[RFC3261]**.

This header field SHALL always contain the following body type descriptors:

"application/sdp;level=1" and "application/x-tia-p25-issi"

In the event that the RFSS supports TIA-102 Supplementary Data Services, this header field SHALL also contain the following body type descriptor:

application/x-tia-p25-supdat

In the event that the RFSS supports TIA-102 Packet Data Services, this header field SHALL also contain the following body type descriptor:

application/x-tia-p25-sndcp

In the event that manufacturer-specific bodies are acceptable, this header field MAY also contain the following body type descriptor:

application/x-tia-p25-issi-<mfid>

where mfid is the manufacturer's identifier as specified in **[102BAAC]**.

mfid = 2HEX-DIGIT

Examples:

Accept: application/sdp;level=1, application/x-tia-p25-issi

¹⁰ The phrases "SDP", "P25-ISSI" and "P25-SNDP bodies", used throughout this document are simply convenient shorthand abbreviations for the literal terms "application/sdp;level=1", "application/x-tia-p25-issi" and "application/x-tia-p25-sndcp". Any use of these convenient shorthand abbreviations, or similarly structured shorthand abbreviations throughout this document does not indicate such use has been approved or mandated by the APCO/NASTD/FED Project 25 Steering Committee, nor does such appearance indicate such use is limited to Project 25 implementations exclusively.

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-issi-44

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp, application/x-tia-p25-issi-44

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-supdat

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-supdat, application/x-tia-p25-issi-44

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp, application/x-tia-p25-supdat

Accept: application/sdp;level=1, application/x-tia-p25-issi,
application/x-tia-p25-sndcp, application/x-tia-p25-supdat,
application/x-tia-p25-issi-44

3.5.2 Allow

The Allow header field indicates all SIP methods supported by the server for ISSI.

This header field SHALL include following SIP methods:

ACK, BYE, CANCEL, INVITE, REGISTER

In the event that the RFSS supports TIA-102 Supplementary Data Services, this header field SHALL also include:

MESSAGE

In the event that the RFSS supports the SIP OPTIONS method, this header field SHALL also include:

OPTIONS

Example:

Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE,
OPTIONS

3.5.3 Call-ID

As stated in [RFC3261], all SIP UAs need a means of guaranteeing that the Call-ID header fields they produce are not inadvertently generated by any other

UA. Any ISSI UA SHOULD NOT generate a Call-ID already generated by itself or by any other UA, unless overridden by method-specific (e.g., INVITE Method specific) behavior, as is the case for SIP call control over the ISSI. Indeed, for any given SU-to-SU call or group call over the ISSI, Call-IDs for the various call segments MAY or MAY NOT be the same.

The construction of the Call-ID is manufacturer-dependent.

The following is an example construction of the Call-ID, provided here for information only:

```

Call-ID      = ( "Call-ID:" / "i:") SP callid CRLF
callid       = timestamp "." id-counter "@p25dr"
timestamp    = 8HEX-DIGIT
id-counter   = 8HEX-DIGIT

```

where

- timestamp is the 8 hexadecimal characters of the 32 LSB bits of the 64 bits timestamp. The 64 bits timestamp is the host setup time relative to "January, 1st 1970 – 0h" expressed in seconds.
- id-counter MAY be chosen using a manufacturer-dependent algorithm. For instance, this field MAY be the 8 hexadecimal characters of the 32-bit integer value set to 0 at the host set-up and incremented by 1 modulo 2^{32} each time the RFSS needs a new CALL-ID identifier for generating a SIP REGISTER or a SIP INVITE.

Example:

```

Call-ID: 431ff9cd.0008c0f3@p25dr
i: 431ff9cd.0008c0f3@p25dr           (Compact form)

```

In the example above, the host has been reset to 01/01/1970-0h + 1 126 169 037 s (=431ff9cd h).

3.5.4 Contact

The Contact header field is a field containing a SIP URI with general characteristics as described in Section 3.4.1.

Contact is defined in the following way:

```

Contact      = ( "Contact:" / "m:") SP contact-id CRLF

contact-id   = "<sip:" (su-id / sg-id / snc-id) "@" domain-name
              numplan-param ">"

```

where

su-id, sg-id, and snc-id are defined in Section 3.4.1.2.
domain-name is defined in Section 3.4.1.1.
numplan-param is defined in Section 3.4.1.3.

Table 14 shows the Contact header field settings for various SIP message types over the ISSI.

Table 14 - Contact header field for ISSI

Message type	ISSI Object	Direction	Object-id	Domain
REGISTER	Unit/Group/ SN Context	Serving RFSS ⇒ Home RFSS	suid / sgid / snc-id	The Serving RFSS of the object
REGISTER	Unit/Group/ SN Context	Home RFSS ⇒ Serving RFSS	suid / sgid / snc-id	The Serving RFSS of the object
2xx Register response	Unit/Group/ SN Context	Serving RFSS ⇒ Home RFSS	suid / sgid / snc-id	The Serving RFSS of the object
2xx Register response	Unit/Group/ SN Context	Home RFSS ⇒ Serving RFSS	suid / sgid / snc-id	The Home RFSS of the object
INVITE	Unit	Calling Serving RFSS ⇒ Calling Home RFSS	calling suid	The Calling Serving RFSS
INVITE	Unit	Calling Home RFSS ⇒ Called Home RFSS	calling suid	The Calling Serving RFSS
INVITE	Unit	Called Home RFSS ⇒ Called Serving RFSS	calling suid	The Calling Serving RFSS
INVITE (in-call roaming)	Unit	Calling Home RFSS ⇒ Calling Serving RFSS	calling suid	The Calling Home RFSS
2xx Invite response	Unit	Called Serving RFSS ⇒ Called Home RFSS	called suid	The Called Serving RFSS
2xx Invite response	Unit	Called Home RFSS ⇒ Calling Home RFSS	called suid	The Called Serving RFSS
2xx Invite response	Unit	Calling Home RFSS ⇒ Calling Serving RFSS	called suid	The Called Serving RFSS
2xx Invite response (in-call roaming)	Unit	Calling Serving RFSS ⇒ Calling Home RFSS	calling suid	The Calling Serving RFSS
INVITE	Group	Serving RFSS ⇒ Home RFSS	sgid	The Serving RFSS
INVITE	Group	Home RFSS ⇒ Serving RFSS	sgid	The Home RFSS
2xx	Group	Home RFSS ⇒ Serving RFSS	sgid	The Home RFSS
2xx	Group	Serving RFSS ⇒ Home RFSS	sgid	The Serving RFSS

Example:

Contact: <sip:45546809AD1213@01.045.00187.p25dr;user=TIA-P25-SU>

m: <sip:45546809AD1213@01.045.00187.p25dr;user=TIA-P25-SU>
(Compact form)

3.5.5 Content-Disposition

The Content-Disposition header field, as generically defined in Section 20.11 of [RFC3261], describes how the SIP message bodies (if any) are to be interpreted by the RFSS receiving the SIP message. As such, this header field is present only if the SIP message has one or more message bodies.

For an SDP body, or any message body as specified in Section 3.5.7, it SHALL be set as follows:

Content-Disposition: session; handling=required

For a manufacturer-specific body, it SHALL be set as follows:

Content-Disposition: session; handling=optional

To ensure interoperability between multi-vendor RFSSs, an RFSS of a given manufacturer SHALL NOT use the “required” qualifier in the content-disposition header field for any manufacturer-specific SIP message body it sends to another RFSS of a different manufacturer.

3.5.6 Content-Length

The Content-length header field indicates the size of the message body in decimal number of octets. It is set as defined in [RFC3261].

Examples:

Content-length: 142
l: 142 (Compact form)

3.5.7 Content-Type

If present (i.e. if a message body is present), this header field indicates the type of the body present in the message.

A TIA-102 ISSI supports the SDP body type, the x-tia-p25-issi body type, the x-tia-p25-supdat body type, and the x-tia-p25-sndcp body type. See [102BACD] for the definition and use of the x-tia-p25-supdat body type when using the MESSAGE method for Supplementary Data.

Each manufacturer MAY add its own body with a name composed of the radical name “x-tia-p25-issi-” followed by the two hexadecimal digits of the manufacturer’s <mfid>. The <mfid> values are defined in [102BAAC-A].

```

man-body    =    "x-tia-p25-issi-" mfid
mfid        =    2HEX-DIGIT

```

If an RFSS receives a manufacturer-specific body with an mfid that it does not support, then the RFSS SHOULD ignore the received manufacturer-specific body, and SHOULD return a GEN_IGNORE_MANUF_BODY 2xx response (refer to Annex A of this document).

All these bodies MAY be present in the same message. In this case, the first Content-type header field in the message SHALL be set as follows in order to specify the literal string, which is used as a boundary delimiter line between the bodies:

```

Content-Type: multipart/mixed; boundary="P25 ISSI body boundary"
c: multipart/mixed; boundary="P25 ISSI body boundary" (Compact form)

```

The boundary delimiter line can be any character string (1-70 characters) beginning with "--" ([RFC2046]). The string SHALL NOT appear in any of the ISSI SIP message bodies but can be different from one message to another. However, in this specification, a unique string is used to bind the bodies in all the ISSI messages:

P25 ISSI body boundary

The boundary delimiter line following the last body part is identical to the previous boundary delimiter lines, with the addition of "--" after the character string.

A Content type indicating a SDP body if present SHALL be set as follows:

```

Content-Type: application/sdp
c: application/sdp (Compact form)

```

A Content type indicating a x-tia-p25-supdat body type if present SHALL be set as follows:

```

Content-Type: application/x-tia-p25-supdat
c: application/x-tia-p25-supdat (Compact form)

```

A Content type indicating a x-tia-p25-sndcp body type if present SHALL be set as follows:

```

Content-Type: application/x-tia-p25-sndcp
c: application/x-tia-p25-sndcp (Compact form)

```

A Content type indicating a x-tia-p25-issi body type if present SHALL be set as follows:

Content-Type: application/x-tia-p25-issi
c: application/x-tia-p25-issi (Compact form)

A Content type indicating a manufacturer-specific body for an mfid of %x07, if present is set as follows:

Content-Type: application/x-tia-p25-issi-07
c: application/x-tia-p25-issi-07 (Compact form)

Table 15 shows the Content header settings for various SIP message types over the ISSI, for SU, SG and SN Context mobility objects.

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Table 15 - Content header field used for ISSI

Message	Direction	Presence of the bodies
REGISTER message for an SU or SG with force or confirm action required	Home RFSS => Serving RFSS	P25-ISSI body
REGISTER message for an SU registration request	Serving RFSS => Home RFSS	P25-ISSI body
200 OK for a REGISTER message requesting SU service profile parameters	Home RFSS => Serving RFSS	P25-ISSI body and up to 14 P25-SNDCP bodies
200 OK for a REGISTER message requesting SG service profile parameters	Home RFSS => Serving RFSS	P25-ISSI body
REGISTER message for a PD SN Context Registration Request (with "SU Presence" parameter value set to "Confirmed")	Serving RFSS => Home RFSS	P25-ISSI body P25-SNDCP body
200 OK for a PD SN Context Registration Request ((with "SU Presence" parameter value set to "Confirmed")	Home RFSS => Serving RFSS	P25-SNDCP body
200 OK for an SU Serving Query Request	Home RFSS => Serving RFSS	P25-ISSI body and up to 14 P25-SNDCP bodies
200 OK for a PD SN Context Serving Query Request	Home RFSS => Serving RFSS	P25-SNDCP body
REGISTER message for a PD SN Context Home Query request (with "force" qualifier)	Home RFSS => Serving RFSS	P25-ISSI body P25-SNDCP body
200 OK for a PD SN Context Home Query request	Serving RFSS => Home RFSS	P25-SNDCP body
INVITE Request for an SU-to-SU Call	Any	SDP and P25-ISSI bodies
200 OK acknowledging an INVITE Request for an SU-to-SU Call	Any	SDP and P25-ISSI bodies
INVITE/Re-INVITE Request for a confirmed group call	Any	SDP and P25-ISSI bodies
INVITE/Re-INVITE Request for an unconfirmed group call	Any	SDP body and optionally P25-ISSI body
200 OK acknowledging an INVITE/Re-INVITE Request for a confirmed group call	Any	SDP and P25-ISSI bodies
200 OK acknowledging an INVITE/Re-INVITE Request for an unconfirmed group call	Any	SDP body and optionally P25-ISSI body
OPTIONS request for an RFSS Service Capability Polling procedure	Polling RFSS => Polled RFSS	P25-ISSI body
200 OK acknowledging an OPTIONS Request for an RFSS Service Capability Polling procedure	Polled RFSS => Polling RFSS	P25-ISSI body

For all the other SIP messages, if there is no manufacturer-specific body, the Content header field is not present.

Example:

```
Content-Type: multipart/mixed; boundary="P25 ISSI body boundary"

--P25 ISSI body boundary
Content-Type: application/sdp
...
--P25 ISSI body boundary
Content-Type: application/x-tia-p25-issi
...
--P25 ISSI body boundary
Content-Type: application/x-tia-p25-issi-18
...
--P25 ISSI body boundary--
```

3.5.8 Cseq

This header field is defined in **[RFC3261]**. It SHALL be present in all SIP messages.

Example:

```
Cseq: 153 INVITE
```

3.5.9 Expires

The Expires header field is defined in **[RFC3261]**.

The usage of the Expire header field is specified in Section 5.

3.5.10 From

The From header field is a field containing a SIP URI with general characteristics as described in section 3.4.1.

From is defined in the following way:

From	=	("From:" / "f:") SP from-spec CRLF
from-spec	=	"<sip:" ident "@" domain-name numplan-param ">" tag-param
ident	=	(su-id / sg-id / snc-id / "TIA-P25-RFSSCapability")
tag-param	=	";tag=" tag-id
tag-id	=	token

where

su-id, sg-id, snc-id, and "TIA-P25-RFSSCapability" are defined in section 3.4.1.2

domain-name is defined in section 3.4.1.1

numplan-param is defined in section 3.4.1.3

token is defined in the **[RFC3261]**.

The construction of the tag ID is manufacturer-dependent. However, a tag calculation method is given below for information only. A new tag SHALL be set on each new Request initiated on a new SIP dialog.

Here is a method for calculation of the tag-id:

The tag parameter of this header field is constructed as follows:

$$\text{tag-id} = \text{hostid} \text{ "." } \text{branch-number-value}$$

hostid is the 8 hexadecimal characters of the 32-bit IPV4 address of the host.
branch-number-value is the 8 hexadecimal characters of a 32-bit integer value, set to 1 for the first branch, and incremented by 1 modulo 2^{32} each time the RFSS tags a SIP dialog.

Example:

From: <sip:000ABF004567@p25dr;user=TIA-P25-SG>;tag=18596ABC.435DABFE

The From header field for the MESSAGE method for Supplementary Data is described in **[102BACD]**.

3.5.10.1 From Header Field for REGISTER, INVITE, CANCEL, ACK, and BYE
The domain name of the From header field is always set to "p25dr" for REGISTER, INVITE, CANCEL, ACK, and BYE transactions.

For a SIP REGISTER, CANCEL, ACK, or BYE transaction, the From field reflects the fully qualified identity of the SG or the SU associated with the procedure.

For a SIP dialog initiated by an INVITE in a group call, the From field reflects the fully qualified identity of the SG associated with the procedure.

For a SIP dialog initiated by an INVITE in an SU-to-SU call, the From field reflects the fully qualified identity of the original Calling SU of the SU-to-SU Call.

Example:

From: <sip:000ABF004567@p25dr;user=TIA-P25-SG>;tag=18596ABC.435DABFE

3.5.10.2 From Header Field for OPTIONS

For a SIP OPTIONS transaction, the From field contains the string "TIA-P25-RFSSCapability" indicating that it is requesting the service capability

information from another RFSS. The domain name of the From header field is set to the RFSS ID of the polling RFSS.

Example - For an RFSS with address 01.002.00003.p25dr, the From header field is:

```
From:
<sip:TIA-P25-RFSSCapability@01.002.00003.p25dr;user=TIA-P25-RFSS>;tag=18
596ABC.435DABFE
```

3.5.10.3 From Header Field for MESSAGE

The From header field format for the MESSAGE method for Supplementary Data and Supplementary Data Monitoring is specified in [102BACD].

3.5.11 Max-Forwards

The Max-Forwards header field is used as described in [RFC3261]. An RFSS SHOULD set the Max-Forwards value in all outgoing SIP messages (e.g., INVITE, ACK, etc.) to 70.

Example:

```
Max-Forwards: 70
```

3.5.12 MIME-Version

This header field SHALL be present if several bodies are included in a message. Otherwise, this header field is optional. If present, this header field SHALL be set as follows:

```
MIME-Version: 1.0
```

3.5.13 Priority

[RFC3261] defines “emergency”, “urgent”, “normal” and “non-urgent” as the priority-values, which SHALL NOT be used in the context of ISSI. [RFC3261] also defines “other-priority” as a “token”, which is used below to define the ISSI priority values:

```
priority      = "Priority:" SP priority-value CRLF
priority-value = value "," ("e" / "a")
value         = integer
```

where

“e” denotes emergency calls

“a” denotes all other calls

This header field defines the priority of the call. The usage of this priority is manufacturer-dependent (i.e., subject to RTP resources preemption rules).

For an INVITE request associated with an emergency call set-up with a priority of *x*, the field SHALL be set as follows:

Priority: *x*;e

For an INVITE request associated with a radio SU call set-up with a priority of *x*, the field SHALL be set as follows:

Priority: *x*;a

Emergency calls are of higher priority than all other calls.

Higher numerical priority means higher call priority.

1 is the lowest priority value.

Example:

Priority: 8;a
Priority: 3;e

3.5.14 Record-Route

Record-Route is defined as follows:

Record-Route = "Record-Route:" SP rec-route *("," rec-route) CRLF
rec-route = "<sip:" rr-radical-name "@" domain-name lr-param ">"
rr-radical-name = ("TIA-P25-U2Uorig" / "TIA-P25-U2Udest")

where

domain-name is defined in Section 3.4.1.1

lr-param is defined in Section 3.4.1.3

The Record-Route header field is mandatory for an SU-to-SU call Set-up and is used in order to force subsequent requests such as SIP ACK to be routed through the Calling Home RFSS and Called Home RFSS. The Record-Route header field is not used for an In-Call roaming.

When a RFSS inserts a Record-Route header field in a request, the field contains SIP URIs with general characteristics as described in Section 3.4.1.

The ISSI domain name SHALL be the domain name of the RFSS which sends the SIP request, as described in Table 16:

Table 16 - Name of the Record-Route parameter values used on the ISSI

Name of the Record-Route parameter values (Used in Table 17 and Table 18 below)	Radical name of the Record-Route header field	ISSI Domain name of the Record-Route header field
U2Uorig Record-Route	TIA-P25-U2Uorig	Domain Name of the Calling Home RFSS
U2Udest Record-Route	TIA-P25-U2Udest	Domain Name of the Called Home RFSS

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Table 17 describes the -Record-Route header fields for an SU-to-SU call setup with two separate Home RFSSs.

Table 17 - Presence of the -Record-Route header fields for an SU-to-SU Call Setup with two separate Home RFSSs.

	Calling Serving => Calling Home (INVITE)	Calling Home => Called Home	Called Home => Called Serving
INVITE Request	None	U2Uorig Record-Route added ¹¹ if a call segment with a Calling Serving RFSS exists (copied from the U2Uorig Route header field received from the Calling Serving RFSS). Otherwise, none.	U2Udest Record-Route added ¹ (copied from the U2Udest Route header field received from the Calling Home RFSS).
	Calling Home => Calling Serving (200 OK)	Called Home => Calling Home	Called Serving => Called Home
18x Response	Copied from the 18x Response received from the Called Home RFSS.	if a call segment with a Called Serving RFSS exists, copied from the 18x Response received from the Called Serving RFSS. Otherwise, copied from the INVITE Request received from the Calling Home RFSS.	Copied from INVITE Request received from the Called Home RFSS.
200 OK	Copied from the 200 OK received from the Called Home RFSS.	if a call segment with a Called Serving RFSS exists, copied from the 200 OK received from the Called Serving RFSS. Otherwise, copied from the INVITE Request received from the Calling Home RFSS.	Copied from INVITE Request received from the Called Home RFSS.

¹¹ Record-Route headers retrieved from the INVITE request of the preceding call segment are copied

Table 18 describes the -Record-Route header fields for an SU-to-SU call setup with co-located Home RFSS.

Table 18 - Presence of the -Record-Route header fields for an SU-to-SU Call Setup with co-located Home RFSS.

	Calling Serving => Co-located Home	Co-located Home => Called Serving
INVITE Request	None	First U2Uorig Record-Route is added ¹² if a call segment with a Calling Serving RFSS exists (copied from the U2Uorig Route header field received from the Calling Serving RFSS). Moreover, U2Udest Record-Route MAY be added. Otherwise, none.
	Co-located Home => Calling Serving	Called Serving => Co-located Home
200 OK	If a call segment with a Called Serving RFSS exists, copied from 200 OK received from the Called Serving RFSS. Otherwise, copied from the INVITE Request received from the Calling Serving RFSS.	Copied from the INVITE Request received from the Co-located Home.

Example:

Record-Route:

<sip:TIA-P25-U2Uorig@05.002.12000.p25dr;lr>

See example in Section 3.10 for the construction on the Record-Route header fields

¹² Record-Route headers retrieved from the INVITE request of the preceding call segment are copied

3.5.15 Route

Route is defined as follows:

```
Route      = "Route:" SP call-route *(", " call-route) CRLF
call-route = "<sip:" r-radical-name "@" domain-name Ir-param ">"
r-radical-name = ("TIA-P25-U2Uorig" / "TIA-P25-U2Udest" / "TIA-P25-Groupcall")
```

where

domain-name is defined in Section 3.4.1.1

Ir-param is defined in Section 3.4.1.3

In this specification, during call setup, the SIP INVITE request from an originating RFSS to a destination RFSS includes the Route header field. This header field contains information that enables the destination RFSS to determine its role for the call (see Section 6.4.1.1). If the destination RFSS discovers that it has not been configured to take on the expected role for the call as indicated by the Route header field, it SHALL return the appropriate SIP failure response to the originating RFSS (see Section 6.4.1.6.2). This enables the detection of system configuration errors.

The Route header field is used for an SU-to-SU call to force routing for a request through the Calling Home RFSS and the Called Home RFSS. The Route header field with a radical name of TIA-P25-U2Uorig indicates that the RFSS that receives the INVITE SHOULD consider itself as the Calling Home RFSS. A radical name of TIA-P25-U2Udest indicates that the RFSS that receives the INVITE SHOULD consider itself as the Called Home RFSS.

The Route header field is used for a Group Call when the Serving RFSS invites the Home RFSS. This field indicates that the RFSS that receives the INVITE SHOULD consider itself as the Home RFSS for the SGID (retrieved from the From header field).

When a RFSS inserts a Route header field in a request, the field contains SIP URIs with general characteristics as described in Section 3.4.1.

The ISSI domain name SHALL be the domain name of the RFSS to which the request SHALL be routed, as described in Table 19:

Table 19 - Name of the Route parameter values used on the ISSI

Name of the Route parameter values (Used in Table 20, Table 21, and Table 22 below)	Radical name of the Route header field	ISSI Domain name of the Route header field
U2Uorig Route	TIA-P25-U2Uorig	Domain Name of the Calling Home RFSS
U2Udest Route	TIA-P25-U2Udest	Domain Name of the Called Home RFSS
Groupcall Route	TIA-P25-Groupcall	Domain Name of the Home RFSS of the group

Table 20 describes the Route header fields for an SU-to-SU call setup with two separate Home RFSSs.

Table 20 - Presence of a Route header field for an SU-to-SU Call Setup with two separate Home RFSSs.

	Calling Serving => Calling Home	Calling Home => Called Home	Called Home => Called Serving
INVITE Request	U2Uorig Route	U2Udest Route	None
ACK Request (confirming receipt of a 200 OK final response) NOTE: The ACK Request confirming receipt of a non-2xx final response does not contain a Route header field.	Copied from the Record-Route header fields of the 200 OK received from the Calling Home RFSS.	If a Call segment with a Calling Serving RFSS exists, copied from the ACK Request received from the Calling Serving RFSS after removing the U2Uorig Route. Otherwise, Copied from the Record-Route header fields of the 200 OK received from the Called Home RFSS.	Copied from the ACK Request received from the Calling Home RFSS after removing the U2Udest Route.

Table 21 describes the Route header fields for an SU-to-SU call setup with co-located Home RFSSs.

Table 21 - Presence of the Route header fields for an SU-to-SU Call Setup with co-located Home RFSS.

	Calling Serving => Co-located Home	Co-located Home => Called Serving
INVITE Request	U2Uorig Route	None
ACK Request (confirming receipt of a 200 OK final response) NOTE: The ACK Request confirming receipt of a non-2xx final response does not contain a Route header field.	Copied from the Record-Route header fields of the 200 OK.	If a Call segment with a Calling Serving RFSS exists, copied from the INVITE Request received from the Calling Serving RFSS after removing the U2Uorig Route. Otherwise, Copied from the Record-Route header fields of the 200 OK received from the Called Serving RFSS.

Table 22 describes the Route header fields for a group call.

Table 22 - Presence of a Route Header Field parameter field for a Group Call

Call Segment of a Group Call	Serving RFSS => Home RFSS	Home RFSS => Serving RFSS
INVITE Request	Groupcall Route	None
ACK Request	None	None

Example:

Route: <sip:TIA-P25-U2Uorig@AF.003.13000.p25dr;lr>

See example in Section 3.10, for the construction of the Route header fields

The Route header field for the MESSAGE method for Supplementary Data is described in [102BACD].

3.5.16 To

The To header field is a field containing a SIP URI with general characteristics as described in Section 3.4.1.

A tag parameter is present when the RFSS has to tag a SIP dialog (refer to Section 19.3 of [RFC3261]).

To is defined in the following way:

To = ("To:" / "t:") SP to-spec CRLF
to-spec = "<sip:" ident "@" domain-name numplan-param ">" [tag-param]
ident = (su-id / sg-id / snc-id / "TIA-P25-RFSSCapability")

where

su-id, sg-id, snc-id, and "TIA-P25-RFSSCapability" are defined in Section 3.4.1.2

domain-name is defined in Section 3.4.1.1

numplan-param is defined in Section 3.4.1.3

tag-param is defined in Section 3.5.10.

When an RFSS initiates a Request on a new SIP dialog, it does not tag the To header field (i.e. tag-param is not present). When an RFSS responds to a

request, it SHALL tag the To field. When a tag-param is retrieved from responses, subsequent messages in the SIP dialog use this tag-param.

3.5.16.1 To Header Field for REGISTER

For a SIP REGISTER transaction, the To field represents the mobility object being registered and is further described in the subsections that follow. In the following subsections, the various cases of mobility management from Section 2.1.3.1.3 are shown in the subsection headings.

3.5.16.1.1 To field for SUID (a(ii)) and SGID registration

A Serving RFSS registering an SUID or SGID SHALL use form:

```
To      =      ("To:" / "t:") SP sd-to-spec [tag] CRLF
sd-to-spec = "<sip:" g-or-u ">"
tag = tag-param
g-or-u = grp / unit
grp = su-id "@p25dr;user=TIA-P25-SU"
unit = sg-id "@p25dr;user=TIA-P25-SG"
```

For example, for SUID 00001.002.000003 and no tag the field would appear as:

```
t: <sip:00001002000003@p25dr;user=TIA-P25-SU>
```

and for SGID 00001.002.0003 and no tag, the field would appear as:

```
t: <sip:000010020003@p25dr;user=TIA-P25-SG >
```

3.5.16.1.2 To field for Group Supplementary Data Interest (a(i))

A Serving RFSS expressing Group Supplementary Data interest SHALL use the following form:

```
To      =      ("To:" / "t:") SP sdg-to-spec CRLF
sdg-to-spec = "<sip:" sg-id "@p25dr;user=TIA-P25-SD-SG" ">" [tag]
```

For example, for interest in Group Supplementary Data group 00001.001.0003, the field would appear as:

```
t: <sip: 000010010003@p25dr;user=TIA-P25-SD-SG >
```

3.5.16.1.3 To field for FNE Group Supplementary Data Interest (c(i))

A Serving RFSS expressing FNE Group Supplementary Data interest SHALL use the following form:

To = ("To:" / "t:") SP fgrp-to-spec CRLF
fgrp-to-spec = "<sip:" sg-id "@p25dr;user=TIA-P25-SD-SGFNE" ">" [tag]

For example, for interest in FNE Group Supplementary Data group 00001.001.0003, the field would appear as:

t: <sip: 000010010003@p25dr;user=TIA-P25-SD-SGFNE >

3.5.16.1.4 To field for FNE Unit Supplementary Data Interest (c(ii),c(iii))

A Serving RFSS expressing FNE Unit Supplementary Data interest SHALL use the following form:

To = ("To:" / "t:") SP funit-to-spec CRLF
funit-to-spec = "<sip:" su-id "@p25dr;user=TIA-P25-SD-SU" ">" [tag]

where the su-id is the SUID of interest. For example, for interest in General FNE supplementary data sent by SUID 00001.002.000003, the field would appear as:

t: <sip: 00001002000003@p25dr;user=TIA-P25-SD-SU>

For interest in General FNE supplementary data for the entire System, the SUID associated with General FNE for that system is used.

3.5.16.1.5 To field for Group SD Monitoring Interest (b(i.1))

A Serving RFSS expressing SD Monitoring Interest in a group SHALL use the following form:

To = ("To:" / "t:") SP gm-to-spec CRLF
gm-to-spec = "<sip:" sg-id "@p25dr;user=TIA-P25-SDMON-SG>" [tag]

For example,

t: <sip:000010010003@p25dr;user=TIA-P25-SDMON-SG>

3.5.16.1.6 To field for FNE Group SD Monitoring Interest (b(i.2))

A Serving RFSS expressing Monitoring Interest in FNE Group Supplementary Data SHALL use the following form:

To = ("To:" / "t:") SP fgm-to-spec CRLF
fgm-to-spec = "<sip:" sg-id "@p25dr;user=TIA-P25-SDMON-SGFNE>" [tag]

For example,

t: <sip:000010010003@p25dr;user=TIA-P25-SDMON-SGFNE>

3.5.16.1.7 To field for Unit SD Monitoring Interest (b(ii), b(iii))

A Serving RFSS expressing SD Monitoring Interest in a unit SHALL use the following form:

To = ("To:" / "t:") SP um-to-spec CRLF
um-to-spec = "<sip:" su-id "@p25dr;user=TIA-P25-SDMON-SU>" [tag]

For example,

t: <sip:00001001000003@p25dr;user=TIA-P25-SDMON-SU>

For monitoring interest in General FNE supplementary data for the entire System, the SUID associated with General FNE for that system is used.

3.5.16.1.8 To field for PD SN Context Registration

A Serving RFSS registering an SN Context for a particular SUID SHALL use the following form:

To = ("To:" / "t:") SP pd-to-spec CRLF
pd-to-spec = "<sip:" snc-id "@p25dr;user=TIA-P25-PD-SNC>" [tag]

For example, for SUID 00001.002.000003, NSAPI 13 and no tag the field would appear as:

t: <sip:00001002000003D@p25dr;user=TIA-P25-PD-SNC>

3.5.16.2 To Header Field for INVITE, CANCEL, ACK, and BYE

For a SIP dialog initiated by an INVITE in a group call, the To field reflects the identity of the SG associated with the procedure.

For a SIP dialog initiated by an INVITE in an SU-to-SU call, the To field reflects the identity of the original Called SU in the SU-to-SU Call.

For a SIP CANCEL, ACK, or BYE transaction, the To field reflects the fully qualified identity of the SG or the SU associated with the procedure.

The To header field for INVITE, CANCEL, ACK, and BYE SHALL be of the form:

To	=	("To:" / "t:") SP to-spec CRLF
to-spec	=	"<sip: ident "@" domain-name numplan-param ">" [tag-param]
ident	=	(su-id / sg-id)
domain-name	=	"p25dr"
numplan-param	=	";user=TIA-P25-SU" / ";user=TIA-P25-SG"

where:

su-id and sg-id are defined in Section 3.4.1.2
domain-name is defined in Section 3.4.1.1 but can only take on the value shown above
numplan-param is defined in Section 3.4.1.3 but can only take on the values shown above
tag-param is defined in Section 3.5.10.

Examples:

To: <sip:000ABF0032457A@p25dr;user=TIA-P25-SU>
To: <sip:000ABF004567@p25dr;user=TIA-P25-SG>;tag=933f9cb2.08cf20cc

3.5.16.3 To Header Field for OPTIONS

For a SIP OPTIONS transaction for RFSS service capability polling, the To field reflects the identity of the RFSS that is being queried for its service capabilities.

The To header field for OPTIONS SHALL be of form:

To	=	("To:" / "t:") SP to-spec CRLF
to-spec	=	"<sip: ident "@" domain-name numplan-param ">" [tag-param]
ident	=	"TIA-P25-RFSSCapability"
numplan-param	=	";user=TIA-P25-RFSS"

where:

domain-name is as defined in Section 3.4.1.1 but can only take on the form of an rfss-name
numplan-param is as defined in Section 3.4.1.3 but can only take on the value shown above
tag-param is defined in Section 3.5.10.
"TIA-P25-RFSSCapability" is defined in Section 3.4.1.2.

Example - For an RFSS requesting the service capabilities of the RFSS with address 01.002.00003.p25dr, the To header field is:

To: <sip:TIA-P25-RFSSCapability@01.002.00003.p25dr;user=TIA-P25-RFSS>

3.5.16.4 To Header Field for MESSAGE

The To header field format for the MESSAGE method for Supplementary Data and Supplementary Data Monitoring is specified in [102BACD].

3.5.17 Via

The Via header field indicates the path taken by the request that SHOULD be followed in routing responses.

The Via header field is a field containing a SIP URI with general characteristics as described in Section 3.4.1.

The Branch parameter is always present.

Via is set as follows:

Via = ("Via:" / "v:") SP via-param *("," via-param) CRLF
via-param = "SIP/2.0/UDP" domain-name branch-param

where

domain-name is defined in Section 3.4.1.1

branch-param is defined in Section 3.4.1.3

As shown in Table 23, the domain name is always the RFSS name that originates the request.

Table 23 - Domain-name of the Via header field

Message	Direction	Domain
Request	RFSS 1 ⇒ RFSS 2	The RFSS 1 name
Response	RFSS 2 ⇒ RFSS 1	The RFSS 1 name

Examples:

Via: SIP/2.0/UDP
34.563.00001.p25dr;branch=z9hG4bK7a83e5750418bce23d5106b4c06cc632.2
Via: SIP/2.0/UDP
3F.876.004F0.p25dr;branch=z9hG4bK23a83e5750d517418bce06b4cc632c06.4

3.5.18 Warning

This optional header field is used in SIP-responses to carry specific P25 Condition Codes for the ISSI application.

The Warning header field contains a 3-digit code, a host name and a warning text. The 1xx and 2xx warnings are used for HTTP, whereas 3xx warnings are SIP-specific. The 399 warning code specifies a "Miscellaneous warning". This code is used in the ISSI application to transmit the specific P25 Condition Codes. The warning text contains both the specific P25 Condition Code and its associated text. The host name is filled with the rfss name where the ISSI condition occurs.

Warning is set as follows for the 399 code:

Warning	=	"Warning: 399" SP warning-content CRLF
warning-content	=	rfss-name "p25dr" SP warning-role
warning-role	=	DQUOTE p25-warning ["." role] DQUOTE
p25-warning	=	p25-warn ["." p25-condition-text]
p25-warn	=	"TIA-P25-ISSI-" p25-condition-code
p25-condition-code	=	2HEX-DIGIT
p25-condition-text	=	qdtex

Where

rfss-name is defined in Section 3.4.1.1

qdtex is as defined in **[RFC3261]**, except that it SHALL NOT contain any colon.

role = "Home" / "Serving" / "CalledHome" / "CalledServ" / "CallingHome" / "CallingServ" / "GroupHome" / "GroupServ" / "FNEhome" / "FNEserv" / "Listening" / "Unknown"

The "role" field is supplied by the RFSS generating the ISSI condition, and reflects its understanding of its role in the subject transaction. The "role" SHALL be included in Warnings generated for Supplementary Data transactions, and MAY be included in all ISSI transactions.

Example:

Warning: 399 5E.7AC.003A1.p25dr "TIA-P25-ISSI-28: Colliding Call: CalledHome"

In addition to the generic SIP responses specified in Annex A of this document, Table 24 summarizes the SIP responses, associated P25 Condition Codes, with mnemonic references used throughout this document, and their specific usage in voice and mobility management services over the ISSI. Note that P25 Condition Codes for Supplementary Data are reserved here, and are defined in **[102BACD]**.

Each mnemonic has a 3-character prefix indicating the applicable TIA-102 services for the referenced SIP response P25 Condition Code:

- MME : TIA-102 Mobility Management Services

- PDE : TIA-102 Packet Data Services
- VCE : TIA-102 Voice Services

When a manufacturer wishes to report an ISSI condition that is not covered by the standard, they SHOULD employ the GEN_MANUF_DISCRETION warning as defined in Annex A of this document and include appropriate text to describe the ISSI condition.

If a P25 Condition Code is Mandatory (according to the “Inclusion of P25 Condition Code in SIP Response” column in Table 24), then it SHALL be included for the specified SIP response. If a P25 Condition Code is Optional, then it MAY be included for the specified SIP response. If a manufacturer chooses to include an optional P25 Condition Code for any specified SIP response, the P25 Condition Code as specified in Table 24 SHALL be used.

IMPORTANT NOTE: Before assigning a new P25 Condition Code to any SIP response, check the list of P25 Condition Codes already defined in Table 24 (see below) and Table 57 (Generic SIP Responses from an RFSS over the ISSI) in this document. This is to avoid duplicate and/or conflicting assignment of P25 Condition Codes.

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Table 24 - P25 Condition Code Summary

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x0A	VCE_NO_RTP	"No RTP Resource"	606 NOT ACCEPTABLE	U2U Call: The Unit to Unit Call MAY NOT be set-up due to a lack of RTP resource	Optional
%x0B	VCE_NO_RF	"No RF Resource"	606 NOT ACCEPTABLE	U2U Call: The Unit to Unit Call MAY NOT be set up due to a lack of RF resource	Optional
%x0C	VCE_VOCODER_MODES_UNACCEPTED	"Vocoder modes unaccepted"	606 NOT ACCEPTABLE	U2U or Group Call: None of the vocoder modes offered by the adjacent RFSS is accepted by this RFSS	Mandatory
%x13	VCE_CALLING_SU_INHIBITED	"Calling SU radio inhibited"	403 FORBIDDEN	U2U Call: The Calling Home RFSS detects that the Calling SU is radio inhibited.	Optional
%x14	VCE_CALLING_SU_NOT_REG	"Calling SU not registered"	403 FORBIDDEN	U2U Call: The Calling Home RFSS detects that the Calling Serving RFSS is not registered for the Calling SUID	Mandatory
%x15	VCE_CALLED_SU_NOT_REG	"Called SU not registered"	403 FORBIDDEN	U2U Call: The Called Home RFSS detects that the Called SUID is not registered	Optional
%x16	VCE_CALLING_SU_NOT_AUTHOR	"Calling SU not authorized"	403 FORBIDDEN	U2U Call: The Calling Home RFSS detects that the Calling SU is not authorized to make this call	Optional
%x17	VCE_CALLED_SU_NOT_AUTHOR	"Called SU not authorized"	403 FORBIDDEN	U2U Call: The Called Home RFSS detects that the Called SU is not authorized to respond to this call	Optional
%x18	MME_SU_REG_DISALLOWED	"SU not authorized"	403 FORBIDDEN	Unit registration: The Home RFSS of the SU disallows the registration of this unit	Optional
%x19	VCE_SG_NOT_REG	"SG not registered"	403 FORBIDDEN	Group Call: SG's Home RFSS detects that the Serving RFSS is not registered for the SGID	Mandatory
%x1A	MME_SG_REG_DISALLOWED	"SG not authorized"	403 FORBIDDEN	Group Registration: The SG's Home RFSS disallows the registration of this group	Optional

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x1B	MME_MUST_CONFIRM_SU_PRESENCE	"SU presence confirmation required"	403 FORBIDDEN	Unit Registration: The Home RFSS of the SU denies the "SU Presence Not Confirmed" registration because the SU is currently registered at a different Serving RFSS. SN Context registration: The Home RFSS of the SU detects that the Serving RFSS is not registered for the SUID	Optional for unit registration Mandatory for SN Context registration
%x1D	VCE_CALLED_SU_INHIBITED	"Called SU radio inhibited"	403 FORBIDDEN	U2U Call: The Called Home RFSS detects that the Called SU is radio inhibited.	Optional
%x1E	VCE_CALLING_SU_UNKNOWN	"Unknown Calling SU"	404 NOT FOUND	U2U Call: the Calling Home RFSS detects the Calling SU is unknown	Optional
%x1F	VCE_CALLED_SU_UNKNOWN	"Unknown Called SU"	404 NOT FOUND	U2U Call: the Called Home RFSS detects the Called SU is unknown	Optional
%x20	VCE_CALLED_HOME_UNKNOWN	"Unknown Called Home RFSS ID"	404 NOT FOUND	U2U Call: The Calling Home RFSS detects that the RFSS ID of the Called Home RFSS is not known	Optional
%x21	MME_SG_UNKNOWN	"Unknown target group"	404 NOT FOUND	Group registration: the Home or the Serving RFSS detects that the SG is unknown	Mandatory only when this ISSI condition is originated from the Serving RFSS.
%x21	VCE_SG_UNKNOWN	"Unknown target group"	404 NOT FOUND	Group Call: the Home or the Serving RFSS detects that the SG is unknown	Mandatory
%x22	MME_SU_UNKNOWN	"Unknown SU"	404 NOT FOUND	Unit Registration: the Home or the Serving RFSS detects that the SU is unknown SN Context registration: the Home or the Serving RFSS detects that the SU is unknown	Mandatory only when this ISSI condition is originated from the Serving RFSS.
%x23	VCE_SU_NOT_REG_BY_CALLED_SERV	"Called Serving RFSS has not registered Called SUID with Called Home RFSS"	404 NOT FOUND	U2U Call: The Called Serving RFSS detects that it has not registered the called SUID with the Called Home RFSS.	Mandatory

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x28	VCE_COLLIDING_CALL	"Colliding Call"	603 DECLINE	U2U Call: The Call is rejected because the same call is proceeding in the opposite direction Group Call: Home RFSS (resp. Serving RFSS) rejects the INVITE request from the Serving RFSS (resp. Home RFSS) because it is already inviting the Serving RFSS (resp. Home RFSS)	Optional
%x29	VCE_CALLED_SU_BUSY	"Called SU is busy"	486 BUSY HERE	U2U Call: The Call is rejected because the Called SU is busy or does not respond.	Optional
%x2A	VCE_CALLED_SU_REFUSE_CALL	"Called SU refuses the call"	486 BUSY HERE	U2U Call: The Call is rejected because the Called SU has cancelled the call while it was waiting for RF resource availability.	Optional
%x2B	VCE_CANNOT_PROTECT_AUDIO	"Protected audio call not granted"	403 FORBIDDEN	U2U or Group Call: The Called Home or SG Home RFSS detects that a request for a protected audio call cannot be granted.	Optional
%x2C	VCE_CANNOT_UNPROTECT_AUDIO	"Unprotected audio call not granted"	403 FORBIDDEN	U2U or Group Call: The Called Home or SG Home RFSS detects that a request for an unprotected audio call cannot be granted.	Optional
%x2D	MME_OBJECT_UNKNOWN	"Unknown Mobility Object"	404 NOT FOUND	Registration of Mobility Object: the Home or the Serving RFSS detects that the mobility object is unknown	Mandatory only when this ISSI condition is originated from the Serving RFSS.
%x2E	MME_OBJECT_REG_DISALLOWED	"Registration of mobility object disallowed"	403 FORBIDDEN	Registration of Mobility Object: Home RFSS of the mobility object disallows registration of this object	Optional
%x2F	MME_SU_AUTHEN_REQUIRED	"SU Requires Authentication"	403 FORBIDDEN	Unit Registration: The Home RFSS of the SU denies the registration because the SU requires Subscriber Authentication	Mandatory
%x50	VCE_NO_RF_AT_CALLED_SERV	"RF resource not available"	183 SESSION PROGRESS	U2U call: the called RFSS has no RF resource	Optional
%x55	MME_PROCESSING_HOME_QUERY	"Query processing in progress"	183 SESSION PROGRESS	Home Query: to inform the Home RFSS that the Serving RFSS is attempting to fulfill the received query request.	Optional
%x56	MME_SU_INFO_NOT_AUTHOR	"SU info not authorized"	403 FORBIDDEN	Serving Query: to inform the Serving RFSS that the Serving RFSS is not authorized to receive information about the subject SU.	Optional

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
%x57	MME_OBJ_NOT_REG_AT_SERV	"Mobility object is not registered at this Serving RFSS"	403 FORBIDDEN	Serving Deregistration: Home RFSS detects that the mobility object is not registered at this Serving RFSS.	Optional
%x70-7F		(Reserved for Supplementary Data use, see [102BACD])			Optional
%x82	PDE_SU_NOT_AUTHOR	"SU not authorized to perform packet data services"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that the SU is not authorized to perform packet data services	Optional
%x83	PDE_STAT_IP_ADDR_INCORRECT	"Incorrect Static IP address"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that the static IP address provided by the SU is incorrect	Optional
%x84	PDE_STAT_IP_ADDR_UNSUPPORTED	"Static IP address not supported"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that static IP address assignment is not supported	Optional
%x85	PDE_DYN_IP_ADDR_POOL_EMPTY	"Dynamic IP address pool empty"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that there are no more available IP addresses for a dynamic IP address assignment	Optional
%x86	PDE_DYN_IP_ADDR_UNSUPPORTED	"Dynamic IP address not supported"	403 FORBIDDEN	SN Context registration: The Home RFSS of the SU detects that dynamic IP address assignment is not supported	Optional

3.6 Status-Code and Reason-Phrase

SIP responses have a Status-Line as their start-line. This status line is composed of a 3-digit integer Status-Code and a Reason-Phrase giving a short textual description of the Status-Code. See Section 7.2 of [RFC3261] for further details. Refer to Section 25.1 of [RFC3261] for an ABNF description of the Status-Code and Reason-Phrase.

The usage of the status codes for ISSI is defined in Section 6.

All the status codes used have the meaning and the syntax as described in [RFC3261].

For ISSI, the reason phrases defined in [RFC3261] are used without any modification or addition. The Warning header field is used to provide additional information regarding the ISSI-specific response.

Examples:

SIP/2.0 180 Ringing
SIP/2.0 403 Forbidden

3.7 SDP Announcements

This section describes the SDP parameters encoding used on the ISSI. The session body is described with SDP as defined in [RFC4566] and [RFC3264].

The RFSS SHOULD simply discard any additional, irrelevant SDP parameters received, and it SHOULD proceed with the call.

3.7.1 SDP offer/answer usage

SDP announcements are used to describe the RTP unicast sessions opened between the RFSS. They are conveyed in the SIP messages. They SHALL be present in the INVITE or re-INVITE requests for the SDP offer and in the associated 200 (OK) responses for the SDP answer but never in the ACK messages.

An RFSS SHALL support one or more of the following vocoder modes:

- Full Rate, with encoding-name of the rtpmap attribute set to "X-TIA-P25-FullRate" (see Section 3.7.2.2.2)
- Half Rate, with encoding-name of the rtpmap attribute set to "X-TIA-P25-HalfRate" (see Section 3.7.2.2.2)
- Native, with encoding-name of the rtpmap attribute set to

“X-TIA-P25-Native” (see Section 3.7.2.2.2)

NOTE: Older RFSS implementations are only capable of offering Full Rate vocoder mode with encoding-name of the rtpmap attribute set to “X-TIA-P25-IMBE” (see Section 3.7.2.2.2).

For backwards compatibility with older RFSS implementations, all RFSSs SHALL support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to “X-TIA-P25-IMBE”.

In the event that the offerer RFSS is aware (e.g., from past interactions) that the answerer RFSS is an RFSS of an older implementation, the SDP offer in an INVITE or re-INVITE request SHOULD, in accordance with local policy, only contain a media format corresponding to the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to “X-TIA-P25-IMBE”.

Otherwise, the SDP offer in an INVITE or re-INVITE request SHALL only contain one or more media formats corresponding to any combination of the following vocoder modes being offered:

- Full Rate, with encoding-name of the rtpmap attribute set to “X-TIA-P25-FullRate”
- Half Rate, with encoding-name of the rtpmap attribute set to “X-TIA-P25-HalfRate”
- Native, with encoding-name of the rtpmap attribute set to “X-TIA-P25-Native”

Media formats in the SDP offer SHALL be listed in the order of preference, with the first format listed being preferred.

The SDP answer in the associated 200 OK response SHALL indicate which particular media format, chosen from all media format(s) being offered, has been accepted.

In the event that none of the media formats listed in the SDP offer is accepted by an answerer RFSS, it SHALL respond with a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: Older RFSS implementations are only capable of responding with a GEN_SDP_PARAMS_UNACCEPTABLE failure response.

Upon receipt of the VCE_VOCODER_MODES_UNACCEPTED or GEN_SDP_PARAMS_UNACCEPTABLE failure response, the offerer RFSS MAY retry the SDP offer in a new INVITE request, for example, offering Full Rate vocoder mode with encoding-name of the rtpmap attribute set to “X-TIA-P25-IMBE”.

As shown in Table 25, 2 types of SDP offer and 2 types of SDP answer exist for

ISSI.

A Type 1 SDP offer or answer is characterized by an even RTP port number between 2 and 65532, inclusive.

A Type 2 SDP offer or answer is characterized by an RTP port number of 65534 (%xFFFE) and an "a=inactive" attribute in the associated media description.

The RTP port number of 65534 SHALL NOT be used to characterize any other type of SDP offer or answer.

NOTE: In older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7.2.2.2), a non-zero RTP port number characterizes a Type 1 SDP offer or answer, whereas a Type 2 SDP offer or answer is characterized by an RTP port number of zero.

When an RFSS is interacting with another RFSS of older implementation:

- It SHALL send SDP offers/answers in accordance with Table 25, and with the type of the SDP offer/answer as characterized in the NOTE above, and
- It SHALL interpret the type of the received SDP offer/answer in accordance with Table 25 and the type characterizations in the NOTE above.

When an RFSS sends an initial INVITE request, it SHALL convey a Type 1 SDP offer. The receiving RFSS MAY acknowledge with a 200 OK containing either a Type 1 SDP answer or a Type 2 SDP answer.

Once a SIP session is set-up, an RFSS MAY send a Re-INVITE conveying a Type 1 SDP offer or a Type 2 SDP offer. The peer RFSS MAY acknowledge by a 200 OK conveying either a Type 1 SDP answer or Type 2 SDP answer for a Type 1 SDP offer, but MAY only acknowledge a Type 2 SDP offer with a Type 2 SDP answer.

Type 2 SDP offer and answer are only used for the group call and SHALL NOT be used for an SU-to-SU call.

Table 25 - SDP offer / answer summary

SIP Message	ISSI Object	Direction	Type of the SDP offer/SDP answer
INVITE	Unit	Calling Serving RFSS ⇔ Calling Home RFSS	Type 1 offer
INVITE	Unit	Calling Home RFSS ⇔ Called Home RFSS	Type 1 offer
INVITE	Unit	Called Home RFSS ⇔ Called Serving RFSS	Type 1 offer
INVITE	Unit	Calling Home RFSS ⇔ Calling Serving RFSS	Type 1 offer
200 OK (INVITE response)	Unit	Called Serving RFSS ⇔ Called Home RFSS	Type 1 answer
200 OK (INVITE response)	Unit	Called Home RFSS ⇔ Calling Home RFSS	Type 1 answer
200 OK (INVITE response)	Unit	Calling Home RFSS ⇔ Calling Serving RFSS	Type 1 answer
200 OK (INVITE response)	Unit	Calling Serving RFSS ⇔ Calling Home RFSS	Type 1 answer
INVITE	Group	Serving RFSS ⇔ Home RFSS	Type 1 offer
200 OK (INVITE response)	Group	Home RFSS ⇔ Serving RFSS	Type 1 or Type 2 answer
INVITE	Group	Home RFSS ⇔ Serving RFSS	Type 1 offer
200 OK (INVITE response)	Group	Serving RFSS ⇔ Home RFSS	Type 1 or Type 2 answer
Re-INVITE	Group	Home RFSS ⇔ Serving RFSS	Type 1 or Type 2 offer
200 OK (Re-INVITE response)	Group	Serving RFSS ⇔ Home RFSS	Type 1 or Type 2 answer
Re-INVITE	Group	Serving RFSS ⇔ Home RFSS	Type 1 or Type 2 offer
200 OK (re-INVITE response)	Group	Home RFSS ⇔ Serving RFSS	Type 1 or Type 2 answer

3.7.2 SDP parameters encoding

The SIP header found at the beginning of the SDP part of the message body is set as defined in Section 3.5.7.

The SDP announcement used in the ISSI is restricted to one session-level part and at least one media-description part. Restrictions applicable to these parts are defined below.

The fields that are not described in the following subsections SHALL NOT be used for the ISSI.

3.7.2.1 Session-level part

The mandatory fields of the session part are version, origin, session name, time, and connection data. These fields are described below.

3.7.2.1.1 The protocol version field

The protocol version is set to 0 as defined in [RFC4566].

version-field = "v=0" CRLF

3.7.2.1.2 The origin field

This field gives the originator of the media session. It is expressed as follows:

origin-field = "o=" username SP session-id SP session-version SP
network-type SP address-type SP address CRLF

[RFC4566] requires that (username, session-id, network-type, address-type, address) forms a globally unique identifier for the SDP session. As (network-type, address-type, address) identifies the host where the session is created, (username, session-id) has to identify the session within the host. A void identifier is used for the username, whereas the session-id distinguishes the sessions.

then

username	SHOULD be set to "-"
session-id	is a numeric field which identifies the session. A timestamp and an id-counter expressed in decimal is suggested.
session-version	the version of the announcement. A timestamp is recommended in [RFC4566] , a timestamp as used in the session-id is suggested.
network-type	is set to "IN"
address-type	is set to "IP4"
address	is the IP address

username	=	"-"
session-id	=	id-counter
id-counter	=	1*DIGIT
session-version	=	timestamp
timestamp	=	1*DIGIT
network-type	=	"IN"
address-type	=	("IP4")
address	=	IP4-address

The IP4-address is described in **[RFC4566]**.

Example:

o=- 43258265 43258376 IN IP4 135.24.98.126

3.7.2.1.3 The session name field

The session name is set to TIA-P25-U2UCall for a unit to unit call, or to TIA-P25-GroupCall for a group call.

session-name-field = ("TIA-P25-U2UCall" / "TIA-P25-GroupCall") CRLF

Example:

s=TIA-P25-GroupCall

3.7.2.1.4 The time field

The session initial time and duration are not managed by SDP. So, unbounded session duration is used.

```
time-field    =    "t=" start-time SP stop-time CRLF
start-time    =    "0"
stop-time     =    "0"
```

Example:

```
t=0 0
```

3.7.2.1.5 The connection field

If only one media is used the connection field can be set at session level with the following parameters:

```
connect-field =    "c=" network-type SP address-type SP connect-address CRLF
network-type  =    "IN"
address-type  =    "IP4"
connect-address =    IP4-address
```

The IP4-address is described in **[RFC4566]**.

Example:

```
c=IN IP4 192.27.36.201
```

3.7.2.2 Media description part

3.7.2.2.1 The media announcement field

The media line used for the description of the ISSI RTP session is described below:

```
media-field =    "m=" media SP port SP transport SP fmt-list CRLF
media       =    "audio"
port        =    1*DIGIT
transport   =    "RTP/AVP"
fmt-list    =    1*(fmt SP) / 1fmt
fmt         =    "100" / "101" / "102"
```

with

```
media    is the media type "audio"
port     is the even RTP port number used, as defined in [RFC4566]
          and [RFC3264], with different port numbers assigned for
```


transport Type 1 SDP offer/answer and Type 2 SDP offer/answer, as
 fmt list specified in Section 3.7.1
 is set to "RTP/AVP"
 is as follows:

In SDP offer: One or more media formats being offered, listed in the order of preference, with the first format listed being preferred; or

In SDP answer: A particular media format, chosen from all media format(s) being offered, that has been accepted

fmt is set to
 "100", for Full Rate vocoder mode; or
 "101", for Half Rate vocoder mode; or
 "102", for Native vocoder mode.

The range 96-127 for the payload type is defined as "dynamic" in [RFC3551]. It is used while the ISSI audio profile is not registered in IANA.

Examples:

m=audio 49280 RTP/AVP 100
 m=audio 49280 RTP/AVP 102 101 100

3.7.2.2.2 Attributes fields

The general form of an rtpmap attribute field is:

attribute-field = "a=" attribute CRLF
 attribute= "rtpmap:" payload-type SP encoding-name "/" clock-rate
 payload-type= "100" / "101" / "102"
 encoding-name= "X-TIA-P25-IMBE" / "X-TIA-P25-FullRate" /
 "X-TIA-P25-HalfRate" / "X-TIA-P25-Native"
 clock-rate = "8000"

The payload-type and encoding-name in the rtpmap attribute of the media description SHALL be set as follows:

Table 26 - payload-type and encoding-name of the rtpmap attribute

ISSI Vocoder Mode	payload-type	encoding-name
Full Rate	100	X-TIA-P25-FullRate
Half Rate	101	X-TIA-P25-HalfRate
Native	102	X-TIA-P25-Native

NOTE: In older RFSS implementations that only support the Full Rate vocoder mode (payload-type of value 100), the encoding-name of the rtpmap attribute is set to "X-TIA-P25-IMBE"

The clock rate parameter is set to 8000.
No encoding parameter is defined.

Examples:

```
a=rtpmap:100 X-TIA-P25-FullRate/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:100 X-TIA-P25-IMBE/8000
```

3.7.3 SDP examples

3.7.3.1 SDP Offer with Multiple Vocoder Modes Being Offered

This example describes a session with the following parameters:

- the session is set up for a group call
- the setup timestamp is 432ed265
- this is a new session, and it is the first version of the session
- IPv4 is used, and the host 192.53.18.122 create the session and handles the data connection on port 49298
- Native, Half Rate, and Full Rate vocoder modes are offered

then the SDP part is set as follows:

```
v=0
o=- 43278265 0 IN IP4 192.53.18.122
s=TIA-P25-GroupCall
c=IN IP4 192.53.18.122
t=0 0
m=audio 49298 RTP/AVP 102 101 100
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

3.7.3.2 SDP Offer in Older RFSS Implementations

This example describes a session with the following parameters:

- the session is set up for a group call
- the setup timestamp is 432ed265
- this is a new session, and it is the first version of the session
- IPv4 is used, and the host 192.53.18.122 create the session and handles the data connection on port 49298
- Full Rate vocoder mode is offered

then the SDP part is set as follows:

```
v=0
o=- 43278265 0 IN IP4 192.53.18.122
s=TIA-P25-GroupCall
c=IN IP4 192.53.18.122
t=0 0
m=audio 49298 RTP/AVP 100
a=rtpmap:100 X-TIA-P25-IMBE/8000
```

3.8 X-TIA-P25-ISSI Body Parameters

This section describes the X-TIA-P25-ISSI application used to carry the User Service Profile (USP) information, Group Service Profile (GSP), RFSS service capability information, and ISSI specific call parameters information.

In any transaction, a given X-TIA-P25-ISSI application MAY only carry the USP information, or the GSP information, or the RFSS service capability information. The RFSS SHALL simply discard any additional, irrelevant profile or parameter information received, and it SHALL proceed with the call. Depending on the implementation, the RFSS MAY generate an error log to locally report this anomaly.

The application/x-tia-p25-issi media type is defined by the following information:

```
Media type name:  application
Media subtype name:  x-tia-p25-issi
Required parameters:  version
'Version' gives the version of the X-TIA-P25-ISSI protocol. The initial
protocol version is set to: "version = 1.0"
Optional parameters:  None
```

The SIP Content-type header field is set as defined in Section 3.5.7.

The X-TIA-P25-ISSI descriptions are entirely textual. The X-TIA-P25-ISSI description consists in one or more lines of text. The order of these lines has no special meaning. The names and parameters of the fields are case-insensitive strings.

3.8.1 Group Service Profile attributes

Table 27 summarizes the available attributes for groups, which are detailed in the following subsections.

Table 27 - Group Service Profile encoding in the ISSI body

Name	Field	Compact form	Meaning	Default
Access Permission	g-access	g-as	Defines whether or not the group is allowed to make non-emergency calls.	Non-emergency Calls enabled.
Announcement Group	g-agroup	g-ag	Defines the announcement subscriber group for the group.	No Default
Priority	g-pri	g-p	Defines the priority of the group.	Subject to Local Policy
Emergency Capable	g-ecap	g-ec	Defines whether or not emergency calls MAY be made on the group.	Emergency Calls Enabled.
Emergency Preemption	g-epreempt	g-ep	Defines whether ruthless or non-ruthless pre-emption is used for emergency calls.	Non-ruthless (next available resource)
Hang Time	g-rfhangt	g-h	Defines the hang-timer for the group.	0 seconds (Transmission trunked)
Confirmed Call Setup Time	g-ccsetupT	g-c	Defines the confirmed call setup timer for the group.	0 seconds (Unconfirmed calls)
Interrupt Mode	g-intmode	g-in	Defines under what conditions talk-group members MAY interrupt each other during a call	Never allowed
Security	g-sec	g-s	Defines the allowable security levels of the call	Clear or Secure
Interconnect Flag	g-ic	g-ic	Defines whether or not the group MAY have PSTN sourced calls	Interconnect calls disallowed
Interconnect Security	g-icsecstart	g-ics	Defines the required security level at the beginning of an interconnect call.	Clear
Manufacturer Specific Parameter	g-man<mfid>...	g-m<mfid>..	Allows for manufacturer specific attributes	None

All the Group Service Profile lines begin with “g-“ .

3.8.1.1 Access Permission (g-access)

The Group Access Permission attribute defines whether or not the group is allowed to make non-emergency calls at the Serving RFSS. It has the following format in ABNF notation:

access-permission = “g-access:” / ”g-as:” boolean CRLF

where “0” is used to indicate that non-emergency calls on the group MAY NOT be placed from the Serving RFSS, and “1” is used to indicate that non-emergency calls MAY be placed.

If the Access Permission attribute is not present, non-emergency calls MAY be placed from the Serving RFSS.

3.8.1.2 Announcement Group (g-agroup)

The Announcement Group attribute defines the announcement subscriber group for the group. It has the following format in ABNF notation:

announcement-group = “g-agroup:” / ”g-ag:” 12HEX-DIGIT CRLF

where the 12HEX-DIGIT represent the fully-qualified hexadecimal form of the AGID.

If no announcement group attribute is included, then the group has no announcement subscriber group.

3.8.1.3 Priority (g-pri)

The Priority attribute defines the suggested priority of the group when queuing group calls on a busy system. The Serving RFSS MAY independently determine the relative priorities of competing groups homed to different systems. However, it SHOULD maintain the relative priorities provisioned by the Home RFSS for competing calls homed to the same system.

NOTE - Operators need to be aware that failing to maintain relative priorities could cause interoperability issues.

The Priority attribute has the following format, in ABNF notation:

group-priority = “g-pri:” / ”g-p:” priority CRLF

priority = “1” / “2” / “3” / “4” / “5” / “6” / “7” / “8” / “9” / “10”

where the priority is restricted to the integer values 1 through 10. Group calls with higher priority values are to be de-queued before group calls with lower priority values.

1 is the lowest priority value.

If no Priority attribute is provided by the Home RFSS, the Serving RFSS MAY assign a priority to the group in accordance with local policy and programming.

3.8.1.4 Emergency Capable (g-ecap)

The Emergency Capable attribute defines whether or not emergency calls can be made on the group. It has the following format, in ABNF notation:

emergency-capable = "g-ecap:" / "g-ec:" boolean CRLF

where a boolean "0" is used to indicate that emergency calls are not allowed, and a "1" is used to indicate that emergency calls are allowed. If the emergency capable attribute is not provided, emergency calls are allowed from the Serving RFSS.

3.8.1.5 Emergency Preemption (g-epreempt)

The Emergency Preemption attribute indicates the type of pre-emption to be applied for emergency calls. Its ABNF format is as follows:

emergency-preemption = "g-epreempt:" / "g-ep:" boolean CRLF

where a boolean "0" indicates that non-ruthless preemption is to be used (next available resource) and a boolean "1" indicates that ruthless preemption is to be used. If the emergency preemption attribute is not provided, non-ruthless preemption is to be used.

3.8.1.6 RF Hang Time (g-rfhangt)

The RF Hang Time attribute specifies the RF hang-timer for the group. It has the following format in ABNF notation:

hang-time = "g-rfhangt:" / "g-h:" rfhangt CRLF

rfhangt = int

where the integer value is the RF hang-timer in tenths of seconds and is limited to the range 0 – 32767. If the RF hang timer attribute is not specified, the RF hang timer is 0 (Transmission Trunked).

The RF hang time has no infinity value.

3.8.1.7 Confirmed Call Setup Time (g-ccsetupT)

The Confirmed Call setup attribute specifies the maximum setup time for a confirmed call on the group. It has the following format in ABNF notation:

```
cc-setup      = "g-ccsetupT:" / "g-c:" ccT CRLF
ccT           = int
```

This attribute SHALL be used to indicate whether calls on the group are confirmed or unconfirmed, as detailed in Section 3.8.5.6, with integer values of the attribute defined as follows:

- A value of 0 indicates that the call is unconfirmed.
- A value in the range 1-32766 specifies the setup time in tenths of seconds and indicates that the call is confirmed.
- A value of 32767 indicates an infinite setup time, and indicates that the call is confirmed.

This attribute MAY also be used for other purposes in accordance with local policy.

3.8.1.8 Interrupt Mode (g-intmode)

The Interrupt Mode attribute specifies the audio interrupt mode of the voice group. It has the following format in ABNF notation:

```
interrupt-mode = "g-intmode:" / "g-in:" imode CRLF
imode          = "0" / "1" / "2"
```

where imode = "0" indicates that audio interruption by SUs within a talk group call is never allowed, "1" indicates that it is allowed based on SU priority, and "2" indicates that it is always allowed. If the Interrupt Mode is not specified, the Interrupt Mode is 0.

3.8.1.9 Group Security Level (g-sec)

The Group Security Level attribute indicates the allowable security levels for the group. It has the following ABNF format:

```
security-level = "g-sec:" / "g-s:" security CRLF
security       = "1" / "2" / "3"
```

where

"1"
"2"

indicates clear mode calls only are allowed
indicates that secure mode calls only are allowed, and

“3” indicates that both clear and secure mode calls are allowed.

If the group security attribute is not provided, both clear and secure mode calls are to be allowed.

3.8.1.10 Interconnect Flag (g-ic)

The Interconnect Flag attribute defines whether or not the group MAY have PSTN sourced calls. It has the following format in ABNF notation:

interconnect = “g-ic:” / ”g-ic:“ boolean CRLF

where “0” indicates that PSTN source calls are not allowed, and “1” indicates that they are. If the interconnect flag is not present, interconnect calls are not allowed.

3.8.1.11 Interconnect Security (g-icsecstart)

The Interconnect Security parameter indicates whether PSTN interconnect calls are to start in the clear or in a secure mode. Its ABNF format is as follows:

Interconnect-security = “g-icsecstart:” / ”g-ics:“ boolean CRLF

where “0” indicates that interconnect calls are to start in the clear, and “1” indicates that they are to start secure. If the interconnect security attribute is not provided, calls are to be started in the clear.

3.8.1.12 Manufacturer Specific (g-man<mfid>)

The Manufacturer Specific attribute allows manufacturer defined profile information that MAY be used for private or experimental purposes. It SHALL have the following radical with the following ABNF notation:

man-grpprof-radical = “g-man” / ”g-m“ mfid “-” octet-string “:“ pvalue CRLF

mfid = 2HEX-DIGIT

where pvalue is defined in Section 25.1 of RFC 3261.

The mfid values are defined in **[102BAAC-A]**

Manufacturer specific attributes that are not understood by an RFSS SHOULD be ignored.

Example:

g-manA4-myparameter:1

3.8.1.13 Example Group Service Profiles

3.8.1.13.1 Fully Specified Example

The following example illustrates a fully specified group service profile using default values (where appropriate) or typical values (where no default is specified).

```
g-access:1
g-agroup:0B4561A27271
g-pri:2
g-ecap:1
g-epreempt:0
g-rfhangt:0
g-ccsetupt:0
g-intmode:0
g-sec:1
g-ic:0
g-icsecstart:0
```

The same list in compact form:

```
g-as:1
g-ag:0B4561A27271
g-p:2
g-ec:1
g-ep:0
g-h:0
g-c:0
g-in:0
g-s:1
g-ic:0
g-ics:0
```

3.8.1.13.2 Minimum Group Service Profile

The minimum group service profile has no attributes, since all attributes are optional. In this case, the Serving RFSS SHALL use implicit default values for all attributes, except those without default values.

3.8.2 User Service Profile attributes

Table 28 summarizes the available attributes for users, which are detailed in the following subsections.

Table 28 - User Service Profile encoding in the ISSI body

Name	Field	Compact form	Meaning	Default
System Access Permission	u-access	u-as	Indicates the type of access allowed to the unit on the Serving RFSS.	Full Access
Duplexity	u-dup	u-d	Indicates whether the unit is full or half-duplex.	Half-duplex
Secure Capable	u-sec	u-s	Indicates whether or not the SU is capable of secure calls.	No
Group Call Capability	u-gcall	u-g	Indicates the types of group calls that the SU MAY make.	All
U2U Call Permissions	u-ucall	u-uc	Indicates the SU's permissions for unit-to-unit calls.	None
U2U Call Priority	u-upri	u-p	Indicates SU's priority in unit to unit calls.	Subject to local policy
Interconnect Permission	u-iccall	u-icc	Indicates the types of interconnect calls the SU MAY participate in.	None
Interconnect Secure	u-icsec	u-ics	Indicates acceptable security modes for interconnect calls	Clear or Secure
Interconnect Priority	u-icpri	u-icp	Priority level when involved in interconnect calls	Subject to local policy
Authentication Type/Capability	u-authtype	u-at	Authentication Type	Unit does NOT support Authentication
Authentication Policy	u-authpol	u-apo	Authentication Policy	Serving RFSS MAY authenticate the unit.
Authentication Parameters	u-authparm	u-apa	Authentication Parameters	None
Availability Check	u-availcheck	u-ac	Indicates the ability of the SU to perform availability check	Both Availability Check and Direct Call
Call Set-Up preference	u-prefsetup	u-prf	Indicates the calling SU preference to make an availability check or a direct call of the Called SU when it sets up an SU-to-SU call.	Availability Check
Radio Inhibit	u-radioinhib	u-rdinh	Indicates whether or not the SU is radio inhibited [102AABD].	Not inhibited
Packet Data Capable	u-packetdata	u-pd	Indicates whether the SU is TIA-102 Packet Data capable	No

Name	Field	Compact form	Meaning	Default
Manufacturer Specific Parameter	u-man<mfid>...	u-m<mfid>...	Allows for manufacturer specific attributes	None

All the User Service Profile lines begin with “u-”.

3.8.2.1 System Access Permission (u-access)

The System Access Permission attribute defines the access permissions for the SU on the Serving RFSS. It has the following format:

system-access = “u-access:” / ”u-as:” access CRLF

access = “0” / “1” / “2”

where access of

“0”

indicates no access is allowed,

“1”

indicates that full access is allowed, and

“2”

indicates that only emergency access is allowed.

If no system access permission is specified, full access is allowed subject to the local policies and programming of the Serving RFSS.

3.8.2.2 Duplexity (u-dup)

The Duplexity attribute indicates whether or not the unit is full duplex. It has the following ABNF format:

duplexity = “u-dup:” / ”u-d:” boolean CRLF

where a boolean value of “0” indicates a half-duplex SU, and a value of “1” indicates a full-duplex SU. If the duplexity attribute is not specified, the unit is half-duplex.

3.8.2.3 Secure Capable (u-sec)

The Secure Capable attribute indicates whether or not the unit is capable of secure (encrypted) operation. It has the following ABNF format:

secure = “u-sec:” / ”u-s:” boolean CRLF

where a boolean value of “0” indicates no secure capability, and a value of “1” indicates an encryption capability. If the secure capable attribute is not specified, the unit is not capable of secure operation.

3.8.2.4 Group Call Permissions (u-gcall)

The Group Call Capability attribute specifies the types of group calls allowed that the given SU MAY initiate. Its format is as follows:

group-permissions = "u-gcall:" / "u-g:" gcap CRLF

gcap = "0" / "1" / "2" / "3"

where gcap of

"0"

indicates no group call permission

"1"

indicates Non-emergency group call permission

"2"

indicates Emergency group call permission, and

"3"

indicates both emergency and non-emergency permission.

If the capability is not specified, the unit is given both emergency and non-emergency permissions.

3.8.2.5 Unit-to-unit Call Permissions (u-ucall)

The Unit-to-unit Call Permissions (u-ucall) indicates the types of unit-to-unit calls in which the SU MAY participate. Its format is:

unitcall-permissions = "u-ucall:" / "u-uc:" ucap CRLF

ucap = "0" / "1" / "2" / "3"

where ucap of

"0"

indicates no unit-to-unit call permission

"1"

indicates receive permission

"2"

indicates permission to initiate a unit-to-unit call

"3"

indicates permission to both receive and initiate a unit-to-unit call.

If no unit-to-unit call permission is provided, the unit has no permission to participate in unit-to-unit calls.

3.8.2.6 Unit-to-unit Call Priority (u-upri)

The unit-to-unit Call Priority attribute specifies the priority to use for the individual unit in unit-to-unit calls. Its format is:

unit-priority = "u-upri:" / "u-p:" priority CRLF

priority = "1" / "2" / "3" / "4" / "5" / "6" / "7" / "8" / "9" / "10"

where the priority is restricted to the integer values 1 through 10. Units with

higher priority are more important than units with lower priority values. 1 is the lowest priority value.

If no Priority attribute is provided by the Home RFSS, the Serving RFSS MAY assign a priority to the unit in accordance with local policy and programming.

3.8.2.7 Interconnect Permissions (u-iccall)

The Interconnect Permissions attribute specifies the permissions of the unit vis-à-vis interconnect (PSTN) calls. It has the following format:

interconnect-permission = "u-iccall:" / "u-icc:" icperm CRLF

icperm = "0" / "1" / "2" / "3"

where icperm of

"0"	indicates that the unit does not have permission to participate in interconnect calls,
"1"	indicates that the unit has permission to be the recipient of an interconnect call,
"2"	indicates that the unit has permission to be the initiator of an interconnect call, and
"3"	indicates that the unit has permission to both initiate and receive interconnect calls.

If no interconnect permission attribute is specified, the unit does not have permission to participate in interconnect calls.

3.8.2.8 Interconnect Security (u-icsec)

The Interconnect Security attribute specifies the acceptable security modes of interconnect calls. Its format is as follows:

ic-security = "u-icsec:" / "u-ics:" icsec CRLF

icsec = "1" / "2" / "3"

where icsec of

"1"	indicates clear mode calls only are allowed
"2"	indicates that secure mode calls only are allowed, and
"3"	indicates that both clear and secure mode calls are allowed.

If no interconnect security attribute is specified, both secure and clear calls are acceptable.

3.8.2.9 Interconnect Call Priority (u-icpri)

The Interconnect Call Priority attribute specifies the priority to use for the individual in interconnect calls. Its format is:

unit-priority = "u-icpri:" / "u-icp:" priority CRLF

priority = "1" / "2" / "3" / "4" / "5" / "6" / "7" / "8" / "9" / "10"

where the priority is restricted to the integer values 1 through 10. Units with higher priority are more important than units with lower priority values. 1 is the lowest priority value.

If no Priority attribute is provided by the Home RFSS, the Serving RFSS MAY assign a priority to the unit in accordance with local policy and programming.

3.8.2.10 Authentication Type/Capability (u-authtype)

The Authentication Type/Capability attribute specifies the type of authentication to be applied to the unit. The ABNF format of the attribute is:

authentication_type = "u-authtype:" / "u-at:" atype CRLF

atype = "0" / "1"

where atype of
"0"

indicates that the unit does not support authentication, and,

"1"

indicates that the unit SHALL be authenticated using link layer authentication as specified in **[102AACE]**, subject to the authentication policy specified in Section 3.8.2.11.

If no authentication type is specified, the unit does not support authentication.

3.8.2.11 Authentication Policy (u-authpol)

The Authentication Policy attribute specifies the authentication policy to be applied to the unit. Its ABNF format is:

authentication_policy = "u-authpol:" / "u-apo:" apol CRLF

apol = "0" / "1" / "2"

where apol of
"0"

indicates that the Serving RFSS MAY authenticate the unit, depending on the

availability of the authentication parameters (u-authparm) and the Serving RFSS local policy.

“1”

indicates that the Serving RFSS SHALL authenticate the unit when the home is reachable. If the home is unreachable, the Serving RFSS MAY authenticate the unit, depending on the availability of the authentication parameters (u-authparm) and the Serving RFSS local policy.

“2”

indicates that the Serving RFSS SHALL always authenticate the unit. If the Serving RFSS does not have authentication parameters (u-authparm) for the unit and is unable to retrieve them from the home, the Serving RFSS SHALL NOT allow the unit to register.

If no authentication policy is specified and the Authentication Type/Capability parameter is present, then the Serving RFSS MAY authenticate the unit, depending on the availability of the authentication parameters (u-authparm) and the Serving RFSS local policy. The Authentication Policy parameter SHALL be ignored if the Authentication Type/Capability attribute indicates that the unit does not support authentication (i.e. has a value of "0" or is not present).

3.8.2.12 Authentication Parameters (u-authparm)

The Authentication Parameters attribute specifies the authentication parameters associated with the authentication type specified in the Authentication Type attribute. Its format is:

authentication-parms = "u-authparm:" / "u-apa:" rs "," ks "," ksprime CRLF

rs = 10 (SP 2HEX-DIGIT)

ks = 16 (SP 2HEX-DIGIT)

ksprime = 16 (SP 2HEX-DIGIT)

where

rs	is the RS parameter described in [102AACE]
ks	is the KS parameter described in [102AACE], and
ksprime	is the KS prime parameter described in [102AACE].

rs, ks and ksprime are interpreted octet-wise from left to right. For example, in

u-authparm: 11 12 13 14 15 16 17 18 19 1a, 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d 3e 3f, 40 41 42 43 44 45 46 47 48 49 4a 4b 4c 4d 4e 4f

RS[0] = 11 and RS[9] = 1a. Likewise, KS[0] = 30 and KS[15] = 3f, and KS'[0] = 40 and KS'[15] = 4f.

If the authentication type attribute is non-zero, the authentication parameter attribute SHALL be supplied.

3.8.2.13 Availability Check Attribute (u-availcheck)

The Availability Check attribute specifies the ability of the SU to support an Availability Check, a direct call or both. It has the following ABNF notation:

availability-check = "u-availcheck:" / "u-ac:" type CRLF

type = "1" / "2" / "3"

where

1 indicates that the SU supports only Availability Check

2 indicates that the SU supports only Direct Call

3 indicates that the SU supports both Availability Check and Direct Call

If the "Availability Check" parameter is missing, the Serving RFSS SHALL assume that the calling SU supports both Availability Check and Direct Call.

3.8.2.14 Call Set-Up Preference Attribute (u-prefsetup)

The Call Set-up Preference attribute indicates the calling SU preference to make an availability check or a direct call of the Called SU when it sets up an SU-to-SU call. It has the following ABNF notation:

call-set-up-preference = "u-prefsetup:" / "u-prf:" type CRLF

type = "0" / "1"

where

0 indicates that the Calling SU prefers Availability Check for the Called SU

1 indicates that the Calling SU prefers Direct Call for the Called SU

If "call setup preference" is not provided by the Home RFSS, the Serving RFSS SHALL assume that the calling SU prefers availability check.

3.8.2.15 Radio Inhibit (u-radinhib)

The Radio Inhibit attribute indicates whether the SU is radio inhibited or not [102AABD]. It has the following ABNF notation:

radio-inhibit = "u-radioinhib:" / "u-rdinh:" type CRLF

type = "0" / "1"

where

0 indicates that the SU is not radio inhibited
1 indicates that the SU is radio inhibited

Upon receipt of a "Radio Inhibit" attribute value "1" as part of an SU's User Service Profile from the Home RFSS, the Serving RFSS SHOULD:

- Consider the SU radio inhibited, and
- Process service requests from the SU in accordance with procedures in [102AABD], and
- Process service requests targeting the SU in accordance with procedures in [102AABD].

If the "Radio Inhibit" attribute is not provided by the Home RFSS, the Serving RFSS SHALL assume that the SU is not radio inhibited.

3.8.2.16 Packet Data Capable (u-packetdata)

The Packet Data Capable attribute indicates whether or not the unit is capable of TIA-102 packet data operation. It has the following ABNF format:

packetdata = "u-packetdata:" / "u-pd:" boolean CRLF

where a boolean value of "0" indicates no TIA-102 packet data capability, and a value of "1" indicates TIA-102 packet data capability. If the Packet Data Capable attribute is not specified, the unit is not capable of TIA-102 packet data operation.

3.8.2.17 Manufacturer Specific Attributes (u-man<mfid>)

The Manufacturer Specific attribute allows manufacturer defined profile information that MAY be used for private or experimental purposes. It SHALL have the following radical with the following ABNF notation:

man-usrprof-radical = "u-man" / "u-m" mfid "-" octet-string ":"
mfid = 2HEX-DIGIT

The mfid values are defined in [102BAAC].

Manufacturer specific attributes that are not understood by an RFSS SHOULD be ignored.

3.8.2.18 Example User Service Profiles

3.8.2.18.1 Fully Specified Profile

The following example illustrates a fully specified user service profile using default values (where appropriate) or typical values (where no default is specified). Note that the unit-to-unit call priority attribute (u-upri) is specified, but would be ignored because unit-to-unit calls are not allowed (u-ucal). Likewise, the interconnect call security (u-icsec) and priority (u-icpri) attributes are specified but would be ignored because interconnect calls are not allowed (u-iccall).

```
u-access:1
u-dup:0
u-sec:0
u-gcall:3
u-ucall:0
u-upri:10
u-iccall:0
u-icsec:1
u-icpri:10
u-authtype:0
u-availcheck:1
u-prefsetup:0
u-radioinhib:0
u-packetdata:0
```

The same list in compact form:

```
u-as:1
u-d:0
u-s:0
u-g:3
u-uc:0
u-p:10
u-icc:0
u-ics:1
u-icp:10
u-at:0
u-ac:1
u-prf:0
u-rdinh:0
u-pd:0
```

3.8.2.18.2 Minimal User Service Profile (With Authentication)

The following example illustrates a minimal user service profile for a unit that requires authentication. Note that all attributes can be defaulted except for the authentication fields.

```

u-authtype:1
u-authpol:2
u-authparm: 11 12 13 14 15 16 17 18 19 1a, 30 31 32 33
34 35 36 37 38 39 3a 3b 3c 3d 3e 3f, 40 41 42 43 44 45 46 47
48 49 4a 4b 4c 4d 4e 4f

```

Or in compact form:

```

u-at:1
u-apo:2
u-apa: 11 12 13 14 15 16 17 18 19 1a, 30 31 32 33 34 35
36 37 38 39 3a 3b 3c 3d 3e 3f, 40 41 42 43 44 45 46 47 48 49
4a 4b 4c 4d 4e 4f

```

3.8.3 RFSS Service Capability Profile attributes

Table 29 summarizes the available attributes for an RFSS, which are detailed in the following subsections. This information MAY be provided in the RFSS service capability polling request message and MAY be returned in the response to an RFSS service capability polling request using the SIP OPTIONS method.

If any given attribute is not present, no assumption SHALL be made about the service capability of the RFSS sending the polling request or response.

All the RFSS Service Capability Profile lines begin with “r-”.

Table 29 - RFSS Service Capability Profile encoding in the ISSI body

Name	Field	Compact form	Meaning
Encrypted SU-to-SU Call Capability	r-ucall-sec	r-us	Defines whether or not the RFSS is capable of encrypted SU-to-SU calls at this time.
Clear SU-to-SU Call Capability	r-ucall-clear	r-uc	Defines whether or not the RFSS is capable of SU-to-SU calls in the clear at this time.
Encrypted Group Call Capability	r-gcall-sec	r-gs	Defines whether or not the RFSS is capable of encrypted unconfirmed group calls at this time.
Clear Group Call Capability	r-gcall-clear	r-gc	Defines whether or not the RFSS is capable of unconfirmed group calls in the clear at this time.
Encrypted Confirmed Group Call Capability	r-gcall-econf	r-ecg	Defines whether or not the RFSS is capable of encrypted confirmed group calls at this time.
Clear Confirmed Group Call Capability	r-gcall-cconf	r-ccg	Defines whether or not the RFSS is capable of clear confirmed group calls at this time.
SD Emergency Alarm Capability	r-sd-ealarm	r-sd-ea	Defines whether or not the RFSS is capable of the supplementary data function "Emergency Alarm"
SD Emergency Alarm Cancel Capability	r-sd-ealarmc	r-sd-ec	Defines whether or not the RFSS is capable of the supplementary data function "Emergency Alarm Cancel"
SD Group Emergency Cancel Capability	r-sd-egroupc	r-sd-gc	Defines whether or not the RFSS is capable of the supplementary data function "Group Emergency Cancel"
SD Call Alert Capability	r-sd-calert	r-sd-ca	Defines whether or not the RFSS is capable of the supplementary data function "Call Alert"
SD Short Message Capability	r-sd-smsg	r-sd-sm	Defines whether or not the RFSS is capable of the supplementary data function "Short Message"
SD Status Query Capability	r-sd-squery	r-sd-sq	Defines whether or not the RFSS is capable of the supplementary data function "Status Query"
SD Status Update Capability	r-sd-supdate	r-sd-su	Defines whether or not the RFSS is capable of the supplementary data function "Status Update"
SD Radio Unit Monitor Capability	r-sd-rum	r-sd-rm	Defines whether or not the RFSS is capable of the supplementary data function "Radio Unit Monitor"
SD Radio Unit Monitor Enhanced Capability	r-sd-rume	r-sd-re	Defines whether or not the RFSS is capable of the supplementary data function "Radio Unit Monitor - Enhanced"
SD Radio Check Capability	r-sd-rcheck	r-sd-rc	Defines whether or not the RFSS is capable of the supplementary data function "Radio Check"

Name	Field	Compact form	Meaning
SD Radio Detach Capability	r-sd-rdetach	r-sd-rd	Defines whether or not the RFSS is capable of the supplementary data function "Radio Detach"
SD Radio Inhibit Capability	r-sd-rinhibit	r-sd-ri	Defines whether or not the RFSS is capable of the supplementary data function "Radio Inhibit"
SD Radio Uninhibit Capability	r-sd-runinhibit	r-sd-ru	Defines whether or not the RFSS is capable of the supplementary data function "Radio Uninhibit"
Packet Data Capability	r-pdata	r-pd	Defines whether or not the RFSS is capable of supporting packet data
Adjacent Site	r-adjsite	r-as	Defines a site that the RFSS has that is adjacent to a site in the other RFSS. Note: zero, one or multiple Adjacent Site attributes can be present.
Unit Roaming	r-roam-unit	r-ru	Defines whether or not the RFSS allows its SUs to roam to the other RFSS.
Group Roaming	r-roam-grp	r-rg	Defines whether or not the RFSS allows its Groups to roam to the other RFSS.
RFSS Known	r-rfss-known	r-rk	Defines whether or not the RFSS is provisioned for the other RFSS.
Vocoder Mode Capability	r-vmodecap	r-vmc	Defines the vocoder modes that the RFSS is capable of.
Manufacturer Specific Parameter	r-man<mfid>...	r-m<mfid>...	Allows for manufacturer specific attributes

3.8.3.1 Encrypted SU-to-SU Call Capability (r-ucall-sec)

The Encrypted SU-to-SU Call Capability attribute defines whether or not the RFSS is capable of encrypted unit to unit calls at this time. It has the following format in ABNF notation:

encrypted-u2u-call-capability = "r-ucall-sec:" / "r-us:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of secure SU-to-SU calls, and "1" is used to indicate that the RFSS is capable of secure SU-to-SU calls.

3.8.3.2 Clear SU-to-SU Call Capability (r-ucall-clear)

The Clear SU-to-SU Call Capability attribute defines whether or not the RFSS is capable of SU-to-SU calls in the clear at this time. It has the following format in ABNF notation:

clear-u2u-call-capability = "r-ucall-clear:" / "r-uc:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of clear SU-to-SU

calls, and “1” is used to indicate that the RFSS is capable of clear SU-to-SU calls.

3.8.3.3 Encrypted Group Call Capability (r-gcall-sec)

The Encrypted Group Call Capability attribute defines whether or not the RFSS is capable of encrypted unconfirmed group calls at this time. It has the following format in ABNF notation:

encrypted-group-call-capability = “r-gcall-sec:” / ”r-gs:” boolean CRLF

where “0” is used to indicate that the RFSS is NOT capable of encrypted unconfirmed group calls, and “1” is used to indicate that the RFSS is capable of encrypted unconfirmed group calls.

3.8.3.4 Clear Group Call Capability (r-gcall-clear)

The Clear Group Call Capability attribute defines whether or not the RFSS is capable of clear unconfirmed group calls at this time. It has the following format in ABNF notation:

clear-group-call-capability = “r-gcall-clear:” / ”r-gc:” boolean CRLF

where “0” is used to indicate that the RFSS is NOT capable of clear unconfirmed group calls, and “1” is used to indicate that the RFSS is capable of clear unconfirmed group calls.

3.8.3.5 Encrypted Confirmed Group Call Capability (r-gcall-econf)

The Encrypted Confirmed Group Call Capability attribute defines whether or not the RFSS is capable of encrypted confirmed group calls at this time. It has the following format in ABNF notation:

enc-conf-group-call-capability = “r-gcall-econf:” / ”r-ecg:” boolean CRLF

where “0” is used to indicate that the RFSS is NOT capable of encrypted confirmed group calls, and “1” is used to indicate that the RFSS is capable of encrypted confirmed group calls.

3.8.3.6 Clear Confirmed Group Call Capability (r-gcall-cconf)

The Clear Confirmed Group Call Capability attribute defines whether or not the RFSS is capable of clear confirmed group calls at this time. It has the following format in ABNF notation:

`clr-conf-group-call-capability = "r-gcall-cconf:" / "r-ccg:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of clear confirmed group calls, and "1" is used to indicate that the RFSS is capable of clear confirmed group calls.

3.8.3.7 SD Emergency Alarm Capability (r-sd-ealarm)

The SD Emergency Alarm Capability attribute defines whether or not the RFSS is capable of the SD Emergency Alarm function at this time. It has the following format in ABNF notation:

`sd-emergency-alarm-capability = "r-sd-ealarm:" / "r-sd-ea:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of the Emergency Alarm function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.8 SD Emergency Alarm Cancel Capability (r-sd-ealarmc)

The SD Emergency Alarm Cancel Capability attribute defines whether or not the RFSS is capable of the SD Emergency Alarm Cancel function at this time. It has the following format in ABNF notation:

`sd-emergency-alarm-cancel-capability = "r-sd-ealarmc:" / "r-sd-ec:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of the Emergency Alarm Cancel function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.9 SD Group Emergency Cancel Capability (r-sd-egroupc)

The SD Group Emergency Cancel Capability attribute defines whether or not the RFSS is capable of the SD Group Emergency Cancel function at this time. It has the following format in ABNF notation:

`sd-group-emergency-cancel capability = "r-sd-egroupc:" / "r-sd-gc:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of the Group Emergency Cancel function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.10 SD Call Alert Capability (r-sd-alert)

The SD Call Alert Capability attribute defines whether or not the RFSS is capable of the SD Call Alert function at this time. It has the following format in ABNF notation:

sd-call-alert-capability = "r-sd-alert:" / "r-sd-ca:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Call Alert function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.11 SD Short Message Capability (r-sd-smsg)

The SD Short Message Capability attribute defines whether or not the RFSS is capable of the SD Short Message function at this time. It has the following format in ABNF notation:

sd-short-message-capability = "r-sd-smsg:" / "r-sd-sm:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Short Message function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.12 SD Status Query Capability (r-sd-squery)

The SD Status Query Capability attribute defines whether or not the RFSS is capable of the SD Status Query function at this time. It has the following format in ABNF notation:

sd-status-query-capability = "r-sd-squery:" / "r-sd-sq:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Status Query function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.13 SD Status Update Capability (r-sd-supdate)

The SD Status Update Capability attribute defines whether or not the RFSS is capable of the SD Status Update function at this time. It has the following format in ABNF notation:

sd-status-update-capability = "r-sd-supdate:" / "r-sd-su:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Status Update function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.14 SD Radio Unit Monitor Capability (r-sd-rum)

The SD Radio Unit Monitor Capability attribute defines whether or not the RFSS is capable of the SD Radio Unit Monitor function at this time. It has the following format in ABNF notation:

`sd-radio-unit-monitor-capability = "r-sd-rum:" / "r-sd-rm:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of the Radio Unit Monitor function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.15 SD Radio Unit Monitor Enhanced Capability (r-sd-rume)

The SD Radio Unit Monitor Capability attribute defines whether or not the RFSS is capable of the SD Radio Unit Monitor Enhanced function at this time. It has the following format in ABNF notation:

`sd-radio-unit-monitor-enhanced-capability = "r-sd-rume:" / "r-sd-re:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of the Radio Unit Monitor Enhanced function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.16 SD Radio Check Capability (r-sd-rcheck)

The SD Radio Check Capability attribute defines whether or not the RFSS is capable of the SD Radio Check function at this time. It has the following format in ABNF notation:

`sd-radio-check-capability = "r-sd-rcheck:" / "r-sd-rc:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of the Radio Check function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.17 SD Radio Detach Capability (r-sd-rdetach)

The SD Radio Detach Capability attribute defines whether or not the RFSS is capable of the SD Radio Detach function at this time. It has the following format in ABNF notation:

`sd-radio-detach-capability = "r-sd-rdetach:" / "r-sd-rd:" boolean CRLF`

where "0" is used to indicate that the RFSS is NOT capable of the Radio Detach function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.18 SD Radio Inhibit Capability (r-sd-rinhibit)

The SD Radio Inhibit Capability attribute defines whether or not the RFSS is capable of the SD Radio Inhibit function at this time. It has the following format in ABNF notation:

sd-radio-inhibit-capability = "r-sd-rinhibit:" / "r-sd-ri:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Inhibit function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.19 SD Radio Uninhibit Capability (r-sd-runinhibit)

The SD Radio Uninhibit Capability attribute defines whether or not the RFSS is capable of the SD Radio Uninhibit function at this time. It has the following format in ABNF notation:

sd-radio-uninhibit-capability = "r-sd-runinhibit:" / "r-sd-ru:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of the Radio Uninhibit function, and "1" is used to indicate that the RFSS is capable of this function.

3.8.3.20 Packet Data Capability (r-pdata)

The Packet Data Capability attribute defines whether or not the RFSS is capable of supporting Packet Data at this time. It has the following format in ABNF notation:

sd-packet-data-capability = "r-pdata:" / "r-pd:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of supporting packet data, and "1" is used to indicate that the RFSS is capable of supporting packet data.

3.8.3.21 Adjacent Site (r-adjsite)

The Adjacent Site attribute defines a site in an RFSS that is adjacent to a site in the other RFSS. It has the following format in ABNF notation:

```
Adjacent-site = "r-adjsite:" / "r-as:" lra SP wacn-id SP sys-id SP
rfss-id SP site-id SP
channel-T SP channel-R SP
serv-class SP signalling CRLF
```

where the parameters represent the values in the ADJ_STS_BCST air interface message as defined in [102AABC] and are defined as follows:

lra	= 2HEX-DIGIT
wacn-id	= 5HEX-DIGIT
sys-id	= 3HEX-DIGIT
rfss-id	= 2HEX-DIGIT
site-id	= 2HEX-DIGIT
channel-T	= 4HEX-DIGIT
channel-R	= 4HEX-DIGIT
serv-class	= 2HEX-DIGIT
signaling	= 1HEX-DIGIT

If an Adjacent Site attribute is included in the profile it SHALL contain all of the parameters. If the channel-R field indicates a value of 0xFFFF, then the indication of the channel-R field SHALL be ignored, and the channel-T field SHALL be used as an implicitly defined channel. Note that "signaling" represents the "CFVA" bits in the ADJ_STS_BCST message.

Zero, one or many of the Adjacent Site attributes MAY be present in the RFSS service profile.

Note that the channels to be sent in the Adjacent Site attributes and the bandplan information needed at the receiving RFSS are coordinated and provisioned by system administrators.

3.8.3.22 Unit Roaming Capability (r-roam-unit)

The Unit Roaming Capability attribute defines whether or not the RFSS is capable of supporting roaming of its SUs to the other RFSS at this time. It has the following format in ABNF notation:

```
unit-roaming-capability = "r-roam-unit:" / "r-ru:" boolean CRLF
```

where "0" is used to indicate that the RFSS is NOT capable of supporting SU roaming to the other RFSS, and "1" is used to indicate that the RFSS is capable of supporting SU roaming to the other RFSS.

3.8.3.23 Group Roaming Capability (r-roam-grp)

The Group Roaming Capability attribute defines whether or not the RFSS is capable of supporting roaming of its Groups to the other RFSS at this time. It has the following format in ABNF notation:

group-roaming-capability = "r-roam-grp:" / "r-rg:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT capable of supporting Group roaming to the other RFSS, and "1" is used to indicate that the RFSS is capable of supporting Group roaming to the other RFSS.

3.8.3.24 RFSS Known Capability (r-rfss-known)

The RFSS Known Capability attribute defines whether or not the RFSS is provisioned for the other RFSS at this time. It has the following format in ABNF notation:

RFSS-known-capability = "r-rfss-known:" / "r-rk:" boolean CRLF

where "0" is used to indicate that the RFSS is NOT provisioned for the other RFSS, and "1" is used to indicate that the RFSS is provisioned for the other RFSS.

3.8.3.25 Vocoder Mode Capability (r-vmodecap)

The Vocoder Mode Capability attribute defines the vocoder modes that the RFSS is capable of. It has the following format in ABNF notation:

vocoder-mode-capability = ("r- vmodecap:" / "r-vmc:") 1*(voc-mode SP) CRLF
voc-mode = "0" / "1" / "2"

where:

- 0 - indicates that the RFSS is capable of Full Rate vocoder mode
- 1 - indicates that the RFSS is capable of Half Rate vocoder mode
- 2 - indicates that the RFSS is capable of Native vocoder mode

3.8.3.26 Manufacturer Specific Attribute (r-man<mfid>)

The Manufacturer Specific attribute allows manufacturer defined profile information that MAY be used for private or experimental purposes. It SHALL have the following radical with the following ABNF notation:

```
man-rfss-radical      = "r-man" / "r-m" mfid "-" octet-string ":" pvalue CRLF
mfid                  = 2HEX-DIGIT
```

where pvalue is defined in Section 25.1 of RFC 3261.

The mfid values are defined in [102BAAC]

Manufacturer specific attributes that are not understood by an RFSS SHOULD be ignored.

Example:

```
r-manA4-myparameter:1
```

3.8.3.27 Example RFSS Service Profiles

The following example illustrates a fully specified RFSS service profile where the RFSS:

- supports SU-to-SU call
- supports clear, secure, confirmed, and unconfirmed group calls
- does not support any supplementary data services
- supports packet data
- has a single site it thinks is adjacent to one of the other RFSS' sites
- supports unit and group roaming
- knows the other RFSS
- supports Half Rate and Full Rate vocoders

```
r-ucall-sec:1
r-ucall-clear:1
r-gcall-sec:1
r-gcall-clear:1
r-gcall-econf:1
r-gcall-cconf:1
r-sd-ealarm:0
r-sd-ealarmc:0
r-sd-egroupc:0
r-sd-calert:0
r-sd-smsg:0
r-sd-squery:0
r-sd-supdate:0
```

```

r-sd-rum:0
r-sd-rume:0
r-sd-rcheck:0
r-sd-rdetach:0
r-sd-rinhibit:0
r-sd-ruinhibit:0
r-pdata:1
r-adjsite:01 00003 002 01 E2 AB23 FFFF 01 3
r-roam-unit:1
r-roam-grp:1
r-rfss-known:1
r-vmodecap:0 1

```

The same list in compact form:

```

r-us:1
r-uc:1
r-gs:1
r-gc:1
r-ecg:1
r-ccg:1
r-sd-ea:0
r-sd-ec:0
r-sd-gc:0
r-sd-ca:0
r-sd-sm:0
r-sd-sq:0
r-sd-su:0
r-sd-rm:0
r-sd-re:0
r-sd-rc:0
r-sd-rd:0
r-sd-ri:0
r-sd-ru:0
r-pd:1
r-as:01 00003 002 01 E2 AB23 FFFF 01 3
r-ru:1
r-rg:1
r-rk:1
r-vmc:0 1

```

3.8.4 ISSI specific register parameter attributes

Table 30 summarizes the available attributes for the ISSI specific register parameter, which are detailed in the following subsections.

Table 30 - ISSI specific register parameters

Name	Field	Compact form	Meaning	Default
Force Register	r-force	r-f	Home Registration Query procedure: force action required for a Serving RFSS.	0: Not Forced
Confirm Register	r-confirm	r-c	Home Registration Query procedure: confirm action required for a Serving RFSS.	0: Not Confirmed
SU Presence	r-presence	r-p	Serving Registration procedure: confirming SU's presence at Serving RFSS	0: Not Confirmed
SU Roaming	r-roaming	r-r	Home Deregistration procedure: indicate whether this procedure is invoked because the SU has roamed away from the Serving RFSS, or in accordance with local policy.	1: Roamed
SU Authentication	r-authentication	r-a	Serving Registration procedure: confirming SU's authentication at Serving RFSS	0: SU Authentication NOT Performed
Emergency Mode Register	r-emergency	r-e	Serving Registration procedure: indicate whether the SU is requesting registration in the emergency mode	0: Non-emergency
SNC Change	snc-change	snc-c	PD SN Context Registration procedure: inform the Home RFSS whether the registration is pertaining to initial SndCP context activation, or pertaining to updated SndCP context parameter values subsequently determined by the Serving RFSS and accepted by the SU.	0: No change

Name	Field	Compact form	Meaning	Default
Supported Vocoder Modes	r-vocodermode	r-vm	Group Registration procedure: Vocoder Modes supported by a Home RFSS or Serving RFSS.	No Default

All these parameters are optional parameters.

3.8.4.1 Force Register

This parameter is used by the Home RFSS in the Home Registration Query procedure. The “Force” qualifier indicates that the Serving RFSS has to re-register the mobility object if it still has interest. See Section 5 for further details.

This parameter has the following ABNF format:

force-register = (“r-force:” / “r-f:”) boolean CRLF

0: Not Forced (Default)

1: Forced

Example:

r-force:1

r-f:0

3.8.4.2 Confirm Register

This parameter is used by the Home RFSS in the Home Registration Query procedure. The “Confirm” qualifier requests that the Serving RFSS confirm its interest for the mobility object. See Section 5 for further details.

This parameter has the following ABNF format:

confirm-register = (“r-confirm:” / “r-c:”) boolean CRLF

0: Not Confirmed (Default)

1: Confirmed

Example:

r-confirm:1

r-c:0

3.8.4.3 SU Presence

This parameter is used by the Serving RFSS in the Registration procedure for an SU. The “SU Presence” qualifier indicates whether or not the Serving RFSS has actually confirmed the presence of the SU at the Serving RFSS immediately prior to sending the Registration request. See Section 5 for further details.

This parameter has the following ABNF format:

su-presence = (“r-presence:” / “r-p:”) boolean CRLF

0: Not Confirmed (Default)

1: Confirmed

Example:

r-presence:1

r-p:0

3.8.4.4 SU Roaming

This parameter is used in the Home Deregistration procedure for an SU or any other mobility object. The “SU Roaming” qualifier indicates whether this procedure is invoked because the SU has roamed away from the Serving RFSS, or in accordance with local policy. See Section 5 for further details.

This parameter has the following ABNF format:

su-roaming = (“r-roaming:” / “r-r:”) boolean CRLF

0: Non-Roamed

1: Roamed (Default)

Example:

r-roaming:1

r-r:0

3.8.4.5 SU Authentication

This parameter is used by the Serving RFSS in the Registration procedure for an SU. The “SU Authentication” qualifier indicates whether or not the Serving RFSS has actually confirmed authentication of the SU at the Serving RFSS prior to sending the Registration request. When SU Authentication is confirmed, it reflects successful SU authentication or successful SU authentication with mutual authentication.

This parameter has the following ABNF format:

su-authentication = (“r-authentication:” / “r-a:”) boolean CRLF

- 0: SU Authentication NOT performed (Default)
- 1: SU Authentication confirmed

Example:

r-authentication:1
r-a:0

3.8.4.6 Emergency Mode Register

This parameter is used in the Serving Registration procedure to indicate whether the SU is requesting registration in the emergency mode.

This parameter has the following ABNF format:

emergency-register = ("r-emergency:" / "r-e:") boolean CRLF

- 0: Non-emergency (Default)
- 1: Emergency

Example:

r-emergency:1
r-e:0

3.8.4.7 SNC Change

This parameter is used by the Serving RFSS in the PD SN Context registration request to inform the Home RFSS whether the procedure is pertaining to initial SND CP context activation request, or pertaining to updated SND CP context parameter values subsequently determined by the Serving RFSS and accepted by the SU.

This parameter has the following ABNF format:

snc-change = ("snc-change:"/"snc-c:") Boolean CRLF

0: No Change, Request pertains to initial SND CP context activation request

1: Change, Request pertains to updated SND CP context parameter values subsequently determined by the Serving RFSS and accepted by the SU

Example:

snc-change:0
snc-c:1

3.8.4.8 Supported Vocoder Modes

This parameter is used in the Group Registration procedure. It is used by a Serving RFSS in group registration requests, and by a Group Home RFSS in group registration responses. The "Supported Vocoder Modes" qualifier indicates the vocoder mode(s) supported by the RFSS for this group, listed in the order of preference, with the first vocoder mode listed being preferred. See Section 5 for further details.

This parameter has the following ABNF format:

supported-vm = ("r-vocodermode:" / "r-vm:") 1*(voc-mode SP) CRLF

voc-mode = "0" / "1" / "2"

- 0: Full Rate vocoder mode
- 1: Half Rate vocoder mode
- 2: Native vocoder mode

If this parameter is not present in a group registration request or response, no assumption can be made on the RFSS's supported vocoder mode.

Examples:

r-vocodermode:2 1 0
r-vm:1

3.8.5 ISSI specific call parameter attributes

These attributes are used in conjunction with an SDP announcement. All these parameters are optional.

Table 31 summarizes the available attributes for the ISSI specific call parameter, which are detailed in the following subsections.

Table 31 - ISSI specific call parameter encoding in the ISSI body

Name	Field	Compact form	Meaning	Values
Initial Transmitter	c-initrans	c-i	Indicates the SU that transmits first in a unit to unit call	0: calling SU 1: called SU (no default value: a value for the calling SU or called SU SHALL be present)
RF Resource Availability	c-resavail	c-r	Indicates if the RF resources are available or not in confirmed group calls or SU-to-SU calls	0: no RF resource (default) 1: RF resource
Preference	c-pref	c-prf	Indicates that the Called Serving RFSS has to perform an availability check with the called SU	0: no availability check 1: availability check (default)
Duplex Mode	c-duplex	c-d	Indicates the mode of the call: half-duplex or full-duplex.	0: half duplex (default) 1: full duplex
In Call Roaming	c-incallroaming	c-icr	Indicates the roaming of the Calling or the Called SU in an SU-to-SU call	0: No roaming (default) 1: An SU is roaming
Group Call Type	c-groupcalltype	c-gct	Indicates whether the call is a confirmed or an unconfirmed group call	0: unconfirmed (default) 1: confirmed
Protected	c-protected	c-prot	Indicates whether the call's audio is requested to be protected.	0: unprotected (default) 1: protected
Console Transmit Request Priority	c-consoletransmitrequestpriority	c-ctxrp	Indicates the priority of the console's transmission request.	All values \$00 through \$FF are allowed, the following values are recommended. The larger the value the higher the priority. \$10: priority 1 (lowest tx priority, default) \$30: priority 2 \$50: priority 3 \$70: priority 4 \$90: priority 5 \$B0: priority 6 \$D0: priority 7 \$F0: priority 8

3.8.5.1 Initial transmitter

This parameter indicates the first transmitter in a call. It is named “c-initrans” or “c-i” in its compact form. Two values are allowed: 0 if the calling SU is chosen to transmit first or 1 if it is the called SU.

This parameter has the following ABNF format:

initial-transmitter = (“c-initrans:” / “c-i:”) boolean CRLF

This parameter is used in a unit to unit call:

- The Called Home RFSS chooses the initial transmitter and sends it in the INVITE message to the Called Serving RFSS.
- When the Called Home RFSS receives the 200 OK response from the Called Serving RFSS, it adds the initial transmitter parameter, and sends it to the Calling Serving RFSS via the Calling Home RFSS. (See Figure 27)

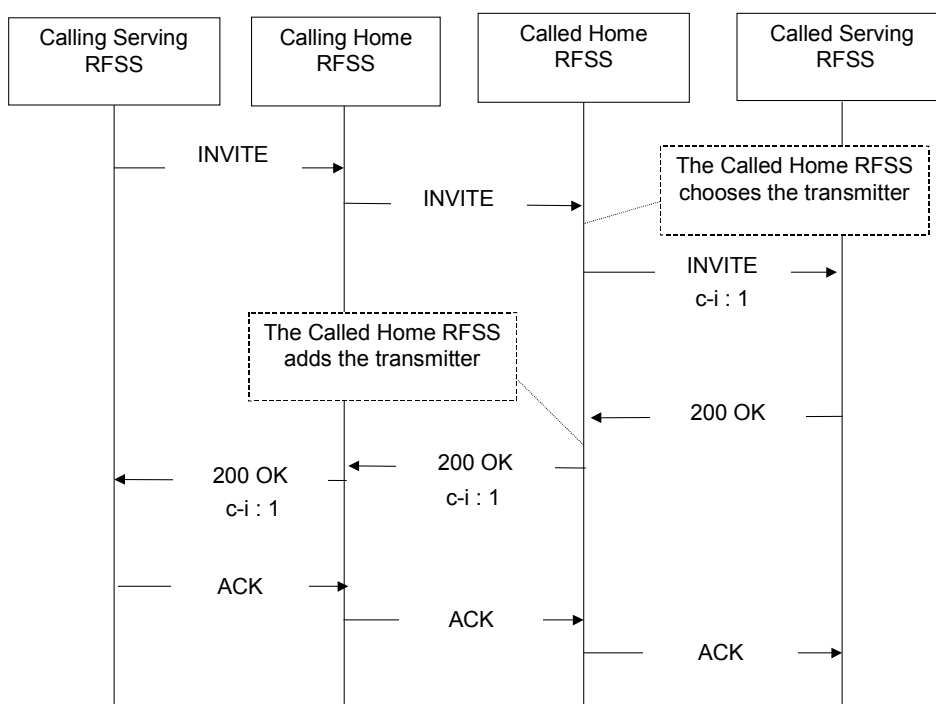


Figure 27 - Use of c-initrans

Example:

c-initrans:1

c-i:0

3.8.5.2 RF Resource Availability

This parameter indicates whether RF resources are available in an RFSS area. A value of "0", which is the default value, indicates that no RF resource is available.

This field SHALL be present in confirmed group calls:

- in the first INVITE message from a Serving RFSS to the Home RFSS.
- in the re-INVITE messages from any Serving RFSS to the Home RFSS when the RF resource allocation state changes.
- in the 200 OK response from each Serving RFSS that provides a Type 1 SDP answer to the Home RFSS's SIP (Re-)INVITE request.

See Figure 28.

This field SHOULD be present in SU-to-SU calls in all the INVITE, 200 OK and ACK messages.

Nevertheless, this field MAY be present in any INVITE, 200 OK or ACK message and MAY be discarded by an RFSS when it is used outside the typical use described above.

It has the following ABNF format:

```
channel-resource-availability = ("c-resavail:" / "c-r:") boolean
CRLF
```

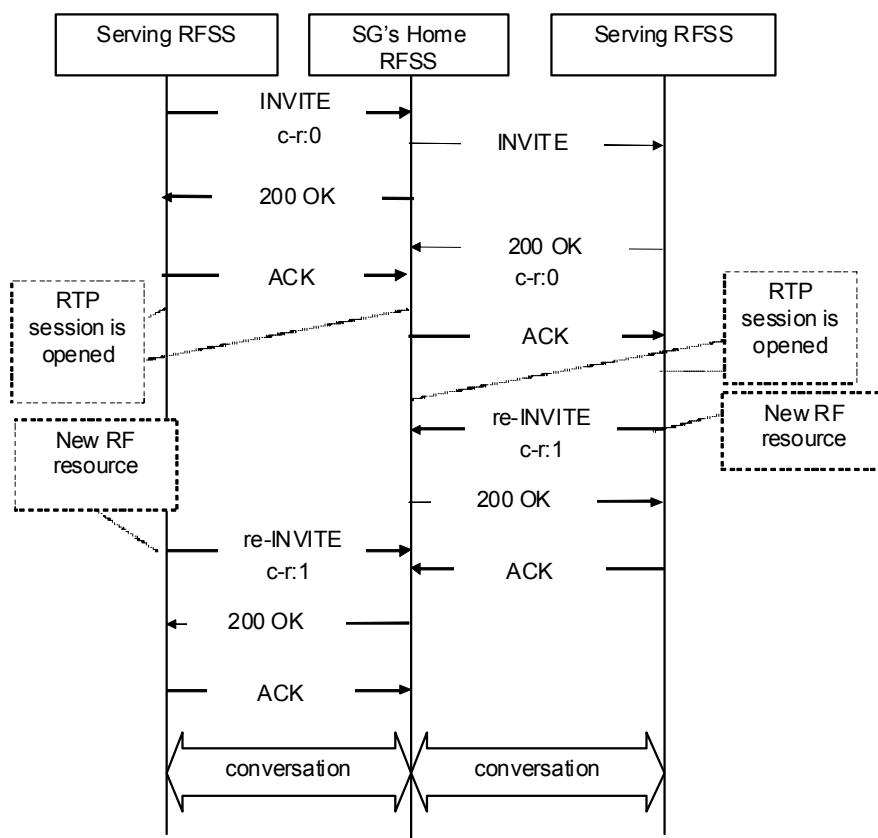


Figure 28 - Use of c-resavail

3.8.5.3 Preference

This parameter indicates to the Called Serving RFSS whether it SHOULD perform an availability check with the called SU prior to granting the request and allocating a traffic channel, or to process the call as a direct call. The called SU is expected to respond to the availability check with a refusal or an acceptance.

The parameter is named “c-pref”, with a compact form “c-prf”. A value of 1 indicates the Called Serving RFSS to perform the availability check. 1 is the default value: perform availability check.

It has the following ABNF format:

preference = (“c-pref:” / “c-prf:”) boolean CRLF

Example

c-pref:0

c-prf:1

The Calling Serving RFSS indicates its preference in the INVITE message with the preference parameter filled with the Call Set-UP Preference attribute (u-prefsetup) of its user profile.

The Called Home RFSS compares the received preference with the Availability Check and the Direct Call attributes of the called SU service profile to set the preference parameter in the INVITE message sent to the Called Serving RFSS:

- if the preference is set to Availability Check (1) and the SU supports Availability Check in its USP, then the preference is set to Availability Check (1).
- if the preference is set to Availability Check (1) and the SU supports only Direct Call in its USP, then the preference is set to Direct Call (0).
- if the preference is set to Direct Call (0) and the SU supports only Availability Check, in its USP, then the preference is set to Availability Check (1).
- if the preference is set to Direct Call (0) and the SU supports Direct Call then the preference is set to Direct Call (0).

Table 32 summarizes the choice made by the Called Home RFSS:

AC0: the SU does not support Availability Check

AC1: the SU supports Availability Check

DC0: the SU does not support Direct Call

DC1: the SU supports Direct Call

Table 32 - Availability Check / Direct Call choice in the Called Home RFSS

Received Preference	Ability of the Called SU		
	AC1 and DC0	AC1 and DC1	AC0 and DC1
AC	AC	AC	DC
DC	AC	DC	DC

Figure 29 illustrates the usage.

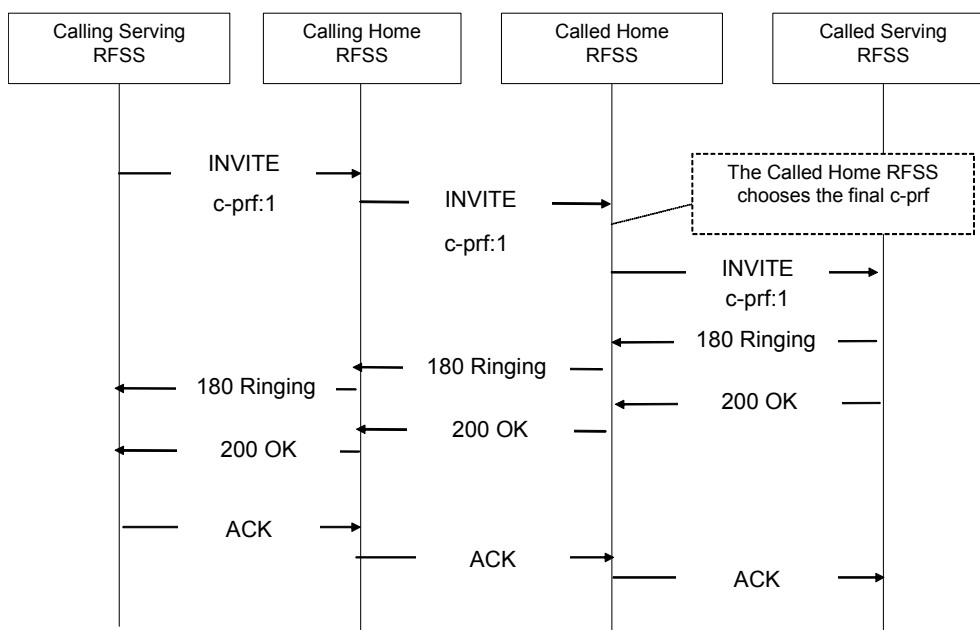


Figure 29 - Use of c-pref

3.8.5.4 Duplex Mode

This parameter defines the mode of the SU-to-SU Call. It is named “c-duplex” with a compact form “c-d”.

This field is set with 0 for half-duplex calls or 1 for full duplex calls.

The default value is 0.

It has the following ABNF format:

duplex-mode = (“c-duplex:” / “c-d:”) boolean CRLF

It is used only in unit to unit calls:

- The Calling Serving RFSS fills this parameter in the INVITE message with the duplex capacity of the Calling SU.
- When the Called Home RFSS receives the INVITE message, it compares the received capacity with the “duplexity” parameter of the Called SU (u-dup), and then deduces the duplex mode of the call:
 - if the capacity received is full duplex and the duplexity in the USP of the Called SU is full duplex, then the duplex mode is set to full duplex
 - in all the other cases the duplex mode is set to half duplex
- The parameter is then put in the INVITE sent to the Called Serving RFSS

- The parameter is also put in all the 200 OK messages acknowledging the INVITE Request to inform the Calling Serving RFSS of the chosen mode.

Table 33 summarizes the choices made by the Called Home RFSS:

FD: Full Duplex

HD: Half Duplex

Table 33 - Duplex Mode choice in the Called Home RFSS

Received Duplex Mode	Duplexity of the Called SU	
	FD	HD
FD	FD	HD
HD	HD	HD

Example:

c-duplex:0

c-d:1

Figure 30 illustrates this usage.

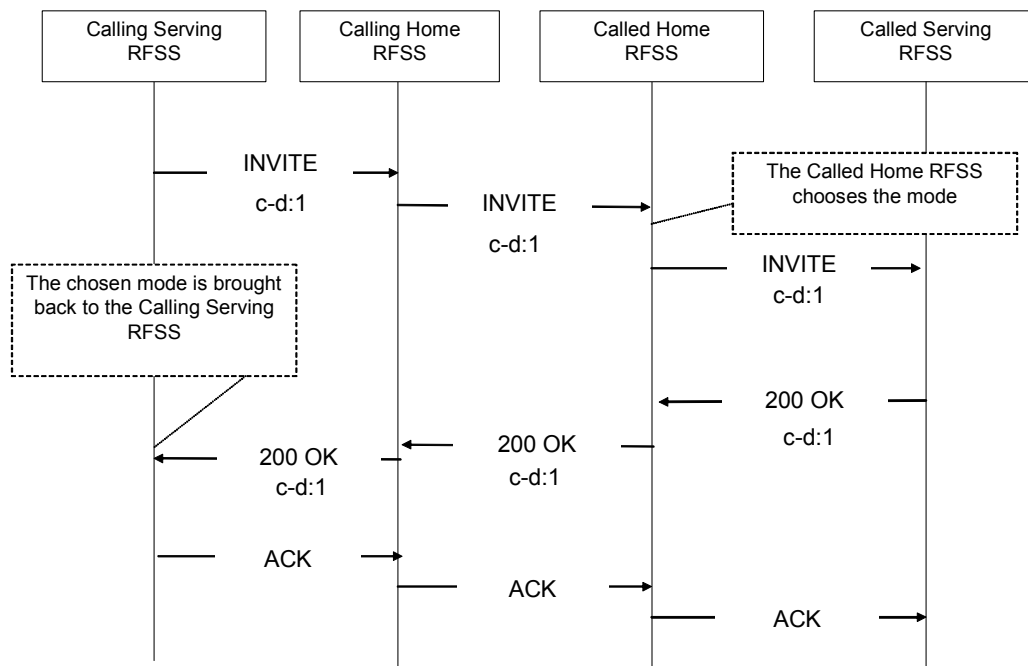


Figure 30 - Use of c-duplex

3.8.5.5 In-call roaming

This parameter indicates the roaming of an SU in an SU-to-SU Call. It is named “c-incallroaming” with a compact form “c-icr”.

It has the following ABNF format:

in-call-roaming = (“c-incallroaming:” / “c-icr:”) boolean CRLF

This field is set to 1 in an INVITE Request sent to set up a call segment as a result of In-Call Roaming, 0 otherwise. The default value is 0.

Figure 31 illustrates this usage.

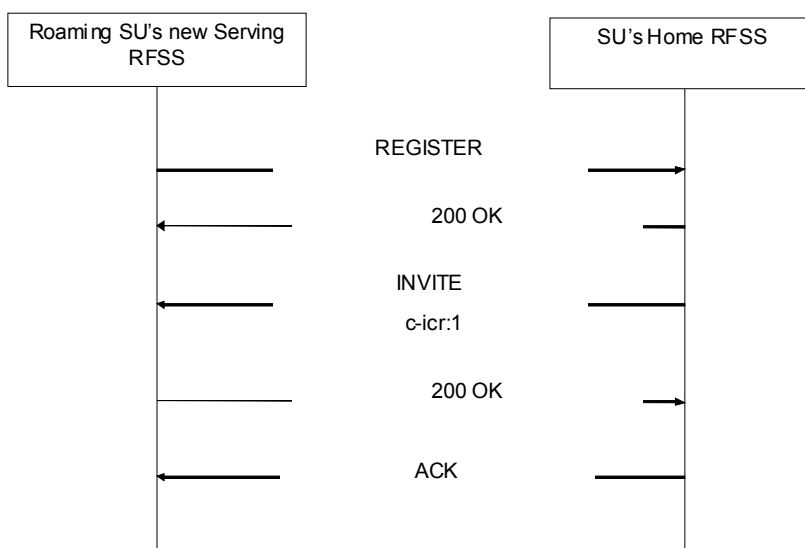


Figure 31 - Use of c-incallroaming

3.8.5.6 Group Call Type

This parameter indicates whether the group call is a confirmed or an unconfirmed group call.

It is named “c-groupcalltype ” with a compact form “c-gct”.

It has the following ABNF format:

group-call-type = (“c-groupcalltype:” / “c-gct:”) boolean CRLF

This field MAY be present only in an initial INVITE Request.

This field with value set to 1 (confirmed) SHALL be present in the INVITE Request from a Serving RFSS to a Home RFSS if the group service profile previously received from the Home RFSS contains the Confirmed Call Setup Time attribute with value greater than zero

This field with value set to 0 (unconfirmed) MAY be present in the INVITE Request from a Serving RFSS to a Home RFSS:

- If the group service profile previously received from the Home RFSS does not contain the Confirmed Call Setup Time attribute, or
- If the group service profile previously received from the Home RFSS contains the Confirmed Call Setup Time attribute with zero value.

For a confirmed group call, this field with value set to 1 (confirmed) SHALL be present in the INVITE Request from a Home RFSS to a Serving RFSS, and SHALL take precedence over the group service profile parameter.

The default value is 0 (unconfirmed).

Figure 32 illustrates this usage.

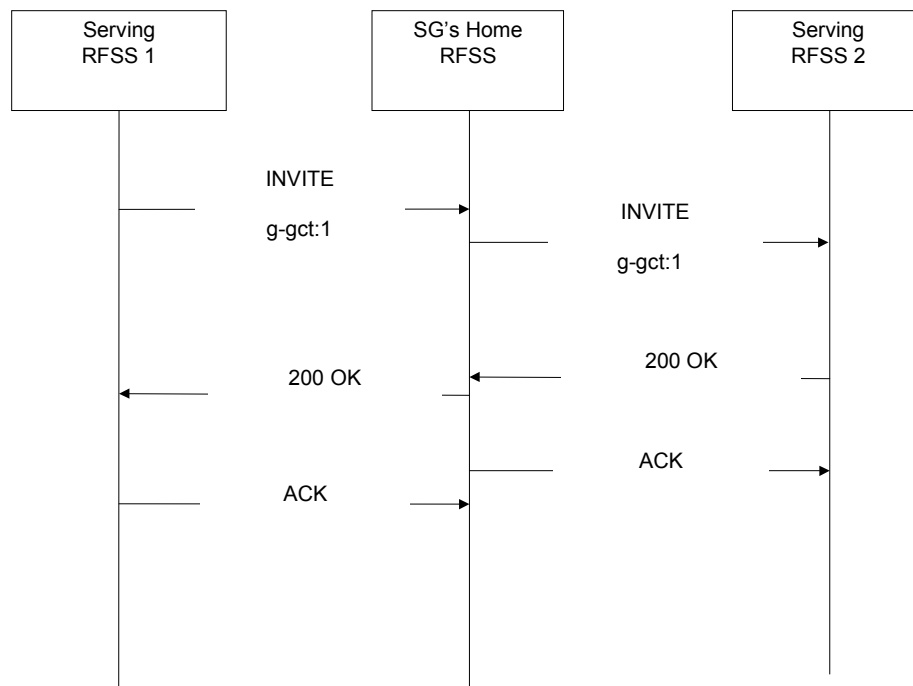


Figure 32 - Use of c-groupcalltype

3.8.5.7 Protected

This parameter defines the audio protection mode of the call. It is named “c-protected” with a compact form “c-prot”.

This field is set to 0 for unprotected (a.k.a. ‘clear’ or ‘unencrypted’) call audio or 1 for protected (a.k.a. ‘secure’ or ‘encrypted’) call audio. The default value is 0.

It has the following ABNF format:

protected= (“c-protected:” / “c-prot:”) boolean CRLF

The Calling Serving RFSS fills this parameter in the INVITE message with the protection mode requested by the Calling SU.

As shown in Figure 33 and Figure 34, when the Called Home RFSS in an SU-to-SU call or the SG Home RFSS in a Group call receives the INVITE message, it determines that the destination SU or SG supports the requested protection mode, and it therefore indicates this protection mode in the INVITE to the Called Serving RFSS(s) and in the 200 OK to the Calling Home RFSS or Calling Serving RFSS.

Examples:

c-protected:0
c-prot:1

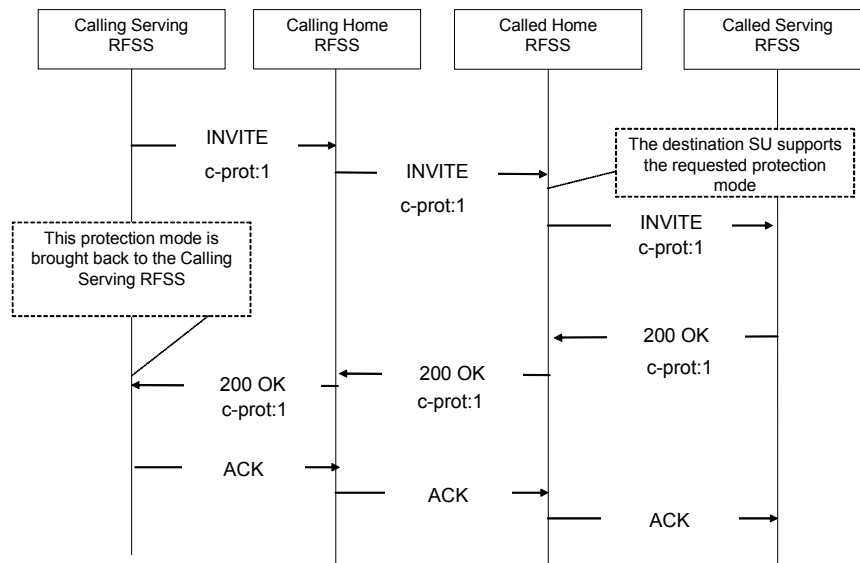


Figure 33 - Use of c-protected in SU-to-SU call

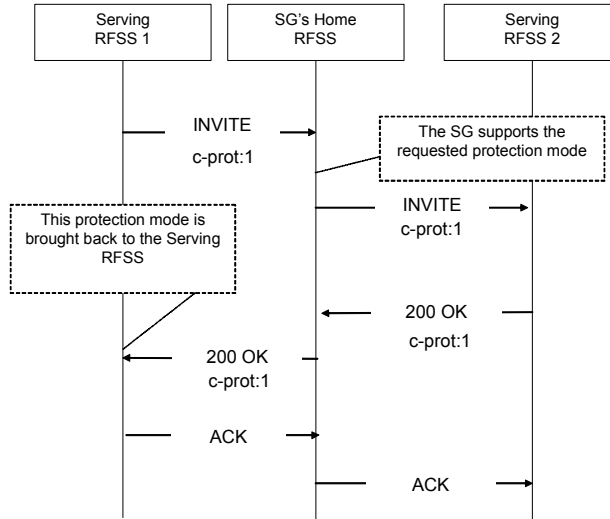


Figure 34 - Use of c-protected in group call

3.8.5.8 Console Transmit Request Priority

The presence of this parameter indicates that the initial call request is a console request, and its value indicates the console's transmit request priority. This parameter is not present for mobile sourced call requests.

Knowing the request is a console transmit request allows the Home RFSS to support the console priority feature. A Home RFSS might receive two competing SIP INVITE requests (one in response to a console request and one in response to a mobile request) and the home uses this indication to give priority to the console request.

The console's requested transmit priority is needed so the Home RFSS can handle multiple simultaneous console INVITE requests. A Home RFSS might receive two or more competing SIP INVITE requests from different consoles and the home uses this indication to give priority in the acquisition of RTP and RFSS resources to the highest priority request.

The parameter is named "c-consoletransmitrequestpriority" with a compact form "c-ctxrp".

It has the following ABNF format:

Console-transmit-request-priority =
("c-consoletransmitrequestpriority" / "c-ctxrp:") 2HEX-DIGIT CRLF

This field MAY be present only in an initial INVITE Request.

This field MAY be present in the INVITE Request from a Serving RFSS to a Home RFSS. When present, the Serving RFSS SHALL set the field value to the priority associated with the console's request to transmit.

The field MAY be present in the INVITE Request from a Home RFSS to a Serving RFSS. When present, the Home RFSS SHALL set the field value to the priority associated with the initial request.

All values \$00 through \$FF are allowed, the values in Table 31 are recommended.

3.9 X-TIA-P25-SNDCP Body Parameters

This section describes the X-TIA-P25-SNDCP application used to carry the SNDCP context parameters.

Only one X-TIA-P25-SNDCP body MAY be carried in a request or a 200 OK response pertaining to a PD SN Context mobility management procedure.

Up to fourteen X-TIA-P25-SNDP bodies MAY be carried in the 200 OK response to a unit registration request also used for SNDP context transfer.

The application/x-tia-p25-sndp media type is defined by the following information:

Media type name: application
Media subtype name: x-tia-p25-sndp
Required parameters: version
'Version' gives the version of the X-TIA-P25-SNDP protocol. The initial protocol version is set to: "version = 1.0"
Optional parameters: None

The SIP Content-type header field is set as defined in section 3.5.7.

The X-TIA-P25-SNDP descriptions are entirely textual. The X-TIA-P25-SNDP description consists in one or more lines of text. The order of these lines has no special meaning. The names and parameters of the fields are case-insensitive strings.

Table 34 summarizes the available attributes specific to PD SN Context mobility management.

Table 34 - ISSI-specific SN Context Mobility Management Parameters¹³

SNDP Data Field [102BAEB] [102BACF]	Field	Compact form	Meaning related to the considered SNDP Context	Default value
SNDP Version	r-snprotv	r-snv	SNDP protocol version	3: SNDP version 3
NSAPI	r-snnsapi	r-sns	SNDP nsapi	1: nsapi 1
NAT	r-snnat	r-snn	Network Address Type	1: IPV4 Dynamic Address
IPV4 Address	r-snipv4ad	r-snipa	IPV4 address	None
DSUT	r-sndsut	r-snds	Data Subscriber Unit Type	1: Alternating Trunked Voice & Data MRC
MDPCO	r-snmdpco	r-snmco	MDP Configuration options	0: PPP
Data Access Control (DAC)	r-sndac	r-snd	PDCH Access Rights	None
PDUPM	r-snpdupm	r-snpp	PDU priority maximum	0: Lowest priority
Ready Timer	r-snreadyt	r-snrt	Ready timer	None
MTU	r-snmtu	r-snm	MTU	4: 1500 bytes
Standby Timer	r-snstandbyt	r-snst	Standby timer	None
PCO Config Protocol	r-snpcoc	r-snpc	PCO Configuration Protocol	0: PPP
PCO Protocol Identity	r-snpcoi	r-snpi	PCO Protocol Identity	\$C023: PAP
PCO Protocol Identity content length	r-snpcoicl	r-snpicl	PCO Protocol Identity content length	0: No Protocol Identity content
PCO Protocol Identity content	r-snpcoic	r-snpic	PCO Protocol Identity content	None
APN Length	r-snapnl	r-snal	Format of the APN coding	0: No APN defined
APN index	r-snapni	r-snai	Value of the APN octet	None
APN string	r-snapns	r-snas	Value of the APN string	None
MSO Length	r-snmsol	r-snml	Length of the MSO Value field	0: MSO Value field not present
MSO Value	r-snmsov	r-snmv	Contents of the MSO Value field	None
Home PDIG Address	r-snhpdigad	r-snhpda	IP address of Home PDIG for an SU	None

¹³ Fields related to compression of SNDP headers and payload are left out of this table because SNDP-based compression is being handled by the Serving RFSS and the SU, but not the Home RFSS.

Table 35 summarizes which Sndcp Context fields are present in the various SN Context mobility Management messages. 1xx provisional messages do not contain any specific SN Context mobility management parameter.

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Table 35 – SNDCP Fields in SN Context Mobility Management Messages¹⁴

SIP Header Field Direction Serving(S) => Home (H) Home (H) => Serving (S)	Where	SU Register-Register S=>H (SNDCP Context Transfer)	SNC Register-Register S => H (with “SU Presence” parameter value of “Confirmed”)	SNC Register-Home Query (with “Force” qualifier) H => S	SNC Register-Home Query (without any qualifier) H => S	SNC Register-Serving Query (without any qualifier) S => H
SNDCP Version	Request	-	M-ISSI	M-ISSI	-	-
SNDCP Version	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
nsapi	Request	-	M-ISSI	M-ISSI	-	-
nsapi	200	T-ISSI	-	-	-	-
NAT	Request	-	M-ISSI	M-ISSI	-	-
NAT	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
IPv4 Address	Request	-	C-ISSI	C-ISSI	-	-
IPv4 Address	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
DSUT	Request	-	M-ISSI	-	-	-
DSUT	200	T-ISSI	-	-	M-ISSI	M-ISSI
MDPCO	Request	-	M-ISSI	-	-	-
MDPCO	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
Data Access Control (DAC)	Request	-	M-ISSI	-	-	-
Data Access Control (DAC)	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
PDUPM	Request	-	M-ISSI	-	-	-
PDUPM	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
Ready Timer	Request	-	M-ISSI	-	-	-
Ready Timer	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
MTU	Request	-	M-ISSI	-	-	-
MTU	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
Standby Timer	Request	-	M-ISSI	-	-	-
Standby Timer	200	T-ISSI	M-ISSI	-	M-ISSI	M-ISSI
PCO Config Protocol	Request	-	C-ISSI	-	-	-

¹⁴ Fields related to compression of SNDCP headers and payload are left out of this table because SNDCP-based compression is being handled by the Serving RFSS and the SU, but not the Home RFSS.

SIP Header Field Direction Serving(S) => Home (H) Home (H) => Serving (S)	Where	SU Register-Register S=>H (SND CP Context Transfer)	SNC Register-Register S => H (with "SU Presence" parameter value of "Confirmed")	SNC Register-Home Query (with "Force" qualifier) H => S	SNC Register-Home Query (without any qualifier) H => S	SNC Register-Serving Query (without any qualifier) S => H
PCO Config Protocol	200	-	C-ISSI	-	-	-
PCO Protocol Identity	Request	-	C-ISSI	-	-	-
PCO Protocol Identity	200	-	C-ISSI	-	-	-
PCO Protocol Identity content length	Request	-	C-ISSI	-	-	-
PCO Protocol Identity content length	200	-	C-ISSI	-	-	-
PCO Protocol Identity content	Request	-	C-ISSI	-	-	-
PCO Protocol Identity content	200	-	C-ISSI	-	-	-
APN Length	Request	-	C-ISSI	C-ISSI	-	-
APN Length	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
APN index	Request	-	C-ISSI	C-ISSI	-	-
APN index	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
APN string	Request	-	C-ISSI	C-ISSI	-	-
APN string	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
MSO Length	Request	-	C-ISSI	C-ISSI	-	-
MSO Length	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
MSO Value	Request	-	C-ISSI	C-ISSI	-	-
MSO Value	200	T-ISSI	C-ISSI	-	C-ISSI	C-ISSI
Home PDIG Address	200	-	M-ISSI	-	-	-

T-ISSI means that if an SNDCP context is to be transferred, the Home RFSS provides in the 200 OK response to the new Serving RFSS's SU Register-Register request, the most current parameter value known to the Home RFSS and known to the SU, or no information if the parameter was not present during SNDCP context activation. C-ISSI and M-ISSI are as defined in Section 1.1 of this document.

The value of each SNDCP parameter depends on the message that carries the parameter:

- In an initial SNC register-register request (i.e. snc-change set to 0), the SNDCP parameter value is either provided by the SU in its SNDCP context activation request if the parameter exists in the request, or by the Serving RFSS.
- In subsequent SNC register-register requests (i.e. snc-change set to 1), the SNDCP parameter value is the one ultimately determined by the Serving RFSS and accepted by the SU.
- In an SNC Register-query with force qualifier, the SNDCP parameter value is the one to be applied in the SNDCP context activation command
- In the 200 OK response to an SNC Home query with no qualifier, the SNDCP parameter value is the last one known to the Serving RFSS that has been accepted by the SU.
- In the 200 OK response to an SNC Serving query, the SNDCP parameter value is the last one known to the Home RFSS that has been accepted by the SU.

The default value for each SNDCP parameter is as shown in Table 34.

3.9.1 SNDCP Version

This parameter is used to convey the protocol version of a SNDCP context activation request or a SNDCP context activation command.

This parameter has the following ABNF format:

r-sndcp-protocol-version= ("r-snprotv:" / "r-snv:") 1HEX-DIGIT CRLF

The hex-digit value is as defined in [102BAEB] for the SNDCP Version field of the SNDCP PDU.

Example:

r-snprotv:2
r-snv:2

3.9.2 SNDCP nsapi

This parameter is used to convey the Network Service Access Point Identifier value for a given SNDCP context transferred between the Home RFSS and the Serving RFSS during unit registration and SN Context registration.

This parameter has the following ABNF format:

```
r-sndcp-nsapi = ( "r-snnsapi:" / "r-sns:" ) 1HEX-DIGIT CRLF
```

The hex-digit value is as defined in [102BAEB] for the NSAPI field of the SNDCP PDU.

3.9.3 NAT

This parameter is the Network Address Type for a given SNDCP context, used to convey the IP address assignment requested and granted during SNDCP context activation.

This parameter has the following ABNF format:

```
r-sndcp-nat = ( "r-snnat:" / "r-snn:" ) 1HEX-DIGIT CRLF
```

The hex-digit value is as defined in [102BAEB] for the NAT field of the SNDCP PDU.

Example:

```
r-snnat:0
r-snn:0
```

3.9.4 IPV4 Address

This parameter is the IPV4 address for a given SNDCP context. The IPV4 address SHALL be present in the PD SN Context registration request if the NAT value is equal to "IPV4 Static Address", otherwise it SHALL be absent. The IPV4 address SHALL be present in the PD SN Context registration response. The IPV4 address SHALL be present in the SU registration response, as part of the SNDCP context transfer, to the New Serving RFSS to which the SU has roamed.

This parameter has the following ABNF format:

```
r-sndcp-ipv4address = ( "r-snipv4ad" / "r-snipa" ) IP4-address CRLF
IP4-address = b1 "." decimal-uchar "." decimal-uchar "." b4
b1 = decimal-uchar
      ;less than "224"; not "0" or "127"
b4 = decimal-uchar
      ;not "0"
decimal-uchar = DIGIT
                / POS-DIGIT DIGIT
```

```

/ ("1" 2*(DIGIT))
/ ("2" ("0"/"1"/"2"/"3"/"4") DIGIT)
/ ("2" "5" ("0"/"1"/"2"/"3"/"4"/"5"))

```

3.9.5 DSUT

This parameter is the Data Subscriber Unit Type for a given SNDCP context.

This parameter has the following ABNF format:

```
r-sndcp-dsut = ( "r-sndsut:" / "r-snds:" ) 1HEX-DIGIT CRLF
```

The hex-digit value is as defined in [102BAEB] for the DSUT field of the SNDCP PDU.

Example:

```

r-sndsut:0
r-snds:0

```

3.9.6 MDPCO

This parameter is used to identify and configure the MRC and MDP capabilities.

This parameter has the following ABNF format:

```
r-sndcp-mdpco = ( "r-snmdpco:" / "r-snmco:" ) 1HEX-DIGIT CRLF
```

The hex-digit value is as defined in [102BAEB] for the MDPCO field of the SNDCP PDU.

Example:

```

r-snmdpco:0
r-snmco:0

```

3.9.7 Data Access Control

This field specifies the DAC value for a given SNDCP context, used by the Home RFSS to control packet data access for groups of SUs.

This parameter has the following ABNF format:

```
r-sndcp-dac = ( "r-sndac:" / "r-snd:" ) 4HEXDIGIT CRLF
```

The 4hex-digit value is as defined in [102BAEB] for the DAC field of the SNDCP PDU.

Example:

r-sndac:0001

r-snd:0001

3.9.8 PDUPM

This field specifies the priority providing differentiated quality of service for IP packet transfer for a given SNDCP context.

This parameter has the following ABNF format:

r-sndcp-pdupm = ("r-snpdupm:" / "r-snpp:") 1HEXDIGIT CRLF

The hex-digit value is as defined in [102BAEB] for the PDUPM field of the SNDCP PDU.

Example:

r-snpdupm:3

r-snpp:3

3.9.9 Ready timer

This field specifies the ready timer value for a given SNDCP context.

This parameter has the following ABNF format:

r-sndcp-ready-timer = ("r-snreadyt:" / "r-snrnt:") 1HEXDIGIT CRLF

The hex-digit value is as defined in [102BAEB] for the ready timer field of the SNDCP PDU.

Example:

r-snreadyt:6

r-snrnt:6

3.9.10 MTU

This field specifies the Maximum Transmission Unit value for a given SNDCP context, indicating the maximum SNDCP message size accepted from higher layer clients.

This parameter has the following ABNF format:

r-sndcp-mtu = ("r-snmtu:" / "r-snm:") 1HEXDIGIT CRLF

The hex-digit value is as defined in [102BAEB] for the MTU field of the SND CP PDU.

Example:

```
r-snmtu:3
r-snm:3
```

3.9.11 Standby timer

This field specifies the standby timer value for a given SND CP context.

This parameter has the following ABNF format:

```
r-sndcp-standby-timer = ( "r-snstandbyt:" / "r-snst:" ) 1HEXDIGIT CRLF
```

The hex-digit value is as defined in [102BAEB] for the standby timer field of the SND CP PDU.

Example:

```
r-snstandbyt:6
r-snst:6
```

3.9.12 PCO Config Protocol

This field specifies the external network protocol options to be transferred to the Home RFSS. The field SHALL NOT be present if the SND CP version for the PD SN Context registration request is equal to 1. This field MAY be present as an option if the SND CP version is equal to 2 or beyond. This field SHALL be present and copied into the PD SN Context registration response if it is present in the PD SN Context registration request. This field SHALL NOT be present in a PD SN Context registration request if the SU presence parameter is set to "Not Confirmed".

This parameter has the following ABNF format:

```
r-sndcp-pco = ( "r-snpcoc:" / "r-snpc:" ) 1HEXDIGIT CRLF
```

The 1hex-digit value is as defined in [102BAEB] for the PCO Configuration Protocol field of the SND CP PDU.

Example:

```
r-snpcoc:0
r-snpc:0
```

3.9.13 PCO Protocol Identity

This field specifies the Protocol Configuration Options (PCO) Protocol Identity as specified in [102BAEB]. This field SHALL be present if the PCO config protocol field is present, otherwise this field SHALL NOT be present.

This parameter has the following ABNF format:

```
r-sndcp-pcoid = ( "r-snpcoi:" / "r-snpi:" ) 4HEX-DIGIT CRLF
```

The 4hex-digit value is as defined in [102BAEB] for the Protocol Identity field of the SND CP PDU.

Example:

```
r-snpcoi:C023
r-snpi:C023
```

3.9.14 Protocol Identity Contents Length

This field specifies the length of the Protocol Identity Contents as specified in [102BAEB]. This field SHALL be present if the PCO config protocol field is present, otherwise this field SHALL NOT be present. Up to three extensions MAY be present

This parameter has the following ABNF format:

```
r-sndcp-pcoidlength = ( "r-snpcoicl:" / "r-snpicl:" ) 2HEX-DIGIT CRLF
```

The 2hex-digit (octet) value is as defined in [102BAEB] for the Length of Protocol Identity Contents field of the SND CP PDU.

Example:

```
r-snpcoicl:00
r-snpicl:00
```

3.9.15 Protocol Identity Contents

This field specifies the Protocol Identity Contents as specified in [102BAEB]. This field SHALL be present only if the Protocol Identity Content Length field is present with non-zero value.

This parameter has the following ABNF format:

```
r-sndcp-pcoidcontents = ( "r-snpcoic:" / "r-snpic:" ) octet-string CRLF
```

The octet-string value is as defined in [102BAEB] for the Protocol Identity Contents field of the SND CP PDU.

3.9.16 APN Length

This field indicates if the Access Point Name is a one-octet index or a string. In case of a string, it indicates the string length. The APN Length SHALL NOT be present if the SMDCP version is V1. The APN Length MAY be present if the SMDCP version is V2 or beyond.

This parameter has the following ABNF format:

```
r-smdcp-apnlength = ( "r-snapnl:" / "r-snal:" ) 2HEX-DIGIT CRLF
```

The 2hex-digit value is as defined in [102BAEB] for the APN Length field of the SMDCP PDU.

Example:

```
r-snapnl:00
r-snal:00
```

3.9.17 APN Index

This field indicates the value of the APN when it is represented by a one-octet index. The APN Index SHALL be present when the APN length is equal to 01.

This parameter has the following ABNF format:

```
r-smdcp-apnindex = ( "r-snapni:" / "r-snai:" ) 2HEX-DIGIT CRLF
```

The 2hex-digit value is as defined in [102BAEB] for the APN Index field of the SMDCP PDU.

Example:

```
r-snapni:00
r-snai:00
```

3.9.18 APN string format

This field indicates the value of the APN string when the APN length is between %x02 and %xFF

This parameter has the following ABNF format:

```
r-smdcp-apnstring = ( "r-snapns:" / "r-snas:" ) DQUOTE apn-string DQUOTE CRLF
apn-string =
apn-network-notalone = *( domainlabel "." ) domainlabel
                        ; not ending by ".cpds"
apn-network-alone =   *( domainlabel "." ) toplabel
```

```

                                ; not ending by ".cpds"
operator-id = *( domainlabel "." ) "cpds"
domainlabel = alphanum / alphanum *( alphanum / "-" ) alphanum
toplabel =   alpha / alpha *( alphanum / "-" ) alphanum
alphanum =   alpha / digit
alpha =      lowalpha / upalpha
lowalpha =   "a" / "b" / "c" / "d" / "e" / "f" / "g" / "h" / "i" /
              "j" / "k" / "l" / "m" / "n" / "o" / "p" / "q" / "r" /
              "s" / "t" / "u" / "v" / "w" / "x" / "y" / "z"
upalpha =    "A" / "B" / "C" / "D" / "E" / "F" / "G" / "H" / "I" /
              "J" / "K" / "L" / "M" / "N" / "O" / "P" / "Q" / "R" /
              "S" / "T" / "U" / "V" / "W" / "X" / "Y" / "Z"
digit =      "0" / "1" / "2" / "3" / "4" / "5" / "6" / "7" /
              "8" / "9"

```

Example:

```

r-snapns:"apnnetwork.us.missouri.kansas.operatorx.cpds"
r-snas:"apnnetwork.us.missouri.kansas.operatorx.cpds"

```

3.9.19 MSO Length

This field indicates whether a Manufacturer Specific Options (MSO) Value field is present, and the length of that field. The MSO Length SHALL NOT be present if the SNDCP version is V1 or V2. The MSO Length MAY be present if the SNDCP version is V3 or beyond.

This parameter has the following ABNF format:

```

r-sndcp-msolength = ( "r-snmsol:" / "r-snml:" ) 2HEX-DIGIT CRLF

```

The 2hex-digit value is as defined in [102BAEB] for the MSO Length field of the SNDCP PDU.

Example:

```

r-snmsol:0
r-snml:0

```

3.9.20 MSO Value

This indicates the contents of the MSO Value field. The MSO VALUE field SHALL be present only if the MSO Length field is present with non-zero value.

This parameter has the following ABNF format:

```

r-sndcp-msovalue = ( "r-snmsov:" / "r-snmv:" ) octet-string CRLF

```

The octet-string value is as defined in [102BAEB] for the MSO Value field of the SNDCP PDU.

3.9.21 Home PDIG Address

This field specifies the IP address in ISSI address realm of a Home Packet Data ISSI Gateway (PDIG) for an SU. The Home PDIG functional entity provides SNDCCP, Home Agent, and other functions to support Packet Data Services in a Home RFSS.

This parameter has the following ABNF format:

```

r-sndccp-hpdigaddress =      ( "r-snhpdigad:" / "r-snhpda:" ) IP4-address CRLF
IP4-address =                b1 "." decimal-uchar "." decimal-uchar "." b4
b1 =                         decimal-uchar
                             ;less than "224"; not "0" or "127"
b4 =                         decimal-uchar
                             ;not "0"
decimal-uchar =              DIGIT
                             / POS-DIGIT DIGIT
                             / ("1" 2*(DIGIT))
                             / ("2" ("0"/"1"/"2"/"3"/"4") DIGIT)
                             / ("2" "5" ("0"/"1"/"2"/"3"/"4"/"5"))

```

3.10 Examples

3.10.1 Mobility Management Examples [Informative]

The following examples illustrate the use of the REGISTER request to register for the various mobility objects defined herein. In the first case (Unit Registration) the 200 OK response is included in the example. In the other cases, for brevity, the 200 OK response is not included. Figure 35 shows the message flow for Registration.

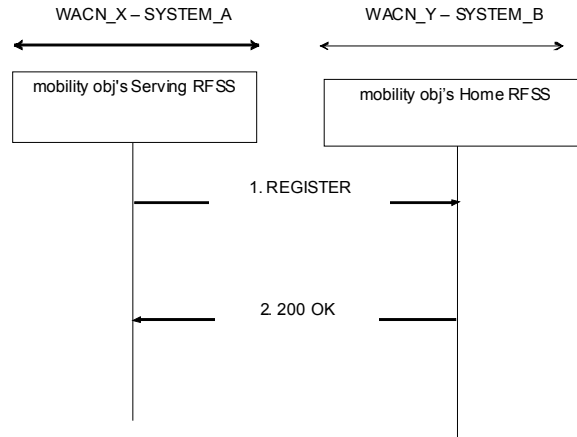


Figure 35 - Registration Example

3.10.1.1 Unit Registration

A unit has roamed into the Serving RFSS domain.

Entity	WACN	Sys.	Id.	SIP Identifier
Home RFSS	99999	111	11	11.111.99999.p25dr
Unit	99999	111	aaaaaa	99999111aaaaaa@p25dr;user=TIA-P25-SU
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS sends the following REGISTER Request to the Home RFSS:

```

REGISTER sip:11.111.99999.p25dr SIP/2.0
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
Max-Forwards: 70
To: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SU>
From: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SU>;tag=12345
Call-ID: 1.1@22.222.99999.p25dr
CSeq: 1 REGISTER
Contact: <sip:99999111aaaaaa@22.222.99999.p25dr;user=TIA-P25-SU>
Expires: 86400
Accept:
application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat,application/x-tia-p25-sndcp
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Length: 0
  
```

The Home RFSS responds with the following 200 OK Final Response to the Serving RFSS:

```

SIP/2.0 200 OK
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
  
```

To: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SU>;tag=98765
 From: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SU>;tag=12345
 Call-ID: 1.1@22.222.99999.p25dr
 CSeq: 1 REGISTER
 Contact: <sip:99999111aaaaaa@22.222.99999.p25dr;user=TIA-P25-SU>
 Expires: 4294967295
 Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
 Content-Length: 0

3.10.1.2 FNE Unit SD Registration

A console or other entity wishes to receive FNE addressed supplementary data sent by a specific unit.

Entity	WACN	Sys.	Id.	SIP Identifier
FNE Home RFSS	99999	111	33	33.111.99999.p25dr
Unit	99999	111	aaaaaa	99999111aaaaaa@p25dr;user=TIA-P25-SD-SU
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the FNE Home RFSS expressing interest in supplementary data for the Unit.

REGISTER sip:33.111.99999.p25dr SIP/2.0
 Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
 Max-Forwards: 70
 To: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SD-SU>
 From: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SD-SU>;tag=12345
 Call-ID: 1.1@22.222.99999.p25dr
 CSeq: 1 REGISTER
 Contact: <sip:99999111aaaaaa@22.222.99999.p25dr;user=TIA-P25-SD-SU>
 Expires: 86400
 Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
 Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
 Content-Length: 0

3.10.1.3 Group SD Registration

A console or other entity wishes to receive group addressed supplementary data. For example, the console might be dispatching the given group. To receive all supplementary data for the group, the entity registers with both the Group Home RFSS and any FNE Home RFSSs in the system (see Section 3.10.1.4). Note that the Group and FNE Home RFSSs might be the same, in which case two register messages need be sent.

Entity	WACN	Sys.	Id.	SIP Identifier
Home RFSS	99999	111	11	11.111.99999.p25dr
Group	99999	111	bbbb	99999111bbbb@p25dr;user=TIA-P25-SD-SG
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the Group Home RFSS expressing interest in supplementary data for the Group.

```
REGISTER sip:11.111.99999.p25dr SIP/2.0
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
Max-Forwards: 70
To: <sip:99999111bbbb@p25dr;user=TIA-P25-SD-SG>
From: <sip:99999111bbbb@p25dr;user=TIA-P25-SD-SG>;tag=12345
Call-ID: 1.1@22.222.99999.p25dr
CSeq: 1 REGISTER
Contact: <sip:99999111bbbb@22.222.99999.p25dr;user=TIA-P25-SD-SG>
Expires: 86400
Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Length: 0
```

3.10.1.4 FNE Group SD Registration

A console or other entity wishes to receive FNE addressed supplementary data sent by SUs affiliated with a specific group.

Entity	WACN	Sys.	Id.	SIP Identifier
FNE Home	99999	111	33	33.111.99999.p25dr
Group	99999	111	bbbb	99999111bbbb@p25dr;user=TIA-P25-SD-SGFNE
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the FNE Home RFSS expressing interest in FNE addressed supplementary data sent by SUs affiliated with the Group.

```
REGISTER sip:33.111.99999.p25dr SIP/2.0
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
Max-Forwards: 70
To: <sip:99999111bbbb@p25dr;user=TIA-P25-SD-SGFNE>
From: <sip:99999111bbbb@p25dr;user=TIA-P25-SD-SGFNE>;tag=12345
Call-ID: 2.1@22.222.99999.p25dr
CSeq: 2 REGISTER
Contact: <sip:99999111bbbb@22.222.99999.p25dr;user=TIA-P25-SD-SGFNE>
Expires: 86400
Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Length: 0
```

3.10.1.5 FNE System SD Registration

A console or other entity wishes to receive all FNE addressed supplementary data for a specific LMR system. The General FNE SUID in this example is \$FFFFFC.

Entity	WACN	Sys.	Id.	SIP Identifier
FNE Home	99999	111	33	33.111.99999.p25dr
Unit	99999	111	FFFFFC	99999111FFFFFC@p25dr;user=TIA-P25-SD-SU
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the FNE Home RFSS expressing interest in supplementary data for the General FNE SUID.

REGISTER sip:33.111.99999.p25dr SIP/2.0
 Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
 Max-Forwards: 70
 To: <sip:99999111FFFFFC@p25dr;user=TIA-P25-SD-SU>
 From: <sip:99999111FFFFFC@p25dr;user=TIA-P25-SD-SU>;tag=12345
 Call-ID: 1.1@22.222.99999.p25dr
 CSeq: 1 REGISTER
 Contact: <sip:99999111FFFFFC@22.222.99999.p25dr;user=TIA-P25-SD-SU>
 Expires: 86400
 Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
 Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
 Content-Length: 0

3.10.1.6 Unit SD Monitoring Registration

A recorder or other entity wishes to monitor supplementary data to and from a specific unit. This case differs from Section 3.10.1.2 only in the “user” parameter.

Entity	WACN	Sys.	Id.	SIP Identifier
Home RFSS	99999	111	11	11.111.99999.p25dr
Unit	99999	111	aaaaaa	99999111aaaaaa@p25dr;user=TIA-P25-SDMON-SU
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the SU’s Home RFSS expressing interest in monitoring supplementary data for the Unit.

REGISTER sip:11.111.99999.p25dr SIP/2.0
 Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
 Max-Forwards: 70
 To: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SDMON-SU>
 From: <sip:99999111aaaaaa@p25dr;user=TIA-P25-SDMON-SU>;tag=12345
 Call-ID: 1.1@22.222.99999.p25dr
 CSeq: 1 REGISTER
 Contact: <sip:99999111aaaaaa@22.222.99999.p25dr;user=TIA-P25-SDMON-SU>
 Expires: 86400
 Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
 Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
 Content-Length: 0

3.10.1.7 Group SD Monitoring Registration

A recorder or other entity wishes to monitor group addressed supplementary data. This case differs from Section 3.10.1.3 only in the “user” parameter.

Entity	WACN	Sys.	Id.	SIP Identifier
Home RFSS	99999	111	11	11.111.99999.p25dr
Group	99999	111	bbbb	99999111bbbb@p25dr;user=TIA-P25-SDMON-SG
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the Group Home RFSS expressing monitoring interest in supplementary data for the Group.

REGISTER sip:11.111.99999.p25dr SIP/2.0
 Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
 Max-Forwards: 70
 To: <sip:99999111bbbb@p25dr;user=TIA-P25-SDMON-SG>
 From: <sip:99999111bbbb@p25dr;user=TIA-P25-SDMON-SG>;tag=12345
 Call-ID: 1.1@22.222.99999.p25dr
 CSeq: 1 REGISTER
 Contact: <sip:99999111bbbb@22.222.99999.p25dr;user=TIA-P25-SDMON-SG>
 Expires: 86400
 Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
 Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
 Content-Length: 0

3.10.1.8 FNE Group SD Monitoring Registration

A recorder or other entity wishes to monitor FNE addressed supplementary data for a specific group. This case differs from Section 3.10.1.4 only in the “user” parameter.

Entity	WACN	Sys.	Id.	SIP Identifier
FNE Home	99999	111	33	33.111.99999.p25dr
Group	99999	111	bbbb	99999111bbbb@p25dr;user=TIA-P25-SDMON-SGFNE
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the FNE Home RFSS expressing monitoring interest in FNE addressed supplementary data for the Group.

REGISTER sip:33.111.99999.p25dr SIP/2.0
 Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
 Max-Forwards: 70
 To: <sip:99999111bbbb@p25dr;user=TIA-P25-SDMON-SGFNE>
 From: <sip:99999111bbbb@p25dr;user=TIA-P25-SDMON-SGFNE>;tag=12345
 Call-ID: 2.1@22.222.99999.p25dr
 CSeq: 2 REGISTER
 Contact: <sip:99999111bbbb@22.222.99999.p25dr;user=TIA-P25-SDMON-SGFNE>
 Expires: 86400
 Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
 Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
 Content-Length: 0

3.10.1.9 FNE System SD Monitoring Registration

A console or other entity wishes to monitor all FNE addressed supplementary data for a specific LMR system. The General FNE SUID in this example is \$FFFFFC. This case differs from Section 3.10.1.4 only in the “user” parameter.

Entity	WACN	Sys.	Id.	SIP Identifier
FNE Home	99999	111	33	33.111.99999.p25dr
Unit	99999	111	FFFFFC	99999111FFFFFC@p25dr;user=TIA-P25-SDMON-SU
Serving RFSS	99999	222	22	22.222.99999.p25dr

The Serving RFSS registers with the FNE Home RFSS expressing monitoring interest in supplementary data for the General FNE SUID.

```
REGISTER sip:33.111.99999.p25dr SIP/2.0
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
Max-Forwards: 70
To: <sip:99999111FFFFFC@p25dr;user=TIA-P25-SDMON-SU>
From: <sip:99999111FFFFFC@p25dr;user=TIA-P25-SDMON-SU>;tag=12345
Call-ID: 1.1@22.222.99999.p25dr
CSeq: 1 REGISTER
Contact: <sip:99999111FFFFFC@22.222.99999.p25dr;user=TIA-P25-SDMON-SU>
Expires: 86400
Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Length: 0
```

3.10.1.10 PD SNC Registration

A unit activates an SN Context into the Serving RFSS domain.

Entity	WACN	Sys.	Id.	SIP Identifier
Home RFSS	99999	111	11	11.111.99999@p25dr
SN Context	99999	111	UID = aaaaaa NSAPI = 1	99999111aaaaaad@p25dr;user=TIA-P25-PD-SN C
Serving RFSS	99999	222	22	22.222.99999@p25dr

The Serving RFSS sends the following REGISTER Request to the Home RFSS:

```
REGISTER sip:11.111.99999.p25dr SIP/2.0
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
Max-Forwards: 70
To: <sip:99999111aaaaaad@p25dr;user=TIA-P25-PD-SNC>
From: <sip:99999111aaaaaad@p25dr;user=TIA-P25-PD-SNC>;tag=12345
Call-ID: 1.1@22.222.99999.p25dr
CSeq: 1 REGISTER
Contact: <sip:99999111aaaaaad@22.222.99999.p25dr;user=TIA-P25-PD-SNC>
Expires: 86400
Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat,application/x-tia-p25-sndcp
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Length: 120
```

r-sns:1
r-snv:3
r-snn:0
r-snipa:30.130.71.115
r-snds:0
r-snmco:0
r-snpp:0
r-snrt:6
r-snst:14
r-snm:4
r-snd:0

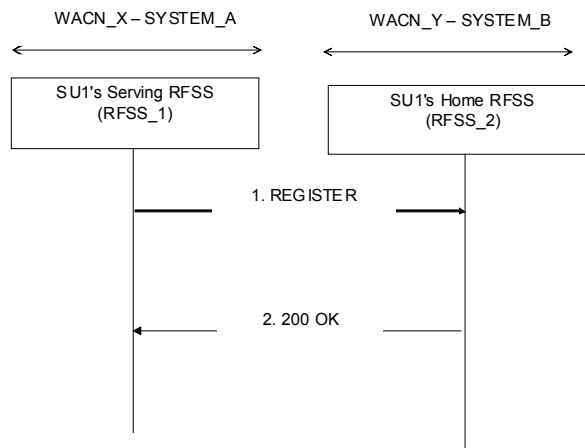
The Home RFSS responds with the following 200 OK Final Response to the Serving RFSS:

SIP/2.0 200 OK
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
To: <sip:99999111aaaaaad@p25dr;user=TIA-P25-PD-SNC>;tag=98765
From: <sip:99999111aaaaaad@p25dr;user=TIA-P25-PD-SNC>;tag=12345
Call-ID: 1.1@22.222.99999.p25dr
CSeq: 1 REGISTER
Contact: <sip:99999111aaaaaad@22.222.99999p25dr;user=TIA-P25-PD-SNC>
Expires: 86400
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Length: 125

r-snv:3
r-snn:0
r-snipa:30.130.71.115
r-snmco:0
r-snpp:0
r-snrt:6
r-snst:14
r-snm:4
r-snd:0
r-snhpda:33.120.52.131

3.10.2 Call Processing Examples [Informative]

Example 1 - SU Registration addressing example (see Figure 36):



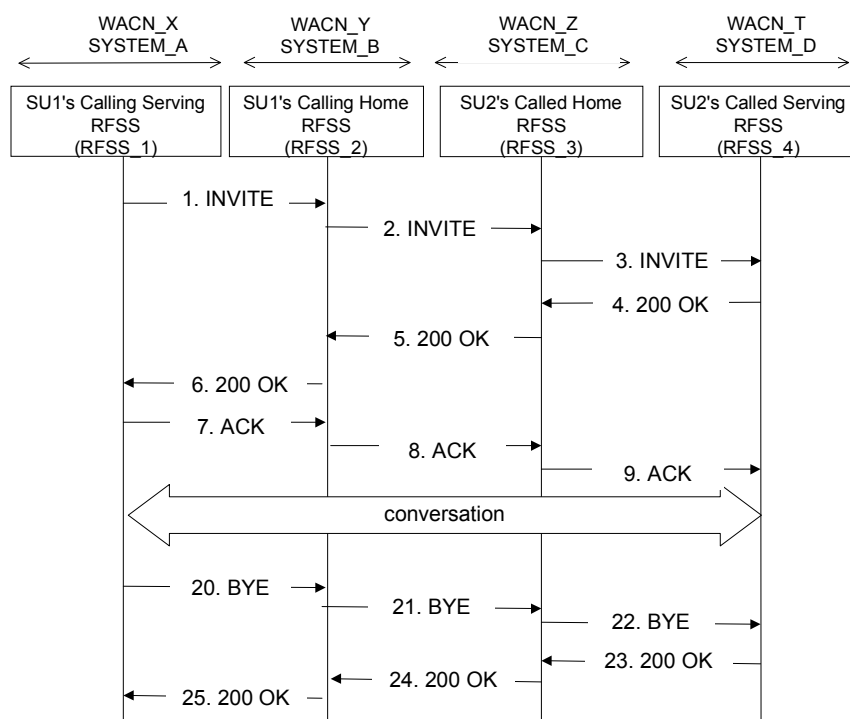
Variable are used in the example for better clarity but SHALL be replaced by the hexadecimal values in the real case	
variable	example
WACN_X	9EF60
WACN_Y	568FA
SYSTEM_A	570
SYSTEM_B	FA1
RFSS_1	6F
RFSS_2	B0
SUID1	568FAFA190FA41

Figure 36 - SU Registration Example

1. REGISTER from SU1's Serving RFSS to the SU1's home system <pre>REGISTER sip:RFSS_2.SYSTEM_B.WACN_Y.p25dr SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK15ADC0.1 Max-Forwards: 70 To: <sip:SUID1@p25dr;user=TIA-P25-SU> From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=16879 Call-ID: 4398c1c2.0008c0f3@p25dr CSeq: 1 REGISTER Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-S U> Expires: 86400 Accept: application/sdp;level=1, application/x-tia-p25-issi, application/x-tia-p25-issi-18 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0</pre> Notes: 1. the SIP URI of the To and From header fields indicate that the SUID1 is related to the SU's numbering plan. This explains the "@p25dr" and "user=TIA-P25-SU" elements. 2. the values SYSTEM_B and WACN_Y are obtained from the SUID1 value analysis. 3. the value RFSS_2 has been provisioned previously in the Serving RFSS_1.	2. OK response from SU1's home system to SU1's Serving RFSS <pre>SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK15ADC0.1 To: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=7A901 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=16879 Call-ID: 4398c1c2.0008c0f3@p25dr CSeq: 1 REGISTER Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SU > Expires: 4294967295 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required Content-Length: ...</pre>
---	---

Example 2 - SU-to-SU Call addressing example (see Figure 37):

The TRYING and RINGING provisional responses are not shown.



Variables are used in the example for better clarity but SHALL be replaced by the hexadecimal values in the real case

variable	example
WACN_X	9EF60
WACN_Y	568FA
WACN_Z	AF501
WACN_T	FED99
SYSTEM_A	570
SYSTEM_B	FA1
SYSTEM_C	0F2
SYSTEM_D	991
RFSS_1	6F
RFSS_2	B0
RFSS_3	BB
RFSS_4	E1
SUID1	568FAFA190FA41
SUID2	AF5010F201FF4E

Figure 37 - SU-to-SU Call example

1. INVITE from SU1's Calling Serving RFSS to SU1's Calling Home RFSS

INVITE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1
 Max-Forwards: 70
 Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 1 INVITE
 Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SU>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-18
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 5;a
 MIME-Version: 1.0
 Content-Disposition: session; handling=required

Notes:

1. This message is sent to the address RFSS_2.SYSTEM_B.WACN_Y.p25dr as provisioned in the RFSS.
2. The Route header field with TIA-P25-U2Uorig tells the RFSS to behave as the "Calling Home RFSS ". The calling SU is identified in the From header field.
3. The Calling Home RFSS forwards the INVITE to the Called Home RFSS on a new SIP dialog.
4. The MIME-Version header field denotes the presence of a P25 body and/or a manufacturer-specific body.

2. INVITE from Calling Home RFSS of SU1 to Called Home RFSS of SU2

INVITE sip:SUID2@ RFSS_3.SYSTEM_C.WACN_Z.p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK78f365.1
 Max-Forwards: 70
 Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 INVITE
 Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SU>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-18
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 5;a
 MIME-Version: 1.0
 Content-Disposition: session; handling=required

Notes:

1. The Route header field referring to the RFSS identified by RFSS_2 is removed. The Record-Route header field referring to the RFSS identified by RFSS_2 is added. The SIP URI provided in the Record-Route will be used later by the SU1's **Calling Serving RFSS**.
2. The Route header field referring to the RFSS identified by RFSS_3 is added. The Route header field with TIA-P25-U2Udest tells the RFSS to behave as the "Called Home RFSS ". The Called SU is identified in the To header field
3. This message is sent to the Home System of SU2.

3. INVITE from SU2's Called Home RFSS to SU2's Called Serving RFSS

INVITE sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK25fea7.1
 Max-Forwards: 70
 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=226fab57
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 INVITE
 Contact: <sip:SUID1@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SU>
 Accept: application/sdp;level=1, application/x-tia-p25-issi,
 application/x-tia-p25-issi-18
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 Priority: 5;a
 MIME-Version: 1.0
 Content-Disposition: session; handling=required
 . . .

Note:

1. The Route header field referring to the RFSS identified by RFSS_3 is removed. The Record-Route header field referring to the RFSS identified by RFSS_3 is added. The SIP URI provided in the Record-Route will be used later by the SU2's Serving RFSS.
2. SU2's Home RFSS retrieves the contact address of SU2's Serving RFSS from its location service, and then sends an INVITE to this RFSS.
3. As no Route header field is present and the radical name of the Request URI is equal to the radical name of the "To" field, the receiving RFSS knows it has to behave as a Called Serving RFSS.

4. OK response from SU2's Called Serving RFSS to SU2's Called Home RFSS

SIP/2.0 200 OK
 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK25fea7.1
 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=226fab57
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 INVITE
 Contact:<sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU>
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 MIME-Version: 1.0
 Content-Disposition: session; handling=required
 . . .

5. OK response from SU2's Called Home RFSS to SU1's Calling Home RFSS

SIP/2.0 200 OK
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK78f365.1
 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 2 INVITE
 Contact:<sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU>
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 MIME-Version: 1.0
 Content-Disposition: session; handling=required
 . . .

6. OK response from SU1's Calling Home RFSS to SU1's Calling Serving RFSS

SIP/2.0 200 OK
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1
 Record-Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Record-Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=000978879
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 1 INVITE
 Contact:<sip:SUID2@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SU>
 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS
 MIME-Version: 1.0
 Content-Disposition: session; handling=required
 . . .

7. ACK from SU1's Calling Serving RFSS to SU1's Calling Home RFSS

ACK sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKeb362754.1
 Max-Forwards: 70
 Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 Route: <sip:TIA-P25-U2Uorig@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=00978879
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 1 ACK
 Content-Length: 0

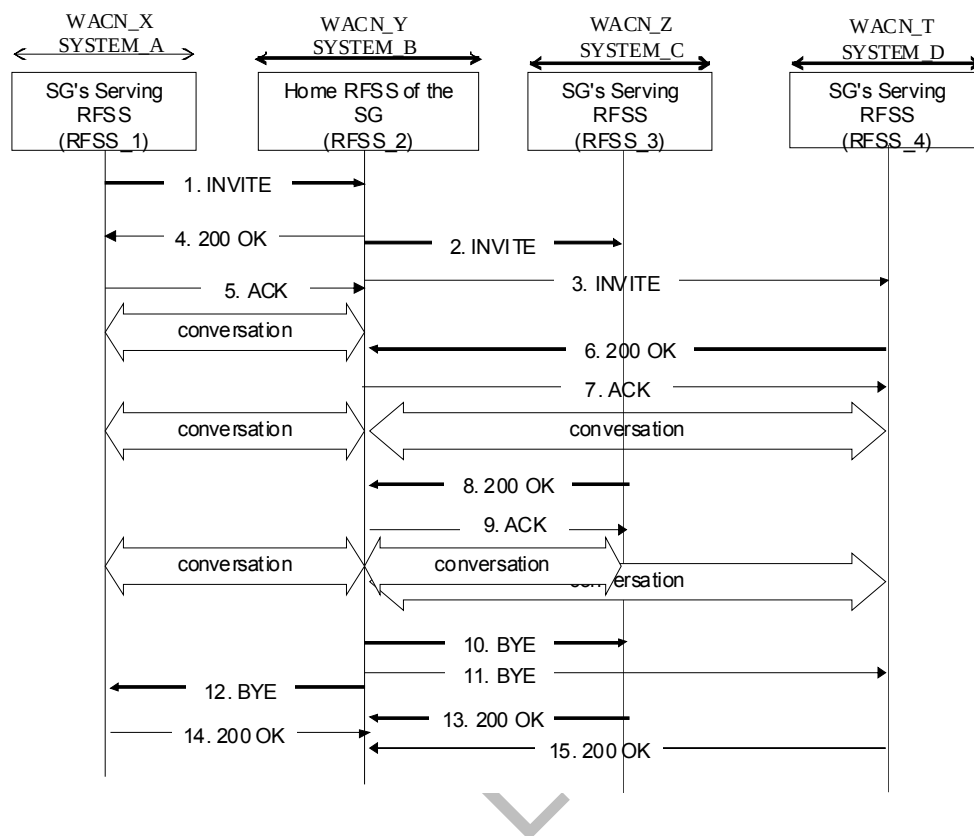
8. ACK from SU1's Calling Home RFSS to SU2's Called Home RFSS

ACK sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bKf37568.1
 Max-Forwards: 70
 Route: <sip:TIA-P25-U2Udest@RFSS_3.SYSTEM_C.WACN_Z.p25dr;lr>
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 1 ACK
 Content-Length: 0

9. ACK to SU2's Called Serving RFSS

ACK sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0
 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bKaf572e.1
 Max-Forwards: 70
 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87
 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=226fab57
 Call-ID: 4300f57f.0008f0ac@p25dr
 CSeq: 1 ACK
 Content-Length: 0

20. BYE from SU1's Calling Serving RFSS to SU1's Calling Home RFSS BYE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK4652c23a.1 Max-Forwards: 70 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=00978879 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 21. BYE from SU1's Calling Home RFSS to SU2's Called Home RFSS BYE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK86573e.1 Max-Forwards: 70 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 22. BYE from SU2's Called Home RFSS to SU2's Called Serving RFSS BYE sip:SUID2@p25dr;user=TIA-P25-SU SIP/2.0 Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK572a8c.1 Max-Forwards: 70 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=2226fab57 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0	23. OK response from SU2's Called Serving RFSS to SU2's Called Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_3.SYSTEM_C.WACN_Z.p25dr;branch=z9hG4bK25fea7.1 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=2269ea87 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=2226fab57 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 24. OK response from the SU2's Called Home RFSS to SU1's Calling Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK78f365.1 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 25. OK response from SU1's Calling Home RFSS to SU1's Calling Serving RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1 To: <sip:SUID2@p25dr;user=TIA-P25-SU>;tag=00978879 From: <sip:SUID1@p25dr;user=TIA-P25-SU>;tag=00016879 Call-ID: 4300f57f.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0
---	--

Example 3 - Unconfirmed Group Call addressing example (see Figure 38):**Figure 38 - Unconfirmed Group Call example**

Variable are used in the example for better clarity but SHALL be replaced by the hexadecimal values in the real case

variable	example
WACN_X	9EF60
WACN_Y	568FA
WACN_Z	AF501
WACN_T	FED99
SYSTEM_A	570
SYSTEM_B	FA1
SYSTEM_C	0F2
SYSTEM_D	991
RFSS_1	6F
RFSS_2	B0
RFSS_3	BB
RFSS_4	E1
SGID	568FAFA190FA

1. INVITE from SG's Serving RFSS_1 to SG's Home RFSS INVITE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1 Max-Forwards: 70 Route: <sip:TIA-P25-Groupcall@RFSS_2.SYSTEM_B.WACN_Y.p25dr;lr> To: <sip:SGID@p25dr;user=TIA-P25-SG> From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_1.SYSTEM_A.WACN_X.p25dr;user=TIA-P25-SG> Accept: application/sdp;level=1, application/x-tia-p25-issi, application/x-tia-p25-issi-18 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Priority: 8;a Content-Disposition: session; handling=required . . . Notes: 1. This message is sent to the address RFSS_2.SYSTEM_B.WACN_Y.p25dr as provisioned in the RFSS 2. The Route header field tells the RFSS to behave as the "Home RFSS of the SG". Addressing a Home RFSS general rule described in the addressing section is applied. 3. During the SG registration process the Home RFSS has stored all the RFSS to contact for this SGID 4. As there is no MIME-Version header field, there is only a SDP body in this message 2. INVITE from SG's Home RFSS to the Serving RFSS_3 INVITE sip:SGID@RFSS_3.SYSTEM_C.WACN_Z;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK0612357.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG> From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_3.SYSTEM_B.WACN_Y.p25dr;user=TIA-P25-SG> Accept: application/sdp;level=1, application/x-tia-p25-issi, application/x-tia-p25-issi-18 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Priority: 8;a Content-Disposition: session; handling=required . . .	Notes: 1. The Route header field referring to the RFSS identified by RFSS_2 is removed. 2. As no Route header field mentioning the "TIA-P25-Groupcall" element is present, the RFSS receiving this knows it has to behave as a "Serving RFSS of the SG". The SG is identified in the To and From header fields. 3. INVITE from SG's Home RFSS to the Serving RFSS_4 INVITE sip:SGID@RFSS_4.SYSTEM_D.WACN_T;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK05478967.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG> From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=22254abd Call-ID: 431ff9cd.00089a44@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_4.SYSTEM_C.WACN_Z.p25dr;user=TIA-P25-SG> Accept: application/sdp;level=1, application/x-tia-p25-issi, application/x-tia-p25-issi-18 Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Priority: 8;a Content-Disposition: session; handling=required . . . 4. OK response from SG's Home RFSS to SG's Serving RFSS_1 SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKac456326.1 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_2.SYSTEM_B.WACN_Y.p25dr;user=TIA-P25-SG> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Disposition: session; handling=required . . .
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5. ACK from SG's Serving RFSS_1 to SG's Home RFSS ACK sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKeb362754.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 1 ACK Content-Length: 0 6. OK response from the SG's Serving RFSS_4 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bK05478967.1 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=2265fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=22254abd Call-ID: 431ff9cd.00089a44@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_4.SYSTEM_D.WACN_T.p25dr;user=TIA-P25-SG> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required . . . 7. ACK from SG's Home RFSS to SG's Serving RFSS_4 ACK sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_2.SYSTEM_B.WACN_Y.p25dr;branch=z9hG4bKf37568.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=2265fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=22254abd Call-ID: 431ff9cd.00089a44@p25dr CSeq: 1 ACK Content-Length: 0	8. OK response from the SG's Serving RFSS_3 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK0612357.1 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 1 INVITE Contact: <sip:SGID@RFSS_3.SYSTEM_C.WACN_Z.p25dr;user=TIA-P25-SG> Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS MIME-Version: 1.0 Content-Disposition: session; handling=required . . . 9. ACK from SG's Home RFSS to SG's Serving RFSS_3 ACK sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bKaf572e.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 1 ACK Content-Length: 0
---	---

10. BYE from SG's Home RFSS to SG's Serving RFSS_3 BYE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK4652c23a.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 Note: The Home RFSS maintains its own Cseq numbering space, which in this example begins with 3. 11. BYE from SG's Home RFSS to SG's Serving RFSS_4 BYE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK86573e.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 431ff9cd.00089a44@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 12. BYE from SG's Home RFSS to SG's Serving RFSS_1 BYE sip:SGID@p25dr;user=TIA-P25-SG SIP/2.0 Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK572a8c.1 Max-Forwards: 70 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0	13. OK response from SG's Serving RFSS_3 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK4652c23a.1 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=11978879 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=1113258f Call-ID: 431ff9cd.00089a43@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 14 OK response from SG's Serving RFSS_1 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK05478967.1 To: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=0069ea87 From: <sip:SGID@p25dr;user=TIA-P25-SG>;tag=00016879 Call-ID: 431ff9cd.0008f0ac@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0 15. OK response from SG's Serving RFSS_4 to SG's Home RFSS SIP/2.0 200 OK Via: SIP/2.0/UDP RFSS_1.SYSTEM_A.WACN_X.p25dr;branch=z9hG4bK1578956.1 To: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1165fabe From: <sip:SGID@p25dr;user=TIA-P25-SU>;tag=1113258f Call-ID: 431ff9cd.00089a44@p25dr CSeq: 2 BYE Allow: ACK, BYE, CANCEL, INVITE, REGISTER, MESSAGE, OPTIONS Content-Length: 0
---	--

3.10.3 RFSS Service Capability Polling Examples [Informative]

The following examples illustrate the use of the SIP OPTIONS method to poll an RFSS for its service capabilities. In the first case (RFSS Service Capability Polling with RFSS Service Capability Profile Returned) the SIP OPTIONS request does not include the polling RFSS' service capability profile, and the 200 OK response includes the polled RFSS' service capability profile. In the second case the SIP OPTIONS request includes the polling RFSS' service capability profile, and the 200 OK response does not include the optional RFSS service capability profile. Figure 39 shows the message flow for RFSS Service Capability Polling.

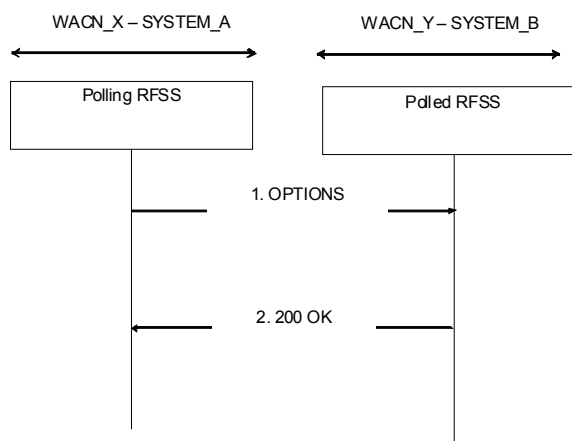


Figure 39 – RFSS Service Capability Polling Example

3.10.3.1 RFSS Service Capability Polling with RFSS Service Capability Profile Included in Response

An RFSS polls a second RFSS for its service capability information. The polled RFSS returns a response including its service capability profile.

Entity	WACN	Sys.	Id.	SIP Identifier
Polled RFSS	99999	111	11	11.111.99999.p25dr
Polling RFSS	99999	222	22	22.222.99999.p25dr

The Polling RFSS sends the following OPTIONS Request to the RFSS being polled:

```

OPTIONS sip:TIA-P25-RFSSCapability@11.111.99999.p25dr;user=TIA-P25-RFSS SIP/2.0
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
Max-Forwards: 70
To: <sip:TIA-P25-RFSSCapability@11.111.99999.p25dr;user=TIA-P25-RFSS>
  
```

From: <sip:TIA-P25-RFSSCapability@22.222.99999.p25dr;user=TIA-P25-RFSS>;tag=12345
Call-ID: 1.1@22.222.99999.p25dr
CSeq: 1 OPTIONS
Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Length: 0

The Polled RFSS responds with the following 200 OK Final Response
(containing the RFSS' service capability profile) to the Polling RFSS:

SIP/2.0 200 OK
Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
To: <sip:TIA-P25-RFSSCapability@11.111.99999.p25dr;user=TIA-P25-RFSS >;tag=98765
From: <sip:TIA-P25-RFSSCapability@22.222.99999.p25dr;user=TIA-P25-RFSS>;tag=12345
Call-ID: 1.1@22.222.99999.p25dr
CSeq: 1 OPTIONS
Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
Content-Type: application/x-tia-p25-issi
MIME-Version: 1.0
Content-Disposition: session; handling=required
Content-Length: 366

r-ucall-sec:1
r-ucall-clear:1
r-gcall-sec:1
r-gcall-clear:1
r-gcall-econf:1
r-gcall-cconf:1
r-sd-ealarm:0
r-sd-ealarmc:0
r-sd-egroupc:0
r-sd-calert:0
r-sd-smsg:0
r-sd-squery:0
r-sd-supdate:0
r-sd-rum:0
r-sd-rume:0
r-sd-rcheck:0
r-sd-rdetach:0
r-sd-rinhibit:0
r-sd-ruinhibit:0
r-pdata:1
r-adsite:01 00003 002 01 E2 AB23 FFFF 01 3
r-roam-unit:1
r-roam-grp:1
r-rfss-known:1

3.10.3.2 RFSS Service Capability Polling with RFSS Service Capability Profile in Request

An RFSS polls a second RFSS for its service capability information, providing its own service capability information to the polled RFSS. The polled RFSS returns a response without the optional service capability profile.

Entity	WACN	Sys.	Id.	SIP Identifier
Polled RFSS	99999	111	11	11.111.99999.p25dr
Polling RFSS	99999	222	22	22.222.99999.p25dr

The Polling RFSS sends the following OPTIONS Request to the RFSS being polled:

OPTIONS sip:TIA-P25-RFSSCapability@11.111.99999.p25dr;user=TIA-P25-RFSS SIP/2.0
 Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1
 Max-Forwards: 70
 To: <sip:TIA-P25-RFSSCapability@11.111.99999.p25dr;user=TIA-P25-RFSS>
 From: <sip:TIA-P25-RFSSCapability@22.222.99999.p25dr;user=TIA-P25-RFSS>;tag=12345
 Call-ID: 1.1@22.222.99999.p25dr
 CSeq: 1 OPTIONS
 Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat
 Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS
 Content-Type: application/x-tia-p25-issi
 MIME-Version: 1.0
 Content-Disposition: session; handling=required
 Content-Length: 366

r-ucall-sec:1
 r-ucall-clear:1
 r-gcall-sec:1
 r-gcall-clear:1
 r-gcall-econf:1
 r-gcall-cconf:1
 r-sd-ealarm:0
 r-sd-ealarmc:0
 r-sd-egroupc:0
 r-sd-calert:0
 r-sd-smsg:0
 r-sd-squery:0
 r-sd-supdate:0
 r-sd-rum:0
 r-sd-rume:0
 r-sd-rcheck:0
 r-sd-rdetach:0
 r-sd-rinhibit:0
 r-sd-ruinhibit:0
 r-pdata:1
 r-adsite:01 00003 002 01 E2 AB23 FFFF 01 3
 r-roam-unit:1
 r-roam-grp:1
 r-rfss-known:1

The Polled RFSS responds with the following 200 OK Final Response (without a service capability profile) to the Polling RFSS:

SIP/2.0 200 OK

Via: SIP/2.0/UDP 22.222.99999.p25dr;branch=z9hG4bK15ADC0.1

To: <sip:TIA-P25-RFSSCapability@11.111.99999.p25dr;user=TIA-P25-RFSS >;tag=98765

From: <sip:TIA-P25-RFSSCapability@22.222.99999.p25dr;user=TIA-P25-RFSS>;tag=12345

Call-ID: 1.1@22.222.99999.p25dr

CSeq: 1 OPTIONS

Accept: application/sdp;level=1,application/x-tia-p25-issi,application/x-tia-p25-supdat

Allow: ACK,BYE,CANCEL,INVITE,REGISTER,MESSAGE,OPTIONS

Content-Length: 0

Limited Use Copy

4. RTP Message Vocabulary for ISSI

This section describes the RTP message formats that SHALL be used over the ISSI. To ensure interoperability in a multi-vendor environment, the RFSS SHALL support RTP primitives, messages and procedures as specified in this document, and in conformance with a RTP version that is compatible with RTP version 2, as defined in [RFC3550].

[RFC3550] defines a control protocol (RTCP) for each RTP connection. RTCP employs the next UDP port above that used for RTP, and is used for such functions as quality feedback, SSRC identifier conflict resolution, and minimal session control. To comply with [RFC3550], each RFSS SHALL allocate a UDP port for RTCP (alongside RTP) for each call, but MAY discard all received RTCP packets without processing.

Figure 40 shows the structure of an ISSI RTP packet which consists of the RTP header and a ISSI payload. The RTP header SHALL be populated as described in Section 4.1. The ISSI payload SHALL consist of a payload header comprising a "Control Octet" and a series of "ISSI Block Headers", and a corresponding series of "ISSI Blocks". The Payload Header SHALL be populated as described in Section 4.2.

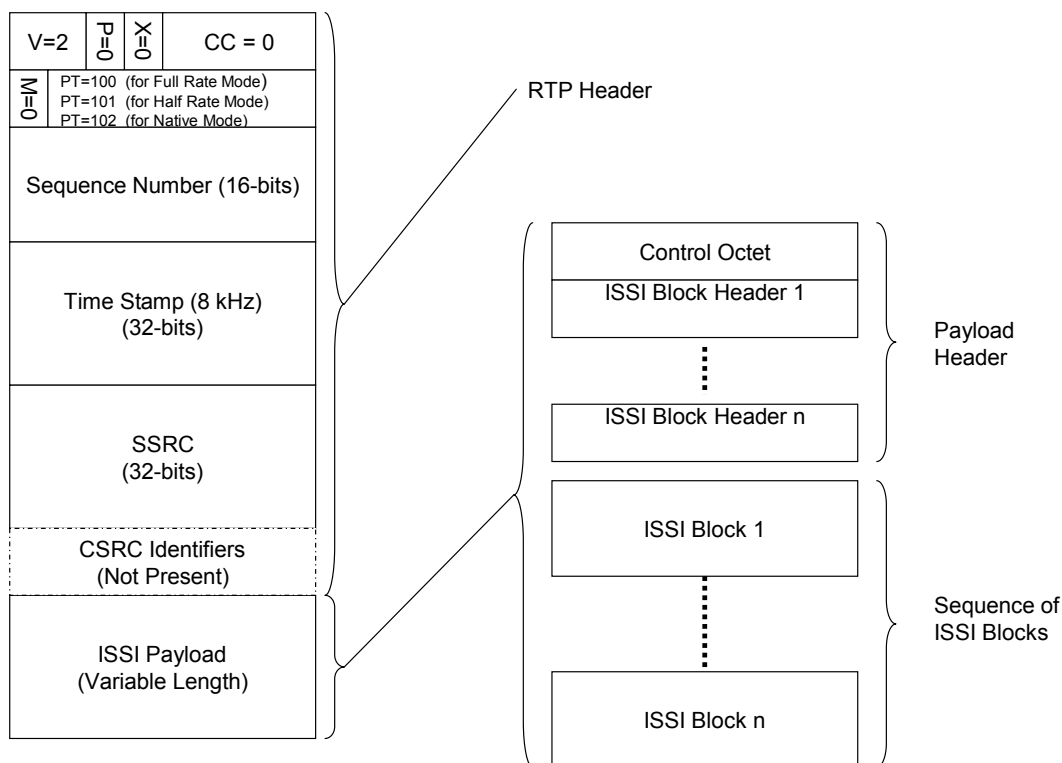


Figure 40 - ISSI RTP Packet Structure

4.1 RTP Fixed Header Field Definitions

Detailed generic definitions of the RTP fixed header fields are contained in [RFC3550].

For voice services over the ISSI, the RTP header fields SHALL be set as follows (all numbers are in decimal):

V (2 bits):	2 (i.e., RTP version 2)
P (1 bit):	0 (i.e., no padding)
X (1 bit):	0 (i.e., no header extension)
CC (4 bits):	0 (i.e. no contributing sources)
M (1 bit):	0
PT (7 bits):	see below

Sequence Number (16 bits):	Increments by 1 for each packet sent and MAY be used by the receiver to identify packet loss. The sequence number SHALL be maintained in accordance with [RFC3550] .
Timestamp (32 bits):	For RTP packets that optionally carry audio (PTT Transmit Request and PTT Transmit Progress), managed in accordance with [RFC3550] with a granularity of 125 microseconds. This RTP header field is not relevant for all other RTP packets and MAY be set to a default value of 0.
SSRC (32 bits):	Identifies the source of the payload. See Below.
CSRC (32 bits):	Never present in ISSI packets

The allowed values for the PT field in the RTP header are 100 (decimal) for Full Rate vocoder mode, 101 (decimal) for Half Rate vocoder mode, and 102 (decimal) for Native vocoder mode. An RFSS MAY alter the proposed value through the use of the SDP “m” and “a” attributes. Please refer to Section 3.7 for further details. The following example illustrates the use of the “m” and “a” fields to select the standard PT value of 100:

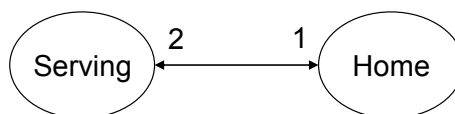
m = audio <port-number> RTP/AVP 100
a = rtpmap: 100 X-TIA-P25-FullRate/8000

The SSRC field is used to identify the source of the payload. For the ISSI the SSRC field SHALL be set depending on the role of the RFSS in the particular call, and, in the case of Unit-to-unit calls, depending on which of the three links of the call the RFSS is employing. The SSRC field SHALL be set as specified in Table 36.

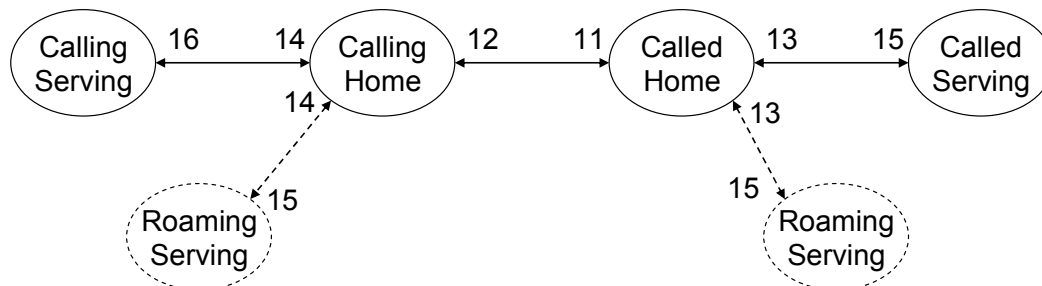
Table 36 - SSRC Values

Call Type	RFSS Role	Link	SSRC
Group	Serving RFSS	N/A	2
Group	Home RFSS	N/A	1
Unit-to-unit	Calling Serving RFSS	To Calling Home RFSS	16
Unit-to-unit	Calling Home RFSS	To Calling Serving RFSS or In-call Roaming Serving RFSS	14
Unit-to-unit	Calling Home RFSS	To Called Home RFSS	12
Unit-to-unit	Called Home RFSS	To Calling Home RFSS	11
Unit-to-unit	Called Home RFSS	To Called Serving or In-call Roaming Serving RFSS	13
Unit-to-unit	Called Serving RFSS	To Called Home RFSS	15
Unit-to-unit	In-call Roaming Serving RFSS	To Called or Calling Home RFSS	15

Figure 41 illustrates the SSRC identifiers required by Table 36.



(a) Group Call



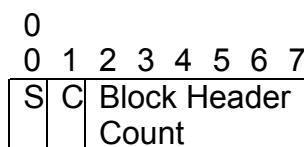
(b) Unit-to-unit Call

Figure 41 - SSRC Identifiers by Call Type

4.2 ISSI Payload Header

4.2.1 Control Octet Field Definitions

The payload header control octet SHALL contain 3 fields as illustrated in Figure 42.

**Figure 42 - Control Octet Format**

The bits of the control octet are defined as follows:

S - Signal bit. Use of the signal bit is described in Section 7.

C – Compact. Indicates whether the Block Headers are compact or verbose. The ISSI SHALL employ verbose payload headers with C=0. Compact payload headers (C=1) SHALL NOT be used over the ISSI.

Block Header Count – The number of Block Headers following the control octet. This is also the number of blocks in the packet.

4.2.2 Block Header Field Definitions

The block headers describe the payload data of the packet. Each block header is a 32-bit word as shown in Figure 43.

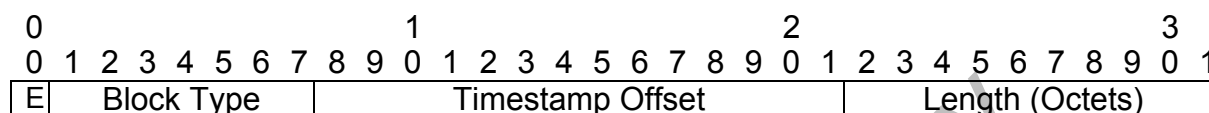


Figure 43 - Block Header

Each block header SHALL contain a bit (E) to identify whether the payload type is a standard IANA type (E set to 0) or a profile specific type (E set to 1). For voice services over the ISSI, the E bit SHALL always be set to 1 (profile specific type).

Each block header SHALL also contain a 7 bit block type. The block type is taken from Table 37 and identifies the format and content of the corresponding block in the sequence of ISSI blocks.

Table 37 - ISSI Block Types

Block Type	Meaning
0	Full Rate Voice
1	Packet Type
2,3,4	Reserved for Fixed Station Interface
5	Full Rate ISSI Header Word
6 - 10	Reserved for Fixed Station Interface and Conventional ISSI
11	RF PTT Control Word
12 - 14	Reserved for Fixed Station Interface
15	Console PTT Control Word
16 - 30	Reserved for Conventional ISSI
31 - 32	Reserved for Fixed Station Interface
33	Half Rate ISSI Header Word
34	Half Rate Voice
35	TalkSpurt Vocoder Mode
36 - 62	Reserved for Future Expansion
63 – 127	Manufacturer Specific

Each block header SHALL contain a 14-bit timestamp offset. This SHALL be in the same units as the RTP header timestamp (125 microseconds). The sum of the RTP timestamp and the offset is the timestamp of each block.

1. For all non-voice, non-manufacturer-specific ISSI blocks (i.e., with Block Type not being 0 or 34 or 63-127), since they are never positioned after a voice block in any given RTP packet, the Timestamp Offset SHALL be set to 0.
2. In the event that a single RTP packet carries one or more Full Rate Voice Blocks (Block Type = 0), per Section 4.4.2 of this document:
 - a. The Timestamp Offset for the first Full Rate Voice Block SHALL be set to 0.
 - b. The Timestamp Offset for the second Full Rate Voice Block, if present, SHOULD be set to 160.
 - c. The Timestamp Offset for the third Full Rate Voice Block, if present, SHOULD be set to 320.
3. For a Half Rate Voice Block (Block Type = 34), the Timestamp Offset SHALL be set to 0.

Each block header SHALL contain a 10-bit length indicator. This SHALL be the number of octets contained in the corresponding block.

For Full Rate Voice, the voice data transmitted over the ISSI between RFSSs comprises:

- Caller Identification
- Information from the CAI Header Data Unit that has been both Golay and Reed-Solomon decoded
- Full Rate voice data that has been both Golay and Hamming decoded
- Encryption Synchronization information that has been Hamming decoded but not Reed-Solomon Decoded, and
- Low Speed Data that has been cyclically decoded

For Half Rate Voice, the voice data transmitted over the ISSI between RFSSs comprises:

- Caller Identification

- Information from the MAC_PTT PDU that has been both Golay and Reed-Solomon decoded
- Half Rate voice data that has been Golay decoded
- Encryption Synchronization information

The following sections define the formats and content of each of the ISSI payload blocks, and the rules for constructing ISSI RTP packets.

4.3 Definition of ISSI Blocks

4.3.1 Full Rate Voice (Block Type = 0)

Full Rate Voice Blocks (shown in Figure 44) carry TIA-102 Common Air Interface (CAI) Full Rate **[102BABA]** voice data representing 20 msec. of speech, along with a variety of additional information that varies with the Frame Type.

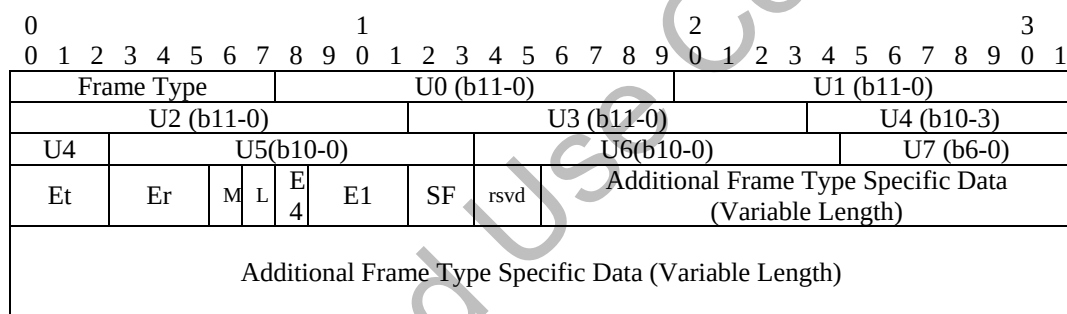


Figure 44 - Full Rate Voice Data

Table 38 summarizes the Full Rate frame types supported by the ISSI, the contents of which are described in the following sub-sections.

Table 38 - Full Rate Frame Types

Name	Frame Type	Description
FR1 (formerly IMBE1)	\$C2	Full Rate Voice 1
FR2 (formerly IMBE2)	\$C3	Full Rate Voice 2
FR3 (formerly IMBE3)	\$C4	Full Rate Voice 3
FR4 (formerly IMBE4)	\$C5	Full Rate Voice 4
FR5 (formerly IMBE5)	\$C6	Full Rate Voice 5
FR6 (formerly IMBE6)	\$C7	Full Rate Voice 6
FR7 (formerly IMBE7)	\$C8	Full Rate Voice 7
FR8 (formerly IMBE8)	\$C9	Full Rate Voice 8
FRFRAME9 (formerly IMBEFRAME9)	\$6A	Full Rate Voice 9 + Low Speed Data
FR10 (formerly IMBE10)	\$CB	Full Rate Voice 10
FR11 (formerly IMBE11)	\$CC	Full Rate Voice 11
FRFRAME12 (formerly IMBEFRAME12)	\$6D	Full Rate Voice 12 + Encryption Sync
FRFRAME13 (formerly IMBEFRAME13)	\$6E	Full Rate Voice 13 + Encryption Sync
FRFRAME14 (formerly IMBEFRAME14)	\$6F	Full Rate Voice 14 + Encryption Sync
FRFRAME15 (formerly IMBEFRAME15)	\$70	Full Rate Voice 15 + Encryption Sync
FRFRAME16 (formerly IMBEFRAME16)	\$71	Full Rate Voice 16 + Encryption Sync
FRFRAME17 (formerly IMBEFRAME17)	\$72	Full Rate Voice 17 + Encryption Sync
FRFRAME12 (formerly IMBEFRAME18)	\$73	Full Rate Voice 18 + Low Speed Data

The Full Rate data is divided into 8 decoded message vectors (U0-U7). The message vectors U0-U6 have been decoded (error correction code removed). Vector U7 is unprotected.

The report count fields SHALL reflect the number of decoding errors detected when the over the air frame was recovered at a receiver and the error correcting code (Golay or Hamming) was decoded. These MAY be used for error mitigation when the voice is decoded. If the number of errors detected is greater than can be indicated in the report field, the field SHALL be set to the maximum value.

E_t SHALL be the total number of errors detected in the frame. This SHALL be the parameter ε_T described by equation 95 in reference [102BABA] limited to a maximum value of 7.

E_r is a scaled version of ε_R described in equation 96 of reference [102BABA]. E_r SHALL be equal to $400 \times \varepsilon_R$ rounded down to the nearest integer and limited to a maximum of 7.

M is the mute frame flag, and L is the lost frame flag. The originating RFSS MAY set M and/or L values for any given frame based upon errors described in sections 7.7 and 7.8 of [102BABA], or in accordance with local policy. When M is set to 1, the audio output can be muted. When M is set to 0 and L is set to 1, the previous audio output can be repeated.

E_4 SHALL be the number of errors detected in vector U4, limited to a maximum of 1.

E_1 SHOULD be the number of errors detected in vector U1, limited to a maximum of 7. If not used, E_1 SHOULD be set to 0.

SF indicates a 2-bit free running super-frame counter that is incremented on the first frame of each super-frame.

In addition to the above 14 octets present on all Full Rate frames, each frame type specifies zero or more additional octets that are appended to the end of the frame. The following subsections describe the additional information carried in each of the frame types.

4.3.1.1 Fully Decoded Voice Frames (\$C2 - \$C9, \$CB, \$CC)

When the ISSI is used to transport fully decoded voice, no additional data is attached to the decoded Full Rate frame.

4.3.1.2 Voice with Low Speed Data (\$6A, \$73)

This frame type (shown in Figure 45) contains two low speed data codewords 0 and 1 for frame type \$6A, or 2 and 3 for frame type \$73 in addition to the Full Rate voice information. The low speed data is accompanied by two 2-bit status words (Si and Sj) to indicate the number of errors decoded at the receiver. They are defined to be:

Si = Status Bit S0 (frame \$6A), Status Bit S2 (frame \$73)

Sj = Status Bit S1 (frame \$6A), Status Bit S3 (frame \$73)

The decimal values of these fields are defined as follows:

00 = no errors

01 = 1 error

02 = 2 errors

03 = erasure – indicates that the decoder detected an uncorrectable combinations of errors in the code word.

0										1										2									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3						
LSD (0 or 2)										LSD (1 or 3)										reserved				Si		Sj			

Figure 45 - Partially Decoded Frames with Low Speed Data

4.3.1.3 Voice with Partially Decoded Encryption Synch (\$6D-\$72)

The Encryption Sync data (shown in Figure 46) consists of 4 encryption sync code words with associated status words in addition to the Full Rate voice. The encryption sync field contains code words:

- 0, 1, 2, 3 for PID/FT \$6D;
- 4, 5, 6, 7 for PID/FT \$6E;
- 8, 9, 10 and 11 for PID/FT \$6F;
- 12, 13, 14 and 15 for PID/FT \$70;
- 16, 17, 18 and 19 for PID/FT \$71 and
- 20, 21, 22 and 23 for PID/FT \$72.

The 7 bit status field indicates the number of bit errors detected when the over the air Hamming code was decoded. It is computed as follows:

$$\text{STATUS} = \text{ST}_3 * 3^{**3} + \text{ST}_2 * 3^{**2} + \text{ST}_1 * 3^{**1} + \text{ST}_0 * 3^{**0}$$

where

ST_n=error status of ES_n

ST_n=0 means no errors

ST_n=1 means 1 error

ST_n=2 means erasure

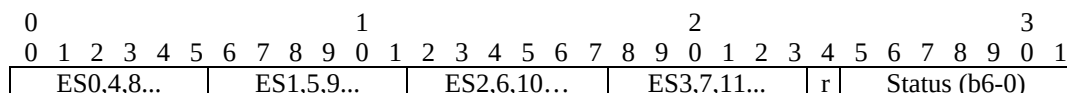


Figure 46 - Partially Decoded Frames with Encryption Synchron

4.3.2 ISSI Packet Type (Block Type = 1)

Figure 47 shows the ISSI Packet Type block payload.

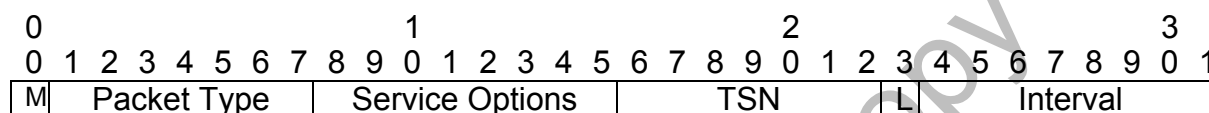


Figure 47 - ISSI Packet Type

For voice services over the ISSI, the RTP Packet Type fields SHALL be set as follows (all numbers are in decimal):

M (1 bit):

The M bit indicates the mute status of a particular talk spurt. The M bit SHALL be processed as described in Section 7.

Packet Type (7 bits):

Identifies the RTP packet type. The following values are defined. All other values are reserved. Procedures related to the use of the various RTP packets are given in Section 7.

- 0 - PTT Transmit Request
- 1 - PTT Transmit Grant,
- 2 - PTT Transmit Progress
- 3 - PTT Transmit End
- 4 - PTT Transmit Start
- 5 - PTT Transmit Mute
- 6 - PTT Transmit Unmute
- 7 - PTT Transmit Wait
- 8 - PTT Transmit Deny
- 9 - PTT Heartbeat
- 10 - PTT Heartbeat Query

Service Options (8 bits):	The service options are taken from the "Service Options" octet (as defined in [102AABC]) in the Unit to Unit Voice Service Request message or Group Voice Service Request message that the RFSS receives from the requesting SU. If unknown, assume 0.
TSN (7 bits):	The Transmission Sequence Number identifies a particular talk spurt in the RTP stream. Use of the TSN is described in Section 7.
L (1-bit):	When set, the Losing Audio Bit identifies a particular talk spurt as being less important (in the opinion of the transmitting RFSS) than some other talk spurt in the RTP stream. Use of the Losing Audio Bit is described in Section 7.
Interval:	Relevant only to the heartbeat and heartbeat query types, the Interval specifies the periodicity, in seconds, at which the sending RFSS wishes to receive heartbeats from the receiving RFSS. For all other Packet Types, the Interval parameter is ignored.

PTT Transmit Request is sent by an SMF to the MMF to initiate a talk spurt. It is sent until the talk spurt ends or until it is explicitly acknowledged with a *PTT Transmit Grant*, or *PTT Transmit Wait*, or *PTT Transmit Deny*, or until the Serving RFSS determines that it has to proceed, thereby switching to *PTT Transmit Progress* packets.

PTT Transmit Grant is sent by the MMF to the SMF in acknowledgment of *PTT Transmit Request*. It indicates the willingness of the MMF to receive a particular talk spurt from the SMF.

PTT Transmit Deny is sent by the MMF to the SMF in acknowledgment of *PTT Transmit Request*. It indicates that the MMF is unwilling to receive a particular talk spurt from the SMF.

PTT Transmit Wait is sent by the MMF to the SMF as a provisional acknowledgement of *PTT Transmit Request*. It indicates that the MMF has not yet determined whether or not to grant the SMF. Upon making its determination, the MMF sends either *PTT Transmit Grant* or *PTT Transmit Deny* as a final acknowledgement of *PTT Transmit Request*.

PTT Transmit Start is sent by a MMF to inform a SMF of the start of a talk spurt. *PTT Transmit Start* is not acknowledged.

PTT Transmit Progress carries voice content. It MAY be sent by either the MMF or the SMF to the other.

PTT Transmit End indicates to a receiving RFSS that a talk spurt is complete. It MAY be sent by either the MMF or the SMF to the other.

PTT Transmit Mute indicates that the sending RFSS is currently unwilling to receive the voice payloads for the indicated talk spurt. It MAY be sent by either the MMF or the SMF to the other.

PTT Transmit Unmute indicates to the receiving RFSS that the sending RFSS wishes to resume receipt of voice payloads for the indicated talk spurt. It MAY be sent by either the MMF or the SMF to the other.

PTT Heartbeat is used to aid in the detection of failed links. Heartbeats are sent periodically by both the MMF and the SMF whether or not there is voice related RTP traffic for the session.

PTT Heartbeat Query provides a mechanism for an RFSS to request that its peer immediately send *PTT Heartbeat*.

4.3.3 Full Rate ISSI Header Word (Block Type = 5)

Figure 48 shows the Full Rate ISSI Header Word data, corresponding roughly to the information conveyed in CAI Header Word (with which a PTT voice message begins), with the addition of the NID from the Header Data Unit.

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
Message Indicator (72-bits)																																							
MFID										AlgID										Key ID																			
										Group ID																				NID (15-8)									
NID (7-0)										SF	VBB		Reserved																										

Figure 48 - Full Rate ISSI Header Word

The Message Indicator, MFID, Group ID and NID are as defined in [102BAAA]. The Group ID field SHALL be set to the 16-bit Group Identity of the SGID.

The SF field is a 2-bit free running superframe counter that is initialized to zero for the ISSI header word and is incremented on the first frame of each superframe.

The VBB (Voice Block Bundling) field is a 2-bit field indicating how many voice blocks are contained in each of the voice-bearing RTP packets that follow this Header Word for this talk spurt. VBB MAY be set to 1 (%01), 2 (%10), or 3 (%11).

4.3.4 RF PTT Control Word (Block Type = 11)

Voice services often need to authorize usage based on the source and allocate resources based on the priority of that source. This information needs to be carried across interfaces such as the Inter-RF Subsystem Interface (ISSI). The RF PTT Control Word carries this required information and is used for all non-console initiated voice services over the ISSI.

Figure 49 shows the RF PTT Control Word

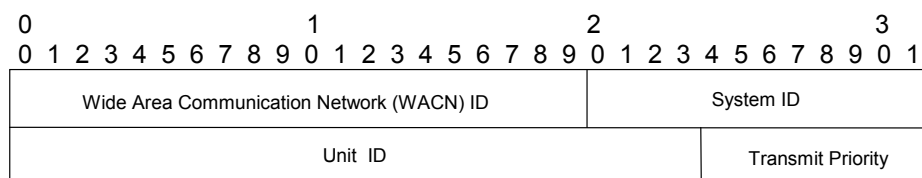


Figure 49 - RF PTT Control Word

For voice services over the ISSI, the PTT control fields SHALL be set as follows (all numbers are in decimal):

- SU Identifier (56 bits):** Fully qualified identifier (comprising the 20-bit WACN ID, 12-bit System ID, and 24-bit Unit ID) of an SU making a request for (or being granted) permission to transmit voice data (e.g., Full Rate frames), or of an SU that is currently transmitting voice data. In the event that the fully qualified SUID is unknown, all three fields (WACN, System and Unit) SHALL be set to 0.
- Transmit Priority (8 bits):** Comprising a Priority Type (bits 7-4) sub-field and a Priority Level (bits 3-0) sub-field, as defined in [T102]. For trunked RFSSs, these sub-field values are set through various

means, e.g., initially according to priority values in the Service Option field of the trunking control channel messages received from subtending SUs [102AABC]. The priority type is set to decimal 2 for preemptive priority talk spurts, 1 for priority talk spurts, and 0 for all other talk spurts. For example, for a Level 3 priority talk spurt, the priority type and priority level are set to 1 and 3, respectively. For conventional RFSSs, the priority type and priority level are set through other means than according to the values in the Service Option field, as there is no such field in the received conventional control messages [102AABG].

4.3.5 Console PTT Control Word (Block Type = 15)

Voice services often need to authorize usage and allocate resources based on the type of source and the priority of the source's request. Also, end points such as consoles need to know the original source of the audio that a console is generating. This information needs to be carried across interfaces such as the Inter-RF Subsystem Interface (ISSI). The Console PTT Control Word enables these functions for console requested talk spurts.

Figure 50 shows the Console PTT Control Word

0	1	2	3
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1			
Console SU Identifier (56 bit SUID)			
Console SU Identifier (continued)			
Physical Console Identifier (56 bit SUID)			
Physical Console Identifier (continued)			
Voice Source Identifier (56 bit SUID)			
Voice Source Identifier (continued)			
Transmit Priority			
Transmission Descriptor			
Console TX Priority			

Figure 50 - Console PTT Control Word

For voice services over the ISSI, the PTT control fields SHALL be set as follows (all numbers are in decimal):

Console SU Identifier (56 bit SUID): Fully qualified identifier
(comprising the 20-bit WACN ID, 12-bit System

ID, and 24-bit Unit ID) of a console SU making a request for (or being granted) permission to transmit voice data (e.g., Full Rate frames), or of a console SU that is currently transmitting voice data. In the event that the fully qualified SUID is unknown, all three fields (WACN, System and Unit) SHALL be set to 0. The default value of this field is the Physical Console Identifier field value.¹⁵

Transmit Priority (8 bits): Comprising a Priority Type (bits 7-4) sub-field and a Priority Level (bits 3-0) sub-field, as defined in [T102]. For trunked RFSSs, these sub-field values are set through various means, e.g., initially according to priority values in the Service Option field of the trunking control channel messages received from subtending SUs [102AABC]. The priority type is set to decimal 2 for preemptive priority talk spurts, 1 for priority talk spurts, and 0 for all other talk spurts. For example, for a Level 3 priority talk spurt, the priority type and priority level are set to 1 and 3, respectively¹⁶.

Physical Console Identifier (56 bit SUID) This is the console's assigned Trunking SUID which is a fully qualified identifier (the console SUID comprising the 20-bit WACN ID, 12-bit System ID, and 24-bit Unit ID) of the console making a request for (or being granted) permission to transmit voice data (e.g., Full Rate frames), or of a console that is currently transmitting voice data. In the event that the fully qualified SUID is unknown or the transmission requestor or source is not a console, all three fields (WACN, System and Unit) SHALL be set to 0.

Transmission Descriptor (8 bits) Used to indicate various attributes associated with the console talk spurt.

¹⁵ Editor's Note: In the future, if this protocol is updated to support the conventional complexity 1 architecture, the Console SU Identifier field can be used to identify the console's assigned SUID for use on a particular conventional resource. In the complexity 1 architecture this field cannot be used to identify a physical console because the same console can be assigned different SUIDs for use on different conventional resources.

¹⁶ Editor's Note: For conventional RFSSs i.e., RFSSs employing conventional complexity 2, the priority type and priority level are set through other means than according to the values in the Service Option field, as there is no such field in the received conventional control messages [102AABG] [102AABC].

\$00: Indicates that the talk spurt is a trunking talk spurt¹⁷

\$01: Indicates that the talk spurt is a conventional talk spurt¹⁸

Voice Source Identifier (56 bit SUID) A fully qualified identifier (an SUID comprising the 20-bit WACN ID, 12-bit System ID, and 24-bit Unit ID) of the talker that is the initial source of the audio. In the event that the fully qualified SUID is unknown or, the source ID is not relevant in the domain of the talk spurt, all three fields (WACN, System and Unit) SHALL be set to 0. An example of where it would not be relevant in the domain of the talk spurt is when the console is patching a conventional talk spurt to a trunking talk group. The conventional SUID will not be relevant on the trunking talk group. The SUID is set to 0 to avoid duplicating an ID that already exists in the trunking system.

Console TX Priority (8 bits) The priority associated with the console's transmission request. All values \$00 through \$FF are allowed, recommended values are:

- \$10: priority 1 (lowest and default value)
- \$30: priority 2
- \$50: priority 3
- \$70: priority 4
- \$90: priority 5
- \$B0: priority 6
- \$D0: priority 7
- \$F0: priority 8

4.3.5.1 General Usage of the Console PTT Control Word Block

An indication that the request is from a console is needed so the Home RFSS can support the console priority feature. Once a call is set up, a participating console MAY request to transmit while an SU is transmitting. The Home RFSS needs to know that the PTT Transmit Request is from a console so it can give priority to the console's transmission request. This indication also enables other parallel consoles to indicate on their user interfaces whether the talk spurt is from a subscriber or a console operator.

¹⁷ Editor's Note: This setting is also used with conventional complexity 2 since the console does not know when trunking groups and units are mapped to conventional groups and units on a particular conventional fixed station.

¹⁸ This setting is for future use in the event that this protocol is updated to support the conventional complexity 1 architecture.

Consoles have multiple levels of transmission priorities. The console's requested transmit priority is needed so the Home RFSS can handle multiple simultaneous console transmission requests. During a call, one console can "take over" from another console when the console making the request is requesting to transmit at a higher priority level. This parameter allows the Home RFSS to perform the arbitration.

4.3.5.2 Transmission Descriptor Field Usage

The Transmission Descriptor is in place for future use. It might be the case that other devices on the network want to know additional information about a console talk spurt. Currently the octet is used to indicate that the talk spurt is a "trunking" or a "conventional" talk spurt.

4.3.5.3 Physical Console Identifier Field Usage

The Physical Console Identifier is needed so consoles know which physical console is making a transmission. A console can support trunking only, or conventional only, or both trunking and conventional. Consoles can be configured with a list of which physically adjacent consoles' talk spurts SHOULD be muted to avoid feed back loops (because the console doing the muting would otherwise play the transmitting console's audio out its speaker, feeding back into the transmitting console's microphone). A method is needed to identify the console that is transmitting on either trunking or conventional resources. It is beneficial to use one identifier for the physical console regardless of which type of talk spurt (trunking or conventional) is being made. The Physical Console Identifier SHOULD be unique across all consoles in all LMR systems. The Console SU Identifier field defaults to the same value as the Physical Console Identifier.¹⁹

4.3.5.4 Console SU Identifier Field Usage

A physical console has by default one Console SU Identifier for the trunking system it is operating on. This field identifies the console in the same manner that the SUID identifies a subscriber radio.²⁰

¹⁹ Editor's Note: The Physical Console Identifier is the same as the console's Trunking SUID for complexity 2 as well. If this protocol is updated to support complexity 1, and if the console only supports conventional, then a Trunking SUID is assigned to the console for use as its Physical Console Identifier. If the console operating in complexity 1 already has a Trunking SUID then the Physical Console Identifier is set equal to that value.

²⁰ Editor's Note: Because this protocol is not used for complexity 1, the Console SU Identifier field defaults to the same value as the Physical Console Identifier. In the complexity 1 architecture a console can have one or more different SUIDs (truncated to WUID on the CAI) associated with the conventional channels that it is dispatching. Therefore, if this protocol is updated to support a complexity 1 type of architecture, the Console SU Identifier field can be the SUID assigned to the console for use on the particular conventional fixed station.

4.3.5.5 Voice Source Identifier Field Usage

The Voice Source Identifier is needed so a console can determine when multiple talk spurts are actually the same talk spurt. For instance, when performing a console patch, the console can re-transmit audio sourced from a different device such as a mobile radio. Parallel consoles can be simultaneously listening to the patching console and to the mobile radio that is the source of the patch audio. These parallel consoles will receive two different streams of audio (the console patch transmit and the source talk spurt) which are actually from the same audio source. The Voice Source Identifier enables the console to know when it is receiving duplicate streams.

4.3.6 Half Rate ISSI Header Word (Block Type = 33)

Figure 51 shows the Half Rate ISSI Header Word data, corresponding roughly to the information conveyed in CAI MAC_PTT (with which a PTT voice message begins).

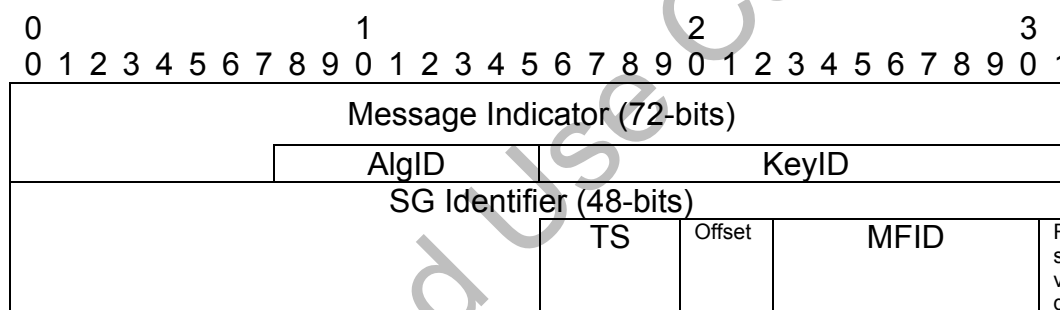


Figure 51 - Half Rate ISSI Header Word

The Message Indicator, AlgID, KeyID, and MFID are as defined in [102BBAC].

The SG Identifier is the fully qualified SGID (comprising the 20-bit WACN ID, 12-bit System ID, and 16-bit Group ID). In the event that the fully qualified SGID is unknown, all three fields (WACN ID, System ID, and Group ID) SHALL be set to 0.

TS is a 4-bit field that indicates the TDMA Time Slot in which the inbound MAC_PTT²¹ was received. This can be used to determine whether the MAC_PTT was received on an FACCH or SACCH [102BBAC]. The TS field value SHOULD be provided. If unknown, the TS field value SHALL be set to %xF (unknown).

²¹ MAC_PTT is defined in [102BBAC].

Offset is a 3-bit field that indicates the value of the Offset field in the inbound MAC_PTT received on this Time Slot. This denotes the number of non-SACCH bursts between the MAC_PTT and the first 4V burst that follows it **[102BBAC]**. The Offset field value **SHOULD** be provided. If unknown, the Offset field value **SHALL** be set to decimal 7 (unknown).

For any given Half Rate talk spurt, at least one Half Rate ISSI Header Word **SHOULD** be sent. A Half Rate ISSI Header Word **MAY** be sent when an inbound MAC_PTT is received. However, to optimize end-to-end delay, a Half Rate ISSI Header Word **SHOULD** be sent as soon as the header word information is available to the RFSS (e.g., upon receipt of the first inbound MAC_PTT).

4.3.7 Half Rate Voice (Block Type = 34)

Half Rate Voice Blocks carry TIA-102 Common Air Interface (CAI) Half Rate (HR) voice data representing 80 msec. of speech (shown in Figure 52), or 40 msec. of speech and Encryption Synchronization Signalling (ESS) information (shown in Figure 53 and Figure 54).

Table 39 summarizes the HR frame types supported by the ISSI, the contents of which are described in the following sub-sections.

Table 39 – Half Rate Frame Types

Name	Frame Type	Description
HR1	\$1	4V - 1
HR2	\$2	4V - 2
HR3	\$3	4V - 3
HR4	\$4	4V - 4
HR5	\$5	4V - U (i.e., 4V-Unknown)
HR6	\$6	2V + ESS

In the event that at least one inbound CAI MAC PTT (with which a PTT voice message begins) has been received, frame types \$1 to \$4 **SHALL** be designated to the four 4-V voice frames in each superframe.

In the event that none of the inbound CAI MAC PTTs has been received, frame type \$5 **SHALL** be designated to all 4-V voice frames in the first superframe.

4.3.7.1 4V Half Rate Voice Block (\$1 - \$5)

The 4V Half Rate Voice data block, corresponding to frame types \$1 to \$5, is as shown in Figure 52.

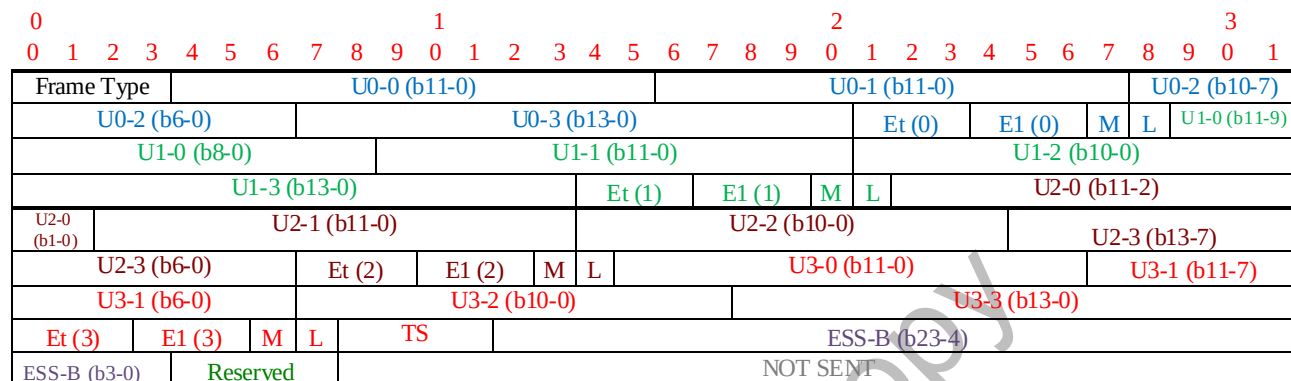


Figure 52 – 4V Half Rate Voice Data

The 4V Half Rate Voice data block contains 4 voice frames.

Each voice frame, of frame number "n", where n= 0, 1, 2, 3, of the Half Rate Voice data is divided into 4 message vectors (Un-0, Un-1, Un-2, Un-3). These message vectors have been decoded (error correction code removed).

Each voice frame also includes error mitigation data as defined in sections C.3 and C.4 in Annex C of this document. The error count fields, Et and E1, SHALL reflect the number of decoding errors detected when the over the air frame was recovered at a receiver and the error correcting code (2 Golay codes) was decoded. These MAY be used for error mitigation when the voice is decoded. If the number of errors detected is greater than can be indicated in the report field, the field SHALL be set to the maximum value.

Et(n) SHALL be the total number of errors detected in the nth frame. This SHALL be the parameter ε_T described by equation 55 in reference [102BABA-1] limited to a maximum value of 7.

E1(n) SHALL be the number of errors detected in the second Golay codeword of the nth frame, $\hat{c}_1$, described by equation 48 in reference [102BABA-1]. This SHALL be set to the value of ε_1 of equation 55 in reference [102BABA-1], limited to a maximum value of 7.

M is the mute frame flag, and L is the lost frame flag. The originating RFSS MAY set M and/or L values for any given frame based upon errors described in sections 5.6 and 5.7 of [102BABA-1], or in accordance with local policy. When M is set to 1, the audio output can be muted. When M is set to 0 and L is set to 1, the previous audio output can be repeated.

TS is a 4-bit field that indicates the TDMA timeslot in which the inbound 4V Half Rate Voice data block was received. The TS field value SHOULD be provided. If unknown, the TS field value SHALL be set to %xF (unknown).

ESS-B is a 24-bit field containing encoded Encryption Synchronization Signal (ESS) information associated with each 4V voice frame, as defined in [102BBAC].

4.3.7.2 2V Half Rate Voice Block (\$6)

The 2V voice data block corresponds to frame type \$6.

The 2V Half Rate Voice data block contains 2 voice frames and additional data.

Each voice frame, of frame number "n", where n= 0, 1, of the Half Rate Voice data is divided into 4 message vectors (Un-0, Un-1, Un-2, Un-3). These message vectors have been decoded (error correction code removed).

The error count fields Et(n) and E1(n) are as specified in Section 4.3.7.1

The M, L, and TS fields are as specified in Section 4.3.7.1.

The EXT field is a 2-bit field that indicates the type of additional data being carried:

- 0 Fully decoded ESS (see Section 4.3.7.2.1)
- 1 ESS-A raw parity bits (see Section 4.3.7.2.2)
- 2 Reserved
- 3 Reserved

4.3.7.2.1 Fully Decoded Encryption Synchronization Signalling Information

In this case (EXT = 0) the inbound ESS information has been successfully decoded by the RFSS. The 2V voice data block with fully decoded ESS information is as shown in Figure 53.

0										1										2										3									
0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9	0	1								
Frame Type		U0-0 (b11-0)														U0-1 (b11-0)														U0-2 (b10-7)									
U0-2 (b6-0)						U0-3 (b13-0)																		Et (0)		E1 (0)		M	L	U1-0 (b11-9)									
U1-0 (b8-0)						U1-1 (b11-0)																		U1-2 (b10-0)															
U1-3 (b13-0)														Et (1)		E1 (1)		M	L	TS		EXT		MI (b71-68)															
MI (b67-36)																																							
MI (b35-4)																																							
MI (b3-0)		ALGID (b7-0)								KeyID (b15-0)																		Reserved											

Figure 53 - 2V Half Rate Voice Data with Fully Decoded ESS

The Message Indicator (MI), Algorithm ID (AlgID), and Key ID (KeyID) are as defined in [102BBAC].

4.3.7.2.2 ESS-A Raw Parity Bits Information

In this case (EXT = 1) the inbound ESS information cannot be decoded by the RFSS. The 2V voice data block with ESS-A raw parity bits information is as shown in Figure 54.



Figure 54 - 2V Half Rate Voice Data with ESS-A Raw Payload

is a 168-bit field, as defined in [102BBAC].

TalkSpurt Vocoder Mode (Block Type = 35)

55 shows the TalkSpurt Vocoder Mode block payload.

0	1	2	3	4	5	6	7
TalkSpurt Vocoder Mode				Reserved			

Figure 55 - TalkSpurt Vocoder Mode

ESS-A is a 168-bit field, as defined in [102BBAC].

4.3.8 TalkSpurt Vocoder Mode (Block Type = 35)

Figure 55 shows the TalkSpurt Vocoder Mode block payload.

0	1	2	3	4	5	6	7
TalkSpurt Vocoder Mode				Reserved			

Figure 55 - TalkSpurt Vocoder Mode

This block type is only included in RTP packets for voice services over the ISSI operating in the Native Vocoder Mode. The field value SHALL be set as follows (all numbers are in decimal):

TalkSpurt Vocoder Mode (4 bits): Identifies the vocoder mode for the current talk spurt. Procedures related to the use of the various talkspurt vocoder modes are given in Section 7.

- 0 – Full Rate
1 – Half Rate
2 to 15 – Reserved

4.3.9 Manufacturer specific data (Block Types 63 – 127)

These packet types are provided so individual manufacturers can carry additional proprietary data within a stream. Equipment conforming to this standard MAY include such blocks however they SHALL follow any standard blocks which are sent in the same packet.

Equipment conforming to this standard SHALL ignore blocks of these types whose MFID they do not understand. However equipment conforming to this standard SHALL appropriately process any standard block types which precede a manufacturer specific block type in the same packet.

Figure 56 shows the manufacturer specific format. The length field specifies the number of octets in this block following the length field. Beyond the mandatory manufacturer's identification (MFID) field and the length, the format of the internals of this packet are left open for any use. Manufacturers are encouraged to propose packet types, which MAY be of general use, to the TIA for inclusion in this profile.

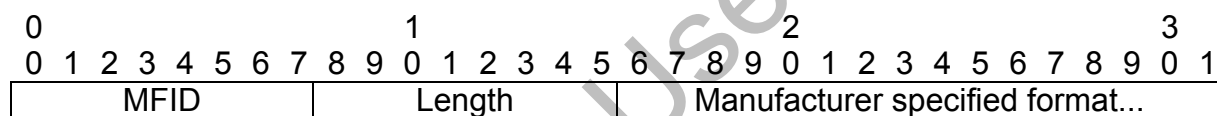


Figure 56 - Manufacturer Specific Format

4.4 Construction of RTP Packets over the ISSI

4.4.1 Basic Construction for All Vocoder Modes

The following basic rules SHALL be applied in the construction of RTP packets transmitted over the ISSI.

1. An ISSI RTP packet SHALL include exactly one ISSI Packet Type block. In the following discussion, the Packet Type from the Packet Type Block is used to identify the packet type (e.g. a PTT Transmit Progress packet has a Packet Type of PTT Transmit Progress (2)).

2. The first ISSI Block in an RTP packet over the ISSI SHALL be an ISSI Packet Type block (Block Type = 1).
3. The second block in PTT Transmit Request, PTT Transmit Start and PTT Transmit Progress packets SHALL be the PTT Control Word Block (Block Type = 11 or 15). Console subsystems SHALL use Block Type 15 when a console is generating the talk spurt. Console subsystems SHALL use Block Type 11 for mobile requested talk spurts (when the console subsystem includes RF sites). All non-console RF subsystems SHALL use Block Type 11.
4. Manufacturer Specific Blocks (Block Types 63 – 127) MAY appear in any RTP packet, and SHALL be the last sequential blocks in the packet.

4.4.2 Specific Construction for Full Rate Vocoder Mode

Figure 57 illustrates the various forms of ISSI RTP packets for Full Rate Vocoder Mode.

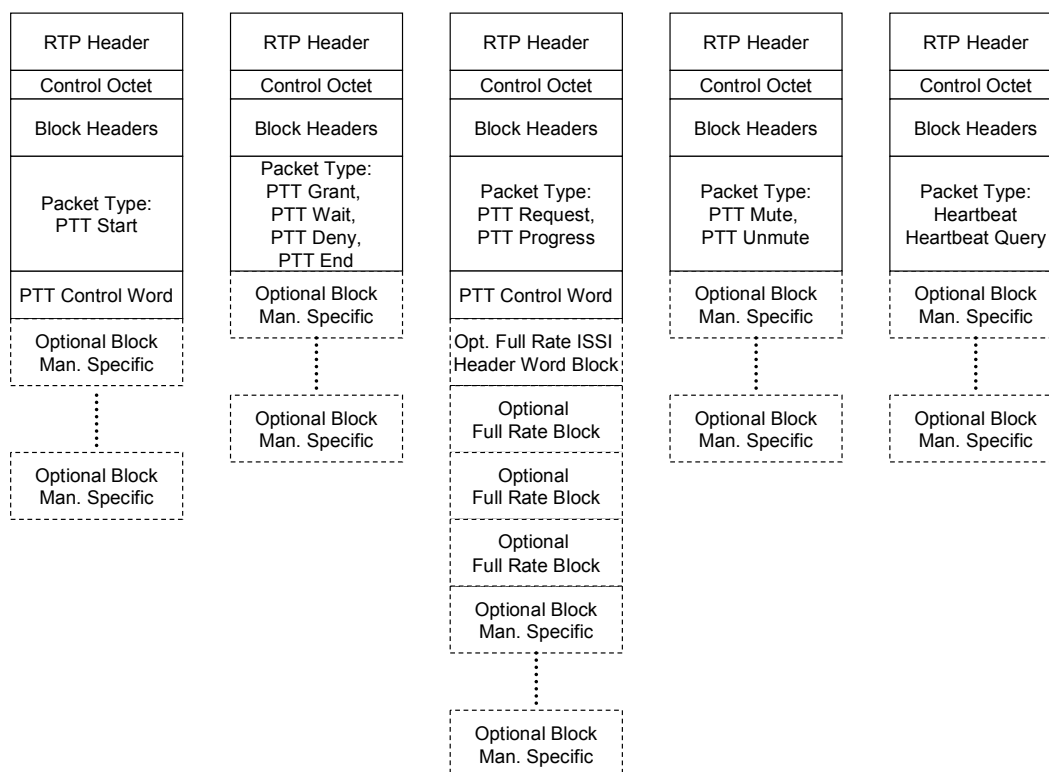


Figure 57 - ISSI RTP Packet Formats for Full Rate Vocoder Mode

In addition to the basic rules specified in Section 4.4.1, the following Full Rate Mode specific rules SHALL be applied in the construction of RTP packets transmitted over the ISSI:

1. Full Rate Voice Blocks (Block Type = 0) and Full Rate ISSI Header Word Blocks (Block Type 5) SHALL only be carried in PTT Transmit Request or PTT Transmit Progress packets.
2. Full Rate Voice Blocks and Full Rate ISSI Header Word Blocks SHALL appear in time order, with the earlier arriving blocks appearing before the later arriving blocks.
3. A single PTT Transmit Request or PTT Transmit Progress packet MAY carry zero to three Full Rate Voice Blocks, representing up to 60 msec. of speech. All of the Full Rate bearing RTP packets in a given talk spurt for a given TSN SHALL contain the same number of Full Rate blocks per RTP packet, with the allowable exception of the last packet in the talk spurt. When multiple Full Rate blocks are bundled into each RTP packet, the RTP packet contents SHALL be aligned with the Full Rate superframe boundaries such that the FR1 (formerly IMBE1) block is always the first Full Rate block within any RTP packet which contains an FR1 (formerly IMBE1) block. I.e., FR18 and FR1 (formerly IMBE18 and IMBE1) SHALL never appear in the same RTP packet.

4.4.3 Specific Construction for Half Rate Vocoder Mode

Figure 58 illustrates the various forms of ISSI RTP packets for Half Rate Vocoder Mode.

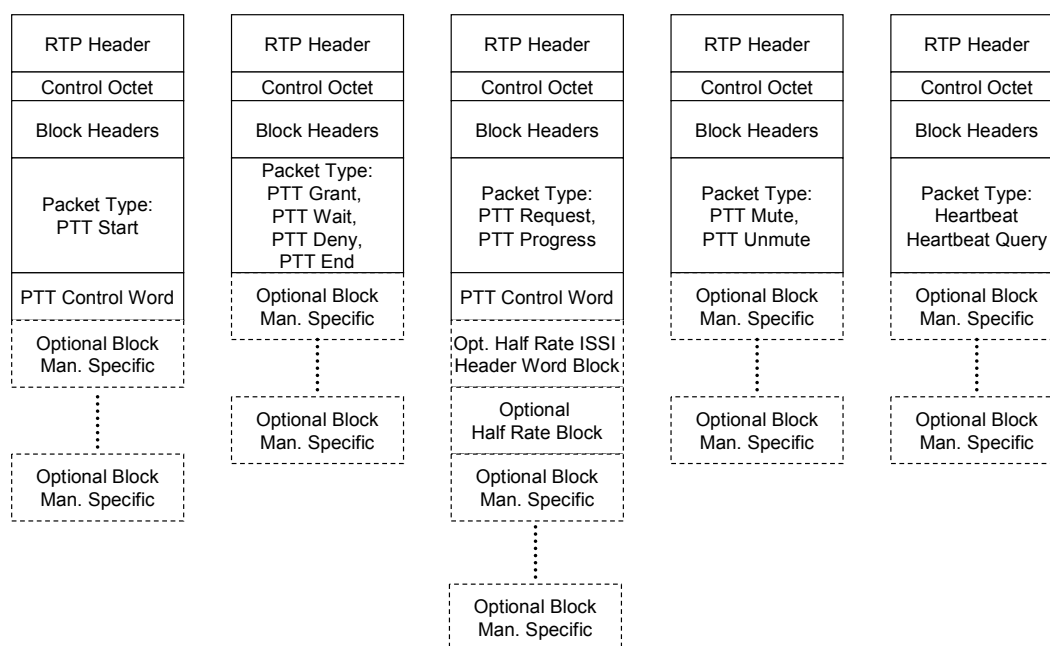


Figure 58 - ISSI RTP Packet Formats for Half Rate Vocoder Mode

In addition to the basic rules specified in Section 4.4.1, the following Half Rate Mode specific rules SHALL be applied in the construction of RTP packets transmitted over the ISSI:

1. Half Rate ISSI Header Word Blocks (Block Type = 33) and Half Rate Voice Blocks (Block Type = 34) SHALL only be carried in PTT Transmit Request or PTT Transmit Progress packets.
2. Half Rate ISSI Header Word Blocks and Half Rate Voice Blocks SHALL appear in time order, with the earlier arriving blocks appearing before the later arriving blocks.
3. A single PTT Transmit Request or PTT Transmit Progress packet MAY carry a 4V Half Rate Voice Block, representing 80 msec. of speech plus ESS information, or a 2V Half Rate Voice Block, representing 40 msec. of speech plus ESS information.

4.4.4 Specific Construction for Native Vocoder Mode

Figure 59 illustrates the various forms of ISSI RTP packets for Native Vocoder Mode.

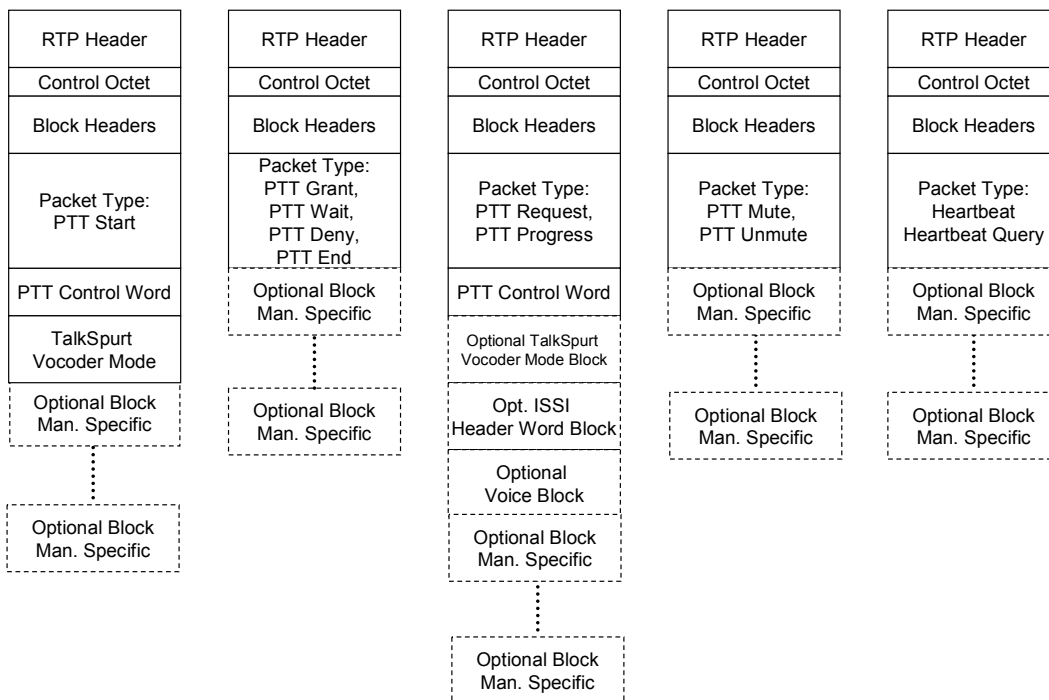


Figure 59 - ISSI RTP Packet Formats for Native Vocoder Mode

In addition to the basic rules specified in Section 4.4.1, the following Native Mode specific rules SHALL be applied in the construction of RTP packets transmitted over the ISSI:

1. The TalkSpurt Vocoder Mode Block (Block Type = 35), which indicates the vocoder mode of any given talk spurt, SHALL be carried in the PTT Transmit Start packets, and any PTT Transmit Request or PTT Transmit Progress packets that do not include voice blocks. Every PTT Transmit Request and PTT Transmit Progress packet SHALL include either one voice block type or a TalkSpurt Vocoder Mode Block.
2. For a Full Rate talk spurt, the Full Rate specific rules as specified in Section 4.4.2 SHALL apply.
3. For a Half Rate talk spurt, the Half Rate specific rules as specified in Section 4.4.3 SHALL apply.

5. Mobility Management

The Mobility Management protocol supports the Unit Mobility Management Function (UMMF) and Group Mobility Management Function (GMMF) described in Section 2.1.

5.1 Definition of Mobility Objects

Mobility Management concerns nine object classes:

1. Subscriber Units are identified by the “TIA-P25-SU” *numplan-param* described in Section 3.4.1.3. A subscriber unit can only be registered on a single Serving RFSS at any particular moment in time. A Serving RFSS that has registered a subscriber unit takes on the role of a Calling or Called Serving RFSS for voice calls and supplementary data transactions involving that unit.
2. Groups, identified by the “TIA-P25-SG” *numplan-param* described in Section 3.4.1.3, can be registered in multiple serving domains at any particular moment in time. A Serving RFSS that has registered a group is allowed to both transmit and receive voice for the given group. Registration of a Group affects voice only, and does not imply a particular role for the Serving RFSS that has registered the group.
3. FNE Unit Supplementary Data, identified by the “TIA-P25-SD-SU” *numplan-param*, can be simultaneously registered by multiple Serving RFSSs. A Serving RFSS that has registered for FNE Unit Supplementary Data with an FNE Home RFSS takes on the role of an FNE Serving RFSS for supplementary data transactions addressed to any General FNE address homed to the FNE Home RFSS and initiated by the given subscriber unit. In addition, a Serving RFSS MAY register for FNE Unit Supplementary Data for a General FNE address, in which case it becomes an FNE Serving RFSS for all transactions addressed to the given General FNE address.
4. Group Supplementary Data, identified by the “TIA-P25-SD-SG” *numplan-param*, can be registered in multiple serving domains at any particular time. A Serving RFSS that has registered for Group Supplementary Data with the Home RFSS of the group takes on the role of a Group Serving RFSS.
5. FNE Group Supplementary Data, identified by the “TIA-P25-SD-SGFNE” *numplan-param*, can be registered in multiple serving domains at any particular time. A Serving RFSS that has registered for FNE Group Supplementary Data with the FNE Home RFSS of a system takes on the role of an FNE Serving RFSS for SD addressed to any General-FNE address homed to that FNE Home RFSS that was originated by a unit affiliated with the given group.

6. Monitoring Interest in Unit Supplementary Data, identified by the “TIA-P25-SDMON-SU” *numplan-param*, can be registered by multiple Serving RFSSs. A Serving RFSS that has registered for Monitoring Interest in Unit Supplementary Data with the Home RFSS of the unit takes on the role of a listening RFSS for SD transactions for which the unit is a participant. In a special case, if the SU identified is the General FNE SUID, the listening RFSS receives all General FNE addressed SD.
7. Monitoring Interest in Group Supplementary Data, identified by the “TIA-P25-SDMON-SG” *numplan-param*, can be registered by multiple Serving RFSSs. A Serving RFSS that has registered for Monitoring Interest in Group Supplementary Data with the Home RFSS of the group takes on the role of a listening RFSS for SD addressed to the Group.
8. Monitoring Interest in FNE Group Supplementary Data, identified by the “TIA-P25-SDMON-SGFNE” *numplan-param*, can be registered by multiple Serving RFSSs. A Serving RFSS that has registered for Monitoring Interest in FNE Group Supplementary Data with an FNE Home RFSS takes on the role of a listening RFSS for SD addressed to any General FNE address homed to that FNE Home RFSS that was originated by a unit affiliated with the given group.
9. PD SN Contexts are identified by the “TIA-P25-PD-SNC” *numplan-param* described in Section 3.4.1.3. An SN Context can only be registered by the Serving RFSS after the associated SU has been successfully unit registered.

Table 40 summarizes the roles taken on by a Serving RFSS that registers each mobility object with the various types of Home RFSSs. For each “Radical” type, the “numplan” column indicates the legal numbering plan parameters, and the “Class” column indicates which mobility object class (from the above 9 classes) is being registered. The “Home RFSS” column indicates the applicable Home RFSSs. The “Serving Role” column indicates the role that the Serving RFSS takes on when he registers the given object with the given Home RFSS.

Table 40 - RFSS Roles

Radical	numplan (TIA-P25-)	Class	Home RFSS	Serving Role	Description
SUID ²²	SU	1	SU	Called	Active participant in unit-to-unit calls and supplementary data to and from the unit.
	SU	1	FNE	Not Allowed	
	SD-SU	3	SU	Not Allowed ²³	
	SD-SU	3	FNE	FNE	Active participant in SD addressed to any General FNE address homed to the RFSS and initiated by the given unit
	SDMON-SU	6	SU	Listening	Monitors All SD to and from the given unit
	SDMON-SU	6	FNE	Not Allowed	
General-FNE ²⁴	SU	1	FNE ²⁵	Not Allowed	
	SD-SU	3	FNE	FNE	Active participant in all SD addressed to the given General FNE address
	SDMON-SU	6	FNE	Listening	Monitors all SD addressed to the given General FNE address
SGID	SG	2	SG	Called	Active participant in group calls for the given group.
	SG	2	FNE	Not Allowed	
	SD-SG	4	SG	Group	Active participant in SD addressed to the given Group
	SD-SG	4	FNE	Not Allowed	
	SD-SGFNE	5	SG	Not Allowed ²³	
	SD-SGFNE	5	FNE	FNE	Active participant in SD addressed to any General FNE address homed to the RFSS and initiated by a unit affiliated with the given group.
	SDMON-SG	7	SG	Listening	Monitors all SD addressed to the given group.
	SDMON-SG	7	FNE	Not Allowed	
	SDMON-SGFNE	8	SG	Not Allowed ²³	
	SDMON-SGFNE	8	FNE	Listening	Monitors all SD addressed to any General FNE address homed to the RFSS and initiated by a unit affiliated with the given group.
SN Context	PD-SNC	9	SU	Serving	Active participant for sending/receiving TIA-102 packet data for the given SN Context.

²² An SUID other than a General FNE SUID.

²³ Unless the RFSS is also home to General FNE, in which case the FNE case applies.

²⁴ A General FNE SUID.

²⁵ Note that the FNE Home and the SU Home are the same in this instance.

5.2 Variations of the SIP REGISTER message

Three variations of the SIP REGISTER are used in the Mobility Management protocol, supporting five procedures. The messages are:

“SIP Register-Register” is used as part of the “Serving Registration” procedure to provide a means for a Serving RFSS to express interest in a mobility object. The Register message is distinguished by the presence of a *contact* header field and a non-zero *expires* header field value. When the mobility object being registered is a group, the SIP REGISTER request and 200 OK response in the Serving Registration procedure can be used by the Serving RFSS and the SG Home RFSS to inform each other of the possible vocoder mode(s) that they could use for the group being registered.

“SIP Register-Query” is sent by a Serving RFSS to query a Home RFSS for service profile information, and by a Home RFSS to confirm the interest of a Serving RFSS in a mobility object.²⁶ It is distinguished by the absence of both the *expires* header field and the *contact* header field. The SIP Register-Query message is used in two procedures described in 5.5, the “Home Query” procedure and the “Serving Query” procedure. The content of the message is consistent in each case, and the particular procedure is determined by the roles of the participating RFSSs pursuant to the mobility object involved.

“SIP Register-Deregister” is used by a Serving RFSS to deregister a mobility object or by a Home RFSS to request deregistration of an SU. The SIP Register-Deregister command is distinguished by the presence of a *contact* header field and a zero *expires* header field value. As with the SIP Register-Query message, the SIP Register-Deregister message is used in two separate procedures, the “Serving Deregistration” procedure and the “Home Deregistration” procedure. Again, the choice of procedure depends on the roles of the RFSSs involved.

For each message type, the content and format of the SIP header fields is as defined in Section 3. Two SIP header fields, *contact* and *expires*, have special meaning to mobility management and are used as defined herein.

The *contact* header field of the SIP REGISTER message SHALL be present in the Register and De-register forms of the request. When present, the “domain-name” of the *contact* header field is the domain-name representation of the Serving RFSS involved in the transaction.

The *contact* field in the 200 OK response to a Register-Query is defined in Section 5.5.

²⁶ Note that RFC 3261 does not describe the usage of a SIP REGISTER message in the way it is being used in this Home Query procedure. I.e., the usage of a SIP REGISTER request to send a registration query about an object from that object's registrar (i.e., the Home RFSS) to a SIP server at one of that object's registered locations (i.e., a Serving RFSS) is not described in RFC 3261. This customized usage has been developed to meet the needs of this ISSI standard.

Table 41 summarizes the values to be used in remaining fields of the five variants of the SIP REGISTER message.

Table 41 - Remaining Field Values in SIP REGISTER

SIP Header Fields	Register-Register	Register-Query		Register-Deregister	
		Serving Query	Home Query	De-registration	Roamed
from/ to	URI, as defined in Section 3, of the object being registered				
contact	Present	None	None	Present	Present
expires	Non-zero	None	None	0	0

Table 42 summarizes the values to be provided in the 200 OK responses to the SIP REGISTER message.

Table 42 - Field Values in 200 OK Response to SIP REGISTER

SIP Header Fields	Response to Register-Register	Register-Query		Register-Deregister	
		Response to Serving Query	Response to Home Query	Response to De-registration	Response to Roamed
from /to	URI, as defined in Section 3, of the object being registered				
contact	Present	See Section 5.5	See Section 5.5	None	None
expires	Non-zero	Non-zero	Non-zero	0	0

5.3 MM-SAP/Message Modeling

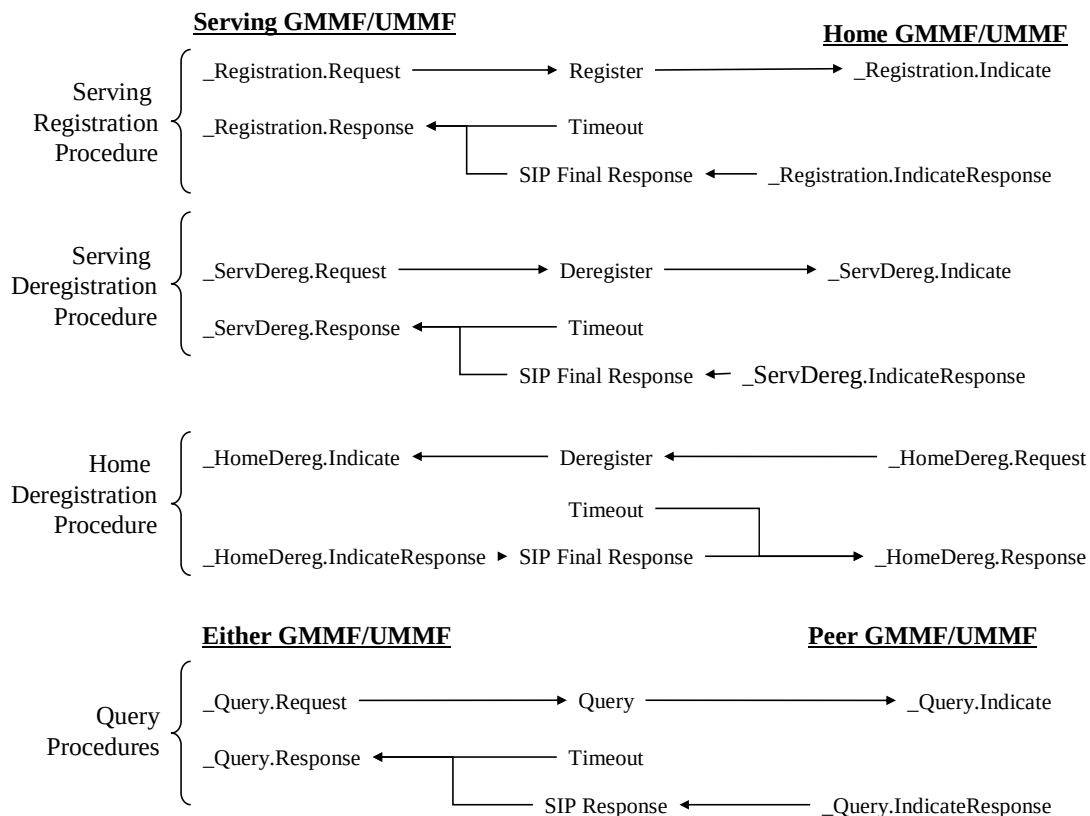


Figure 60 - MM SAP/Message Model

Figure 60 illustrates the mapping from Mobility Management SAP primitives to the messages of the Mobility Management Protocol²⁷. In each case, a Request primitive results in the transmission of a mobility management message, and the subsequent issuing of an Indication to the peer control function, and the IndicateResponse of the control function results in the transmission of a SIP Final Response (200 OK, 4xx, 5xx, 6xx), or a SIP Provisional Response (183 Session Progress), with status passing to the initiating control function.

²⁷ In Figure 60, the leading “MM” has been removed for conciseness.

5.4 State Based Summary

Figure 61 illustrates a state model of a Serving RFSS performing mobility management functions for an object. This model is intended as an aid to describing the protocol and is not to be interpreted as suggesting or requiring any particular implementation. Conditions causing a transition are illustrated as text, and actions that occur during the transition by text following an indicating “/”. Again, the leading “MM” is removed from primitives for conciseness.

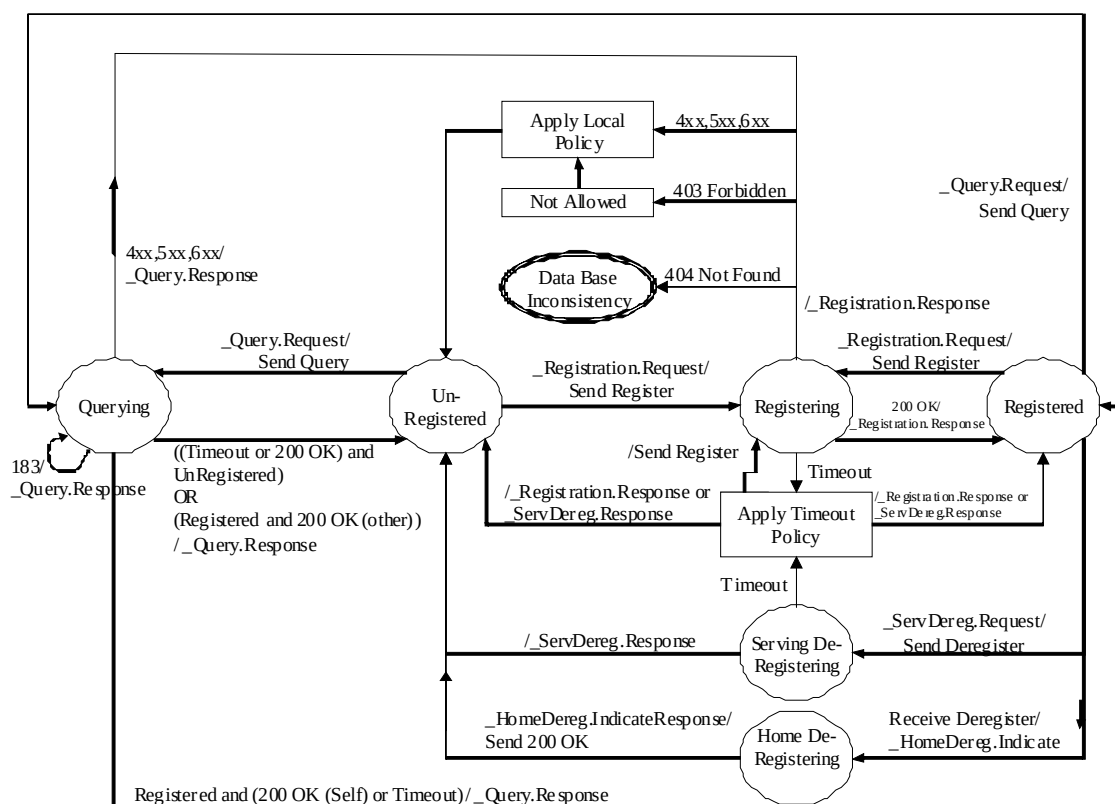


Figure 61 - Serving Mobility Function State Model

Objects begin in the “Unregistered” state. Upon receipt of a registration request from the control function (such as an SU registration request, or a group affiliation request), the Serving RFSS enters the “Registering” state, in which it is attempting to complete the SIP Registration transaction. Alternatively, upon experiencing a query request (such as an SU registration that requires the link layer authentication parameters for that SU), the RFSS enters the Querying state in which it attempts to complete a query transaction.

In both the Querying and the Registering states, SIP errors result in the application of local policies for dealing with the specific errors encountered.

In the Registering state, timeouts result in the application of a local timeout policy that can result in a return to the Un-registered state, a transition to the Registered state²⁸, or a return, after a suitable delay, to the registering state. 200 OK responses received in the Registering state complete successful registration transactions.

From the Registered state, MM_Query.Requests, MM_Registration.Requests (such as the expiration of the registration timer), and MM_Deregistration.Requests (such as an SU de-registration request), cause transitions to the corresponding states.

In the Registered state, receipt of a SIP Deregistration message from the Home RFSS will result in an MM_HomeDereg.Indicate primitive to the Controlling Function and transition to the Home Deregistering State. The Controlling Function will return a MM_HomeDereg.IndicateResponse primitive, which will trigger the SIP response to the Deregistration message and a transition to the Un-Registered state.

In the Serving Deregistering state, the RFSS attempts to complete a de-registration transaction with the Home RFSS. Again, timeouts are handled in a policy specific way. In this case, however, both 4xx, 5xx and 6xx responses and the 200 OK transaction result in a transition to the Un-registered state.

Transitions from the Querying state, are dependent on the previous state, always returning to the Un-Registered state when the previous state is Un-registered (sometimes following application of local policy), but returning to the Registered state only when the query successfully confirms the registration or times-out.

Figure 61 does not illustrate any transitions that occur as a result of receiving a Query from the Home RFSS. All five states SHOULD respond to home queries by propagating an MM_Query.Indication to the control function and, likewise, propagating the MM_Query.IndicationResponse to the initiating RFSS, for subsequent forwarding to the initiating RFSS control function via MM_Query.Response. Queries do not, in and of themselves, cause state transitions in the mobility object.

²⁸ Note that a transition to the “Registered” state can occur, for example, if the object was previously in the “Registered” state and a SIP timeout occurred. The timeout, in this instance, would be indicative of a failed link between the Serving and Home RFSSs and local policy could dictate the maintenance of the current registration.

5.5 Mobility Management Procedures

Five Mobility Management procedures are defined herein:

1. **Serving Registration (SIP REGISTER-Register)** – the Serving Registration procedure is used by a Serving RFSS to register a mobility object (e.g., SU).

The Serving Registration mobility management procedure is applicable to all nine classes of mobility objects. Note that in all cases where multiple entities can express interest through Serving Registration in a single mobility object (Group addressed SD, General FNE addressed SD, and SD Monitoring), each Serving RFSS expresses interest once for all interested entities at that RFSS.

When the Serving Registration procedure is used to register a Class 2 object (Group addressed voice) the Serving RFSS and the SG Home RFSS MAY use the procedure to inform each other of the vocoder mode(s) that they support for the group being registered. Possible supported modes include Full Rate, Half Rate, and Native mode.

2. **Serving Query (SIP REGISTER-Query)** – the Serving Query procedure is used by a Serving RFSS to query a Home RFSS:

- a. For service profile information for an SU (class 1) mobility object, or
- b. For the most current SMDCP context information (known to be agreed between the Serving RFSS and the SU) stored in the Home RFSS for an SN Context (class 9) mobility object.

Serving Query is not applicable to Group, supplementary data and supplementary data monitoring objects (classes 2-8).

3. **Home Query (SIP REGISTER-Query)** – the home query procedure is used by a Home RFSS:

- a. To query a Serving RFSS to determine whether or not it still has interest in an SU (class 1) or Group (class 2) mobility object. Home Query has two mutually exclusive qualifiers, for a total of three functions. The “force” qualifier requests that the Serving RFSS, either deny interest or, if it still has interest, re-register the mobility object via the Serving Registration procedure. In the case where the mobility object is an SU, the force qualifier requires that the SU be forced to re-register and re-affiliate with the Serving RFSS. The force qualifier MAY be used by a Home RFSS to, for example, withdraw permission for a mobility object to operate within a previously granted serving domain. The “confirm” qualifier requests that the Serving RFSS actively confirm its need for the subject

mobility object.

- b. To initiate SNDCP context activation for an SN Context (class 9) mobility object, by sending a Home Query with the “force” qualifier to the Serving RFSS, causing it to send a Context Activation Command to the SU.
- c. To retrieve the most current SNDCP context information stored in the Serving RFSS for an SN Context (class 9) mobility object. Home Query without any qualifier is used in this case.

The Home Query procedure is not applicable to supplementary data and supplementary data monitoring objects (classes 3-8).

4. **Serving Deregistration (SIP REGISTER-Deregister)** – the serving deregistration procedure is initiated by a Serving RFSS to remove interest in a mobility object.

The Serving Deregistration procedure is applicable to all nine classes of mobility objects.

5. **Home Deregistration (SIP REGISTER-Deregister)** – the home deregistration procedure is initiated by a Home RFSS:

- a. To inform a Serving RFSS that an SU (class 1) that was previously operating in its domain has moved to a different domain, or
- b. To deregister any of the nine classes of mobility objects from a Serving RFSS for reasons dependent upon local policy.

The “SU Roaming” qualifier is used to distinguish between these two types of home deregistration.

The deregistration of an SU via the home deregistration procedure MAY require that the Serving RFSS deregister groups that were affiliated with the SU if no other demand for the group exists within that RFSS.

SIP REGISTER messages SHALL be transmitted in accordance with the progressive timeout retransmission protocol for unreliable transport specified in **[RFC3261]** (Sections 17.1.2 and 17.2.2), subject to the following qualifications. The following “general requirements” apply to all five mobility management procedures specified in this section.

- 1. The value of T1 and T2 (the base timers) SHALL be established by administrative agreement between system operators responsible

for the communicating RFSSs. The default value for the TIA-102 ISSI SHALL be as given in Table 58.

2. In the event that a Serving RFSS receives a SIP REGISTER request for an object not present at the Serving RFSS:
 - a. For an SU mobility object, the Serving RFSS SHALL return an MME_SU_UNKNOWN failure response.
 - b. For an SG mobility object, the Serving RFSS SHALL return an MME_SG_UNKNOWN failure response.
 - c. For any other mobility object, the Serving RFSS SHALL return an MME_OBJECT_UNKNOWN failure response.
3. In the event that a Home RFSS receives a SIP REGISTER request for an object not homed to the RFSS:
 - a. For an SU mobility object, the Home RFSS SHALL return an MME_SU_UNKNOWN failure response.
 - b. For an SG mobility object, the Home RFSS SHALL return an MME_SG_UNKNOWN failure response.
 - c. For any other mobility object, the Home RFSS SHALL return an MME_OBJECT_UNKNOWN failure response.
4. In the event that a Home RFSS receives a SIP REGISTER request pertaining to a Mobility Management procedure that it does not support, it SHALL return a GEN_PROC_UNSUPPORTED failure response, as specified in Annex A of this document.
5. In the event that a Serving RFSS receives a SIP REGISTER request pertaining to a Mobility Management procedure that it does not support, it SHALL return a GEN_PROC_UNSUPPORTED failure response, as specified in Annex A of this document.
6. Specific responses found in the following procedures SHALL be mandatory in preference to other responses implied or specified in **[RFC3261]**.
7. An RFSS MAY generate additional SIP responses of types 4xx, 5xx and 6xx, as defined in Annex A of this document, but never in preference to those specified herein.
8. With the exception of 183 Session Progress, no 1xx or 3xx responses SHOULD be generated in response to a SIP REGISTER request. When received, a 1xx response MAY be ignored, and a

3xx response SHOULD terminate the SIP transaction.

9. Contrary to **[RFC3261]** (Section 10.3, item 8), an RFSS responding to a SIP REGISTER request SHALL NOT enumerate all bindings for the given object in 200 OK responses. It SHALL, instead, only return the current binding between the RFSS initiating the request and the mobility object's Home RFSS.
10. Following a timeout, a request SHALL NOT be retried for at least RetryTime seconds.

5.5.1 Serving Registration Procedure

The Serving Registration procedure is invoked by a Serving RFSS when it determines that it has interest in services associated with a mobility object. A typical trigger for a Serving RFSS to register an SU is the registration of the SU at a site within the serving domain of the RFSS. Group affiliation is a typical trigger for the Serving Registration of a group (although it should be noted that only the first affiliation to a particular group need be registered by any particular Serving RFSS). SNDCP context activation is a typical trigger for the Serving Registration of an SN Context. The Serving Registration procedure SHALL proceed as follows:

1. The Serving RFSS SHALL send a SIP REGISTER-Register message.
 - a. When the mobility object being registered is an SU or SN Context, the Serving RFSS MAY include an SU Presence parameter in the SIP REGISTER-Register message:
 - i. If the Serving Registration procedure is being invoked as a result of SU registration at the Serving RFSS, the Serving RFSS SHOULD include the SU Presence parameter with a value of "Confirmed" in the SIP REGISTER-Register message. A Serving RFSS SHALL NOT set the SU Presence parameter value to "Confirmed" if the Serving RFSS has not confirmed the presence of the SU being registered immediately prior to sending the SIP REGISTER-register message.
 - ii. If the Serving Registration procedure is being invoked as a result of an SU's initial SNDCP context activation request at the Serving RFSS, the Serving RFSS SHALL include in the SIP REGISTER-Register message:
 1. The SU Presence parameter with a value of "Confirmed", and

2. The SNC Change parameter with a value of “No Change”, and
 3. A P25-SNDCP body containing pertinent SNDCP context parameters received from the SU, as specified in Section 3.9.
- iii. If the Serving Registration procedure is being invoked because the Serving RFSS wishes to update the Home RFSS on subsequent changes in SNDCP context parameter values for an existing SNDCP context, the Serving RFSS SHALL include in the SIP REGISTER-Register message:
1. the SU Presence parameter with a value of “Confirmed”, and
 2. The SNC Change parameter with a value of “Change”, and
 3. A P25-SNDCP body containing updated SNDCP context parameter values subsequently determined by the Serving RFSS and accepted by the SU, as specified in Section 3.9.
- iv. If the Serving Registration procedure is being invoked because the Serving RFSS wishes to refresh the SIP Register timer for an SN Context registration request, or in accordance with local policy, the Serving RFSS SHALL include the SU Presence parameter, with a value set to “Not Confirmed”, in the SIP REGISTER-Register message. The SIP REGISTER-Register message SHALL NOT include the SNC Change parameter.
- b. When the mobility object being registered is an individual SU, the Serving RFSS MAY include the SU Authentication parameter in the SIP REGISTER-Register message. If the Serving Registration procedure is being invoked as a result of SU registration with successful SU authentication at the Serving RFSS, the Serving RFSS SHALL include the SU Authentication parameter with a value of “SU Authentication Confirmed” in the SIP REGISTER-Register message. A Serving RFSS SHALL NOT set the SU Authentication parameter value to “SU Authentication Confirmed” if the Serving RFSS has not successfully authenticated the SU being registered prior to sending the SIP REGISTER-register message.
- c. When the mobility object being registered is a group (SG), the Serving RFSS MAY include the Supported Vocoder Modes parameter in the SIP REGISTER-Register message. The supported

vocoder modes are listed in order of preference as specified in Section 3.8.4.8.

2. Upon receipt of a SIP REGISTER-register the Home RFSS SHALL behave as follows:
 - a. In the event that the Home RFSS wishes to allow the object to be registered,
 - i. it SHALL respond with a 200 OK message. The 200 OK message SHALL feature an "expires" header field indicating its registration expiration interval set by the Home RFSS, and the body SHALL include service profile information for the requested object.
 1. For SU registration, the Home RFSS SHOULD set the "Expires" header field to a value greater than 14400 (decimal).

NOTE: The value 14400 in seconds represents the SU's minimum (default) WUID validity timer value of 4 hours, per [AABD]. By setting the "Expires" header field to a value greater than 14400, the Home RFSS ensures that SIP registration for the SU over the ISSI expires later than the SU's minimum (default) WUID validity timer expiration. This helps minimize unconfirmed registration renewals from overriding SU-initiated registration in race conditions, while providing protection against excessive system loading.
 2. For other mobility objects, the Home RFSS MAY set the "Expires" header field in accordance with local policy to facilitate timely database recovery in the event of a Home RFSS failure.
 - ii. Furthermore, the Home RFSS SHALL update its location information for the mobility object based on the *contact* header field of the received SIP REGISTER message.
 - iii. Furthermore, in the event that the object is a subscriber unit, and the subscriber was previously registered with a different Serving RFSS:
 1. The Home RFSS SHALL initiate a de-registration request procedure to the previous Serving RFSS, if any.

2. The 200 OK returned to the New Serving RFSS SHALL include a P25-SNDCP body containing the context parameters for all active contexts. If there are no active contexts then the P25-SNDCP body is not present.
- iv. Furthermore, in the event that the object is a group (SG):
 1. If the received SIP REGISTER-Register message includes the Supported Vocoder Modes parameter, the Home RFSS SHOULD update its supported vocoder mode information associated with the Serving RFSS and the SG mobility object.
 2. The 200 OK message returned to the Serving RFSS MAY include the Supported Vocoder Modes parameter. The supported vocoder modes are listed in order of preference as specified in Section 3.8.4.8.
 - v. Furthermore, in the event that the object is an SN Context, and the received SIP REGISTER-Register message includes the SNC Change parameter with a value of "No Change", the Home RFSS SHALL verify the received parameters and determine the values of parameters, if any, to be returned to the Serving RFSS. If the received parameters are acceptable, the Home RFSS SHALL return a 200 OK message to the Serving RFSS. The 200 OK message SHALL include a P25-SNDCP body that contains a Home Packet Data ISSI Gateway (PDIG) Address and other pertinent SNDCP parameters as specified in Section 3.9.
 - vi. Furthermore, in the event that the object is an SN Context, and the received SIP REGISTER-Register message includes the SNC Change parameter with a value of "Change", the Home RFSS SHALL verify the received parameters. If the received parameters are acceptable, the Home RFSS SHALL update the stored parameter values for that object accordingly and return a 200 OK message to the Serving RFSS.
 - vii. Following the Serving Registration of the object, INVITE messages for voice services SHALL be accepted from and propagated to the Serving RFSS in accordance with the call setup procedures.
- b. In the event that the Home RFSS wishes to disallow registration of the object at the Serving RFSS:

- i. For an SU mobility object:
 - 1. The Home RFSS SHALL return an MME_SU_AUTHEN_REQUIRED failure response in the event of the following:
 - a. The Home RFSS requires authentication of the SU, and
 - b. The SU Authentication parameter in the SIP REGISTER-register message from the Serving RFSS is set to "SU Authentication NOT performed" or the SU Authentication parameter is absent in the SIP REGISTER-register message from the Serving RFSS.
 - 2. The Home RFSS SHALL return an MME_MUST_CONFIRM_SU_PRESENCE failure response in the event of the following:
 - a. The SIP REGISTER-register message from the Serving RFSS does not contain the SU Presence parameter or that it contains the SU Presence parameter with a value of "Not Confirmed", and
 - b. The Home RFSS has detected that the SU is currently registered at a different Serving RFSS.
 - 3. In all other cases where the Home RFSS wishes to disallow Serving Registration of the SU, the Home RFSS SHALL return an MME_SU_REG_DISALLOWED failure response.
- ii. For an SG mobility object, the Home RFSS SHALL return an MME_SG_REG_DISALLOWED failure response.
- iii. For an SN Context mobility object:
 - 1. In the event that the Home RFSS detects that the SU is not authorized to perform packet data services, it SHALL return a PDE_SU_NOT_AUTHOR failure response.

2. In the event that the Home RFSS detects that the static IP address provided by the SU is incorrect, it SHALL return a PDE_STAT_IP_ADDR_INCORRECT failure response.
 3. In the event that the Home RFSS detects that static IP address assignment is not supported, it SHALL return a PDE_STAT_IP_ADDR_UNSUPPORTED failure response.
 4. In the event that the Home RFSS detects that there are no more available IP addresses for a dynamic IP address assignment, it SHALL return a PDE_DYN_IP_ADDR_POOL_EMPTY failure response.
 5. In the event that the Home RFSS detects that dynamic IP address assignment is not supported, it SHALL return a PDE_DYN_IP_ADDR_UNSUPPORTED failure response.
 6. In all other cases where the Home RFSS wishes to disallow Serving Registration of the SN Context mobility object, the Home RFSS SHALL return an MME_OBJECT_REG_DISALLOWED failure response.
- iv. For any other mobility object, the Home RFSS SHALL return an MME_OBJECT_REG_DISALLOWED failure response.
3. Upon receipt of any response to the SIP REGISTER message, the Serving RFSS SHALL behave as follows:
 - a. In the event that the response is a 200 OK message, the object has been successfully registered. The Serving RFSS MAY schedule periodic re-registrations of the object to occur before the time specified in the *expires* header field of the 200 OK response has elapsed. If the 200 OK message is the response to a Serving Registration request for an SG mobility object, and if the 200 OK contains an ISSI body with a Group Service Profile which includes an Announcement Group attribute, then the Serving RFSS SHALL initiate a Serving Registration request for the announcement subscriber group indicated in the profile unless the Serving RFSS already has registered interest in the announcement subscriber

group. If the 200 OK message is the response to a Serving Registration request for an SG mobility object, and if the 200 OK contains the Supported Vocoder Modes parameter, then the Serving RFSS SHOULD update its supported vocoder mode information associated with the Home RFSS and the SG mobility object.

- b. In the event that the response is an MME_SU_UNKNOWN or MME_SG_UNKNOWN or MME_OBJECT_UNKNOWN failure response (as specified in the “general requirements” above), the Home RFSS is asserting that it is not the home for the given object. There is disagreement among system data bases that needs to be resolved. The Serving RFSS SHOULD discontinue any attempt to register the object with the Home RFSS.
- c. In the event that the response is one of the following failure messages:
 - i. MME_MUST_CONFIRM_SU_PRESENCE
 - ii. MME_SU_REG_DISALLOWED
 - iii. MME_SG_REG_DISALLOWED
 - iv. MME_OBJECT_REG_DISALLOWED
 - v. PDE_SU_NOT_AUTHOR
 - vi. PDE_STAT_IP_ADDR_INCORRECT
 - vii. PDE_STAT_IP_ADDR_UNSUPPORTED
 - viii. PDE_DYN_IP_ADDR_POOL_EMPTY
 - ix. PDE_DYN_IP_ADDR_UNSUPPORTED

the Home RFSS has denied registration of the object at the Serving RFSS. The Serving RFSS SHALL discontinue any attempt to register the object until it receives a subsequent registration trigger. If appropriate, the Serving RFSS MAY notify the source of the registration trigger (e.g. an SU attempting to register or affiliate) of the error condition. Furthermore, the Serving RFSS SHOULD initiate de-registration procedures for other objects (such as groups) that were registered as a result of interest from the forbidden object. (For example, it would deregister any groups for which the SU was the only affiliated member.)

- d. In the event that the response is an MME_SU_AUTHEN_REQUIRED failure response, the Home RFSS has denied registration of the SU at the Serving RFSS due to failure of the Serving RFSS to authenticate the SU. The Serving RFSS SHALL act in accordance with local policy to authenticate the SU or deny the SU registration.

- e. Other 4xx, 5xx and 6xx messages SHALL be interpreted in accordance with the SIP RFC and local policy.
4. If the SIP Serving Registration procedure terminates due to a SIP timeout (Section 17.1.2 of [RFC3261]), the Home RFSS is unreachable. The Serving RFSS SHALL take appropriate error recovery steps in accordance with its security and other policies, and MAY retry the request at a later time.

5.5.2 Serving Query

A Serving RFSS MAY query the Home RFSS of any applicable mobility object (as specified in Section 5.5) via the Serving Registration Query procedure at any time before or after Serving Registration of the mobility object with the Home RFSS.

A Serving RFSS that supports authentication and requires prior knowledge of the Service Profile information of a SU mobility object, SHALL query the Home RFSS using the Serving Registration Query procedure prior to authentication of the SU mobility object and subsequent Serving Registration of the SU mobility object with the Home RFSS.

A Serving RFSS that requires knowledge of the most current SMDCP context information stored in the Home RFSS for an SN Context (class 9) mobility object MAY query the Home RFSS of the object via the Serving Registration Query procedure. A typical trigger for a Serving Registration Query might be the need to re-synchronize databases in accordance with local policy.

The procedure SHALL proceed as follows:

1. The Serving RFSS SHALL send a SIP REGISTER-Query message to the Home RFSS of the mobility object.
2. A Home RFSS that receives an error free SIP REGISTER-Query message SHALL behave as follows:
 - a. In the event that the Serving RFSS is not authorized to receive information about the subject SU, return an MME_SU_INFO_NOT_AUTHOR failure response.
 - b. Otherwise:
 - i. For a Serving Query pertaining to an SU (class 1) mobility object, respond with a 200 OK message containing the requested service profile information. The 200 OK message SHALL also include a P25-SMDCP body containing pertinent

SNDCP context parameters (as specified in Section 3.9) for all currently registered SN Contexts.

- ii. For a Serving Query pertaining to an SN Context (class 9) mobility object, respond with a 200 OK message that includes a P25-SNDCP body containing the most current SNDCP context information (as specified in Section 3.9) stored in the Home RFSS for the object.

Moreover, the 200 OK responses SHALL contain the *contact* header field if the mobility object is currently registered with the querying Serving RFSS. If the mobility object is not registered with the querying Serving RFSS, the Home RFSS SHALL NOT include the *contact* header field.

3. In the event that the Serving RFSS fails to receive a response to the SIP REGISTER-Query message (i.e. experiences a SIP timeout in accordance with [RFC3261], Section 17.1.2), the Home RFSS is unreachable. The Serving RFSS MAY retry the request at a later time.
4. In the event that the Serving RFSS receives a an MME_SU_INFO_NOT_AUTHOR or a GEN_PROC_UNSUPPORTED failure response to the SIP REGISTER-Query message, it SHALL discontinue any attempt to register the object unless it receives a subsequent registration trigger, and, if appropriate, MAY notify the source of the registration trigger (e.g. an SU attempting to register or affiliate) of the error condition.
5. In the event that the Serving RFSS receives a an MME_SU_UNKNOWN or MME_OBJECT_UNKNOWN failure response (as specified in the “general requirements” above) to the SIP REGISTER-Query message, the Home RFSS is asserting that it is not the home for the given object. There is a disagreement among system data bases that needs to be resolved. The Serving RFSS SHOULD discontinue any attempt to register the object with the Home RFSS.
6. In the event that the Serving RFSS receives some other 4xx, 5xx or 6xx error response to the SIP REGISTER-Query message, it SHALL behave in accordance with local policy.

5.5.3 Home Query

A Home RFSS that wishes to confirm the registration of any applicable mobility object (as specified in Section 5.5) at a particular Serving RFSS MAY query the Serving RFSS via the Home Registration Query procedure. The Home Registration Query SHALL proceed as follows:

1. The Home RFSS SHALL send a SIP REGISTER-query message to the Serving RFSS. In the event that the Home Query has the “force” qualifier set and pertains to an SN Context (class 9) mobility object, the REGISTER-query message SHALL include a P25-SNDCP body containing pertinent SNDCP context parameters as specified in Section 3.9.
2. Upon the Serving RFSS receiving the SIP-REGISTER-query message,
 - a. If the Serving RFSS has no interest in the given object, such as the SU is no longer registered in this RFSS:
 - i. For an SU mobility object, the Serving RFSS SHALL return an MME_SU_UNKNOWN failure response.
 - ii. For an SG mobility object, the Serving RFSS SHALL return an MME_SG_UNKNOWN failure response.
 - iii. For any other mobility object, the Serving RFSS SHALL return an MME_OBJECT_UNKNOWN failure response.
 - b. Subsequent to determining that the request is satisfactory, and in the absence of the conditions described above, the Serving RFSS,
 - i. In the event that the “force” qualifier is set, SHALL respond with 200 OK with the *contact* header field reflecting its own domain, and, subsequent to the completion of the query procedure, SHALL:
 1. if the object is an SU, force registration, re-affiliation, and authentication if applicable, of the SU on the air link (resulting in an ISSI registration trigger). If the SU does not respond to the registration within a time that is dependent on the SU and airlink involved, the Serving RFSS SHALL deregister the SU with the Home RFSS.
 2. If the object is an SN Context, send a Context Activation Command to the SU on the air link, therefore triggering a Context Activation Request from the SU.

3. For any other object, initiate an ISSI registration (i.e., Serving Registration) procedure for the object.
- ii. in the event that the confirm qualifier is set, SHALL take such actions as MAY be required to actively determine whether or not the object is currently required by the RFSS, employing the MME_PROCESSING_HOME_QUERY 183 provisional response to notify the Home RFSS that it is attempting to fulfill the request, and SHALL subsequently reply with the 200 OK response with a *contact* header field reflecting its own domain, or, if the object is no longer required.
 1. For an SU mobility object, the Serving RFSS SHALL return an MME_SU_UNKNOWN failure response, and SHALL actively query for the presence of the SU within the domain of the Serving RFSS.
 2. For an SG mobility object, the Serving RFSS SHALL return an MME_SG_UNKNOWN failure response, and MAY actively confirm the affiliation status of at least one SU associated with the group.
 3. For any other mobility object, the Serving RFSS SHALL return an MME_OBJECT_UNKNOWN failure response.
 - iii. In the event that neither the confirm qualifier nor the force qualifier is set,:
 1. For a Home Query pertaining to an SN Context (class 9) mobility object, the Serving RFSS SHALL respond with a 200 OK message that includes a P25-SNDCP body containing the most current SNDCP context information stored in the Serving RFSS for the object.
 2. For a Home Query pertaining to any other mobility object, the Serving RFSS MAY take the steps to confirm the registration as defined above, or MAY confirm the registration through some other mechanism consistent with local policy. It SHALL subsequently reply with the 200 OK response with a *contact* header field reflecting its own domain, or, if the object is not required:
 - a. For an SU mobility object, the Serving RFSS SHALL return an MME_SU_UNKNOWN failure response.

- b. For an SG mobility object, the Serving RFSS SHALL return an MME_SG_UNKNOWN failure response.
 - c. For any other mobility object, the Serving RFSS SHALL return an MME_OBJECT_UNKNOWN failure response
- iv. In any event:
 - 1. For an SU mobility object, if the Serving RFSS determines that the unit is not present within its domain, it SHALL respond with an MME_SU_UNKNOWN failure response, and behave as if the unit has been deregistered.
 - 2. For an SG mobility object, if the Serving RFSS determines that it has no interest in the indicated group, it SHALL respond with an MME_SG_UNKNOWN failure response, and behave as if the group has been deregistered.
 - 3. For any other mobility object, if the Serving RFSS determines that it has no interest in the indicated object, it SHALL return an MME_OBJECT_UNKNOWN failure response, and behave as if the object has been deregistered.

5.5.4 Serving Deregistration

A Serving RFSS that no longer has interest in a mobility object SHALL notify the Home RFSS of that fact via the Serving Deregistration procedure. Typically, a Serving RFSS SHOULD, for example, de-register a group when no more SUs are affiliated with the group. Prior to initiating the Serving Deregistration procedure, the Serving RFSS SHOULD terminate all active TIA-102 services (e.g., tear-down any active call legs that exist between itself and the Home RFSS and that also involve the subject mobility object). The Serving Deregistration procedure SHALL proceed as follows:

1. The Serving RFSS SHALL send a SIP REGISTER-deregister message to the Home RFSS of the mobility object.
2. Upon receipt of a SIP-Register-deregister message, the Home RFSS SHALL behave as follows:

- a. In the event that a Home RFSS receives a SIP REGISTER-deregister message for a mobility object not believed to be active within the indicated domain, it SHALL return an MME_OBJ_NOT_REG_AT_SERV failure response.

NOTE: In older RFSS implementations, a 405 METHOD NOT ALLOWED failure response is returned instead.

- b. Otherwise, the Home RFSS MAY respond with a 4xx, 5xx or 6xx message if appropriate, but
 - c. SHALL respond with 200 OK, if the mobility object is currently thought to be active within the serving domain.
 - d. Furthermore, the Home RFSS SHALL delete location information pertaining to the referenced mobility object and Serving RFSS pair.
3. Upon receipt of a 200 OK, 4xx, 5xx, or 6xx response to the SIP-Register-deregister message, the Serving RFSS SHALL consider the transaction complete.
 4. In the event that the Register-Deregister request experiences a SIP timeout per Section 17.1.2 of **[RFC3261]**, the Serving RFSS SHALL apply a local Timeout policy that MAY include attempting the de-registration at a later time, or MAY behave as if the 200 OK response was received.
 5. In any event, the Home RFSS SHALL discontinue routing of associated traffic or cancel services for the mobility object to the requesting RFSS, and SHALL immediately initiate call-leg teardown procedures as described in Section 6.6.5 for any calls for those objects that are no longer valid.

5.5.5 Home Deregistration

A Home RFSS MAY request a Serving RFSS to deregister a mobility object by executing the Home Deregistration procedure. Prior to initiating the procedure, the Home RFSS SHOULD terminate all active TIA-102 services (e.g., tear down any active call legs (using the procedures described in Sections 6.5.3 and 6.5.4) that exist between the Home RFSS and the Serving RFSS and exclusively involve the de-registering mobility object). The Home Deregistration procedure SHALL proceed as follows:

1. The Home RFSS SHALL send a SIP REGISTER-deregister²⁹ message to the Serving RFSS.
2. In the event that the Home RFSS initiates the Home Deregistration procedure upon detecting that an SU has roamed from the Serving RFSS, the REGISTER-deregister message MAY include the "SU Roaming" qualifier, with value set to "Roamed". Upon receiving this SIP REGISTER-deregister message, the Serving RFSS :
 - a. MAY respond with a 4xx, 5xx or 6xx error message except where contraindicated above (i.e. it SHALL NOT respond with an error message simply because it does not believe that the referenced unit is not currently within its serving domain).
 - b. Otherwise, it SHALL reply with 200 OK regardless of whether or not it believes that the indicated SU is still within its domain. It SHALL thereafter act as if the unit was deregistered via the serving deregistration procedure. It MAY subsequently check for the SU's presence and invoke the ISSI registration (i.e., Serving Registration) procedure if it finds that the SU is present.
3. In the event that the Home RFSS initiates the Home Deregistration procedure not because an SU has roamed from the Serving RFSS, but in accordance with local policy, the REGISTER-deregister message SHALL include the "SU Roaming" qualifier, with value set to "Non-Roamed". Upon receiving the SIP REGISTER-deregister message, the Serving RFSS SHALL reply with 200 OK. The Serving RFSS also informs the SU accordingly. It SHALL thereafter act as if the unit was deregistered via the serving deregistration procedure.
4. Upon receipt of a De-registration request for a mobility object with an established session, the Serving RFSS SHALL terminate the session (e.g., initiate call-leg teardown procedures as described in Sections 6.5.3

²⁹ Note that RFC 3261 does not describe the usage of a SIP REGISTER message in the way it is being used in this Home Deregistration procedure. I.e., the usage of a SIP REGISTER message to request de-registration of an object from that object's registrar (i.e., the Home RFSS) to a SIP server at one of that object's registered locations (i.e., a Serving RFSS) is not described in RFC 3261. This customized usage has been developed to meet the needs of this ISSI standard.

and 6.5.4). It MAY also delete all mobility management information pertaining to the mobility object.

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6. Call Control

As stated previously, RF Resources, for the purposes of a console, is to be construed throughout the document as referring to analogous resources (e.g. talk-paths or link bandwidth) of a console. No change has been made to this document relating to the availability of RF Resources. When consoles are considered, RF resources refer to the console's similar, limited availability resources.

The ISSI Call Control protocol is applicable to both Group Call and SU-to-SU Call services. Differences in these call services are implemented in the Controlling Function entity described in the Architecture section (Section 2.1). The Call Control Protocol Element has a Service Access Point to the Controlling Function, and utilizes an underlying SIP protocol as described in **[RFC3261]** to set up and tear down ISSI calls between RFSSs. Figure 62 shows the protocol layering and its SAPs.

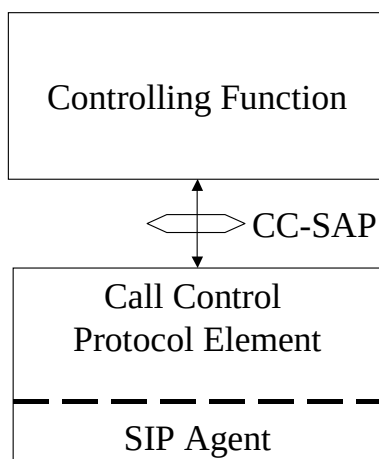


Figure 62 - ISSI Call Control Protocol

The following sections describe the Call Control procedures in detail:

- Section 6.1 describes call control primitives as building blocks for the state models
- Section 6.2 describes SIP transactions used by the ISSI
- Section 6.3 describes state models for the protocol
- Section 6.4 describes basic SIP functionality utilized by the protocol
- Section 6.5 describes procedures for SU-to-SU call processing
- Section 6.6 describes procedures for Group Call processing.

6.1 CC-SAP

The Call Control Service Access Point (CC-SAP) describes primitives that the Call Control Function can use to drive the protocol. These primitives are described in detail in the Architecture section (see Section 2.1).

Figure 63 illustrates the mapping from Call Control SAP primitives to the messages of the Call Control Protocol.

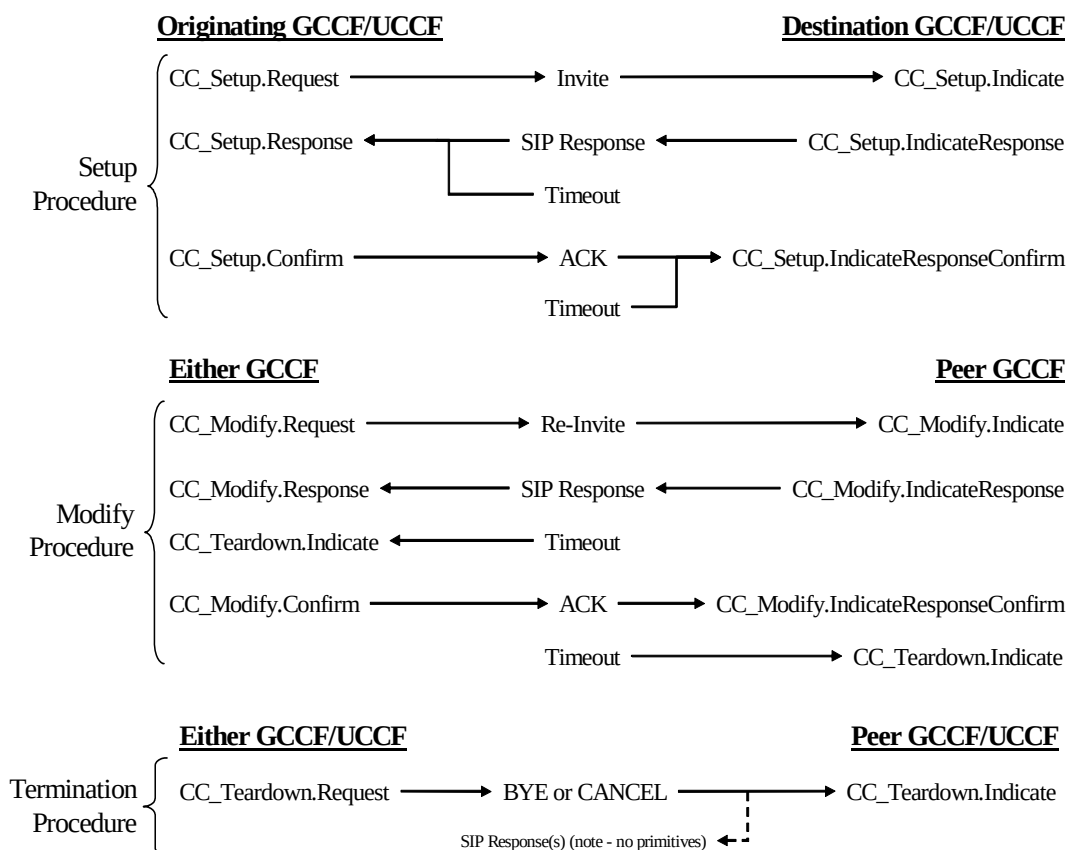


Figure 63 - CC-SAP/Message Model

Note that in the Termination Procedure, the SIP Agent determines whether a BYE or a CANCEL is necessary, depending on the state of the SIP dialog, as explained in Section 6.4.2. The SIP Agent also handles the timeout or the SIP Final Response in the termination case; no **CC_Teardown.IndicateResponse** primitive is necessary between the Controlling Function and the Call Control Protocol Element to trigger the SIP response, and no **CC_Teardown.Response** is necessary when the final response is received. When necessary, the SIP Agent also handles waiting to send the BYE or CANCEL message until a necessary provisional or final SIP response or SIP ACK is received,

Each primitive, depending on the context of its use, has a set of parameters in order to be able to handle the appropriate SIP transaction. These parameters are described in Table 43. Not every parameter in the SIP Messages is listed in the table; the table describes the parameters that have a direct effect on the call control state models and procedures. Other parameters, like ID and System information, are also necessary but are not explicitly described. See the SIP Messages and Parameters Section (Section 3) for a complete list of SIP message parameters.

Some of the primitives have “Status” parameters that indicate “Success” or “Failure”. Note that “Success” in these cases denotes a successful transaction, not necessarily a successful call set-up.

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Table 43 - Parameters of the CC-SAP primitives

	SU-to-SU Call	Group Call
CC_Setup.Request	Preference (AC/DC) Duplex Mode (Half/Full) Priority Protected (Y/N) InCallRoaming (Y/N) TX(Y/N) Offered VM (Full Rate/Half Rate/Native)	RF State (RF OK or RF KO) Call Type (Confirmed or Unconfirmed) Priority Protected (Y/N) Offered VM (Full Rate/Half Rate/Native)
CC_Setup.Indicate	Preference (AC/DC) Duplex Mode (Half/Full) Priority Protected (Y/N) In-Call Roaming (Y/N) TX(Y/N) Offered VM (Full Rate/Half Rate/Native)	RF State (RF OK or RF KO) Call Type (Confirmed or Unconfirmed) Priority Protected (Y/N) Offered VM (Full Rate/Half Rate/Native)
CC_Setup.IndicateResponse	Status_IR(Ringing/Progress/Success/Failure) Duplex Mode (Half/Full) Protected (Y/N) TX(Y/N) Answered VM (Full Rate/Half Rate/Native)	Status_IR(Success/Failure) Protected (Y/N) RTP State (RTP OK or RTP KO) RF State (RF OK or RF KO) Answered VM (Full Rate/Half Rate/Native)
CC_Setup. Response	Status_R(Ringing/Progress/Success/Failure) Duplex Mode (Half/Full) Protected (Y/N) TX(Y/N) Answered VM (Full Rate/Half Rate/Native)	Status_R(Success/Failure) Protected (Y/N) RTP State(RTP OK or RTP KO) RF State (RF OK or RF KO) Answered VM (Full Rate/Half Rate/Native)
CC_Setup. Confirm	None	None
CC_Setup.IndicateResponseConfirm	Status_IRC(Success/Failure)	Status_IRC(Success/Failure)
CC_Teardown.Request	None	None
CC_Teardown.Indicate	None	None
CC_Modify.Request	Not Applicable	RF State (RF OK or RF KO) RTP State (RTP OK or RTP KO) Priority Offered VM (Full Rate/Half Rate/Native)
CC_Modify.Indicate	Not Applicable	RF State (RF OK or RF KO) RTP State (RTP OK or RTP KO) Priority Offered VM (Full Rate/Half Rate/Native)
CC_Modify.IndicateResponse	Not Applicable	Status_IR(Success/Failure) RTP State (RTP OK or RTP KO) Answered VM (Full Rate/Half Rate/Native)
CC_Modify.Response	Not Applicable	Status_R(Success/Failure) RTP State (RTP OK or RTP KO) Answered VM (Full Rate/Half Rate/Native)

	SU-to-SU Call	Group Call
CC_Modify.Confirm	Not Applicable	None
CC_Modify.IndicateResponseConfirm	Not Applicable	None

6.1.1 Answered VM

The Answered VM parameter is related to the SDP media description (See Section 3.7.2.2) conveyed in the 200 OK. For group calls and SU-to-SU calls this parameter indicates which vocoder mode (Half Rate, Full Rate, or Native) the respondent has chosen to be allowed for the talk spurts on the call segment to the other RFSS. In the case where the Offered mode indicates that the offering RFSS could support a combination of modes, the answering RFSS responds with a single mode.

This parameter is only present when RTP resources are available and is significant in the following cases:

- For a CC_Setup.IndicateResponse(success, RTP OK) exerted by the Controlling Function at a destination RFSS.
- For a CC_Setup.Response(success, RTP OK) received at the Controlling Function of the originating RFSS
- For a CC_Modify.IndicateResponse(success, RTP OK) exerted by the Controlling Function at a destination RFSS (group call only)
- For a CC_Modify.Response(success, RTP OK) received at the Controlling Function of the originating RFSS (group call only)

6.1.2 Call Type

The Call Type (Confirmed or Unconfirmed) parameter is related to the Call Set-Up Group Call Type Attribute (see Section 3.8.5.6). It indicates if the group call is confirmed or unconfirmed.

6.1.3 Duplex Mode

The Duplex Mode (Half or Full) parameter is related to the Call Set-Up Duplex Mode Attribute (see Section 3.8.5.4). It indicates if the call is full- or half-duplex.

6.1.4 In-Call Roaming

The In-Call Roaming (Y/N) parameter is related to the In Call Roaming attribute (see Section 3.8.5.5). This parameter indicates if the SIP INVITE for SU-to-SU call setup is due to in-call roaming.

6.1.5 Offered VM

The Offered VM parameter is related to the SDP media description (See Section 3.7.2.2) conveyed in the INVITE. For group calls and SU-to-SU calls, this parameter indicates which vocoder mode (Full Rate, Half Rate, or Native) the initiator can support for talk spurts on the call segment to the other RFSS. When the offering RFSS offers a combination of modes it expects the answering RFSS to pick just one of them in its answer.

This parameter is only present when RTP resources are available and is significant in the following cases:

- For a CC_Setup.Request exerted by the Controlling Function at an originating RFSS
- For a CC_Setup.Indicate received at the Controlling Function of the destination RFSS
- For a CC_Modify.Request(RTP OK) exerted by the Controlling Function at an originating RFSS (group call only)
- For a CC_Modify.Indicate(RTP OK) received at the Controlling Function of the destination RFSS (group call only)

6.1.6 Preference

The Preference (AC/DC) parameter is related to the Call Set-Up Preference Attribute (see Section 3.8.5.3). This parameter indicates if an Availability Check is to be done (AC) or not (DC – Direct Call).

6.1.7 Priority

The Priority parameter is related to the SIP Priority header field priority. This parameter indicates the priority of the call. The possible values for this parameter are given in the SIP Messages and Procedures, Section 3.5.13.

6.1.8 RF State

The RF State parameter is related to the ISSI parameter conveyed in the INVITE, Re-INVITE, 200 OK, and 183 Session Progress messages (See RF Resource availability parameter in Section 3.8.5.2).

This parameter is only present for confirmed group calls and is significant in the following cases:

- For a CC_Setup.Request exerted by the Controlling Function at a Serving RFSS
- For a CC_Setup.Indicate received at the Controlling Function of the Home RFSS
- For a CC_Setup.IndicateResponse(success, RTP OK) exerted by the Controlling Function at a Serving RFSS

- For a CC_Setup.Response(success, RTP OK) received at the Controlling Function of the Home RFSS
- For a CC_Modify.Request(RTP OK) exerted by the Controlling Function at a Serving RFSS
- For a CC_Modify.Indicate(RTP OK) received at the Controlling Function of the Home RFSS
- For a CC_Modify.IndicateResponse(success, RTP OK) exerted by the Controlling Function at a Serving RFSS
- For a CC_Modify.Response(success, RTP OK) received at the Controlling Function of the Home RFSS

This parameter indicates the availability of RF resources at the Serving RFSS.

6.1.9 RTP State

The RTP State parameter is related to the type of the SDP Offer and SDP answer conveyed respectively in the SIP INVITE, SIP Re-INVITE and SIP 200 OK Messages (See SDP Announcement in Section 3.7). This parameter indicates the availability of RTP resources at the RFSS.

6.1.10 Status_IR

The Status_IR parameter triggers the Call Control Protocol Element to send:

- a SIP 180 RINGING (Status_IR="Ringing"),
- a SIP 183 Session Progress (Status_IR = "Progress"),
- a SIP 200 OK (Status_IR = "Success"), or
- a SIP 4xx, 5xx or 6xx error (Status_IR = "Failure").

6.1.11 Status_IRC

The Status_IRC parameter allows the Call Control Protocol Element to report to the Controlling Function when:

- a SIP ACK has been received (Status_IRC = "Success") or
- a SIP Timeout has occurred (Status_IRC = "Failure").

6.1.12 Status_R

The Status_R parameter allows the Call Control Protocol Element to report to the Controlling Function when:

- a SIP 180 RINGING has been received (Status_R="Ringing"),
- a SIP 183 SESSION PROGRESS has been received (Status_R = "Progress"),
- a SIP 200 OK has been received (Status_R = "Success") or
- a SIP Timeout has occurred or a 4xx, 5xx or 6xx error has been received (Status_R = "Failure") for a CC_Setup.Response or

- a 4xx,5xx, or 6xx other than 481 or 408 has been received (Status_R = "Failure") for CC_Modify.Response.

6.1.13 TX

The TX (Y/N) parameter is related to the Call Set-Up Transmitter Attribute (see Section 3.8.5.1). It indicates if the RFSS has the initial transmitter for an SU-to-SU call.

6.1.14 Protected

The Protected (Y/N) parameter is related to the Call Set-Up Protected Attribute (see Section 3.8.5.7). It indicates if the call is placed in protected mode or not.

6.2 SIP Transactions

SIP transactions are used by the protocol to communicate call setup information between RFSSs. These transactions abstract some of the SIP functionality described in [RFC3261]. The behavior of the protocol when using these SIP transactions is described in Section 6.4, and the SIP messages used are summarized below:

- SIP INVITE
- SIP 200 OK
- SIP ACK
- SIP BYE
- SIP 100 TRYING
- SIP 180 RINGING
- SIP 183 SESSION PROGRESS
- SIP 4xx/5xx/6xx
- SIP CANCEL
- SIP Re-INVITE

Parameters for these SIP messages are described in Section 3.

In some circumstances, a SIP timer MAY expire. This event MAY also generate primitives from the call control module to the controlling function.

6.3 State Models for ISSI Voice Services

The ISSI Voice Services Call Control Protocol Element is described by four state models:

- SU-to-SU Call Originating RFSS Setup/Teardown State Model
- SU-to-SU Call Destination RFSS Setup/Teardown State Model
- Group Call Originating RFSS Setup/Teardown State Model
- Group Call Destination RFSS Setup/Teardown State Model

For group calls, after a SIP dialog has been successfully established between 2 RFSSs, RTP sessions MAY or MAY NOT be established depending on RTP resource availability. RTP session status is represented by the three states:

- RTP Session Open
- No RTP Resource at Local RFSS
- No RTP Resource at Peer RFSS

During the lifetime of the call, the call segment MAY transition among the three states as RTP resource availability changes. These transitions require sub-state behavior descriptions, and are described in details in separate sections.

Figure 64 illustrates how the Originating and Destination State Models are used in the various RFSS roles for SU-to-SU Calls. The Originating RFSS is the RFSS that sends the SIP Invite to initiate call setup. During call setup, the Calling Serving RFSS initiates the call to the Calling Home RFSS, the Calling Home initiates to the Called Home RFSS, and the Called Home RFSS initiates to the Called Serving RFSS. When In-Call Roaming occurs, the Home RFSSs initiate call setup to the new (roamed to) Serving RFSSs.

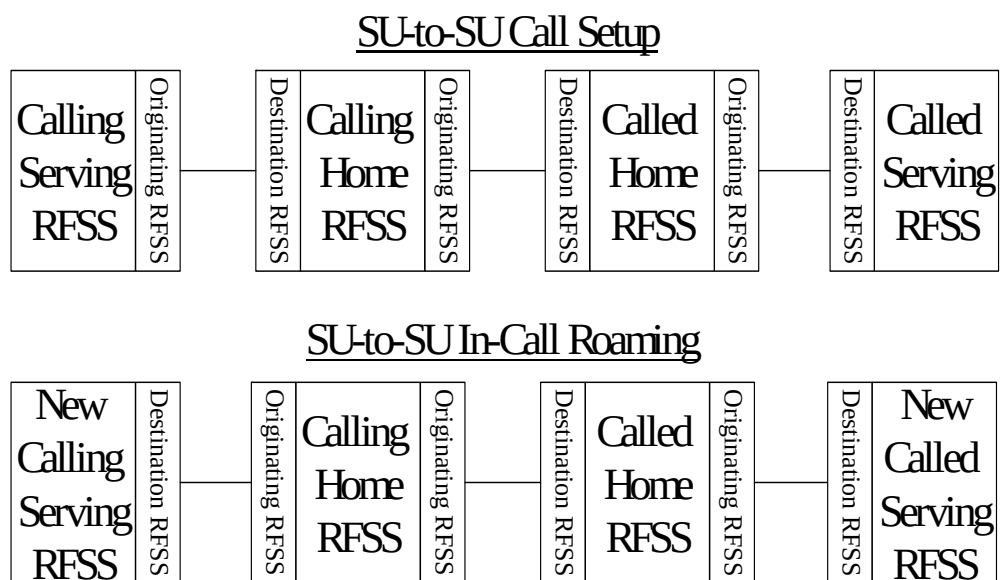


Figure 64 - SU-to-SU RFSS Roles

Figure 65 illustrates how the Originating and Destination State Models are used in the various RFSS roles for Group Calls. A Serving RFSS MAY initiate call setup to the group's Home RFSS, which initiates call setup to all other Serving RFSSs for the group. Note that, in the case where a group call is initiated at the group's Home RFSS, the Home RFSS will be the Originating RFSS for all Serving RFSSs.

Group Call Setup

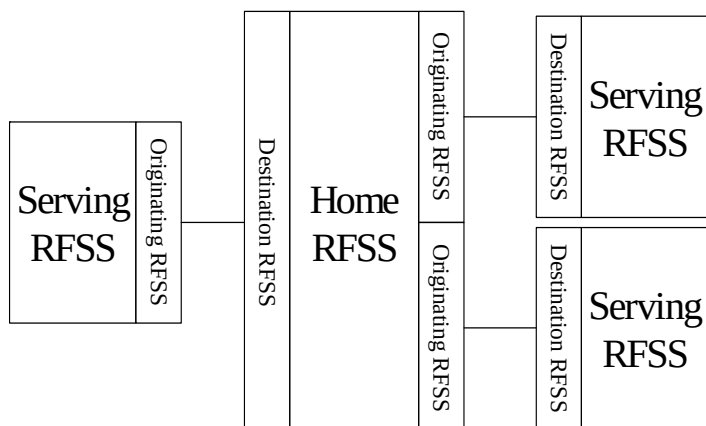


Figure 65 - Group Call RFSS Roles

In the following state models, each Transition is triggered by an Event, and MAY result in some Outputs. The notation, <Event>/<Output> is used in describing the transitions in all state diagrams. Note that failures in the Controlling Function, or in communication between the Controlling Function and the Call Control Protocol Element, that result in missing primitive triggers to the Call Control Protocol Element are beyond the scope of this specification. Note also that the state models in this section do not cover functions of the SIP protocol handled by the SIP transaction layer as described in Section 17 of [RFC3261]. This includes SIP retry and timeout processing, and the generation of SIP ACK in response to 4xx, 5xx, and 6xx SIP responses.

6.3.1 State Model: SU-to-SU Call - Originating RFSS

Figure 66 shows the Su-to-SU Call Originating RFSS State Model.

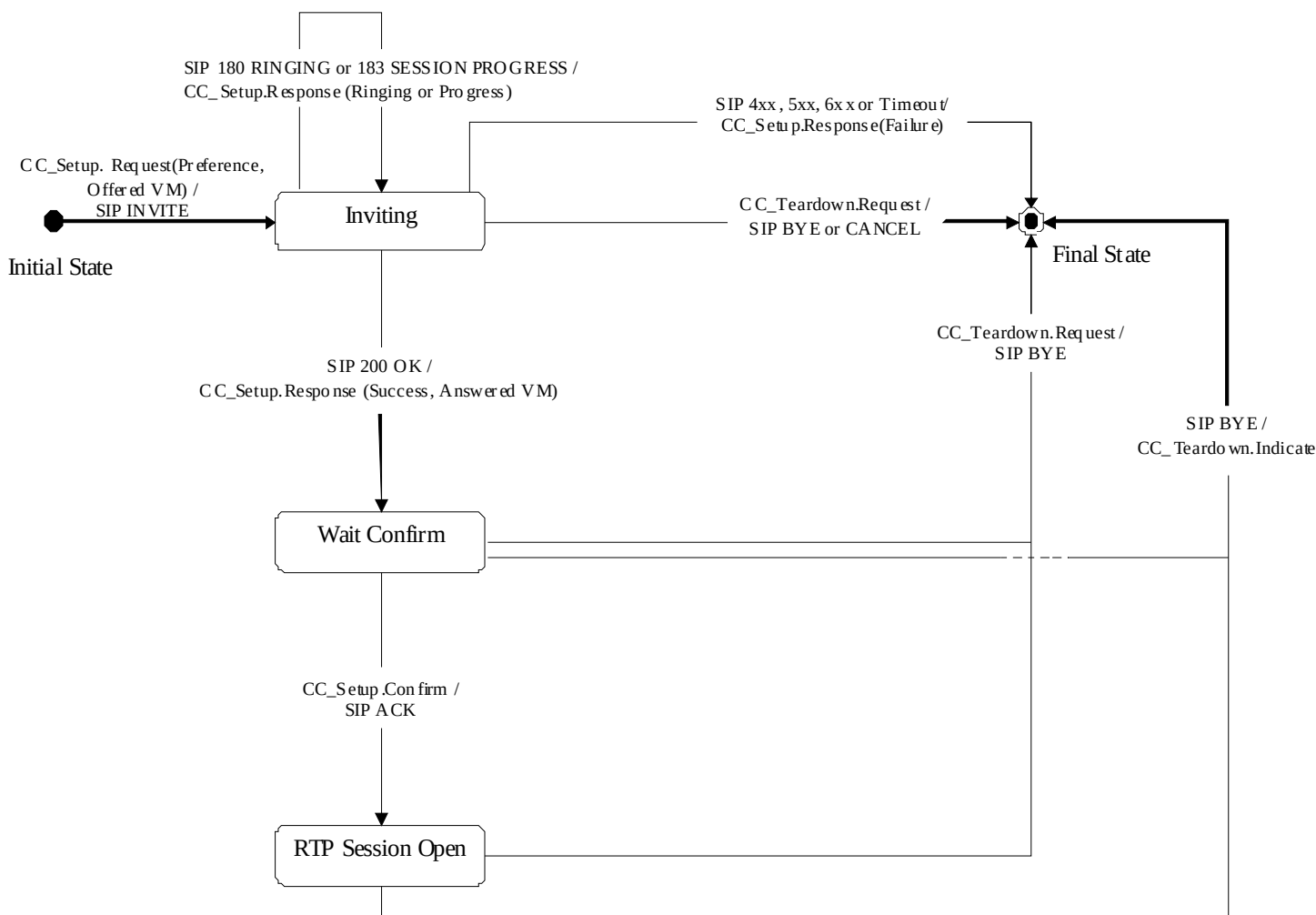


Figure 66 - SU-to-SU Call Originating RFSS State Model

The following sections describe the states, events, and actions in this model.

6.3.1.1 Initial State

The initial state is a “dummy” state that each instance of the state machine enters temporarily upon creation. The initial state is immediately exited as part of processing the event that stimulated creation of the instance of the state machine. In the case of the Originating RFSS state model, the event which can

cause creation of an instance of the state machine is exertion of a CC_Setup.Request primitive by the Controlling Function

6.3.1.1.1 CC_Setup.Request Event

The Controlling Function exerts a CC_Setup.Request primitive when:

1. as the Calling Serving RFSS, it receives an SU-to-SU Call Request that causes it to initiate the call to the Calling Home RFSS
2. as a Calling Home RFSS, it receives a CC_Setup.Indicate primitive from the Call Control Protocol Element that causes it to initiate the call to the Called Home RFSS.
3. as a Called Home RFSS, it receives a CC_Setup.Indicate primitive from the Call Control Protocol Element that causes it to initiate the call to the Called Serving RFSS.
4. as a Home RFSS, it receives a MM_Registration.Indicate primitive for a party involved in an SU-to-SU call (In-Call Roaming) that causes it to initiate the call to the new Serving RFSS

The Controlling Function exerts the primitive after validating the request and reserving RTP resources for the call. As a Calling Serving RFSS the Controlling Function asserts the Offered VM parameter taking into account its own capabilities and the Calling SU's possible mode(s). As a Called Home RFSS or Calling Home RFSS the Controlling Function asserts the Offered VM parameter taking the offered capabilities and its own capabilities for the call segment into account.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP INVITE transaction to the appropriate Destination RFSS, and move to the *Inviting* state.

6.3.1.2 Inviting State

In this state, the Originating RFSS waits for provisional or final responses to the SIP INVITE.

6.3.1.2.1 SIP 180 RINGING Event

This event occurs when the SIP Agent receives a SIP 180 RINGING message from the Destination RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.Response(Ringing) primitive, and remains in the *Inviting* state.

The Controlling Function:

1. when it is a Home RFSS during call setup, forwards the RINGING indication via a CC_Setup.IndicateResponse primitive to its other call segment.

2. when it is a Home RFSS during in-call roaming, ignores the RINGING indication.
3. when it is the Calling Serving RFSS, performs the appropriate signaling to the Calling party to indicate the RINGING state of the call.

6.3.1.2.2 SIP 183 SESSION PROGRESS Event

This event occurs when the SIP Agent receives a SIP 183 SESSION PROGRESS message from the Destination RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.Response(Progress) primitive, and remains in the *Inviting* state.

The Controlling Function:

1. when it is a Home RFSS during call setup, forwards the Progress indication via a CC_Setup.IndicateResponse primitive to its other call segment.
2. when it is a Home RFSS during in-call roaming, ignores the Progress indication.
3. when it is the Calling Serving RFSS, performs the appropriate signaling to the Calling party to indicate the queued state of the call.

6.3.1.2.3 SIP 200 OK Event

This event occurs when the SIP Agent receives a SIP 200 OK message from the Destination RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.Response(Success, Answered VM) primitive, and moves to the *Wait Confirm* state.

The Controlling Function:

1. when it is a Home RFSS during call setup, forwards the OK indication via a CC_Setup.IndicateResponse primitive to its other call segment.
2. when it is a Home RFSS during in-call roaming, exerts a CC_Setup.Confirm primitive back to the Call Control Protocol Element to complete the call setup with the Serving RFSS.
3. when it is the Calling Serving RFSS, takes the necessary steps to use an agreed upon vocoder mode and allocate RF resources for the call if it hasn't already done so.

6.3.1.2.4 SIP 4xx, 5xx, 6xx, or Timeout Event

This event occurs when the SIP Agent receives a SIP 4xx, 5xx, or 6xx final response from the Destination RFSS, or when the SIP Agent times out waiting for the SIP final response.

Editor's note: To ensure backwards compatibility, it is expected that the offer/answer architecture will utilize the 606 response to indicate that none of the offered vocoder modes is supported by the answering RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.Response(Failure) primitive, and terminates the call, moving to the *Final* state.

The Controlling Function:

1. when it is a Home RFSS during call setup, forwards the failure indication via a CC_Setup.IndicateResponse primitive to its other call segment, and tears down the call.
2. when it is a Home RFSS during in-call roaming, forwards the failure indication via a CC_Teardown.Request primitive to its other call segment and tears down the call.
3. when it is the Calling Serving RFSS, performs the appropriate signaling to the Calling party to indicate that the call is torn down, and tears down the call.

6.3.1.2.5 CC_Teardown.Request Event

The Controlling Function exerts a CC_Teardown.Request primitive when:

1. the RFSS deallocates the RTP resources assigned to the call for any reason.
2. the RFSS decides to terminate the call for any other reason, due to local policy
3. as the Calling Serving RFSS, the Calling party cancels the call request.
4. as a Home RFSS, it receives a CC_Teardown.Indicate primitive from the Call Control Protocol Element that it forwards to its other call segment.
5. as a Home RFSS during in-call roaming, it receives a CC_Setup.Response(Failure) primitive from the Call Control Protocol Element that it forwards to its other call segment.
6. as a Home RFSS, it receives a MM_Registration.Indicate or MM_Deregistration.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call to the old Serving RFSS
7. as a Calling Serving RFSS, it receives a MM_Roaming.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call.

Note that the Controlling Function releases any allocated RTP resources and tears down the call when the Controlling Function exerts this primitive.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert any primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate either a SIP BYE or a SIP CANCEL transaction to the Destination RFSS. The SIP

Agent determines which transaction to send based on the state of the SIP dialog, as described in Section 6.4.2.

The Call Control Protocol Element terminates the call, moving to the *Final* state. The SIP Agent handles any SIP response to the SIP BYE or SIP CANCEL message as described in Section 6.4.2.

6.3.1.3 Wait Confirm State

In this state, the Originating RFSS waits for the Controlling Function to allocate RF resources and confirm the completion of the call setup.

6.3.1.3.1 CC_Setup.Confirm Event

The Controlling Function exerts a CC_Setup.Confirm primitive when:

1. as the Calling Serving RFSS, it allocates RF resources to the call, and performs the appropriate signaling to the call party to indicate that the call is active.
2. as a Home RFSS during call setup, it receives a CC_Setup.IndicateResponseConfirm primitive from the Call Control Protocol Element that it forwards to its other call segment.
3. as a Home RFSS during in-call roaming, it receives a CC_Setup.Response primitive that SHOULD cause it to complete call setup for the new Serving RFSS

When this primitive is exerted, the Controlling Function SHOULD also tell the Transmission Control protocol to establish the RTP session for the call segment by exerting a TC.Create primitive (see Section 7).

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP ACK transaction to the Destination RFSS, and move to the *RTP Session Open* state.

6.3.1.3.2 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Destination RFSS.

When the SIP BYE is received, the Call Control Protocol Element, indicates the call teardown to the Controlling Function by exerting a CC_Teardown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 200 OK to the Destination RFSS in response to the SIP BYE message as described in Section 6.4.4.

The Controlling Function:

1. releases RTP resources allocated to the call.

2. when it is a Home RFSS, forwards the teardown indication via a CC_Teardown.Request primitive to its other call segment, and tears down the call.
3. when it is the Serving RFSS, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.1.3.3 CC_Teardown.Request Event

The Controlling Function exerts a CC_Teardown.Request primitive when:

1. the RFSS deallocates the RTP resources assigned to the call for any reason.
2. the RFSS decides to terminate the call for any other reason, due to local policy.
3. as the Calling Serving RFSS, the Calling party cancels the call request.
4. as a Home RFSS, it receives a CC_Teardown.Indicate primitive from the Call Control Protocol Element that it forwards to its other call segment.
5. as a Home RFSS, it receives a MM_Registration.Indicate or MM_Deregistration.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call to the old Serving RFSS
6. as a Calling Serving RFSS, it receives a MM_Roaming.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call.
7. as a Calling Serving RFSS, it is unable to allocate RF resources to the call for any reason (such as a resource allocation timeout).

Note that the Controlling Function releases any allocated RTP and RF resources and tears down the call when the Controlling Function exerts this primitive.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert any response primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP BYE transaction to the Destination RFSS.

The Call Control Protocol Element terminates the call, moving to the *Final* state. The SIP Agent handles any SIP response to the SIP BYE message as described in Section 6.4.4.

6.3.1.4 RTP Session Open State

In this state, the call is in progress and the RFSS waits for it to be torn down.

6.3.1.4.1 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Destination RFSS.

When the SIP BYE is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_Teardown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 200 OK to the Destination RFSS in response to the SIP BYE message as described in Section 6.4.4.

The Controlling Function:

1. tells the Transmission Control protocol to terminate the RTP session for this call segment by exerting a TC.Destroy primitive (see Section 7.1).
2. releases RTP resources allocated to the call.
3. when it is a Home RFSS, forwards the teardown indication via a CC_Teardown.Request primitive to its other call segment, and tears down the call.
4. when it is the Serving RFSS, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.1.4.2 CC_Teardown.Request Event

The Controlling Function exerts a CC_Teardown.Request primitive when:

1. the RFSS deallocates the RTP resources assigned to the call for any reason.
2. it receives a TC_Error.Indication which shows that the RTP heartbeat timeout has expired as described in Section 7, indicating that there is loss of RTP connectivity.
3. the RFSS decides to terminate the call for any other reason, due to local policy.
4. as the Calling Serving RFSS, the Calling party cancels the call request.
5. as a Home RFSS, it receives a CC_Teardown.Indicate primitive from the Call Control Protocol Element that it forwards to its other call segment.
6. as a Home RFSS, it receives a MM_Registration.Indicate or MM_Deregistration.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call to the old Serving RFSS
7. as a Calling Serving RFSS, it receives a MM_Roaming.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call.
8. as a Calling Serving RFSS, it deallocates the RF resources assigned to the call for any reason.
9. as a Called Home RFSS, its Hang Timer expires.

Note that the Controlling Function tells the Transmission Control protocol to terminate the RTP session for this call segment by exerting a TC.Destroy primitive (see Section 7), releases any allocated RTP and RF resources, and tears down the call when the Controlling Function exerts this primitive.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert any response primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP BYE transaction to the Destination RFSS.

The Call Control Protocol Element terminates the call, moving to the *Final* state. The SIP Agent handles any SIP response to the SIP BYE message as described in Section 6.4.4.

6.3.1.5 Final State

The *Final* state is the terminal state of the call. If the Originating RFSS has any outstanding SIP transactions, such as waiting for the 200 (OK) response to a BYE message sent by the Originating RFSS, the SIP Agent handles it without any further interaction with the Call Control layer.

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6.3.2 State Model: SU-to-SU Call - Destination RFSS

Figure 67 shows the SU-to-SU Call Setup/Teardown Destination RFSS State Model.

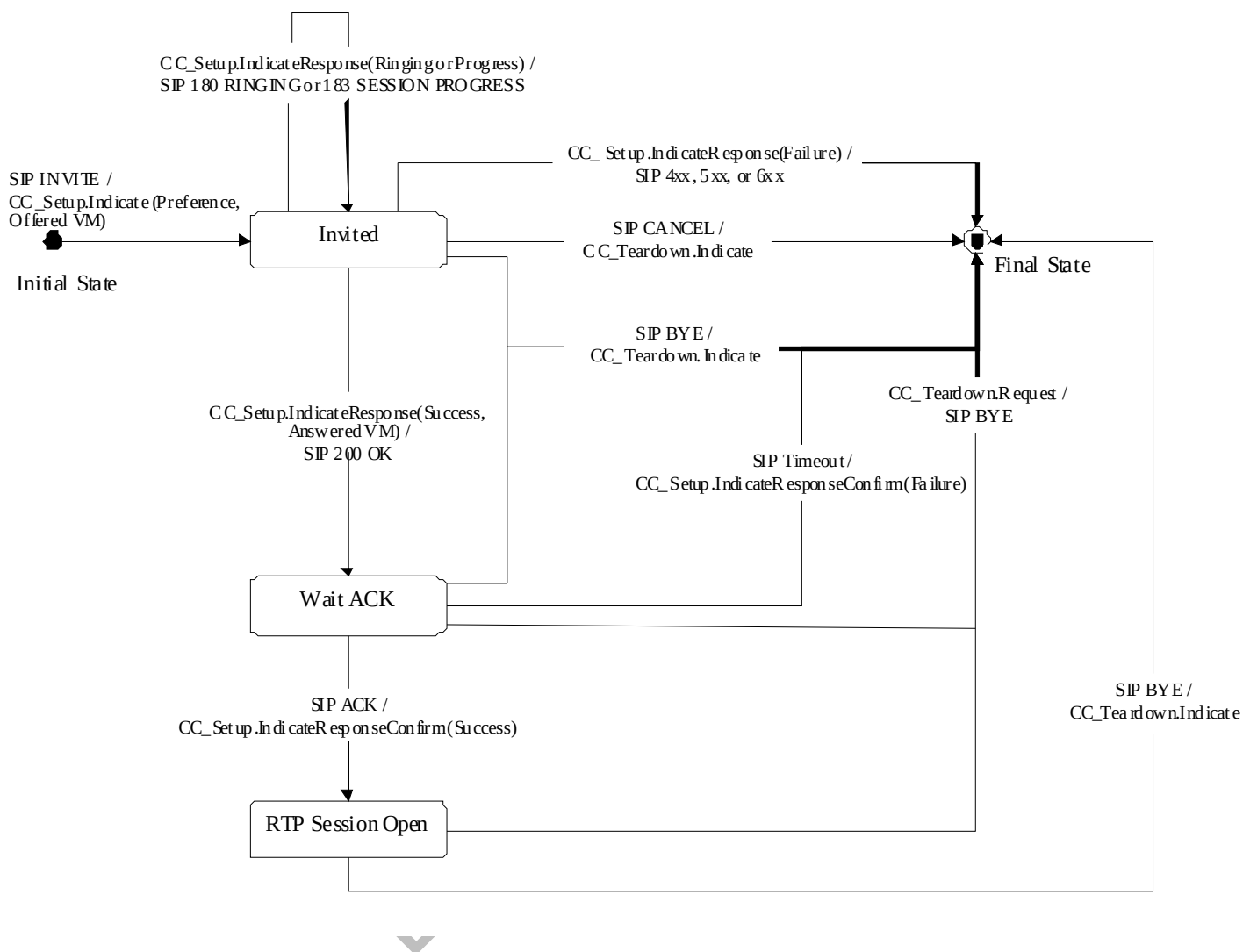


Figure 67 - SU-to-SU Call Destination RFSS State Model

The following sections describe the states, events, and actions in this model.

6.3.2.1 Initial State

The initial state is a “dummy” state that each instance of the state machine enters temporarily upon creation. The initial state is immediately exited as part of processing the event that stimulated creation of the instance of the state machine. In the case of the Destination RFSS state model, the event which can cause creation of an instance of the state machine is reception of a SIP INVITE message by the SIP Agent.

6.3.2.1.1 SIP INVITE Event

This event occurs when the SIP Agent receives a SIP Invite message from the Originating RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.Indicate primitive, and moves to the *Invited* state.

The Controlling Function:

1. allocates RTP resources for the call.
2. when it is a Called Serving RFSS during call setup, it determines what vocoder mode(s) to agree to use on the talk spurts associated with the call segment based on the mode(s) offered, its own capabilities, and the mode it requires for the Called SU.
3. when it is a Called Serving RFSS during call setup, and Availability Check is not required for the call (see Section 3.8.2.13), attempts to allocate RF resources for the call. Successful allocation at this time results in exertion of a CC_Setup.IndicateResponse(Success) primitive to the Call Control Protocol Element, otherwise the RF resource allocation are queued, and a CC_Setup.IndicateResponse(Progress) primitive is exerted.
4. when it is a Called Serving RFSS during call setup, and Availability Check is required for the call (see Section 3.8.2.13), performs the appropriate signaling to the called party to execute the Availability Check.
5. when it is a Serving RFSS during in-call roaming, allocates RF resources for the call.
6. when it is a Home RFSS during call setup, validates the call parties, then initiates the call on the other call segment by exerting a CC_Setup.Request primitive. The vocoder mode(s) offered in the CC_Setup.Request primitive are based on those offered in the CC_Setup.Indicate and the Home RFSS's own capabilities. Validation failure will result in exertion of a CC_Setup.IndicateResponse(Failure) primitive.

6.3.2.2 Invited State

6.3.2.2.1 CC_Setup.IndicateResponse(Ringing) Event

The Controlling Function exerts this primitive when:

1. as the Called Serving RFSS, it receives an indication from the called party that the called party is available.
2. as a Home RFSS during call setup, it receives a CC_Setup.Response(Ringing) primitive that it forwards to its other call segment.

When this primitive is exerted, the Call Control Protocol Element will send a SIP 180 RINGING message to the Originating RFSS, and remain in the *Invited* state.

6.3.2.2.2 CC_Setup.IndicateResponse(Progress) Event

The Controlling Function exerts this primitive when:

1. as the Called Serving RFSS, it attempts to allocate RF resources (either when it receives the SIP INVITE without Availability Check or when it receives an Availability Check response indicating the called party accepts the call) but none are currently available.
2. as a Home RFSS during call setup, it receives a CC_Setup.Response(Progress) primitive that it forwards to its other call segment.

When this primitive is exerted, the Call Control Protocol Element will send a SIP 183 SESSION PROGRESS message to the Originating RFSS, and remain in the *Invited* state.

6.3.2.2.3 CC_Setup.IndicateResponse(Success, Answered VM) Event

The Controlling Function exerts this primitive when:

1. as the Called Serving RFSS, when RTP and RF resources are successfully allocated to the call and the allowed vocoder mode has been chosen.
2. as a Home RFSS during call setup, it receives a CC_Setup.Response(Success) primitive that it forwards to its other call segment.

When this primitive is exerted, the Call Control Protocol Element will send a SIP 200 OK message to the Originating RFSS, and move to the *Wait ACK* state.

6.3.2.2.4 CC_Setup.IndicateResponse(Failure) Event

The Controlling Function exerts this primitive when:

1. the RFSS deallocates the RTP resources assigned to the call for any reason.
2. the RFSS locally detects an error after analysis of the received INVITE.

3. the RFSS decides to terminate the call for any other reason, due to local policy.
4. the RFSS cannot support any of the offered vocoder modes.
5. as the Serving RFSS, a call party cancels the call request.
6. as the Called Serving RFSS, it receives an indication from the called party that the called party does not accept the call.
7. as a Home RFSS during call setup, it receives a CC_Setup.Response(Failure) primitive that it forwards to its other call segment.
8. as a Home RFSS, validation of the call parties fails when a SIP INVITE is received and the call is terminated.
9. as a Home RFSS, it receives a MM_Registration.Indicate or MM_Deregistration.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call to the old Serving RFSS
10. as a Serving RFSS, it receives a MM_Roaming.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call.

Note that the Controlling Function releases any allocated RTP resources and tears down the call when the Controlling Function exerts this primitive.

When this primitive is exerted, the Call Control Protocol Element will send a SIP 4xx, 5xx, or 6xx message to the Originating RFSS (see Section 6.4.1.6), and terminate the call, moving to the *Final* state.

6.3.2.2.5 SIP CANCEL Event

This event occurs when the SIP Agent receives a SIP CANCEL message from the Originating RFSS.

When the SIP CANCEL is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_Teardown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 487 message to the Originating RFSS in response to the SIP INVITE and SIP 200 OK in response to the SIP CANCEL message as described in Section 6.4.2.1.

The Controlling Function:

1. releases RTP resources allocated to the call
2. when it is a Home RFSS, forwards the teardown indication via a CC_Teardown.Request primitive to its other call segment, and tears down the call.
3. when it is the Serving RFSS, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.2.2.6 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Originating RFSS.

When the SIP BYE is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a `CC_Teardown.Indicate` primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 200 OK to the Originating RFSS in response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. releases RTP resources allocated to the call
2. when it is a Home RFSS, forwards the teardown indication via a `CC_Teardown.Request` primitive to its other call segment, and tears down the call.
3. when it is the Serving RFSS, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.2.3 Wait ACK State

In this state, the Destination RFSS is waiting for the SIP ACK message from the Originating RFSS to complete call setup.

6.3.2.3.1 SIP ACK Event

This event occurs when the SIP Agent receives a SIP ACK message from the Originating RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a `CC_Setup.IndicateResponseConfirm(Success)` primitive, and moves to the *RTP Session Open* state.

The Controlling Function:

1. tells the Transmission Control protocol to establish an RTP session for the call segment by exerting a `TC.Create` primitive (see Section 7).
2. when it is a Serving RFSS, performs the appropriate signaling to the call party to indicate that the call is active.
3. when it is a Home RFSS during call setup, forwards the setup indication to its other call segment via a `CC_Setup.Confirm` primitive.
4. when it is the Called Home RFSS, starts the Hang Timer. The Hang Timer is reset whenever voice activity is detected for a call, and if it expires the call is torn down.

6.3.2.3.2 SIP Timeout Event

This event occurs when the SIP Agent times out waiting for the SIP ACK.

When the SIP Timeout occurs, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a `CC_Setup.IndicateResponseConfirm(Failure)` primitive, and terminates the call, moving to the *Final* state. The SIP Agent sends a SIP BYE message to the Originating RFSS and handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. releases RTP resources allocated to the call
2. when it is a Home RFSS, forwards the teardown indication via a `CC_Teardown.Request` primitive to its other call segment, and tears down the call.
3. when it is the Serving RFSS, performs the appropriate signaling to the call party to indicate that the call is torn down, releases RF resources allocated to the call, and tears down the call.

6.3.2.3.3 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Originating RFSS.

When the SIP BYE is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a `CC_Teardown.Indicate` primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 200 OK to the Originating RFSS in response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. releases RTP resources allocated to the call
2. when it is a Home RFSS, forwards the teardown indication via a `CC_Teardown.Request` primitive to its other call segment, and tears down the call.
3. when it is the Serving RFSS, performs the appropriate signaling to the call party to indicate that the call is torn down, releases RF resources allocated to the call, and tears down the call.

6.3.2.3.4 CC_Teardown.Request Event

The Controlling Function exerts a `CC_Teardown.Request` primitive when:

1. the RFSS deallocates the RTP resources assigned to the call for any reason.
2. the RFSS decides to terminate the call for any other reason, due to local policy.
3. as the Serving RFSS, a call party cancels the call request.
4. as a Home RFSS, it receives a `CC_Teardown.Indicate` primitive from the Call Control Protocol Element that it forwards to its other call segment.

5. as a Home RFSS, it receives a MM_Registration.Indicate or MM_Deregistration.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call to the old Serving RFSS
6. as a Serving RFSS, it receives a MM_Roaming.Indicate primitive for a party involved in an SU-to-SU call that causes it to tear down the call.
7. as a Serving RFSS, it deallocates the RF resources assigned to the call for any reason.

Note that the Controlling Function releases any allocated RTP and RF resources and tears down the call when the Controlling Function exerts this primitive.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert any response primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP BYE transaction to the Originating RFSS. The SIP Agent will send a SIP BYE message to the Originating RFSS when the ACK is received or when SIP Timeout occurs, and will handle any SIP response to the SIP BYE message as described in Section 6.4.4.1. The Call Control Protocol Element terminates the call, moving to the *Final* state.

6.3.2.4 RTP Session Open State

The behavior of the Call Control Protocol Element for the SU-to-SU Call Destination RFSS in these states is identical to the behavior of the SU-to-SU Call Originating RFSS, and is described in Section 6.3.1.4.

6.3.2.5 Final State

The *Final* state is the terminal state of the call. If the Destination RFSS has any outstanding SIP transactions, such as waiting for the 200 (OK) response to a BYE message sent by the Destination RFSS, the SIP Agent handles it without any further interaction with the Call Control layer.

6.3.3 State Model: Group Call - Originating RFSS

Figure 68 shows the Group Call Originating RFSS State Model.

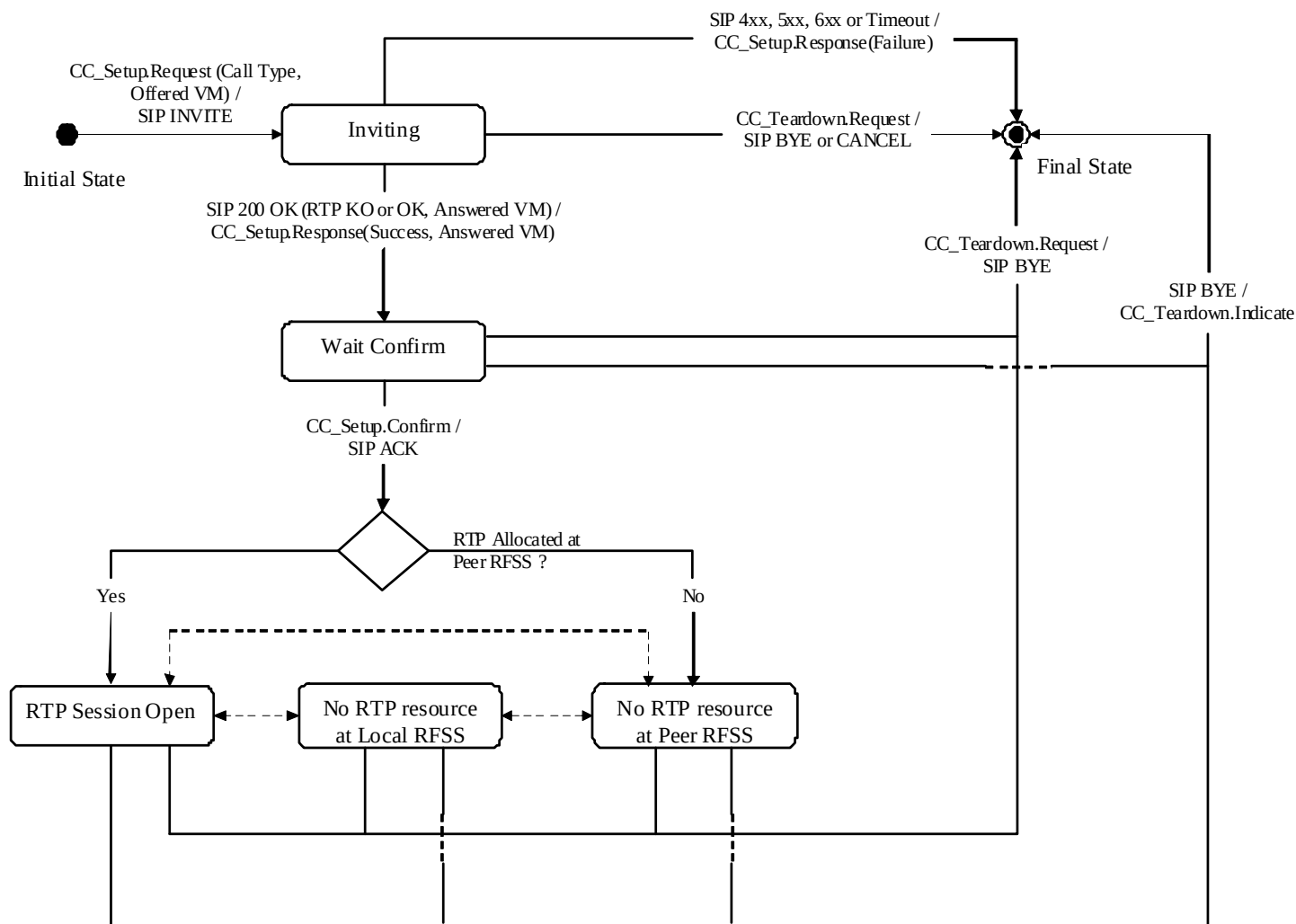


Figure 68 - Group Call Originating RFSS State Model

The following sections describe the states, events, and actions in this model.

6.3.3.1 Initial State

The initial state is a “dummy” state that each instance of the state machine enters temporarily upon creation. The initial state is immediately exited as part of processing the event that stimulated creation of the instance of the state machine. In the case of the Originating RFSS state model, the event which can

cause creation of an instance of the state machine is exertion of a `CC_Setup.Request` primitive by the Controlling Function

6.3.3.1.1 CC_Setup.Request Event

The Controlling Function exerts a `CC_Setup.Request` primitive when:

1. as the Serving RFSS, it receives a Group Call Request that causes it to initiate the call to the Home RFSS
2. as a Home RFSS, it receives a `CC_Setup.Indicate` primitive from the Call Control Protocol Element that causes it to initiate the call to the Serving RFSSs.
3. as a Home RFSS, it receives a `MM_Registration.Indicate` primitive that would require an RFSS to be added to an active call.
4. as a Home RFSS, it determines that it can bring an RFSS into an active call that has previously dropped out of a call for some reason.

The Controlling Function exerts the primitive after validating the request, reserving RTP resources, determining what vocoder mode(s) to offer, and, for Confirmed Group Call, attempting to reserve RF resources for the call. For Unconfirmed Group Call, RF resource allocation MAY occur at any time after this point in the call depending on local policy. For a more complete discussion of RF resource allocation for confirmed and unconfirmed group calls, see the SIP Messages and Parameters Section 3.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP INVITE transaction to the appropriate Destination RFSS, and move to the *Inviting* state.

6.3.3.2 Inviting State

In this state, the Originating RFSS waits for provisional or final responses to the SIP INVITE.

6.3.3.2.1 SIP 200 OK Event

This event occurs when the SIP Agent receives a SIP 200 OK message from the Destination RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a `CC_Setup.Response(Success, Answered VM)` primitive, and moves to the *Wait Confirm* state.

The Controlling Function:

1. exerts a `CC_Setup.Confirm` primitive to the Call Control Protocol Element to continue call setup.

2. when it is a Serving RFSS, tells Transmission Control to establish an RTP Session for the call segment by exerting a TC.Create primitive (see Section 7).

6.3.3.2.2 SIP 4xx, 5xx, 6xx, or Timeout Event

This event occurs when the SIP Agent receives a SIP 4xx, 5xx, or 6xx final response from the Destination RFSS, or when the SIP Agent times out waiting for the SIP final response.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.Response(Failure) primitive, and terminates the call, moving to the *Final* state.

The Controlling Function:

1. when it is a Home RFSS, possibly takes action to bring the Serving RFSS back into the call (see Section 6.6.1.3 for the action (if any) to take).
2. when it is a Serving RFSS, releases any RTP resources allocated to the call and, when the RF Hang Timer expires, releases any RF resources allocated to the call, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.3.2.3 CC_Teardown.Request Event

The Controlling Function exerts a CC_Teardown.Request primitive when:

1. the RFSS decides to terminate the call for any reason, due to local policy
2. the Home RFSS receives a MM_Deregistration.Indicate primitive that indicates that a group is no longer active on a Serving RFSS

Note that when the Controlling function exerts this primitive:

1. if the primitive was exerted due to local policy, the Controlling Function on the Home RFSS can, again according to local policy, try to re-establish the SIP session for this call segment by exerting a CC_Setup.Request primitive to the Call Control Protocol Element.
2. if the group is no longer active on the Serving RFSS, the Home RFSS Controlling Function will release RTP resources allocated for the call segment.
3. the Controlling Function for a Serving RFSS releases any allocated RTP resources and, when its RF Hang Timer expires, releases RF resources and tears down the call.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert any primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate either a SIP BYE or a SIP CANCEL transaction to the Destination RFSS. The SIP

Agent determines which transaction to send based on the state of the SIP dialog, as described in Section 6.4.2.

The Call Control Protocol Element terminates the call, moving to the *Final* state. The SIP Agent handles any SIP response to the SIP BYE or SIP CANCEL message as described in Section 6.4.2.

6.3.3.3 Wait Confirm State

In this state, the Originating RFSS waits for the Controlling Function to confirm the completion of the call setup.

6.3.3.3.1 CC_Setup.Confirm Event

The Controlling Function exerts a CC_Setup.Confirm primitive when it receives a CC_Setup.Response primitive from the Call Control Protocol Element.

When this primitive is exerted, if the peer RFSS has RTP resources allocated, the Controlling Function SHOULD also tell the Transmission Control protocol to establish the RTP session for the call segment by exerting a TC.Create primitive (see Section 7).

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP ACK transaction to the Destination RFSS. The next state depends on the state of RTP resource allocation on the Destination RFSS, which was indicated by the RTP parameter received in the SIP 200 OK message. If the Destination RFSS has RTP resources allocated, then the Call Control Protocol Element will move to the *RTP Session Open* state. If the Destination RFSS does not have RTP resources allocated, then the Call Control Protocol Element will move to the *No RTP Resource at Peer RFSS* state.

6.3.3.3.2 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Destination RFSS.

When the SIP BYE is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_Teardown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 200 OK to the Destination RFSS in response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. when it is a Home RFSS, takes action to bring the Serving RFSS back into the call by exerting a CC_Setup.Request to the Call Control Protocol Element.
2. when it is the Serving RFSS, releases any RTP resources allocated to the call and, when its RF Hang Timer expires, releases any RF resources

allocated to the call, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.3.3.3 CC_TearDown.Request Event

The Controlling Function exerts a CC_TearDown.Request primitive when:

1. the RFSS decides to terminate the call for any reason, due to local policy.
2. the RFSS receives a MM_Deregistration.Indicate primitive that indicates that a group is no longer active on a Serving RFSS

Note that when the Controlling function exerts this primitive:

1. if the primitive was exerted due to local policy, the Controlling Function on the Home RFSS can, again according to local policy, try to re-establish the SIP session for this call segment by exerting a CC_Setup.Request primitive to the Call Control Protocol Element.
2. if the group is no longer active on the Serving RFSS, the Home RFSS Controlling Function releases RTP resources allocated for the call segment.
3. the Controlling Function for a Serving RFSS releases any allocated RTP resources and, when its RF Hang Timer expires, releases RF resources and tears down the call.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert a any response primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP BYE transaction to the Destination RFSS.

The Call Control Protocol Element terminates the call, moving to the *Final* state. The SIP Agent handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

6.3.3.4 RTP Session Open, No RTP resource at Local RFSS, and No RTP resource at Peer RFSS States

In these states, the call is in progress and the RFSS waits for it to be torn down. Each of these states has sub-state behavior that is described in Sections 6.3.5, 6.3.6, and 6.3.7.

6.3.3.4.1 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Destination RFSS.

When the SIP BYE is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_TearDown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will

handle sending SIP 200 OK to the Destination RFSS in response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. if it has an RTP session open, tells the Transmission Control protocol to terminate the RTP session for this call segment by exerting a TC.Destroy primitive (see Section 7).
2. when it is a Home RFSS, takes action to bring the Serving RFSS back into the call by exerting a CC_Setup.Request to the Call Control Protocol Element.
3. when it is the Serving RFSS, releases any RTP resources allocated to the call and, when its RF Hang Timer expires, releases any RF resources allocated to the call, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.3.4.2 CC_Teardown.Request Event

The Controlling Function exerts a CC_Teardown.Request primitive when:

1. it receives a TC_Error.Indication from the Transmission Control protocol, which shows that the RTP heartbeat timeout has expired as described in Section 7, indicating that there is loss of RTP connectivity.
2. the RFSS decides to terminate the call for any reason, due to local policy.
3. as a Home RFSS, it receives a MM_Deregistration.Indicate primitive that indicates that a group is no longer active on a Serving RFSS
4. as a Home RFSS, its Hang Timer expires.

Note that when the Controlling Function exerts this primitive:

1. if it has an RTP session open, it tells the Transmission Control protocol to terminate the RTP session for this call segment by exerting a TC.Destroy primitive (see Section 7).
2. if the primitive was exerted due to local policy, the Controlling Function on the Home RFSS can, again according to local policy, try to re-establish the SIP session for this call segment by exerting a CC_Setup.Request primitive to the Call Control Protocol Element.
3. if the group is no longer active on the Serving RFSS, the Home RFSS Controlling Function releases RTP resources allocated for the call segment.
4. if its Hang Timer has expired, the Controlling Function on the Home RFSS terminates all call segments, releases any allocated RTP and RF resources and terminates the call.
5. the Controlling Function for a Serving RFSS releases any allocated RTP resources and, when its RF Hang Timer expires, releases RF resources and tears down the call.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert any response primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP BYE transaction to the Destination RFSS.

The Call Control Protocol Element terminates the call, moving to the *Final* state. The SIP Agent handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

6.3.3.5 Final State

The *Final* state is the terminal state of the call. If the Originating RFSS has any outstanding SIP transactions, such as waiting for the 200 (OK) response to a BYE message sent by the Originating RFSS, the SIP Agent handles it without any further interaction with the Call Control layer.

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6.3.4 State Model: Group Call - Destination RFSS

Figure 69 shows the Group Call Setup/Teardown Destination RFSS State Model.

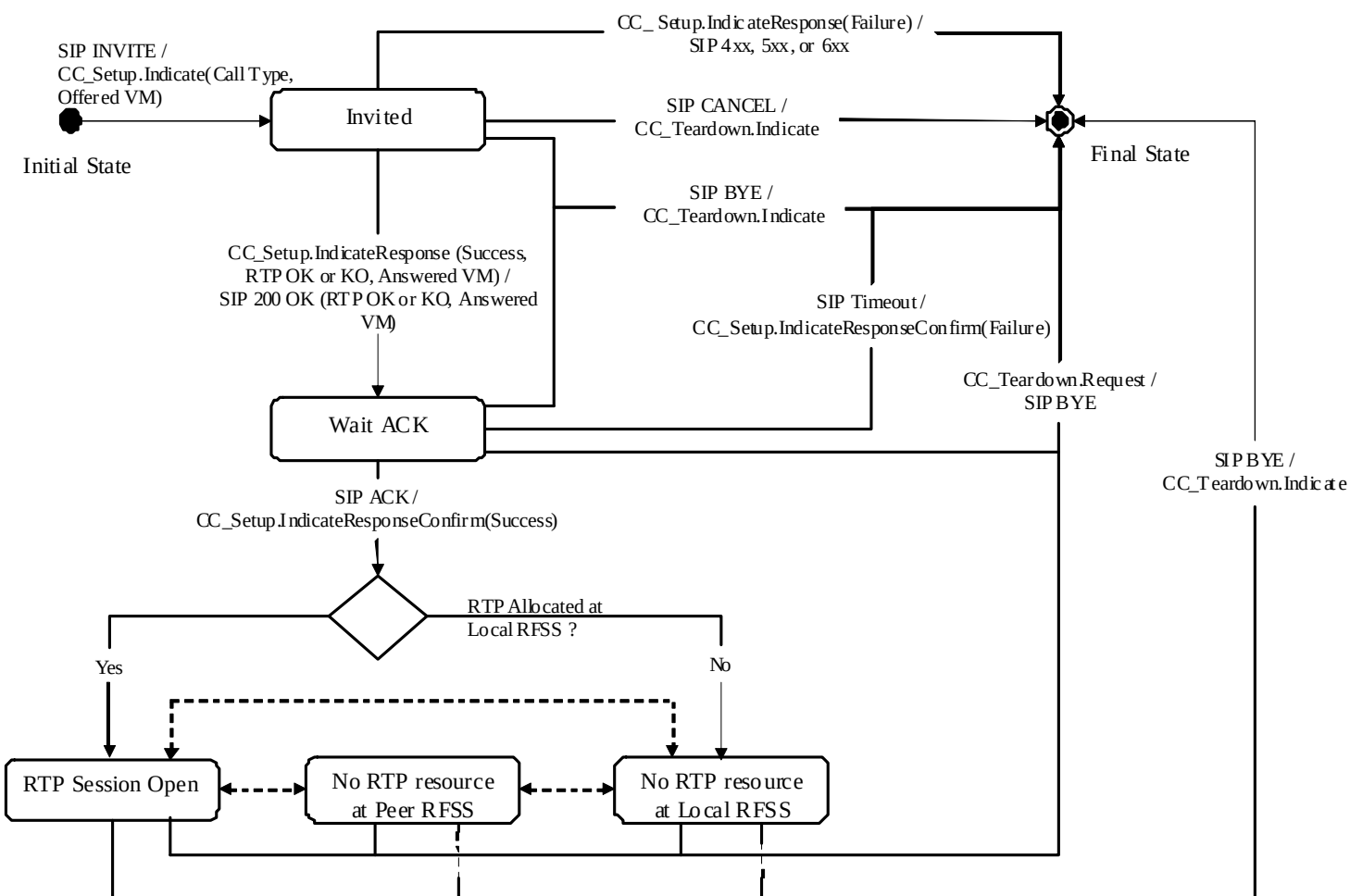


Figure 69 - Group Call Destination RFSS State Model

The following sections describe the states, events, and actions in this model.

6.3.4.1 Initial State

The initial state is a “dummy” state that each instance of the state machine enters temporarily upon creation. The initial state is immediately exited as part of processing the event that stimulated creation of the instance of the state machine. In the case of the Destination RFSS state model, the event which can cause creation of an instance of the state machine is reception of a SIP INVITE message by the SIP Agent.

6.3.4.1.1 SIP INVITE Event

This event occurs when the SIP Agent receives a SIP Invite message from the Originating RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.Indicate primitive, and moves to the *Invited* state.

The Controlling Function:

1. attempts to allocate RTP resources for the call.
2. for Confirmed Group call, attempts to allocate RF resources for the call. For Unconfirmed Group Call, RF resources allocation MAY occur at any time from this point in the call until the PTT Transmit Grant/Start is received from the Transmission Control protocol (see Section 7), depending on the manufacturer's preference.
3. when it is a Serving RFSS, and it can accept one of the offered vocoder modes it replies by exerting a CC_Setup.IndicateResponse(Success) primitive to the Call Control function. If none of the offered vocoder modes is acceptable, then a CC_Setup.IndicateResponse (Failure) primitive is exerted.
4. when it is a Home RFSS during call setup, validates the call parties, and determines which vocoder mode to accept. If validation is successful, it replies by exerting a CC_Setup.IndicateResponse(Success) primitive to the Call Control Protocol Element, allocates RTP resources for the other call segments, then initiates the call on the other call segments by exerting CC_Setup.Request primitives. In the case of validation failure, or if none of the offered vocoder modes is acceptable then a CC_Setup.IndicateResponse(Failure) primitive is exerted.

6.3.4.2 Invited State

6.3.4.2.1 CC_Setup.IndicateResponse(Success, RTP OK or KO, Answered VM) Event

The Controlling Function exerts this primitive when:

1. as the Serving RFSS, it receives a CC_Setup.Indicate from the Call Control Protocol Element.
2. as a Home RFSS, it successfully validates the call parties in the CC_Setup.Indicate.

Note that the Controlling Function might not have allocated RTP resources when it exerts this primitive.

When this primitive is exerted, the Call Control Protocol Element will send a SIP 200 OK (RTP OK or KO, Answered VM) message to the Originating RFSS, and move to the *Wait ACK* state.

6.3.4.2.2 CC_Setup.IndicateResponse(Failure) Event

The Controlling Function exerts this primitive when:

1. an RFSS locally detects an error after analysis of the received INVITE.
2. the RFSS decides to terminate the call for any other reason, due to local policy.
3. as a Home RFSS, it receives a MM_Deregistration.Indicate primitive that indicates that a group is no longer active on a Serving RFSS
4. as a Home RFSS, validation of the call parties when a SIP INVITE is received fails and the call is terminated.
5. as a Home RFSS, none of the offered vocoder modes for the call is acceptable and the call is terminated.
6. as a Serving RFSS, none of the offered vocoder modes is acceptable and the call segment is terminated.

Note that when the Controlling function exerts this primitive:

1. if the primitive was exerted due to local policy, the Controlling Function on the Home RFSS can, again according to local policy, try to re-establish the SIP session for this call segment by exerting a CC_Setup.Request primitive to the Call Control Protocol Element.
2. if the group is no longer active on the Serving RFSS, the Home RFSS Controlling Function will release RTP resources allocated for the call segment.
3. the Controlling Function on a Serving RFSS releases any allocated RTP and, when its RF Hang Timer expires, releases RF resources and tears down the call.

When this primitive is exerted, the Call Control Protocol Element will send a SIP 4xx, 5xx, or 6xx message to the Originating RFSS (see Section 6.4.1.6), and terminate the call, moving to the *Final* state.

6.3.4.2.3 SIP CANCEL Event

This event occurs when the SIP Agent receives a SIP CANCEL message from the Originating RFSS.

When the SIP CANCEL is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_Teardown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 487 message to the Originating RFSS in response to the SIP

INVITE and SIP 200 OK in response to the SIP CANCEL message as described in Section 6.4.2.1.

The Controlling Function:

1. when it is a Home RFSS, takes action to bring the Serving RFSS back into the call by exerting a CC_Setup.Request to the Call Control Protocol Element.
2. when it is the Serving RFSS, releases any RTP resources allocated to the call and, when its RF Hang Timer expires, releases any RF resources allocated to the call, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.4.2.4 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Originating RFSS.

When the SIP BYE is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_Teardown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 200 OK to the Originating RFSS in response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. when it is a Home RFSS, takes action to bring the Serving RFSS back into the call by exerting a CC_Setup.Request to the Call Control Protocol Element.
2. when it is the Serving RFSS, releases any RTP resources allocated to the call and, when its RF Hang Timer expires, releases any RF resources allocated to the call, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.4.3 Wait ACK State

In this state, the Destination RFSS is waiting for the SIP ACK message from the Originating RFSS to complete call setup.

6.3.4.3.1 SIP ACK Event

This event occurs when the SIP Agent receives a SIP ACK message from the Originating RFSS.

The Call Control Protocol Element indicates this to the Controlling Function by exerting a CC_Setup.IndicateResponseConfirm (Success) primitive. The next state depends on the state of RTP resource allocation on the Destination RFSS, which was indicated by the RTP parameter in the CC_Setup.IndicateResponse primitive. If the Destination RFSS has RTP resources allocated, then the Call Control Protocol Element will move to the *RTP Session Open* state. If the

Destination RFSS does not have RTP resources allocated, then the Call Control Protocol Element will move to the *No RTP Resource at Local RFSS* state.

The Controlling Function:

1. if the local RFSS has RTP resources allocated, tells the Transmission Control protocol to establish an RTP session for this call segment by exerting a TC.Create primitive (see Section 7).
2. when it is the Home RFSS, starts the Hang Timer. The Hang Timer is reset whenever there is voice activity, and if it expires the call is torn down.

6.3.4.3.2 SIP Timeout Event

This event occurs when the SIP Agent times out waiting for the SIP ACK.

When the SIP Timeout occurs, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_Setup.IndicateResponseConfirm (Failure) primitive, and terminates the call, moving to the *Final* state. The SIP Agent sends a SIP BYE message to the Originating RFSS and handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. when it is a Home RFSS, takes action to bring the Serving RFSS back into the call by exerting a CC_Setup.Request to the Call Control Protocol Element.
2. when it is the Serving RFSS, releases any RTP resources allocated to the call and, when its RF Hang Timer expires, releases any RF resources allocated to the call, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.4.3.3 SIP BYE Event

This event occurs when the SIP Agent receives a SIP BYE message from the Originating RFSS.

When the SIP BYE is received, the Call Control Protocol Element indicates the call teardown to the Controlling Function by exerting a CC_Teardown.Indicate primitive, and terminates the call, moving to the *Final* state. The SIP Agent will handle sending SIP 200 OK to the Originating RFSS in response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. when it is a Home RFSS, takes action to bring the Serving RFSS back into the call by exerting a CC_Setup.Request to the Call Control Protocol Element.

2. when it is the Serving RFSS, releases any RTP resources allocated to the call and, when its RF Hang Timer expires, releases any RF resources allocated to the call, performs the appropriate signaling to the call party to indicate that the call is torn down, and tears down the call.

6.3.4.3.4 CC_Teardown.Request Event

The Controlling Function exerts a CC_Teardown.Request primitive when:

1. the RFSS decides to terminate the call for any reason, due to local policy.
2. as a Home RFSS, it receives a MM_Deregistration.Indicate primitive that indicates that a group is no longer active on a Serving RFSS

Note that when the Controlling function exerts this primitive:

1. if the primitive was exerted due to local policy, the Controlling Function on the Home RFSS can, again according to local policy, try to re-establish the SIP session for this call segment by exerting a CC_Setup.Request primitive to the Call Control Protocol Element.
2. if the group is no longer active on the Serving RFSS, the Home RFSS Controlling Function will release RTP resources allocated for the call segment.
3. the Controlling Function on a Serving RFSS releases any allocated RTP and, when its RF Hang Timer expires, releases RF resources and tears down the call.

The Controlling Function SHOULD consider exertion of this primitive sufficient to tear down the call in the Call Control Protocol Element. The Call Control Protocol Element will not exert any response primitive in reply.

When this primitive is exerted, the Call Control Protocol Element will initiate a SIP BYE transaction to the Originating RFSS. The SIP Agent will send a SIP BYE message to the Originating RFSS when the ACK is received or when SIP Timeout occurs, and will handle any SIP response to the SIP BYE message as described in Section 6.4.4.1. The Call Control Protocol Element terminates the call, moving to the *Final* state.

6.3.4.4 RTP Session Open, No RTP resource at Local RFSS, and No RTP resource at Peer RFSS States

The behavior of the Call Control Protocol Element for the Group Call Destination RFSS in these states is identical to the behavior of the Group Call Originating RFSS, and is described in Section 6.3.3.4.

6.3.4.5 Final State

The *Final* state is the terminal state of the call. If the Destination RFSS has any outstanding SIP transactions, such as waiting for the 200 (OK) response to a BYE message sent by the Destination RFSS, the SIP Agent handles it without any further interaction with the Call Control layer.

6.3.5 Sub-state Model: Group Call - RTP Session Open State

Once an RFSS moves into the *RTP Session Open* state, call segment characteristic MAY be modified upon the RFSS's own initiative or upon the initiative of the Peer RFSS. The sub-state model diagram in Figure 70 below focuses on sub-states that exist when call segment characteristics are modified.

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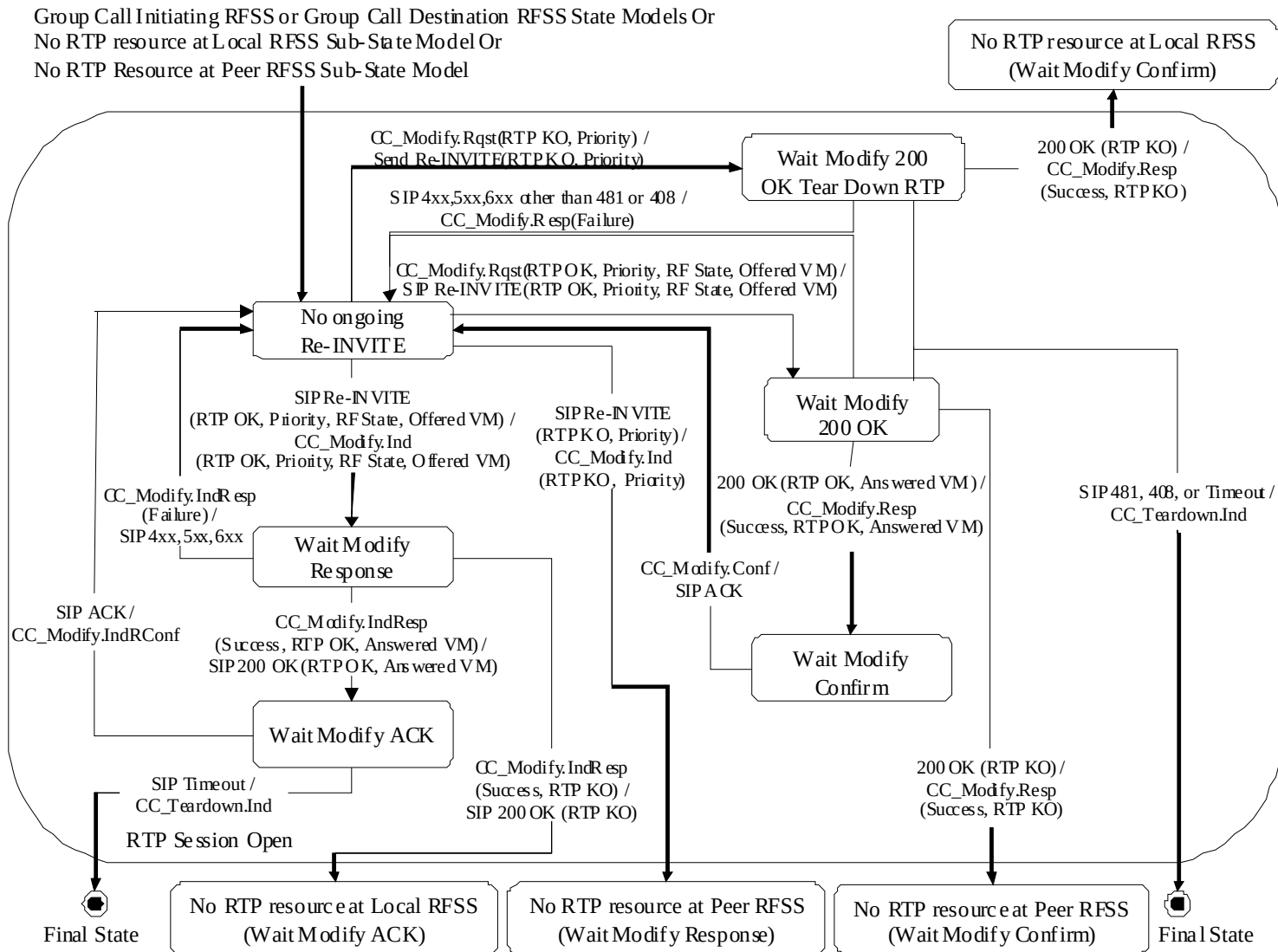


Figure 70 - Group Call Sub-State Model - RTP Session Open

This sub-state model shows:

- The 6 possible sub-states when the RFSS is in the *RTP Session Open* State.
- Transitions from the Group Call Originating RFSS and Group Call Destination RFSS State Models, No RTP resource at Local RFSS Sub-state Model, and No RTP resource at Peer RFSS Sub-state Model.
- Transitions to sub-states of the “No RTP resource at Local RFSS” Sub-state Model and “No RTP resource at Peer RFSS” Sub-state Model.

This sub-state model does not show:

- The transition to the *Final* State from any sub-state upon receipt of CC_Teardown.Request or SIP BYE events. These events appear only in the Group Call Originating RFSS and Group Call Destination RFSS State Models (see Section 6.3.3.4).
- Colliding Re-INVITE situation (see Section 6.6.3).

6.3.5.1 RTP Session Open – No Ongoing Re-INVITE Sub-state

This sub-state is reached for a group call segment:

1. At the Originating RFSS, after successful Group Call Segment Setup, or
2. At the Destination RFSS, after successful Group Call Segment Setup, or
3. At an RFSS, after successful setup of a RTP Session subsequent to a SIP Re-INVITE procedure.

When the Call Control Protocol Element is in the *RTP Session Open – No ongoing Re-INVITE* state for a group call segment, it MAY at any time:

1. Receive a SIP Re-INVITE request from the Peer RFSS, or
2. Receive a CC_Modify.Request from the Controlling Function requesting modification of the characteristics of the group call segment

6.3.5.1.1 SIP Re-INVITE (RTP OK, Priority, RF State, Offered VM) Event

The SIP Re-INVITE(RTP OK, Priority, RF State, Offered VM) message from a Peer RFSS conveys the same SDP parameters as the one currently in use in a type 1 SDP Offer (see Section 3.7.1 for SDP Offer/Answer Usage). Basically, this means the Peer RFSS retains its allocated RTP resource for serving the RTP session. This message has three parameters besides 'RTP OK'.

1. The 'Priority' parameter which can be higher than the one previously used to set up the RTP session (see Section 3.5.13 for the SIP Priority header field)
2. The 'RF State' resource allocation state parameter MAY be present if the RFSS is the Home RFSS for a confirmed group call (see Section 3.8.5.2 for the RF Resource availability parameter in the ISSI SIP body).
3. The Offered VM parameter which can be a different vocoder mode than the one the group call segment is currently allowed to use (see the SDP media description Section 3.7.2.2 for the offered vocoder modes)

Upon receipt of the SIP Re-INVITE(RTP OK, Priority, RF State, Offered VM) message, the Call Control Protocol Element exerts a CC_Modify.Indicate(RTP OK, Priority, *RF State*, Offered VM) primitive to the RFSS's Controlling Function and moves to the *RTP Session Open – Wait Modify Response* state for this group call segment.

The Controlling Function:

1. validates the contents of the Re-INVITE message. If the Re-INVITE message is successfully validated, the Controlling Function responds with

a CC_Modify.IndicateResponse(Success, RTP OK, Answered VM) primitive, otherwise the Controlling Function responds with a CC_Modify.IndicateResponse(Failure) primitive. If a change in vocoder mode is offered and accepted, the Answered VM parameter indicates the accepted vocoder mode, and the Controlling Function takes necessary action based on local policy.

2. if the new priority is higher than the current priority, raises the priority of the call. At the Home RFSS, the Controlling Function propagates the new priority to other call segments by exerting a CC_Modify.Request primitive.
3. at the Home RFSS, for a confirmed group call, collects the RF State (if any) and determines whether conditions are met to start the confirmed group call.

6.3.5.1.2 SIP Re-INVITE (RTP KO, Priority) Event

The SIP Re-INVITE (RTP KO, Priority) message from a Peer RFSS conveys a type 2 SDP offer (see Section 3.7 on SDP Offer/Answer Usage). Basically, this means the Peer RFSS intends to release the RTP resource serving the RTP session.

Upon receipt of the SIP Re-INVITE(RTP KO, Priority) message, the Call Control Protocol Element exerts a CC_Modify.Indicate primitive to the RFSS's Controlling Function and moves to the *No RTP resource at Peer RFSS – Wait Modify Response* state for this group call segment.

The Controlling Function:

1. validates the contents of the Re-INVITE message. If the Re-INVITE message is successfully validated, the Controlling Function responds with a CC_Modify.IndicateResponse(Success, RTP KO) primitive, otherwise the Controlling Function responds with a CC_Modify.IndicateResponse(Failure) primitive.
2. exerts a TC.Destroy primitive to the Transmission Control protocol to tear down the RTP session for this group call segment.
3. if the new priority is higher than the current priority and status_IR is equal to "Success", raises the priority of the call. At the Home RFSS, the Controlling Function propagates the new priority to other call segments by exerting a CC_Modify.Request primitive.

6.3.5.1.3 CC_Modify.Request(RTP OK, Priority, RF State, Offered VM) Event

The CC_Modify.Request(RTP OK, Priority, RF State, Offered VM) primitive is exerted by the Controlling Function when:

1. a Serving RFSS wants to promote the priority of the Call, or
2. a Home RFSS wants to promote the priority of the Call, or
3. a Serving RFSS wants to report the RF resource allocation state to the Home RFSS in a confirmed group call, or
4. a Serving RFSS wants to change the allowed vocoder mode of the group call segment, or
5. a Home RFSS wants to change the allowed vocoder mode of the group call segment, or
6. any combination of Bullets 1, 3 and 4, or
7. bullet 2 and bullet 5 combined

Upon receipt of this event, the Call Control Protocol Element triggers a SIP Re-INVITE request to the Peer RFSS and moves to the *RTP session open – Wait Modify 200 OK* state for this call segment.

6.3.5.1.4 CC_Modify.Request(RTP KO, Priority) Event

The CC_Modify.Request(RTP KO, Priority) primitive is exerted by the Controlling Function when a Serving RFSS or a Home RFSS wants to tear down the RTP session but wants to maintain the SIP session.

Upon receipt of this event, the Call Control Protocol Element triggers a SIP Re-INVITE request to the Peer RFSS and moves to the *RTP Session Open – Wait Modify 200 OK Tear Down RTP* state for this call segment.

6.3.5.2 RTP Session Open – Wait Modify Response Sub-state

6.3.5.2.1 CC_Modify.IndicateResponse(Success, RTP OK, Answered VM) Event

The CC_Modify.IndicateResponse(Success, RTP OK, Answered VM) primitive is exerted by the Controlling Function when it has received a CC_Modify.Indicate(RTP OK, Priority, RF State, Answered VM).

Upon receiving the CC_Modify.IndicateResponse(Success, RTP OK, Answered VM) while in the *Wait Modify Response* sub-state, the Call Control Protocol Element sends a SIP 200 OK (RTP OK, Answered VM) message to the Peer RFSS and enters the *RTP Session Open – Wait Modify ACK* state.

6.3.5.2.2 CC_Modify.IndicateResponse(Success, RTP KO) Event

The CC_Modify.IndicateResponse(Success, RTP KO) primitive is exerted by the Controlling Function when the RTP resource assigned to this group call segment is preempted given the new characteristics of the call.

Upon receiving the CC_Modify.IndicateResponse(Success, RTP KO) while in the *Wait Modify Response* sub-state, the Call Control Protocol Element sends a SIP 200 OK (RTP KO) message to the Peer RFSS and enters the *No RTP Resource at Local RFSS – Wait Modify ACK* state.

6.3.5.2.3 CC_Modify.IndicateResponse(Failure) Event

The CC_Modify.IndicateResponse(Failure) primitive is exerted by the Controlling Function when it locally detects an error after analysis of the received Re-INVITE.

Upon receiving the CC_Modify.IndicateResponse(Failure) while in the *Wait Modify Response* sub-state, the Call Control Protocol Element returns the appropriate SIP 4xx, 5xx or 6xx error response (see Section 6.4.1.6) and moves to the *RTP Session Open – No Ongoing Re-Invite* state.

6.3.5.3 RTP Session Open – Wait Modify 200 OK Sub-state

6.3.5.3.1 SIP 200 OK (RTP OK, Answered VM) Event

Upon receiving the SIP 200 OK(RTP OK, Answered VM) message while in the *RTP Session Open – Wait Modify 200 OK* state, the Call Control Protocol Element exerts a CC_Modify.Response(Success, RTP OK, Answered VM) primitive to the Controlling Function and moves to the *RTP Session Open – Wait Modify Confirm* state.

The Controlling Function:

1. updates the priority of the Call if it was changed by the CC_Modify.Request.
2. updates the RF resource allocation state if it was present in the CC_Modify.Request.
3. updates the group call segment's vocoder mode if it was changed by the CC_Modify.Request and takes additional action as required by local policy.
4. exerts a CC_Modify.Confirm primitive to the Call Control Protocol Element.

6.3.5.3.2 SIP 200 OK (RTP KO) Event

Upon receiving the SIP 200 OK(RTP KO) message while in the *RTP Session Open – Wait Modify 200 OK* state, the Call Control Protocol Element exerts a CC_Modify.Response(Success, RTP KO) primitive to the Controlling Function and moves to the *No RTP resource at Peer RFSS – Wait Modify Confirm* state.

The Controlling Function:

1. Releases the RTP resource assigned to the group call segment and exerts a TC.Destroy primitive to the Transmission Control protocol (see Section 7) to tear down the RTP session.
2. Exerts a CC_Modify.Confirm primitive to the Call Control Layer.

6.3.5.3.3 SIP 481, 408, or Timeout Event

Upon receiving a SIP Timeout, SIP 481 or SIP 408 error response, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the Controlling Function and moves to the *Final* state.

The Controlling Function:

1. tears down the RTP session by exerting a TC.Destroy primitive to the Transmission Control protocol and considers the group call segment to be torn down.
2. if it is the Home RFSS, tries to setup this group call segment again under conditions described in Section 6.3.3.

6.3.5.3.4 SIP 4xx, 5xx, 6xx other than 481 or 408 Event

Upon receiving a SIP 4xx, 5xx or 6xx error response other than 481 or 408, the Call Control Protocol Element exerts a CC_Modify.Response(Failure) primitive to the Controlling Function and moves to the *RTP Session Open – No Ongoing Re-Invite* state.

The Controlling Function:

1. considers the state of the group call segment to be the same as it was before generating the CC_Modify.Request.
2. tries to change the group call segment characteristics again by exerting CC_Modify Request upon its local policy.

6.3.5.4 RTP Session Open – Wait Modify 200 OK Tear Down RTP Sub-state

6.3.5.4.1 SIP 200 OK (RTP KO) Event

Upon receiving the SIP 200 OK(RTP KO) message while in the *RTP Session Open – Wait Modify 200 OK to tear down RTP session* state, the Call Control Protocol Element exerts a CC_Modify.Response(Success, RTP KO) primitive to the Controlling Function and moves to the *No RTP resource at Local RFSS – Wait Modify Confirm* state.

The Controlling Function:

1. releases the RTP resources assigned to the group call segment and exerts a TC.Destroy primitive to the Transmission Control protocol (see Section 7) to tear down the RTP session for this call segment.
2. Exerts a CC_Modify.Confirm primitive to the Call Control protocol.

6.3.5.4.2 SIP 481, 408, or Timeout Event

Upon receiving a SIP Timeout, SIP 481 or SIP 408 error response, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the Controlling Function and moves to the *Final* state.

The Controlling Function:

1. tears down the RTP session by exerting a TC.Destroy primitive to the Transmission Control protocol and considers the group call segment to be torn down.
2. if it is the Home RFSS, tries to setup this group call segment again under conditions described in Section 6.3.3.

6.3.5.4.3 SIP 4xx, 5xx, 6xx other than 481 or 408 Event

Upon receiving a SIP 4xx, 5xx or 6xx error response other than 481 or 408, the Call Control Protocol Element exerts a CC_Modify.Response(Failure) primitive to the Controlling Function and moves to the *RTP Session Open – No Ongoing Re-Invite* state.

The Controlling Function:

1. considers the state of the group call segment to be the same as it was before generating the CC_Modify.Request.
2. tries to change the group call segment characteristics again by exerting CC_Modify.Request upon its local policy.

6.3.5.5 RTP Session Open – Wait Modify ACK Sub-state

6.3.5.5.1 SIP ACK Event

Upon receiving a SIP ACK message while in the *RTP Session Open – Wait Modify ACK* state, the Call Control Protocol Element exerts a

CC_Modify.IndicateResponseConfirm primitive to the Controlling Function and moves the *RTP Session Open – No Ongoing Re-Invite* state.

The Controlling Function is finished with the call segment characteristic modification procedure when this is received.

6.3.5.5.2 SIP Timeout Event

Upon receiving a SIP Timeout Event while in the *RTP Session Open – Wait Modify ACK* state, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the Controlling Function and moves to the *Final* state. The SIP Agent sends a SIP BYE message to the Peer RFSS and handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function:

1. tears down the RTP session by exerting a TC.Destroy primitive to the Transmission Control protocol and considers the group call segment to be torn down.
2. if it is the Home RFSS, tries to setup this group call segment again under conditions described in Section 6.3.3.

6.3.5.6 RTP Session Open – Wait Modify Confirm Sub-state

6.3.5.6.1 CC_Modify.Confirm Event

The CC_Modify.Confirm primitive is exerted by the Controlling Function when it receives a CC_Modify.Response primitive from the Call Control Protocol Element.

Upon receiving a “CC_Modify.Confirm” while in the “RTP Session Open – Wait Modify Confirm” state, the Call Control sends a SIP ACK to the Peer RFSS and returns to the *RTP Session Open – No Ongoing Re-Invite* state.

6.3.6 Sub-state Model: Group Call - No RTP Resource at Peer RFSS State

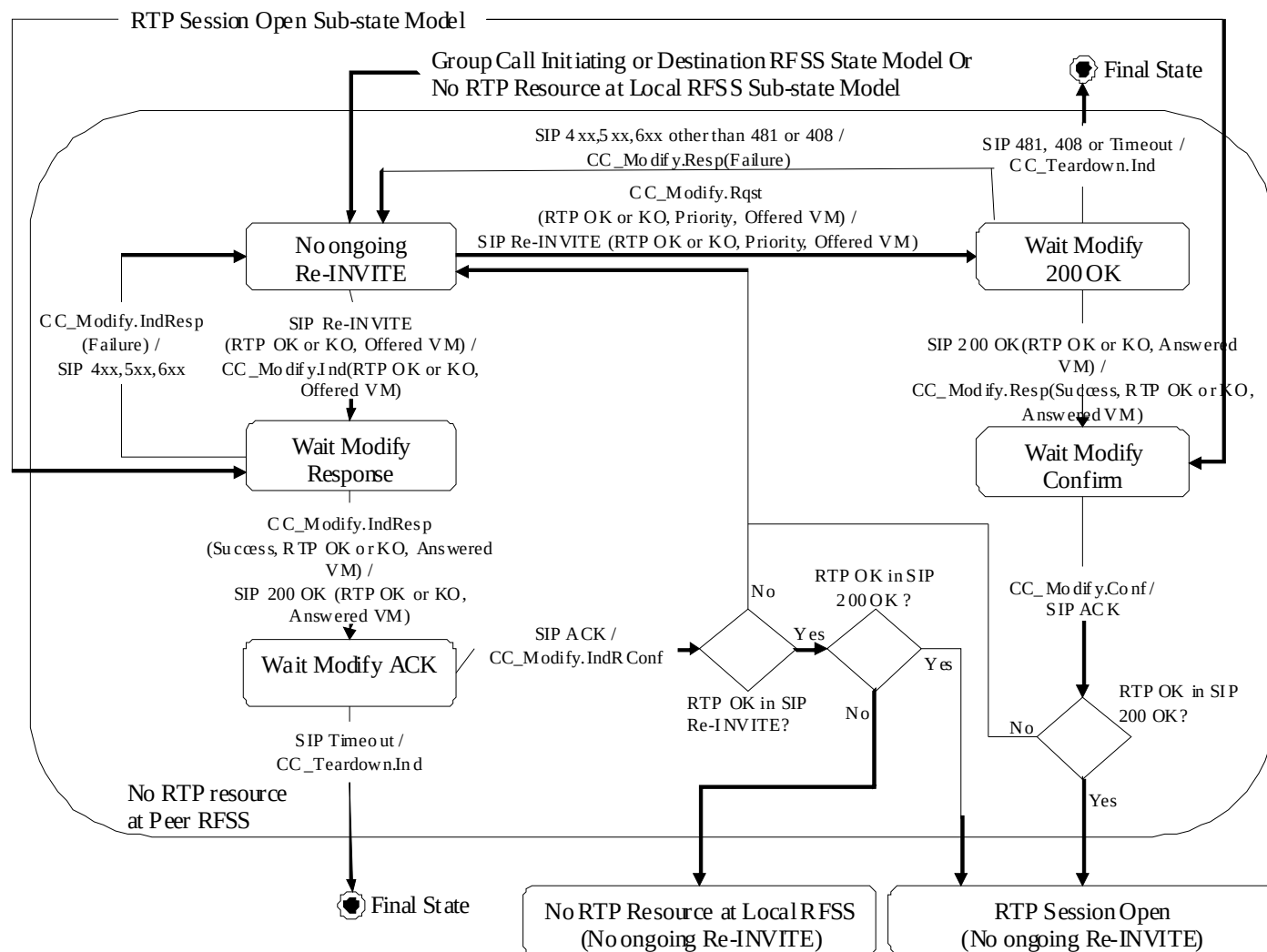


Figure 71 - RFSS Sub-State Model - No RTP Resource at Peer RFSS State

The sub-state model in Figure 71 shows:

- The 5 possible sub-states when the Call Control Protocol Element is in the *No RTP resource at Peer RFSS* state.
- Transitions from the Group Call Originating RFSS and Group Call Destination RFSS State Models, the RTP Session Open Sub-state Model and the No RTP Resource at Local RFSS Sub-state Model.
- Transitions to sub-states of the RTP Session Open Sub-state Model and the No RTP Resource at Local RFSS Sub-state Model.

This sub-state model does not show the transition to the *Final* State from any sub-state upon receipt of CC_Teardown.Request or SIP BYE events. These events appear only in the Group Call Originating RFSS and Group Call Destination RFSS State Models (see Section 6.3.3.4).

6.3.6.1 No RTP Resource at Peer RFSS – No Ongoing Re-INVITE Sub-state

This sub-state is reached for a group call segment:

1. At the Originating RFSS, after successful Group Call Segment Setup, or
2. At the Destination RFSS, after successful Group Call Segment Setup

When the Call Control Protocol Element is in the *RTP Session Open – No ongoing Re-INVITE* state for a group call segment, it MAY at any time:

1. Receive a SIP Re-INVITE request from the Peer RFSS, or
2. Receive a CC_Modify.Request from the Controlling Function requesting modification of the characteristics of the group call segment.

6.3.6.1.1 SIP Re-INVITE(RTP OK or RTP KO, Priority, RF State, Offered VM) Event

The SIP Re-INVITE(RTP OK or RTP KO, Priority, RF State, Offered VM) message represents receipt of either a SIP Re-INVITE (RTP OK, Priority, RF State, Offered VM) message or a SIP Re-INVITE(RTP KO, Priority) message.

1. If RTP resources are allocated at Peer RFSS:
The SIP Re-INVITE(RTP OK, Priority, RF State, Offered VM) message from a Peer RFSS conveys a type 1 SDP Offer (see Section 3.7 on SDP Offer/Answer Usage). Basically, this means the Peer RFSS has allocated its RTP resource to serve the RTP session. This message has three parameters besides 'RTP OK':
 - a. The Priority which can be higher than the one previously used to set up the RTP session (see Section 6.6.2 for SIP Priority header field definition).
 - b. The RF resource allocation state parameter (see Section 3.8.5.2 for the RF Resource availability parameter in the ISSI SIP body), which MAY be present when the receiving RFSS is the Home RFSS for a confirmed group call.
 - c. The Offered VM parameter which can be a different vocoder mode than the one the group call segment is currently allowed to use (see the SDP media description Section 3.7.2.2 for the offered vocoder modes)
2. If RTP resources are not allocated at Peer RFSS:
The SIP Re-INVITE(RTP KO, Priority) message from a Peer RFSS conveys a type 2 SDP Offer. This means the Peer RFSS is sending a SIP heartbeat request. The Priority is indicated but not the RF Resource allocation state.

Upon receipt of the SIP Re-INVITE(RTP OK or RTP KO, Priority, RF State, Offered VM) message, the Call Control Protocol Element exerts a CC_Modify.Indicate(RTP OK or KO, Offered VM) primitive to the RFSS's Controlling Function and moves to the *No Resource at Peer RFSS – Wait Modify Response* state for this group call segment.

The Controlling Function:

1. acknowledges with a CC_Modify.IndicateResponse(Failure) if it detects an error after analysis of the received Re-INVITE.
2. if the RTP State parameter in the primitive is equal to RTP OK
 - a. if it does not have RTP resources allocated, attempts to allocate RTP resources, taking into account the received priority.
 - b. if it is the Home RFSS, takes into account the RF resource allocation state if it is present.
 - c. acknowledges:
 - i. with a CC_Modify.IndicateResponse(Success, RTP OK, Answered VM), if RTP resources are allocated and it accepts the offered vocoder mode.
 - ii. with a CC_Modify.IndicateResponse(Success, RTP KO), if RTP resources are not allocated.
3. if the RTP State parameter in the primitive is equal to RTP KO, acknowledges with a CC_Modify.IndicateResponse(Success, RTP KO).

6.3.6.1.2 CC_Modify.Request(RTP OK or RTP KO, Priority, RF State, Offered VM) Event

The CC_Modify.Request(RTP OK, Priority, RF State, Offered VM) primitive from the Controlling Function is exerted when an RFSS has RTP resources allocated to a call segment and:

1. a Serving RFSS wants to promote the priority of the Call, or
2. a Home RFSS wants to promote the priority of the Call, or
3. a Serving RFSS wants to report the RF resource allocation state to the Home RFSS in a confirmed group call, or
4. a Serving RFSS wants to change the allowed vocoder mode of the group call segment, or
5. a Home RFSS wants to change the allowed vocoder mode of the group call segment, or
6. any combination of bullet 1, bullet 3, and bullet 4, or.
7. bullet 2 and bullet 5 combined.

The CC_Modify.Request(RTP KO, Priority) primitive from the Controlling Function is exerted when an RFSS does not have RTP resources allocated to a call segment and :

1. the RFSS wants to promote the priority of the Call, or
2. the Home RFSS wants to send a SIP heartbeat to the Serving RFSS.

After reception of this event, the Call Control Protocol Element triggers a SIP Re-INVITE request to the Peer RFSS and transitions to the *No RTP Resource at Peer RFSS – Wait Modify 200 OK* state for this call segment.

6.3.6.2 No RTP Resource at Peer RFSS – Wait Modify Response Sub-state

This sub-state is reached for a group call segment:

1. From the *No RTP resource at Peer RFSS – No Ongoing Re-Invite* state, after the transition described in Section 6.3.6.1.1, or
2. From the *RTP Session Open – No Ongoing Re-Invite* state, after the transition described in Section 6.3.5.1.2.

6.3.6.2.1 CC_Modify.IndicateResponse(Success, RTP OK or RTP KO, Answered VM) Event

1. The CC_Modify.IndicateResponse(Success, RTP OK, Answered VM) primitive will be exerted by the Controlling Function in response to a CC_Modify.Indicate(RTP OK, Priority, RF State, Offered VM) primitive if it has RTP resources allocated to a call segment.
2. The CC_Modify.IndicateResponse(Success, RTP KO) primitive will be exerted by the Controlling Function in response to a CC_Modify.Indicate(RTP OK, Priority, RF State, Offered VM) primitive if it does not have RTP resources allocated to a call segment.
3. The CC_Modify.IndicateResponse(Success, RTP KO) primitive will be exerted by the Controlling Function in response to a CC_Modify.Indicate(RTP KO, Priority, RF State) primitive because a type 2 answer is required to a type 2 SDP Offer (see Section 3.7 on SDP offer/answer usage).

Upon receiving the CC_Modify.IndicateResponse(Success, RTP OK or RTP KO, Answered VM) while in the *Wait Modify Response* sub-state, the Call Control Protocol Element sends a SIP 200 OK message to the Peer RFSS and moves to the *No RTP Resource at Peer RFSS – Wait Modify ACK* state.

6.3.6.2.2 CC_Modify.IndicateResponse(Failure) Event

The CC_Modify.IndicateResponse(Failure) primitive is exerted by the Controlling Function when it locally detects a failure upon analysis of the received Re-INVITE.

Upon receiving the CC_Modify.IndicateResponse(Failure) while in the *Wait Modify Response* sub-state, the Call Control Protocol Element returns an appropriate SIP 4xx, 5xx or 6xx error response and moves to the *No RTP resource at Peer RFSS – No Ongoing Re-Invite* state.

6.3.6.3 No RTP Resource at Peer RFSS – Wait Modify 200 OK Sub-state

6.3.6.3.1 SIP 200 OK (RTP OK or RTP KO, Answered VM) Event

Upon receiving the SIP 200 OK(RTP OK or RTP KO, Answered VM) message while in the *No RTP resource at Peer RFSS – Wait Modify 200 OK*, the Call Control Protocol Element exerts a CC_Modify.Response(Success, RTP OK or RTP KO, Answered VM) primitive to the Controlling Function and moves to the *No RTP at Peer RFSS – Wait Modify Confirm* state.

The Controlling Function:

1. exerts a CC_Modify.Confirm primitive to the Call Control Protocol Element.
2. if CC_Modify.Response(Success, RTP OK, Answered VM) is received, exerts a TC.Create primitive to the Transmission Control (see Section 7).

6.3.6.3.2 SIP 481, 408, or Timeout Event

Upon receiving a SIP Timeout, a SIP 481 error response, or a SIP 408 error response, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the Controlling Function and moves to the *Final* state.

The Controlling Function:

1. considers the group call segment to be torn down.
2. if it is the Home RFSS, tries to setup this group call segment again under conditions described in Section 6.3.3.

6.3.6.3.3 SIP 4xx, 5xx, 6xx other than 481 or 408 Event

Upon receiving a SIP 4xx, 5xx or 6xx error response other than SIP 481 or SIP 408, the Call Control Protocol Element exerts a CC_Modify.Response(Failure) primitive to the Controlling Function and moves to the *No RTP Resource at Peer RFSS – No Ongoing Re-Invite* state.

The Controlling Function:

1. considers the state of the group call segment to be the same as it was before generating the CC_Modify.Request.
2. tries to change the group call segment characteristics again by exerting CC_Modify Request upon its local policy.

6.3.6.4 No RTP Resource at Peer RFSS – Wait Modify ACK Sub-state

6.3.6.4.1 SIP ACK Event

Upon receiving a SIP ACK message while in the *No RTP Resource at Peer RFSS – Wait Modify ACK* sub-state, the Call Control Protocol Element exerts a CC_Modify.IndicateResponseConfirm primitive to the Controlling Function and:

1. If RTP OK was indicated in the SIP Re-INVITE from the Peer RFSS
 - a. If RTP OK was indicated in the SIP 200 OK by the Local RFSS, the Call Control Protocol Element moves to the *RTP Session Open – No Ongoing Re-Invite* state and exerts a TC.Create primitive to the Transmission Control protocol to establish the RTP session for this call segment.
 - b. Otherwise, the Call Control Protocol Element moves to the *No RTP Resource at Local RFSS – No Ongoing Re-Invite* state.
2. Otherwise, if RTP KO was indicated in the SIP Re-INVITE by the Peer RFSS, the Call Control Protocol Element moves to the *No RTP Resource at Peer RFSS – No Ongoing Re-Invite* state.

The Controlling Function is finished with the call segment characteristic modification procedure when this is received.

6.3.6.4.2 SIP Timeout Event

Upon receiving a SIP Timeout Event while in the *No RTP Resource at Peer RFSS – Wait Modify ACK* state, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the Controlling Function and moves to the *Final* state. The SIP Agent sends a SIP BYE message to the Peer RFSS and handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function :

1. considers the group call segment to be torn down.
2. if it is the Home RFSS, tries to setup this group call segment again under conditions described in Section 6.3.3.

6.3.6.5 No RTP Resource at Peer RFSS – Wait Modify Confirm Sub-state

This sub-state is reached for a group call segment:

1. from the *No RTP resource at Peer RFSS – Wait Modify 200 OK* state, after the transition described in Section 6.3.6.3.1, or
2. from the *RTP Session Open – Wait Modify 200 OK* state, after the transition described in Section 6.3.5.3.2.

6.3.6.5.1 Event CC_Modify.Confirm

The CC_Modify.Confirm primitive is exerted by the Controlling Function when it receives a CC_Modify.Response primitive from the Call Control Protocol Element.

Upon receiving a CC_Modify.Confirm primitive while in the *RTP Session Open – Wait Modify Confirm* state, the Call Control Protocol Element sends a SIP ACK message to the Peer RFSS and:

1. If RTP OK was indicated in the SIP 200 OK by the Peer RFSS, RTP resources are allocated at the Local RFSS, so the Call Control Protocol Element moves to the *RTP Session Open – No Ongoing Re-Invite* state.
2. Otherwise, if RTP KO was indicated in the SIP 200 OK by the Peer RFSS, the Call Control Protocol Element moves to the *No RTP Resource at Peer RFSS – No Ongoing Re-Invite* state.

6.3.7 Sub-state Model: Group Call - No RTP Resource at Local RFSS State

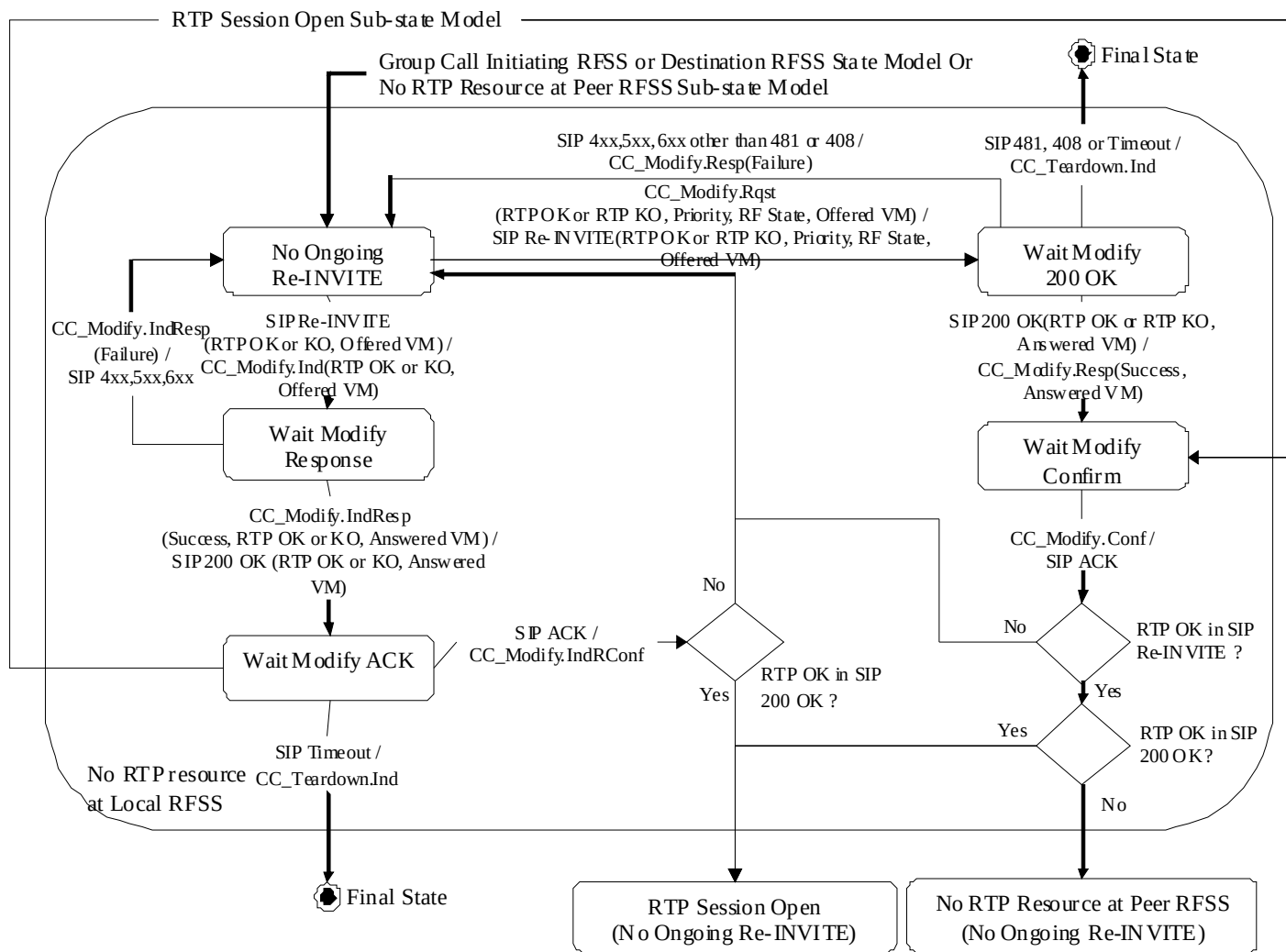


Figure 72 - RFSS Sub-State Model - No RTP Resource at Local RFSS State

The sub-state model in Figure 72 shows:

- The 5 possible sub-states when the Call Control Protocol Element is in the *No RTP resource at Local RFSS* state.
- Transitions from the Group Call Originating RFSS and Group Call Destination RFSS State Models, the *RTP Session Open* Sub-state Model, and the *No RTP Resource at Peer RFSS* Sub-state Model.
- Transitions to sub-states of the *RTP Session Open* Sub-state Model and the *No RTP resource at Peer RFSS* Sub-state Model.

This sub-state model does not show:

- The transition to the *Final* State from any sub-state upon receipt of CC_Teardown.Request or SIP BYE events. These events appear only in the Group Call Originating RFSS and Group Call Destination RFSS State Models (see Section 6.3.3.4).
- Colliding Re-INVITE situation (see Section 6.6.3).

6.3.7.1 No RTP Resource at Local RFSS – No Ongoing Re-INVITE Sub-state

This sub-state is reached for a group call segment:

1. at the Originating RFSS, after successful Group Call Segment Setup, or
2. at the Destination RFSS, after successful Group Call Segment Setup.

When the Call Control Protocol Element is in the *RTP Session Open – No Ongoing Re-INVITE* state for a group call segment, it MAY at any time:

1. receive a SIP Re-INVITE request from the Peer RFSS, or
2. receive a CC_Modify.Request from the Controlling Function requesting modification of the characteristics of the group call segment.

6.3.7.1.1 SIP Re-INVITE(RTP OK or KO, Priority, RF State, Offered VM) Event

The SIP Re-INVITE(RTP OK or RTP KO, Priority, RF State, Offered VM) message represents receipt of either a SIP Re-INVITE (RTP OK, Priority, RF State, Offered VM) message or a SIP Re-INVITE(RTP KO, Priority) message.

1. If RTP resources are allocated at Peer RFSS:
The SIP Re-INVITE(RTP OK, Priority, RF State, Offered VM) message from a Peer RFSS conveys a type 1 SDP Offer (see Section 3.7 on SDP Offer/Answer Usage). Basically, this means the Peer RFSS has allocated its RTP resource to serve the RTP session. This message has three parameters besides 'RTP OK':
 - a. The Priority which can be higher than the one previously used to set up the RTP session (see Section 6.4.3 for SIP Priority header field definition).
 - b. The RF resource allocation state parameter (see Section 3.8.5.2 for the RF Resource availability parameter in the ISSI SIP body), which MAY be present when the receiving RFSS is the Home RFSS for a confirmed group call.
 - c. The Offered VM parameter which can be a different vocoder mode than the one the group call segment is currently allowed to use (see the SDP media description Section 3.7.2.2 for the offered vocoder modes)
2. If RTP resources are not allocated at Peer RFSS:
The SIP Re-INVITE(RTP KO, Priority) message from a Peer RFSS conveys a type 2 SDP Offer. This means the Peer RFSS (Home RFSS) is sending a SIP heartbeat request. The Priority is indicated but not the RF

Resource allocation state, nor does the Peer RFSS request to modify the vocoder mode.

Upon receipt of the SIP Re-INVITE(RTP OK or RTP KO, Priority, RF State, Offered VM) message, the Call Control Protocol Element exerts a CC_Modify.Indicate(RTP OK or KO, Priority, RF State, Offered VM) primitive to the RFSS's Controlling Function and moves to the *No Resource at Local RFSS – Wait Modify Response* state for this group call segment.

The Controlling Function:

1. acknowledges with a CC_Modify.IndicateResponse(Failure) if it detects an error after analysis of the received Re-INVITE.
2. if the RTP State parameter in the primitive is equal to RTP OK
 - a. attempts to allocate RTP resources, taking into account the received priority.
 - b. if it is the Home RFSS, takes into account the RF resource allocation state if it is present.
 - c. takes into account the Offered VM parameter if it is different than the current vocoder mode allowed on the group call segment.
 - d. acknowledges:
 - i. with a CC_Modify.IndicateResponse(Success, RTP OK, Answered VM), if RTP resources are allocated.
 - ii. with a CC_Modify.IndicateResponse(Success, RTP KO), if RTP resources are not allocated.
3. if the RTP State parameter in the primitive is equal to RTP KO, acknowledges with a CC_Modify.IndicateResponse(Success, RTP KO).

6.3.7.1.2 CC_Modify.Request(RTP OK or RTP KO, Priority, RF State, Offered VM)) Event

The CC_Modify.Request(RTP OK, Priority, RF State, Offered VM)) primitive from the Controlling Function is exerted when the RFSS has RTP resources allocated to the call segment. In addition:

1. a Serving RFSS MAY want to promote the priority of the Call, or
2. a Home RFSS MAY want to promote the priority of the Call, or
3. a Serving RFSS needs to report the RF resource allocation state to the Home RFSS in a confirmed group call, or
4. a Serving RFSS wants to change the allowed vocoder mode of the group call segment, or
5. a Home RFSS wants to change the allowed vocoder mode of the group call segment, or
6. any combination of bullet 1, bullet 3, and bullet 4.
7. bullet 2 and bullet 5 combined.

The CC_Modify.Request(RTP KO, Priority) primitive from the Controlling Function is exerted when a Home RFSS wants to send a SIP heartbeat to the Serving RFSS or promote the priority of the call.

After reception of this event, the Call Control triggers a SIP Re-INVITE request to the Peer RFSS and transitions to the *No RTP Resource at Local RFSS – Wait Modify 200 OK* state for this call segment.

6.3.7.2 No RTP Resource at Local RFSS – Wait Modify Response Sub-state

6.3.7.2.1 CC_Modify.IndicateResponse(Success, RTP OK or RTP KO, Answered VM) Event

1. The CC_Modify.IndicateResponse(Success, RTP OK, Answered VM) primitive will be exerted by the Controlling Function in response to a CC_Modify.Indicate(RTP OK, Priority, RF State, Offered VM) primitive if it has RTP resources allocated to a call segment and it agrees with one of the offered vocoder mode(s).
2. The CC_Modify.IndicateResponse(Success, RTP KO) primitive will be exerted by the Controlling Function in response to a CC_Modify.Indicate(RTP OK, Priority, RF State, Offered VM) primitive if it does not have RTP resources allocated to a call segment.
3. The CC_Modify.IndicateResponse(Success, RTP KO) primitive will be exerted by the Controlling Function in response to a CC_Modify.Indicate(RTP KO, Priority, RF State, Offered VM) primitive because a type 2 answer is required to a type 2 SDP Offer (see Section 3.7 on SDP offer/answer usage).

Upon receiving the CC_Modify.IndicateResponse(Success, RTP OK or RTP KO, Answered VM) while in the *Wait Modify Response* sub-state, the Call Control Protocol Element sends a SIP 200 OK message to the Peer RFSS and enters the *No RTP Resource at Local RFSS – Wait Modify ACK* state.

6.3.7.2.2 CC_Modify.IndicateResponse(Failure) Event

The CC_Modify.IndicateResponse(Failure) primitive is exerted by the Controlling Function when it locally detects an error upon analysis of the received Re-INVITE.

Upon receiving the CC_Modify.IndicateResponse(Failure) while in the *Wait Modify Response* sub-state, the Call Control Protocol Element returns an appropriate SIP 4xx, 5xx or 6xx error response and moves to the *No RTP resource at Local RFSS – No Ongoing Re-Invite* state.

6.3.7.3 No RTP Resource at Local RFSS – Wait Modify 200 OK Sub-state

6.3.7.3.1 SIP 200 OK (RTP OK or RTP KO, Answered VM) Event

Upon receiving the SIP 200 OK(RTP OK or RTP KO, Answered VM) message while in the *No RTP Session at Local RFSS – Wait Modify 200 OK* state, the Call

Control Protocol Element exerts a CC_Modify.Response(Success, RTP OK or RTP KO, Answered VM) primitive to the Controlling Function and moves to the *No RTP at Local RFSS – Wait Modify Confirm* state.

The Controlling Function:

1. exerts a CC_Modify.Confirm primitive to the Call Control Protocol Element
2. if CC_Modify.Response(Success, RTP OK, Answered VM) is received and the RTP session was not set-up before this re-invite procedure, exerts a TC.Create primitive to the Transmission Control (see Section 7).

6.3.7.3.2 SIP 481, 408, or Timeout Event

Upon receiving a SIP Timeout, a SIP 481 or a SIP 408 error response, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the Controlling Function and moves to the *Final* state.

The Controlling Function :

1. considers the group call segment to be torn down.
2. if it is the Home RFSS, tries to setup this group call segment again under conditions described in Section 6.3.3.

6.3.7.3.3 SIP 4xx, 5xx, 6xx other than 481 or 408 Event

Upon receiving a SIP 4xx, 5xx or 6xx error response other than 481 or 408, the Call Control Protocol Element exerts a CC_Modify.Response(Failure) to the Controlling Function and moves to the *No RTP resource at Local RFSS – No Ongoing Re-Invite* state

The Controlling Function:

1. considers the state of the group call segment to be the same as it was before generating the CC_Modify.Request.
2. tries to change the group call segment characteristics again by exerting CC_Modify Request upon its local policy.

6.3.7.4 No RTP Resource at Local RFSS – Wait Modify ACK Sub-state

This sub-state is reached for a group call segment:

1. from the *No RTP resource at Local RFSS – Wait Modify Response* state, after the transition described in Section 6.3.7.2.1, or
2. from the *RTP Session Open – Wait Modify Response* state, after the transition described in Section 6.3.5.2.2.

6.3.7.4.1 SIP ACK Event

Upon receiving a SIP ACK message while in the *No RTP Resource at Local RFSS – Wait Modify ACK* state, the Call Control Protocol Element exerts a `CC_Modify.IndicateResponseConfirm` primitive to the Controlling Function and:

1. If RTP OK was indicated in the SIP 200 OK by the Local RFSS, the Call Control Protocol Element moves to the *RTP Session Open – No Ongoing Re-Invite* state and exerts a `TC.Create` primitive to the Transmission Control protocol to establish the RTP session for this call segment.
2. Otherwise, if RTP KO was indicated in the SIP 200 OK by the Local RFSS, the Call Control Protocol Element moves to the *No RTP Resource at Local RFSS – No Ongoing Re-Invite* state.

The Controlling Function is finished with the call segment characteristic modification procedure when this is received.

6.3.7.4.2 SIP Timeout Event

Upon receiving a SIP Timeout while in the *No RTP Resource at Local RFSS – Wait Modify ACK* state, the Call Control Protocol Element exerts a `CC_Teardown.Indicate` primitive to the Controlling Function and moves to the *Final* state. The SIP Agent sends a SIP BYE message to the Peer RFSS and handles any SIP response to the SIP BYE message as described in Section 6.4.4.1.

The Controlling Function :

1. considers the group call segment to be torn down.
2. if it is the Home RFSS, tries to setup this group call segment again under conditions described in Section 6.3.3.

6.3.7.5 No RTP Resource at Local RFSS – Wait Modify Confirm Sub-state

This sub-state is reached for a group call segment:

1. from the *No RTP resource at Local RFSS – Wait Modify 200 OK* state, after the transition described in Section 6.3.7.3.1, or
2. from the *RTP Session Open – Wait Modify 200 OK to tear down RTP Session* state, after the transition described in Section 6.3.5.4.1.

6.3.7.5.1 CC_Modify.Confirm Event

The `CC_Modify.Confirm` primitive is exerted by the Controlling Function when it receives a `CC_Modify.Response` primitive from the call control protocol.

Upon receiving a `CC_Modify.Confirm` primitive while in the *No RTP Resource at Local RFSS – Wait Modify Confirm* state, the Call Control Protocol Element sends a SIP ACK message to the Peer RFSS and:

1. If RTP OK was indicated in the SIP Re-INVITE by the Local RFSS

- a. If RTP KO was indicated in the SIP 200 OK by the Peer RFSS, the Call Control Protocol Element moves to the *RTP Session Open – No Ongoing Re-Invite* state.
- b. Otherwise, the Call Control Protocol Element moves to the *No RTP Resource at Peer RFSS – No Ongoing Re-Invite* state.
2. Otherwise, if RTP KO was indicated in the SIP Re-INVITE by the Local RFSS, the Call Control Protocol Element moves to the *No RTP Resource at Local RFSS – No Ongoing Re-Invite* state.

6.4 Basic ISSI SIP Control Procedures

Basic ISSI SIP control procedures specify generic RFSS behavior for initiation and termination of a SIP session between two RFSSs over the ISSI, as well as subsequent initiation and termination of an RTP session within the SIP session. For SU-to-SU and group call control over the ISSI, these basic control procedures apply, subject to qualifications detailed in Sections 6.5 and 6.6 of this document.

An originating RFSS is an RFSS that sends a SIP request. A destination RFSS is an RFSS that returns SIP responses to a request from an originating RFSS.

In the context of the following procedures, an RTP resource is a port number assigned by an RTP resource manager, plus other manufacturer-specific resources (e.g., processing and memory resources) required to support an RTP session.

6.4.1 Initiating an ISSI SIP Session

When an RFSS desires to initiate a SIP session with another RFSS over the ISSI, it sends an INVITE request to the other RFSS, according to the procedures in the following subsections.

The originating RFSS SHALL allocate an RTP resource before initiating an INVITE request to a destination RFSS.

6.4.1.1 INVITE Request

The originating RFSS SHALL create and send an INVITE request to the destination RFSS, following the procedures in Section 13.2.1 of [RFC3261], with SIP header fields, SDP body, and possibly an ISSI-specific body as specified in Section 3 of this document.

The originating RFSS SHALL retransmit the INVITE request following the procedures, for unreliable transports of INVITE requests over UDP, in Section 17.1.1 of [RFC3261]. The following qualifications apply:

1. In the event that a response is not received upon expiry of Timer B (as defined in **[RFC3261]**), the originating RFSS SHALL terminate the INVITE request and SHALL destroy all stored information related to this SIP session.
2. In the event that a 3xx, 4xx, 5xx, or 6xx error response is received, the originating RFSS SHALL terminate the INVITE request and SHALL destroy all stored information related to this SIP session.

The destination RFSS SHALL process the INVITE request following the procedures in Section 13.3.1 of **[RFC3261]**.

Upon receipt of an INVITE request, the destination RFSS determines its role based on the information in the Request URI, the Route header field, the From header field, and the To header field, as shown in Table 44.

Table 44 - How an RFSS determines its role upon receipt of a SIP INVITE Request

Information in Received INVITE Request	Expected Role of Destination RFSS
Route header field is present and the radical name is equal to "TIA-P25-U2Uorig"	Calling Home RFSS (SU-to-SU Call Setup)
Route header field is present and the radical name is equal to "TIA-P25-U2Udest"	Called Home RFSS (SU-to-SU Call Setup)
Request URI and To header field encode the same SUID number.	Called Serving RFSS (SU-to-SU Call Setup and In-Call Roaming)
Request URI and From header field encode the same SUID number.	Calling Serving RFSS (In-Call Roaming)
Request URI and To header field encode the same SGID number.	Serving RFSS of the SG (Group Call Setup)
Route header field is present and the radical name is equal to "TIA-P25-GroupCall"	Home RFSS of the SG (Group Call Setup)

6.4.1.2 100 (TRYING) Provisional Response

Upon receipt of an INVITE request, the destination RFSS MAY return a 100 (TRYING) provisional response to the preceding SIP entity, indicating that the INVITE has been received, as specified in Section 21 of **[RFC3261]**. The 100 (TRYING) response SHALL contain SIP header fields as specified in Section 3 of this document.

Upon receipt of a 100 (TRYING) response to its INVITE request, the originating RFSS SHALL stop retransmissions of the INVITE request, following the procedures in Section 21.1.1 of **[RFC3261]**.

6.4.1.3 180 (RINGING) Provisional Response

For SU-to-SU calls only, following a 100 (TRYING) response, the destination RFSS MAY also return a 180 (RINGING) provisional response to indicate that the called SU has been alerted. The 180 (RINGING) response SHALL contain SIP header fields as specified in Section 3 of this document.

6.4.1.4 183 (SESSION PROGRESS) Provisional Response

For SU-to-SU calls only, following a 180 (RINGING) response, if any, the destination RFSS MAY send a VCE_NO_RF_AT_CALLED_SERV 183 provisional response to indicate that the called radio site currently does not have any RF resource to serve the SU-to-SU call. This provisional response SHALL contain SIP header fields as specified in Section 3 of this document.

6.4.1.5 200 (OK) Final Response

The destination RFSS SHALL return a 200 (OK) final response to the originating RFSS to indicate that the INVITE request has been granted, following the procedures in Section 13.3.1.4 of **[RFC3261]**, with SIP header fields, SDP body, and possibly an ISSI-specific body as specified in Section 3 of this document. The destination RFSS SHALL retransmit the 200 (OK) response following the procedures (for unreliable transports over UDP) in Section 13.3.1.4 of **[RFC3261]**.

The originating RFSS SHALL process the 200 (OK) response, following the procedures in Section 13.2.2.4 of **[RFC3261]**, and SHALL consider the SIP session successfully set up.

Before receiving the 200 (OK) response, the originating RFSS MAY cancel the set up of the corresponding SIP session, following procedures in Section 6.4.2 of this document.

After receiving this 200 (OK) message, the originating RFSS SHALL send either an ACK request in the event that it considers the SIP session successfully set up, or a BYE request otherwise.

In the event that, while awaiting the associated ACK request from the originating RFSS, a SIP timeout occurs, the destination RFSS SHALL return a BYE request to terminate the SIP session between the originating and destination RFSSs, following procedures in Section 6.4.4 of this document.

6.4.1.6 Other Final Responses

Under certain situations (e.g., when the destination RFSS is currently not willing or able to set up additional SIP sessions), the destination RFSS MAY reject the INVITE request and return failure responses, following procedures in Section 13.3.1.3 of [RFC3261], with qualifications as detailed in the following subsections.

The failure responses SHALL contain SIP header fields as defined in Section 3 of this document.

3xx, 4xx, 5xx, 6xx failure responses, other than the general call control failure responses described in Section 6.4.1.6.1 and the specific failure responses in Section 6.4.1.6.2, SHALL NOT be sent by the destination RFSS.

6.4.1.6.1 General Call Control Failure Responses

Further to the generic SIP failure responses specified in Annex A of this document, Table 45 shows additional general failure responses for rejecting a call control related SIP request over the ISSI:

Table 45 - Additional General Call Control Failure Responses Sent by an RFSS Over the ISSI

Failure Response	General Call Control Related Cause
487 (REQUEST TERMINATED)	The destination RFSS receives a CANCEL request that matches an existing INVITE Request.
491 (REQUEST PENDING)	The destination RFSS receives a re-INVITE request while awaiting a final response on its own re-INVITE request sent within the same dialog.

The originating RFSS SHALL process all 4xx, 5xx, and 6xx final responses, following the procedures in Section 13.2.2.3 of [RFC3261].

6.4.1.6.2 Specific Failure Responses

Specific failure responses for a group call are listed in Table 46.

Table 46 - Specific group call failure responses sent by an RFSS over the ISSI

Failure Response	Specific cause in the context of a group call
403 (FORBIDDEN) (with specific P25 Condition Code as defined in Section 3.5.18)	The Home RFSS detects that the Serving RFSS is not registered for this SGID (VCE_SG_NOT_REG). OR The Home RFSS detects that a request for a protected audio call cannot be granted (VCE_CANNOT_PROTECT_AUDIO) OR The Home RFSS detects that a request for an unprotected audio call cannot be granted (VCE_CANNOT_UNPROTECT_AUDIO)
404 (NOT FOUND) (with specific P25 Condition Code as defined in Section 3.5.18)	The Home RFSS cannot find the SGID in its Home Location Register (VCE_SG_UNKNOWN). OR The Serving RFSS has not registered this SGID with any Home RFSS (VCE_SG_UNKNOWN)
603 (DECLINE) (with specific P25 Condition Code as defined in Section 3.5.18)	The Home RFSS or the Serving RFSS rejects the INVITE request to set up the group call segment due to colliding call setup requests, as specified in Section 6.6.3 of this document (VCE_COLLIDING_CALL).
606 (NOT ACCEPTABLE) (with specific P25 Condition Code as defined in Section 3.5.18)	The Home RFSS detects that none of the vocoder mode(s) offered by the Serving RFSS is accepted by the Home RFSS (VCE_VOCODER_MODES_UNACCEPTED). OR The Serving RFSS detects that none of the vocoder mode(s) offered by the Home RFSS is accepted by the Serving RFSS (VCE_VOCODER_MODES_UNACCEPTED).

Specific failure responses for an SU-to-SU call are listed in Table 47.

Table 47 - Specific SU-to-SU call failure responses sent by an RFSS over the ISSI

Failure Response	Specific cause in the context of an SU-to-SU call
403 (FORBIDDEN) (with specific P25 Condition Codes as defined in Section 3.5.18)	The Calling Home RFSS detects that the Calling Serving RFSS is not registered for the Calling SUID (VCE_CALLING_SU_NOT_REG), OR The Called Home RFSS detects that Called SUID is not registered (VCE_CALLED_SU_NOT_REG). OR The Calling Home RFSS detects that the Calling SU is not authorized to make this call (VCE_CALLING_SU_NOT_AUTHOR) OR The Calling Home RFSS detects that the Calling SU is radio inhibited (VCE_CALLING_SU_INHIBITED). OR The Called Home RFSS detects that the Called SU is not authorized to receive this call (VCE_CALLED_SU_NOT_AUTHOR) OR The Called Home RFSS detects that the Called SU is radio inhibited (VCE_CALLED_SU_INHIBITED). OR The Called Home RFSS detects that a request for a protected audio call cannot be granted (VCE_CANNOT_PROTECT_AUDIO) OR The Called Home RFSS detects that a request for an unprotected audio call cannot be granted. (VCE_CANNOT_UNPROTECT_AUDIO)
404 (NOT FOUND) (with specific P25 Condition Codes as defined in Section 3.5.18)	The Calling Home RFSS cannot find the Calling SUID in its Home Location Register (VCE_CALLING_SU_UNKNOWN).. OR The Calling Home RFSS detects that the address of the Called Home RFSS is unknown (VCE_CALLED_HOME_UNKNOWN), OR The Called Home RFSS cannot find the Called SUID in its Home Location Register (VCE_CALLED_SU_UNKNOWN). OR The Called Serving RFSS has not registered the Called SUID with the Called Home RFSS (VCE_SU_NOT_REG_BY_CALLED_SERV).
486 (BUSY) (with specific P25 Condition Codes as defined in Section 3.5.18)	The Called Serving RFSS or the Called Home RFSS detects that the Called SU is busy or does not respond (VCE_CALLED_SU_BUSY) OR The Called SU cancels the SU-to-SU call while it was waiting for RF Resource availability (VCE_CALLED_SU_REFUSE_CALL)
603 (DECLINE) (with specific P25 Condition Code as defined in Section 3.5.18)	The Called Serving RFSS, the Called Home RFSS, the Calling Home RFSS or the Calling Serving RFSS rejects the INVITE request to set up the SU-to-SU calls due to simultaneous call setup requests from both SUs, as specified in Section 6.5.2 of this document (VCE_COLLIDING_CALL)
606 (NOT ACCEPTABLE) (with specific P25 Condition Code as defined in Section 3.5.18)	The Called Serving RFSS, the Called Home RFSS or the Calling Home RFSS detects that there is no available RTP resource to support the RTP session (VCE_NO_RTP)

Failure Response	Specific cause in the context of an SU-to-SU call
606 (NOT ACCEPTABLE) (with specific P25 Condition Code as defined in Section 3.5.18)	The Called Serving RFSS detects that RF resource, according to local policy, is not available within a reasonable timeframe (e.g., upon expiry of a local timer) (VCE_NO_RF).
606 (NOT ACCEPTABLE) (with specific P25 Condition Code as defined in Section 3.5.18)	The Called Serving RFSS, the Called Home RFSS or the Calling Home RFSS detects that none of the vocoder mode(s) offered by the adjacent RFSS is accepted by this RFSS (VCE_VOCODER_MODES_UNACCEPTED).

6.4.1.7 ACK Request

To confirm reception of a 200 (OK) or any other final response, the originating RFSS SHALL return an ACK request to the destination RFSS, with SIP header fields as specified in Section 3 of this document, and following the general guidelines in Section 17.1.1.3 of [RFC3261].

If the ACK request corresponds to a 200 (OK) response that carries a Type 1 SDP response (as defined in Section 3.7), then sending and receiving the ACK request create transmission contexts at the originating and destination RFSSs, respectively.

Upon receipt of an ACK request for the 200 (OK) response, the destination RFSS SHALL consider the SIP session successfully set up.

6.4.2 Canceling a SIP Session Setup (INVITE) Request

According to Sections 9, 12.1, 15, and 19.3 of [RFC3261], to cancel an INVITE request, an originating RFSS MAY send a:

1. CANCEL request after receiving any 1xx provisional response and before receiving any final response for the INVITE request, following procedures in Section 6.4.2.1 of this document, or
2. BYE request after receiving a 100 (TRYING) response with a tag parameter in the "To" header field, or any 101-199 provisional response, following procedures in Section 6.4.4.1 of this document.

6.4.2.1 CANCEL Method

6.4.2.1.1 CANCEL Request

The originating RFSS SHALL generate a CANCEL Request following the procedures in Section 9.1 of [RFC3261], with SIP header fields as specified in Section 3 of this document.

The destination RFSS SHALL process the CANCEL request following the procedures in Section 9.2 of [RFC3261].

6.4.2.1.2 200 OK Final Response

The destination RFSS SHALL return a 200 (OK) final response if the INVITE Request has been successfully cancelled following the procedures in Section 9.2 of [RFC3261], with SIP header fields as specified in Section 3 of this document.

The originating RFSS SHALL consider the CANCEL Request successfully processed by the destination RFSS when it receives the 200 (OK) response.

6.4.2.1.3 Other Final Responses

Other than the 200 (OK) response, only general call control failure responses as specified in Section 6.4.1.6.1 of this document are expected as possible final responses to the CANCEL request.

6.4.3 Subsequent Modification of ISSI SIP Session Characteristics

After a SIP session is set up, the originating or the destination RFSS MAY send a Re-INVITE request to modify the SDP characteristics of the SIP session.

6.4.3.1 Re-INVITE Request

The Re-INVITE Request is only supported for group calls.

A Re-INVITE request contains a new SDP offer which MAY have the same characteristics as the most previously answered SDP offer.

The originating RFSS SHALL create and send a re-INVITE request to the destination RFSS, following the procedures in Sections 14.1 of [RFC3261], with SIP header fields, SDP body, and possibly an ISSI-specific body as specified for the INVITE request in Section 3 of this document.

Procedures for retransmitting the re-INVITE request and associated qualifications are as specified for the INVITE request in Section 6.4.1.1 of this document.

The destination RFSS SHALL process the re-INVITE request following the procedures in Section 14.2 of **[RFC3261]**.

In the event that the Re-INVITE Request is denied by a 481 (CALL/TRANSACTION DOES NOT EXIST), a 408 (Request Timeout) or SIP Timeout, the group call segment SHALL be considered torn down.

In the event that the Re-INVITE Request is denied by other 3xx, 4xx, 5xx or 6xx failure responses, the group call segment characteristics remain unchanged as if no re-INVITE request has been issued. Subject to local policy, the originating RFSS MAY:

1. Retry the SIP re-INVITE request, modified as required per failure response received.

OR

2. Send a BYE request to the destination RFSS to terminate the SIP session, following procedures in Section 6.4.4 of this document.

6.4.3.2 100 (TRYING) Provisional Response

The procedures in Section 6.4.1.2 of this document apply.

6.4.3.3 200 (OK) Final Response

The destination RFSS SHALL return a 200 (OK) response to the originating RFSS to indicate that the re-INVITE request has been granted. The procedures in Section 6.4.1.5 of this document apply.

6.4.3.4 Other Final Responses

Other final responses for the Re-INVITE request are as specified for the INVITE Request in Section 6.4.1.6 of this document.

6.4.3.5 ACK Request

The procedures in Section 6.4.1.7 of this document apply.

6.4.4 Terminating an ISSI SIP Session

After a SIP session over the ISSI is successfully set up between two RFSSs (i.e., a 200 (OK) response to an INVITE request has been received), one of the RFSSs MAY initiate termination of the SIP session by sending a BYE request to the other RFSS over the ISSI, according to the procedures in the following subsections.

6.4.4.1 BYE Method

6.4.4.1.1 BYE Request

The originating RFSS SHALL create the BYE request following the procedures in Section 15.1.1 of **[RFC3261]**, with SIP header fields as specified in Section 3 of this document, and send it to the destination RFSS. The originating RFSS SHALL retransmit the BYE request following the procedures (for unreliable transports of non-INVITE requests over UDP) in Section 17.1.2 of **[RFC3261]**.

The destination RFSS SHALL process the BYE request following the procedures in Section 15.1.2 of **[RFC3261]**.

6.4.4.1.2 200 (OK) Final Response

The destination RFSS SHALL return a 200 (OK) final response to the originating RFSS to indicate that the BYE request has been granted, and that the SIP session between the originating RFSS and the destination RFSS is being terminated. The destination RFSS SHALL also destroy all stored information related to this SIP session.

Upon receipt of a 200 (OK) response to its BYE request, the originating RFSS SHALL stop retransmissions of the BYE request, following the procedures in Section 17.1.2 of **[RFC3261]**. The originating RFSS SHALL also destroy all stored information related to this SIP session.

6.4.4.1.3 Other Final Responses

6.4.4.1.3.1 General Failure Responses

General call control failure responses for a SIP BYE Request are as specified for the INVITE message in Section 6.4.1.6.1 of this document.

6.4.4.1.3.2 Specific Failure Responses

Specific failure response for a SIP BYE Request as specified in Section 6.4.1.6.2 of this document.

6.5 Call Control Procedures for ISSI SU-to-SU Voice Services

Seven types of call control procedures are applicable to SU-to-SU voice services over the ISSI, as specified in the subsections below:

1. ISSI SU-to-SU Voice Call Setup
2. ISSI Support for Roaming of the Called SU
3. ISSI Support for Roaming of the Calling SU
4. ISSI SU-to-SU Voice Call Release by Calling Serving RFSS
5. ISSI SU-to-SU Voice Call Release by Calling Home RFSS
6. ISSI SU-to-SU Voice Call Release by Called Home RFSS
7. ISSI SU-to-SU Voice Call Release by Called Serving RFSS

Only the first three procedure types involve SIP INVITE requests.

In the event that a Home RFSS receives a SIP INVITE request pertaining to a call control procedure that it does not support, it SHALL return a GEN_PROC_UNSUPPORTED failure response as specified in Annex A of this document.

In the event that a Serving RFSS receives a SIP INVITE request pertaining to a call control procedure that it does not support, it SHALL return a GEN_PROC_UNSUPPORTED failure response as specified in Annex A of this document.

6.5.1 General

In its generalized, most elaborate configuration, an SU-to-SU call involves an RFSS serving the Calling SU (Calling Serving RFSS), the Calling Home RFSS, the Called Home RFSS, and an RFSS serving the Called SU (Called Serving RFSS). An SU-to-SU call therefore comprises 3 call segments:

- a. Call Segment 1: Between the Calling Serving RFSS and Calling Home RFSS.
- b. Call Segment 2: Between the Calling Home RFSS and Called Home RFSS.
- c. Call Segment 3: Between the Called Home RFSS and Called Serving RFSS.

Prior to making any SU-to-SU call, the Calling Serving RFSS SHALL successfully register the Calling SU with the Calling Home RFSS.

In the event that the Calling SU is served by the Home RFSS of the SU itself, then the Calling Serving RFSS behavior is not involved over the ISSI. Similarly, in the event that the Called SU is served by the Home RFSS of the SU itself,

then the Called Serving RFSS behavior is not involved over the ISSI. Moreover, the Calling and Called SUs MAY have the same Home RFSS, in which case the Calling Home RFSS coincides with the Called Home RFSS. As such, an SU-to-SU call MAY, in its simplest configuration, involve only one RFSS, the Home RFSS of the calling and called SUs.

The following procedures pertain to the generalized configuration that involves all four aforementioned RFSSs. In this configuration, 3 independent SIP dialogs are initiated using 3 consecutive SIP INVITE requests, one for each call segment. The three dialogs have distinctly different tags in the To and From sip header fields, and MAY have the same or different Call-IDs. Provisional (except 100 (TRYING)) and final responses from the Called Serving RFSS in Call Segment 3 are propagated back to Call Segments 2 and 1 as provisional and final responses to the SIP INVITE requests in these call segments. The RTP session between the Calling Serving RFSS and Called Serving RFSS is established by bridging together the component RTP sessions in all 3 call segments. Procedures for simpler configurations can be readily derived from these generalized procedures.

Upon receipt of a SIP INVITE request, an RFSS determines its role based on the values in the Route header field, the Request URI, and the To and/or the From header fields, as specified in Section 6.4 of this document.

6.5.2 ISSI SU-to-SU Voice Call Setup

6.5.2.1 Actions Required at Calling Serving RFSS

Upon receipt of an endpoint trigger (e.g., upon receipt of an SU-to-SU voice service request over the air interface), the Calling Serving RFSS MAY initiate an SU-to-SU call over the ISSI to the called SU depending on availability of RTP resource. The Calling Serving RFSS SHALL allocate RTP resource and initiate a SIP INVITE request to the Calling Home RFSS to ultimately establish an RTP session with it, according to the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Calling Serving RFSS, listed in the order of preference, with the first format listed being preferred.
2. The SIP INVITE request MAY include a "Preference" parameter in its ISSI-specific body, as specified in Section 3.8.5.3 of this document, indicating whether the Called Serving RFSS SHOULD perform an availability check with the Called SU prior to granting the INVITE request, or that Called Serving RFSS SHOULD handle this as a direct call without prior availability check.

3. The SIP INVITE request MAY include a “Duplex Mode” parameter in its ISSI-specific body, as specified in Section 3.8.5.4 of this document, indicating whether the Called Serving RFSS SHOULD handle this as a full duplex or half duplex call.
4. The SIP INVITE request MAY include a “Protected” parameter in its ISSI-specific body, as specified in Section 3.8.5.7 of this document, indicating whether the call’s audio is requested to be protected.
5. In the event that, while awaiting a final response on the SIP INVITE request, the Calling Serving RFSS receives any 1xx provisional responses, the Calling Serving RFSS SHALL map the SIP response into appropriate information which it then sends over the air interface to the Calling SU. Notably,
 - a. A SIP 180 Ringing Message SHALL entail informing the Calling SU that the called SU has been alerted.
 - b. A VCE_NO_RF_AT_CALLED_SERV 183 provisional response SHALL entail informing the Calling SU that no RF resource is available at the Called SU’s radio site.
6. In the event that, while awaiting a final response on the SIP INVITE request, the Calling Serving RFSS detects that the Calling SU is canceling the call, the Calling Serving RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document.
7. In the event that, while awaiting a final response on the SIP INVITE Request, the Calling Serving RFSS detects that the RTP resource has been preempted, the Calling Serving RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document.
8. In the event that, while awaiting a final response on the SIP INVITE Request, the Calling Serving RFSS detects that the SU-to-SU Call has to be released due to local policy, the Calling Serving RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document.
9. In the event that, while awaiting a final response on its SIP INVITE request, the Calling Serving RFSS receives a SIP INVITE request from the Calling Home RFSS to set up a call between the Called SU and Calling SU, the Calling Serving RFSS SHALL process the received SIP INVITE request as follows:
 - a. In the event that the Calling Serving RFSS’s SIP INVITE request has higher priority than the received SIP INVITE request, the Calling Serving RFSS SHALL return a VCE_COLLIDING_CALL failure response to the Calling Home RFSS.

- b. For equal priority SIP INVITE requests, in the event that the hexadecimal value of the fully qualified Calling SUID is greater than that of the fully qualified Called SUID, then the Calling Serving RFSS SHALL return a VCE_COLLIDING_CALL failure response to the Calling Home RFSS.

10. In the event that:

- a. A VCE_CALLING_SU_NOT_REG failure response is received, indicating that the Calling SU is not registered with the Calling Home RFSS, or
- b. A VCE_CALLING_SU_UNKNOWN failure response is received, indicating that the Calling SU is not known to the Calling Home RFSS

then, the Calling Serving RFSS SHALL consider the SU-to-SU call setup to have failed and SHOULD, before attempting a new SIP INVITE request, initiate registration of the Calling SU with its Home RFSS, following the procedures in Section 5 of this document.

11. In the event that a VCE_VOCODER_MODES_UNACCEPTED or a GEN_SDP_PARAMS_UNACCEPTABLE failure response is received from the Calling Home RFSS:

- a. The Calling Serving RFSS MAY retry the INVITE request, with the set of offered SDP media formats modified as required per failure response received (see also Section 3.7), or
- b. The Calling Serving RFSS MAY consider the SU-to-SU call setup to have failed, and an appropriate failure response is returned to the Calling SU.

12. In the event that the Calling Serving RFSS receives any other SIP 3xx, 4xx, 5xx, 6xx failure responses, the Calling Serving RFSS SHALL consider the SU-to-SU call setup to have failed, and an appropriate failure response is returned to the calling SU.

13. When the Calling Serving RFSS receives the 200 OK message, with an "RF Resource Availability" parameter value invariably indicating that the Called Serving RFSS has available RF resource to support the call:

- a. The Calling Serving RFSS SHALL consider the Called Serving RFSS to be ready to start the SU-to-SU call for the Called SU.
- b. The Calling Serving RFSS SHALL wait until RF resource is granted at the Calling SU's radio site to support the call, before it returns an

SIP ACK Request, with an “RF Resource Availability” parameter value invariably indicating that the Calling Serving RFSS has available RF resource to support the call, to the Calling Home RFSS .

- c. The Calling Serving RFSS SHALL determine, from the “Initial Transmitter” parameter value in the received 200 (OK) response, whether or not the calling SU has been chosen by the Called Home RFSS to transmit first, and notify the calling SU accordingly.
14. In the event that the “Duplex Mode” parameter, as specified in Section 3.8.5.4 of this document, is present in the received 200 (OK) response’s ISSI-specific body, then the Calling Serving RFSS SHALL handle the call according to the received “Duplex Mode” value (be it half duplex or full duplex).
 15. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received 200 (OK) response’s ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, then the Calling Serving RFSS SHALL handle the call initially according to the received “Protected” value (be it protected or unprotected).

6.5.2.2 Actions Required at Calling Home RFSS

The Calling Home RFSS SHALL process the SIP INVITE request from the Calling Serving RFSS as follows:

1. In the event that the Calling Home RFSS detects that the Calling SU is not registered, it SHALL return a VCE_CALLING_SU_NOT_REG failure response.
2. In the event that the Calling Home RFSS detects that the Calling SU is radio inhibited, it SHALL return a VCE_CALLING_SU_INHIBITED failure response to the Calling Serving RFSS.
3. In the event that:
 - a. The RFSS ID of the Called Home RFSS is not known to the Calling Home RFSS, the Calling Home RFSS SHALL return a VCE_CALLED_HOME_UNKNOWN failure response.
 - b. The Calling SU is not known to the Calling Home RFSS , the Calling Home RFSS SHALL return a VCE_CALLING_SU_UNKNOWN failure response.

4. In the event that the Calling SU is not authorized to make this call to the Called SU, the Calling Home RFSS SHALL return a VCE_CALLING_SU_NOT_AUTHOR failure response.
5. In the event that the Calling Home RFSS detects that none of the SDP media formats offered by the Calling Serving RFSS is accepted by the Calling Home RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

6. In the event that the Calling Home RFSS does not have available RTP resources to set up the RTP sessions with the Calling Serving RFSS and with the Called Home RFSS, it SHALL return a VCE_NO_RTP failure response to the Calling Serving RFSS.

In the event that the Calling Home RFSS has not rejected the received SIP INVITE request, and that it has available RTP resources to set up the RTP sessions with the Calling Serving RFSS and with the Called Home RFSS, it SHALL initiate an SIP INVITE request to the Called Home RFSS, following the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Calling Home RFSS, listed in the order of preference, with the first format listed being preferred, also taking into account the set of SDP media formats contained in the SIP INVITE request it received from the Calling Serving RFSS.
2. In the event that the "Preference" parameter, as specified in Section 3.8.5.3 of this document, is present in the received SIP INVITE request's ISSI-specific body, the Calling Home RFSS SHALL include this "preference" parameter and its received value in its SIP INVITE request to the Called Home RFSS.
3. In the event that the "Duplex Mode" parameter, as specified in Section 3.8.5.4 of this document, is present in the received SIP INVITE request's ISSI-specific body, the Calling Home RFSS SHALL include this "Duplex Mode" parameter and its value in its SIP INVITE request to the Called Home RFSS.

4. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received SIP INVITE request’s ISSI-specific body, the Calling Home RFSS SHALL include this “Protected” parameter and its value in its SIP INVITE request to the Called Home RFSS.
5. In the event that the Calling Home RFSS receives a SIP CANCEL request from the Calling Serving RFSS, it SHALL process the received SIP CANCEL request, following the procedures in Section 6.4.2.1 of this document. And, in the event that the SIP CANCEL request is received prior to receiving any final response from the Called Home RFSS on the SIP INVITE request, the Calling Home RFSS SHALL send a SIP CANCEL request to the Called Home RFSS, following the procedures in Section 6.4.2.1 of this document.
6. In the event that the Calling Home RFSS receives a SIP BYE request from the Calling Serving RFSS, it SHALL process the received SIP BYE request and it SHALL send a SIP BYE request to the Called Home RFSS, following the procedures in Section 6.4.4.1 of this document.
7. In the event that while awaiting a final response on the SIP INVITE Request, the Calling Home RFSS detects that the Calling SU is radio inhibited, the Calling Home RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document and SHALL return a VCE_CALLING_SU_INHIBITED failure response to the Calling Serving RFSS.
8. In the event that while awaiting a final response on the SIP INVITE Request, the Calling Home RFSS detects that the RTP resources allocated to this SU-to-SU call have been preempted, the Calling Home RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document and SHALL return a VCE_NO_RTP failure response to the Calling Serving RFSS.
9. In the event that while awaiting a final response on the SIP INVITE Request, the Calling Home RFSS detects that the SU-to-SU Call has to be released due to local policy, the Calling Home RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document and return the appropriate 4xx, 5xx, 6xx failure responses to the Calling Serving RFSS.
10. In the event that, while awaiting a final response on its SIP INVITE request, the Calling Home RFSS receives a SIP INVITE request from the Called Home RFSS to set up the same SU-to-SU call in the opposite direction (colliding call situation), the Calling Home RFSS SHALL process the received SIP INVITE request as follows:

- a. In the event that the SIP INVITE request from the Calling Home RFSS has higher priority than that of the received SIP INVITE request, the Calling Home RFSS SHALL return a VCE_COLLIDING_CALL failure response to the Called Home RFSS .
 - b. For equal priority SIP INVITE requests, in the event that the hexadecimal value of the fully qualified Calling SUID is greater than that of the fully qualified Called SUID, then the Calling Home RFSS SHALL return a VCE_COLLIDING_CALL failure response to the Called Home RFSS .
11. In the event that, while awaiting a final response on the SIP INVITE request, a 101-199 provisional response is received from the Called Home RFSS, the Calling Home RFSS SHALL propagate the provisional response to the Calling Serving RFSS, unless the provisional response is received while the Calling Home RFSS is awaiting a response from a SIP CANCEL or BYE request that it has just sent to the Called Home RFSS, in which case the provisional response is simply discarded.
12. In the event that a VCE_VOCODER_MODES_UNACCEPTED or a GEN_SDP_PARAMS_UNACCEPTABLE failure response is received from the Called Home RFSS:
- a. The Calling Home RFSS MAY retry the INVITE request, with the set of offered SDP media formats modified as required per failure response received (see also Section 3.7), or
 - b. The Calling Home RFSS MAY propagate the same failure response to the Calling Serving RFSS.
- NOTE: If the Calling Home RFSS is of an older RFSS implementation that only supports the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), it possibly returns a GEN_SDP_PARAMS_UNACCEPTABLE failure response or another SIP failure response to the Calling Serving RFSS instead.
13. In the event that any other SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE request is received from the Called Home RFSS, the Calling Home RFSS SHALL propagate the same failure response to the Calling Serving RFSS.
14. In the event that a SIP 200 (OK) response is received from the Called Home RFSS for the SIP INVITE request, the Calling Home RFSS SHALL

return a SIP 200 (OK) response to the Calling Serving RFSS, with the following qualifications:

- a. The SIP 200 (OK) response SHALL convey a Type 1 SDP answer (as defined in Section 3.7 of this document), corresponding to a Type 1 SDP offer received in the INVITE Request from the Calling Serving RFSS. The Type 1 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Calling Serving RFSS in its SIP INVITE request, has been accepted by the Calling Home RFSS, also taking into account the SDP media format contained in the SIP 200 (OK) response from the Called Home RFSS.
 - b. The SIP 200 (OK) response SHALL include an "RF Resource Availability" parameter, with value copied from the received 200 (OK) response.
 - c. The 200 (OK) response SHALL include an "Initial Transmitter" parameter, with value copied from the received 200 (OK) response.
 - d. In the event that the received 200 (OK) response includes the "Duplex Mode" parameter, then the outgoing 200 (OK) response SHALL also include the "Duplex Mode" parameter, with value copied from the received 200 (OK) response.
 - e. In the event that the received 200 (OK) response includes the "Protected" parameter, then the outgoing 200 (OK) response SHALL also include the "Protected" parameter, with value copied from the received 200 (OK) response.
15. Upon receiving an SIP ACK request from the Calling Serving RFSS confirming receipt of a 200 OK final response, the Calling Home RFSS SHALL propagate the SIP ACK request to the Called Home RFSS. The SIP ACK request SHALL include a "RF Resource Availability" parameter with value copied from the received ACK request.

6.5.2.3 Actions Required at Called Home RFSS

The Called Home RFSS SHALL process the SIP INVITE request from the Calling Home RFSS as follows:

1. In the event that the Called Home RFSS detects that the Called SU is not registered, it SHALL return a VCE_CALLED_SU_NOT_REG failure response to the Calling Home RFSS.

2. In the event that the Called Home RFSS detects that the Called SU is radio inhibited, it SHALL return a VCE_CALLED_SU_INHIBITED failure response to the Calling Home RFSS.
3. In the event that the Called SU is not authorized to receive this call from the Calling SU, the Called Home RFSS SHALL return VCE_CALLED_SU_NOT_AUTHOR failure response.
4. In the event that the Called SU is not known by the Called Home RFSS, the Called Home RFSS SHALL return a VCE_CALLED_SU_UNKNOWN failure response.
5. In the event that the Called Home RFSS detects that none of the SDP media formats offered by the Calling Home RFSS is accepted by the Called Home RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

6. In the event that the Called Home RFSS does not have available RTP resources to set up RTP sessions with the Calling Home RFSS and with the Called Serving RFSS, it SHALL return a VCE_NO_RTP failure response to the Calling Home RFSS.

In the event that the Called Home RFSS has not rejected the received SIP INVITE request, and that it has available RTP resources to set up the RTP sessions with the Called Serving RFSS and with the Calling Home RFSS, it SHALL initiate a SIP INVITE request to the Called Serving RFSS, following the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Called Home RFSS, listed in the order of preference, with the first format listed being preferred, also taking into account the set of SDP media formats contained in the SIP INVITE request it received from the Calling Home RFSS.
2. The SIP INVITE request SHALL include an "Initial Transmitter" parameter in its ISSI-specific body, as defined in Section 3.8.5.1 of this document, with value indicating its choice of whether the calling SU or called SU SHOULD transmit first

3. In the event that the “Preference” parameter, as specified in Section 3.8.5.3 of this document, is present in the received INVITE request’s ISSI-specific body, the Called Home RFSS SHALL include this “preference” parameter in its SIP INVITE request to the Called Serving RFSS with value set as follows:
 - a. In the event that the Called SU supports the received preference (be it “Availability Check or “Direct Call”), then the outgoing SIP INVITE’s preference SHOULD be set to the received preference.
 - b. In the event that the Called SU does not support the received preference, then the outgoing SIP INVITE’s preference SHALL be set to the call mode that the Called SU supports (be it “Availability Check or “Direct Call”).
4. In the event that the “Duplex Mode” parameter, as specified in Section 3.8.5.4 of this document, is present in the received INVITE request’s ISSI-specific body, the Called Home RFSS SHALL include this “Duplex Mode” parameter in its SIP INVITE request to the Called Serving RFSS with value set as follows:
 - a. In the event that the received duplex mode and the Called SU’s duplexity are both equal to Full Duplex, then the outgoing SIP INVITE’s “Duplex Mode” SHALL be set to Full Duplex.
 - b. In all other cases, the outgoing SIP INVITE’s “Duplex Mode” SHALL be set to Half Duplex.
5. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received INVITE request’s ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, the Called Home RFSS SHALL determine whether the destination SU supports the value of this parameter:
 - a. In the event that the destination SU supports this parameter value, then the Called Home RFSS SHALL include the parameter value in its SIP INVITE request to the Called Serving RFSS.
 - b. In the event that the destination SU does not support this parameter value, then the Called Home RFSS SHALL return a VCE_CANNOT_PROTECT_AUDIO or VCE_CANNOT_UNPROTECT_AUDIO failure response, as appropriate, to the Calling Home RFSS.
6. In the event that the Called Home RFSS receives a SIP CANCEL request from the Calling Home RFSS, it SHALL process the received SIP

CANCEL request, following the procedures in Section 6.4.2.1 of this document. And, in the event that the SIP CANCEL request is received prior to receiving any final response from the Called Serving RFSS on the INVITE request, the Called Home RFSS SHALL send a SIP CANCEL request to the Called Serving RFSS, following the procedures in Section 6.4.2.1 of this document.

7. In the event that the Called Home RFSS receives a SIP BYE request from the Calling Home RFSS, it SHALL process the received SIP BYE request, following the procedures in Section 6.4.4.1 of this document, and it SHALL send a SIP BYE request to the Called Serving RFSS, following the procedures in Section 6.4.4.1 of this document.
8. In the event that while awaiting a final response on the SIP INVITE Request, the Called Home RFSS detects that the Called SU is radio inhibited, the Called Home RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document, and SHALL return a VCE_CALLED_SU_INHIBITED failure response to the Calling Home RFSS.
9. In the event that while awaiting a final response on the SIP INVITE Request, the Called Home RFSS detects that the RTP resource has been preempted, the Called Home RFSS SHALL cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document, and SHALL return a VCE_NO_RTP failure response to the Calling Home RFSS.
10. In the event that while awaiting a final response on the SIP INVITE Request, the Called Home RFSS detects that the SU-to-SU Call has to be released due to local policy, the Called Home RFSS MAY cancel its SIP INVITE request, following the procedures in Section 6.4.2 of this document and return the appropriate 4xx, 5xx, 6xx error code to the Calling Home RFSS.
11. In the event that, while awaiting a final response on its SIP INVITE request, the Called Home RFSS receives a SIP INVITE request from the Called Serving RFSS to set up a call involving both SUs in the opposite direction (Colliding Call situation), the Called Home RFSS SHALL process the received SIP INVITE request as follows:
 - a. In the event that the Called SU Home RFSS's SIP INVITE request has higher priority than that of the received SIP INVITE request, the Called Home RFSS SHALL return a VCE_COLLIDING_CALL failure response to the Called Serving RFSS.
 - b. For equal priority SIP INVITE requests, in the event that the hexadecimal value of the fully qualified Calling SUID is greater than that of the fully qualified Called SUID, then the Called Home RFSS

SHALL return a VCE_COLLIDING_CALL failure response to the Called Serving RFSS.

12. In the event that, while awaiting a final response on the SIP INVITE request, a 101-199 provisional response is received from the Called Serving RFSS, the Called Home RFSS SHALL propagate the provisional response to the Calling Home RFSS, unless the provisional response is received while the Called Home RFSS is awaiting a response from a SIP CANCEL or BYE request that it has just sent to the Called Serving RFSS, in which case the provisional response is simply discarded.
13. In the event that a VCE_VOCODER_MODES_UNACCEPTED or a GEN_SDP_PARAMS_UNACCEPTABLE failure response is received from the Called Serving RFSS:
 - a. The Called Home RFSS MAY retry the INVITE request, with the set of offered SDP media formats modified as required per failure response received (see also Section 3.7), or
 - b. The Called Home RFSS MAY propagate the same failure response to the Calling Home RFSS.

NOTE: If the Called Home RFSS is of an older RFSS implementation that only supports the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), it possibly returns a GEN_SDP_PARAMS_UNACCEPTABLE failure response or another SIP failure response to the Calling Home RFSS instead.

14. In the event that any other SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE request is received from the Called Serving RFSS, the Called Home RFSS SHALL return the same response to the Calling Home RFSS.
15. In the event that a SIP 200 (OK) response for the SIP INVITE request is received from the Called Serving RFSS, the Called Home RFSS SHALL return a SIP 200 (OK) response to the Calling Home RFSS, conveying a Type 1 SDP answer corresponding to the SDP offer received in the INVITE Request from the Calling Home RFSS (as defined in Section 3.7 of this document), with the following qualifications:
 - a. The SIP 200 (OK) response SHALL include a Type 1 SDP answer (as defined in Section 3.7 of this document) that indicates which particular media format, chosen from the set of SDP media formats offered by the Calling Home RFSS in its SIP INVITE request, has been accepted by the Called Home RFSS, also taking into account

the SDP media format contained in the SIP 200 (OK) response from the Called Serving RFSS.

- b. The SIP 200 (OK) response SHALL include an "RF Resource Availability" parameter with value copied from the received 200 (OK) response.
 - c. The 200 (OK) response SHALL include an "Initial Transmitter" parameter, as defined in 3.8.5.1 of this document, with value indicating its choice of whether the calling SU or called SU SHOULD transmit first.
 - d. In the event that the INVITE request sent to the Called Serving RFSS contains a "Duplex Mode" parameter, then the 200 (OK) response SHALL also contain the "Duplex Mode" parameter, with value copied from the outgoing INVITE request
 - e. In the event that the INVITE request sent to the Called Serving RFSS contains a "Protected" parameter, then the 200 (OK) response SHALL also contain the "Protected" parameter, with value copied from the outgoing INVITE request.
16. Upon receiving an SIP ACK request from the Calling Home RFSS confirming receipt of a 200 OK final response, the Called Home RFSS SHALL send a SIP ACK request to the Called Serving RFSS, and SHALL start a Tu2uhangtime timer that is reinitialized every time the Called Home RFSS detects voice activity on the RTP sessions of the SU-to-SU call. The SIP ACK request SHALL include a "RF Resource Availability" parameter with value copied from the received ACK request.

6.5.2.4 Actions Required at Called Serving RFSS

The Called Serving RFSS SHALL process the SIP INVITE request from the Called Home RFSS according to the procedures in Section 6.4.1 of this document, with the following qualifications:

- 1. In the event that the Called Serving RFSS detects that none of the SDP media formats offered by the Called Home RFSS is accepted by the Called Serving RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

2. In the event that the Called Serving RFSS receives the SIP INVITE request while awaiting a final response on a SIP INVITE request it has previously sent to the Called Home RFSS to set up the same SU-to-SU call in the opposite direction (colliding call situation), the Called Serving RFSS SHALL process the received SIP INVITE request as follows:
 - a. In the event that the Called Serving RFSS's SIP INVITE request has higher priority than the received SIP INVITE request, the Called Serving RFSS SHALL return a VCE_COLLIDING_CALL failure response to the Called Home RFSS.
 - b. For equal priority SIP INVITE requests, in the event that the hexadecimal value of the fully qualified Called SUID is greater than that of the fully qualified Calling SUID, then the Called Serving RFSS SHALL return a VCE_COLLIDING_CALL failure response to the Called Home RFSS.
3. In the event that the Called Serving RFSS does not have available RTP resources to set up the associated RTP session, it SHALL return a VCE_NO_RTP failure response to the Called Home RFSS.
4. In the event that the "Preference" parameter, as specified in Section 3.8.5.3 of this document, is present in the received INVITE request's ISSI-specific body, with value indicating "Availability Check", then the Called Serving RFSS SHALL perform availability check on the called SU.
5. In the event that the "Duplex Mode" parameter, as specified in Section 3.8.5.4 of this document, is present in the received INVITE request's ISSI-specific body, then the Called Serving RFSS SHALL handle the call according to the received "Duplex Mode" value (be it half duplex or full duplex).
6. In the event that the "Protected" parameter, as specified in Section 3.8.5.7 of this document, is present in the received INVITE request's ISSI-specific body, or if the parameter is not present and takes on the implied default value of "unprotected", then the Called Serving RFSS SHALL handle the call initially according to the received "Protected" value (be it protected or unprotected).
7. In the event that:
 - a. The Called SU is busy or does not respond, the Called Serving RFSS SHALL return a VCE_CALLED_SU_BUSY failure response to the Called Home RFSS.

- b. The Called SU refuses the call, the Called Serving RFSS SHALL return a VCE_CALLED_SU_REFUSE_CALL failure response to the Called Home RFSS.
- 8. In the event that the Called SU is alerted, the Called Serving RFSS SHOULD return a SIP 180 (RINGING) provisional response to the Called Home RFSS.
- 9. In the event that RF resource is temporarily unavailable at the Called SU's radio site to support the call, the Called Serving RFSS SHOULD return a VCE_NO_RF_AT_CALLED_SERV 183 provisional response to the Called Home RFSS.
- 10. In the event that RF resource, according to local policy, is not available within a reasonable timeframe (e.g., upon expiry of a local timer), the Called Serving RFSS SHALL return a VCE_NO_RF failure response to the Called Home RFSS.
- 11. In the event that the Called Serving RFSS has not registered the called SUID with the Called Home RFSS, the Called Serving RFSS SHALL return a VCE_SU_NOT_REG_BY_CALLED_SERV failure response to the Called Home RFSS.
- 12. In the event that RTP resource is available and allocated, and RF resource is available and ready to be granted at the Called SU's radio site to support the call, the Called Serving RFSS SHALL return a SIP 200 (OK) response to the Called Home RFSS, with the following qualifications:
 - a. The SIP 200 (OK) response SHALL convey a Type 1 SDP answer corresponding to the Type 1 SDP offer received in the SIP INVITE Request from the Called Home RFSS. The Type 1 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Called Home RFSS in its SIP INVITE request, has been accepted by the Called Serving RFSS.
 - b. The 200 (OK) response SHALL include an "RF Resource Availability" parameter with value invariably indicating that the Called Serving RFSS has available RF resource to support the call.

The Called Serving RFSS SHALL process a SIP CANCEL request from the Called Home RFSS according to the procedures in Section 6.4.2.1 of this document.

The Called Serving RFSS SHALL process a SIP BYE request from the Called Home RFSS according to the procedures in Section 6.4.4.1 of this document.

The Called Serving RFSS SHALL process a SIP ACK request from the Called Home RFSS confirming receipt of a 200 OK final response, according to the procedures in Section 13.3 of [RFC3261], with the following qualifications:

1. When the Called Serving RFSS receives the SIP ACK message, it SHALL consider the SU-to-SU Call to have started for the Calling SU and SHOULD also start the call for the Called SU (notably, when the Called SU is a radio SU, it SHOULD grant the RF resource to the Called SU)
2. The Called Serving RFSS SHALL determine, from the “Initial Transmitter” parameter value in the received INVITE request, whether or not the called SU has been chosen by the Called Home RFSS to transmit first, and notify the called SU accordingly.

6.5.3 ISSI Support for Roaming of the Called SU

6.5.3.1 Actions Required at New Called Serving RFSS

In the event that the Called SU roams to a new Called Serving RFSS, the new Called Serving RFSS SHALL request the Called Home RFSS to register the Called SU, following procedures in Section 5 of this document.

The new Called Serving RFSS SHALL process the INVITE request from the Called Home RFSS according to procedures in Section 6.5.2.4 of this document, with the following qualifications:

1. The new Called Serving RFSS SHALL determine, from the “Initial Transmitter” parameter value in the received INVITE request, whether or not the called SU has been chosen by the Called Home RFSS to transmit first, and notify the called SU accordingly.
2. As the “p25c-incallroaming” parameter in the INVITE request is set to a value that indicates in-call roaming, No 180 (RINGING) OR 183 (SESSION PROGRESS) provisional response SHOULD be sent back to the Called Home RFSS.
3. When RF resource becomes available, the Called Serving RFSS SHALL return a 200 (OK) response to the Called Home RFSS. The 200 (OK) response SHALL include an “RF Resource Availability” parameter indicating that the Called Serving RFSS has available RF resource for the call. The Type 1 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Called Home RFSS in its SIP INVITE request, has been accepted by the Called Serving RFSS.

4. In the event that RF resource, according to local policy, is not available within a reasonable timeframe (e.g., upon expiry of a local timer), the Called Serving RFSS SHALL return a VCE_NO_RF failure response.

6.5.3.2 Actions Required at Called Home RFSS

The Called Home RFSS SHALL register the Called SU as requested by the new Called Serving RFSS, following procedures in Section 5 of this document.

Upon successful registration of the Called SU in the new Called Serving RFSS:

1. In the event that the SU-to-SU call is set up and still in progress, the Called Home RFSS SHALL send an SIP INVITE request to the new Called Serving RFSS to establish a call segment between the two RFSSs, following the procedures in Section 6.4.1 of this document, with the following qualifications:
 - a. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Called Home RFSS, listed in the order of preference, with the first format listed being preferred, also taking into account the SDP media format that the Called Home RFSS currently uses for talk spurt transmission with the Calling Home RFSS.
 - b. The SIP INVITE request SHALL include an "Initial Transmitter" parameter in its ISSI-specific body, as defined in Section 3.8.5.1 of this document, with value indicating its choice of whether the calling SU or called SU SHOULD transmit first.
 - c. The SIP INVITE request SHALL include a "p25c-incallroaming" parameter with value set to indicate in-call roaming,
2. In the event that the SIP INVITE request is denied by a 3xx, 4xx, 5xx, or 6xx failure response, then, subject to local policy, the Called Home RFSS MAY retry the INVITE request, modified as required per failure response received, or it MAY send a BYE request to the Calling Home RFSS to terminate the SU-to-SU call.
3. In the event that the SU-to-SU call set-up is in progress, the Called Home RFSS SHALL terminate the previous SU-to-SU Call.

Upon successful establishment of a call segment with the new Called Serving RFSS, the Called Home RFSS SHALL bridge together the RTP sessions it has with the new Called Serving RFSS and with the Calling Home RFSS. The Called Home RFSS SHALL also send a SIP BYE request to the previous Called

Serving RFSS to tear down the call segment between them, following the procedures in Section 6.4.4 of this document.

Upon successful teardown of the call segment between the Called Home RFSS and the previous Called Serving RFSS, the Called Home RFSS SHALL deregister the Called SU from the previous Called Serving RFSS, following procedures in Section 5 of this document.

6.5.3.3 Actions Required at Previous Called Serving RFSS

The previous Called Serving RFSS SHALL process the SIP BYE request from the Called Home RFSS, following procedures in Section 6.4.4 of this document.

The previous Called Serving RFSS SHALL process the deregistration request from the Called Home RFSS, following procedures in Section 5 of this document.

6.5.4 ISSI Support for Roaming of the Calling SU

The procedures in Section 6.5.3 of this document apply, with the following qualifications:

1. “calling” and “called” are transposed.
2. In the event that the SIP INVITE request is denied by a 3xx, 4xx, 5xx, or 6xx failure response, then, subject to local policy, the Calling Home RFSS MAY retry the INVITE request, modified as required per failure response received, or it MAY send BYE requests to the Calling Serving RFSS and Called Home RFSS to terminate the SU-to-SU call.

6.5.5 ISSI SU-to-SU Voice Call Release by Calling Serving RFSS

The Calling Serving RFSS MAY initiate termination of an SU-to-SU voice call over the ISSI in the event that:

1. The Calling SU terminates the call
2. The Calling Serving RFSS needs to preempt the RTP resource so it can be used for another RTP session, or
3. The Calling Serving RFSS needs to preempt the RF resource so it can be used for another call, or
4. The Calling Serving RFSS detects that it has lost RTP connectivity with the Calling Home RFSS (see Section 7 of this document), or

5. While awaiting a response for a SIP CANCEL request, the Calling Serving RFSS receives, from the Calling Home RFSS, a 200 (OK) for the SIP INVITE request.
6. Local policy requires that the call be terminated.

Procedures in the following subsections apply.

6.5.5.1 Actions Required at Calling Serving RFSS

The Calling Serving RFSS SHALL send a SIP BYE request to the Calling Home RFSS, following the procedures in Section 6.4.4 of this document.

6.5.5.2 Actions Required at Calling Home RFSS

The Calling Home RFSS SHALL process the received SIP BYE request and send a SIP BYE request to the called SU's Home RFSS, following the procedures in Section 6.4.4 of this document.

6.5.5.3 Actions Required at Called Home RFSS

The Called Home RFSS SHALL process the received SIP BYE request and SHALL send a SIP BYE request to the Called Serving RFSS, following the procedures in Section 6.4.4 of this document.

6.5.5.4 Actions Required at Called Serving RFSS

The Called Serving RFSS SHALL process the received SIP BYE request, following the procedures in Section 6.4.4 of this document.

6.5.6 ISSI SU-to-SU Voice Call Release by Calling Home RFSS

The Calling Home RFSS MAY initiate termination of an SU-to-SU voice call over the ISSI in the event that:

1. It needs to preempt the RTP resources so they can be used for another RTP session, or
2. It detects that it has lost RTP connectivity with the Calling Serving RFSS (see Section 7 of this document), or
3. It detects that it has lost RTP connectivity with the Called Home RFSS (see Section 7 of this document).

4. While awaiting a response for a SIP CANCEL request, the Calling Home RFSS receives, from the Called Home RFSS, a 200 (OK) for the SIP INVITE request.
5. Local policy requires that the call be terminated.

Procedures in the following subsections apply.

6.5.6.1 Actions Required at Calling Home RFSS

The Calling Home RFSS SHALL send a SIP BYE request to the Calling Serving RFSS, and to the Called Home RFSS, following the procedures in Section 6.4.4 of this document.

6.5.6.2 Actions Required at Called Home RFSS

The Called Home RFSS SHALL process the received SIP BYE request and send a SIP BYE request to the Called Serving RFSS, following the procedures in Section 6.4.4 of this document.

6.5.6.3 Actions Required at Called Serving RFSS

The Called Serving RFSS SHALL process the received SIP BYE request, following the procedures in Section 6.4.4 of this document.

6.5.6.4 Actions Required at Calling Serving RFSS

The Calling Serving RFSS SHALL process the received SIP BYE request, following the procedures in Section 6.4.4 of this document.

6.5.7 ISSI SU-to-SU Voice Call Release by Called Home RFSS

The Called Home RFSS MAY initiate termination of an SU-to-SU voice call over the ISSI in the event that:

1. The Tu2uhangtime timer expires, or
2. Any of the situations in Section 6.5.6 of this document, with “calling” and “called” transposed, occurs.

The procedures in Section 6.5.6 of this document, with “calling” and “called” transposed, apply.

6.5.8 ISSI SU-to-SU Voice Call Release by Called Serving RFSS

The Called Serving RFSS MAY initiate termination of an SU-to-SU voice call over the ISSI in the event that:

1. The Called SU terminates the call
2. The Called Serving RFSS needs to preempt the RTP resources so they can be used for another RTP session, or
3. The Called Serving RFSS needs to preempt the RF resource so it can be used for another call, or
4. The Called Serving RFSS detects that it has lost RTP connectivity with the Called Home RFSS (see Section 7 of this document).
5. Local policy requires that the call be terminated.

Procedures in the following subsections apply.

6.5.8.1 Actions Required at Called Serving RFSS

The Called Serving RFSS SHALL send a SIP BYE request to the Called Home RFSS, following the procedures in Section 6.4.4 of this document.

6.5.8.2 Actions Required at Called Home RFSS

The Called Home RFSS SHALL process the received SIP BYE request and send a SIP BYE request to the Calling Home RFSS, following the procedures in Section 6.4.4 of this document.

6.5.8.3 Actions Required at Calling Home RFSS

The Calling Home RFSS SHALL process the received SIP BYE request and send a SIP BYE request to the Calling Serving RFSS, following the procedures in Section 6.4.4 of this document.

6.5.8.4 Actions Required at Calling Serving RFSS

The Calling Serving RFSS SHALL process the received SIP BYE request, following the procedures in Section 6.4.4 of this document.

6.6 Call Control Procedures for ISSI Group Voice Services

Seven types of call control procedures are applicable to group voice services over the ISSI, as specified in the subsections below:

1. ISSI Group Voice Call Setup
2. Subsequent Modification of ISSI SIP Session Characteristics
3. Handling (re-)INVITE Request Collisions
4. ISSI SIP Heartbeat
5. ISSI SIP Session Termination by Home RFSS
6. ISSI SIP Session Termination by Serving RFSS
7. ISSI Group Voice Call Release

Only the first four procedure types involve SIP INVITE or re-INVITE requests.

In the event that a Home RFSS, or a Serving RFSS, receives a SIP INVITE or re-INVITE request pertaining to a call control procedure that it does not support, it SHALL respond with a GEN_PROC_UNSUPPORTED failure response as specified in Annex A of this document.

6.6.1 ISSI Group Voice Call Setup

6.6.1.1 General

A group call MAY be an unconfirmed or a confirmed group call.

For an unconfirmed group call, an RFSS initiates the group call over the ISSI upon receipt of an endpoint trigger (e.g., upon receipt of a group voice service request over the air interface as an end user pushes the PTT button on an SU affiliated with the group). The originating RFSS MAY be the Home RFSS or a Serving RFSS. If the originating RFSS is not the Home RFSS, the RFSS initiates the group call by sending a SIP INVITE request to the Home RFSS. If the Home RFSS accepts this SIP INVITE request, it would set up a call segment with the originating RFSS, and invite other Serving RFSSs in the group to join the call by sending individual SIP INVITE requests to them. If the originating RFSS is the Home RFSS, it directly invites other Serving RFSSs in the group to join the call. The Home RFSS bridges together individual RTP sessions (one for each call segment) and controls the transmission of RTP packets among the RFSSs in the group call. The unconfirmed group call is started over the ISSI as soon as the first SIP session and associated RTP session are successfully established between the Home RFSS and a Serving RFSS in the group, and talk spurt transmission over the ISSI MAY immediately begin.

For a confirmed group call, an RFSS initiates the group call over the ISSI upon receipt of an endpoint trigger, and the confirmed group call is started over the ISSI as soon as the first SIP session and associated RTP session are

successfully established between the Home RFSS and a Serving RFSS in the group, similar to unconfirmed group call. However, talk spur transmission does not begin until the Home RFSS determines that sufficient RF resources are available for the confirmed group call, or upon the expiry of a timer associated with the confirmed group call (refer to the procedures in the following subsections for details). Similar to unconfirmed calls, the originating RFSS MAY be the Home RFSS or a Serving RFSS.

For both unconfirmed and confirmed group calls, the procedures in the following subsections apply.

Upon receipt of an INVITE Request, the destination RFSS determines its role based on the values in the Route header field, the Request URI, and the To and/or the From header fields as specified in Section 6.4.1.1 of this document.

In a group call, the SIP messages within a SIP dialog SHALL have the same From and To fields. Each SIP dialog is distinguished by a Call-ID, To tag, and From tag. Various SIP dialogs in a group call MAY have different Call-IDs.

6.6.1.2 Group Voice Call Setup Initiated by Serving RFSS

In the event that the originating RFSS is the Home RFSS itself, procedures in Section 6.6.1.3 apply. Otherwise, the following procedures apply.

6.6.1.2.1 Actions Required at Serving RFSS

The Serving RFSS SHALL allocate RTP resources, and initiate a SIP INVITE request to the group's Home RFSS to ultimately establish an RTP session with the Home RFSS, according to the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Serving RFSS, listed in the order of preference, with the first format listed being preferred, which MAY take into account information on the Home RFSS's set of supported SDP media formats that the Serving RFSS MAY have received during group registration over the ISSI (see Section 5).
2. The SIP INVITE request MAY include a "Group Call Type" parameter (as defined in Section 3.8.5.6) that indicates whether it is for a confirmed or unconfirmed group call.
3. The SIP INVITE request MAY include a "Protected" parameter (as defined in Section 3.8.5.7) that indicates whether the call's audio is requested to be protected.

4. For a confirmed group call, the SIP INVITE request SHALL include an “RF Resource Availability” parameter (as defined in Section 3.8.5.2) that indicates whether the Serving RFSS has available RF resource to support the confirmed group call. In the event that RF resource is currently unavailable, the Serving RFSS SHALL send a re-INVITE request, with appropriate “RF Resource Availability” parameter setting, to the Home RFSS when RF resource is later available, following the procedures in Section 6.4.3 of this document.
5. The Serving RFSS MAY cancel its SIP INVITE request, following procedures in Section 6.4.2 of this document.
6. Upon receipt of a VCE_SG_NOT_REG or VCE_SG_UNKNOWN failure response, the Serving RFSS SHALL consider itself not registered to the group and SHOULD, before attempting a new INVITE request, initiate registration with the Home RFSS, following procedures in Section 5 of this document.
7. In the event that, while awaiting a response on the SIP INVITE request, the Serving RFSS receives a SIP INVITE or re-INVITE request from the Home RFSS, the Serving RFSS SHALL handle the received SIP INVITE or re-INVITE request, according to procedures in Section 6.6.3 of this document.
8. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received SIP 200 OK ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, then the Serving RFSS SHALL handle the call initially according to the received “Protected” value (be it protected or unprotected).
9. In the event that a SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE is received, the Serving RFSS MAY retry the SIP INVITE request subject to local policy.

6.6.1.2.2 Actions Required at Home RFSS

The Home RFSS SHALL process the received SIP INVITE request, according to procedures in Section 6.4.1 of this document, with the following qualifications:

1. In the event that the Serving RFSS initiating the SIP INVITE request is not registered to the group, the Home RFSS SHALL return a VCE_SG_NOT_REG failure response.
2. In the event that the group is not known to the Home RFSS, the Home RFSS SHALL return a VCE_SG_UNKNOWN failure response response.

3. In the event that the Home RFSS detects that none of the SDP media formats offered by the Serving RFSS is accepted by the Home RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

4. The Home RFSS SHALL process any received SIP CANCEL or BYE request following the procedures in Section 6.4.2.1 or 6.4.4.1 of this document.
5. In the event that the Home RFSS is able to allocate RTP resource to set up the RTP session with the originating RFSS, it SHALL return a SIP 200 (OK) response that carries a Type 1 SDP answer, and it SHALL proceed to invite other Serving RFSSs registered to the Home RFSS to join the call, according to the procedures in the following subsections, with the following qualification:
 - a. The Type 1 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Serving RFSS in its SIP INVITE request, has been accepted by the Home RFSS, which MAY take into account information on the sets of SDP media formats supported by other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).
6. In the event that the Home RFSS is unable to allocate RTP resource, it SHOULD return a SIP 200 (OK) response that carries a Type 2 SDP answer, and SHOULD send a re-INVITE request that carries a Type 1 SDP offer to the Serving RFSS when RTP resource is later available, following the procedures in Section 6.4.3 of this document. In this situation, and if the group call has not been already set-up with another Serving RFSS, the Home RFSS SHOULD NOT proceed to invite other Serving RFSSs registered with the Home RFSS for that group. The following qualification apply:
 - a. The Type 2 SDP answer in the SIP 200 (OK) response SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Serving RFSS in its SIP INVITE request, has been accepted by the Home RFSS, which MAY take into account information on the sets of SDP media formats supported by other Serving RFSSs in the same group that the

Home RFSS MAY have received during their group registrations over ISSI (see Section 5).

7. In the event that the “Protected” parameter, as specified in Section 3.8.5.7 of this document, is present in the received INVITE request’s ISSI-specific body, or if the parameter is not present and takes on the implied default value of “unprotected”, the Home RFSS SHALL determine whether the SG supports the value of this parameter:
 - a. In the event that the SG supports this parameter value, then the Home RFSS SHALL include the parameter value in its 200 OK response to the Serving RFSS, and in the SIP INVITE requests to the other Serving RFSSs for the SG. (see Section 6.6.1.3).
 - b. In the event that the SG does not support this parameter value, then the Home RFSS SHALL return a VCE_CANNOT_PROTECT_AUDIO or VCE_CANNOT_UNPROTECT_AUDIO failure response, as appropriate, to the Serving RFSS.

6.6.1.3 Call Segment Setup Initiated by Home RFSS

The Home RFSS MAY initiate setup of one or more call segments with the Serving RFSSs that are already registered to the group, under the following situations:

1. The Home RFSS initiates the group call and therefore directly invites Serving RFSSs in the group to join the call.
2. A Serving RFSS initiates the group call, and, upon successful setup of a call segment between the Home and Serving RFSS, the Home RFSS invites other Serving RFSSs in the group to join the call.

6.6.1.3.1 Actions Required at Home RFSS

The Home RFSS SHALL allocate RTP resources for RTP sessions with other Serving RFSSs within the group that are successfully registered with the Home RFSS, but have not yet been invited to join the group call. Throughout the group call, the Home RFSS SHOULD, with allocated RTP resources, initiate SIP INVITE or re-INVITE requests to any Serving RFSS in the group that is not already participating in the call.

The Home RFSS SHALL then initiate SIP INVITE requests to these Serving RFSSs, to ultimately establish one-to-one RTP sessions with them, following the procedures in Section 6.4.1 of this document, with the following qualifications:

1. The SIP INVITE request SHALL include a Type 1 SDP offer (as defined in Section 3.7 of this document) that contains one or more media formats supported by the Home RFSS, listed in the order of preference, with the first format listed being preferred, taking into account SDP media format(s) that the Home RFSS MAY already be using for talk spurt transmissions with some other Serving RFSSs in the group. The Type 1 SDP offer MAY also take into account information on the sets of SDP media formats supported by this and other Serving RFSSs in the same group that the Home RFSS MAY have received during their group registrations over ISSI (see Section 5).
2. The SIP INVITE request MAY include a "Group Call Type" parameter (as defined in Section 3.8.5.6) that indicates whether it is for a confirmed or unconfirmed group call.
3. The SIP INVITE request MAY include a "Protected" parameter (as defined in Section 3.8.5.7) that indicates whether the call's audio is requested to be protected. (see Section 6.6.1.2.2)
4. In the event that, while awaiting a final response on the SIP INVITE request to a given Serving RFSS, the Home RFSS receives a SIP INVITE or re-INVITE request from the Serving RFSS, the Home RFSS SHALL handle the received SIP INVITE or re-INVITE request, according to procedures in Section 6.6.3 of this document. In the event that the Home RFSS originated SIP INVITE request does not have precedence, the Home RFSS SHALL wait for a SIP final response from each remaining Serving RFSS in the group (i.e., any Serving RFSS in the group except the Serving RFSS whose SIP INVITE request has precedence) that has not yet returned any final response to the SIP INVITE from the Home RFSS, following the procedures in Section 6.4.1.1 of this document. Subsequently, the Home RFSS:
 - a. SHOULD send a SIP re-INVITE request to each remaining Serving RFSS in the group that has returned a 2xx final response to the SIP INVITE from the Home RFSS, following the procedures in Section 6.4.3.1 of this document, to update the session parameters to match those of the new call initiated by the given Serving RFSS.
 - b. SHALL initiate SIP INVITE requests to those Serving RFSSs in the group that have not yet joined the group call, following the procedures in Section 6.6.1.3 of this document.

5. Any time after receiving a SIP 1xx provisional response and before receiving a final response from the Serving RFSS, the Home RFSS MAY cancel its SIP INVITE request, following procedures in Section 6.4.2 of this document.
6. Upon receipt of a VCE_SG_UNKNOWN failure response, the Home RFSS SHALL consider the Serving RFSS no longer registered to the group and SHOULD remove this RFSS from its list of Serving RFSSs for the group.
7. In the event that a SIP 3xx, 4xx, 5xx, or 6xx response for the SIP INVITE is received, the Home RFSS MAY retry the SIP INVITE request subject to local policy.

For an unconfirmed group call, talk spurt transmission MAY start as soon as the first SIP session and associated RTP session for a group call have been successfully established between the Home RFSS and a Serving RFSS.

For a confirmed group call, after successful setup of the first SIP session and RTP session, the Home RFSS SHALL control talk spurt transmission according to the following procedures:

1. Upon receipt of the first PTT Transmit Request from a Serving RFSS, the Home RFSS SHALL check whether sufficient number of RFSSs have RF resources available to support the confirmed group call.
2. In the event that sufficient RF resources are available, the Home RFSS SHALL grant the first PTT Transmit Request the permission to transmit talk spurts.
3. In the event that insufficient RF resources are available, the Home RFSS SHALL start a Tgchstartconfirmed timer.
 - a. In the event that sufficient RF resources later become available, the Home RFSS SHALL stop the Tgchstartconfirmed timer and grant one of the received PTT Transmit Requests the permission to transmit, using the selection criteria as specified in Section 7.9.1 of this document.
 - b. Upon expiry of the Tgchstartconfirmed timer, the Home RFSS SHOULD grant one of the received PTT Transmit Requests the permission to transmit talk spurts, using the selection criteria as specified in Section 7.9.1 of this document.

4. Depending on local policy, the Home RFSS MAY or MAY NOT check RF resource availability , by repeating the above steps, before granting each subsequent PTT Transmit Request the permission to transmit.

Moreover, as soon as the first SIP session for a confirmed or unconfirmed group call have been successfully established between the Home RFSS and a Serving RFSS, the Home RFSS SHALL start a Tgchhangtime timer that is reinitialized every time the Home RFSS detects voice activity on the RTP sessions for the group call.

As each RTP session is established, the Home RFSS SHALL bridge it to other RTP session(s) established earlier, and SHALL control the ensuing talk spurts among the various RFSSs participating in the group call, following the procedures in Section 7 of this document.

6.6.1.3.2 Actions Required at Serving RFSS

The Serving RFSS SHALL process the received SIP INVITE request, according to the procedures in Section 6.4.1 of this document, with the following qualifications:

1. In the event that the group is not known to the Serving RFSS, the Serving RFSS SHALL return a VCE_SG_UNKNOWN failure response.
2. In the event that the Serving RFSS detects that none of the SDP media formats offered by the Home RFSS is accepted by the Serving RFSS, it SHALL return a VCE_VOCODER_MODES_UNACCEPTED failure response.

NOTE: For older RFSS implementations that only support the Full Rate vocoder mode with encoding-name of the rtpmap attribute set to "X-TIA-P25-IMBE" (see Section 3.7), a GEN_SDP_PARAMS_UNACCEPTABLE failure response is returned instead.

3. In the event that the Serving RFSS is able to allocate RTP resource to set up the RTP session with the Home RFSS, it SHALL return a SIP 200 (OK) response that carries a Type 1 SDP answer. The Type 1 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Home RFSS in its SIP INVITE request, has been accepted by the Serving RFSS.
4. In the event that the Serving RFSS is unable to allocate RTP resource, it SHOULD return a SIP 200 (OK) response that carries a Type 2 SDP answer, and SHOULD send a re-INVITE request that carries a Type 1 SDP offer to the Home RFSS when RTP resource is later available,

following the procedures in Section 6.4.3 of this document. The following qualifications apply:

- a. The Type 2 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Home RFSS in its SIP INVITE request, has been accepted by the Serving RFSS.
 - b. The subsequent Type 1 SDP offer SHALL indicate which particular media format, chosen from the set of SDP media formats originally offered by the Home RFSS in its SIP INVITE request, is now offered by the Serving RFSS, also taking into account the media format in the Type 2 SDP answer that the Serving RFSS previously sent to the Home RFSS.
5. For a confirmed group call, in the event that the 200 (OK) conveys a Type 1 SDP answer, it SHALL include an "RF Resource Availability" parameter (as defined in Section 3.8.5.2) that indicates whether the Serving RFSS has available RF resource to support the confirmed group call. The Type 1 SDP answer SHALL indicate which particular media format, chosen from the set of SDP media formats offered by the Home RFSS in its SIP INVITE request, has been accepted by the Serving RFSS. In the event that RF resource is currently unavailable, the Serving RFSS SHALL send a re-INVITE request, with appropriate "RF Resource Availability" parameter setting, to the Home RFSS when RF resource is later available, following the procedures in Section 6.4.3 of this document.
 6. In the event that the "Protected" parameter, as specified in Section 3.8.5.7 of this document, is present in the received INVITE request's ISSI-specific body, or if the parameter is not present and takes on the implied default value of "unprotected", then the Serving RFSS SHALL handle the call initially according to the received "Protected" value (be it protected or unprotected).
 7. The Serving RFSS SHALL process any received SIP CANCEL or BYE request following the procedures in Sections 6.4.2.1 and 6.4.4.1 of this document.

6.6.2 Subsequent Modification of ISSI SIP Session Characteristics

After a SIP session is being set up between the Home RFSS and a Serving RFSS, either RFSS MAY request to modify the characteristics of the SIP session by sending a Re-INVITE request to the other RFSS, following the procedures in Section 6.4.3 of this document.

Specifically, a SIP Re-INVITE request MAY be sent under the following situations:

1. Offer a change in the SDP media format for talk spurt transmission between the two RFSSs, using Type 1 SDP offer or Type 2 offer as defined in Section 3.7 of this document.
2. Offer a change in the RTP resource availability for setting up or tearing down of an RTP session between the two RFSSs, using Type 1 SDP offer or Type 2 SDP offer as defined in Section 3.7 of this document.
3. Offer a change in the call priority, per the SIP header field "Priority" as specified in Section 3.5.13 of this document.
4. Offer a change in the RF resource availability state (i.e., whether RF resource is available to support the confirmed group call) after a SIP session is set up for a confirmed call, per the "RF Resource Availability" attribute specified in Section 3.8.5.2 of this document.
5. Synchronize the SIP states between the Home RFSS and Serving RFSS when there is no RTP session (i.e., employing SIP heartbeats as described in Section 6.6.4).
6. Combination of (1) through (4) above.

6.6.3 Handling (re-)INVITE Request Collisions

In the event that a Serving RFSS and the Home RFSS simultaneously send SIP INVITE or re-INVITE requests to each other for a given group call segment, procedures in the following subsections are used to handle the SIP request collision.

6.6.3.1 Handling Simultaneous INVITE Requests

1. In the event that the colliding SIP INVITE requests are of equal priority, then the Home RFSS's SIP INVITE request SHALL have precedence over the Serving RFSS's SIP INVITE request.
2. In the event that the colliding SIP INVITE requests have different priorities, then the SIP INVITE request with higher priority SHALL have precedence over the other SIP INVITE request.

The RFSS whose SIP INVITE request has precedence SHALL reject the received SIP INVITE request by sending a VCE_COLLIDING_CALL to the other RFSS,

The RFSS whose SIP INVITE request does not have precedence MAY cancel its own SIP INVITE request by sending a SIP CANCEL request to the other RFSS.

6.6.3.2 Handling Simultaneous re-INVITE Requests

In the event that a Serving RFSS and the Home RFSS simultaneously send SIP re-INVITE requests to each other, the Home RFSS SHALL return a 491 (REQUEST PENDING) to the Serving RFSS, and vice versa.

The Serving RFSS and Home RFSS SHALL each start a timer with different timer values set according to procedures in Section 14.1 of [RFC3261], and SHALL each reattempt its re-INVITE request when its timer expires.

6.6.3.3 Handling Simultaneous INVITE Request and Re-INVITE Request

In the event that an RFSS (be it Serving RFSS or Home RFSS) sends an INVITE request to the other RFSS (be it Home RFSS or Serving RFSS), while the other RFSS sends a Re-INVITE request to the RFSS, then the INVITE request SHALL have precedence over the re-INVITE request.

The RFSS whose SIP INVITE request has precedence SHALL reject the received SIP re-INVITE request by sending a VCE_COLLIDING_CALL to the other RFSS,

The RFSS whose SIP re-INVITE request does not have precedence MAY cancel its own SIP re-INVITE request by sending a SIP CANCEL request to the other RFSS.

6.6.4 ISSI SIP Heartbeat

SIP heartbeats, in the form of SIP re-INVITE requests, are sent by the Home RFSS to synchronize the SIP states between the Home RFSS and Serving RFSS when there is no RTP session. A SIP heartbeat is a SIP re-INVITE carrying a Type 2 SDP offer. The dialog identification is the same as for the original dialog, per Section 14 of [RFC3261]. The following procedure applies:

1. Upon successful setup of a SIP session with a Serving RFSS (i.e., upon sending or receiving a SIP ACK request for a 200 OK final response to the SIP INVITE request), the Home RFSS SHALL start a tgchconfirmsip timer.
2. The Home RFSS SHALL stop the timer under the following situations:
 - a. Upon successful termination of the SIP session, or
 - b. Upon successful setup of the associated RTP session, or

- c. In the event that, while awaiting availability of RTP resources at the Serving RFSS for setting up an impending RTP session, RTP resources at the Home RFSS has been preempted.
3. The Home RFSS SHALL restart the timer under the following situations:
 - a. Upon successful termination of the associated RTP session, or
 - b. Upon sending a SIP ACK request that confirms receipt of a 200 OK final response to its re-INVITE request to the Serving RFSS.
4. Upon expiry of the timer, the Home RFSS SHALL send a re-INVITE request to the Serving RFSS over the ISSI, in an attempt to initiate an RTP session, following the procedures in Section 6.4.3 of this document.
5. Selectively repeat the above steps, as appropriate.

6.6.5 ISSI SIP Session Termination by Home RFSS

The Home RFSS MAY initiate termination of a specific SIP session over the ISSI under the following situations:

1. When the Home RFSS's Transmission Control detects that it has lost RTP connectivity with the Serving RFSS (see Section 7 of this document), or
2. Upon expiry of the Tgchhangtime timer, or
3. Other situations as defined by local policy.

To terminate the SIP session, the Home RFSS sends a SIP BYE request to the Serving RFSS, following the procedures in Section 6.4.4 of this document.

6.6.6 ISSI SIP Session Termination by Serving RFSS

A Serving RFSS MAY initiate termination of a specific SIP session over the ISSI, therefore terminating the associated SIP session and RTP session with the Home RFSS, under the following situations:

1. When it detects that no more SUs are affiliated with the group, or
2. When the Serving RFSS's Transmission Control detects that it has lost RTP connectivity with the Home RFSS (see Section 7 of this document), or
3. Other situations as defined by local policy.

To terminate the SIP session, the Serving RFSS sends a SIP BYE request to the Home RFSS, following the procedures in Section 6.4.4 of this document.

A Serving RFSS MAY only terminate a SIP session. This action does not trigger the release of the group voice call.

6.6.7 ISSI Group Voice Call Release

In the event that no voice activity is detected for a group call before the expiry of the Tgchhangtime timer, the Home RFSS SHALL initiate a group voice call release (thus terminating all associated SIP and RTP sessions) over the ISSI by sending a BYE request to all Serving RFSSs involved, following the procedures in Section 6.4.4 of this document.

The ISSI group voice call is released when all SIP sessions associated with the group call have been terminated.

7. Push-to-Talk (PTT) Management

This section describes the procedures for the management of talk spurts within an RTP session. Throughout this section, the behaviors of the "MMF" and "SMF" are described without reference to the ISSI Control Function described in Section 2. For instance, when the MMF arbitrates a request, it makes a request to the control function (in accordance with the models of Section 2), which returns a result. The procedures sections in particular use this shorthand for the sake of brevity.

These procedures comprise the following control functions:

- Request, by a Serving RFSS, for permission to transmit RTP voice payload packets
- Queuing, Granting or Denying, by a Home RFSS of permission to transmit
- Initiation, by a Home RFSS, of outbound talk spurts
- Propagation of voice by a Home RFSS
- Management of Losing Audio by a given RFSS
- Termination of PTT transmission from a given RFSS
- Muting (per processing of the M bit as defined in Section 4.3.2 of this document) of undesired audio by a Home or Serving RFSS

Having established a call segment via the procedures described in Sections 6.5 and 6.6, both parties commence arbitration of individual PTTs (also known as "talk spurts") within the context of the call using RTP. During SIP Call Setup, the Home RFSS informs the Serving RFSS of the UDP port that it intends to use for the session, and, likewise, the Serving RFSS informs the Home RFSS of the port that it intends to use for the session. During SIP Call Setup the Home RFSS and Serving RFSS also negotiate the ISSI Vocoder Mode for the RTP session as either Full Rate, Half Rate or Native Vocoder Mode.

In accordance with Section 4, every RTP packet on the ISSI contains one and only one ISSI Packet Type block which appears as the first block of the packet. These together with the S-bit (from the Control Octet field of the ISSI Payload Header) and SSRC identifiers (from the RTP header) allow all of the required PTT control functions of the ISSI to be implemented.

7.1 SAP Primitive Model

Figure 73, Figure 74, and Figure 75 illustrate the mapping of Transmission Control SAP primitives to messages of the protocol³⁰.

As can be seen from Figure 73, the SpurtStart and normal SpurtRqst primitives map directly to PTT Transmit Start and PTT Transmit Request, respectively, and result in corresponding indications at the peer RFSS. Also, the SpurtRqst Indication Response and Decision Request primitives from the MMF map through PTT Transmit Grant, PTT Transmit Deny and PTT Transmit Wait into the Decision Indication at the SMF. Note that the Decision Request only occurs following a previous deferral of a decision. Other primitives (e.g. an unstopable request or marking a stream as unstopable) cause the protocol to “self-generate” a primitive response (e.g. SelfGrant). Finally, a request timeout causes the generation of a Decision Timeout indication, which, depending on policy, can cause a SelfGrant or Deny decision indication.

³⁰ For conciseness, throughout this section abbreviated forms of the primitive names are used, dropping the prefix (TC) and reducing the suffix as follows: “.Rqst” for “.Request”, “.Resp” for “.Response”, “.Ind” for “.Indicate”, “.IndResp” for “.IndicateResponse”. The remaining suffixes are not used in this protocol.

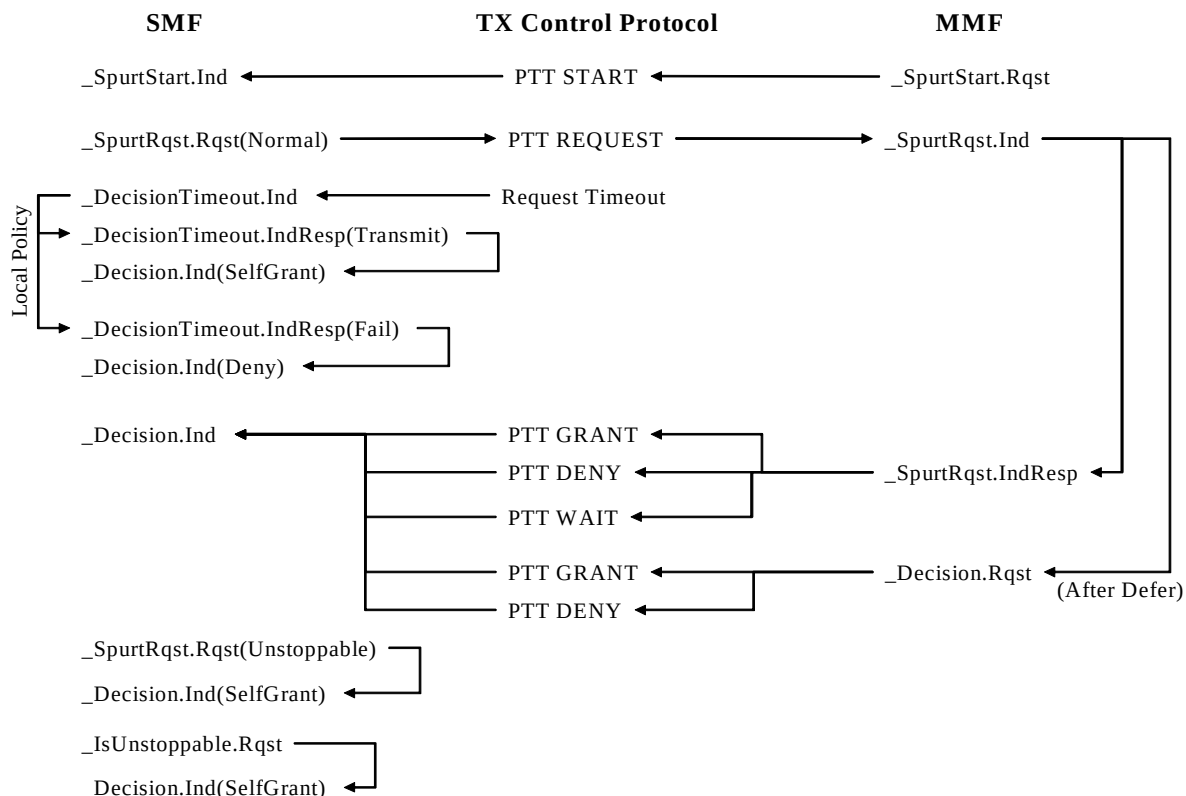


Figure 73 - Asymmetric TC-SAP Primitive Mappings

Figure 74 shows additional asymmetric primitive mappings. Here an audio request from the SMF can result in either a PTT Transmit Request or PTT Transmit Progress being generated by the protocol, depending on the internal state of the protocol. At the MMF, the resulting PTT Transmit Request message always generates a spurt request indication and an audio indication. In contrast, PTT Transmit Progress in certain cases generate a Spurt Request indication at the MMF (if all previous requests have been dropped by the network, for example), and also causes an Audio Indication.

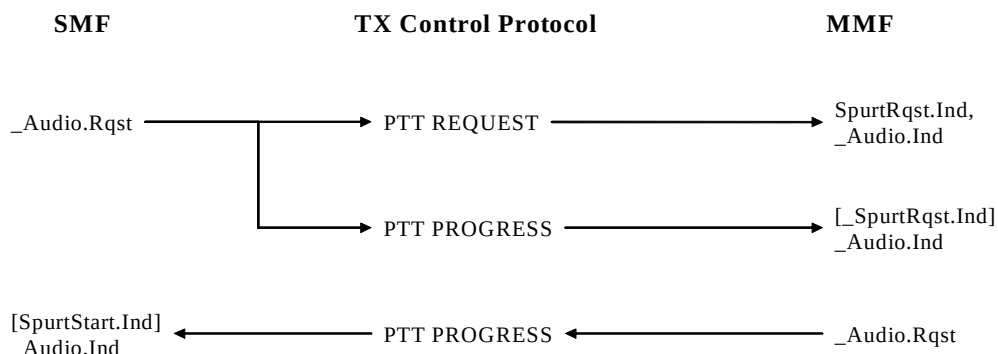


Figure 74 - Additional Asymmetric TC-SAP Primitive Mappings

In the opposite direction, the MMF audio request causes a PTT Transmit Progress message, which in certain cases generate a Spurt Start Indication at the SMF, and also causes an Audio Indication. Note that this discussion does not take “mute” functions into account, which prevent voice bearing audio requests from being sent as progress messages.

Finally, Figure 75 illustrates the symmetric primitive mappings, whereby a mute request results in PTT Transmit Mute or PTT Transmit Unmute messages that subsequently result in Mute indications, or a request to change the losing audio state causes the setting or clearing of the losing audio bit in the next normally generated PTT Transmit Progress or PTT Transmit Request message.

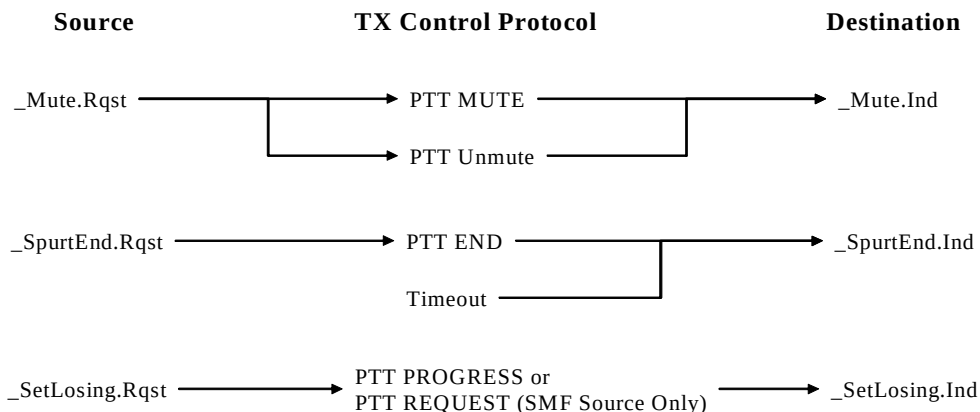


Figure 75 - Symmetric TC-SAP Primitive Mappings

7.2 State Model

The following paragraphs represent the protocol requirements in terms of two related sets of state models. The SMF Transmission and MMF Transmission state models refer, respectively, to talk spurts initiated by an SMF and the MMF. Likewise the SMF Receive and MMF Receive state models refer to talk spurt received by the SMF and MMF. A separate but related set of two state models describe the behavior of the RFSSs with respect to the mute and unmute functions.

In the following figures, circles represent states and squares represent particular important messages sent while transitioning between states. Diamonds represent critical decision points corresponding to "IndicationResponse" primitives. Lines are labeled with the conditions causing the transitions, and, following a "/" with the primitive invoked as a result of the transition or with the message sent as a result of the transition, or both. When a line is unlabeled, the transition is always taken.

Table 48 summarizes the transition conditions, their particular meanings, and the models in which they are used.

Table 48 - Summary of Transitions

Transition	Meaning	Applicable Models
Audio to Send	Audio from the Control Function is ready to send (<code>_Audio.Rqst</code>).	MMF Transmission SMF Transmission
Change in Losing Audio State	The Control Function has promoted the talk spurt to the preferred audio, or demoted it to losing (<code>_SetLosing.Rqst</code>)	MMF Transmission SMF Transmission
Deny	Receipt of PTT Transmit Deny message	SMF Transmission
Deny Decision	A decision to Deny a request, such as, for example, due to the call already being granted to another talk spurt (<code>_Decision.Rqst(Deny)</code>)	MMF Receive
Deny Trigger	A locally significant event that causes a Deny (<code>_Decision.Rqst(Deny)</code>).	MMF Receive
End Loss Timeout	Expiration of the protocol EndLossTimeout which is used to detect the end of a talk spurt when PTT Transmit End is lost.	MMF Receive SMF Receive Mute Receive Mute Transmit
End Trigger	Determination that the talk spurt has ended, such as receipt of Terminator Data Units over the CAI (<code>_SpurtEnd.Rqst</code>)	MMF Transmission SMF Transmission
Grant	Receipt of PTT Transmit Grant	SMF Transmission
Grant Decision	Decision to Grant the access to the given talk spurt (<code>_SpurtRqst.IndResp(Grant)</code>)	MMF Receive
Grant Trigger	A locally significant event that causes a grant (<code>_Decision.Rqst(Grant)</code>)	MMF Receive
Mute	Receipt of PTT Transmit Mute	Mute Transmit
Mute or Unmute for Unknown TSN	Receipt of PTT Transmit Mute or Unmute for previously unknown TSN, as would happen with a delayed packet	Mute Transmit
Mute Trigger	A locally significant event that causes a mute, such as the determination that some new talk spurt is more important and there is no demand for losing audio. (<code>_Mute.Rqst(Mute)</code>)	Mute Receive
Progress or PTT Progress	Receipt of a PTT Transmit Progress packet	MMF Receive SMF Receive
PTT End	Receipt of a PTT Transmit End packet	MMF Receive SMF Receive
PTT Request	Receipt of a PTT Transmit Request packet	MMF Receive

Transition	Meaning	Applicable Models
PTT Start	Receipt of a PTT Transmit Start packet	SMF Receive
Receive M=0	Receipt of a packet with the mute bit equal to 0	Mute Receive
Receive M=1	Receipt of a packet with the mute bit equal to 1	Mute Receive
Request Timeout	Expiration of the RequestTimeout such as would occur due to packet loss on the network.	SMF Transmission
Response for Inactive TSN	Receipt of PTT Transmit Grant, PTT Transmit Deny or PTT Transmit Wait for a talk spurt that is not currently active (e.g. one that has ended)	SMF Transmission
TMute-Progress Timeout	Expiration of the TMuteProgress timer that defines the periodicity of transmission heartbeats while muted.	Mute Transmit
Tx Trigger	Receipt of a locally significant Tx trigger such as a PTT voice request from an SU (<code>_SpurtRqst.Rqst(Normal)</code> or <code>_SpurtStart.Rqst</code>)	MMF Transmission SMF Transmission
Tx Trigger becomes Unstoppable	A locally significant event corresponding to an SU transitioning from a requesting to a transmitting state, such as might occur when an SU has spent too much time in queue and needs to be granted the channel. (<code>_IsUnstoppable.Rqst</code>)	SMF Transmission
Unmute	Receipt of a PTT Transmit Unmute packet	Mute Transmit
Unmute Trigger	Locally significant determination that a talk spurt has to be unmuted, such as when a higher priority spurt has ended. (<code>_Mute.Rqst(Unmute)</code>)	Mute Receive
Unstoppable Tx Trigger	Receipt of a locally significant transmission trigger for a unit that cannot be interrupted, such as for a conventional talk spurt (<code>_SpurtRqst.Rqst(Unstoppable)</code>)	SMF Transmission
Wait	Receipt of PTT Transmit Wait	SMF Transmission
Wait Decision	Determination that a decision has to be deferred, such as during confirmed call setup prior to the confirmation of resources (<code>_SpurtRqst.IndResp(Deferred)</code>)	MMF Receive

Transition	Meaning	Applicable Models
Wait Timeout	Expiration of the WaitTimeout which indicates that the SMF has to retry a PTT Transmit Request to stimulate a decision by the MMF.	SMF Transmission

Each state model described (transmitting, receiving, mute/unmute, etc.) is for a single particular talk spurt (i.e. a specific transmission sequence number).

7.2.1 Transmission State Model

Figure 76 illustrates a state model for a SMF initiating a talk spurt to an MMF.

Two types of transmission triggers (Tx Triggers) are received by a SMF. A “normal” trigger (Tx Trigger) is one that can be rejected, such as those that occur due to a PTT Transmit Request from an SU on a trunked system. An “Unstoppable” trigger is one that cannot be rejected without loss of audio, such as the initiation of a talk spurt by an SU on a conventional system. Under certain circumstances, a normal trigger can convert to an “Unstoppable” trigger, due to excessive queuing times or some other local policy.

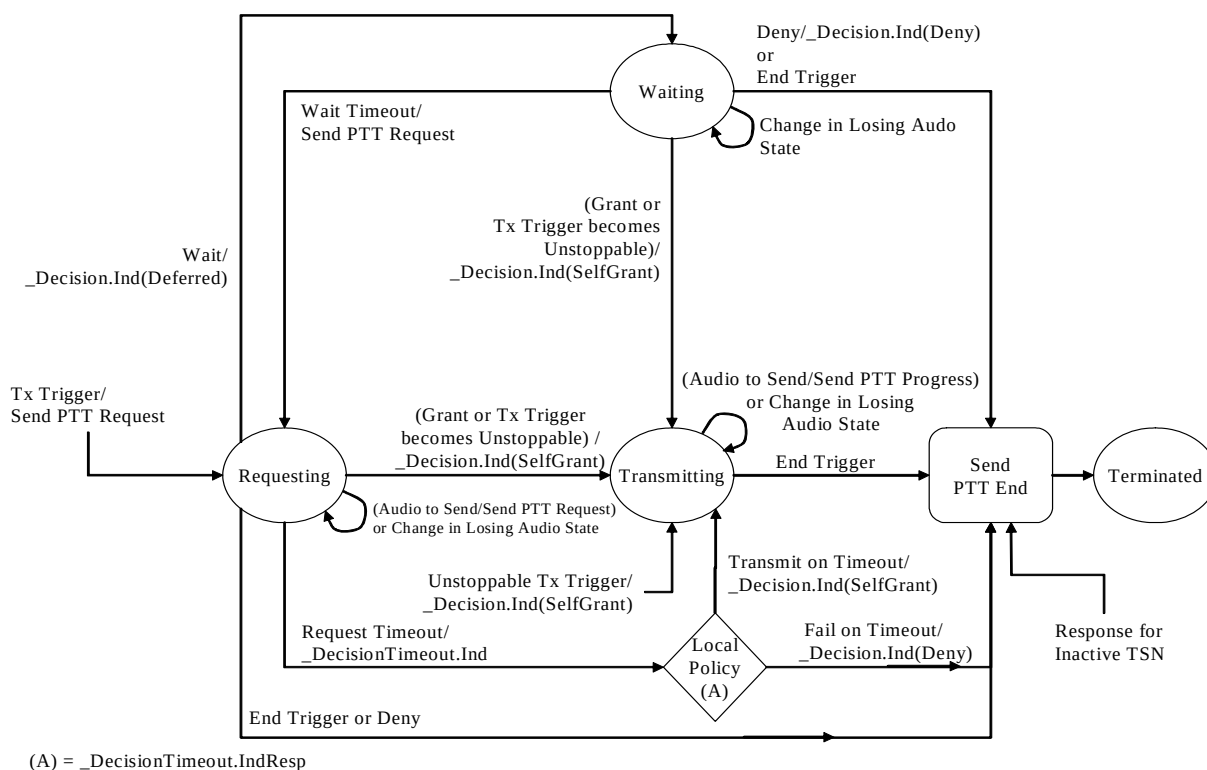


Figure 76 - SMF Transmission State Model

Upon receipt of either type of Tx Trigger, a transmission state instance will be created, starting in either the Requesting or Transmitting state, depending on the type of trigger. In the Requesting state, the SMF transmits PTT Transmit Requests and remains until it receives either a response to a PTT Transmit Request (Grant, Wait or Deny), or determines that it can no longer wait for a response and has to proceed with the talk spurt transmission ("Tx Trigger becomes Unstoppable"), or it decides to terminate the request ("Request Timeout"). If a Request Timeout occurs, the SMF makes a transition to the Terminated state or the Transmitting state, depending on local policy. An End Trigger (e.g. receipt of Terminator Data Units on the CAI) or Deny response likewise transitions to the Terminated state. A grant of the PTT Transmit Request causes a transition to Transmitting state, as does the determination that the talk spurt is "unstoppable". Finally, a "Wait" response to the request indicates that the MMF has not yet made a decision about the request, and causes a transition to the Waiting state.

In the Waiting State, the SMF waits for a final grant or deny from the MMF, periodically retrying the request. Otherwise, the Waiting state is like the Requesting state, with similar transitions to Transmitting and Terminated.

In the Transmitting state, PTT Transmit Progress packets are sent to the MMF. In response to an End Trigger, the SMF proceeds to the Ending state to terminate the talk spurt.

If the initial trigger is unstoppable, the process begins in the Transmitting state.

Transitions to the Terminated state pass through “Send PTT Transmit End” where a sequence of PTT Transmit End packets is sent to the MMF. PTT Transmit End packets are not acknowledged, so upon completion of the sequence, the SMF transitions to the Terminated state. Upon entry to the Terminated state, the TSN can be recycled in accordance with local policy. Note that PTT Transmit Ends are also sent whenever the SMF receives responses (Grant, Deny, Wait) for any inactive TSN.

Figure 77 illustrates a similar state model for transmissions from the MMF to a SMF. Upon receipt of a Tx Trigger (_SpurtStart.Rqst), the MMF sends a burst of PTT Transmit Start messages, informing the SMF that a talk spurt is beginning. It then immediately enters the Transmitting state.

In the transmitting state, the MMF sends PTT Transmit Progress packets to the SMF until the spurt is terminated. It then sends PTT Transmit End and transitions to the terminated state.

Note that the SMF transmission model differs from the MMF only in the addition of the *Requesting* and *Waiting* states. Essentially, the MMF is always “unstoppable” with regard to the SMF.

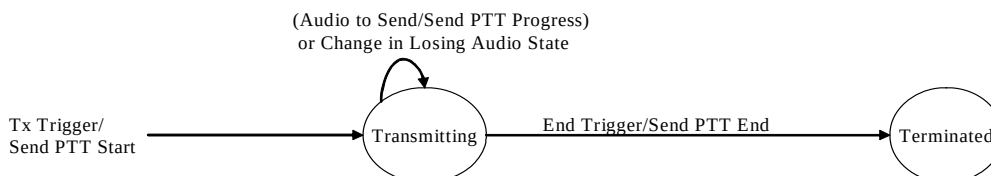


Figure 77 - MMF Transmission State Model

7.2.2 Receive State Models

Figure 78 illustrates the state model for an SMF receiving a talk spurt from an MMF. An SMF would create an instance of a receive state, and enter *Receiving* upon receipt of a PTT Transmit Start or PTT Transmit Progress message with a new transmission sequence number. In the *Receiving* state, the SMF processes PTT Transmit Progress packets in accordance with local policy, and transitions to “Done” upon receipt of PTT Transmit End, or when a sufficient time has passed without receiving any Progress packets.

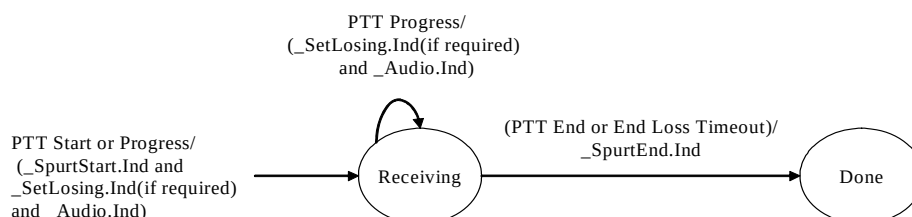


Figure 78 - SMF Receive State Model

The MMF Receive State Model, shown in Figure 79, is the most complicated of the four as it has to accommodate arbitration of PTT Transmit Requests. The MMF creates a state instance when it receives a PTT Transmit Request or PTT Transmit Progress with a new Transmission Sequence Number from an SMF. These correspond to a “normal” Tx Trigger, and an “unstoppable” TX trigger respectively in the SMF.

The *Receiving* and *Done* states are identical to those of the SMF with the exception that a PTT Transmit Request while in *Receiving* is responded to with PTT Transmit Grant. This portion of the state table is entered as a result of receipt of a PTT Transmit Progress packet either following arbitration or without arbitration as a result of an “unstoppable” trigger. It is also entered following a grant of a normal trigger.

The mute and unmute states described in Section 7.2.3 are particularly important here to allow the Home RFSS to mute talk spurts that are unneeded (losing audio, for example) but cannot be stopped.

When the process is instantiated as a result of a PTT Transmit Request, however, the MMF arbitrates the request, and enters the *Receiving*, *Deny* or *Wait* states depending on its decision.

Upon entry into the *Wait* state, the MMF sends PTT Transmit Wait, informing the SMF that it has not yet made a decision. This causes a corresponding transition of the SMF process to the “Waiting” state, in which, as described above, it

periodically resends the request. Upon making a decision, the MMF enters the Receiving or Deny State. It is also possible for the SMF to unilaterally decide to commence the transmission of progress packets. In this case, the MMF transitions to the receiving side of the table to decide whether to mute or accept the talk spurt.

One of the primary functions of an MMF is to fan-out talk spurts received from SMFs to all other SMFs that have interest in the talk spurt. Thus, when a MMF enters the *Receiving* state for one talk spurt, it

- Often initiates other talk spurts with other SMFs (e.g. distributing incoming audio packets received for this state machine to other SMF),
- Can change the “losing-audio” marking for some other existing talk spurts to other SMF.

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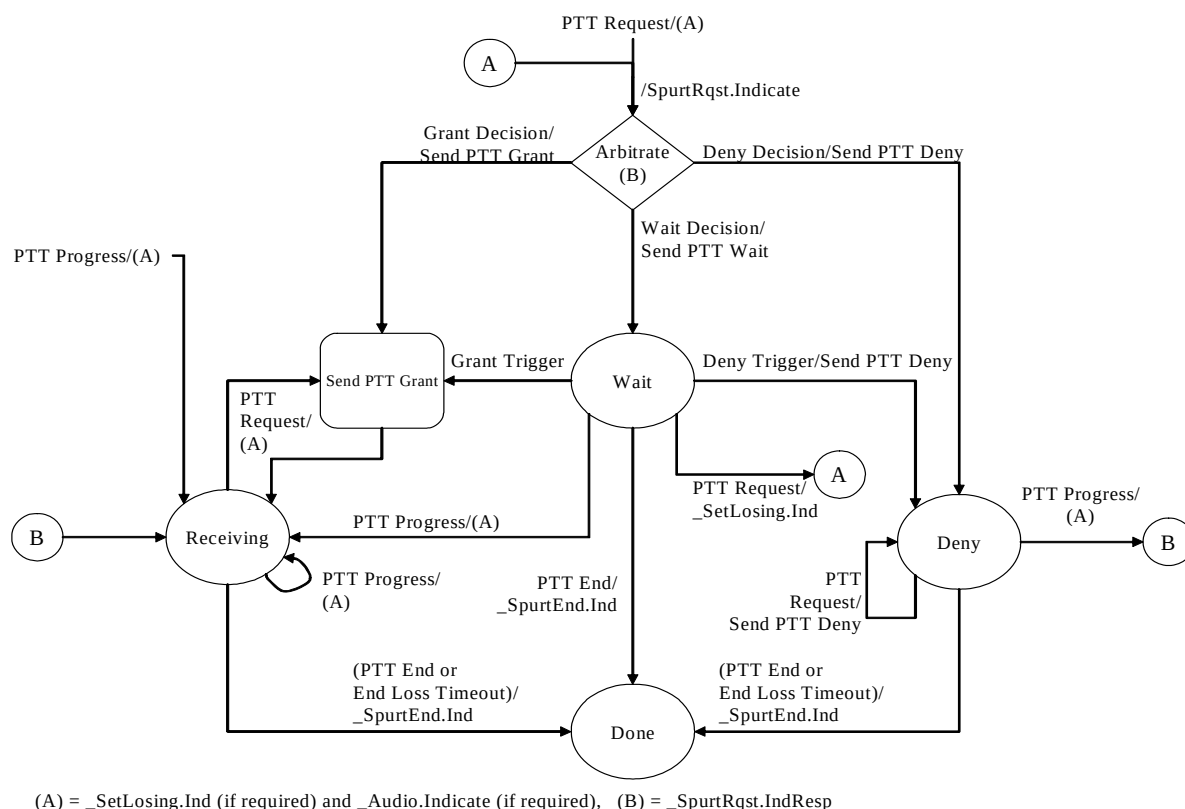


Figure 79 - MMF Receive State Model

When a MMF enters the “Done” state, the state machine will then be deleted, and it can change the “losing-audio” marking for some other existing talk spurts to other SMF.

7.2.3 State Models for the Management of Voice Content

The reception of voice content is managed through the use of the PTT Transmit Mute and PTT Transmit Unmute messages, and the “M” bit of the ISSI Packet Type payload. Figure 80 and Figure 81 illustrate receive and transmit state models, respectively, for the management of voice content. Herein, the receive state model applies to the receiver of the talk spurt that is sending the mute/unmute messages, and the transmit model applies to the transmitter of the talk spurt that is receiving the mute/unmute messages.

The Mute/Unmute state machine in Figure 80 begins in the *Unmute* state. A Mute Trigger (the determination that the RFSS doesn't want to receive voice payload for this talk spurt) causes a transition to the *Mute* state that includes the transmission of a PTT Transmit Mute message to the transmitting RFSS. While in the *Mute* state, the receiving RFSS sends PTT Transmit Mute whenever it receives a message with M=0 (unmuted). It also expects to receive periodic heartbeat packets with the correct TSN and mute (M) state. On receipt of PTT Transmit End, or the expiration of the End Loss timeout, the talk spurt is done.

Upon receiving an Unmute trigger, the RFSS sends a single instance of PTT Transmit Unmute and returns to the un-muted state. Again, receipt of the wrong M state causes a retransmission of the Unmute message.

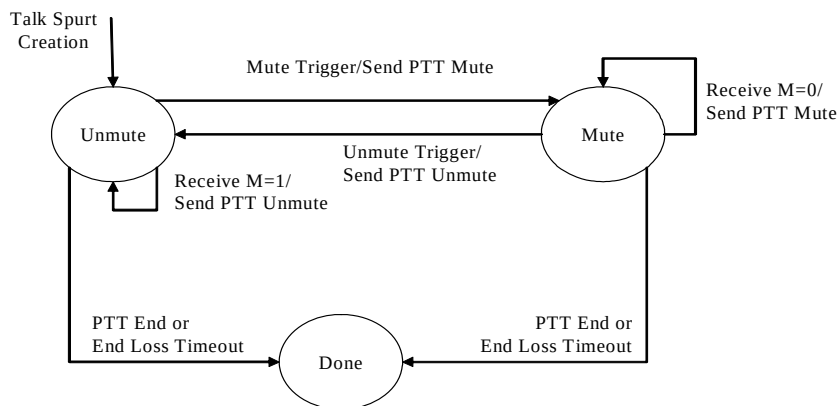


Figure 80 - RFSS Mute Function Receive State Model

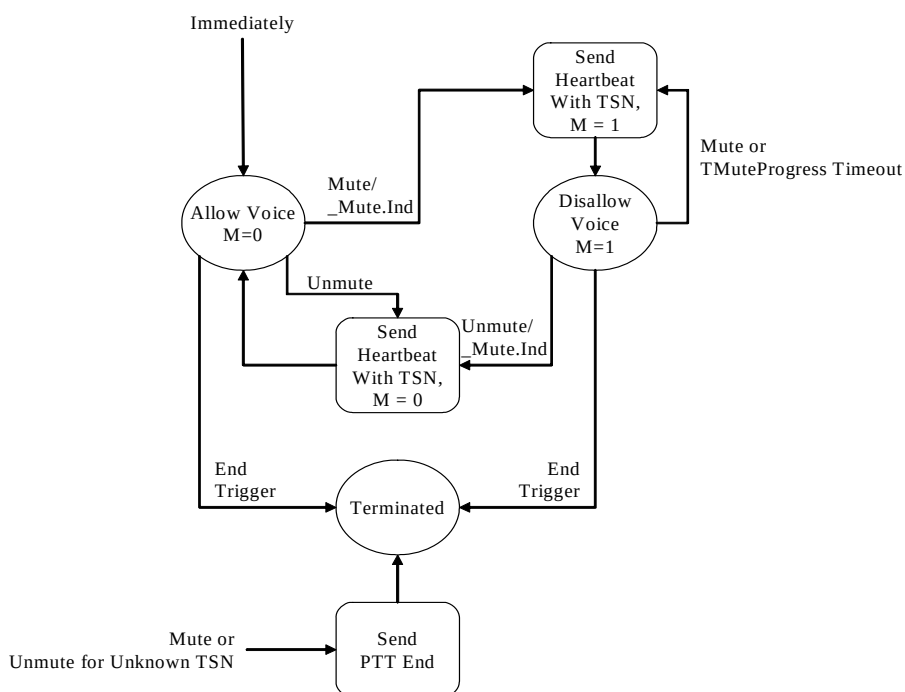


Figure 81 - RFSS Mute Function Transmit State Model

The Transmit state model for muting is shown in Figure 81. It is entered immediately upon creation of the talk spurt. Initially, voice is allowed, and all packets (for the talk spurt) from the transmitting RFSS to the receiving RFSS have M=0. Upon receipt of a mute command, a heartbeat is sent with the TSN of the talk spurt and M=1. Voice is disallowed for the talk spurt until the RFSS receives Unmute which it responds to with a similar heartbeat. As with the other transmit state models, receipt of a command (Mute or Unmute) for an unknown TSN results in the transmission of PTT Transmit End.

7.3 Management of Header Fields

All of the fields of the RTP header are managed in accordance with [RFC3550]. However several of these fields, the SSRC field, the Sequence Number and the Timestamp, require detailed specification for the purposes of the ISSI.

The SSRC field SHALL always contain the SSRC of the RFSS sending the RTP packet (i.e. an MMF SHALL always use its own SSRC identifier, and an SMF SHALL always use its own SSRC identifier as determined by the role of the

RFSS in the call based on the information in the Request URI, the Route header field, the From header field and the To header field of the SIP INVITE Request. For detail, see Sections 4.1 and 6.4.1.1.)

The informative S bit provides a quick indication on the PTT transmission status, whether it is at the transmit request or transmit end phase, or somewhere in between. The S bit starts with value 0 during the PTT Transmit Request phase. The S bit toggles to 1 during the PTT Transmit Grant and PTT Transmit Progress phases. The S bit returns to 0 during the PTT Transmit End phase. It SHALL be set as defined in Table 49.

Table 49 - Summary of Packet Types

ISSI Packet Type	Sent By	Response	S-bit
PTT Transmit Request	SMF	PTT Transmit Grant, PTT Transmit Deny, PTT Transmit Wait	0
PTT Transmit Start	MMF	None	1
PTT Transmit Grant	MMF	None	1
PTT Transmit Deny	MMF	None	0
PTT Transmit Wait	MMF	None	0
PTT Transmit Progress	Both	None	1
PTT Transmit End	Both	None	0
PTT Transmit Mute	Both	Heartbeat (non-zero TSN)	1
PTT Transmit Unmute	Both	Heartbeat (non-zero TSN)	1
PTT Heartbeat	Both	None	0
PTT Heartbeat Query	Both	Heartbeat (Zero TSN)	0

Three primary call types are supported on the ISSI: Unconfirmed Group Calls (UGC), Confirmed Group Calls (CGC), and SU-to-SU Calls (SUC). Table 50 summarizes the typical use of the various packet types in these calls, but does not restrict their use. The table entries indicate the transmitter of the packet (MMF or SMF or Both), or N/A if the packet is not typically used for the given type of call.

Table 50 - Summary of Typical Use

ISSI Packet Type	UGC	CGC	SUC	Notes
PTT Transmit Request	SMF	SMF	SMF	
PTT Transmit Start	MMF	MMF	MMF	
PTT Transmit Grant	MMF	MMF	MMF	
PTT Transmit Deny	MMF	MMF	N/A	Not typically used in SU calls because SU calls are treated as full duplex.
PTT Transmit Wait	N/A	MMF	N/A	
PTT Transmit Progress	Both	Both	Both	
PTT Transmit End	Both	Both	Both	
PTT Transmit Mute	Both	Both	N/A	Not typically used in SU calls because SU calls are treated as full duplex.
PTT Transmit Unmute	Both	Both	N/A	Not typically used in SU calls because SU calls are treated as full duplex.
PTT Heartbeat	Both	Both	Both	
PTT Heartbeat Query	Both	Both	Both	

The RTP header sequence number SHALL be maintained in accordance with [RFC3550] Section 5.1, that is, it SHALL be initialized to an unpredictable value at the beginning of the call, and SHALL be incremented by one for each RTP packet sent, regardless of the transmission sequence number (TSN) of the packet. For example, if the RTP sequence number for a Serving RFSS is 7, and it is sending two talk spurts (A & B) for a group to the Home RFSS, with alternating packets, the RTP sequence numbers would be A=7, B=8, A=9, B=10, and so on.

An RFSS SHALL process all RTP packets related to a particular talk spurt (i.e. with the same TSN) in order of their RTP header sequence number. If a late arriving packet cannot be processed in order, it SHALL be silently discarded.

For a group call, the Home RFSS (MMF) SHALL act as an RTP mixer, re-sequencing and time-stamping incoming RTP packets so as to present a consistent stream to each SMF. For SU-to-SU call, the called Home RFSS SHALL perform the re-sequencing and re-numbering of RTP packets from the called RFSS to the calling RFSS. The Calling Home RFSS SHALL perform the re-sequencing and re-numbering of RTP packets from the calling RFSS to the called RFSS. This is to accommodate roaming.

7.4 Management of Transmission Sequence Numbers

Upon initiation of a talk spurt transmission on a call segment, a transmitting RFSS SHALL choose a transmission sequence number (TSN) consistent with the restrictions in the following paragraphs.

The Transmission Sequence Number applicable to talk spurts originating at a Serving RFSS traveling to a Home RFSS SHALL be even and greater than 0. The transmission sequence number applicable to talk spurts originating at a Home RFSS traveling to a Serving RFSS SHALL be odd.

For each new talk spurt, the initiating RFSS SHALL select a currently unused Transmission Sequence Number. The following is a recommended algorithm for selection of the Transmission Sequence Number:

```

if (all valid transmission sequence numbers are active) FAIL;
else
{
  do
  {
    TSN = (TSN + 2) mod 64;
    if (TSN == 0) TSN = TSN + 2
  }
  until an unused TSN is found
}

```

Other methods for selecting the TSN are possible, (including, for example, least recently used) and an RFSS SHALL NOT assume that its peer assigns TSNs sequentially. All RTP packets, in whatever direction, pursuant to the same talk spurt SHALL have the same transmission sequence number³¹.

For SU-to-SU calls, the transmission sequence number applicable to talk spurts originating in the Calling Home RFSS and traveling to the Called Home RFSS SHALL be even and greater than 0. Those originating from the Called Home RFSS SHALL be odd.

The transmission sequence number for *Heartbeat* packets employed for connection maintenance SHALL always be 0. The transmission sequence number 0 SHALL NOT be used with regular voice packets (e.g. *Progress* and *Request*). The TSN for *Heartbeat Query* SHALL always be 0. The transmission sequence number for *Heartbeat* packets used to keep alive a muted talk spurt SHALL be the TSN for the talk spurt (see Section 7.8.1).

Within the following procedures, a Calling Home RFSS in a unit-to-unit call SHALL behave in accordance with the procedures for a MMF in its transactions

³¹ Note that this does not imply that the Transmission Sequence Number is the same on all segments of a call. In general, the Transmission Sequence Number can be different on each different call segment.

with the Calling Serving RFSS, and SHALL behave in accordance with the procedures for an SMF in its transactions with the Called Home RFSS. The Called Home RFSS SHALL always behave in accordance with the procedures for an MMF.

7.5 Losing Audio Processing

At any particular time, a call MAY carry 0, 1 or more than one talk spurt. A talk spurt is uniquely identified on a call segment by the Transmission Sequence Number described in Section 7.4.

Each talk spurt also has a volatile characteristic known as the “losing audio flag”. When an RFSS forwards more than one talk spurt to a peer, it SHALL set the losing audio flag on packets associated with all streams except the preferred one. Upon receiving a talk spurt from an SMF, depending on the current state of the call, the MMF MAY re-mark the losing-audio bit before distributing the talk spurts to other SMFs.

At any particular time, only one stream in each direction of an RTP session MAY be the preferred talk spurt (i.e., the one with the losing audio flag cleared). For transmission over an air interface, an RFSS SHOULD prefer talk spurts with the losing audio flag clear.

When an MMF begins receiving a new talk spurt (due to issuing a PTT Transmit Grant as in Section 7.9.1 or due to receiving unsolicited PTT Transmit Progress messages as in Section 7.9.6), or when a talk spurt is completed (as in Section 7.9.7), it MAY select a new preferred talk spurt, and set the losing audio flag on existing talk spurts appropriately.

7.6 Transmission Vocoder Mode

As previously stated, during SIP Call Setup the Home RFSS and Serving RFSS negotiate the ISSI Vocoder Mode for the RTP session as either Full Rate, Half Rate or Native Vocoder Mode.

In Native Mode, the Vocoder Mode MAY vary from talk spurt to talk spurt, but SHALL remain the same for any given talk spurt. The TalkSpurt Vocoder Mode block SHALL be included in the packet only if a voice frame is not included in the packet. When a voice frame is included in a packet, the Vocoder Mode for the talk spurt is implied by the format of the voice frame. When there is no voice frame in a packet, the TalkSpurt Vocoder Mode block indicates which Vocoder Mode (Full Rate or Half Rate) the talk spurt is using.

In Full Rate and Half Rate modes, the Vocoder Mode that is determined during SIP Call Setup SHALL be used for every talk spurt in the RTP session. The TalkSpurt Vocoder Mode block SHALL NOT be included in Full Rate and Half Rate modes.

7.7 Handling Service Options

The Service Options in the ISSI Packet Type block SHALL be used by the SMF and the MMF to maintain the state of a talk spurt.

- The protected state of a talk spurt SHALL depend on the Service Options “P” bit.
- The emergency status of a talk spurt SHALL depend on the Service Options “E” bit.
- The priority of a talk spurt SHALL depend on the Service Options “Priority level” bits.

7.8 Sending RTP Packets

7.8.1 Sending Heartbeats

RFSSs SHALL send an RTP packet that includes a Heartbeat ISSI packet type and a TSN of 0 on all active RTP sessions at a nominal rate of once in **Heartbeat** seconds. Heartbeats are sent whether or not voice is active on the RTP session.

An RFSS MAY send a Heartbeat Query at any time. A MMF SHALL send a Heartbeat Query immediately upon establishment of an RTP session.

Heartbeats with non-zero TSNs are sent as part of the Mute and Unmute functions, described in Sections 7.9.8 and 7.9.9.

7.8.2 Initiation of a Serving Transmission

7.8.2.1 Requesting Permission for a Transmission (PTT Transmit Request)

A SMF MAY, at any time, request permission to transmit RTP voice payload packets to an MMF. Upon sending the initial PTT Transmit Request for a talk spurt, the SMF SHALL set the **RequestTimeout** to its default value.

If the transmit priority of the talk spurt is lower than or equal to that of the most recently transmitting RFSS, then, in order to allow for loss or delayed receipt of PTT Transmit End packets, and to optimize bandwidth utilization over the ISSI, an RFSS SHOULD only request permission to transmit if:

- 1) **MinKeybackTime** seconds have elapsed since the receipt of the first (of up to **Rend**) PTT Transmit End packets received from the MMF.

OR,

- 2) The RFSS has not received any non-heartbeat RTP packet for the session for more than **EndLossTimeout** seconds.

To ensure that the PTT Transmit Request reaches the MMF despite possible packet losses over the IP-based backbone, the RFSS SHALL continue to send PTT Transmit Request packets to the MMF at a nominal rate of once in **Trequest** seconds until one of the following events occurs:

1. The SMF receives PTT Transmit Grant from the MMF, in which case it SHALL behave as described in Section 7.9.3.
2. The SMF receives PTT Transmit Deny from the MMF, in which case it SHALL behave as described in Section 7.9.4.
3. The SMF receives PTT Transmit Wait from the MMF, in which case it SHALL behave as described in Section 7.9.5.
4. The **RequestTimeout** expires, in which case it SHALL behave as described in Section 7.10.1.
5. The SMF receives an end trigger, in which case it SHALL behave as described in Section 7.8.5.

An MMF SHALL NOT send PTT Transmit Request to a SMF. Talk spurts outbound from a MMF to an SMF are initiated via PTT Transmit Start.

PTT Transmit Request packets MAY include voice content, which SHALL be processed in accordance with Section 7.11.

7.8.2.2 Initiating a Talk Spurt without a-Priori Permission (Unstoppable Triggers)
In the event that an SMF receives a transmission trigger for which the voice stream cannot be discontinued at the source (an “unstoppable” trigger), it MAY start a talk spurt to the MMF by sending a PTT Transmit Progress packet as described in Section 7.8.4. This mechanism MAY be appropriate when the source of the talk spurt is an SU on a conventional channel, during an SU-to-SU call, and at other times.

The request mechanism of Section 7.8.2.1 is the preferred mechanism for an SMF to initiate a group talk spurt.

7.8.3 Initiation of a Home Talk Spurt (PTT Transmit Start)

An MMF SHALL initiate a talk spurt by sending **Rstart** instances of PTT Transmit Start at a nominal rate of once in **Tstart** seconds. If the MMF has PTT Transmit Progress packets available to send, it MAY simultaneously begin to send PTT Transmit Progress packets as described in Section 7.8.4 after sending at least one PTT Transmit Start packet and before completing the sequence of sending **Rstart** instances of PTT Transmit Start. Otherwise, the MMF SHALL begin to send PTT Transmit Progress packets after all PTT Transmit Start packets are sent.

7.8.4 Sending PTT Transmit Progress Packets

An SMF that has received a PTT Transmit Grant or that has initiated a talk spurt without permission, or an MMF that has sent a PTT Transmit Start SHALL send PTT Transmit Progress packets containing voice payload at a rate governed by the payload, if it has not received a PTT Transmit Mute as described in Section 7.9.8.

When the RFSS is operating in Native Vocoder Mode for the call segment, the voice payload SHALL be left in the format in which it was received. When operating in Full Rate or Half Rate Vocoder Mode for a call segment, the RFSS MAY convert the format of the source voice for inclusion in the PTT Transmit Progress packets if the source voice is not in the same format as required by the call segment.

An SMF using PTT Transmit Progress as a means of initiating a talk spurt without permission MAY send up to **Rstart** instances of PTT Transmit Progress containing at most a header payload before sending PTT Transmit Progress packets containing voice payload.

7.8.5 Terminating a Talk Spurt

Upon receipt of an End Trigger, an RFSS SHALL terminate the talk spurt by sending PTT Transmit End packets **Rend** times at a nominal rate of once in **Tend** seconds and SHALL also cancel all pending timeouts.

Following the transmission of the PTT Transmit End packets, the RFSS MAY tear down the indicated call-segment as described in Section 6, in accordance with local policies and procedures.

7.8.6 Muting a Talk Spurt

Upon receipt of a Mute Trigger, or upon detection of a M-bit conflict as described in Section 7.9.11, an RFSS SHALL mute the talk spurt by sending a single PTT Transmit Mute packet.

7.8.7 Unmuting a Talk Spurt

Upon receiving an Unmute trigger, an RFSS SHALL send a PTT Transmit Unmute packet. It SHALL thereafter resend PTT Transmit Unmute packets at a nominal rate of once in **Tunmute** seconds until one of the following events occurs:

- 1) The RFSS receives PTT Transmit End for the talk spurt, at which time it SHALL behave as described in Section 7.9.7.
- 2) The RFSS receives a packet for the talk spurt with the "M" bit set to 0, at which time it SHALL reset the **EndLossTimeout** to **TEndLoss**.

7.9 Processing of RTP Packets

RTP packets received by an RFSS SHALL be processed as described in the following sections.

Out of sequence RTP packets SHOULD NOT be processed. Furthermore, an RFSS that receives an RTP packet with an incorrect transmission sequence number or invalid packet type SHALL act in accordance with local policy that MAY include:

- 1) silently discarding the packet or
- 2) tearing down the call-segment

Except as specified hereafter, it SHALL NOT respond as if the packet was correctly numbered.

A transmission sequence number is incorrect when:

- 1) An RFSS that generates even-numbered transmission sequence numbers for its talk spurts on a particular call-segment (i.e. an SMF) receives
 - a. a PTT Transmit Start, PTT Transmit Progress or PTT Transmit End packet with an even transmission sequence number, or
 - b. a PTT Transmit Grant, PTT Transmit Deny, PTT Transmit Wait, PTT Transmit Mute or PTT Transmit Unmute packet with an odd transmission sequence number.
- 2) Or when an RFSS that generates odd-numbered transmission sequence numbers for its talk spurts on a particular call-segment (an MMF) receives
 - a. a PTT Transmit Request, PTT Transmit Progress or PTT Transmit End packet with an odd transmission sequence number, or

- b. a PTT Transmit Mute or Transmit Unmute packet with an even transmission sequence number.

The packet type is invalid when:

- 1) A Home RFSS in a group call, or a Called Home RFSS in an SU-to-SU call (an MMF) receives a PTT Transmit Start, PTT Transmit Grant, PTT Transmit Deny, or PTT Transmit Wait packet or when
- 2) A Calling Home RFSS in a SU-to-SU call receives PTT Transmit Request on its call-segment with the Called Home RFSS (i.e. when acting as an SMF) or when
- 3) Any Serving RFSS (SMF) receives PTT Transmit Request.

Moreover, an SMF that receives PTT Transmit Grant, PTT Transmit Deny, PTT Transmit Wait, PTT Transmit Mute or PTT Transmit Unmute for an inactive talk spurt (i.e., with no corresponding PTT Transmit Request) SHALL reply with PTT Transmit End as described in Section 7.8.5.

An MMF or SMF that receives PTT Transmit End or a PTT Heartbeat (with a non-zero TSN) for an unknown talk spurt SHALL silently discard the packet.

7.9.1 Processing PTT Transmit Request

An MMF that receives a PTT Transmit Request SHALL behave as follows:

- 1) It SHALL arbitrate the request in accordance with local policies and procedures governed by the guidelines below.
- 2) In the event that the MMF wishes to allow the talk spurt to proceed, it SHALL Grant the talk spurt by sending a single instance of the PTT Transmit Grant message. It SHALL also initiate talk spurts to other SMFs that it determines SHOULD be joined to the transmission, by sending PTT Transmit Start messages as described in Section 7.8.3 for each. The MMF MAY also select a new preferred talk spurt and set the losing audio flag on existing talk spurt appropriately, as described in Section 7.5.
- 3) In the event that the MMF wishes to prevent the talk spurt from proceeding, it SHALL deny the talk spurt by sending a single transmission of the PTT Transmit Deny message.

- 4) In the event that the MMF wishes to defer making a decision about the talk spurt (e.g., Confirmed group call processing), it SHALL reply with a single transmission of the PTT Transmit Wait message. Subsequently, the MMF SHALL reply with a single instance of PTT Transmit Grant or PTT Transmit Deny, as described above.
- 5) It SHALL process any voice content in accordance with Section 7.11.

The decision by a MMF to Grant, Deny or Defer a decision on a talk spurt SHALL be made in accordance with the following guidelines:

- 1) If the talk spurt has already been denied, the MMF SHALL Deny the talk spurt.
- 2) If the talk spurt has already been granted, the MMF SHALL Grant the talk spurt.
- 3) If a Wait decision has been made, and no subsequent decision has been made, the MMF SHALL continue to defer the decision.
- 4) If no talker is active for the session, the MMF MAY grant the talk spurt immediately.
- 5) If no talker is active for the session, the MMF MAY reserve judgment for Telection seconds, and make a decision based on all requests received in that time.
- 6) A console-initiated talk spurt SHOULD be preferred over an SU initiated talk spurt.
- 7) An emergency talk spurt SHOULD be preferred over any other talk spurt, regardless of priority.
- 8) A strictly higher priority talk spurt SHOULD be preferred over a lower priority talk spurt.
- 9) A talk spurt with voice payload SHOULD be preferred over a talk spurt without voice payload.
- 10) An earlier arriving request SHOULD be preferred over a later arriving request.
- 11) A strictly higher priority console initiated transmission request SHOULD be preferred over a lower priority console transmission.

- 12) If the protected state of the talk spurt (indicated in the Service Options field of the ISSI Packet Type block) is not supported by the destination SG, the MMF SHALL Deny the talk spurt.

7.9.2 Processing PTT Transmit Start

An SMF that receives a PTT Transmit Start packet SHALL apply local policy to determine whether or not it wishes to receive subsequent PTT Transmit Progress packets associated with the talk spurt and SHALL take such steps as MAY be required by local policy to prepare for the reception of said packets. It SHALL also set the **EndLossTimeout** to **TFirstPacketTime**.

7.9.3 Processing PTT Transmit Grant

A SMF receiving a PTT Transmit Grant SHALL stop sending PTT Transmit Request packets and SHALL cancel the **RequestTimeout**, and SHALL begin to send PTT Transmit Progress packets as described in Section 7.8.4, taking such other actions (such as granting an RF channel to a requesting SU) as MAY be required by local policy.

7.9.4 Processing PTT Transmit Deny

Upon receipt of a PTT Transmit Deny, an SMF that has requested permission to transmit as described in Section 7.8.2.1 and has not received a response, or has received a PTT Transmit Wait response as described in Section 7.9.5, SHALL terminate the talk spurt as described in Section 7.8.5. An SMF that is sending PTT Transmit Progress packets as described in Section 7.8.4 MAY terminate the talk spurt as described in Section 7.8.5.

It SHALL furthermore take other actions in accordance with local policy to notify the source of the Transmit Trigger that it has been denied access in accordance with the procedures for the particular source.

7.9.5 Processing PTT Transmit Wait

A SMF receiving a PTT Transmit Wait after requesting permission to transmit as described in Section 7.8.2.1 and not having received a response SHALL stop sending PTT Transmit Request packets and SHALL cancel the **RequestTimeout**, and SHALL set the **WaitTimeout** to its default value. After receiving a response, or while transmitting PTT Transmit Progress packets, the SMF SHALL silently discard PTT Transmit Wait.

It SHALL furthermore take other actions in accordance with local policy to notify the source of the Transmit Trigger that it is awaiting a final response (grant or denial) for its PTT Transmit Request, in accordance with the procedures for the particular source.

7.9.6 Processing PTT Transmit Progress

An RFSS receiving a PTT Transmit Progress packet for an unknown talk spurt SHALL begin to process PTT Transmit Progress packets as it would if it had issued a PTT Transmit Grant (for an MMF) or had received a PTT Transmit Start (for an SMF). An MMF in this condition SHALL also initiate talk spurts to other SMFs that it determines SHOULD be joined to the talk spurt, by sending a PTT Transmit Start as described in Section 7.8.3 for each. The MMF MAY also select a new preferred talk spurt and set the losing audio flag on existing talk spurts appropriately, as described in Section 7.5.

Upon receiving a PTT Transmit Progress packet, voice content SHALL be processed in accordance with Section 7.11 and the RFSS SHALL reset the **EndLossTimeout** to **TEndLoss** seconds.

7.9.7 Processing PTT Transmit End

PTT Transmit End signifies the end of a talk spurt. Upon receipt of a PTT Transmit End,

- 1) An SMF SHALL cancel all active timeouts.
- 2) An MMF, if it has not previously done so, SHALL terminate the talk spurts, associated with the talk spurt that has ended, that the MMF initiated to other SMFs using PTT Transmit Start, by sending PTT Transmit End messages to those SMFs as described in Section 7.8.5. The MMF MAY select a new preferred talk spurt, and set the losing audio flag on existing talk spurts appropriately, as described in Section 7.5.
- 3) An RFSS MAY, in accordance with local policies and procedures, tear down the call segment as described in Section 6.

7.9.8 Processing PTT Transmit Mute

Upon receipt of a PTT Transmit Mute, an RFSS SHALL reply with a single Heartbeat packet with the "M" bit set to 1 and the TSN of the talk spurt, and SHALL discontinue the forwarding of voice content to the muting RFSS. All subsequent messages for the talk spurt (prior to subsequent receipt of PTT Transmit Unmute) SHALL have the M bit set to 1.

Until it receives a PTT Transmit Unmute, the RFSS SHALL send Heartbeat messages with the "M" bit set to 1 and the TSN of the talk spurt at a rate of once in **TMuteProgress** seconds.

7.9.9 Processing of PTT Transmit Unmute

Upon receipt of a PTT Transmit Unmute packet, an RFSS SHALL reply with a single Heartbeat packet with the “M” bit set to zero and the TSN of the talk spurt and SHALL resume forwarding of voice content to the unmuting RFSS. All subsequent messages for the talk spurt (prior to receipt of a subsequent PTT Transmit Mute) SHALL have the M bit set to 0.

7.9.10 Processing Heartbeats

An RFSS that fails to receive a Heartbeat or Heartbeat Query for a session for greater than four times **Theartbeat** seconds MAY tear down the call segment in accordance with the procedures of Section 6.

Theartbeat SHALL initially be set to its default value. Upon receipt of a Heartbeat or Heartbeat query, **Theartbeat** SHALL be set to the value contained in the Interval field of the message.

Upon receipt of a Heartbeat Query, an RFSS SHALL immediately respond with a Heartbeat.

A receiving RFSS that has muted its peer’s talk spurt SHALL, upon receipt of a heartbeat for the talk spurt, reset the **EndLossTimeout** to **TMutedEndLoss** seconds.

7.9.11 Processing of the M-bit

A receiving RFSS that has muted a talk spurt and subsequently receives a packet with the M bit set to 0, or subsequently receives a packet with voice content, SHALL resend the PTT Transmit Mute message.

A receiving RFSS that has unmuted a talk spurt and subsequently receives a packet with the M bit set to 1, SHALL resend the PTT Transmit Unmute message.

7.10 Processing of Timeouts

7.10.1 Processing of the RequestTimeout

A **RequestTimeout** occurs when a SMF has sent a number of PTT Transmit Request packets for a talk spurt and received no answer from the MMF. In the event of a **RequestTimeout**, the SMF SHALL behave in accordance with local policy. Such policy MAY include acting as if PTT Transmit Deny was received or acting as if PTT Transmit Grant was received, depending on the particular circumstances.

7.10.2 Processing of the WaitTimeout

A **WaitTimeout** occurs when, after having received a PTT Transmit Wait in response to PTT Transmit Request, no PTT Transmit Grant or PTT Transmit Deny is subsequently received within the given time. In the event of a **WaitTimeout**, a SMF SHALL re-request permission to transmit, as described in Section 7.8.2.1.

7.10.3 Processing of the EndLossTimeout

An **EndLossTimeout** occurs when no RTP packets are received for a talk spurt for a period of time. In the event of an **EndLossTimeout**, the RFSS SHALL behave as if a PTT Transmit End was received for the talk spurt, as described in Section 7.9.7.

7.10.4 Other Timeouts

Other timeouts referred to as part of the Transmission Control Protocol are described in other sections:

- See Section 7.8.2.1 for **MinKeyback**
- See Sections 7.8.1 and 7.9.10 for **Theartbeat**
- See Section 7.9.8 for **TMuteProgress**

7.11 Processing of Voice Content

An MMF that receives voice content SHALL forward it to SMFs that have been joined to the call, if a PTT Transmit Mute has not been received from an SMF as described in Section 7.9.8.

When the MMF is operating in Native Vocoder Mode for the call segment to the SMF, the forwarded voice payload SHALL be left in the format in which it was received. When operating in Full Rate or Half Rate Vocoder Mode for a call segment to an SMF, the MMF MAY convert the format of the received voice before forwarding it to the SMF if the source voice is not in the same format as required by the call segment to the SMF.

An SMF that receives voice content SHALL forward it to appropriate end-points within its serving domain, in accordance with its local policies and procedures.

8. RFSS Service Capability Polling

8.1 RFSS Service Capability Polling SAP Primitive Model

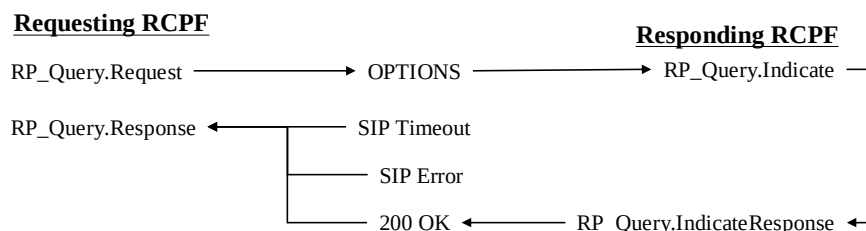


Figure 82 - RP SAP/Message Model

Figure 82 illustrates the mapping from RFSS Service Capability Polling SAP primitives to the messages of the SIP protocol. At the Polling RFSS, the controlling function employs the RP_Query.Request primitive to initiate a SIP OPTIONS transaction. When this is received by the Polled RFSS, it causes an RP_Query.Indicate primitive to the responding control function.

The RP_Query.IndicateResponse of the responding control function results in the transmission of a SIP 200 OK, which carries any detailed service capability parameters provided by the control function. When the 200 OK is received at the Polling RFSS, it results in an RP_Query.Response(Success) primitive to the requesting control function, again carrying any detailed service capability parameters provided by the Polled RFSS.

If a SIP timeout occurs, or if the Polled RFSS replies with a SIP Error, the RP_Query.Response(Failure) primitive is provided to the requesting control function.

8.2 RFSS Service Capability Polling State Model

The following state models are intended as an aid to describing the protocol and are not to be interpreted as suggesting or requiring any particular implementation. Conditions causing a transition are illustrated as text, and actions that occur during transitions are indicated by text following an indicating “/”.

8.2.1 RP Requesting State Model

Figure 83 illustrates a state model for a Polling RFSS handling an RFSS service capability polling request. Receipt by the protocol layer of the RP_Query.Rqst primitive initiates the process and causes the transmission of a SIP OPTIONS Request to the Polled RFSS. The SIP Transaction can complete because the OPTIONS request was successfully processed by the Polled RFSS, or as a result of an unsuccessful SIP Final Response or SIP Timeout. Unsuccessful processing causes an RP_Query.Resp(Failure) to be indicated to the controlling function, and ends the transaction. Successful processing of the request causes an RP_Query.Resp(Success) at the Polling RFSS, carrying any detailed service capability parameters provided in the SIP 200 OK message.

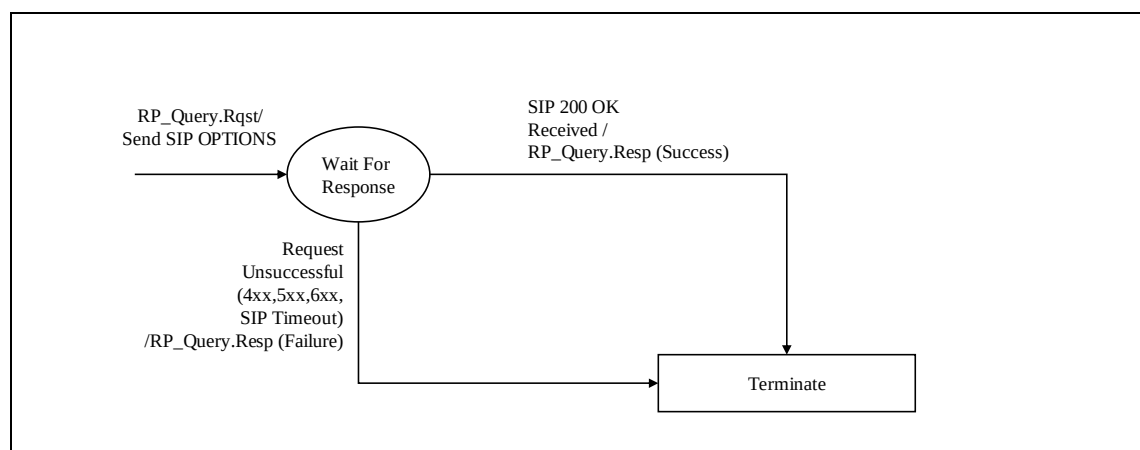


Figure 83 – RP Requesting State Model

The following tables summarize the states and transitions of the RFSS Service Capability Polling Requesting state model.

Table 51 - RP Requesting State Summaries

State	Description
Wait for Response	The RFSS is waiting for response to its SIP OPTIONS request.
Terminate	The State Process is terminated

Table 52 - RP Requesting State Transitions

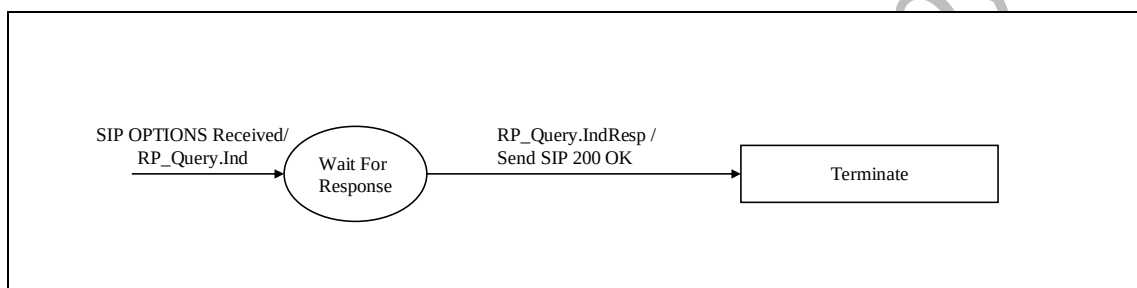
Transition	Condition
RP_Query.Rqst	Received RP_Query.Rqst from the controlling function.
Request Unsuccessful	An unsuccessful SIP final response (4xx,5xx,6xx) was received for the SIP OPTIONS request, or a SIP Timeout Occurred during the transmission of the request.
SIP 200 OK Received	A SIP 200 OK was received from the Polled RFSS.

Table 53 - RP Requesting Transition Actions

Action	Behavior
Send SIP OPTIONS	Send a SIP OPTIONS request to the Polled RFSS.
RP_Query.Resp (Failure)	Indicate unsuccessful request to the controlling function.
RP_Query.Resp (Success)	Indicate successful response to the controlling function.

8.2.2 RP Responding State Model

Figure 84 illustrates a state model for a Polled RFSS handling an RFSS service capability polling request. Receipt of the SIP OPTIONS request initiates the process and causes an RP_Query.Indicate primitive to the controlling function. When the controlling function responds with an RP_Query.IndicateResponse primitive, the SIP 200 OK message is sent, carrying any detailed service capability parameters provided by the controlling function.

**Figure 84 – RP Responding State Model**

The following tables summarize the states and transitions of the RFSS Service Capability Polling Responding state model.

Table 54 - RP Responding State Summaries

State	Description
Wait for Response	The protocol layer is waiting for response to its RP_Query.Indicate from the controlling function.
Terminate	The State Process is terminated

Table 55 - RP Responding State Transitions

Transition	Condition
RP_Query.IndResp	Received RP_Query.IndResp from the controlling function.
SIP OPTIONS Received	A SIP OPTIONS request was received from the Polling RFSS.

Table 56 - RP Responding Transition Actions

Action	Behavior
RP_Query.Ind	Indicate receipt of the SIP OPTIONS request to the controlling function.
Send SIP 200 OK	Send a successful response to the Polling RFSS.

8.3 RFSS Service Capability Polling Procedures

A single RFSS Service Capability Polling procedure is defined herein.

1. **RFSS Service Capability Polling** – the RFSS service capability polling procedure is used by a polling RFSS to query another RFSS for its presence and optional RFSS service capability profile, and optionally to provide the polling RFSS service capability profile to the polled RFSS. The trigger for the procedure is manufacturer dependent.

The RFSS service capability polling procedure SHALL proceed as follows:

1. The polling RFSS SHALL send a SIP OPTIONS message to the RFSS it wishes to poll. The SIP OPTIONS message MAY contain a body carrying the RFSS' service capability profile as defined in section 3.8.3 of this document.

SIP OPTIONS messages SHALL be transmitted in accordance with the progressive timeout retransmission protocol for unreliable transport specified in **[RFC3261]** (Sections 17.1.2 and 17.2.2), subject to the following qualifications:

- a. The value of T1 and T2 (the base timers) SHALL be established by administrative agreement between system operators responsible for the communicating RFSSs. The default value for the TIA-102 ISSI SHALL be as given in Table 58.
- b. According to **[RFC3261]** the timer F SHOULD be set to 64xT1 (Section 17.1.2.2). If a manufacturer does not want to send retries for the OPTIONS request the timer F MAY be set to 1xT1 (or whatever the manufacturer desires).
- c. Specific responses found in the following procedures SHALL be mandatory in preference to other responses implied or specified in **[RFC3261]**.
- d. A polled RFSS SHOULD NOT generate 1xx or 3xx responses in response to a SIP OPTIONS request, and a polling RFSS MAY ignore any 1xx or 3xx responses received for a SIP OPTIONS request.

2. Upon receipt of a SIP OPTIONS message the RFSS SHALL respond with
 - a. a 200 OK message. The 200 OK message MAY contain a body carrying the RFSS' service capability profile as defined in section 3.8.3 of this document.
 - b. or a SIP Final Error (4xx, 5xx, or 6xx) response message, as defined in Annex A of this document.
3. Upon receipt of any response to the SIP OPTIONS message, the polling RFSS SHALL behave as follows:
 - a. In the event that the response is a 200 OK, a 403 FORBIDDEN, a 405 METHOD NOT ALLOWED, or a 501 NOT IMPLEMENTED message, the polling RFSS SHOULD consider the polled RFSS as being available although this behavior is manufacturer dependent.
 - b. Other 4xx, 5xx and 6xx messages SHALL be interpreted in accordance with **[RFC3261]** and local policy.
4. If the RFSS Service Capability Request procedure terminates due to a SIP timeout (Section 17.1.2 of **[RFC3261]**), the polled RFSS is unreachable. The polling RFSS SHALL take appropriate error recovery steps in accordance with its security and other policies, and MAY retry the request at a later time. Following a SIP timeout, a polling RFSS SHOULD wait for at least RetryTime seconds before retrying a request.

NOTE: To avoid excessive SIP OPTIONS requests from being sent to the polled RFSS, the polling RFSS retries a request without waiting for at least RetryTime seconds only in exceptional circumstances.

Annex A (Normative) – Generic SIP Responses Over the ISSI

Table 57 shows generic SIP responses from an RFSS over the ISSI, along with applicable P25 Condition Codes and mnemonic references used throughout this document. These responses are applicable to all SIP methods used by the ISSI.

Each mnemonic has a 3-character prefix GEN, indicating that the referenced SIP response is applicable to all TIA-102 services using any SIP method over the ISSI.

If a P25 Condition Code is Mandatory (according to the “Inclusion of P25 Condition Code in SIP Response” column in Table 57), then it SHALL be included for the specified SIP response. If a P25 Condition Code is Optional, then it MAY be included for the specified SIP response. If a manufacturer chooses to include an optional P25 Condition Code for any specified SIP response, the P25 Condition Code as specified in Table 57 SHALL be used.

IMPORTANT NOTE: Before assigning a new P25 Condition Code to any SIP response, check the list of P25 Condition Codes already defined in Table 57 (see below) and Table 24 (P25 Condition Code Summary) in this document. This is to avoid duplicate and/or conflicting assignment of P25 Condition Codes.

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Table 57 - Generic SIP Responses from an RFSS over the ISSI

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
(none)		(none)	400 (BAD REQUEST)	The destination RFSS detects that: • A SIP header field has malformed syntax, or • A mandatory or conditional SIP header field is missing, or • A mandatory or conditional SIP header field has missing, disallowed, or inconsistent field value(s).	(none)
(none)		(none)	405 (METHOD NOT ALLOWED)	The SIP method specified in the Request-Line is understood by the destination RFSS, but is not allowed or supported for the address identified by the Request-URI. The response SHALL include an Allow header field containing a list of valid SIP methods for the indicated address.	(none)
(none)		(none)	408 (REQUEST TIMEOUT)	The destination RFSS is not able to generate a response in time.	(none)
(none)		(none)	415 (UNSUPPORTED MEDIA TYPE)	The destination RFSS does not understand the mandatory body type(s) contained in the SIP request for the requested SIP method. The response SHALL contain an Accept header field listing all body types that the destination RFSS understands.	(none)
(none)		(none)	416 (UNSUPPORTED URI SCHEME)	The destination RFSS detects an unsupported SIP URI scheme in the request URI.	(none)
(none)		(none)	481 (TRANSACTION DOES NOT EXIST)	The destination RFSS receives a request that does not match an existing dialog or transaction.	(none)

P25-condition-code	Mnemonic	P25-condition-text	SIP Response	Usage	Inclusion of P25 Condition Code in SIP Response
(none)		(none)	488 (NOT ACCEPTABLE HERE)	The destination RFSS detects that: - An ISSI body attribute has malformed syntax, or - A mandatory ISSI body attribute is missing, or - A mandatory ISSI body attribute has missing, disallowed, or inconsistent field value(s).	(none)
(none)		(none)	501 (NOT IMPLEMENTED)	The destination RFSS does not recognize the SIP request method and is not capable of supporting it for any SU or SG.	(none)
(none)		(none)	503 (SERVICE UNAVAILABLE)	The destination RFSS is temporarily overloaded and therefore unable to process the request.	(none)
(none)	GEN_SDP_PARAMS_UNACCEPTABLE	(none)	606 (NOT ACCEPTABLE)	The destination RFSS detects that: - An SDP parameter has malformed syntax, or - A mandatory SDP parameter is missing or misordered, or - A mandatory SDP parameter has missing, disallowed, or inconsistent field value(s).	(none)
%x1C	GEN_PROC_UNSUPPORTED	Requested procedure not supported	403 FORBIDDEN	Any requested procedure	Optional
%x64	GEN_IGNORE_NON_ISSI_HEADER	Non-ISSI header(s) ignored	2xx	Any	Optional
%x65	GEN_IGNORE_MANUF_BODY	Manufacturer specific body ignored	2xx	Any	Optional
%xFF	GEN_MANUF_DISCRETION	Manufacturer's Discretion	Any	Non-standardized warning.	Optional

In addition to the generic SIP responses in Table 57, various ISSI messages and procedures standards MAY specify other allowable TIA-102 service specific SIP responses.

In the event that an RFSS receives a SIP 3xx response, or any SIP failure response that is not defined in Table 57, nor specified as an allowable TIA-102 service specific SIP failure response, as might be the case if the response is from another RFSS implemented in accordance with a different version of the ISSI messages and procedures standards, the RFSS SHOULD treat this response as a valid failure response, although the resulting RFSS behavior SHALL be manufacturer-specific.

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Annex B (Normative) – Table of Timers and Constants

Table 58 shows the timers and constants used in this document.

Table 58 – Timers and Constants

Timer	Purpose	Default Value	Range
EndLossTimeout	Packet loss timeout.	Variable	0.1 to 30 sec
MinKeybackTime	Preferred Minimum time between the end of a previous transmission and the transmission of an subsequent PTT Transmit Request.	60 msec.	0 to 1 sec
Rend	Number of retransmissions for PTT Transmit End packets.	4	1 to 5
RequestTimeout	PTT Transmit Request Timeout.	1 sec	0.1 to 3 sec
RetryTime	Minimum time to wait following a timeout before retrying the request.	10 sec	1 to 30 sec
Rstart	Number of retransmissions for PTT Transmit Start packets.	4	1 to 5
T1	Standard value of the SIP T1 timer for use in the ISSI.	500 msec. ³²	(Per RFC 3261)
T2	Standard value of the SIP T2 timer for use in the ISSI.	4 sec. ³³	(Per RFC 3261)
Telection	(Optional) Arbitration window.	100 msec.	Implementation-dependent
Tend	Retransmission Time for PTT Transmit End packets.	20 msec.	20 to 100 ms
TEndLoss	Normal EndLossTimeout.	200 msec.	0.1 to 30 sec
TFirstPacketTime	EndLoss Timeout for the first packet of a talk spurt.	500 msec.	0.1 to 3 sec

³² SIP Default per RFC 3261: 500 msec

³³ SIP Default per RFC 3261: 4 sec.

Timer	Purpose	Default Value	Range
Tgchconfirmsip	Maximum time interval which when exceeded will entail sending of a re-INVITE message from the Home RFSS to the Serving RFSS to synchronize the SIP states between the Home RFSS and Serving RFSS when there is no RTP session.	30 seconds	10 seconds to 5 minutes
Tgchhangtime	Maximum time interval (i.e., ISSI hang time) during which there is no voice activity over the ISSI among RFSSs participating in a group call, which when exceeded will entail group call release.	30 seconds	0 to infinity
Tgchstartconfirmed	Maximum time interval the Home RFSS waits before starting a confirmed group call.	5 seconds	0 to infinity
Theartbeat	Heartbeat interval.	10 sec.	5 – 20 sec
TMutedEndLoss	EndLossTimeout when the RFSS has muted the talk spurt.	2 sec.	0.5 to 4 sec
TMuteProgress	Rate at which a Muted RFSS sends Heartbeat messages with “M” bit set to 1 and the TSN of the talk spurt.	500 msec.	0.25 to 1 sec
Trequest	Retransmission time for PTT Transmit Request packets.	500 msec.	0.1 to 1 sec
Tstart	Retransmission Time for PTT Transmit Start packets.	360 msec.	0.1 to 1 sec

Timer	Purpose	Default Value	Range
Tu2uhangtime	Maximum time interval during which there is no voice activity among RFSSs participating in an SU-to-SU call, which when exceeded will entail SU-to-SU call release.	30 seconds	0 to infinity
Tunmute	Unmute Retransmission Interval.	250 msec	0.1 to 1 sec
WaitTimeout	Time to wait following receipt of PTT Transmit Wait before retrying a request.	5 sec	1 to 10 sec

Annex C (Informative) – TIA-102 Vocoder Error Mitigation Data³⁴

C.1 Introduction

DVSI has reviewed the Vocoder Specification (TIA-102BABA) to identify certain channel error mitigation data for the Full-Rate P25 Vocoder. The purpose is to enable the transport of compressed voice data and error mitigation data over the ISSI and/or Fixed Station Interfaces currently being formulated. DVSI has also conducted a similar review of the proposed Half-Rate Vocoder to identify which error mitigation parameters would be useful in the event such a Half-Rate Vocoder is used within P25. Finally, DVSI has identified supplemental error mitigation data that may be used by an Enhanced Vocoder (either Full-Rate or Half-Rate) to provide improved error mitigation.

In preparing this information, DVSI has adopted the view that the calculations of Chapter 7 in the Vocoder Specification **[102BABA]**, including Equations [95-98] are computed as part of the FEC decoder, while the calculations of Chapter 9 are computed as part of the Voice Decoder. The error mitigation data identified in this Annex is the data that needs to be transmitted between the FEC Decoder and the Voice Decoder to enable this division. The information is valid and identical for both soft-decision and hard-decision FEC decoding.

The objective in this review has been to identify a set of data that enables full error mitigation functionality while using as few bits as possible for transport. To facilitate this, DVSI has attempted to determine the number of bits required for each error mitigation parameter. In some cases, the number of bits required to represent the full range of the parameter is greater than the number of bits required to represent the more limited range used by the error mitigation algorithm. In these cases, this Annex recommends that only the minimum number of bits required by the error mitigation algorithm be used for transport. For example, the parameter e_T , which is the total errors in the frame computed according to Equation 95 in the Vocoder Specification, has a range of [0, 30] if soft-decision decoding is used in the FEC. However only values in the range of [0,7] are required by the error mitigation algorithm, and hence this Annex only assigns 3 bits to this parameters. Values that exceed 7 would be limited to 7 in this case.

³⁴ The text in this Annex was sourced, with minor editorial modifications, from a contribution, entitled “P25 Vocoder Error Mitigation Data”, that John C. Hardwick, of Digital Voice Systems, Inc., submitted to the APCO Interface Committee Project 25 Vocoder Task Group, 11 October 2005. Appearance of “P25” throughout this Annex indicates the item has been adopted or may be adopted by the APCO/NASTD/FED Project 25 Steering Committee for use in its Standard.

C.2 Full-Rate Vocoder Error Mitigation Data

For the Full-Rate P25 Vocoder, the identified Error Mitigation Data is as follows:

Frame_Repeat_Flag (LOST):	1 bit
Mute_Frame_Flag (MUTE):	1 bit
eT :	3 bits -- see Note 1
e4:	1 bits – see Note 2
<u>eR:</u>	<u>3 bits – see Note 3</u>
Total:	9 bits

C.3 Half-Rate Error Mitigation Data

For the proposed Half-Rate P25 Vocoder, the identified Error Mitigation Data is as follows:

Frame_Repeat_Flag (LOST):	1 bit
Mute_Frame_Flag (MUTE):	1 bit
<u>eT :</u>	<u>3 bits -- see Note 1</u>
Total:	5 bits

C.4 Enhanced Vocoder Supplemental Error Mitigation Data

For DVSI's Enhanced P25 Vocoders, performance in noisy channel conditions is improved if the supplementation error mitigation parameter e1 is included using 3 bits. DVSI recommends this addition.

C.5 Notes

Note 1: eT should be limited to the range [0, 7]

Note 2: e4 should be limited to the range [0, 1]

Note 3: eR should be scaled by 400 and then converted to the largest integer in the range [0, 7] that doesn't exceed the value, i.e. $\text{floor}(400 \cdot eR)$.

Annex D (Informative) – SU-to-SU Call Flows

In the message sequence charts below, the triggering and resulting primitives from and to the Controlling Function are shown only in the endpoint Serving RFSSs, in order to preserve readability on the diagrams.

D.1 Availability Check, RF Resources Available

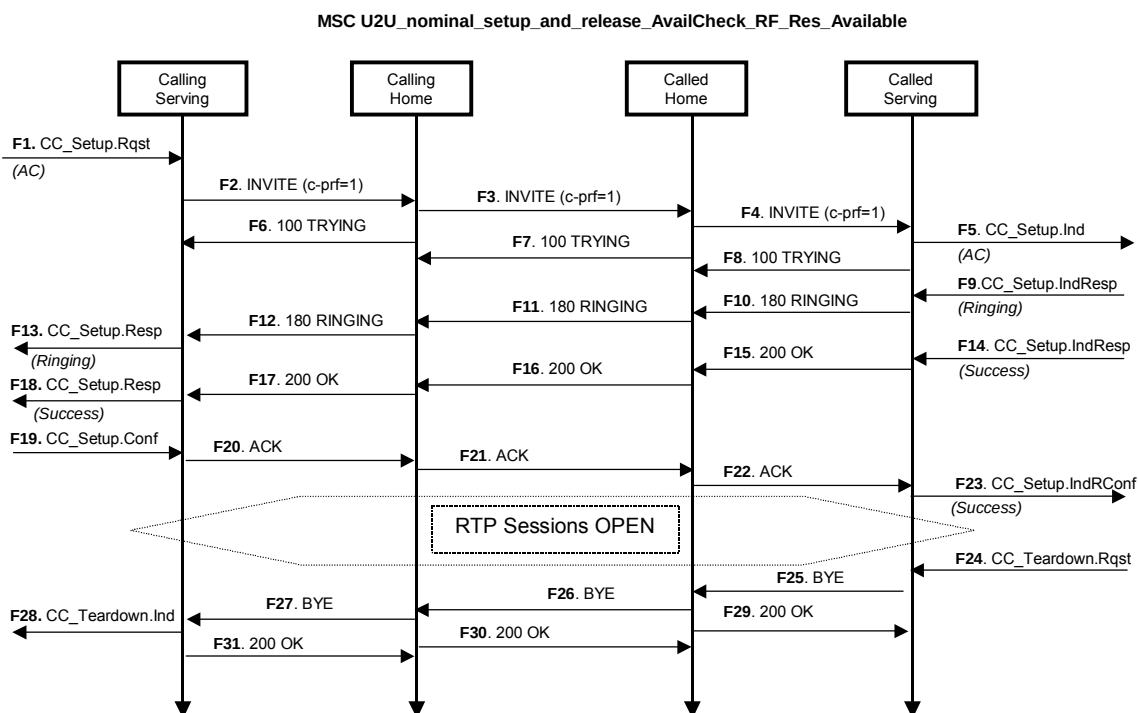


Figure 85 - MSC
U2U_nominal_setup_and_release_AvailCheck_RF_Res_Available

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

F1. As shown in Figure 85, the Calling SU triggers an SU-to-SU Call Setup. RTP Resources are allocated and a CC_Setup.Request(AC) primitive is exerted on the Call Control Protocol Element at the Calling Serving RFSS. The Preference

parameter specifies that the SU-to-SU Call is preferred to be set up with an availability check.

F2. SIP INVITE is sent to the Calling Home RFSS.

F3. Two RTP Resources are allocated. The SIP INVITE is propagated to the Called Home RFSS.

F4. Two RTP Resources are allocated. The Called Home RFSS verifies that the Called SU supports Availability Check. The SIP INVITE is propagated to the Called Serving RFSS.

F5. A CC_Setup.Indicate(AC) primitive is exerted to the Controlling Function of the Called Serving RFSS. The Called Serving RFSS proceeds with an Availability Check to the Called SU.

F6, F7, F8. SIP 100 TRYING is possibly sent at each call segment following SIP RFC 3261 requirements.

F9. The Called SU is ringing. The Controlling Function at the Called Serving RFSS exerts a CC_Setup.IndicateResponse(Ringing) to the Call Control Protocol Element.

F10, F11, F12. The SIP180 RINGING is propagated from the Called Serving RFSS to the Calling Serving RFSS.

F13. The CC_Setup.Response(Ringing) primitive is exerted to the Controlling Function at the Calling Serving RFSS.

F14. The Called SU accepts the Call and an RF resource is successfully allocated at the Called Site. The Controlling Function at the Called Serving RFSS exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element.

F15, F16, F17. The SIP 200 OK is propagated from the Called Serving RFSS to the Calling Serving RFSS.

F18. The Call Control Protocol Element at the Calling Serving RFSS exerts a CC_Setup.Response(Success) primitive to the Controlling Function, indicating that the Called Serving RFSS is ready to start the Call.

F19. Calling Serving RFSS has an RF resource allocated for the call. The Controlling Function at the Calling Serving RFSS exerts a CC_Setup.Confirm primitive to the Call Control Protocol Element.

F20, F21, F22. The Calling Serving RFSS sends the SIP ACK message and it is propagated to the Called Serving RFSS.

F23. A CC_Setup.IndicateResponseConfirm(Success) primitive is exerted to the Controlling Function of the Called Serving RFSS. This enables the Controlling Function to start the call for the Called SU (i.e. to grant the RF Channel to a Called radio SU).

F24. The Called SU releases the call. A CC_Teardown.Request is exerted to the Call Control Protocol Element at the Called Serving RFSS.

F25, F26, F27. A SIP BYE message is sent and propagated from the Called Serving RFSS to the Calling Serving RFSS.

F28. Upon receipt of the BYE message, a CC_Teardown.Indicate primitive is exerted to the Controlling Function at the Calling Serving RFSS. The Calling Serving RFSS releases the call for the Calling SU.

F29, F30, F31. The SIP BYE is acknowledged by a 200 OK at each Call Segment..

D.2 Availability Check, No Called RF resource

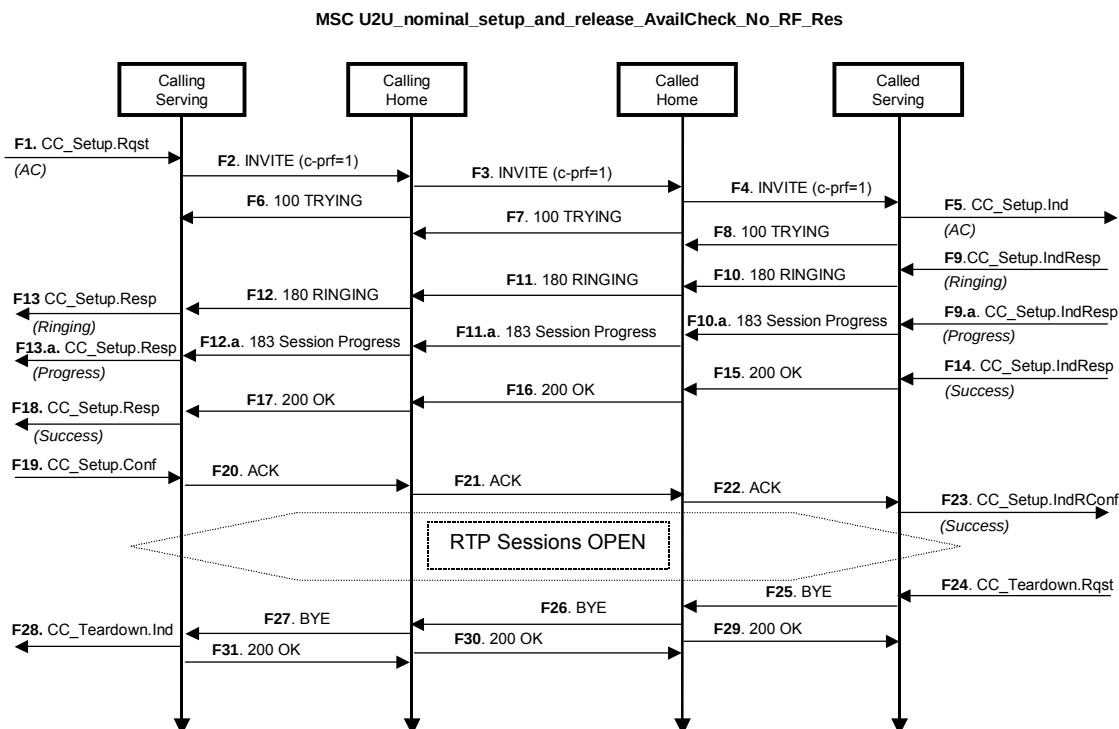


Figure 86 - MSC U2U_nominal_setup_and_release_AvailCheck_No_RF_Res

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

This MSC, as shown in Figure 86, describes the same scenario described in Section D.1, except that the Called Serving RFSS temporarily lacks available RF resources.

F1 – F13. The Calling Serving RFSS begins setting up an SU-to-SU call as described in Section D.1.

F9.a. An RF resource is not available at the Called Site. The Controlling Function at the Called Serving RFSS exerts a CC_Setup.Response(Progress) to the Call Control Protocol Element.

F10.a, F11.a, F12.a. The SIP 183 Session Progress is propagated from the Called Serving RFSS to the Calling Serving RFSS to indicate that an RF resource is temporarily unavailable at the Called Site.

F13.a. The CC_Setup.Response(Progress) primitive is exerted to the Controlling Function at the Calling Serving RFSS.

F14 – F31. When an RF resource becomes available, call setup and teardown proceeds as in the normal case in Section D.1.

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D.3 Direct call, No Calling RF Resource

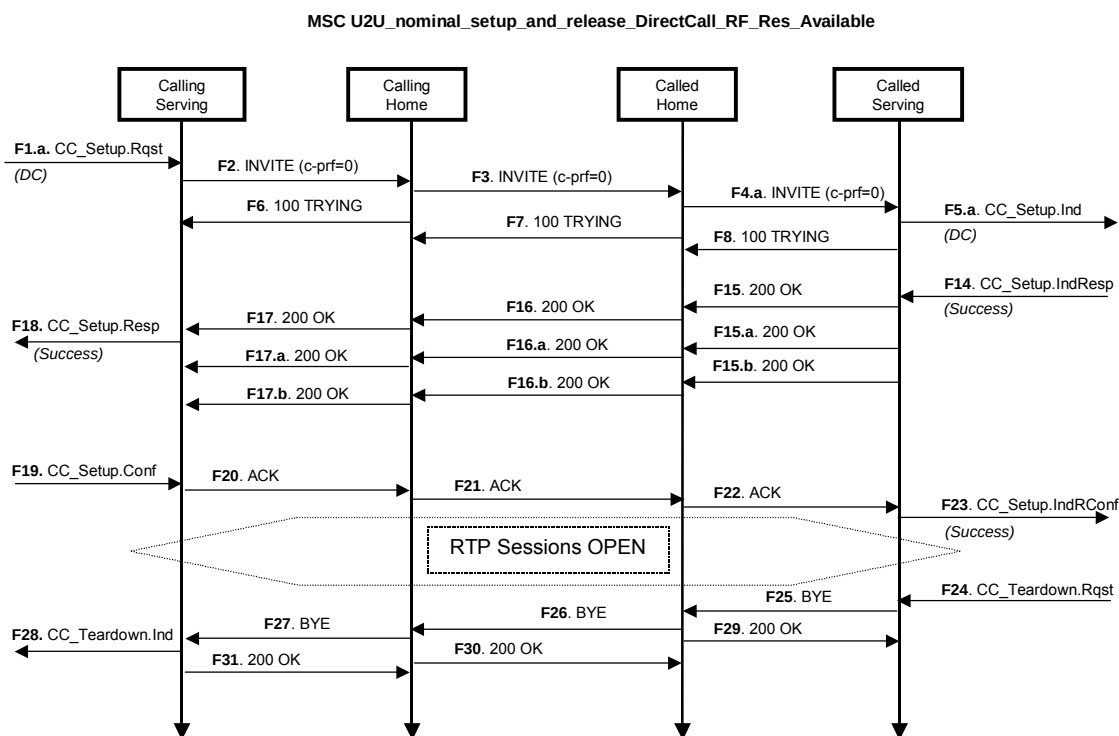


Figure 87 - MSC
U2U_nominal_setup_and_release_DirectCall_RF_Res_Available

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

This MSC, as shown in Figure 87, describes the same scenario described in Section D.1 except that the Called Serving RFSS proceeds with a Direct Call instead of an Availability Check, and there is a temporary lack of RF Resources at the Calling Serving RFSS.

F1.a. The Calling SU triggers an SU-to-SU Call Setup. RTP Resources are allocated and a CC_Setup.Request(DC) primitive is exerted to the Call Control Protocol Element at the Calling Serving RFSS. The Preference parameter

specifies the SU-to-SU Call is preferred to be set up without an availability check (Direct Call).

F2, F3. The SIP Invite is propagated to the Called Home RFSS.

F4.a. Two RTP Resources are allocated. The Called Home RFSS verifies that the Called SU supports Direct Call. The SIP INVITE is propagated to the Called Serving RFSS.

F5.a. A CC_Setup.Indicate(DC) primitive is exerted to the Controlling Function of the Called Serving RFSS. The Called Serving RFSS proceeds with a Direct Call to the Called SU (no availability check is performed).

F6 – F18. Call setup proceeds as in the normal case in Section D.1.

F15.a, F15.b The SIP 200 OK is re-transmitted until a SIP ACK message is received (or a SIP Timeout occurs) if the SIP ACK is delayed due to a temporary lack of RF resources at the Calling Site.

F16.a, F16.b – F17.a, F17.b. The SIP 200 OK re-transmission is propagated to the Calling Home RFSS and Calling Serving RFSS, respectively.

F19 – F31. When an RF resource becomes available, call setup and teardown proceeds as in the normal case in Section D.1.

D.4 Direct Call, No Calling or Called RF Resource

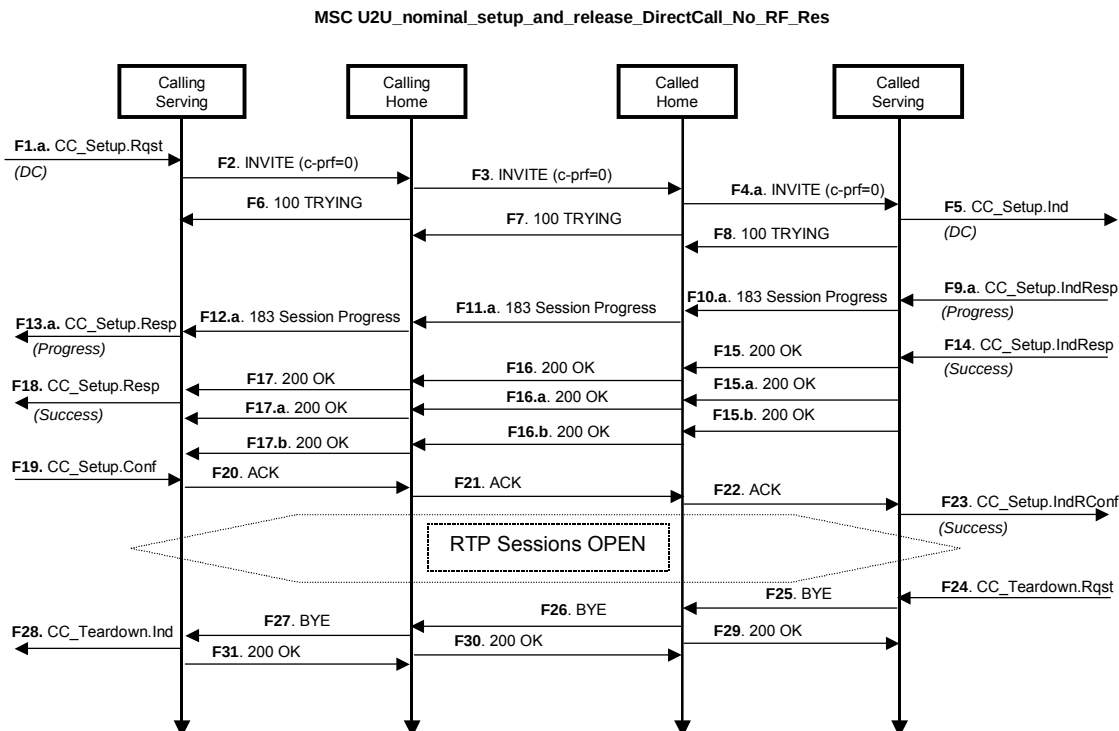


Figure 88 - MSC U2U_nominal_setup_and_release_DirectCall_No_RF_Res

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

This MSC, as shown in Figure 88, describes the same scenario described in Section D.3 (F1.a – F8) except that there is also a temporary lack of RF Resources at the Called Serving RFSS, which results in F9.a, F10.a, F11.a, F12.a, and F13.a, as described in Section D.2.

F14 – F31. When an RF resource becomes available, call setup and teardown proceeds as described in Section D.3.

D.5 Called SU Denial

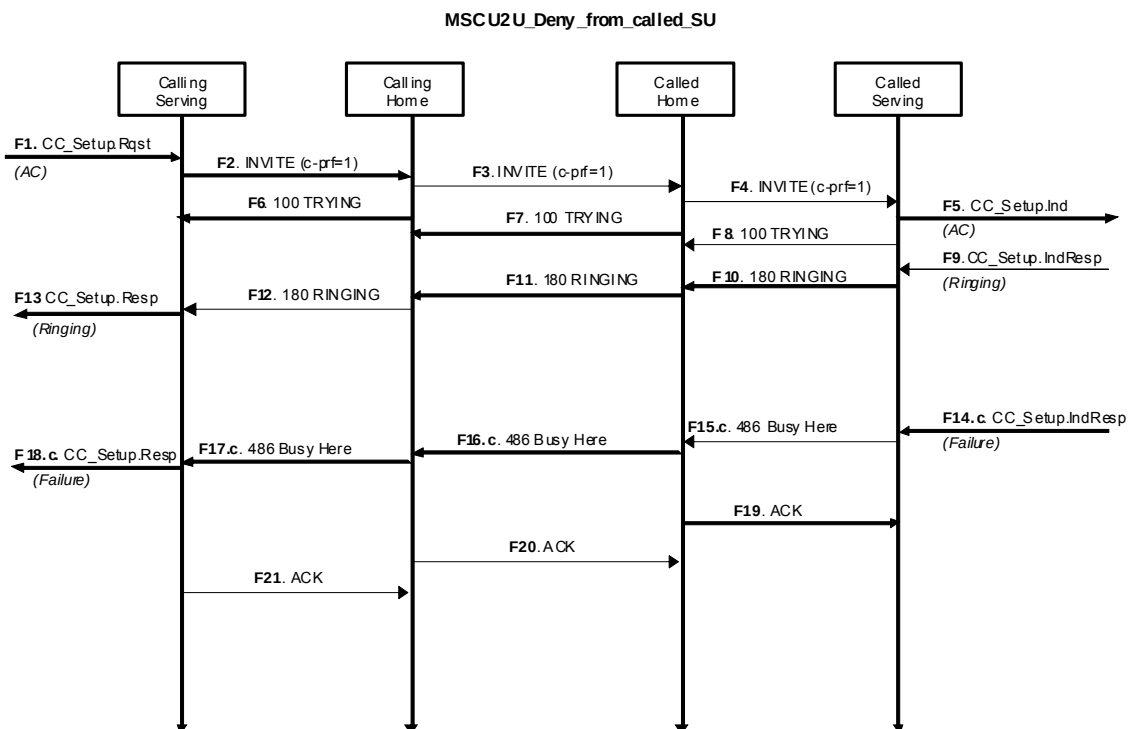


Figure 89 - MSC U2U_Deny_from_called_SU

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

F1 – F13. As shown in Figure 89, the Calling Serving RFSS starts an SU-to-SU Call with Availability Check as described in Section D.1.

F14.c. The Called SU denies the call. The Controlling Function at the Called Serving RFSS exerts a CC_Setup.Response(Failure) primitive to the Call Control Protocol Element.

F15.c, F16.c, F17.c. The SIP 486 'Busy Here' is propagated from the Called Serving RFSS to the Calling Serving RFSS to indicate that Called SU has refused the call.

F18.c. The CC_Setup.Response (Failure) primitive is exerted to the Controlling Function at the Calling Serving RFSS. The Controlling Function terminates the call with the Calling SU.

F19, F20, F21. The Called Home RFSS, Calling Home RFSS, and Calling Serving RFSS return a SIP ACK to confirm receipt of the SIP 486 'Busy Here'.

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D.6 Calling SU Cancellation

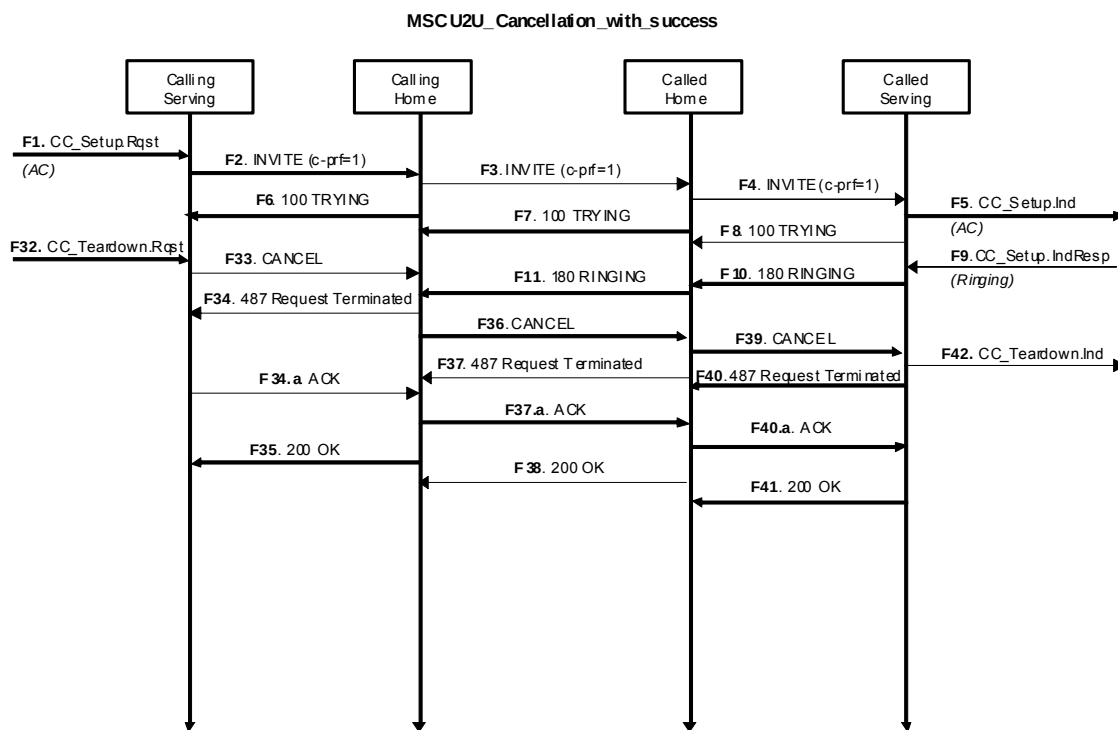


Figure 90 - MSC U2U_Cancellation_with_success

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

F1 – F11. As shown in Figure 90, the Calling Serving RFSS starts an SU-to-SU Call with Availability Check as described in Section D.1.

F32. The Calling SU triggers an SU-to-SU Call Cancellation. The CC_Teardown.Request primitive is exerted to the Call Control Protocol Element at the Calling Serving RFSS.

F33. A SIP CANCEL message is sent to the Calling Home RFSS.

F34. A SIP 487 'Request Terminated' terminates the SIP INVITE Request.

F34.a A SIP ACK confirms receipt of the SIP 487 'Request Terminated'.

F35. A SIP 200 OK positively acknowledges the SIP CANCEL message.

F36. The SIP CANCEL message is propagated to the Called Home RFSS.

F37. A SIP 487 'Request Terminated' terminates the SIP INVITE Request.

F37.a A SIP ACK confirms receipt of the SIP 487 'Request Terminated'.

F38. A SIP 200 OK positively acknowledges the SIP CANCEL message.

F39. The SIP CANCEL message is propagated to the Called Serving RFSS.

F40. A SIP 487 'Request Terminated' terminates the SIP INVITE Request.

F40.a A SIP ACK confirms receipt of the SIP 487 'Request Terminated'.

F41. A SIP 200 OK positively acknowledges the SIP CANCEL message.

F42. Upon receipt of the SIP CANCEL message at the Called Serving RFSS, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive. The Controlling Function terminates the call with the Called SU.

D.7 Cancellation Failure

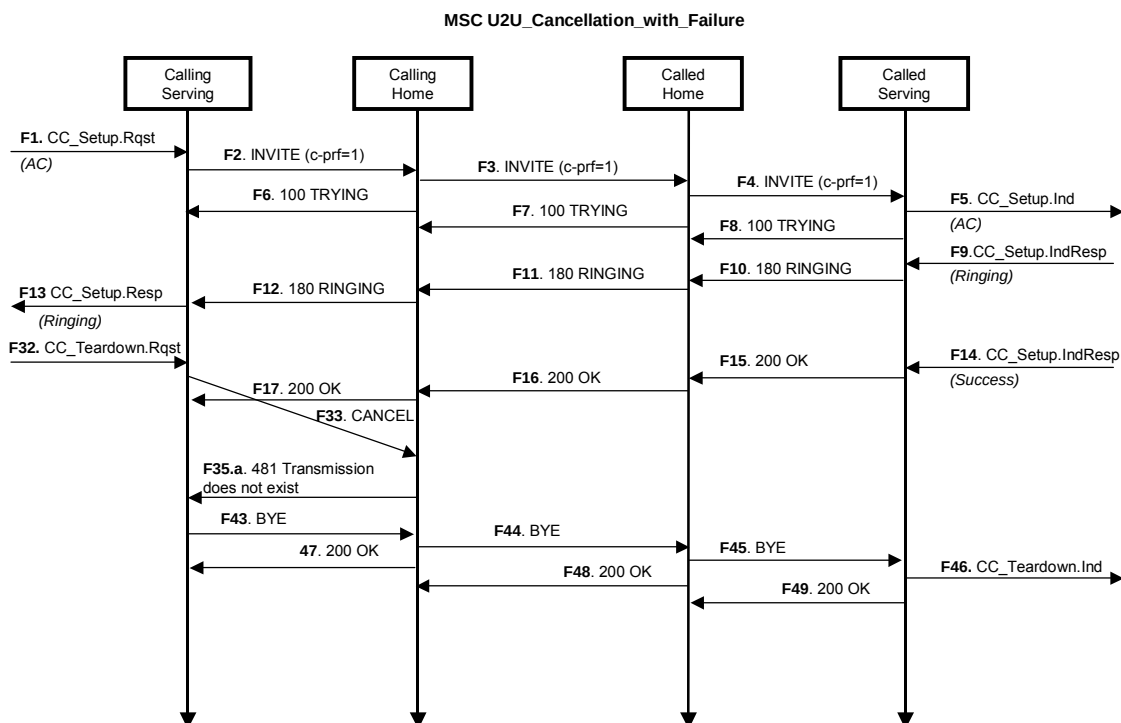


Figure 91 - MSC U2U_Cancellation_with_Failure

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

F1 – F16. As shown in Figure 91, the Calling Serving RFSS starts an SU-to-SU Call with Availability Check as described in Section D.1.

F32. The Calling SU triggers an SU-to-SU Call Cancellation. The CC_Teardown.Request primitive is exerted to the Call Control Protocol Element at the Calling Serving RFSS.

F33. A SIP CANCEL message is sent to the Calling Home RFSS.

F17. A SIP 200 OK message is received from the Calling Home RFSS. Because a CC_Teardown.Request has been received, the CC_Setup.Response(Success) is NOT sent to the Controlling Function.

F35.a. Because the SIP 200 OK has already been received, a SIP 481 'Transaction Does Not Exist' message is sent to negatively acknowledge the SIP CANCEL message.

F43, F44, F45. The SIP BYE request is generated by the Calling Serving RFSS and propagated to the Called Serving RFSS.

F46. Upon receipt of the SIP BYE message at the Called Serving RFSS, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive. The Controlling Function terminates the call with the Called SU.

F47, F48, F49 The SIP BYE is acknowledged by a 200 OK at each Call Segment.

D.8 Set Up Failure

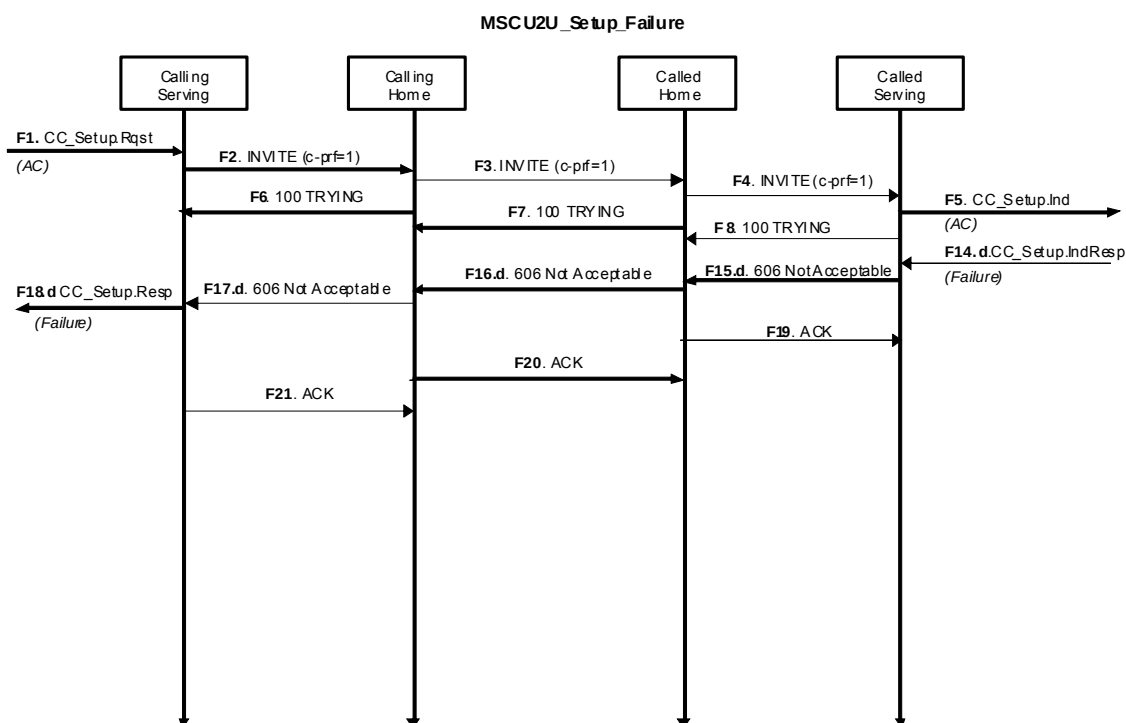


Figure 92 - MSC U2U_Setup_Failure

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

F1 – F8. As shown in Figure 92, the Calling Serving RFSS starts an SU-to-SU Call with Availability Check as described in Section D.1.

F14.d. The Called Serving RFSS refuses the call because it lacks RTP resources. The Controlling Function at the Called Serving RFSS exerts a CC_Setup.IndicateResponse(Failure) to the Call Control Protocol Element.

F15.d, F16.d, F17.d. The SIP 606 'Not Acceptable' message is propagated from the Called Serving RFSS to the Calling Serving RFSS to indicate that the call cannot be setup due to a lack of RTP resources.

F18.d. The CC_Setup.Response(Failure) primitive is exerted to the Controlling Function at the Calling Serving RFSS. The Controlling Function terminates the call with the Calling SU.

F19, F20, F21. The Called Home RFSS, Calling Home RFSS, and Calling Serving RFSS return a SIP ACK message to confirm receipt of the SIP 606 'Not Acceptable' message.

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D.9 Release By Home

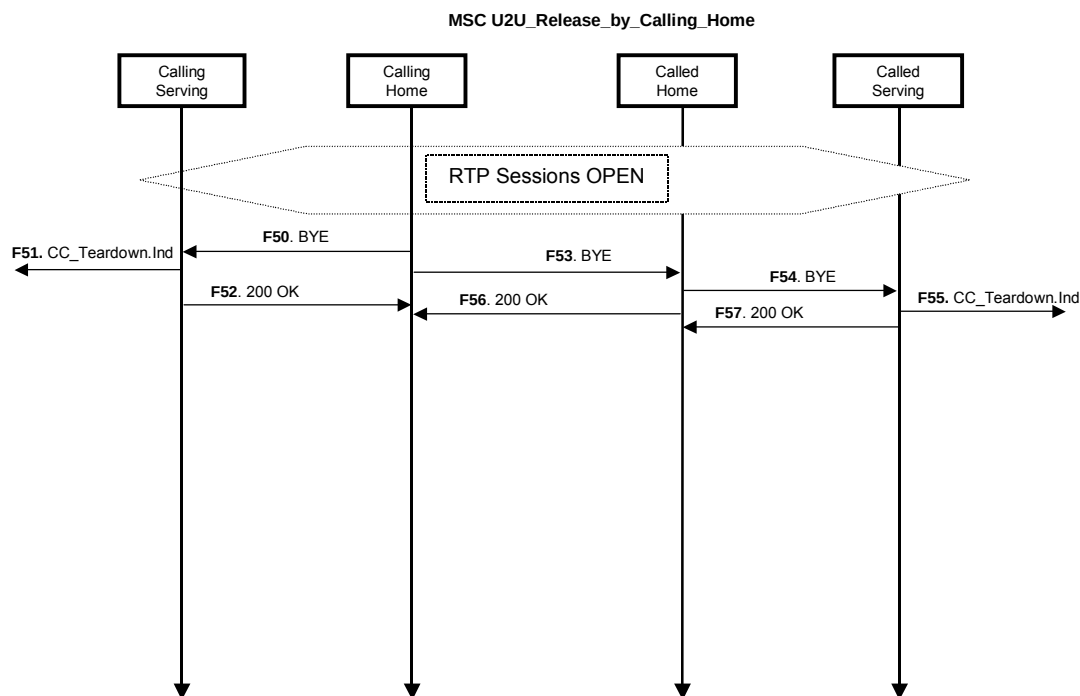


Figure 93 - MSC U2U_Release_by_Calling_Home

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.
- An SU-to-SU call has been successfully set up from the Calling SU to the Called SU.

F50, F53. As shown in Figure 93, the Calling Home RFSS triggers a call termination and sends out a SIP BYE message to both the Calling Serving RFSS and the Called Home RFSS.

F51. Upon receipt of the SIP BYE message at the Calling Serving RFSS, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the

Controlling Function. The Controlling Function terminates the call with the Calling SU.

F52. The Calling Serving RFSS acknowledges the SIP BYE message with a SIP 200 OK.

F54. The BYE Request is propagated to the Called Serving RFSS

F55. Upon receipt of the SIP BYE message at the Called Serving RFSS, the Call Control Protocol Element exerts a CC_Teardown.Indicate primitive to the Controlling Function. The Controlling Function terminates the call with the Called SU.

F56, F57. The SIP BYE message is acknowledged with a SIP 200 OK at each Call Segment.

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D.10 Roaming

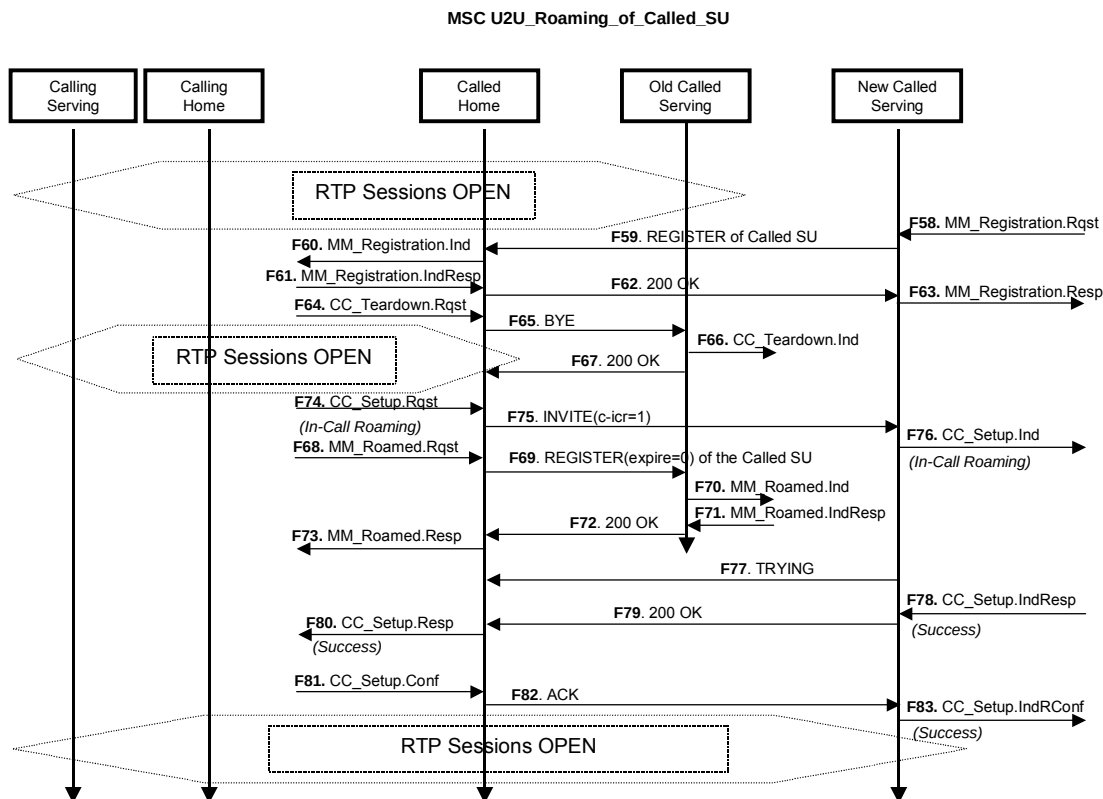


Figure 94 - MSC U2U_Roaming_of_Called_SU

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.
- An SU-to-SU call has been successfully set up from the Calling SU to the Called SU.

F58. As shown in Figure 94, an MM_Registration.Request is exerted by the Call Control Protocol Element at the new Called Serving RFSS when the Called SU registers in the new Called Serving RFSS.

F59. A SIP REGISTER message is sent to the Called Home RFSS.

F60. Upon receipt of the SIP REGISTER message, an MM_Registration.Indication primitive is exerted to the Controlling Function at the Called Home RFSS.

F61. The Controlling Function acknowledges with an MM_Registration.IndicateResponse primitive.

F62. A SIP 200 OK message acknowledging the REGISTER message is sent to the new Called Serving RFSS.

F63. Upon receipt of the SIP 200 OK message, the Call Control Protocol Element exerts an MM_Registration.Response primitive to the Controlling Function of the new Called Serving RFSS.

F64. The Controlling Function at the Called Home RFSS exerts a CC_Teardown.Request primitive to tear down the Call Segment with the old Called Serving RFSS.

F65. A SIP BYE message is sent to the old Called Serving RFSS. At this stage, RTP sessions are open between the Calling Serving RFSS, the Calling Home RFSS, and the Called Home RFSS only.

F66. Upon receipt of the SIP BYE message, the CC_Teardown.Indicate primitive is exerted by the Call Control Protocol Element.

F67. A SIP 200 OK is sent by the old Called Serving RFSS to acknowledge the SIP BYE message.

F68. The Controlling Function at the Called Home RFSS exerts a MM_Roamed.Request primitive to notify the old Called Serving RFSS that the Called SU has roamed to a new Called Serving RFSS.

F69. A SIP REGISTER(expire=0) message is sent to the old Calling Serving RFSS.

F70. Upon receipt of the SIP REGISTER message, a MM_Roamed.Indicate primitive is exerted to the Controlling Function at the old Called Serving RFSS.

F71. The Controlling Function acknowledges with an MM_Roamed.IndicateResponse primitive.

F72. A SIP 200 OK acknowledging the SIP REGISTER Message is sent to the Called Home RFSS.

F73. Upon receipt of the SIP 200 OK message, the Call Control Protocol Element exerts an MM_Roamed.Response to the Controlling Function of the Called Home RFSS.

F74. Just after the **F61** event, the Controlling Function exerts a `CC_Setup.Request(In-Call Roaming)` primitive to set up a new call segment with the new Called Serving RFSS.

F75. A SIP INVITE(c-icr=1) message is sent from the Called Home RFSS to the new Called Serving RFSS.

F76. Upon receipt of the SIP INVITE message, the Call Control Protocol Element exerts a `CC_Setup.Indicate(In-Call Roaming)` primitive to the Controlling Function.

F77. A SIP 100 TRYING message is optionally sent from the new Called Serving RFSS to the Called Home RFSS as defined by RFC 3261.

F78. The Controlling Function at the new Called Serving RFSS allocates an RTP and RF resources and exerts a `CC_Setup.IndicateResponse(Success)` primitive to the Call Control Protocol Element.

F79. A SIP 200 OK message is sent from the new Called Serving RFSS to the Called Home RFSS.

F80. Upon receipt of the SIP 200 OK message at the Called Home RFSS, the Call Control Protocol Element exerts a `CC_Setup.Response(Success)` primitive to the Controlling Function.

F81. The Controlling Function at the Called Home RFSS bridges the new RTP Session set up between it and the new Called Serving RFSS with the RTP Session set up with the Calling Home RFSS, and exerts a `CC_Setup.Confirm` primitive to the Call Control Protocol Element.

F82. A SIP ACK message is sent from the Called Home RFSS to the Called Serving RFSS.

F83. Upon receipt of the SIP ACK message at the Called Serving RFSS, the Call Control Protocol Element exerts a `CC_Setup.IndicateResponseConfirm(Success)` primitive to the Controlling Function. This causes the Controlling Function to start the call for the Called SU (i.e. to grant the RF Channel to a Called radio SU).

D.11 Colliding Invites

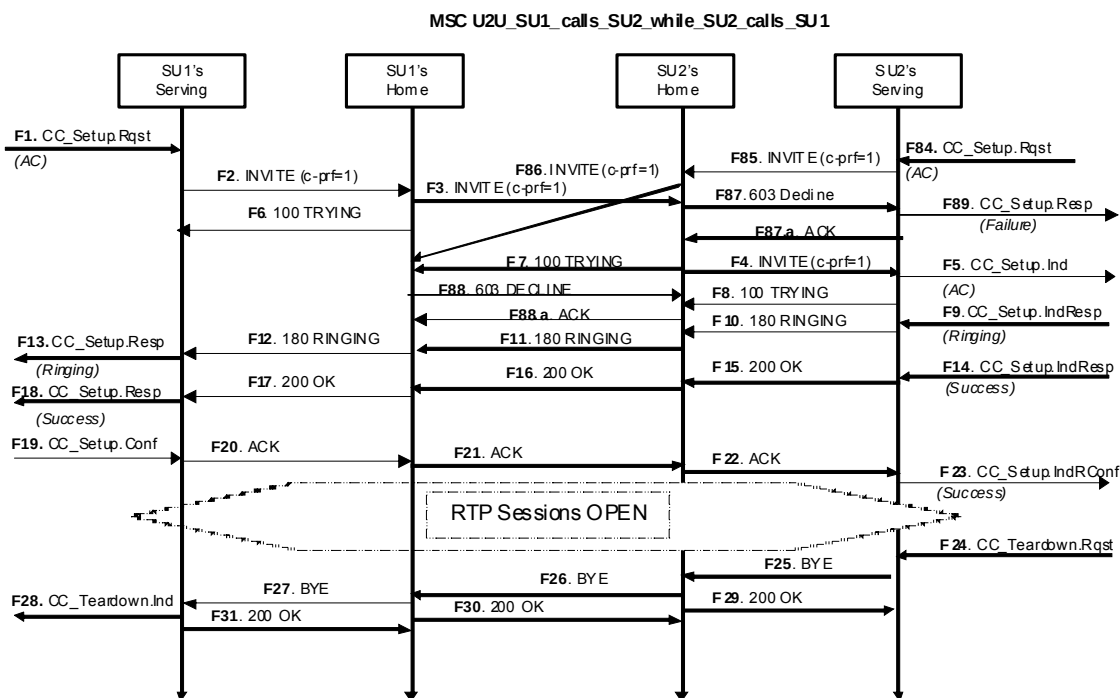


Figure 95 - MSC U2U_SU1_calls_SU2_while_SU2_calls_SU1

Initial Conditions:

- The Called Serving RFSS has successfully registered the Called SU with the Called Home RFSS.
- The Calling Serving RFSS has successfully registered the Calling SU with the Calling Home RFSS.

F1 – F23. As shown in Figure 95, the Calling Serving RFSS starts an SU-to-SU Call with Availability Check as described in Section D.1.

F84. In the meantime, SU2, located at the Called Serving RFSS, tries to make an SU-to-SU Call to SU1. The Controlling Function of SU2's Serving RFSS exerts a CC_Setup.Request(DC, RF OK or RF KO) to set up the call.

F85. SU2's Serving RFSS sends a SIP INVITE to SU2's Home RFSS.

F86. SU2's Home RFSS sends a SIP INVITE to SU1's Home RFSS to set up the SU-to-SU Call.

F87. Upon receipt of a SIP INVITE from SU1's Home RFSS indicating that SU1 is trying to make a call to SU2, SU2's Home RFSS compares both SU addresses and, as SUID(SU2) is strictly lower than SUID(SU1), SU2's Home RFSS declines the call originated by SU2 by sending a SIP 603 Decline message to SU2's Serving RFSS.

F87.a. SU2's Serving RFSS confirms receipt of the SIP 603 Decline message by returning a SIP ACK message to SU2's Home RFSS.

F88. Upon receipt of an SIP INVITE from SU2's Home RFSS indicating that SU2 is trying to make a call to SU1, SU1's Home RFSS compares both SU addresses and, as SUID(SU2) is strictly lower than SUID(SU1), SU1's Home RFSS declines the call originated by SU2 by sending a SIP 603 Decline message to SU2's Home RFSS.

F88.a. SU2's Home RFSS confirms receipt of the SIP 603 Decline message by returning a SIP ACK message to SU1's Home RFSS.

F89. Upon receipt of the SIP 603 Decline message at SU2's Serving RFSS, the Call Control Protocol Element exerts a CC_Setup.Response(Failure) primitive.

F24 – F31. Eventually, the call from SU1 to SU2 is torn down normally as described in Section D.1.

Annex E (Informative) – Group Call Flows

In the message sequence charts below, the triggering and resulting primitives from and to the Controlling Function are shown in all of the RFSSs.

E.1 Initiated by Home RFSS

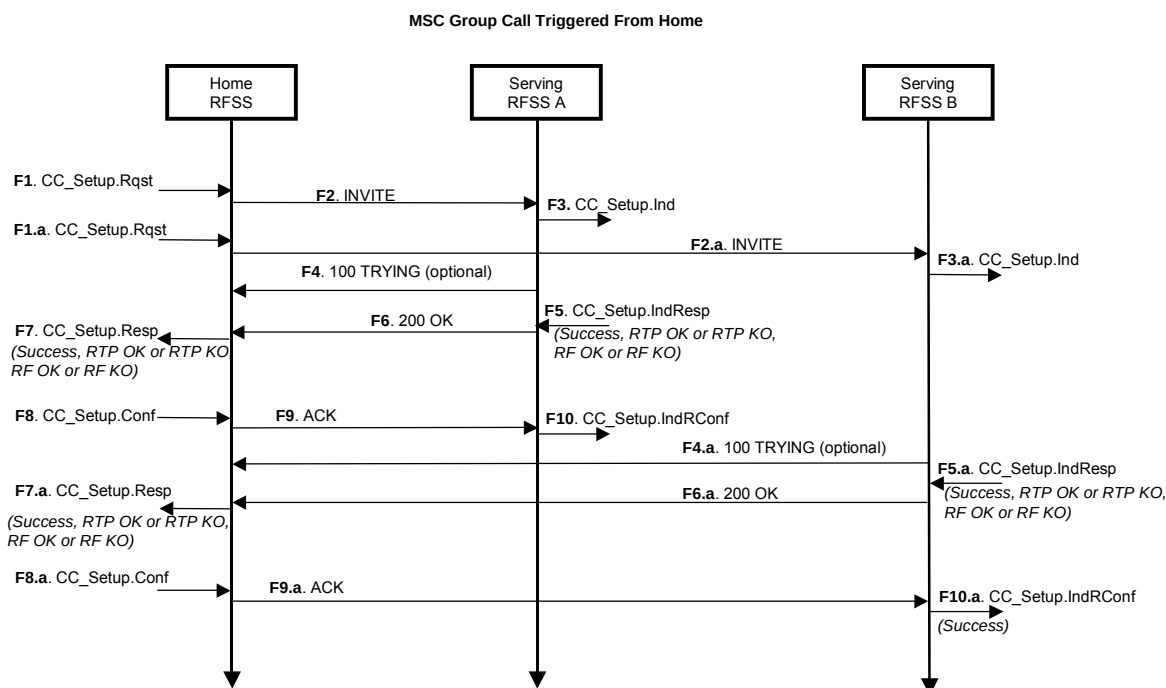


Figure 96 - MSC Group Call Triggered from Home

Initial Conditions:

- RFSS A and RFSS B are registered with a group.
- An SU initiates a Group Call at the group's Home RFSS.

F1, F1.a. As shown in Figure 96, the Home RFSS Controlling Function allocates RTP resources for both call segments, and exerts two CC_Setup.Request primitives to the Call Control Protocol Element in order to setup group call segments with the Serving RFSSs.

F2, F2.a SIP INVITE messages are sent to the Serving RFSSs.

F3, F3.a. Upon receipt of the SIP INVITE message at an RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function.

F4, F4.a. SIP 100 TRYING messages are optionally sent to the Home RFSS following SIP RFC 3261 requirements.

F5, F5.a. The Controlling Function at the Serving RFSSs attempts to allocate RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success, RTP OK or KO, RF OK or KO) to the Call Control Protocol Element at the Serving RFSS. The RF OK or KO parameter is significant only for a confirmed group call with RTP resources allocated.

F6, F6.a. The Serving RFSSs send SIP 200 OK messages to the Home RFSS.

F7, F7.a. Upon receipt of the SIP 200 OK messages at the Home RFSS, the Call Control Protocol Element exerts CC_Setup.Response(Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F8, F8.a. The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Home RFSS.

F9, F9.a. The Home RFSS sends the ACK messages to the Serving RFSSs.

F10, F10.a. Upon receipt of the ACK messages at the Serving RFSSs, the Call Control Protocol Element exerts a CC_Setup.IndicateResponseConfirm(Success) primitive to the Controlling Function.

E.2 Initiated by Serving RFSS

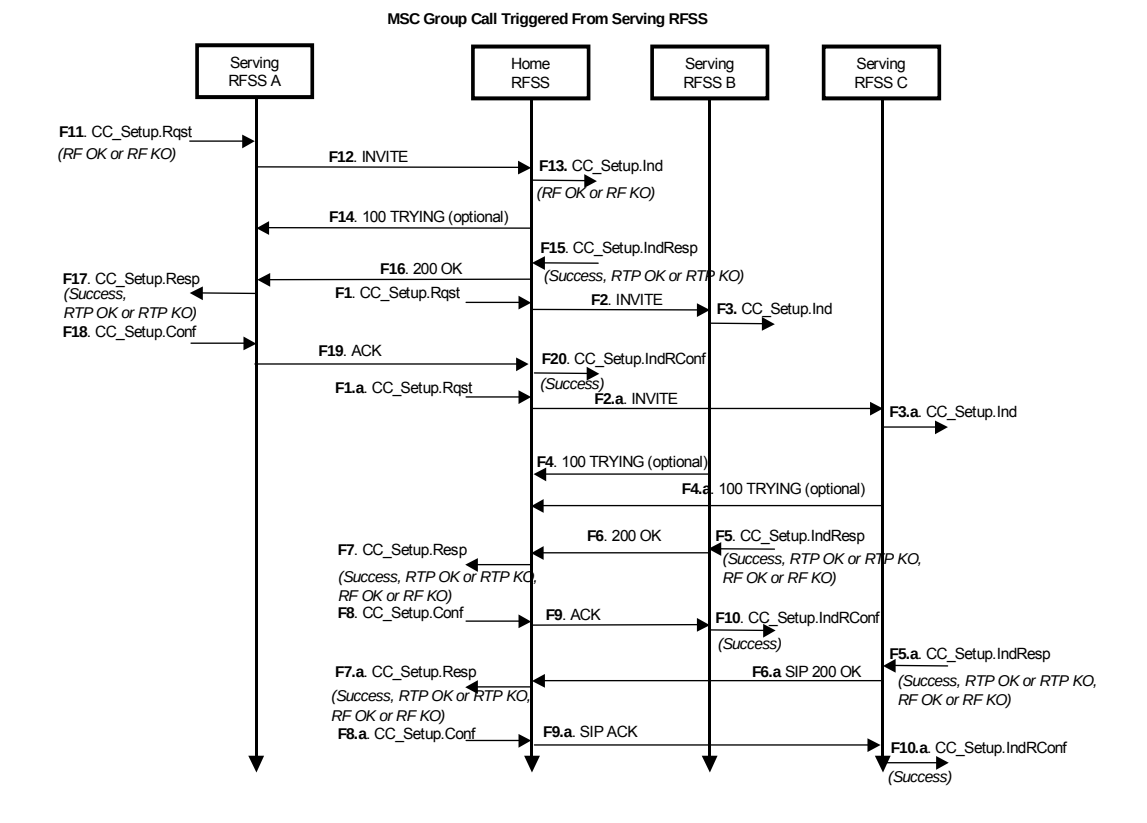


Figure 97 - MSC Group Call Triggered from Serving RFSS

Initial Conditions:

- RFSS A, RFSS B, and RFSS C are registered with a group.
- An SU initiates a Group Call at Serving RFSS A.

F11. As shown in Figure 97, RFSS A's Controlling Function allocates an RTP resource and exerts a CC_Setup.Request(RF OK or RF KO) primitive to the Call Control Protocol Element in order to set up a group call segment with the Home RFSS. The RF OK or RF KO parameter is significant only for a confirmed group call, and specifies whether RFSS A has allocated RF resources to serve the confirmed group call.

F12. A SIP INVITE message is sent to the Home RFSS.

F13. Upon receipt of the SIP INVITE message at the Home RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function.

F14. A SIP 100 TRYING message is optionally sent to the initiating Serving RFSS following SIP RFC 3261 requirements.

F15. The Controlling Function at the Home RFSS attempts to allocate RTP resources for the call segment with the initiating Serving RFSS and exerts a CC_Setup.IndicateResponse(Success, RTP OK or RTP KO) to the Call Control Protocol Element at the Home RFSS.

F16. The Home RFSS sends a SIP 200 OK message to the initiating Serving RFSS.

F1, F1.a – F10, F10a. The Home RFSS allocates RTP resources and sets up call segments with the other Serving RFSSs as described in Section E.1.

F17. Upon receipt of the SIP 200 OK message at initiating Serving RFSS, the Call Control Protocol Element exerts a CC_Setup.Response(Success, RTP OK or RTP KO) primitive to the Controlling Function.

F18. The Controlling Function exerts a CC_Setup.Confirm primitive to the Call Control Protocol Element at the initiating Serving RFSS.

F19. The initiating Serving RFSS sends the SIP ACK message to the Home RFSS.

F20.. Upon receipt of the ACK message at the Home RFSS, the Call Control Protocol Element exerts a CC_Setup.IndicateResponseConfirm(Success) primitive to the Controlling Function.

E.3 Initiated by Two Serving RFSSs

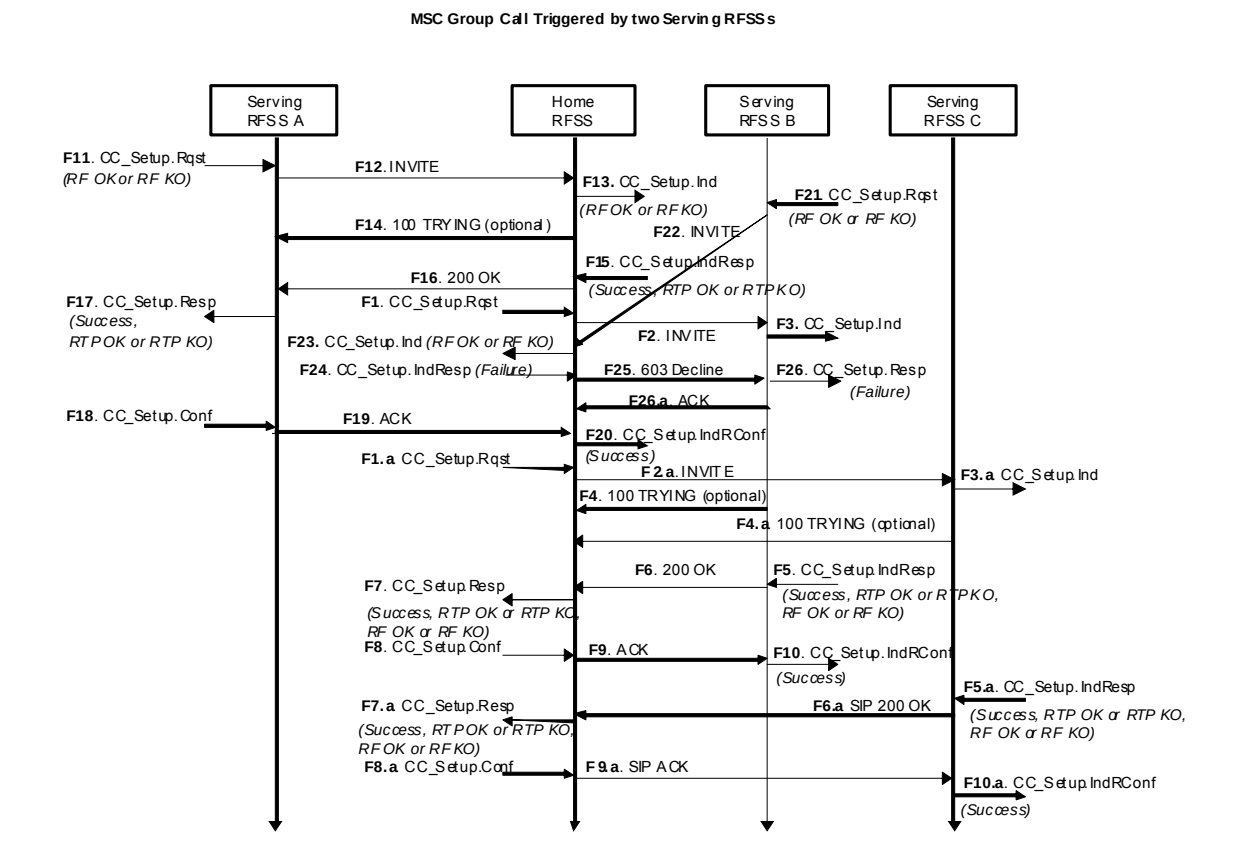


Figure 98 - MSC Group Call Triggered by two Serving RFSSs

Initial Conditions:

- RFSS A, RFSS B, and RFSS C are registered with a group.
- An SU initiates a Group Call at Serving RFSS A.
- An SU initiates a Group Call at Serving RFSS B.

F11 – F20. As shown in Figure 98, a group call segment is set up between initiating Serving RFSS A and the Group Home RFSS as described in Section E.2.

F21. Serving RFSS B's Controlling Function allocates an RTP resource and exerts a CC_Setup.Request(RF OK or RF KO) primitive to the Call Control Protocol Element in order to set up a group call segment with the Home RFSS. The RF OK or RF KO parameter is significant only for a confirmed group call and specifies whether the RFSS has allocated RF resources to serve the confirmed group call.

F22. A SIP INVITE message is sent from Serving RFSS B to the Home RFSS. The Home RFSS receives the SIP INVITE (**F21**) from Serving RFSS B, while it is processing the SIP INVITE (**F12**) from Serving RFSS A.

F23.. Upon receipt of the SIP INVITE message at the Home RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function.

F24. The Home RFSS Controlling Function detects a colliding INVITE situation and, as both the requests have same priority, it exerts a CC_Setup.IndicateResponse(Failure) to refuse the INVITE request from Serving RFSS B.

F25. A SIP 603 Decline is sent to Serving RFSS B.

F26. Upon receipt of the 603 Decline message at Serving RFSS B, the Call Control Protocol Element sends out the CC_Setup.Resp(Failure) event to the Controlling Function.

F26.a. Serving RFSS B confirms receipt of the 603 Decline message by returning a SIP ACK message to the Home RFSS.

F1, F1.a – F10, F10a. The Home RFSS allocates RTP resources and sets up call segments with the other Serving RFSSs as described in Section E.1.

E.4 No RTP Resources at Home RFSS

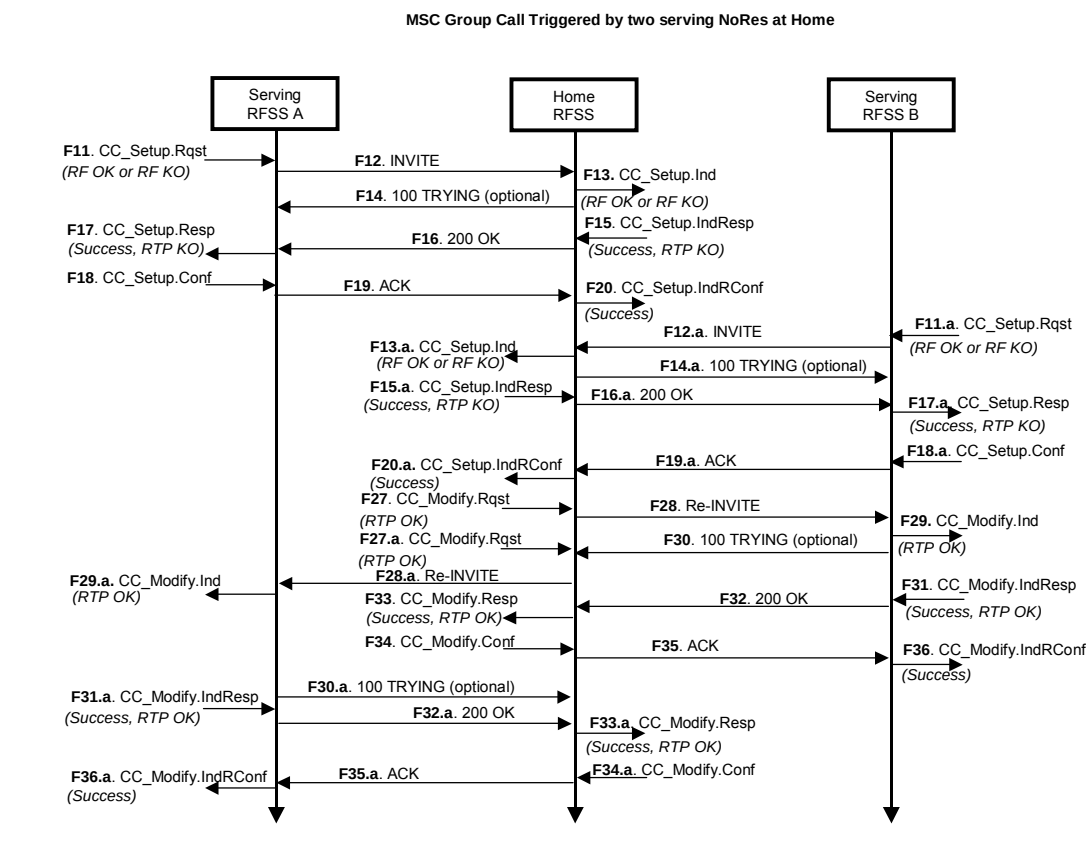


Figure 99 - MSC Group Call Triggered by two Serving RFSS with No RTP Resource At Home

Initial Conditions:

- RFSS A and RFSS B are registered with a group.
- An SU initiates a Group Call at Serving RFSS A.
- An SU initiates a Group Call at Serving RFSS B.

F11, F11.a – F20, F20.a. As shown in Figure 99, the SIP call segment setup is completed between the Home RFSS and Serving RFSSs A and B, as described in Section E.2. Note that the CC_Setup.IndicateResponse (F15, F15.a) and CC_Setup.Response(F17, F17.a) primitives say “RTP KO”, indicating that the Home RFSS does not have RTP resources allocated to the call segments.

F27, F27.a. The Home RFSS has newly available RTP resources. The Controlling Function exerts a CC_Modify.Request(RTP OK) primitive in order to setup an RTP session with the Serving RFSSs.

F28, F28.a. SIP Re-INVITE messages are sent to the Serving RFSSs.

F29, F29.a. Upon receipt of the SIP Re-INVITE message at RFSS A and B, the Call Control Protocol Elements there exert CC_Modify.Indicate(RTP OK) primitives to their Controlling Functions.

F30, F30.a. SIP 100 TRYING messages are optionally sent to the Home RFSS following SIP RFC 3261 requirements.

F31, F31.a. The Controlling Functions either have RTP resources allocated or they don't. They exert CC_Setup.IndicateResponse(Success, RTP OK or RTP KO, RF OK or RF KO) primitives to the Call Control Protocol Elements at the Serving RFSSs. The RF OK or KO parameter is significant only for a confirmed group call with RTP resources allocated.

F32, F32.a. The Serving RFSSs send SIP 200 OK messages to the Home RFSS.

F33, F33.a. Upon receipt of the SIP 200 OK messages at Home RFSS, the Call Control Protocol Element exerts CC_Setup.Resp(success, RTP OK or RTP KO, RF OK or RF KO) primitives to the Controlling Function.

F34, F34.a. The Controlling Function exerts CC_Modify.Conf primitives to the Call Control Protocol Element at the Home RFSS.

F35, F35.a. The Home RFSS sends SIP ACK messages to the Serving RFSSs.

F36, F36.a. Upon receipt of the SIP ACK messages at the Serving RFSSs, the Call Control Protocol Elements there exert CC_Setup.IndicateResponseConfirm(Success) primitives to their Controlling Functions.

E.5 NO RTP Resources at Serving RFSS

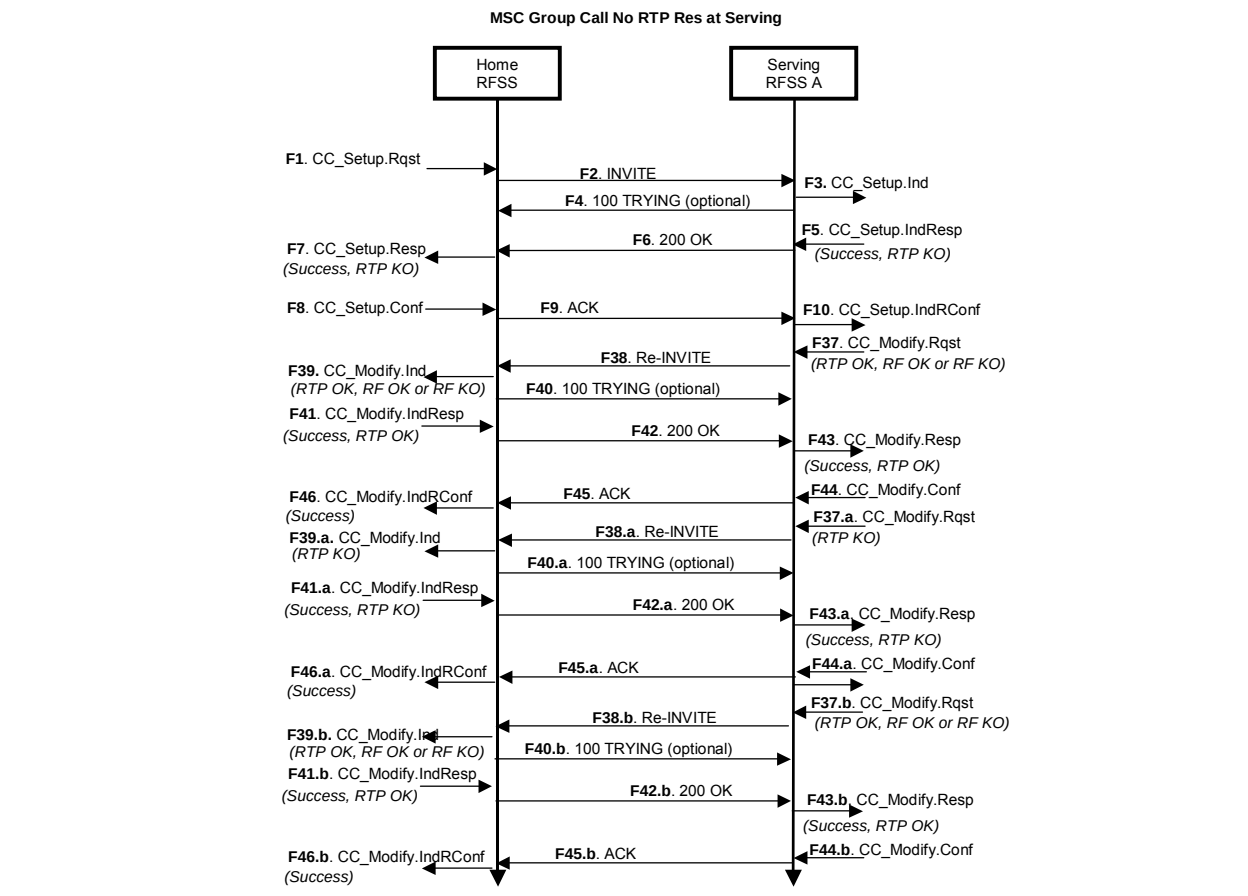


Figure 100 - MSC Group Call No RTP Resource At Serving RFSS

Initial Conditions:

- A Serving RFSS is registered with a group.
- An SU initiates a Group Call at the group's Home RFSS.

F1 – F10. As shown in Figure 100, the Home RFSS sets up a group call segment with the Serving RFSS as described in Section E.1. Note that the Serving RFSS does not have RTP resources allocated to the call segment, as indicated by the RTP KO parameter in the CC_Setup.IndicateReponse (F5) and CC_Setup.Reponse (F7) primitives.

F37. The Controlling Function at the Serving RFSS successfully allocates RTP resources to the call segment. It exerts a CC_Modify.Request (RTP OK, RF OK or RF KO) primitive to the Call Control Protocol Element. The RF OK or KO parameter is significant only for a confirmed group call.

F38. A SIP Re-INVITE message is sent to the Home RFSS

F39. Upon receipt of the SIP Re-INVITE message at Home RFSS, the Call Control Protocol Element exerts a CC_Modify.Indicate(RTP OK, RF OK or RF KO) primitive to the Controlling Function.

F40. A SIP 100 TRYING message is optionally sent to the Serving RFSS following SIP RFC 3261 requirements.

F41. The Controlling Function at the Home RFSS has RTP resources allocated to the call segment, and it exerts a CC_Modify.IndicateResponse(Success, RTP OK) primitive to the Call Control Protocol Element at the Home RFSS.

F42. The Home RFSS sends the SIP 200 OK message to the Serving RFSS.

F43. Upon receipt of the SIP 200 OK message at the Serving RFSS, the Call Control Protocol Element exerts a CC_Modify.Response(Success, RTP KO) primitive to the Controlling Function.

F44. The Controlling Function exerts a CC_Modify.Conf primitive to the Call Control Protocol Element at the Serving RFSS.

F45. The Serving RFSS sends the SIP ACK message to the Home RFSS.

F46. Upon receipt of the SIP ACK message at the Home RFSS, the Call Control Protocol Element exerts a CC_Modify.IndicateResponseConfirm(Success) event to the Controlling Function.

F37.a - F46.a. RTP resources are preempted at the Serving RFSS. The same Re-INVITE procedure is triggered except that the RTP State parameter in the CC_Modify.Request (**F37.a**) and CC_Modify.Indicate (**F39.a**) primitives indicates that the Serving RFSS wishes to tear down the RTP session, and the Home RFSS responds to the type 2 SDP offer with an RTP State parameter indicating RTP KO in the CC_Modify.IndicateResponse (**F41.a**) and CC_Modify.Response (**F43.a**) primitives.

F37.b - F46.b. RTP resources are newly available at the Serving RFSS. The same Re-INVITE procedure as the one described by **F37 to F46** description is executed.

E.6 RTP Loss

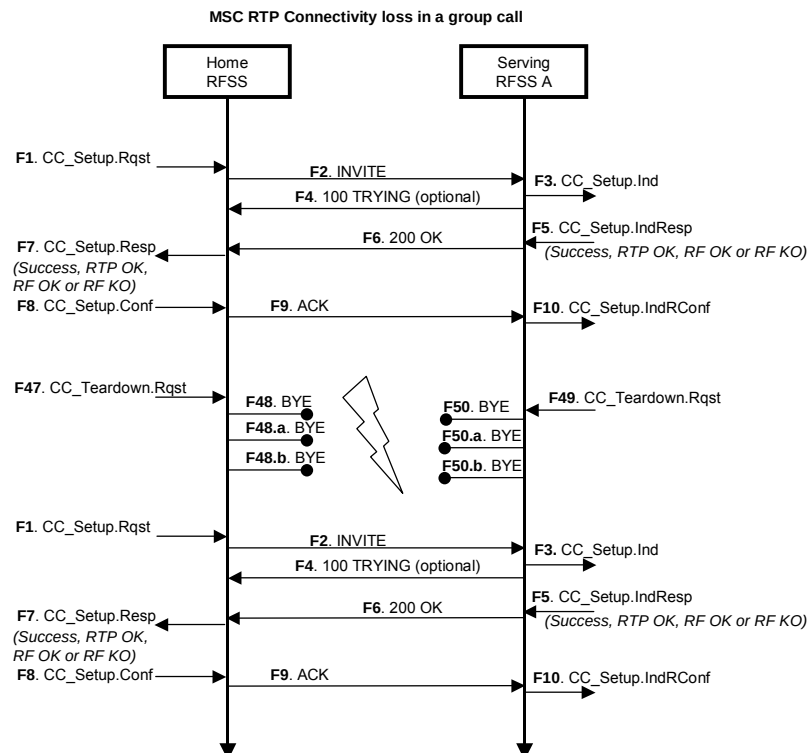


Figure 101 - MSC connectivity loss in a group call

Initial Conditions:

- A Serving RFSS is registered with a group.
- An SU initiates a Group Call at the group's Home RFSS.

F1 – F10. As shown in Figure 101, the Home RFSS sets up a group call segment with the Serving RFSS as described in Section E.1.

F47, F49. A loss of RTP connectivity occurs. The Controlling Functions at both the Home RFSS and the Serving RFSS detect it and exert CC_Teardown.Request primitives to their Call Control Protocol Elements.

F48, F48.a, F48.b. The Home RFSS sends BYE requests but, because network connectivity between RFSSs has failed, it does not receive any acknowledgment. In the end (SIP timer F expiry), the Home RFSS considers the group call segment torn down.

F50, F50.a, F50.b. The Serving RFSS sends BYE requests but, because network connectivity between RFSSs has failed, it does not receive any acknowledgment. In the end (SIP timer F expiry), the Home RFSS considers the group call segment torn down.

F1 - F10. The Home RFSS decides to setup the group call segment again because the group call is still active. The INVITE procedure described in Section E.1 is executed again.

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E.7 Confirmed Call Initiated by Home RFSS

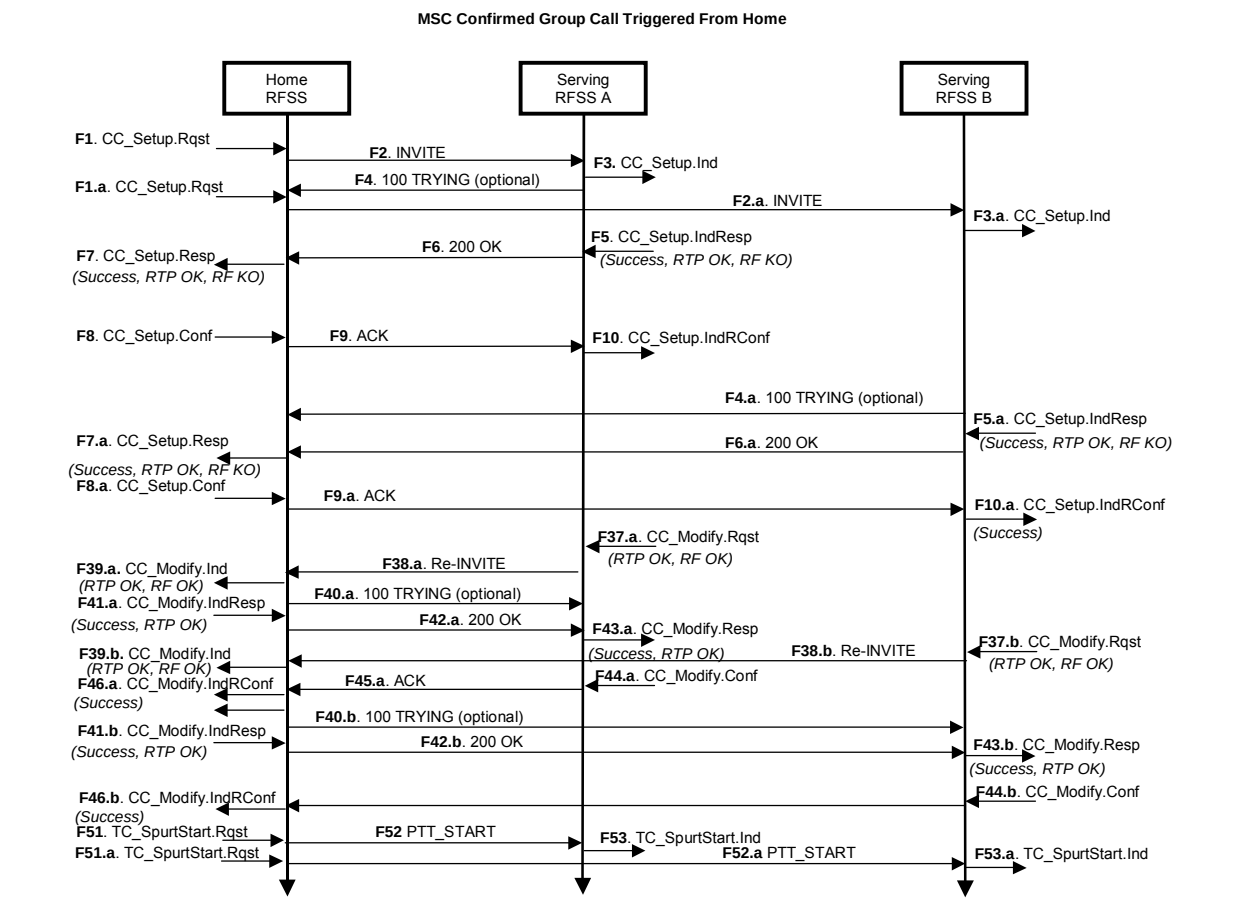


Figure 102 - MSC Confirmed Group Call triggered from Home

Initial Conditions:

- RFSS A and RFSS B are registered with a group.
- An SU initiates a Group Call at the group's Home RFSS.

F1, F1.a – F10, F10.a. As shown in Figure 102, the Home RFSS allocates RTP resources and sets up group call segments with both Serving RFSS A (F1 – F10) and Serving RFSS B (F1.a - F10.a) as described in Section E.1. Both Serving RFSS A and Serving RFSS B have RTP resources but do not have RF resources, as indicated in the CC_Setup.IndicateResponse (F5, F5.a) and CC_Setup.Response (F7, F7.a) primitives.

F37.a - F46.a. Serving RFSS A Re-INVITES the Home RFSS using the procedure described in Section E.5 because it has allocated RF resources. The CC_Modify.Request primitive (F37.a) indicates RF OK.

F37.b - F46.b. Serving RFSS B Re-INVITES the Home RFSS using the procedure described in Section E.5 because it has allocated RF resources. The CC_Modify.Request primitive (**F37.b**) indicates RF OK.

F51, F51.a. After completing the Re-INVITE procedure that indicates that all necessary RF resources are available, the Controlling Function at the Home RFSS decides to start the confirmed group call by exerting a TC_SpurStart.Request primitive to the Transmission Control Protocol Element.

F52, F52.a. The PTT_Start message is sent to Serving RFSSs to start the Talk Spurt.

F53, F53.a. The Transmission Control Protocol Element at the Serving RFSSs exerts a TC_SpurStart.Indicate primitive to the Controlling Function.

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E.8 Confirmed Call Initiated by Serving RFSS

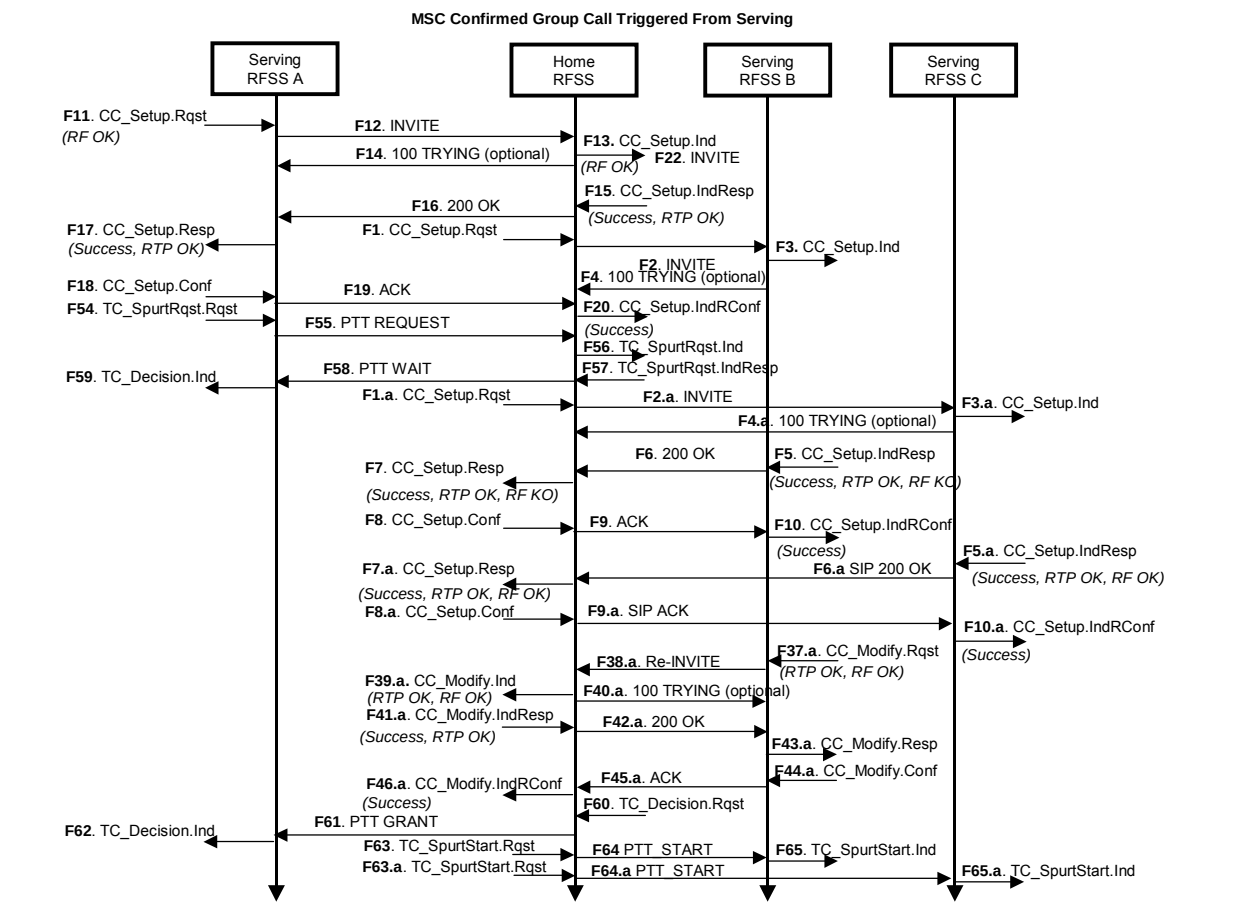


Figure 103 - MSC Confirmed Group Call triggered from Serving

Initial Conditions:

- RFSS A, RFSS B, and RFSS C are registered with a group.
- An SU initiates a Group Call at Serving RFSS A.

F11 – F20. As shown in Figure 103, Serving RFSS A sets up a group call segment with the Home RFSS as described in Section E.2.

F1, F1.a – F10, F10.a. The Home RFSS allocates RTP resources and sets up group call segments with both Serving RFSS B (**F1 - F10**) and Serving RFSS C (**F1.a - F10.a**) using the procedure described in Section E.1. Serving RFSS A and Serving RFSS C have allocated RF resources, indicated in the CC_Setup.Request (**F11**) from RFSS A and the CC_Setup.IndicateResponse (**F5.a**) from RFSS C, but Serving RFSS B does not have RF resources, indicated in its CC_Setup.IndicateResponse (**F5**).

F54. When an RTP session is successfully setup with the Home RFSS, the Controlling Function at Serving RFSS A exerts a TC_SpurRqst.Request primitive to the Transmission Control Protocol Element.

F55. The PTT_REQUEST is sent to the Home RFSS.

F56. Upon receipt of the PTT_REQUEST message, the Transmission Control Protocol Element at the Home RFSS exerts a TC_SpurRqst.Indicate primitive to the Controlling Function. The Controlling Function starts the Tgchstartconfirm timer.

F57. The Controlling Function determines that conditions are not met to start the confirmed group call and exerts a TC_SpurRqst.IndicateResponse(Wait) primitive to the Transmission Control Protocol Element in order to return a PTT_WAIT to Serving RFSS A.

F58. The PTT_WAIT is sent to the Serving RFSS A.

F59. Upon receipt of the PTT_WAIT message, the Transmission Control Protocol Element exerts a TC_Decision.Indicate primitive to the Controlling Function.

F37.a – F46.a. When Serving RFSS B allocates RF resources, it Re-INVITES the Home RFSS using the procedure described in Section E.5. The CC_Modify.Request primitive (**F37.a**) indicates RF OK.

F60, F60.a. The Controlling Function at the Home RFSS determines that conditions are met to start the confirmed group call. The Controlling Function stops the Tgchstartconfirm timer and:

- exerts a TC_Decision.Request primitive to the Transmission Control Protocol Element in order to return a PTT_GRANT to Serving RFSS A
- exerts TC_SpurRqst.Start primitives to the Transmission Control Protocol Element in order to send a PTT_START to both Serving RFSS B and Serving RFSS C.

F61, F61.a. The PTT_Start message is sent to Serving RFSSs to start the Talk Spurt.

F62, F62.a. The Transmission Control Protocol Element at the Serving RFSSs exerts a TC_SpurStart.Indicate primitive to the Controlling Function.

Annex F (Informative) – SU-to-SU Call Flows Involving Multiple Vocoder Modes

The following message sequence charts provide examples on how the Calling and Called RFSSs agree upon vocoder mode for each call segment of a SU-to-SU call.

F.1 Mixed Mode SU-to-SU Call

The following call flow addresses the case where all the RFSSs support mixed vocoder modes (i.e. both FullRate and HalfRate) but do not support conversion between Full Rate (FR) and Half Rate (HR) vocoder modes. Therefore a common vocoder mode needs to be agreed upon by all participating RFSSs (i.e. Calling Serving RFSS, Calling Home RFSS, Called Home RFSS and Called Serving RFSS).

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F.1.1.1 HalfRate Offered SU-to-SU Call

In this example, the offered vocoder mode (i.e. HalfRate) is the only vocoder mode offered by the Calling Serving RFSS 1. The offer is accepted by the other three RFSSs.

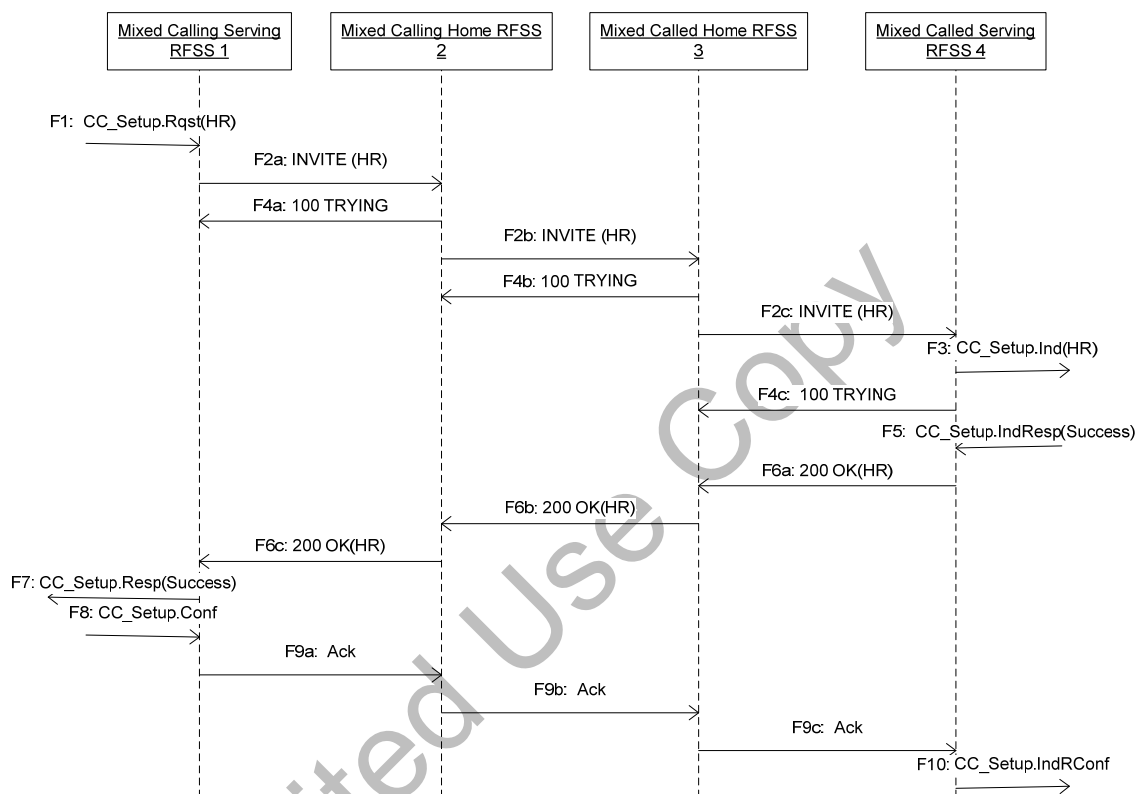


Figure 104 - HalfRate Offered SU-to-SU

F1: The Calling Serving RFSS 1 Controlling Function allocates RTP resources for a call segment, and exerts a CC_Setup.Request primitive to the Call Control Protocol Element in order to set up SU-to-SU call segments with the Calling Home RFSS 2. The HR mode is chosen and included in the call set up request.

F2a: A SIP Invite message is sent from the Calling Serving RFSS 1 to the Calling Home RFSS 2. In this case, the mode of the call in the Home RFSS is HalfRate.

m=audio 49280 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F2b: A SIP Invite message is sent from the Calling Home RFSS 2 to Called Home RFSS 3. In this case, the mode of the call is HalfRate.

m=audio 37220 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F2c: A SIP Invite message is sent from the Called Home RFSS 3 to Called Serving RFSS 4. In this case, the mode of the call is HalfRate.

m=audio 76420 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F3: Upon receipt of the SIP INVITE message at an RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in the HalfRate mode.

F4: SIP 100 TRYING is possibly sent at each call leg following the procedures of Section 6.4.1.2.

F5: The Called Serving RFSS 4 agrees with the offered VM = HR. The Controlling Function at the Serving RFSS allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F6a: Called Serving RFSS 4 send SIP 200 OK message to Called Home RFSS 3. In this case, the offered vocoder mode of the call was HalfRate and was accepted.

m=audio 33300 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F6b: Called Home RFSS 3 send SIP 200 OK message to Calling Home RFSS 2. In this case, the offered vocoder mode of the call was HalfRate and was accepted.

m=audio 33400 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F6c: Calling Home RFSS 2 send SIP 200 OK message to Calling Serving RFSS 1. In this case, the offered vocoder mode of the call was HalfRate and was accepted.

m=audio 33600 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F7: Upon receipt of the SIP 200 OK message, the Call Control Protocol Element exerts CC_Setup.Response(Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F8: The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Calling Serving RFSS 1.

F9a, F9b, F9c: Each RFSS sends the ACK message to peer RFSS.

F10: A CC_Setup.IndicateResponseConfirm(Success) primitive is exerted to the Controlling Function of the Called Serving RFSS 4. This enables the Controlling Function to start the call for the Called SU (i.e. to grant the RF Channel to a Called radio SU).

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F.1.2 Mixed Mode Offered SU-to-SU Call

In this example, both FullRate and HalfRate vocoder modes are offered by the Calling Serving RFSS 1. The Called Home RFSS 3 decides to limit the proposed vocoder modes to HalfRate. The Called Serving RFSS 4 accepts the offer to use HalfRate. The Called Home RFSS 3 responds to the Calling Home RFSS 2 with the selected vocoder mode. The choice is accepted by the other two RFSSs.

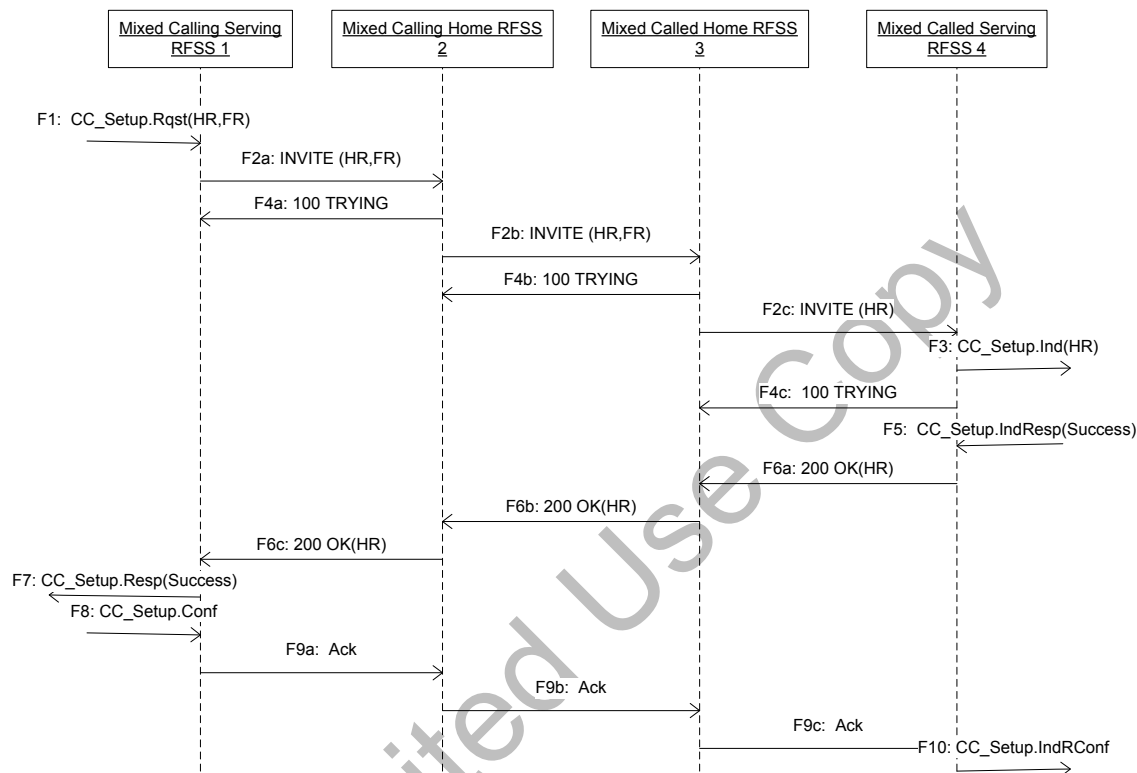


Figure 105 - Mixed Mode Offered SU-to-SU Call

F1: The Calling Serving RFSS 1 Controlling Function allocates RTP resources for a call segment, and exerts a CC_Setup.Request primitive to the Call Control Protocol Element in order to set up SU-to-SU call segments with the Calling Home RFSS 2. The HR mode is preferred and FR mode is allowed. Both VM's are included in the call set up request.

F2a: A SIP Invite message is sent from the Calling Serving RFSS 1 to the Calling Home RFSS 2. In this case, the preferred mode of the call in the Home RFSS is HalfRate. The FullRate mode is offered also.

```

m=audio 49280 RTP/AVP 101 100
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
  
```

F2b: A SIP Invite message is sent from the Calling Home RFSS 2 to Called Home RFSS 3. In this case, Both HalfRate and FullRate modes are included as received from the Calling Serving RFSS 1.

```
m=audio 37220 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F2c: The Called Home RFSS 3 decides to limit the offered modes to HalfRate. A SIP Invite message is sent from the Called Home RFSS 3 to Called Serving RFSS 4. In this case, the mode of the call is HalfRate.

```
m=audio 76420 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F3: Upon receipt of the SIP INVITE message at an RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in the HalfRate mode.

F4: SIP 100 TRYING is possibly sent at each call leg following the procedures of Section 6.4.1.2.

F5: The Called Serving RFSS 4 agrees with the offered VM = HR. The Controlling Function at the Serving RFSS allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F6a: Called Serving RFSS 4 sends SIP 200 OK message to Called Home RFSS 3. In this case, the offered vocoder mode of the call was HalfRate and was accepted.

```
m=audio 33300 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F6b: Called Home RFSS 3 sends SIP 200 OK message to Calling Home RFSS 2. In this case, the offered vocoder mode of the call was HalfRate and was accepted.

```
m=audio 33400 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F6c: Calling Home RFSS 2 sends SIP 200 OK message to Calling Serving RFSS 1. In this case, the offered vocoder mode of the call was HalfRate and was accepted.

```
m=audio 33600 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F7: Upon receipt of the SIP 200 OK message, the Call Control Protocol Element exerts CC_Setup.Response(Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F8: The Controlling Function exerts CC_Setup.Confirm primitives to the

Call Control Protocol at the Calling Serving RFSS 1.

F9a, F9b, F9c: Each RFSS sends the ACK message to peer RFSS.

F10: A CC_Setup.IndicateResponseConfirm(Success) primitive is exerted to the Controlling Function of the Called Serving RFSS 4. This enables the Controlling Function to start the call for the Called SU (i.e. to grant the RF Channel to a Called radio SU).

Limited Use Copy

F.2 SU-to-SU Call – First Offer is Declined

The following call flow shows the case where the offered VM is declined. In this case, none of the RFSSs support conversion between FR and HR VMs. The Calling Serving RFSS 1 can support FR and HR VM's. The Called Serving RFSS 4 is only capable of FullRate vocoder mode and therefore declines the offer. If the Calling Serving RFSS 1 is willing to assign a channel using the FullRate vocoder mode instead of HalfRate, the RFSS can try a second INVITE to the Calling Home RFSS 2 with FullRate instead of HalfRate. The example includes the optional second INVITE process culminating in the call proceeding.

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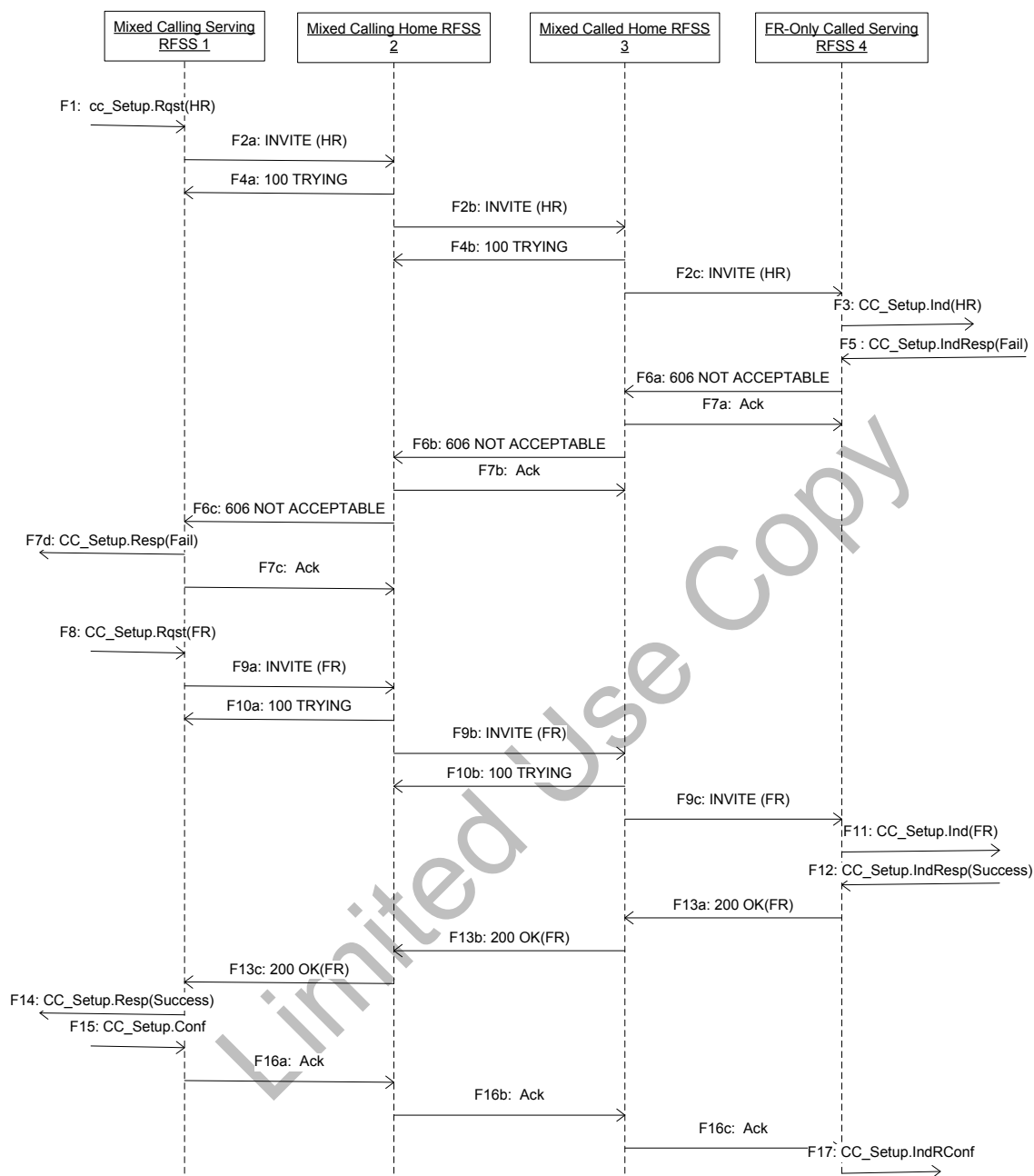


Figure 106 - SU-to-SU Call - First VM Offered is declined by Calling Serving RFSS

F1: The Calling Serving RFSS 1 Controlling Function allocates RTP resources for a call segment, and exerts a CC_Setup.Request primitive to the Call Control Protocol Element in order to set up SU-to-SU call segments with the Calling Home RFSS 2. The HR mode is chosen and included in the call set up request.

F2a: A SIP Invite message is sent from the Calling Serving RFSS 1 to the Calling Home RFSS 2. In this case, the mode of the call in the Home RFSS is HalfRate.

m=audio 51212 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F2b: A SIP Invite message is sent from the Calling Home RFSS 2 to Called Home RFSS 3. In this case, the mode of the call is HalfRate.

m=audio 33330 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F2c: A SIP Invite message is sent from the Called Home RFSS 3 to Called Serving RFSS 4. In this case, the mode of the call is HalfRate.

m=audio 12340 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000

F3: Upon receipt of the SIP INVITE message at an RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in the HalfRate mode.

F4: SIP 100 TRYING is possibly sent at each call leg following the procedures of Section 6.4.1.2.

F5: The Called Serving RFSS 4 does not agree with the offered VM = HR. The Controlling Function at the Serving RFSS exerts a CC_Setup.IndicateResponse(Fail) to the Call Control Protocol Element at the Serving RFSS.

F6a: None of the vocoder modes offered by the Called Home RFSS 3 are accepted by the Called Serving RFSS 4, the Called Serving RFSS 4 responds with a 606 Not Acceptable message indicating the offered vocoder mode is not supported by the Serving RFSS at this time.

F6b, F6c: The Called Home RFSS 3 and Calling Home RFSS 2 see the 606 Not Acceptable and forward the result accordingly.

F7: The Calling Serving RFSS 1 sees the offer was declined. If the only vocoder mode the RFSS is willing to assign is HR, then the call flow ends at this point. If the Calling Serving RFSS 1 is willing to assign a FR channel, then it can initiate a second INVITE with FR instead of HR. In this case, the call flow proceeds with a second INVITE being initiated by a second CC_Setup.Request primitive to the Call Control Protocol Element in order to set up SU-to-SU call segments with the Calling Home RFSS 2. The FR mode is chosen and included in the call set up request.

F7a, F7b, F7c: The Calling Serving RFSS 1, Calling Home RFSS 2 and Called Home RFSS 3 acknowledge the 606 Not Acceptable to stop the re-transmission timer in the peer RFSS.

F8-F17: The second set of INVITE messages are forwarded between the participating RFSSs, accepted and the call proceeds.

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F.3 Native Mode SU-to-SU Call

The following call flow addresses the case where the Called Home RFSS 3 supports conversion between FR and HR Vocoder Modes (VM). The Called Home RFSS 3 offers all three vocoder modes (i.e. Native, FullRate and HalfRate) to the Called Serving RFSS 4 and agrees to a VM for the Call Segment without forcing the other Calling RFSSs to agree on that particular VM. If both Called parties support rate conversion then the agreed VM for the call segment will be Native. Otherwise the VM will be either FR or HR and the Called Home RFSS 3 will convert the audio as required before forwarding to the Calling Home RFSS 2.

In this particular example, the Calling Serving RFSS 1 only offers HR VM and the Called Serving RFSS 4 is only capable of FR mode, so the Called Home RFSS 3 uses its capability to do the rate conversion to complete the call.

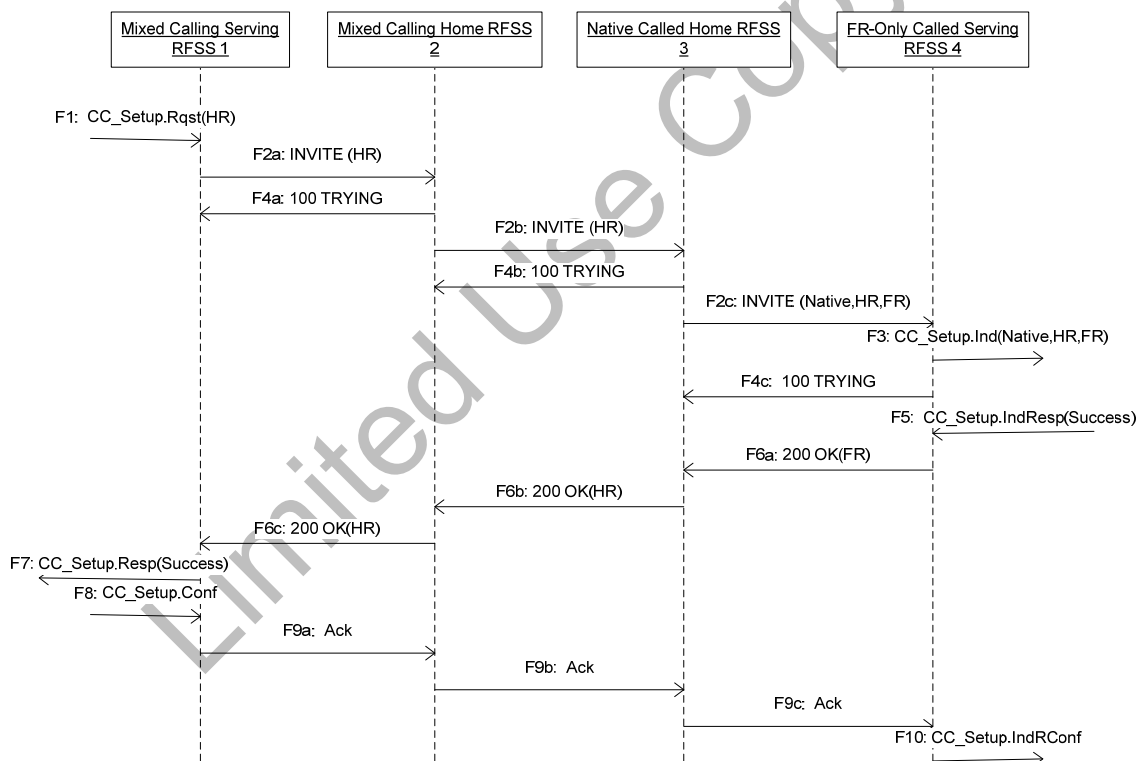


Figure 107 - Native Mode SU-to-SU Call

F1: The Calling Serving RFSS 1 Controlling Function allocates RTP resources for a call segment, and exerts a CC_Setup.Request primitive to the Call Control Protocol Element in order to set up SU-to-SU call segments with the Calling Home RFSS 2. The HR mode is chosen and included in the call set up request.

F2a: SIP INVITE message is sent from the Calling Serving RFSS 1 to the Calling Home RFSS 2. In this case, the mode of the call is HalfRate.

```
m=audio 49280 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F2b: SIP INVITE message is sent from the Calling Home RFSS 2 to the Called Home RFSS 3. In this case, the mode of the call is HalfRate.

```
m=audio 8810 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F2c: SIP INVITE message is sent from the Called Home RFSS 3 to the Called Serving RFSS 4. In this case, the preferred mode of the call segment is Native. If Native cannot be supported then HalfRate is preferred over FullRate. FullRate will be acceptable.

```
m=audio 50280 RTP/AVP 102 101 100
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F3: Upon receipt of the SIP INVITE message at an RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in all three modes with FullRate least preferred but the Called Serving RFSS 4 wants to assign FullRate mode to the call.

F4a, F4b, F4c: SIP 100 TRYING is possibly sent at each call leg following the procedures of Section 6.4.1.2.

F5: The Called Serving RFSS 4 chooses the offered VM = FullRate. The Controlling Function at the Serving RFSS allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F6a: The Called Serving RFSS 4 sends SIP 200 OK (indicating the offered FR Mode was accepted).

```
m=audio 20200 RTP/AVP 100
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F6b: The Called Home RFSS 3 sends SIP 200 OK to the Calling Home RFSS 2. In this case, the offered vocoder mode of the call from the Calling Home RFSS 2 included HalfRate and was accepted. The Called Home RFSS 3 will convert FR from the Called Serving RFSS 4 to HR mode as required to complete the call.

```
m=audio 30300 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F6c: The Calling Home RFSS 2 sends SIP 200 OK to the Calling Serving RFSS 1. In this case, the offered vocoder mode of the call from the Calling Serving RFSS 1 was HalfRate and was accepted.

```
m=audio 40400 RTP/AVP 101  
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F7: Upon receipt of the SIP 200 OK messages, the Call Control Protocol Element exerts CC_Setup.Response(Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F8: The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Calling Serving RFSS 1.

F9a, F9b, F9c: Each RFSS sends the ACK message to peer RFSS.

F10: A CC_Setup.IndicateResponseConfirm(Success) primitive is exerted to the Controlling Function of the Called Serving RFSS 4. This enables the Controlling Function to start the call for the Called SU (i.e. to grant the RF Channel to a Called radio SU).

The Call proceeds and the Called Home RFSS 3 converts voice from the Calling Home RFSS 2 to FR and voice from the Called Serving RFSS 4 to HR.

Annex G (Informative) – Group Call Flows Involving Multiple Vocoder Modes

The following message sequence charts provide examples on how the Home and Serving RFSSs agree upon vocoder mode for each call segment of a group call. The example system is comprised of four RFSSs: a Home RFSS for the SGID and three Serving RFSSs as depicted in Figure TDB2-1. The three Serving RFSSs span the spectrum as follows:

- o Serving RFSS 2 supports both FR and HR VMs
- o Serving RFSS 3 supports FR VM only
- o Serving RFSS 4 supports HR VM only.

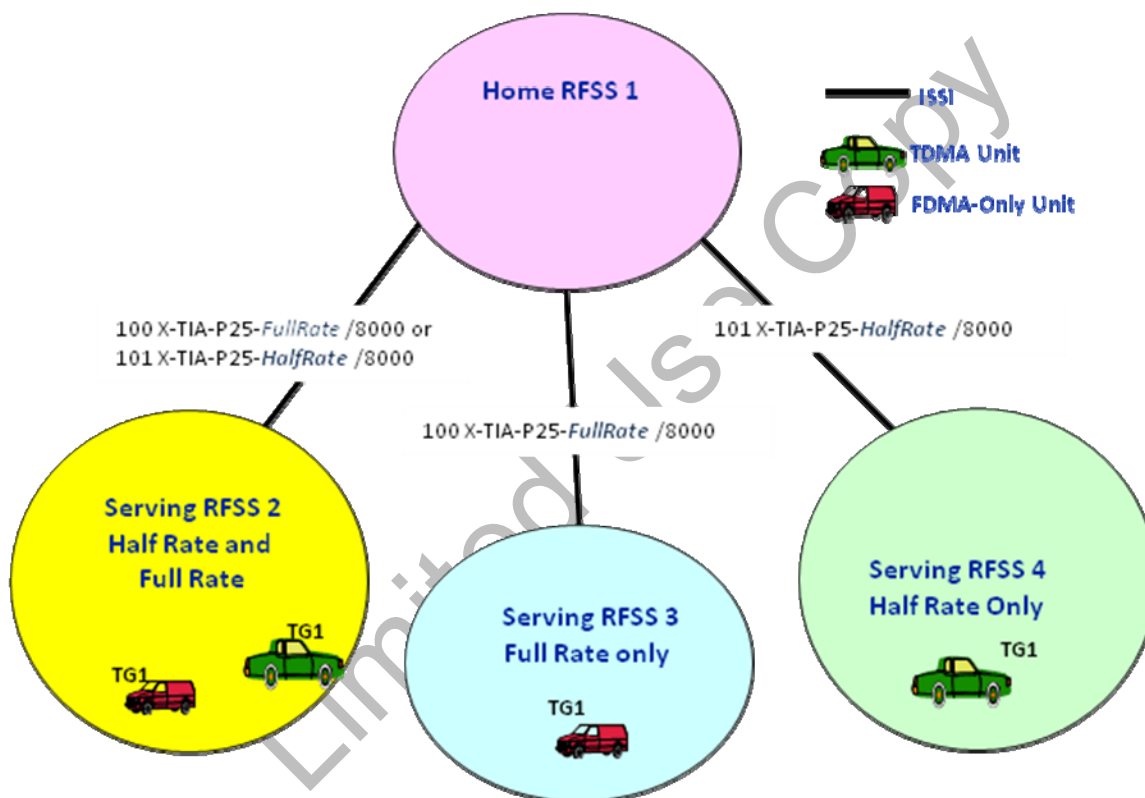


Figure 108 - Block Diagram of the Example System

G.1 Mixed Mode Group Call 1

The following call flow addresses the case where the Home RFSS supports Mixed vocoder modes (i.e. both FullRate and HalfRate) but does not support conversion between Full Rate (FR) and Half Rate (HR) vocoder modes. Therefore the Mixed Home RFSS 1 selects the vocoder mode of the call for each call segment and negotiates a common mode for all call segments. If the Home RFSS 1 chooses to change the VM during a call, the Home RFSS 1 RE-INVITE's each Serving RFSS to change the VM. If a given Serving RFSS cannot support the change in VM, then that Serving RFSS drops out of the call. In this case, none of the RFSSs support conversion between FR and HR VMs.

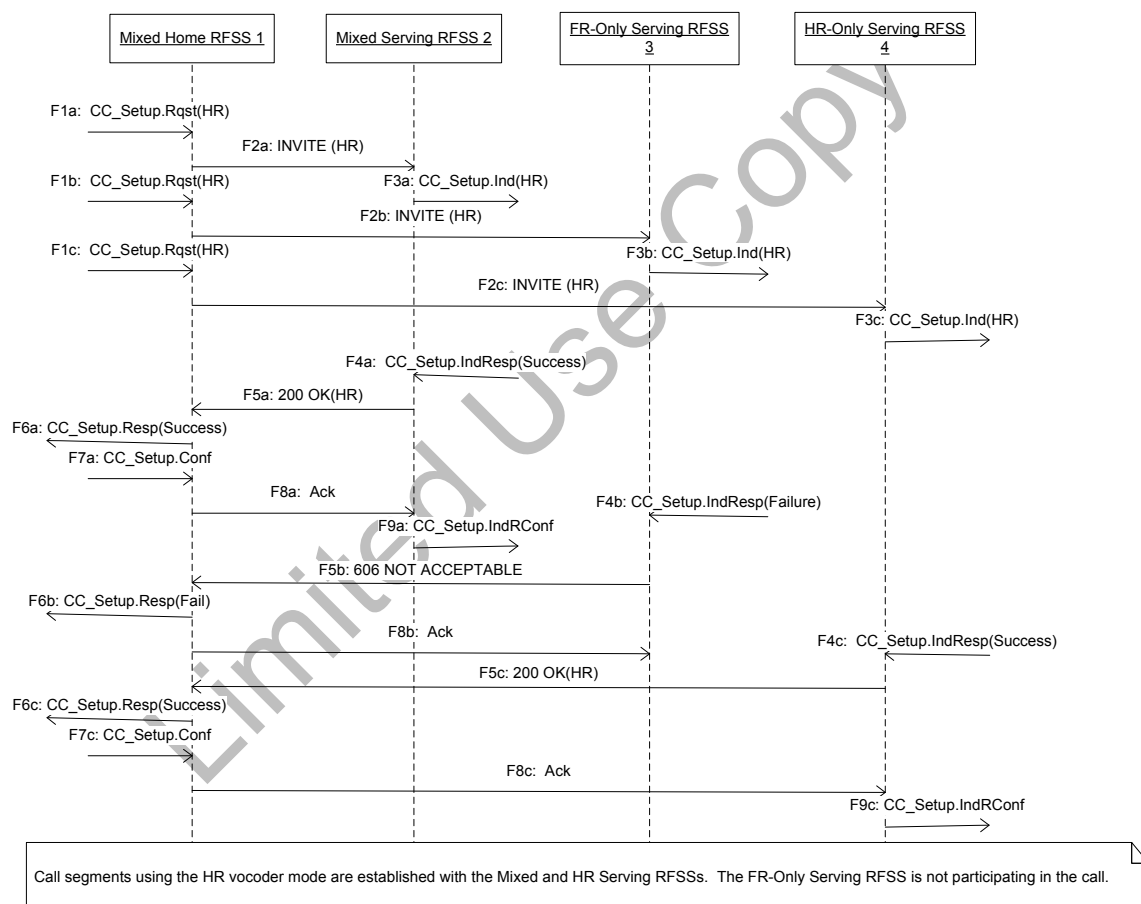


Figure 109 - Mixed Mode Group Call Triggered from Home RFSS

Initial Conditions:

- RFSS 2, RFSS 3 and RFSS 4 are registered with a group.
- An SU initiates a Group Call at the group's Home RFSS 1.

F1a, F1b, F1c: As shown in Figure F-2, the Home RFSS 1 Controlling Function allocates RTP resources for each call segment, and exerts three

CC_Setup.Request primitives to the Call Control Protocol Element in order to set up group call segments with the Serving RFSSs. The HR mode is chosen and included in the call set up request.

F2: A SIP Invite message is sent from the Home RFSS 1 to the Serving RFSSs. In this case, HalfRate is the only mode offered.

F2a:

```
m=audio 49280 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F2b:

```
m=audio 49380 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F2c:

```
m=audio 49480 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F3a, F3b, F3c: Upon receipt of the SIP INVITE message at each Serving RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in the HalfRate mode. No additional VMs are being offered.

F4a: The Mixed Serving RFSS 2 agrees with the offered VM = HR. The Controlling Function at the Serving RFSS 2 allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F4b: The FR-Only Serving RFSS 3 cannot accept HR. FR was not included in the list of offered VMs, so Call Control indicates the call set up is not accepted.

F4c: The HR-Only Serving RFSS 4 agrees with the offered VM = HR. The Controlling Function at the Serving RFSS 4 allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F5a: Serving RFSS 2 sends SIP 200 OK. In this case, the offered vocoder mode of the call from the Home RFSS 1 was HalfRate and was accepted.

```
m=audio 20200 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F5b: None of the vocoder modes offered by the Home RFSS 1 are accepted by Serving RFSS 3. Serving RFSS 3 responds with a '606 Not Acceptable' message indicating the offered vocoder mode is not supported by the Serving RFSS at this time.

F5c: Serving RFSS 4 sends SIP 200 OK. In this case, the offered vocoder mode of the call from the Home RFSS 1 was HalfRate and was accepted.

```
m=audio 6660 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```


F6a, F6c: Upon receipt of the SIP 200 OK messages at the Home RFSS 1, the Call Control Protocol Element exerts CC_Setup.Response(Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F6b: Upon receipt of the SIP '606 Not Acceptable' message at the Home RFSS 1, the Call Control Protocol Element exerts CC_Setup.Response(Fail) primitives to the Controlling Function.

F7a, F7c: The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Home RFSS 1. ³⁵

F8a, F8b, F8c: The Home RFSS 1 sends the ACK messages to the Serving RFSSs.

F9a, F9c: Upon receipt of the ACK messages at the Serving RFSSs, the Call Control Protocol Element exerts a CC_Setup.IndicateResponseConfirm(Success) primitive to the Controlling Function.

Call segments using the HR vocoder mode are established with the Mixed and HR Serving RFSSs (i.e. RFSSs 2 and 4). The FR-Only Serving RFSS 3 is not participating in the call.

³⁵ F7b is not included since upon indication of CC_SETUP.Response(Failure), the state machines in section 6.3.3 do not result in a CC_SETUP.Conf. At this point the SIP protocol itself takes care of the ACK to SIP 4xx-6xx failure responses.

G.2 Mixed Mode Group Call 2 (Home RFSS supports RC)

The following call flow addresses the case where the Home RFSS 1 supports rate conversion (RC) between FR and HR Vocoder Modes (VM). The Native Home RFSS 1 offers all three vocoder modes (i.e. Native, FullRate and HalfRate) to each Serving RFSS and agrees to a VM for each Call Segment without forcing the other Serving RFSSs to agree on the same VM. If both parties support rate conversion then the agreed VM for the call segment will be Native (shown in the next call flow). Otherwise the VM will be either FR or HR and the Home RFSS 1 will convert the audio as required before forwarding to a given Serving RFSS.

In this case, The Home RFSS 1 supports conversion between FR and HR VMs but none of the other RFSSs support rate conversion. The result is a single vocoder mode is agreed upon for each call segment and the FR-Only Serving RFSS 3 is included in the call.

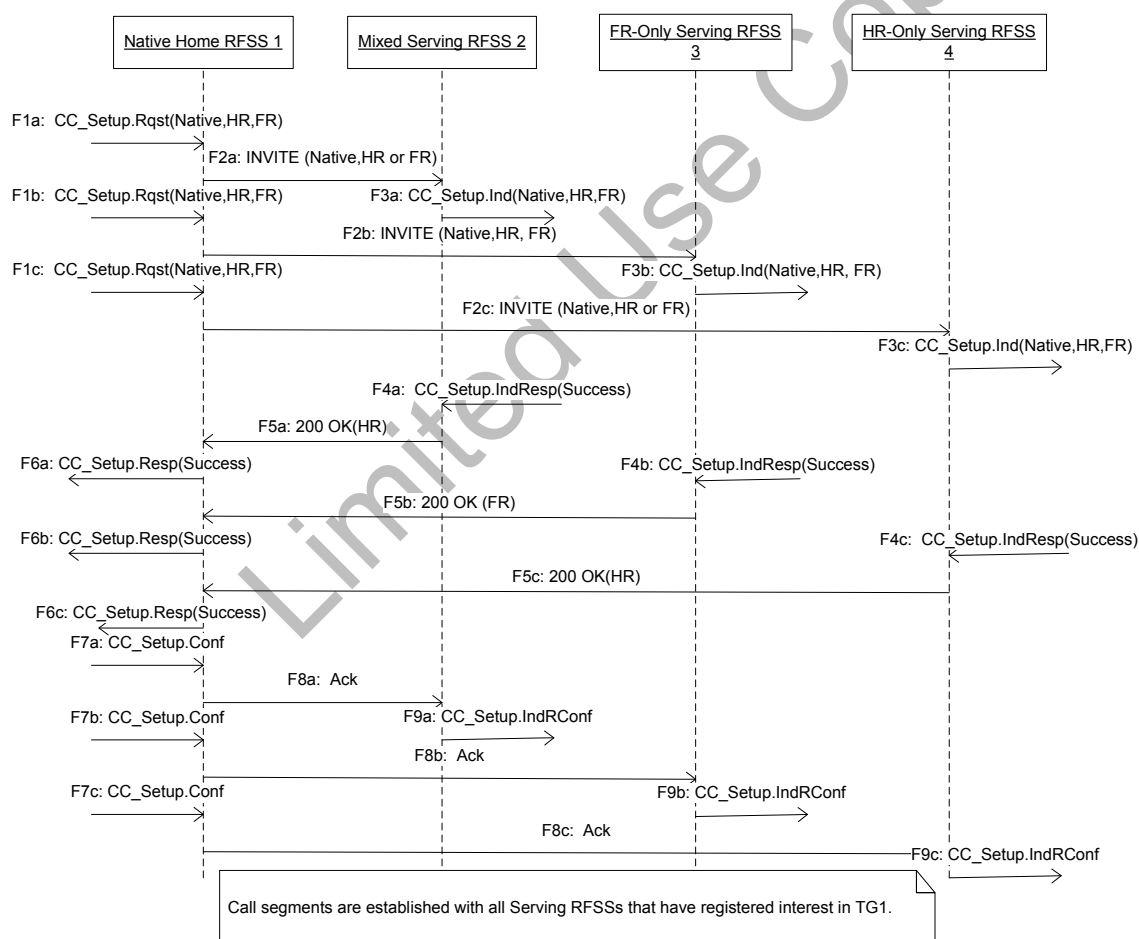


Figure 110 - Mixed Mode Group Call (Home RFSS supports RC)

F1a, F1b, F1c: The Home RFSS 1 Controlling Function allocates RTP resources for each call segment, and exerts three CC_Setup.Request primitives to the Call Control Protocol Element in order to set up group call segments with the Serving RFSSs. The Native mode is chosen as Preferred and included in the call set up request. Use of HalfRate and FullRate is allowed in that order.

F2: SIP Invite messages are sent from the Home RFSS 1. In this case, the preferred mode of the call segment is Native. If Native cannot be supported then HalfRate is preferred over FullRate. FullRate will be acceptable.

F2a:

```
m=audio 49280 RTP/AVP 101
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F2b:

```
m=audio 49380 RTP/AVP 101
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F2c:

```
m=audio 49480 RTP/AVP 101
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F3a, F3b, F3c: Upon receipt of the SIP INVITE message at an RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. All three possible vocoder modes are being offered. If Native is not supported, HR is preferred over FR.

F4a, F4c: The Mixed and HR-Only Serving RFSSs do not support the preferred VM = Native but do agree with the offered VM = HR. The Controlling Functions at the Serving RFSSs allocate RTP and RF resources and exert a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F4b: The FR-Only Serving RFSS 3 sees FR is included in the list of offered VMs, so Call Control indicates the call set up is accepted with VM = FR.

F5a, F5c: Serving RFSS 2 (Mixed) and Serving RFSS 4 (HR-Only) send SIP 200 OK messages. In this case, the HalfRate vocoder mode was offered and accepted.

F5a

```
m=audio 8080 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F5c

```
m=audio 8086 RTP/AVP 101
```

a=rtpmap:101 X-TIA-P25-HalfRate/8000

F5b: Serving RFSS 3 (FR-Only) finds 100 FullRate in the offered VMs and chooses FR. A SIP 200 OK message is returned to agree with the FR mode.

m=audio 43210 RTP/AVP 100

a=rtpmap:100 X-TIA-P25-FullRate/8000

F6a, F6b, F6c: Upon receipt of the SIP 200 OK messages at the Home RFSS 1 from each Serving RFSS, the Call Control Protocol Element exerts CC_Setup.Response (Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F7: The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Home RFSS 1.

F8a, F8b, F8c: The Home RFSS 1 sends the ACK messages to the Serving RFSSs.

F9a, F9b, F9c: Upon receipt of the ACK messages at the Serving RFSSs, the Call Control Protocol Element exerts a CC_Setup.IndicateResponseConfirm(Success) primitive to the Controlling Function.

G.3 Native Mode Group Call (Home & Serving RFSS2 support RC)

The following call flow addresses the case where the Home RFSS 1 supports conversion between FR and HR Vocoder Modes (VM). The Native Home RFSS 1 offers all three vocoder modes (i.e. Native, FullRate and HalfRate) to each Serving RFSS and agrees to a VM for each Call Segment without forcing the other Serving RFSSs to agree on the same VM. In this case, The Home RFSS 1 and Serving RFSS 2 both support conversion between FR and HR VMs but the other two RFSSs do not. The result is the Native vocoder mode is agreed upon for the call segment between Home RFSS 1 and Serving RFSS 2. Both of the other RFSSs are included in the call also.

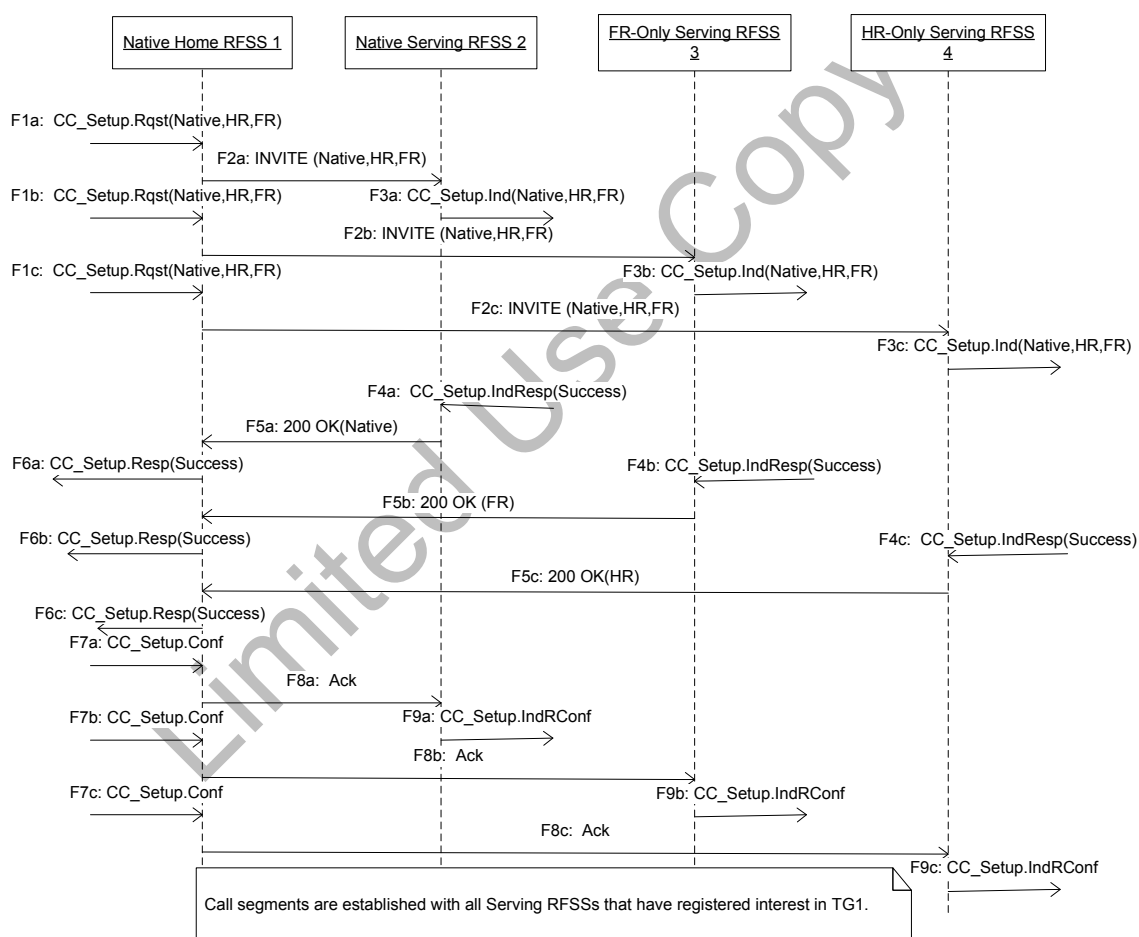


Figure 111 - Native Mode Group Call (Home RFSS supports RC)

F1a, F1b, F1c: The Home RFSS 1 Controlling Function allocates RTP resources for each call segment, and exerts three CC_Setup.Request primitives to the Call Control Protocol Element in order to set up group call segments with the Serving

RFSSs. The Native mode is chosen and included in the call set up request. HalfRate and FullRate VMs are allowed in that order.

F2: SIP Invite messages are sent from the Home RFSS 1. In this case, the preferred mode of the call segment is Native. If Native cannot be supported then HalfRate is preferred over FullRate. FullRate will be acceptable.

F2a:

```
m=audio 49280 RTP/AVP 101
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F2b:

```
m=audio 49380 RTP/AVP 101
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F2c:

```
m=audio 49480 RTP/AVP 101
a=rtpmap:102 X-TIA-P25-Native/8000
a=rtpmap:101 X-TIA-P25-HalfRate/8000
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F3a, F3b, F3c: Upon receipt of the SIP INVITE message at an RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in the Native mode. HalfRate and FullRate VM's are allowed in that order.

F4a: The Native Serving RFSS 2 agrees with the offered Native VM. The Controlling Function at the Serving RFSS allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F4b: The FR-Only Serving RFSS 3 cannot accept Native or HR but FR was included in the list of offered VMs, so Call Control indicates the call set up is accepted and specified the FR mode.

F4c: The HR-Only Serving RFSS 4 chooses the offered VM = HR. The Controlling Function at the Serving RFSS allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F5a: Native Serving RFSS 2 sends SIP 200 OK. In this case, the preferred vocoder mode of the call from the Home RFSS 1 was Native and was accepted.

```
m=audio 12220 RTP/AVP 102
a=rtpmap:102 X-TIA-P25-Native/8000
```

F5b: The FR-Only Serving RFSS 3 finds 100 FullRate in the offered VMs and chooses FR. Serving RFSS 3 sends SIP 200 OK.

```
m=audio 30300 RTP/AVP 100
a=rtpmap:100 X-TIA-P25-FullRate/8000
```

F5c: The HR-Only Serving RFSS 4 finds 101 HalfRate in the offered VMs and chooses HR. Serving RFSS 4 sends SIP 200 OK.

```
m=audio 7640 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F6a, F6b, F6c: Upon receipt of the SIP 200 OK messages at the Home RFSS 1 from each Serving RFSS, the Call Control Protocol Element exerts CC_Setup.Response (Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F7: The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Home RFSS 1.

F8a, F8b, F8c: The Home RFSS 1 sends the ACK messages to the Serving RFSS.

F9a, F9b, F9c: Upon receipt of the ACK messages at the Serving RFSSs, the Call Control Protocol Element exerts a CC_Setup.IndicateResponseConfirm(Success) primitive to the Controlling Function.

Call segments are established with all Serving RFSSs that have registered interest in TG1.

G.4 Mixed Mode Group Call (Serving RFSS Initiates Group Call)

Similar to the first call flow the Home RFSS 1 in this example supports Mixed vocoder modes (i.e. both FullRate and HalfRate) but does not support conversion between Full Rate (FR) and Half Rate (HR) vocoder modes. Therefore the Mixed Home RFSS 1 selects the vocoder mode of the call for each call segment and negotiates a common mode for all call segments. In this case, the Mixed Serving RFSS 2 initiates the call instead of the Home RFSS 1. The offered VM is HalfRate. The Home RFSS 1 agrees to the offered VM and initiates sessions with the other Serving RFSSs.

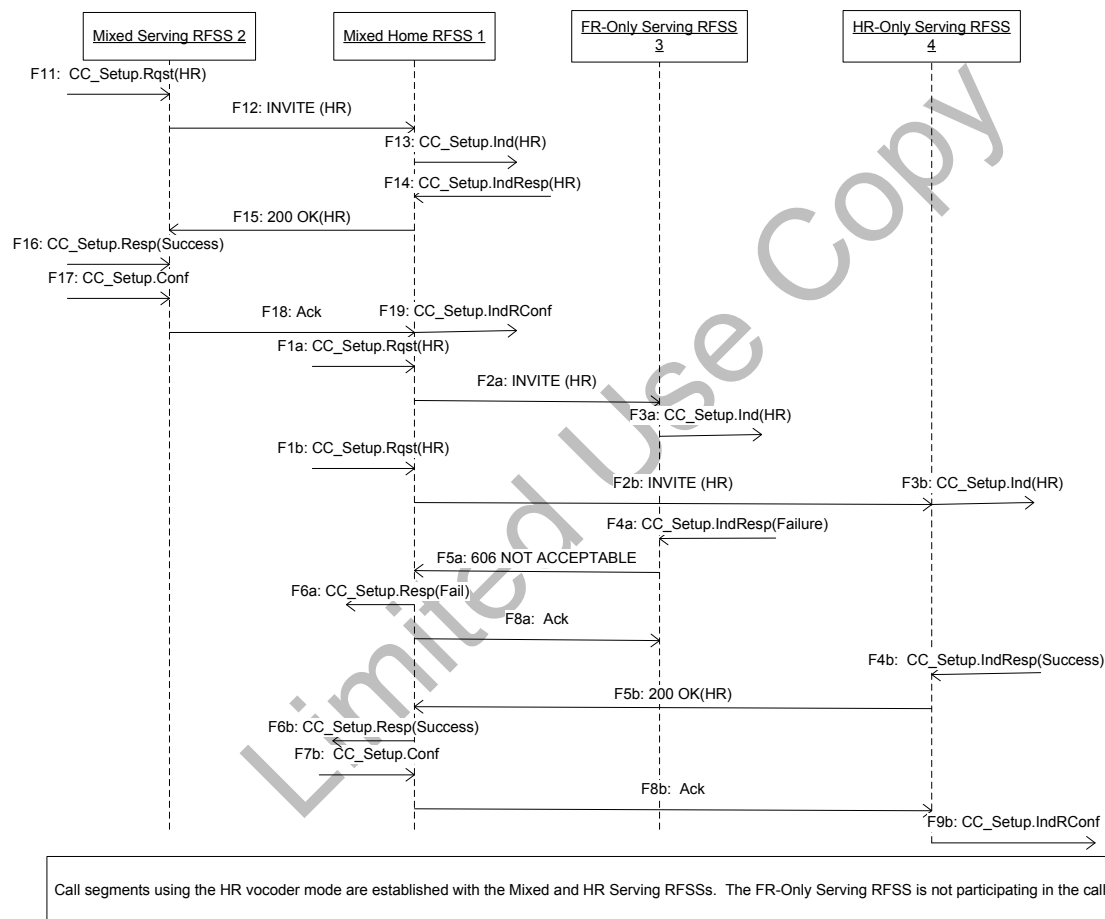


Figure 112 - Mixed Mode Group Call Triggered from Mixed Serving RFSS 2

Initial Conditions:

- ☐ RFSS 2, RFSS 3 and RFSS 4 are registered with a group.
- ☐ An SU initiates a Group Call at the Mixed Serving RFSS 2.

F11: As shown in Figure F-5, the Mixed Serving RFSS 2 Controlling Function allocates RTP resources for a call segment, and exerts a CC_Setup.Request

primitive to the Call Control Protocol Element in order to set up a group call segment with the Group Home RFSS 1. The HR mode is chosen and included in the call set up request.

F12: A SIP Invite message is sent from the Serving RFSS 2 to the SGID Home RFSS 1. In this case, HalfRate is the only mode offered.

```
m=audio 49280 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F13: Upon receipt of the SIP INVITE message at the Home RFSS 1, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in the HalfRate mode. No additional VMs are being offered.

F14: The Mixed Home RFSS 1 agrees with the offered VM = HR. The Controlling Function at the Home RFSS 1 allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F15: The Home RFSS 1 sends SIP 200 OK. In this case, the offered vocoder mode of the call from the Serving RFSS was HalfRate and was accepted.

```
m=audio 20200 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F16: Upon receipt of the SIP 200 OK messages at the Serving RFSS 2, the Call Control Protocol Element exerts CC_Setup.Response(Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F17: The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Serving RFSS 2.

F18: The Serving RFSS 2 sends the ACK messages to the Home RFSS 1.

F1a, F1b: As shown in Figure F-5, the Home RFSS 1 Controlling Function allocates RTP resources for each call segment, and exerts two CC_Setup.Request primitives to the Call Control Protocol Element in order to set up group call segments with the Serving RFSSs. The HR mode is chosen and included in the call set up request.

F2: SIP Invite messages are sent from the Home RFSS 1 to each Serving RFSS. In this case, HalfRate is the only mode offered.

F2a:

```
m=audio 44440 RTP/AVP 101
a=rtpmap:101 X-TIA-P25-HalfRate/8000
```

F2b:

```
m=audio 44450 RTP/AVP 101
```

a=rtpmap:101 X-TIA-P25-HalfRate/8000

F3a, F3b: Upon receipt of the SIP INVITE message at each Serving RFSS, the Call Control Protocol Element exerts a CC_Setup.Indicate primitive to the Controlling Function. The call is being offered in the HalfRate mode. No additional VMs are being offered.

F4a: The FR-Only Serving RFSS 3 cannot accept HR. FR was not included in the list of offered VMs, so Call Control indicates the call set up is not accepted.

F4b: The HR-Only Serving RFSS 4 agrees with the offered VM = HR. The Controlling Function at the Serving RFSS allocates RTP and RF resources and exerts a CC_Setup.IndicateResponse(Success) to the Call Control Protocol Element at the Serving RFSS.

F5a: None of the vocoder modes offered by the Home RFSS 1 are accepted by Serving RFSS 3, the Serving RFSS responds with a '606 Not Acceptable' message indicating the offered vocoder mode is not supported by the Serving RFSS at this time.

F5b: The Serving RFSS 4 sends SIP 200 OK. In this case, the offered vocoder mode of the call from the Home RFSS 1 was HalfRate and was accepted.

m=audio 6660 RTP/AVP 101

a=rtpmap:101 X-TIA-P25-HalfRate/8000

F6a: Upon receipt of the SIP '606 Not Acceptable' messages at the Home RFSS 1, the Call Control Protocol Element exerts CC_Setup.Response(Fail) primitives to the Controlling Function.

F6b: Upon receipt of the SIP 200 OK messages at the Home RFSS 1, the Call Control Protocol Element exerts CC_Setup.Response(Success, RTP OK or KO, RF OK or KO) primitives to the Controlling Function.

F7a, F7b: The Controlling Function exerts CC_Setup.Confirm primitives to the Call Control Protocol at the Home RFSS 1.

F8a, F8b: The Home RFSS 1 sends the ACK messages to the Serving RFSSs.

F9b: Upon receipt of the ACK messages at the Serving RFSSs, the Call Control Protocol Element exerts a CC_Setup.IndicateResponseConfirm(Success) primitive to the Controlling Function.

Call segments using the HR vocoder mode are established with the Mixed and HR Serving RFSSs. The FR-Only Serving RFSS 3 is not participating in the call.

Annex H (informative) – Future Considerations

H.1 ISSI SIP and RTP Security Services

ISSI SIP messages traversing the IP-based backbone network between RFSSs could be subject to common security threats, whereby attackers can potentially read and modify unprotected packets in the network, therefore stealing TIA-102 services, eavesdropping on, and even disrupting system communications. In addition an attacker could spoof a valid user either stealing TIA-102 services or inject malicious traffic. To counter the threats, we need to consider implementing security services in RFSSs that:

- Preserve the confidentiality and integrity of messaging,
- Prevent replay attacks or message spoofing,
- Provide for the authentication and privacy of the participants in a voice service session,
- Prevent denial-of-service attacks, and
- Provide for the confidentiality, integrity, and authentication of SIP and RTP message bodies (i.e., SDP).

H.2 ISSI SIP and RTP Compression Services

Especially when transmitting over low speed links, we need to consider using suitable compression schemes that would pare down individual SIP message and RTP packet lengths considerably, therefore significantly reducing the overall voice services call setup time and transmission delay.

H.3 IP-Based Backbone Bandwidth Reservation

RFC 2474 defines a replacement header field, called the DS (Differentiated Services) field, superseding the existing definitions of the IPv4 TOS (Type of Service) octet. This new field includes a 6-bit DSCP (Differentiated Services Codepoint) which could be flexibly mapped to particular forwarding treatment, or per-hop behavior (PHB), at each Differentiated Services compliant network node (e.g., routers, etc.) along its path, for efficient routing of SIP messages and RTP packets to their destinations, according to the urgency of the call request (e.g., preemptive priority call). Appropriate assignment and mapping of DSCPs for ISSI applications are for further study.

H.4 Mixed IANA and ISSI Payload Carrying Trunked, Conventional, and Additional Data

Each block header will contain a bit (E) to identify if the payload type is a standard IANA type (E set to 0) or a profile specific type (E set to 1).

Each block header will contain a 7 bit block profile type. The profile type could be one of the standard RTP profile types (with E set to 0) or, it could be a profile specific type (E set to 1). Occasionally, a mixed IANA and ISSI payload can be utilized. The RTP stream is directed from a single sender to a single receiver talk group. This talk group, while logically a single entity, could contain multiple types of radio equipment. For instance, in the case of a fire being responded to by multiple jurisdictions, there could be repeaters and consoles, analog repeaters and consoles, and virtual radio PC endpoints. All of these devices need to operate as a single communications system for the duration of the fire. The TIA-102 (IMBE) vocoder is unable to carry the signaling tones used by conventional systems. Therefore, the conventional system needs to use a codec such as G.711. But, converting IMBE to G.711 and back to IMBE could cause significant voice degradation. The only feasible means of tying the two radio technologies together is to carry both codecs. By carrying the audio in redundant form, a single talk spurt can be used to feed the entire group guaranteeing that the entire group will receive all of the same information. The bearer plane of a dispatch radio system needs to contain additional information in addition to the voice itself. This information provides details on the original radio source and destination group as well as ancillary data and encryption information. The profile specific block types provide for carrying this additional data in addition to the voice in single or redundant form.

H.5 Refreshing Service Profile Information

Group and user service profile information (e.g., priority, authentication parameter values, etc.) are subject to change from time to time. A mechanism needs to be in place for the Home RFSS to update the Serving RFSS with the most current service profile information.

Annex I (informative) – Console Functionality

This annex provides additional detail about the specific console features supported by the ISSI.

The ISSI is extended to enable the console to:

- Perform group registration for the groups that it wants to dispatch
- Participate in trunking group voice calls
- Participate in trunking unit to unit voice calls.

In Figure 113, ISSI support for consoles enables the console subsystem to interact with the other RFSSs and participate in trunking voice calls. The console subsystem is a type of RFSS. With that abstraction, enabling console participation in trunking voice calls over the ISSI follows rather easily. To fully understand the principles described herein, the reader is assumed to have knowledge of the ISSI as defined in this document.

As stated above, the ISSI is extended to support consoles over the trunking console subsystem interface. There are additional system features which a console might use. The system features which are NOT part of the ISSI support for consoles are listed below. These might be considered for future revisions of the Trunked Console Subsystem Interface:

- Console patch – both audio patch and re-grouped patch
- Console ability to rekey a subscriber radio
- Console/logging device ability to express monitor interest in group and unit-to-unit calls
- System Wide call

Consoles perform automatic keying (OTAR) via data link independent OTAR as described in **[102AACA]**.

The console acts as a data host by employing Ed as described in **[102BAEB]**.

As described in **[102BACD]**, console subsystems act as Serving RFSSs with regards to supplementary data transport and monitoring.

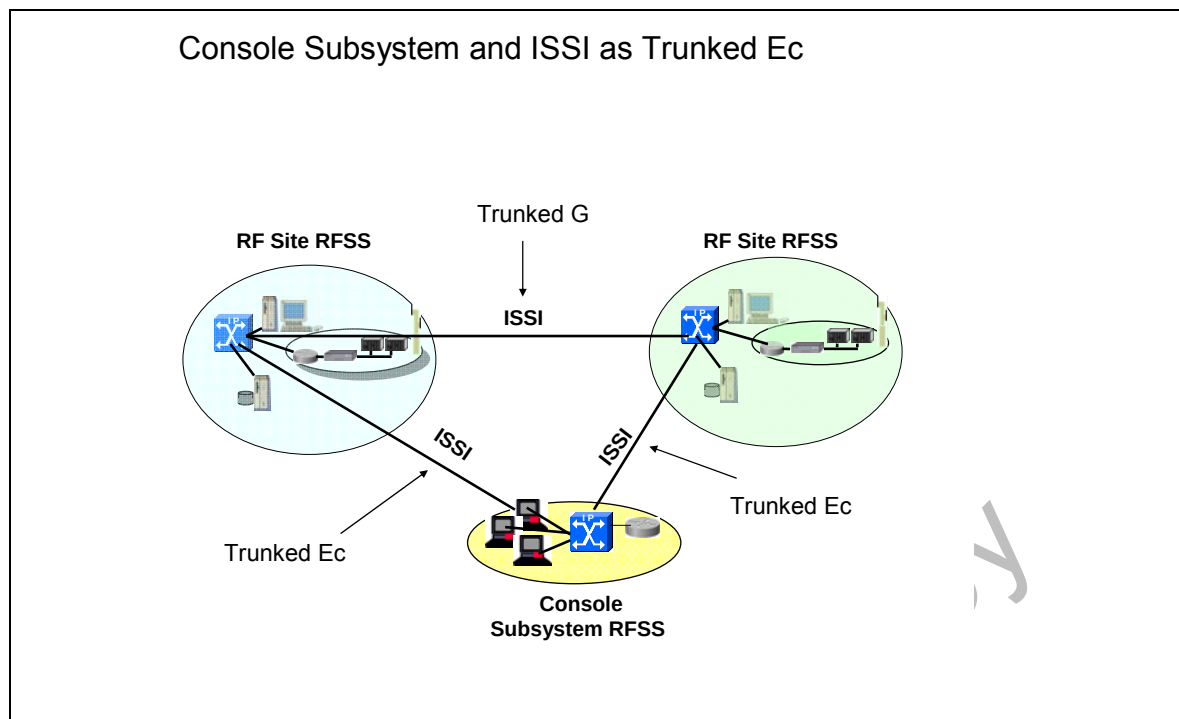


Figure 113 - Console Subsystem and ISSI Architecture

ISSI is extended to support the following console specific features:

I.1 Console Priority

A console talk spurt is given priority over a mobile talk spurt on a group. Both users are allowed to transmit at the same time on the group, but the mobile's audio will not be repeated. The console's audio is the audio that is repeated for the group. The mobile's audio is routed to the infrastructure as losing audio (losing audio is already supported by the ISSI). Other parallel consoles are able to listen to both sources of audio for the group, the console and the mobile.

I.2 Console Take Over by another Console

Consoles have multiple levels of transmit priority associated with their requests to transmit. The priority level chosen are based on the console position and the type of talk spurt the dispatcher is requesting. The console positions are allowed to be assigned priority levels such as “supervisor” and “non-supervisor”. There are many ways to assign priority levels to a console, two examples include basing them on the physical console position or basing them on the current user that is logged in to the console. Supervisor requests are in general higher priority than non-supervisor requests. The type of talk spurt varies from an emergency call talk spurt to a normal (non-emergency) talk spurt. Typically emergency call transmission requests are higher priority than non-emergency transmission requests. These console transmission priorities are not the same as the priority included in the over-the-air transmissions.

A console is able to “take over” the floor for a group from another console if the requesting console is requesting at a higher priority than the currently transmitting console. It’s similar to the console priority feature but between consoles. The losing console’s audio is routed to the infrastructure as losing audio.

I.3 Console Display of Transmission Source Type

For group voice and unit-to-unit calls, the interface supports information transfer intended to allow a console to indicate whether an active talk spurt is being sourced from a mobile or a console.

I.4 Console Mutes its Speakers to Prevent Audio Feedback

Consoles that are physically adjacent to each other possibly create feedback loops if a console’s transmit audio comes out of an adjacent console’s speaker and feeds back into the transmitting console’s microphone. It is possible to prevent the feedback if a console knows the source of each talk spurt and whether the source is a console physically adjacent to it. In this case the console will prevent the suspect audio from being played in its speakers.

I.5 Console Knowledge of When it is Receiving Duplicate Audio

Consoles are able to receive the same audio on multiple resources when the resources are part of patch group. If the console knows the source of each talk spurt it is listening to it is able to only play one copy of the audio to the user. Poor audio quality possibly results if the console attempts to play out multiple copies of the same talk spurt.

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